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# Abelian Varieties

— *EYNTKA* Series —

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## Part I

# Complex Abelian Varieties





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An abelian variety wears three faces at once: analytically a complex torus  $\mathbb{C}^g/\Lambda$ , cohomologically a polarized integral Hodge structure of weight one, and algebraically a projective group variety. This Part develops the first two faces in full, working entirely over the complex numbers. The torus description turns every question into linear algebra over a lattice refined by a complex structure: polarizations become positive Hermitian forms, line bundles become theta functions, and the dual variety and the endomorphism algebra become explicit constructions on the pair  $(V, \Lambda)$ .

The one fact the analytic viewpoint cannot supply from within is that a polarizable torus is genuinely projective; that is the work of the theta functions in Chapter 4. Everything else is a computation with  $(V, \Lambda)$ , and Part II rebuilds the same class of objects over an arbitrary field, where no lattice is available and the theorem of the cube takes its place.



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# Complex Tori

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An elliptic curve over  $\mathbb{C}$  is two things at once: a projective cubic carved out by a Weierstrass equation, and a quotient  $\mathbb{C}/\Lambda$  of the complex plane by a lattice. The second description is the transparent one. The group law is ordinary addition, the torsion points are the obvious fractions of periods, and the whole object is a compact complex Lie group of the simplest imaginable kind. This chapter takes that second description as the starting point for everything that follows and asks what survives in higher dimension: replace  $\mathbb{C}$  by  $\mathbb{C}^g$  and  $\Lambda$  by a lattice of rank  $2g$ , and one obtains a *complex torus*  $X = \mathbb{C}^g/\Lambda$ , a compact complex Lie group of dimension  $g$ .

Complex tori are easy to write down and easy to map between: a homomorphism is nothing more than a  $\mathbb{C}$ -linear map of the universal covers carrying one lattice into the other, so the entire category is governed by linear algebra over  $\mathbb{Z}$  and  $\mathbb{C}$ . What is not easy, and what occupies the rest of Part I, is the question of which complex tori are *algebraic*, that is, embed into projective space. For  $g = 1$  the answer is every one of them; for  $g \geq 2$  it is a delicate condition on the lattice, and most tori fail it. The present chapter develops the linear-algebraic anatomy of a complex torus, its endomorphisms, and the Hodge structure on its first cohomology, and closes by making the algebraicity question precise, so that chapters 2 and 3 can answer it.

## 1.1 Complex Tori and Their Morphisms

The starting object of the entire analytic theory is the simplest quotient of a complex vector space by a discrete subgroup: a complex torus. Its transparency is the whole point. A complex torus is built from a  $g$ -dimensional vector space and a lattice of rank  $2g$  inside it, and every question about the torus that the first chapters will ask reduces, sooner or later, to a question about that lattice together with the complex structure on the ambient space. This section fixes the object, establishes that it is a compact connected complex Lie group, and then settles the first structural question one can ask: what are the maps between complex tori? The answer, that every holomorphic homomorphism is

induced by a  $\mathbb{C}$ -linear map of the ambient vector spaces carrying one lattice into the other, is the linearization principle that makes the category of complex tori governable by linear algebra over  $\mathbb{Z}$ .

The reader who has met the  $g = 1$  case in [4] will recognize everything here as the higher-dimensional shadow of the picture  $E = \mathbb{C}/\Lambda$ : a compact Riemann surface of genus one, a complex Lie group, whose endomorphisms are exactly the complex numbers scaling the lattice into itself. The passage from  $\mathbb{C}$  to  $\mathbb{C}^g$  costs nothing conceptually and buys the whole subject.

A lattice in a real vector space is a discrete subgroup of maximal possible rank. Recording this precisely is the first order of business, because the rank condition, that the lattice has rank  $2g$  in a space of real dimension  $2g$ , is exactly what makes the quotient compact.

**Definition 1.1.1: Full Lattice**

Let  $V = \mathbb{C}^g$ , regarded as a real vector space of dimension  $2g$ . A *full lattice* in  $V$  is a discrete subgroup  $\Lambda \subseteq V$  of rank  $2g$  over  $\mathbb{Z}$ ; equivalently,  $\Lambda = \mathbb{Z}\lambda_1 \oplus \cdots \oplus \mathbb{Z}\lambda_{2g}$  for real-linearly independent vectors  $\lambda_1, \dots, \lambda_{2g} \in V$ , so that  $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = V$ . The vectors  $\lambda_1, \dots, \lambda_{2g}$  form a  $\mathbb{Z}$ -basis of  $\Lambda$ .

The two conditions in the definition, discreteness and rank  $2g$ , are the ones that matter. Discreteness alone permits a subgroup of smaller rank, such as  $\mathbb{Z} \subseteq \mathbb{C}$ ; the rank condition forces the lattice to span  $V$  over  $\mathbb{R}$ , so that the quotient  $V/\Lambda$  is compact rather than a cylinder or a line. Every full lattice in  $V$  is the image of the standard lattice  $\mathbb{Z}^{2g} \subseteq \mathbb{R}^{2g}$  under a real-linear isomorphism  $\mathbb{R}^{2g} \xrightarrow{\sim} V$ , and two  $\mathbb{Z}$ -bases of the same  $\Lambda$  differ by an element of  $\mathrm{GL}_{2g}(\mathbb{Z})$ , exactly as in the  $g = 1$  case where the change-of-basis group is  $\mathrm{GL}_2(\mathbb{Z})$ .

The quotient of  $V$  by a full lattice is the object of study.

**Definition 1.1.2: Complex Torus**

Let  $V = \mathbb{C}^g$  and let  $\Lambda \subseteq V$  be a full lattice. The *complex torus* of dimension  $g$  associated to  $(V, \Lambda)$  is the quotient group

$$X = V/\Lambda$$

equipped with the quotient topology and the quotient complex structure. Its points are the cosets  $v + \Lambda$ ; the projection  $\pi: V \rightarrow X$ ,  $\pi(v) = v + \Lambda$ , is the quotient map.

A complex torus is at once a group, a topological space, and a complex manifold, and the content of the definition is that these three structures are compatible. The group structure is the additive one inherited from  $V$ : the sum of  $v + \Lambda$  and  $w + \Lambda$  is  $(v + w) + \Lambda$ , well defined because  $\Lambda$  is a subgroup. The complex structure comes from  $V$  through the covering  $\pi$ , which is a local biholomorphism, so that a function on an open set of  $X$  is holomorphic precisely when its pullback along  $\pi$  is holomorphic on the corresponding open set of  $V$ . That these two structures cohere, that the group operations are holomorphic, is the assertion that  $X$  is a complex Lie group, which is the next proposition.

The name “torus” is topological. As a real manifold,  $X$  is diffeomorphic to the product of  $2g$  circles: choosing a  $\mathbb{Z}$ -basis  $\lambda_1, \dots, \lambda_{2g}$  of  $\Lambda$  identifies  $V$  with  $\mathbb{R}^{2g}$  carrying  $\Lambda$  to  $\mathbb{Z}^{2g}$ , and  $\mathbb{R}^{2g}/\mathbb{Z}^{2g} = (\mathbb{R}/\mathbb{Z})^{2g} \cong (S^1)^{2g}$ . The complex structure is the additional datum: the same real torus  $(S^1)^{2g}$  carries many inequivalent complex structures once  $g \geq 1$ , coming from the many ways a rank- $2g$  lattice can sit inside  $\mathbb{C}^g$ , and distinguishing them is one of the threads of this chapter.

**Proposition 1.1.3: A Complex Torus Is a Compact Connected Complex Lie Group**

Let  $X = V/\Lambda$  be a complex torus of dimension  $g$ . Then  $X$  is a compact, connected, abelian complex Lie group of complex dimension  $g$ : the multiplication  $m: X \times X \rightarrow X$ ,  $m(x, y) = x + y$ , and the inversion  $\iota: X \rightarrow X$ ,  $\iota(x) = -x$ , are holomorphic, and  $X$  is compact and connected.

**Proof:**

*Step 1: The complex structure and dimension.* The projection  $\pi: V \rightarrow X$  is a covering map of topological spaces, since  $\Lambda$  acts on  $V = \mathbb{C}^g$  by translations, properly discontinuously and freely (discreteness of  $\Lambda$  gives, for each  $v$ , a neighborhood  $U$  with  $(U + \lambda) \cap U = \emptyset$  for all nonzero  $\lambda \in \Lambda$ , and  $\pi|_U$  a homeomorphism onto its image). Declaring  $\pi|_U$  to be a biholomorphism transports the complex structure of  $V$  to  $X$ ; the transition maps between two such charts are translations  $v \mapsto v + \lambda$  of  $V$ , which are biholomorphic, so the charts are compatible and  $X$  is a complex manifold of complex dimension  $g$ . This is the standard construction of the quotient complex structure ([16]).

*Step 2: Compactness and connectedness.* Fix a  $\mathbb{Z}$ -basis  $\lambda_1, \dots, \lambda_{2g}$  of  $\Lambda$  and consider the closed fundamental parallelopete

$$P = \left\{ \sum_{j=1}^{2g} t_j \lambda_j \mid 0 \leq t_j \leq 1 \right\}$$

Since every  $v \in V$  is congruent modulo  $\Lambda$  to a point of  $P$ , the restriction  $\pi|_P: P \rightarrow X$  is surjective. As  $P$  is compact (a continuous image of the cube  $[0, 1]^{2g}$ ) and  $\pi$  is continuous,  $X = \pi(P)$  is compact. Connectedness is immediate:  $V$  is connected and  $\pi$  is a continuous surjection, so its image  $X$  is connected.

*Step 3: The group operations are holomorphic.* The addition  $\tilde{m}: V \times V \rightarrow V$ ,  $(v, w) \mapsto v + w$ , is holomorphic, and it descends to  $m: X \times X \rightarrow X$  because  $\tilde{m}$  carries  $\Lambda \times \Lambda$  into  $\Lambda$ . To see  $m$  is holomorphic near a point  $(x, y)$ , lift to points  $v, w \in V$  with  $\pi(v) = x$ ,  $\pi(w) = y$ ; on a suitable product of chart neighborhoods,  $m$  reads in local coordinates as  $\tilde{m}$  followed by a chart of  $X$ , hence is holomorphic. The same argument applied to  $\tilde{\iota}: V \rightarrow V$ ,  $v \mapsto -v$ , which carries  $\Lambda$  into  $\Lambda$ , shows  $\iota$  is holomorphic. Commutativity of  $m$  is inherited from the commutativity of addition in  $V$ . Thus  $X$  is an abelian complex Lie group, completing the proof.

The proposition is the reason complex tori are the natural first examples in the analytic theory of abelian varieties: they are group objects in the category of compact complex manifolds, produced by nothing more than a quotient. The group is written additively throughout, with identity element  $0 = 0 + \Lambda$  and inverse  $-x$ ; this matches the  $g = 1$  convention of [4], where the group law on  $E = \mathbb{C}/\Lambda$  is the descended addition of  $\mathbb{C}$ . The Lie algebra of  $X$  is  $V$  itself, with the trivial (abelian) bracket, and the exponential map  $\text{Lie}(X) = V \rightarrow X$  is simply the projection  $\pi$ ; this is the analytic incarnation of the fact that a complex torus is its own “linearization”, the observation that governs the study of endomorphisms in the next section.

The first concrete instance is the  $g = 1$  torus, which is an elliptic curve; it is the running example that every later construction specializes to.

**Example 1.1.4: The Elliptic Curve as a Complex Torus**

Take  $g = 1$ , so  $V = \mathbb{C}$ , and let  $\tau \in \mathbb{H}$  lie in the upper half-plane. The lattice

$$\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau = \{m + n\tau : m, n \in \mathbb{Z}\}$$

is a full lattice in  $\mathbb{C}$ : the generators 1 and  $\tau$  are  $\mathbb{R}$ -linearly independent precisely because  $\text{Im}\tau > 0$ . The quotient

$$E = \mathbb{C}/\Lambda$$

is a complex torus of dimension 1. By proposition 1.1.3 it is a compact connected abelian complex Lie group, and as a Riemann surface it has genus 1: the holomorphic 1-form  $dz$  on  $\mathbb{C}$  is  $\Lambda$ -invariant, so it descends to a nowhere-vanishing holomorphic 1-form on  $E$ , forcing the genus to be 1.

The content that goes beyond the group and manifold structure is that  $E$  is algebraic. The Weierstrass uniformization theorem provides an isomorphism of complex Lie groups

$$\mathbb{C}/\Lambda \xrightarrow{\sim} E(\mathbb{C}), \quad z + \Lambda \mapsto [\wp(z) : \wp'(z) : 1]$$

between the torus and the complex points of the projective plane cubic  $y^2 = 4x^3 - g_2x - g_3$ , where  $\wp$  is the Weierstrass function of  $\Lambda$  and  $g_2, g_3$  are the associated invariants. The construction of  $\wp$ , the proof that this map is a biholomorphism, and the verification that it is a group homomorphism are carried out in [16] and [4]; they are not reproduced here. What matters for the present chapter is that in dimension one the complex torus and the algebraic curve are the same object, and that this coincidence, automatic when  $g = 1$ , becomes a nontrivial condition, the Riemann relations, once  $g \geq 2$ .

Products of complex tori are again complex tori, and the smallest higher-dimensional example is the square of an elliptic curve. It will recur throughout the book as the concrete non-simple abelian surface against which general constructions are tested.

**Example 1.1.5: The Product  $E \times E$** 

Let  $E = \mathbb{C}/\Lambda$  be the elliptic curve of example 1.1.4, with  $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$ . Set

$$V = \mathbb{C}^2, \quad \Lambda' = \Lambda \times \Lambda = \{(m_1 + n_1\tau, m_2 + n_2\tau) : m_i, n_i \in \mathbb{Z}\}$$

Then  $\Lambda'$  is a full lattice in  $\mathbb{C}^2$ : a  $\mathbb{Z}$ -basis is

$$(1, 0), \quad (\tau, 0), \quad (0, 1), \quad (0, \tau)$$

four vectors, real-linearly independent because  $\{1, \tau\}$  is  $\mathbb{R}$ -linearly independent in  $\mathbb{C}$ . The associated complex torus is

$$X = V/\Lambda' = (\mathbb{C}/\Lambda) \times (\mathbb{C}/\Lambda) = E \times E$$

a complex torus of dimension 2. Its period matrix, the  $2 \times 4$  matrix whose columns are the chosen  $\mathbb{Z}$ -basis expressed in the standard  $\mathbb{C}$ -basis of  $\mathbb{C}^2$ , is

$$\Pi = \begin{pmatrix} 1 & \tau & 0 & 0 \\ 0 & 0 & 1 & \tau \end{pmatrix}$$

As a group,  $X$  is  $E \times E$  with coordinatewise addition, and as a complex manifold it is the product of  $E$  with itself. This torus is the prototype of a non-simple abelian surface: it is isogenous to a product, it carries the diagonal and antidiagonal copies of  $E$  as subtori, and its endomorphism ring is the ring  $M_2(\text{End}(E))$  of  $2 \times 2$  matrices over  $\text{End}(E)$ , a point developed in section 1.2.

With the object in hand, the natural question is what its morphisms are. A morphism of complex tori should respect both structures they carry: the complex manifold structure (holomorphy) and the group structure (homomorphism). The following definition names the maps that respect both.

**Definition 1.1.6: Homomorphism of Complex Tori**

Let  $X = V/\Lambda$  and  $X' = V'/\Lambda'$  be complex tori. A *homomorphism*  $f: X \rightarrow X'$  is a map that is simultaneously holomorphic and a homomorphism of abelian groups, so that  $f(x + y) = f(x) + f(y)$  for all  $x, y \in X$ . An *isomorphism* is a homomorphism admitting a holomorphic inverse homomorphism. When  $X = X'$ , a homomorphism  $X \rightarrow X$  is an *endomorphism*, and the endomorphisms form a ring  $\text{End}(X)$  under pointwise addition and composition.

A homomorphism automatically preserves the identity,  $f(0) = 0$ . This is the constraint that distinguishes a homomorphism from an arbitrary holomorphic map of the underlying manifolds, and the next result shows exactly how much it constrains: a holomorphic homomorphism is a  $\mathbb{C}$ -linear map of the universal covers that carries one lattice into the other, nothing more. The endomorphism ring  $\text{End}(X)$  is the central invariant of an abelian variety; the linearization theorem below is what makes it computable, by identifying  $\text{End}(X)$  with a ring of matrices preserving a lattice.

The mechanism behind the theorem is the interplay of two facts. First,  $V$  is the universal cover of  $X$  (it is simply connected, being a vector space, and  $\pi: V \rightarrow X$  is a covering), so a based map out of  $X$  lifts uniquely to a map out of  $V$ . Second, holomorphy plus the homomorphism property force the lift to be not just holomorphic but  $\mathbb{C}$ -linear, because its complex derivative is a  $\Lambda$ -periodic holomorphic function on the compact torus, hence constant. The clean statement separates the general holomorphic map (a translate of a homomorphism) from the homomorphism itself.

**Proposition 1.1.7: Holomorphic Maps of Tori Are Linear**

Let  $X = V/\Lambda$  and  $X' = V'/\Lambda'$  be complex tori, with projections  $\pi: V \rightarrow X$  and  $\pi': V' \rightarrow X'$ .

1. Every holomorphic homomorphism  $f: X \rightarrow X'$  lifts to a unique  $\mathbb{C}$ -linear map  $F: V \rightarrow V'$  satisfying  $F(\Lambda) \subseteq \Lambda'$  and  $\pi' \circ F = f \circ \pi$ . Conversely, every  $\mathbb{C}$ -linear  $F: V \rightarrow V'$  with  $F(\Lambda) \subseteq \Lambda'$  descends to a holomorphic homomorphism  $f: X \rightarrow X'$ . These two operations are mutually inverse, so

$$\text{Hom}(X, X') \cong \{F \in \text{Hom}_{\mathbb{C}}(V, V') : F(\Lambda) \subseteq \Lambda'\}$$

an isomorphism of abelian groups; for  $X = X'$  it is an isomorphism of rings  $\text{End}(X) \cong \{F \in \text{End}_{\mathbb{C}}(V) : F(\Lambda) \subseteq \Lambda\}$ .

2. Every holomorphic map  $h: X \rightarrow X'$  (not assumed a homomorphism) is a translate of a homomorphism: there is a unique homomorphism  $f: X \rightarrow X'$  and a unique point  $c = h(0) \in X'$  with  $h = t_c \circ f$ , where  $t_c$  is translation by  $c$ . In particular  $h$  is a homomorphism if and only if  $h(0) = 0$ .

**Proof:**

The two parts are proved in turn; part (2) is deduced from part (1) at the end.

1. *Existence and uniqueness of the lift.* Since  $V'$  is simply connected and  $\pi': V' \rightarrow X'$  is a covering map, the composite  $f \circ \pi: V \rightarrow X'$  lifts through  $\pi'$ : because  $V$  is simply connected, the lifting criterion for covering spaces ([16]) produces a continuous  $F: V \rightarrow V'$  with  $\pi' \circ F = f \circ \pi$ , and this lift is unique once we normalize  $F(0) = 0$  (using  $f(0) = 0$ , which holds as  $f$  is a homomorphism). The map  $F$  is holomorphic: locally  $\pi$  and  $\pi'$  are biholomorphisms and  $f$  is holomorphic, so  $F = (\pi')^{-1} \circ f \circ \pi$  is holomorphic on each chart.

Next,  $F$  is a homomorphism of the additive group  $V$ . For fixed  $w \in V$ , consider

$$g_w(v) = F(v + w) - F(v) - F(w)$$

Applying  $\pi'$  and using that  $f$  is a homomorphism together with  $\pi' \circ F = f \circ \pi$ , one computes  $\pi'(g_w(v)) = f(\pi(v + w)) - f(\pi(v)) - f(\pi(w)) = 0$  in  $X'$ , so  $g_w(v) \in \Lambda'$  for all  $v$ . Thus  $g_w: V \rightarrow \Lambda'$  is a continuous map from the connected space  $V$  into the discrete set  $\Lambda'$ , hence constant; its value at  $v = 0$  is  $g_w(0) = F(w) - F(0) - F(w) = 0$ . Therefore  $g_w \equiv 0$ , that is,  $F(v + w) = F(v) + F(w)$  for all  $v, w$ , so  $F$  is additive.

*The lift is  $\mathbb{C}$ -linear.*  $F$  is holomorphic and additive; additivity gives  $\mathbb{R}$ -linearity (a continuous additive map of real vector spaces is  $\mathbb{R}$ -linear). It remains to promote  $\mathbb{R}$ -linearity to  $\mathbb{C}$ -linearity, and this is where holomorphy enters. Because  $F$  is  $\mathbb{R}$ -linear its differential  $dF$  is constant, equal to  $F$  itself under the identification of a linear map with its derivative. Concretely, each component of  $F$  is a holomorphic function on  $V = \mathbb{C}^g$  whose first-order Taylor expansion is exact, so the Jacobian matrix  $\partial F_i / \partial z_j$  is a constant matrix and the antiholomorphic derivatives  $\partial F_i / \partial \bar{z}_j$  vanish identically by the Cauchy-Riemann equations. A real-linear map whose antiholomorphic part vanishes is  $\mathbb{C}$ -linear. Hence  $F \in \text{Hom}_{\mathbb{C}}(V, V')$ , and  $F(\Lambda) \subseteq \Lambda'$  because for  $\lambda \in \Lambda$  we have  $\pi'(F(\lambda)) = f(\pi(\lambda)) = f(0) = 0$ , so  $F(\lambda) \in \Lambda'$ .

*The converse and the bijection.* Conversely, let  $F \in \text{Hom}_{\mathbb{C}}(V, V')$  satisfy  $F(\Lambda) \subseteq \Lambda'$ . Then  $F$  carries cosets to cosets: if  $v - v' \in \Lambda$  then  $F(v) - F(v') = F(v - v') \in \Lambda'$ , so the rule  $f(v + \Lambda) = F(v) + \Lambda'$  is well defined. It is a group homomorphism because  $F$  is additive, and holomorphic because  $F$  is holomorphic and  $\pi, \pi'$  are local biholomorphisms; it satisfies  $\pi' \circ F = f \circ \pi$  by construction. The two assignments  $f \mapsto F$  (lift) and  $F \mapsto f$  (descent) are mutually inverse: the lift of the descent of  $F$  is  $F$  by uniqueness of lifts, and the descent of the lift of  $f$  is  $f$  since both agree after composing with the surjection  $\pi$ . Both operations are additive in their argument, so the bijection  $\text{Hom}(X, X') \cong \{F : F(\Lambda) \subseteq \Lambda'\}$  is an isomorphism of abelian groups. When  $X = X'$ , composition of homomorphisms corresponds to composition of the linear lifts, and the identity endomorphism corresponds to  $\text{id}_V$ , so the bijection is a ring isomorphism.

2. *A general holomorphic map is a translate of a homomorphism.* Let  $h: X \rightarrow X'$  be holomorphic and set  $c = h(0) \in X'$ . Let  $t_{-c}: X' \rightarrow X'$  be translation by  $-c$ , a biholomorphism of  $X'$  (its lift is the translation  $v' \mapsto v' - \gamma$  of  $V'$  for any  $\gamma \in V'$  with  $\pi'(\gamma) = c$ ). Then  $f := t_{-c} \circ h: X \rightarrow X'$  is holomorphic and satisfies  $f(0) = h(0) - c = 0$ . It remains to see that a holomorphic map  $f$  of tori with  $f(0) = 0$  is a homomorphism; granting



this,  $h = t_c \circ f$  is the asserted decomposition, and  $c$  is forced by evaluating at 0, giving uniqueness of both  $c$  and  $f = t_{-c} \circ h$ .

To prove  $f$  is a homomorphism, lift  $f$  to a holomorphic map  $F: V \rightarrow V'$  with  $F(0) = 0$  and  $\pi' \circ F = f \circ \pi$ , exactly as in the first paragraph of part (1); this used only that  $V, V'$  are simply connected covers and  $f(0) = 0$ , not that  $f$  is a homomorphism. Fix  $\lambda \in \Lambda$  and consider  $\delta_\lambda(v) = F(v + \lambda) - F(v)$ . Applying  $\pi'$  gives  $\pi'(\delta_\lambda(v)) = f(\pi(v + \lambda)) - f(\pi(v)) = 0$ , since  $\pi(v + \lambda) = \pi(v)$ ; thus  $\delta_\lambda(v) \in \Lambda'$  for all  $v$ . As  $\delta_\lambda$  is continuous on the connected space  $V$  with values in the discrete set  $\Lambda'$ , it is constant. Differentiating the identity  $F(v + \lambda) = F(v) + \delta_\lambda$  shows the Jacobian satisfies  $dF(v + \lambda) = dF(v)$  for all  $\lambda \in \Lambda$ , so each entry of  $dF$  is a  $\Lambda$ -periodic holomorphic function on  $V$ . Such a function descends to a holomorphic function on the compact connected torus  $X$ , hence is constant by the maximum principle ([16]). A holomorphic map  $F$  with constant Jacobian and  $F(0) = 0$  is  $\mathbb{C}$ -linear, given by  $v \mapsto dF \cdot v$ . In particular  $F$  is additive, so  $f$  is a homomorphism, completing the proof.

The proposition is the organizing principle of the whole analytic theory. It says that the seemingly transcendental category of complex tori and holomorphic homomorphisms is equivalent to a purely linear-algebraic category: objects are pairs  $(V, \Lambda)$  of a complex vector space with a full lattice, and morphisms are  $\mathbb{C}$ -linear maps carrying lattice into lattice. Two structures are being remembered at once, the complex-linear structure on  $V$  and the integral structure  $\Lambda$ , and a homomorphism must respect both. This is why the subject sits at the meeting point of complex analysis and the arithmetic of lattices: the complex structure is analytic, the lattice is arithmetic, and a morphism is exactly a map compatible with the two.

The mechanism of the proof deserves a second look, because it recurs. The key step, that the lift  $F$  is  $\mathbb{C}$ -linear, splits into two independent facts. That  $F$  is additive is topology: a continuous map into a discrete group from a connected space is constant, applied to the “defect”  $F(v + w) - F(v) - F(w)$ . That  $F$  is then  $\mathbb{C}$ -linear, not just  $\mathbb{R}$ -linear, is complex analysis: the antiholomorphic derivative of a holomorphic map vanishes. Part (2) isolates the same periodic-derivative argument in its cleanest form, where the derivative of the lift is a holomorphic function on the compact torus and so must be constant, since the only holomorphic functions on a compact connected complex manifold are the constants. This last fact, that  $X$  carries no nonconstant global holomorphic functions, is a recurring obstruction: it is the reason abelian varieties are studied through their line bundles and meromorphic functions rather than through holomorphic functions, a theme taken up in chapter 2.

For the elliptic curve of example 1.1.4, the proposition recovers a computation the reader has already seen in [4]: the endomorphism ring of  $E = \mathbb{C}/\Lambda$  is

$$\text{End}(E) \cong \{\alpha \in \mathbb{C} : \alpha\Lambda \subseteq \Lambda\}$$

the set of complex numbers scaling the lattice into itself, since a  $\mathbb{C}$ -linear endomorphism of  $V = \mathbb{C}$  is multiplication by a scalar  $\alpha \in \mathbb{C}$  and the lattice condition reads  $\alpha\Lambda \subseteq \Lambda$ . For a generic  $\tau$  this set is just  $\mathbb{Z}$ , acting by  $\alpha = n$ , and  $\text{End}(E) = \mathbb{Z}$ ; for special  $\tau$  satisfying a quadratic equation over  $\mathbb{Q}$  it is an order in an imaginary quadratic field, the case of complex multiplication. The higher-dimensional generalization of this scalar description, the analytic and rational representations of an endomorphism, is the subject of section 1.2.

One consequence of proposition 1.1.7 organizes the philosophy of the chapter. The category of complex tori is governed by linear algebra over  $\mathbb{Z}$ , refined by a complex structure. An isomorphism  $X \cong X'$  is a  $\mathbb{C}$ -linear isomorphism  $V \xrightarrow{\sim} V'$  carrying  $\Lambda$  onto  $\Lambda'$ ; fixing  $V = V' = \mathbb{C}^g$ , this means the

two lattices are related by an element of  $\mathrm{GL}_g(\mathbb{C})$ , and the moduli of complex tori of dimension  $g$  is the space of full lattices in  $\mathbb{C}^g$  modulo this action. All of the finer structure the book develops, polarizations, the Néron-Severi group, theta functions, isogenies, sits on top of this linear-algebraic skeleton, adding a positivity condition (the Riemann relations) that most complex tori fail. The reader should hold this contrast in mind against the algebraic picture that Part II builds: there a complex torus is replaced by a projective group scheme over an arbitrary field, with no lattice and no complex structure available, and the linear-algebraic control of the present section is replaced by the geometric rigidity of the theorem of the cube. Over  $\mathbb{C}$  the two descriptions agree, and moving between them is where the subject's leverage lies; but the agreement is a theorem, not a definition, and it holds only for the tori that satisfy the algebraicity condition of chapter 3.

### Exercise 1.1.1

1. Let  $V = \mathbb{C}$  and let  $\tau \in \mathbb{C}$ . Show that  $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$  is a full lattice in  $\mathbb{C}$  (definition 1.1.1) if and only if  $\tau \notin \mathbb{R}$ . When  $\tau \notin \mathbb{R}$ , exhibit an explicit real-linear isomorphism  $\mathbb{R}^2 \xrightarrow{\sim} \mathbb{C}$  carrying  $\mathbb{Z}^2$  onto  $\Lambda$ .
2. Consider the subgroup  $\Lambda = \mathbb{Z} \oplus \mathbb{Z}i \oplus \mathbb{Z}(1+i)/2 \subseteq \mathbb{C}$ , that is,  $\Lambda = \{m + ni + k(1+i)/2 : m, n, k \in \mathbb{Z}\}$ . Determine whether  $\Lambda$  is discrete, and whether it has rank 2 over  $\mathbb{Z}$ . Is  $\Lambda$  a full lattice in  $\mathbb{C}$ ? Justify each answer.
3. Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ , be the elliptic curve of example 1.1.4. Using proposition 1.1.7, determine  $\mathrm{End}(E)$  explicitly in the case  $\tau = i$ : find every  $\alpha \in \mathbb{C}$  with  $\alpha\Lambda \subseteq \Lambda$ , and identify the resulting ring.
4. Let  $X = V/\Lambda$  be a complex torus of dimension  $g$  and let  $n$  be a positive integer. Consider the multiplication-by- $n$  map  $[n]: X \rightarrow X$ ,  $x \mapsto nx$ .
  1. Show that  $[n]$  is a homomorphism of complex tori and identify its  $\mathbb{C}$ -linear lift  $F: V \rightarrow V$  under proposition 1.1.7.
  2. Show that  $[n]$  is surjective with kernel  $\ker[n] = \frac{1}{n}\Lambda/\Lambda$ , and deduce that  $\ker[n] \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$  as an abelian group.
5. Prove that on a complex torus  $X = V/\Lambda$  every global holomorphic function is constant. Then use this fact to give a direct proof of the following special case of proposition 1.1.7: if  $h: X \rightarrow X'$  is a holomorphic map of complex tori with  $h(0) = 0$ , and its lift  $F: V \rightarrow V'$  (normalized by  $F(0) = 0$ ) is known to exist, then  $F$  is  $\mathbb{C}$ -linear.

## 1.2 The Analytic and Rational Representations

An endomorphism of a complex torus  $X = V/\Lambda$  is, by proposition 1.1.7, nothing more exotic than a  $\mathbb{C}$ -linear map of the covering space  $V$  that carries the lattice  $\Lambda$  into itself. This double life is the source of everything that makes the endomorphism theory of complex tori tractable: a single endomorphism can be read either on  $V$ , where it is a  $\mathbb{C}$ -linear operator on a  $g$ -dimensional complex vector space, or on  $\Lambda$ , where it is a  $\mathbb{Z}$ -linear operator on a free abelian group of rank  $2g$ . These two readings are the analytic and the rational representation. Neither sees the whole endomorphism on its own, but together they constrain it tightly, and the comparison between them is the mechanism that forces the endomorphism ring of a complex torus to be small and arithmetically rigid. This

section sets up the two representations, proves the comparison relating them, and extracts from it the characteristic-polynomial identity that will govern degrees of isogenies and, in Part II, the whole arithmetic of endomorphisms.

Throughout,  $X = V/\Lambda$  is a complex torus of dimension  $g$ , with  $V = \mathbb{C}^g$  and  $\Lambda \subset V$  a full lattice (definition 1.1.2).

The endomorphisms of  $X$  form a ring under composition and pointwise addition, and it is this ring, rather than any single endomorphism, that carries the arithmetic content. Composition is composition of maps; the sum of two endomorphisms  $f, h$  is the map  $x \mapsto f(x) + h(x)$ , which is again a homomorphism because  $X$  is abelian. The zero endomorphism is the constant map to the origin, and the identity is the identity map. The ring and its rationalization are recorded below.

**Definition 1.2.1: Endomorphism Ring and Endomorphism Algebra**

Let  $X = V/\Lambda$  be a complex torus. The *endomorphism ring* of  $X$  is

$$\text{End}(X) = \{ f: X \rightarrow X \mid f \text{ holomorphic, } f(0) = 0 \}$$

with addition induced pointwise from the group law of  $X$  and multiplication given by composition. The *endomorphism algebra* of  $X$  is the  $\mathbb{Q}$ -algebra

$$\text{End}^0(X) = \text{End}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$$

Requiring  $f(0) = 0$  is what makes  $\text{End}(X)$  a ring: by proposition 1.1.7 the holomorphic maps  $X \rightarrow X$  fixing the origin are exactly the group homomorphisms, and a general holomorphic map is such a homomorphism followed by a translation. Discarding the translations leaves a set closed under both composition and addition. Passing to  $\text{End}^0(X)$  discards the finer integral structure and keeps only the  $\mathbb{Q}$ -algebra, which is the object Albert's classification will eventually pin down in chapter 6; the integral ring  $\text{End}(X)$  sits inside it as an order. The rank of  $\text{End}(X)$  as an abelian group, equivalently  $\dim_{\mathbb{Q}} \text{End}^0(X)$ , will turn out to be at most  $2g^2$ , and the two representations below are what bound it.

The first representation lifts an endomorphism to the universal cover. By proposition 1.1.7 every  $f \in \text{End}(X)$  is induced by a unique  $\mathbb{C}$ -linear map  $F: V \rightarrow V$  satisfying  $F(\Lambda) \subseteq \Lambda$ ; conversely such an  $F$  descends to an endomorphism of  $X$ . Reading  $f$  through  $F$  is the analytic representation.

**Definition 1.2.2: Analytic Representation**

The *analytic representation* of  $\text{End}(X)$  is the ring homomorphism

$$\rho_a: \text{End}(X) \longrightarrow \text{End}_{\mathbb{C}}(V)$$

sending  $f \in \text{End}(X)$  to the unique  $\mathbb{C}$ -linear map  $\rho_a(f) = F: V \rightarrow V$  with  $F(\Lambda) \subseteq \Lambda$  inducing  $f$ . Extending scalars,  $\rho_a$  denotes also its  $\mathbb{Q}$ -linear extension  $\text{End}^0(X) \rightarrow \text{End}_{\mathbb{C}}(V)$  and its  $\mathbb{R}$ -linear extension to  $\text{End}(X) \otimes_{\mathbb{Z}} \mathbb{R}$ .

The analytic representation records how an endomorphism moves the tangent directions at the origin:  $\rho_a(f)$  is exactly the derivative of  $f$  at 0, viewed as a  $\mathbb{C}$ -linear operator on the tangent space  $V = T_0X$ . It is injective, since an endomorphism is determined by its lift, so  $\text{End}(X)$  embeds into the  $g^2$ -dimensional complex algebra  $\text{End}_{\mathbb{C}}(V) \cong M_g(\mathbb{C})$ . What this embedding does *not* do is bound the

size of  $\text{End}^0(X)$ : an injection of  $\text{End}(X)$  into a complex vector space forces neither finite generation of  $\text{End}(X)$  as an abelian group nor finite-dimensionality of  $\text{End}^0(X)$  as a  $\mathbb{Q}$ -algebra, because a  $\mathbb{Q}$ -subspace of a finite-dimensional real vector space can be infinite-dimensional over  $\mathbb{Q}$  (already  $\mathbb{Q}(\pi) \subset \mathbb{C} = M_1(\mathbb{C})$  in the case  $g = 1$ ), and a subgroup of a vector space need not be discrete. Both bounds instead come from the rational representation defined below, which lands in  $M_{2g}(\mathbb{Z})$ : its additive group  $\mathbb{Z}^{4g^2}$  is finitely generated, so  $\text{End}(X)$  has finite rank, whence  $\text{End}^0(X) = \text{End}(X) \otimes \mathbb{Q}$  is finite-dimensional over  $\mathbb{Q}$ . What the analytic representation cannot see is exactly this lattice: it forgets that  $F$  preserves  $\Lambda$ , and so it forgets the integrality that will do the real work. Choosing a  $\mathbb{C}$ -basis of  $V$  identifies  $\rho_a(f)$  with a matrix in  $M_g(\mathbb{C})$ , whose entries are the complex numbers by which  $f$  scales the coordinates  $z_1, \dots, z_g$ .

The second representation forgets the complex structure entirely and remembers only the action on the lattice. Since  $f$  is a group homomorphism carrying  $\Lambda$  to itself, restricting the lift  $F$  to  $\Lambda$  gives a  $\mathbb{Z}$ -linear endomorphism of a free abelian group of rank  $2g$ .

### Definition 1.2.3: Rational Representation

The *rational representation* of  $\text{End}(X)$  is the ring homomorphism

$$\rho_r: \text{End}(X) \longrightarrow \text{End}_{\mathbb{Z}}(\Lambda) \cong M_{2g}(\mathbb{Z})$$

sending  $f \in \text{End}(X)$  to the restriction  $\rho_r(f) = F|_{\Lambda}: \Lambda \rightarrow \Lambda$  of its analytic lift. Extending scalars gives the  $\mathbb{Q}$ -linear extension  $\rho_r: \text{End}^0(X) \rightarrow \text{End}_{\mathbb{Q}}(\Lambda \otimes \mathbb{Q})$  and the  $\mathbb{R}$ -linear extension  $\rho_r \otimes \mathbb{R}: \text{End}(X) \otimes_{\mathbb{Z}} \mathbb{R} \rightarrow \text{End}_{\mathbb{R}}(\Lambda \otimes \mathbb{R})$ , where  $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = V$  as real vector spaces.

The name “rational” reflects the object it naturally lives on: after tensoring with  $\mathbb{Q}$ , the representation acts on the  $2g$ -dimensional  $\mathbb{Q}$ -vector space  $\Lambda \otimes \mathbb{Q}$ , and  $\rho_r$  itself is the integral representation of  $\text{End}(X)$  on the rank- $2g$  lattice  $\Lambda$  that this action packages. Fixing a  $\mathbb{Z}$ -basis  $\lambda_1, \dots, \lambda_{2g}$  of  $\Lambda$  presents  $\rho_r(f)$  as an integer matrix in  $M_{2g}(\mathbb{Z})$ . This is the source of every integrality constraint on  $\text{End}(X)$ : whatever  $f$  does, it does it as an integer matrix of size  $2g$ , so its trace and determinant are integers and its characteristic polynomial has integer coefficients. The analytic representation, by contrast, produces a complex matrix of half the size and carries no such constraint on its own. The comparison between the two is what transfers the integrality from  $\rho_r$  to the eigenvalues seen by  $\rho_a$ .

Before comparing them, it helps to ground both representations on the endomorphism every complex torus possesses.

### Example 1.2.4: Multiplication by $n$

For  $n \in \mathbb{Z}$ , the multiplication-by- $n$  map  $[n]: X \rightarrow X$ ,  $x \mapsto nx$ , is an endomorphism: it fixes the origin and is a homomorphism because  $X$  is abelian. Its analytic lift is scalar multiplication by  $n$  on  $V$ , so

$$\rho_a([n]) = n \cdot \text{id}_V = n I_g \in \text{End}_{\mathbb{C}}(V), \quad \rho_r([n]) = n \cdot \text{id}_{\Lambda} = n I_{2g} \in \text{End}_{\mathbb{Z}}(\Lambda)$$

The analytic representation is a  $g \times g$  scalar matrix and the rational representation a  $2g \times 2g$  scalar matrix, both with the single eigenvalue  $n$ . Consequently the characteristic polynomial of  $\rho_a([n])$  is  $(t - n)^g$  and that of  $\rho_r([n])$  is

$$\det(t I_{2g} - n I_{2g}) = (t - n)^{2g}$$

The kernel of  $[n]$  on  $X$  is the  $n$ -torsion  $X[n] = \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$ , a group of order  $n^{2g}$ . This integer  $n^{2g}$  is the *analytic degree* of  $[n]$ , and it is exactly  $|\det \rho_r([n])| = |n^{2g}|$ , the absolute value of

the determinant of the rational representation. In Part II this coincidence becomes a theorem: for any isogeny the degree equals the determinant of the rational representation, and the case  $f = [n]$  yields the degree formula  $\deg[n] = n^{2g}$  obtained there by the theorem of the cube.

The computation for  $[n]$  exhibits in miniature the pattern the comparison proposition will establish in general: the characteristic polynomial of the rational representation is a product of a degree- $g$  polynomial and its complex conjugate, here a literal square  $((t - n)^g)^2$  because  $n$  is real, and its roots come in the pairs one expects when a complex  $g$ -dimensional picture is doubled into a real  $2g$ -dimensional one. For  $[n]$  the two “halves” coincide because  $n$  is real; the content of the next result is that in general the second half is the complex conjugate of the first.

The two representations are related through a single linear-algebra fact: the real vector space underlying  $\Lambda \otimes \mathbb{R}$  is  $V$  itself, and  $V$  carries a complex structure. An  $\mathbb{R}$ -linear operator on  $V$  that happens to be  $\mathbb{C}$ -linear is determined, after complexifying, by how it acts on the holomorphic and antiholomorphic directions, and these two actions are conjugate. This is what ties  $\rho_r$  to  $\rho_a$  and its conjugate.

#### Proposition 1.2.5: Comparison of the Analytic and Rational Representations

Let  $X = V/\Lambda$  be a complex torus of dimension  $g$ . Complexifying the rational representation gives an isomorphism

$$\rho_r \otimes_{\mathbb{Z}} \mathbb{C} \cong \rho_a \oplus \bar{\rho}_a$$

of complex representations of  $\text{End}(X)$  on the  $2g$ -dimensional complex vector space  $V \oplus \bar{V}$ , where  $\bar{\rho}_a$  denotes the complex-conjugate representation  $f \mapsto \overline{\rho_a(f)}$ . Before complexifying, the underlying real statement is that  $\rho_r \otimes_{\mathbb{Z}} \mathbb{R}$  is the realification  $(\rho_a)_{\mathbb{R}}$  of the analytic representation, that is, the representation  $f \mapsto \rho_a(f)_{\mathbb{R}}$  on the  $2g$ -dimensional real vector space  $\Lambda \otimes \mathbb{R} = V$ ; the splitting into  $\rho_a$  and its conjugate emerges only after tensoring this real representation with  $\mathbb{C}$ .

#### Proof:

*Step 1: Identify the real space the rational representation acts on.* Since  $\Lambda$  is a full lattice,  $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = V$  as real vector spaces (definition 1.1.2); this is the defining property of a full lattice, that its real span is all of  $V$ . For  $f \in \text{End}(X)$  the operator  $\rho_r(f)$  is by definition the restriction of the  $\mathbb{C}$ -linear lift  $F = \rho_a(f)$  to  $\Lambda$ , so under  $\Lambda \otimes \mathbb{R} = V$  the  $\mathbb{R}$ -linear extension of  $\rho_r(f)$  is precisely  $F$  regarded as an  $\mathbb{R}$ -linear operator on the  $2g$ -dimensional real vector space  $V$ . Thus  $\rho_r \otimes_{\mathbb{Z}} \mathbb{R}$  is the representation  $f \mapsto \rho_a(f)_{\mathbb{R}}$ , where  $(\cdot)_{\mathbb{R}}$  denotes an operator viewed as  $\mathbb{R}$ -linear by forgetting the complex structure. It therefore suffices to compare, for a single  $\mathbb{C}$ -linear operator  $F$  on  $V$ , its realification  $F_{\mathbb{R}}$  with the pair  $(F, \bar{F})$ .

*Step 2: Complexify and split into holomorphic and antiholomorphic parts.* Complexify the real vector space  $V$  to  $V \otimes_{\mathbb{R}} \mathbb{C}$ . The complex structure  $J$  on  $V$ , that is, multiplication by  $i$ , is an  $\mathbb{R}$ -linear operator with  $J^2 = -\text{id}$ , so on  $V \otimes_{\mathbb{R}} \mathbb{C}$  it has the two eigenvalues  $\pm i$ , and

$$V \otimes_{\mathbb{R}} \mathbb{C} = V^{1,0} \oplus V^{0,1}$$

is the decomposition into the  $(+i)$ -eigenspace  $V^{1,0}$  and the  $(-i)$ -eigenspace  $V^{0,1}$  of  $J$ . Each summand has complex dimension  $g$ . The two natural  $\mathbb{C}$ -linear maps

$$V^{1,0} \xrightarrow{\sim} V, \quad V^{0,1} \xrightarrow{\sim} \bar{V}$$

are isomorphisms:  $V^{1,0}$  is  $\mathbb{C}$ -linearly isomorphic to  $(V, J)$  via  $v \otimes 1 - Jv \otimes i \mapsto v$ , and  $V^{0,1}$  is  $\mathbb{C}$ -linearly isomorphic to the conjugate space  $\bar{V} = (V, -J)$  via  $v \otimes 1 + Jv \otimes i \mapsto v$ . These are the standard holomorphic and antiholomorphic identifications for the underlying real space of a complex vector space.

*Step 3: A  $\mathbb{C}$ -linear operator acts on the two parts as  $F$  and  $\bar{F}$ .* Let  $F: V \rightarrow V$  be  $\mathbb{C}$ -linear, and let  $F_{\mathbb{R}}$  be  $F$  regarded as  $\mathbb{R}$ -linear on  $V$ . Its complexification  $F_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$  is  $\mathbb{C}$ -linear on  $V \otimes_{\mathbb{R}} \mathbb{C}$  and commutes with  $J$ , because  $F$  commutes with  $J$  precisely because  $F$  is  $\mathbb{C}$ -linear ( $F(Jv) = F(iv) = iF(v) = JF(v)$ ). An operator commuting with  $J$  preserves each eigenspace of  $J$ , so  $F_{\mathbb{R}} \otimes \mathbb{C}$  preserves both  $V^{1,0}$  and  $V^{0,1}$  and splits as

$$F_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} = (F_{\mathbb{R}} \otimes \mathbb{C})|_{V^{1,0}} \oplus (F_{\mathbb{R}} \otimes \mathbb{C})|_{V^{0,1}}$$

Under the isomorphism  $V^{1,0} \cong V$  of Step 2 the restriction to  $V^{1,0}$  is  $F$  itself (this is the holomorphic direction, on which the complex structure agrees with that of  $V$ ). Under  $V^{0,1} \cong \bar{V}$  the restriction to  $V^{0,1}$  is  $F$  acting on the conjugate space, which is by definition  $\bar{F}$ : if  $F$  has matrix  $A$  in a  $\mathbb{C}$ -basis of  $V$ , then  $F$  acting on  $\bar{V}$  has matrix  $\bar{A}$ . Therefore

$$F_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \cong F \oplus \bar{F}$$

as  $\mathbb{C}$ -linear operators on  $V \oplus \bar{V}$ .

*Step 4: Assemble the representations.* Steps 1 and 3 hold naturally in  $f$ : the identifications in Step 2 are chosen once for the vector space  $V$  and do not depend on  $f$ , so applying Step 3 to  $F = \rho_a(f)$  for every  $f \in \text{End}(X)$  simultaneously gives an isomorphism of representations. Combining with Step 1, which identifies  $(\rho_r \otimes \mathbb{R})(f)$  with  $\rho_a(f)_{\mathbb{R}}$ , complexification yields

$$\rho_r \otimes_{\mathbb{Z}} \mathbb{C} = (\rho_r \otimes_{\mathbb{Z}} \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C} \cong \rho_a \oplus \bar{\rho}_a$$

as complex representations on  $V \oplus \bar{V}$ , a  $2g$ -dimensional complex vector space. Before this final complexification, Step 1 identifies  $\rho_r \otimes_{\mathbb{Z}} \mathbb{R}$  with the realification  $(\rho_a)_{\mathbb{R}}$  on the  $2g$ -dimensional real space  $V$ ; the conjugate splitting is precisely what the passage to  $\mathbb{C}$  produces, completing the proof.

The comparison is the structural heart of the section. The rational representation is real, of dimension  $2g$ , and carries all the integrality; the analytic representation is complex, of dimension  $g$ , and carries the geometry. The proposition says that once the rational representation is complexified, it becomes the analytic representation doubled by its conjugate. Every subsequent statement about how  $\text{End}(X)$  is constrained is a shadow of this single fact. In particular the eigenvalues of  $\rho_r(f)$  are exactly the eigenvalues of  $\rho_a(f)$  together with their complex conjugates, which is the precise sense in which the  $2g$  real dimensions split into  $g$  holomorphic and  $g$  antiholomorphic ones. For a self-map that is already defined over  $\mathbb{R}$ , such as  $[n]$ , the conjugate half is identical to the original half, which is why  $\rho_r([n])$  and  $\rho_a([n])$  shared the single eigenvalue  $n$  in example 1.2.4.

The eigenvalue statement is best packaged as an identity of characteristic polynomials, since that is the form in which it feeds the degree theory of isogenies and, in Part II, the study of Frobenius over finite fields. Recall that for a  $\mathbb{C}$ -linear operator  $F$  with matrix  $A$  in some basis, the characteristic polynomial is  $P(t) = \det(tI - A)$ , and  $\bar{P}(t)$  denotes the polynomial obtained by conjugating the coefficients, so that  $\bar{P}(t) = \det(tI - \bar{A})$ .

**Proposition 1.2.6: Characteristic Polynomial of the Rational Representation**

Let  $X = V/\Lambda$  be a complex torus of dimension  $g$  and  $f \in \text{End}(X)$ . Let

$$P_a(t) = \det(t I_g - \rho_a(f)) \in \mathbb{C}[t]$$

be the characteristic polynomial of the analytic representation of  $f$ , a monic polynomial of degree  $g$ , and let

$$P_r(t) = \det(t I_{2g} - \rho_r(f)) \in \mathbb{Z}[t]$$

be that of the rational representation. Then

$$P_r(t) = P_a(t) \overline{P_a(t)}$$

Consequently  $P_r$  is a monic polynomial of degree  $2g$  with integer coefficients whose roots are the eigenvalues of  $\rho_a(f)$  together with their complex conjugates, occurring in conjugate pairs.

**Proof:**

*Step 1: The characteristic polynomial is unchanged by complexification.* The characteristic polynomial of  $\rho_r(f)$  computed over  $\mathbb{Z}$  (equivalently over  $\mathbb{Q}$ , or over  $\mathbb{R}$ ) is unchanged when the operator is regarded as acting on the complexification  $\Lambda \otimes \mathbb{C}$ , since  $\det(t I - M)$  depends only on the matrix  $M$  and not on the field over which one computes the determinant. Thus  $P_r(t) = \det(t I_{2g} - (\rho_r(f) \otimes \mathbb{C}))$ .

*Step 2: Apply the comparison.* By proposition 1.2.5,  $\rho_r(f) \otimes \mathbb{C}$  is conjugate, as a  $\mathbb{C}$ -linear operator, to  $\rho_a(f) \oplus \overline{\rho_a(f)}$  acting on  $V \oplus \bar{V}$ . Conjugate operators have equal characteristic polynomials, and the characteristic polynomial of a direct sum is the product of the characteristic polynomials of the summands. Therefore

$$P_r(t) = \det(t I_g - \rho_a(f)) \cdot \det(t I_g - \overline{\rho_a(f)}) = P_a(t) \overline{P_a(t)}$$

where the second factor is  $\overline{P_a(t)}$  because conjugating the matrix  $A$  of  $\rho_a(f)$  conjugates every coefficient of  $\det(t I_g - A)$ .

*Step 3: Integrality and the conjugate-pair structure.* Both  $P_a$  and  $\overline{P_a}$  are monic of degree  $g$ , so their product  $P_r$  is monic of degree  $2g$ . Since  $\rho_r(f)$  is represented by an integer matrix in  $M_{2g}(\mathbb{Z})$  in any  $\mathbb{Z}$ -basis of  $\Lambda$ , its characteristic polynomial  $P_r(t) = \det(t I_{2g} - \rho_r(f))$  has coefficients in  $\mathbb{Z}$ , so  $P_r \in \mathbb{Z}[t]$ . Finally, if  $\alpha_1, \dots, \alpha_g$  are the roots of  $P_a$ , that is, the eigenvalues of  $\rho_a(f)$ , then the roots of  $\overline{P_a}$  are  $\overline{\alpha_1}, \dots, \overline{\alpha_g}$ , so the multiset of roots of  $P_r$  is  $\{\alpha_1, \dots, \alpha_g, \overline{\alpha_1}, \dots, \overline{\alpha_g}\}$ , which is stable under complex conjugation and hence pairs each root with its conjugate, as we sought to show.

The identity  $P_r = P_a \overline{P_a}$  is the engine that converts the analytic picture into arithmetic. On the one hand  $P_r$  has integer coefficients, forced by the rational representation; on the other its roots are the analytic eigenvalues  $\alpha_1, \dots, \alpha_g$  and their conjugates. Together these say that the eigenvalues of  $\rho_a(f)$  are algebraic integers stable, as a set-with-conjugates, under complex conjugation, and that their symmetric functions satisfy integrality relations coming from  $\mathbb{Z}[t]$ . Two invariants of  $f$  acquire clean expressions: the trace of the rational representation is  $\text{tr} \rho_r(f) = \text{tr} \rho_a(f) + \text{tr} \overline{\rho_a(f)} = 2 \text{Re}(\text{tr} \rho_a(f))$ , an



integer, and its determinant is  $\det \rho_r(f) = \det \rho_a(f) \cdot \overline{\det \rho_a(f)} = |\det \rho_a(f)|^2 \geq 0$ . The nonnegativity of  $\det \rho_r(f)$  is not a formality: when  $f$  is an isogeny the determinant is strictly positive, so the rational representation is orientation-preserving on  $\Lambda \otimes \mathbb{R} = V$ , which is the analytic reason the degree of an isogeny, computed in Part II as this determinant, is a positive integer rather than a nonzero integer of either sign. When  $f$  is not an isogeny the determinant is zero and the operator is degenerate. For  $f = [n]$  one recovers  $P_r(t) = (t - n)^g \cdot \overline{(t - n)^g} = (t - n)^{2g}$  and  $\det \rho_r([n]) = |n^g|^2 = n^{2g}$ , matching example 1.2.4.

The two representations together explain why the endomorphism algebras of complex tori, and of abelian varieties, are so constrained. The analytic representation embeds  $\text{End}^0(X)$  into  $M_g(\mathbb{C})$ , exhibiting each endomorphism as a  $g$ -dimensional complex operator; the rational representation embeds it into  $M_{2g}(\mathbb{Q})$ , which forces  $\text{End}(X)$  to have finite rank and hence bounds  $\dim_{\mathbb{Q}} \text{End}^0(X)$ , and, through proposition 1.2.6, forces every element to satisfy a monic integer polynomial of degree  $2g$ , so every element of  $\text{End}(X)$  is an algebraic integer and  $\text{End}(X)$  is an order in the finite-dimensional  $\mathbb{Q}$ -algebra  $\text{End}^0(X)$ . An endomorphism cannot scale the holomorphic tangent directions by arbitrary complex numbers: the eigenvalues it produces must assemble, with their conjugates, into the roots of an integer polynomial. This integrality is exactly the rigidity that will let Albert's classification, taken up in chapter 6, enumerate the possible endomorphism algebras of a simple abelian variety, cutting the list down to four types of division algebra with involution rather than the continuum a  $g$ -dimensional complex operator might suggest.

### Exercise 1.2.1

1. Let  $X = V/\Lambda$  have dimension  $g = 2$ , and let  $f = [3] \in \text{End}(X)$  be multiplication by 3. Following example 1.2.4, write down the matrices  $\rho_a(f)$  and  $\rho_r(f)$ , the two characteristic polynomials  $P_a(t)$  and  $P_r(t)$ , and verify directly that  $P_r(t) = P_a(t) \overline{P_a(t)}$  as asserted by proposition 1.2.6. What is the order of the kernel  $X[3]$ ?
2. Suppose  $g = 1$  and  $f \in \text{End}(X)$  has analytic representation the  $1 \times 1$  matrix  $\rho_a(f) = (\alpha)$  for some  $\alpha \in \mathbb{C}$ . Using proposition 1.2.6, express  $P_r(t)$ ,  $\text{tr} \rho_r(f)$ , and  $\det \rho_r(f)$  in terms of  $\alpha$ . Deduce that  $\alpha$  is an algebraic integer, and show that if  $f$  is an isogeny then  $\det \rho_r(f) > 0$ .
3. The analytic representation  $\rho_a: \text{End}(X) \rightarrow \text{End}_{\mathbb{C}}(V)$  is injective, embedding  $\text{End}(X)$  into the  $g^2$ -dimensional algebra  $M_g(\mathbb{C})$ . Explain why this injectivity alone does not bound  $\dim_{\mathbb{Q}} \text{End}^0(X)$ , and then explain precisely which property of the rational representation  $\rho_r$  supplies the missing bound. Illustrate the failure of the analytic bound in the case  $g = 1$  with an explicit  $\mathbb{Q}$ -subspace of  $M_1(\mathbb{C}) = \mathbb{C}$  that is infinite-dimensional over  $\mathbb{Q}$ .
4. Let  $F: V \rightarrow V$  be a  $\mathbb{C}$ -linear operator on a complex vector space of dimension  $g$ , and let  $F_{\mathbb{R}}$  be  $F$  regarded as an  $\mathbb{R}$ -linear operator on the underlying  $2g$ -dimensional real space. Prove that

$$\det(F_{\mathbb{R}}) = |\det(F)|_{\mathbb{C}}^2$$

Deduce, via proposition 1.2.5, that  $\det \rho_r(f) = |\det \rho_a(f)|^2 \geq 0$  for every  $f \in \text{End}(X)$ .

5. Prove that every  $f \in \text{End}(X)$  satisfies a monic polynomial of degree  $2g$  with integer coefficients, and hence is an algebraic integer inside  $\text{End}^0(X)$ . Then show that the eigenvalues  $\alpha_1, \dots, \alpha_g$  of  $\rho_a(f)$  are algebraic integers whose set, together with the conjugates  $\overline{\alpha_1}, \dots, \overline{\alpha_g}$ , is stable under complex conjugation. (Use proposition 1.2.6 and the fact that  $\rho_r(f)$  is an integer matrix.)



6. A complex torus  $X$  has *complex multiplication* by an imaginary quadratic field  $K = \mathbb{Q}(\sqrt{-d})$  (with  $d > 0$  squarefree) when  $g = 1$  and there is  $f \in \text{End}(X)$  whose analytic representation  $\rho_a(f) = (\alpha)$  has  $\alpha \in K \setminus \mathbb{Q}$ . Using proposition 1.2.6, show that the characteristic polynomial  $P_r(t)$  of  $\rho_r(f)$  is the minimal polynomial of  $\alpha$  over  $\mathbb{Q}$ , an irreducible quadratic in  $\mathbb{Z}[t]$ , and that  $\rho_r(f) \in M_2(\mathbb{Z})$  has no rational eigenvalue. Contrast this with the case  $f = [n]$  of example 1.2.4, where  $\rho_r$  is scalar.

### 1.3 Homology, Cohomology, and the Hodge Structure on $H^1$

A complex torus  $X = V/\Lambda$  (definition 1.1.2) is built from two pieces of linear data: the lattice  $\Lambda$ , a free abelian group of rank  $2g$ , and the complex structure on the ambient space  $V = \mathbb{C}^g$ . This section shows that both pieces are visible in cohomology, and that they interact there in a way that turns out to control the entire theory. The lattice is the topology: the singular homology and cohomology of  $X$  are exterior algebras manufactured from  $\Lambda$  alone, with no reference to the complex structure at all. The complex structure reappears one level up, as a splitting of the first cohomology into holomorphic and antiholomorphic parts. That splitting is a Hodge structure of weight one, and the passage from a complex torus to its weight-one Hodge structure is the bridge to Hodge theory that organizes everything downstream: a polarization, the subject of chapter 3, is exactly a piece of extra structure on this Hodge structure, and the entire complex-analytic theory of Part I can be read as the study of that extra structure.

The topological facts are quickly settled because  $X$  is a product of circles. As a smooth manifold,  $X = V/\Lambda \cong (S^1)^{2g}$ : choosing a  $\mathbb{Z}$ -basis  $\lambda_1, \dots, \lambda_{2g}$  of  $\Lambda$  identifies  $V$  with  $\mathbb{R}^{2g}$  carrying  $\Lambda = \mathbb{Z}^{2g}$ , and  $\mathbb{R}^{2g}/\mathbb{Z}^{2g}$  is the  $2g$ -fold product of  $\mathbb{R}/\mathbb{Z} = S^1$ . Everything topological about  $X$  is therefore the topology of a torus.

#### Proposition 1.3.1: First Homology of a Complex Torus

Let  $X = V/\Lambda$  be a complex torus of dimension  $g$ . The quotient map  $V \rightarrow X$  induces a canonical isomorphism of abelian groups

$$\Lambda \xrightarrow{\sim} H_1(X, \mathbb{Z})$$

sending a lattice vector  $\lambda \in \Lambda$  to the homology class of the loop  $t \mapsto t\lambda$ ,  $t \in [0, 1]$ . In particular  $H_1(X, \mathbb{Z})$  is free of rank  $2g$ .

#### **Proof:**

The universal cover of  $X$  is the projection  $\pi: V \rightarrow V/\Lambda = X$ , a covering map because  $\Lambda$  acts on  $V$  freely, properly discontinuously, by translations. Since  $V \cong \mathbb{R}^{2g}$  is contractible, it is the universal cover, and the deck-transformation group is  $\Lambda$ ; hence  $\pi_1(X, 0) \cong \Lambda$ , the isomorphism sending  $\lambda \in \Lambda$  to the homotopy class of the loop obtained by projecting any path in  $V$  from 0 to  $\lambda$ . The straight path  $t \mapsto t\lambda$  is one such path, so  $\lambda$  corresponds to the loop  $t \mapsto \pi(t\lambda)$ .

Because  $\Lambda$  is abelian, the fundamental group  $\pi_1(X, 0) \cong \Lambda$  equals its own abelianization. By the Hurewicz theorem ([14]), the Hurewicz map  $\pi_1(X, 0) \rightarrow H_1(X, \mathbb{Z})$  is the abelianization map, hence here an isomorphism. Tracing the two identifications,  $\lambda \in \Lambda$  maps to the homology class

of the loop  $t \mapsto \pi(t\lambda)$ , which is the assertion. As  $\Lambda$  is free abelian of rank  $2g$ , so is  $H_1(X, \mathbb{Z})$ , completing the proof.

The content of this proposition is that the lattice is not an auxiliary choice but an intrinsic invariant of the torus:  $\Lambda \cong H_1(X, \mathbb{Z})$  is recovered from  $X$  as a topological space, before any mention of the complex structure. The identification is the analytic origin of a theme that runs through the whole book. Under the analytic representation of section 1.2, an endomorphism of  $X$  acts on  $\Lambda$ ; that action is now revealed to be the action on  $H_1(X, \mathbb{Z})$ , and its  $\ell$ -adic shadow is the action on the Tate module studied in Part II. The lattice is where the arithmetic lives.

Higher homology and cohomology follow from the product structure. Rather than re-derive the (co)homology of a product of circles, the computation is imported from algebraic topology and applied.

### Theorem 1.3.2: Cohomology of a Complex Torus

Let  $X = V/\Lambda$  be a complex torus of dimension  $g$ , and write  $\Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$  for the dual lattice. For every  $n \geq 0$  there is a canonical isomorphism

$$H^n(X, \mathbb{Z}) \cong \bigwedge^n \Lambda^\vee$$

and, taken together, these assemble into an isomorphism of graded rings

$$H^\bullet(X, \mathbb{Z}) \cong \bigwedge^\bullet \Lambda^\vee$$

from the singular cohomology ring of  $X$  (cup product) to the exterior algebra on  $\Lambda^\vee$  (wedge product). In particular  $H^n(X, \mathbb{Z})$  is free of rank  $\binom{2g}{n}$ , and  $H^1(X, \mathbb{Z}) \cong \Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$ .

#### Proof:

As a smooth manifold  $X \cong (S^1)^{2g}$ , the  $2g$ -fold product of circles. The integral cohomology ring of a single circle is  $H^\bullet(S^1, \mathbb{Z}) = \mathbb{Z}[e]/(e^2)$ , the exterior algebra on one degree-one generator  $e$ . By the Künneth theorem for a finite product of spaces whose integral cohomology is free of finite rank in each degree ([14]), the cross product induces a ring isomorphism

$$H^\bullet(X, \mathbb{Z}) \cong \bigotimes_{j=1}^{2g} H^\bullet(S^1, \mathbb{Z}) = \bigotimes_{j=1}^{2g} (\mathbb{Z} \oplus \mathbb{Z}e_j)$$

where the tensor product is the graded-commutative tensor product of graded rings. Because each factor is the exterior algebra  $\bigwedge(\mathbb{Z}e_j)$  on one generator and the tensor product of exterior algebras is the exterior algebra on the direct sum of the generating modules, the right-hand side is  $\bigwedge^\bullet(\bigoplus_{j=1}^{2g} \mathbb{Z}e_j)$ . The generators  $e_1, \dots, e_{2g}$  are the degree-one classes dual to the  $2g$  circle factors, so  $\bigoplus_j \mathbb{Z}e_j = H^1(X, \mathbb{Z})$  and the isomorphism reads  $H^\bullet(X, \mathbb{Z}) \cong \bigwedge^\bullet H^1(X, \mathbb{Z})$ .

It remains to identify  $H^1(X, \mathbb{Z})$  with  $\Lambda^\vee$ . Cohomology in degree one is dual to homology in degree one for a space with free homology: the universal-coefficients theorem ([14]) gives  $H^1(X, \mathbb{Z}) \cong \text{Hom}(H_1(X, \mathbb{Z}), \mathbb{Z})$ , the Ext term vanishing because  $H_0(X, \mathbb{Z}) = \mathbb{Z}$  is free. Combined with the identification  $H_1(X, \mathbb{Z}) \cong \Lambda$  of proposition 1.3.1, this yields  $H^1(X, \mathbb{Z}) \cong \text{Hom}(\Lambda, \mathbb{Z}) =$

$\Lambda^\vee$ . Substituting into the Künneth isomorphism gives  $H^n(X, \mathbb{Z}) \cong \bigwedge^n \Lambda^\vee$  in each degree and the graded-ring statement overall. The rank of  $\bigwedge^n \Lambda^\vee$  is  $\binom{2g}{n}$  since  $\Lambda^\vee$  is free of rank  $2g$ , as we sought to show.

The cohomology ring of a complex torus is thus the exterior algebra on a single rank- $2g$  lattice, with no input beyond the group  $\Lambda$ . Two consequences are worth isolating. First, the total dimension  $\sum_n \dim H^n(X, \mathbb{Q}) = 2^{2g}$ , and the Poincaré polynomial is  $(1+t)^{2g}$ , recording that a torus is as topologically simple as a compact manifold of its dimension can be. Second, and more important for what follows, degree one carries all the information: every cohomology group is a wedge power of  $H^1(X, \mathbb{Z})$ , so any structure placed on  $H^1$  propagates to the whole ring by functoriality of  $\bigwedge$ . The complex structure will now supply exactly such a structure on  $H^1$ .

To see where the complex structure enters, pass to complex coefficients. By theorem 1.3.2 and the universal-coefficients theorem,

$$H^1(X, \mathbb{C}) \cong \text{Hom}(\Lambda, \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{C} \cong \text{Hom}_{\mathbb{Z}}(\Lambda, \mathbb{C}) = \text{Hom}_{\mathbb{R}}(V, \mathbb{C})$$

the last equality because  $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = V$ , so a  $\mathbb{Z}$ -linear map  $\Lambda \rightarrow \mathbb{C}$  extends uniquely to an  $\mathbb{R}$ -linear map  $V \rightarrow \mathbb{C}$ . The space  $\text{Hom}_{\mathbb{R}}(V, \mathbb{C})$  of  $\mathbb{R}$ -linear functionals on  $V$  is where the complex structure asserts itself: a functional  $f: V \rightarrow \mathbb{C}$  that is  $\mathbb{R}$ -linear need be neither  $\mathbb{C}$ -linear nor  $\mathbb{C}$ -antilinear, and the two extremes span the whole space. This is the splitting that defines the Hodge structure.

The holomorphic side of the splitting has an independent description as the space of global holomorphic 1-forms, which merits a name of its own. The torus  $X$  inherits from  $V = \mathbb{C}^g$  the standard holomorphic coordinates  $z_1, \dots, z_g$ : these are not functions on  $X$  (they are not  $\Lambda$ -periodic), but their differentials  $dz_1, \dots, dz_g$  are  $\Lambda$ -invariant, since translation by a lattice vector is a translation in  $V$  and  $d(z_k + c) = dz_k$  for any constant  $c$ . Hence each  $dz_k$  descends to a global holomorphic 1-form on  $X$ , and likewise each  $d\bar{z}_k$  descends to a global antiholomorphic 1-form.

#### Definition 1.3.3: The Hodge Structure on $H^1$

Let  $X = V/\Lambda$  be a complex torus. The *Hodge decomposition* of  $H^1(X, \mathbb{C}) = \text{Hom}_{\mathbb{R}}(V, \mathbb{C})$  (the identification derived above) is the direct-sum decomposition

$$H^1(X, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$$

into

- $H^{1,0} = V^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$ , the  $\mathbb{C}$ -linear functionals on  $V$ , called the *holomorphic part*; and
- $H^{0,1} = \overline{V^*} = \text{Hom}_{\bar{\mathbb{C}}}(V, \mathbb{C})$ , the  $\mathbb{C}$ -antilinear functionals on  $V$ , called the *antiholomorphic part*.

The subspace  $H^{1,0}$  is identified with the space  $H^0(X, \Omega_X^1)$  of global holomorphic 1-forms on  $X$ , and  $H^{0,1} = \overline{H^{1,0}}$  is its complex conjugate. This decomposition makes  $H^1(X, \mathbb{Z})$  an integral *Hodge structure of weight one*.

The decomposition is the linear-algebra statement that an  $\mathbb{R}$ -linear map  $V \rightarrow \mathbb{C}$  splits uniquely into a  $\mathbb{C}$ -linear part and a  $\mathbb{C}$ -antilinear part. That the two summands are complex conjugates of one another,  $H^{0,1} = \overline{H^{1,0}}$ , and that they have equal dimension  $g$ , is the defining feature of a weight-one

Hodge structure: the language of Hodge structures, developed in general in [7], is here entirely explicit and requires no transcendental input. A complex torus is, from this vantage, nothing but a weight-one Hodge structure wearing a geometric disguise, and the correspondence between the two is the reason abelian varieties and Hodge theory are the same subject viewed from two sides. The next proposition proves that the decomposition of definition 1.3.3 is correct as stated, including the identification of  $H^{1,0}$  with holomorphic 1-forms.

**Proposition 1.3.4: Explicit Hodge Decomposition**

Let  $X = V/\Lambda$  with  $V = \mathbb{C}^g$  and holomorphic coordinates  $z_1, \dots, z_g$ . Then:

1. Every  $\mathbb{R}$ -linear functional  $f: V \rightarrow \mathbb{C}$  is uniquely the sum  $f = f' + f''$  of a  $\mathbb{C}$ -linear functional  $f' \in V^*$  and a  $\mathbb{C}$ -antilinear functional  $f'' \in \overline{V^*}$ , giving  $H^1(X, \mathbb{C}) = V^* \oplus \overline{V^*} = H^{1,0} \oplus H^{0,1}$  with  $\dim_{\mathbb{C}} H^{1,0} = \dim_{\mathbb{C}} H^{0,1} = g$ .
2. The classes of  $dz_1, \dots, dz_g$  form a basis of  $H^{1,0}$  and the classes of  $d\bar{z}_1, \dots, d\bar{z}_g$  form a basis of  $H^{0,1}$ ; the identification  $H^{1,0} = H^0(X, \Omega_X^1)$  sends a  $\mathbb{C}$ -linear functional to the holomorphic 1-form it defines.
3. The two summands are exchanged by complex conjugation:  $\overline{H^{1,0}} = H^{0,1}$ .

**Proof:**

1. *The splitting.* Let  $J: V \rightarrow V$  denote multiplication by  $i$ , the complex structure on  $V$ . For an  $\mathbb{R}$ -linear  $f: V \rightarrow \mathbb{C}$ , define

$$f'(v) = \frac{1}{2}(f(v) - i f(iv)), \quad f''(v) = \frac{1}{2}(f(v) + i f(iv))$$

Then  $f = f' + f''$  by inspection. The functional  $f'$  is  $\mathbb{C}$ -linear:  $f'(iv) = \frac{1}{2}(f(iv) - i f(-v)) = \frac{1}{2}(f(iv) + i f(v)) = i f'(v)$ , using  $f(-v) = -f(v)$ . The functional  $f''$  is  $\mathbb{C}$ -antilinear: the same computation gives  $f''(iv) = \frac{1}{2}(f(iv) + i f(v)) = -i f''(v)$ . For uniqueness, suppose  $f = g' + g''$  with  $g'$   $\mathbb{C}$ -linear and  $g''$   $\mathbb{C}$ -antilinear; then  $f' - g' = g'' - f''$  is simultaneously  $\mathbb{C}$ -linear and  $\mathbb{C}$ -antilinear, hence satisfies both  $h(iv) = ih(v)$  and  $h(iv) = -ih(v)$ , forcing  $h = 0$ . Thus the decomposition  $\text{Hom}_{\mathbb{R}}(V, \mathbb{C}) = V^* \oplus \overline{V^*}$  is direct. A  $\mathbb{C}$ -linear functional on  $V = \mathbb{C}^g$  is determined by its values on a  $\mathbb{C}$ -basis, so  $\dim_{\mathbb{C}} V^* = g$ ; likewise  $\dim_{\mathbb{C}} \overline{V^*} = g$ , consistent with  $\dim_{\mathbb{C}} H^1(X, \mathbb{C}) = 2g$ .

2. *The holomorphic forms.* A global holomorphic 1-form on  $X$  pulls back along  $\pi: V \rightarrow X$  to a  $\Lambda$ -invariant holomorphic 1-form  $\omega$  on  $V$ . Writing  $\omega = \sum_{k=1}^g h_k(z) dz_k$  with  $h_k$  holomorphic,  $\Lambda$ -invariance forces each  $h_k(z + \lambda) = h_k(z)$ , so each  $h_k$  is a  $\Lambda$ -periodic holomorphic function on  $V$ , hence bounded (its values are attained on the compact fundamental domain), hence constant by Liouville's theorem ([16]). Thus  $\omega = \sum_k c_k dz_k$  with  $c_k \in \mathbb{C}$ : the global holomorphic 1-forms are exactly the constant-coefficient combinations of  $dz_1, \dots, dz_g$ , and  $\dim_{\mathbb{C}} H^0(X, \Omega_X^1) = g$  with basis  $dz_1, \dots, dz_g$ . A constant form  $\sum_k c_k dz_k$  is precisely a  $\mathbb{C}$ -linear functional  $V \rightarrow \mathbb{C}$ , namely  $v \mapsto \sum_k c_k z_k(v)$ ; this is the identification  $H^0(X, \Omega_X^1) = V^* = H^{1,0}$ . Under the de Rham comparison  $H^1(X, \mathbb{C}) \cong H_{\text{dR}}^1(X)$  ([14]), the class of a translation-invariant closed 1-form is its own de Rham class, and the invariant forms  $dz_1, \dots, dz_g, d\bar{z}_1, \dots, d\bar{z}_g$  are closed and represent a basis of  $H^1(X, \mathbb{C})$ ; the holomorphic ones span  $H^{1,0}$  and the antiholomorphic ones span  $H^{0,1}$ .

3. *Conjugation.* Complex conjugation on the coefficient field  $\mathbb{C}$  acts on  $H^1(X, \mathbb{C}) = H^1(X, \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{C}$  through the second factor, fixing the real structure  $H^1(X, \mathbb{R})$ . On functionals it sends  $f$  to  $\bar{f}: v \mapsto \overline{f(v)}$ . If  $f$  is  $\mathbb{C}$ -linear then  $\bar{f}(iv) = \overline{f(iv)} = \overline{if(v)} = -i \overline{f(v)} = -i \bar{f}(v)$ , so  $\bar{f}$  is  $\mathbb{C}$ -antilinear; thus conjugation carries  $V^* = H^{1,0}$  onto  $\bar{V}^* = H^{0,1}$ , and  $\overline{H^{1,0}} = H^{0,1}$ . Concretely  $\overline{dz_k} = d\bar{z}_k$ , exchanging the two bases of part (2), completing the proof.

The proposition makes the Hodge decomposition of a complex torus entirely computable: no harmonic analysis, no elliptic operators, no appeal to the general Hodge theorem is needed, because the harmonic representatives of  $H^1$  are visibly the translation-invariant forms  $dz_k$  and  $d\bar{z}_k$ . This is the sense in which the complex torus is the simplest nontrivial example in Hodge theory, and why the whole subject of this book can be developed by explicit linear algebra where the general Kähler theory of [7] would require the harmonic projection. The data of the Hodge structure amounts to the position of a single  $g$ -dimensional subspace  $H^{1,0} \subset H^1(X, \mathbb{C})$ , a subspace transverse to its own conjugate ( $H^{1,0} \oplus \overline{H^{1,0}} = H^1(X, \mathbb{C})$ ); recovering the complex structure on  $V$  from this subspace is the inverse construction, and a polarization (chapter 3) is a further alternating form making the subspace “positive.”

The wedge-power structure of theorem 1.3.2 propagates the weight-one decomposition to every degree. Applying  $\bigwedge^n$  to  $H^1(X, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$  splits  $H^n(X, \mathbb{C})$  into pieces  $H^{p,q} = \bigwedge^p H^{1,0} \otimes \bigwedge^q H^{0,1}$  with  $p + q = n$ , the Hodge decomposition of weight  $n$ . Weight two is the first place where this structure carries geometry the reader will use directly, because the class of a line bundle lives in  $H^2$ ; the next result records the integral degree-two picture and isolates the piece that will become the Néron-Severi group.

**Proposition 1.3.5:  $H^2$  as Alternating Forms on the Lattice**

Let  $X = V/\Lambda$  be a complex torus. There is a canonical isomorphism

$$H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee \cong \{ E: \Lambda \times \Lambda \rightarrow \mathbb{Z} \mid E \text{ is } \mathbb{Z}\text{-bilinear and alternating} \}$$

identifying a degree-two class with an alternating integral form on  $\Lambda$ . Extending scalars to  $\mathbb{C}$  and using definition 1.3.3, this decomposes as

$$H^2(X, \mathbb{C}) = H^{2,0} \oplus H^{1,1} \oplus H^{0,2}, \quad H^{p,q} = \bigwedge^p H^{1,0} \otimes \bigwedge^q H^{0,1}$$

with  $\dim_{\mathbb{C}} H^{2,0} = \dim_{\mathbb{C}} H^{0,2} = \binom{g}{2}$  and  $\dim_{\mathbb{C}} H^{1,1} = g^2$ . A class  $E \in H^2(X, \mathbb{R})$ , viewed as an alternating  $\mathbb{R}$ -bilinear form on  $V$ , lies in  $H^{1,1}$  if and only if it satisfies the invariance  $E(iv, iw) = E(v, w)$  for all  $v, w \in V$ .

**Proof:**

*Step 1: The integral identification.* By theorem 1.3.2,  $H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee$ . An element of  $\bigwedge^2 \Lambda^\vee$  is an alternating  $\mathbb{Z}$ -bilinear form on  $\Lambda$ : the pairing  $\bigwedge^2 \Lambda^\vee \rightarrow \text{Hom}(\bigwedge^2 \Lambda, \mathbb{Z})$  sending  $\varphi \wedge \psi$  to the form  $(\lambda, \mu) \mapsto \varphi(\lambda)\psi(\mu) - \varphi(\mu)\psi(\lambda)$  is an isomorphism onto the alternating forms, because  $\Lambda$  is free of finite rank. This is the stated identification  $H^2(X, \mathbb{Z}) \cong \{E: \Lambda \times \Lambda \rightarrow \mathbb{Z} \text{ alternating}\}$ .

*Step 2: The Hodge decomposition of  $H^2$ .* Wedging the decomposition  $H^1(X, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$

with itself,  $\bigwedge^2 H^1(X, \mathbb{C}) = \bigwedge^2 H^{1,0} \oplus (H^{1,0} \otimes H^{0,1}) \oplus \bigwedge^2 H^{0,1}$ , which are the summands  $H^{2,0}$ ,  $H^{1,1}$ ,  $H^{0,2}$  respectively. Since  $\dim_{\mathbb{C}} H^{1,0} = \dim_{\mathbb{C}} H^{0,1} = g$ , the outer summands have dimension  $\dim \bigwedge^2 \mathbb{C}^g = \binom{g}{2}$  and the middle summand has dimension  $g \cdot g = g^2$ . As a check,  $2\binom{g}{2} + g^2 = g(g-1) + g^2 = 2g^2 - g = \binom{2g}{2}$ , the rank of  $H^2$ .

*Step 3: The (1, 1) condition.* An alternating  $\mathbb{R}$ -bilinear form  $E$  on  $V$  extends  $\mathbb{C}$ -bilinearly to  $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$ , which decomposes as  $V^{1,0} \oplus V^{0,1}$  under  $J = \text{multiplication by } i$  (the  $\pm i$ -eigenspaces of  $J \otimes 1$ ). The form  $E$ , as an element of  $\bigwedge^2 V_{\mathbb{C}}^*$ , lies in the  $(1, 1)$ -piece exactly when it pairs  $V^{1,0}$  with  $V^{0,1}$  and annihilates  $\bigwedge^2 V^{1,0}$  and  $\bigwedge^2 V^{0,1}$ . A direct computation identifies this with the eigenvalue condition under  $J$ :  $E$  is of type  $(1, 1)$  if and only if  $E(Jv, Jw) = E(v, w)$  for all  $v, w$ , that is,  $E(iv, iw) = E(v, w)$ . Indeed, on the  $(2, 0) \oplus (0, 2)$  part  $J$  acts by  $E(Jv, Jw) = -E(v, w)$  (each of  $v, w$  contributing a factor  $i$  from the same isotropic subspace), while on the  $(1, 1)$  part the two factors  $i$  and  $-i$  cancel to give  $E(Jv, Jw) = E(v, w)$ ; the condition  $E(iv, iw) = E(v, w)$  therefore cuts out exactly  $H^{1,1} \cap H^2(X, \mathbb{R})$ , as we sought to show.

The last clause is the technical heart of Part I stated in advance. An integral class  $E \in H^2(X, \mathbb{Z})$  that is of type  $(1, 1)$  is an alternating integral form on  $\Lambda$  satisfying  $E(iv, iw) = E(v, w)$ , and by the correspondence  $H \mapsto E = \text{Im} H$ , with inverse  $H(v, w) = E(iv, w) + iE(v, w)$ , such a form is exactly the imaginary part of a Hermitian form  $H$  on  $V$ . These are the classes that arise as first Chern classes of line bundles: the group of them is the Néron-Severi group  $\text{NS}(X)$ , developed in chapter 2, and the requirement that the Hermitian form  $H$  be positive definite is the Riemann relation that singles out the algebraic tori in chapter 3. Thus the abstract statement “ $H^2(X, \mathbb{Z}) = \text{alternating forms on } \Lambda$ , with a distinguished  $(1, 1)$  part” is the frame into which the entire theory of polarizations will be poured. In degree two the integral Lefschetz  $(1, 1)$ -theorem holds on any complex torus: every integral  $(1, 1)$  class in  $H^2$  is algebraic, because the Appell-Humbert theorem of chapter 2 will realize it as the first Chern class of an actual line bundle on  $X$ . (This is a statement about divisors only; in higher codimension the analogous integral Hodge conjecture can fail for abelian varieties, and nothing here bears on it.)

The general picture is best seen first in the case  $g = 1$ , where the lattice has rank 2, the cohomology is that of a 2-torus, and every subspace is one-dimensional.

### Example 1.3.6: The Hodge Structure of an Elliptic Curve

Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ , the standard elliptic-curve lattice ([4]). Here  $g = 1$ ,  $V = \mathbb{C}$  with coordinate  $z$ , and the topological facts of proposition 1.3.1 and theorem 1.3.2 read:

- $H_1(E, \mathbb{Z}) \cong \Lambda = \mathbb{Z} \cdot 1 \oplus \mathbb{Z} \cdot \tau$ , generated by the two loops  $\gamma_1: t \mapsto t$  and  $\gamma_\tau: t \mapsto t\tau$ ,  $t \in [0, 1]$ , the sides of the fundamental parallelogram.
- $H^1(E, \mathbb{Z}) \cong \Lambda^\vee = \mathbb{Z}\alpha \oplus \mathbb{Z}\beta$ , where  $\alpha, \beta$  is the dual basis:  $\alpha(1) = 1$ ,  $\alpha(\tau) = 0$ ,  $\beta(1) = 0$ ,  $\beta(\tau) = 1$ .
- $H^2(E, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee = \mathbb{Z}(\alpha \wedge \beta)$ , the fundamental class, and  $H^n(E, \mathbb{Z}) = 0$  for  $n > 2$ .

The Hodge decomposition of definition 1.3.3 is

$$H^1(E, \mathbb{C}) = H^{1,0} \oplus H^{0,1} = \mathbb{C} \cdot dz \oplus \mathbb{C} \cdot d\bar{z}$$

a splitting of a 2-dimensional space into two lines. The single holomorphic 1-form  $dz$  spans

$H^{1,0} = H^0(E, \Omega_E^1)$ , consistent with  $\dim H^0(E, \Omega_E^1) = g = 1$  (a genus-one curve has a one-dimensional space of holomorphic differentials), and  $d\bar{z} = d\bar{z}$  spans  $H^{0,1}$ .

The relation between the two bases is recorded by the *periods*: the numbers 1 and  $\tau$  that generate  $\Lambda$  also fix the position of the Hodge line  $H^{1,0}$  inside  $H^1(E, \mathbb{C})$ . The form  $dz$  is a class in  $H^1(E, \mathbb{C}) = H^1(E, \mathbb{Z}) \otimes \mathbb{C}$ ; expressing it in the integral basis  $\alpha, \beta$  amounts to integrating  $dz$  over the loops  $\gamma_1, \gamma_\tau$ . The period integrals are

$$\int_{\gamma_1} dz = \int_0^1 dt = 1, \quad \int_{\gamma_\tau} dz = \int_0^1 \tau dt = \tau$$

so that, under the pairing of  $H^1(E, \mathbb{C})$  with  $H_1(E, \mathbb{Z})$ , the class of  $dz$  is  $\alpha + \tau\beta$ . The row vector  $(1 \ \tau)$  of its periods is the period matrix  $\Pi$  of  $E$ , the  $1 \times 2$  matrix whose columns record the  $\mathbb{Z}$ -basis  $1, \tau$  of  $\Lambda$  in the  $\mathbb{C}$ -basis  $\{1\}$  of  $V = \mathbb{C}$ , and the ratio of periods  $\tau = (\int_{\gamma_\tau} dz) / (\int_{\gamma_1} dz)$  recovers the point of the upper half-plane classifying  $E$  up to isomorphism ([4]). The positivity  $\text{Im}\tau > 0$  is the  $g = 1$  Riemann relation: it is equivalent to  $\frac{i}{2} \int_E dz \wedge d\bar{z} > 0$ , the statement that the natural Hermitian form is positive, which is the condition studied in general in chapter 3.

The elliptic curve exhibits the entire structure of this section in miniature: the lattice is  $H_1$ , its dual is  $H^1$ , the top wedge is the fundamental class in  $H^2$ , and the Hodge line  $H^{1,0} = \mathbb{C}dz$  tilts inside  $H^1(E, \mathbb{C})$  by an amount recorded in the period  $\tau$ . Moving that line, that is, varying  $\tau$  over the upper half-plane, varies the complex structure while leaving the lattice and its cohomology fixed; this is the first appearance of a period domain and of moduli, and in dimension  $g$  the Hodge line becomes a  $g$ -dimensional subspace whose variation is governed by the Siegel upper half-space of chapter 3.

The unifying statement of the section is that the passage  $X \mapsto (H^1(X, \mathbb{Z}), H^{1,0})$  is a functor from complex tori to weight-one Hodge structures, and that it loses nothing: a complex torus and its weight-one Hodge structure carry the same information, because the complex structure on  $V$  is recovered from the position of  $H^{1,0} \subset H^1(X, \mathbb{C})$ , and  $\Lambda = H_1(X, \mathbb{Z})$  is recovered as the integral lattice. This is the precise form of the assertion in the chapter opening that a complex torus is governed by linear algebra, and it is the bridge on which the rest of the book is built. Everything added in Chapters 2 through 6 is extra structure on this Hodge structure: a line bundle is a class in the integral  $(1, 1)$  part of  $H^2$ ; a polarization is such a class whose Hermitian form is positive; the endomorphism algebra is the algebra of Hodge-structure endomorphisms.

### Exercise 1.3.1

1. Let  $X = V/\Lambda$  be a complex torus of dimension  $g$ . Using theorem 1.3.2, compute the rank of  $H^n(X, \mathbb{Z})$  for each  $n$  and the total rank  $\sum_n \text{rank } H^n(X, \mathbb{Z})$ . Verify directly that the alternating sum  $\sum_n (-1)^n \text{rank } H^n(X, \mathbb{Z})$  vanishes when  $g \geq 1$ , and interpret this as the Euler characteristic of  $X$ .
2. Let  $f: V \rightarrow \mathbb{C}$  be the  $\mathbb{R}$ -linear functional on  $V = \mathbb{C}^2$  (with coordinates  $z_1 = x_1 + iy_1$ ,  $z_2 = x_2 + iy_2$ ) given by  $f(z_1, z_2) = \bar{z}_1 + iz_2$ . Using the formulas of proposition 1.3.4(1), write  $f = f' + f''$  with  $f' \in H^{1,0}$  and  $f'' \in H^{0,1}$ , and express each of  $f', f''$  as a constant-coefficient combination of  $dz_1, dz_2, d\bar{z}_1, d\bar{z}_2$ .
3. Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ , as in example 1.3.6. The antiholomorphic form  $d\bar{z}$  spans  $H^{0,1}$ . Compute the periods  $\int_{\gamma_1} d\bar{z}$  and  $\int_{\gamma_\tau} d\bar{z}$  over the two generating loops, express the



- class of  $d\bar{z}$  in the integral basis  $\alpha, \beta$  of  $H^1(E, \mathbb{Z})$ , and confirm that  $\overline{d\bar{z}} = d\bar{z}$  is consistent with the relation  $\overline{H^{1,0}} = H^{0,1}$  of proposition 1.3.4(3).
4. Let  $X = V/\Lambda$  have dimension  $g = 2$ . Using proposition 1.3.5, give the dimensions of  $H^{2,0}$ ,  $H^{1,1}$ ,  $H^{0,2}$  and check they sum to  $\text{rank } H^2(X, \mathbb{Z})$ . Then, taking  $V = \mathbb{C}^2$  with the standard complex structure  $J$ , exhibit one nonzero alternating  $\mathbb{R}$ -bilinear form  $E$  on  $V$  that is of type  $(1, 1)$  and one that is not, justifying each by the criterion  $E(iv, iw) = E(v, w)$ .
  5. Prove that the assignment  $\lambda \mapsto [t \mapsto \pi(t\lambda)]$  of proposition 1.3.1 is a group homomorphism  $\Lambda \rightarrow H_1(X, \mathbb{Z})$  directly, that is, without invoking the Hurewicz theorem: show that for  $\lambda, \mu \in \Lambda$  the loop  $t \mapsto \pi(t(\lambda + \mu))$  is homologous to the concatenation of the loops for  $\lambda$  and  $\mu$ . (You may use that in  $H_1$  the class of a concatenation is the sum of the classes, and that homotopic loops are homologous.)
  6. Let  $H$  be a Hermitian form on  $V$  (linear in the first variable, antilinear in the second, with  $\overline{H(v, w)} = H(w, v)$ ), and set  $E = \text{Im}H$ . Prove that  $E$  is an alternating  $\mathbb{R}$ -bilinear form satisfying  $E(iv, iw) = E(v, w)$ , so that by proposition 1.3.5 it represents a class of type  $(1, 1)$  in  $H^2(X, \mathbb{R})$ . Prove conversely that  $H$  is recovered from  $E$  by  $H(v, w) = E(iv, w) + iE(v, w)$ , and that this  $H$  is Hermitian. (This is the correspondence recorded after proposition 1.3.5.)

## 1.4 Which Complex Tori Are Algebraic?

The preceding sections have made a complex torus  $X = V/\Lambda$  (definition 1.1.2) into a thoroughly computable object: a compact complex Lie group whose morphisms are linear algebra over  $\mathbb{Z}$  (proposition 1.1.7) and whose cohomology is the exterior algebra of a lattice (theorem 1.3.2). Nothing so far has required that  $X$  be anything more than a complex manifold. The question that organizes the rest of Part I is whether  $X$  is also an algebraic object, that is, whether it can be realized as a projective variety cut out by polynomial equations. This section makes the question precise, settles it completely in dimension one, and exhibits an explicit two-dimensional torus for which the answer is no.

The stake is the following. A complex torus is a quotient, and quotients are cheap: any lattice of rank  $2g$  in  $\mathbb{C}^g$  produces one, and the space of lattices is  $2g^2$ -complex-dimensional. Projective varieties, by contrast, are constrained: their meromorphic functions form a field of transcendence degree equal to the dimension, and the ambient projective space forces relations among coordinates. To ask that a torus be projective is to ask that its meromorphic function field be as large as possible. The tension between the ease of writing down tori and the scarcity of projective ones is the phenomenon this section names, and the criterion that resolves it, the Riemann relations, is the subject of Chapter 3.

### Definition 1.4.1: Abelian Variety (Provisional)

A complex torus  $X = V/\Lambda$  is called *algebraic*, or an *abelian variety*, if there exists a holomorphic embedding  $X \hookrightarrow \mathbb{P}^N$  into some complex projective space, that is, a holomorphic map that is an isomorphism onto a closed complex submanifold of  $\mathbb{P}^N$ .

This definition is provisional in two respects, both resolved later. It is stated in the analytic category, in terms of a holomorphic embedding into  $\mathbb{P}^N$ ; that a compact complex submanifold of projective space is automatically the zero locus of homogeneous polynomials, hence a projective variety in the



algebraic sense, is Chow's theorem, cited to [16]. And the criterion for such an embedding to exist is not yet visible from the definition. The refined definition, in terms of the existence of a polarization, is definition 3.2.1, and theorem 3.2.6 records the equivalence of the two formulations. The name *abelian variety* anticipates that the group structure carried by  $X$  (proposition 1.1.3) becomes, once  $X$  is projective, a morphism of varieties: an abelian variety is a projective group.

Why insist on projectivity rather than some weaker form of algebraicity? The answer is that projectivity is exactly the condition under which the group law and the geometry interact well. A projective variety is proper, so the image of any morphism out of it is closed; a connected group variety that is proper is automatically commutative and its group law is uniquely determined by the identity, facts developed scheme-theoretically in Part II. None of this holds for a general complex torus regarded only as a complex manifold. Projectivity is what promotes a compact complex Lie group to an object of algebraic geometry.

The one-dimensional case is settled by machinery already in hand, and settled affirmatively for every torus without exception.

#### Proposition 1.4.2: One-Dimensional Tori Are Elliptic Curves

Every one-dimensional complex torus  $X = \mathbb{C}/\Lambda$  is an abelian variety. Explicitly, the map

$$\varphi: \mathbb{C}/\Lambda \longrightarrow \mathbb{P}^2, \quad z \longmapsto \begin{cases} [\wp(z) : \wp'(z) : 1] & z \notin \Lambda \\ [0 : 1 : 0] & z \in \Lambda \end{cases}$$

built from the Weierstrass function  $\wp$  of  $\Lambda$  and its derivative is a holomorphic embedding whose image is the smooth plane cubic

$$Y^2Z = 4X^3 - g_2(\Lambda)XZ^2 - g_3(\Lambda)Z^3 \quad (1.1)$$

with  $g_2(\Lambda) = 60 \sum_{\lambda \neq 0} \lambda^{-4}$  and  $g_3(\Lambda) = 140 \sum_{\lambda \neq 0} \lambda^{-6}$ .

#### Proof:

The Weierstrass function  $\wp$  of the lattice  $\Lambda$  is the  $\Lambda$ -periodic meromorphic function

$$\wp(z) = \frac{1}{z^2} + \sum_{\lambda \in \Lambda \setminus \{0\}} \left( \frac{1}{(z - \lambda)^2} - \frac{1}{\lambda^2} \right)$$

with a double pole at each lattice point and no others; its construction, convergence, and the differential equation

$$\wp'(z)^2 = 4\wp(z)^3 - g_2(\Lambda)\wp(z) - g_3(\Lambda) \quad (1.2)$$

are developed in full in [16] and applied to elliptic curves in [4]. Only (1.2) and the pole orders of  $\wp$  and  $\wp'$  are used here.

By (1.2), for  $z \notin \Lambda$  the point  $(\wp(z), \wp'(z))$  satisfies the affine equation  $Y^2 = 4X^3 - g_2X - g_3$ , so  $\wp(z)$  lies on the projective curve (1.1). Both  $\wp$  and  $\wp'$  are  $\Lambda$ -periodic, so  $\wp$  is well-defined on the quotient  $\mathbb{C}/\Lambda$ . At a lattice point  $\wp$  has a double pole and  $\wp'$  a triple pole, and the local expansions  $\wp(z) \sim z^{-2}$ ,  $\wp'(z) \sim -2z^{-3}$  near  $z = 0$  give

$$[\wp(z) : \wp'(z) : 1] = [z^{-2} : -2z^{-3} : 1] = [z : -2 : z^3] \longrightarrow [0 : 1 : 0]$$

as  $z \rightarrow 0$ , so the assignment  $\varphi(0) = [0 : 1 : 0]$  extends  $\varphi$  holomorphically across the lattice points. Away from  $\Lambda$ ,  $\varphi$  is a composition of the meromorphic functions  $\wp, \wp'$  with the standard chart of  $\mathbb{P}^2$ , hence holomorphic, so  $\varphi$  is holomorphic on all of  $\mathbb{C}/\Lambda$ .

The cubic (1.1) is smooth because the lattice discriminant  $g_2^3 - 27g_3^2$  is nonzero for a genuine lattice ([16]), so the polynomial  $4X^3 - g_2X - g_3$  has distinct roots. That  $\varphi$  is a biholomorphism onto this smooth cubic is the uniformization theorem, proved in full as the isomorphism of Riemann surfaces  $\mathbb{C}/\Lambda \cong E(\mathbb{C})$  in [4]. Injectivity uses that  $\wp$  has degree two and  $\wp'$  is odd, so the fibers of  $\wp$  are the pairs  $\{a, -a\}$  separated by the value of  $\wp'$ ; surjectivity uses that  $\wp$  attains every value; and a bijective holomorphic map of compact Riemann surfaces is biholomorphic. A biholomorphism onto a smooth cubic in  $\mathbb{P}^2$  is in particular a holomorphic embedding, so  $X = \mathbb{C}/\Lambda$  is an abelian variety in the sense of definition 1.4.1, completing the proof.

The content of proposition 1.4.2 is that the two descriptions of an elliptic curve, the transcendental quotient  $\mathbb{C}/\Lambda$  and the algebraic plane cubic (1.1), are the same object. The Weierstrass function is precisely the nonconstant meromorphic function whose existence witnesses algebraicity: it and its derivative generate the field of  $\Lambda$ -periodic meromorphic functions, and that field is  $\mathbb{C}(X)$ , of transcendence degree one. Every one-dimensional torus carries such a function, and so every one is projective. The purpose of the rest of Part I is to explain why this uniformity breaks down in higher dimension, and the first sign of the breakdown is that in dimension two or more, a torus may have no nonconstant meromorphic functions at all.

The right invariant to track is the size of the meromorphic function field. For a compact complex manifold  $X$  of dimension  $g$ , the field  $\mathcal{M}(X)$  of meromorphic functions is a finitely generated extension of  $\mathbb{C}$  whose transcendence degree, the *algebraic dimension*  $a(X) := \text{trdeg}_{\mathbb{C}} \mathcal{M}(X)$ , satisfies  $0 \leq a(X) \leq g$ . This bound and the finite generation are theorems of Siegel; the detailed development for complex tori, including the equivalences below, is in Birkenhake and Lange, *Complex Abelian Varieties*. The two extremes are the cases of interest. At the top,  $a(X) = g$  holds exactly when  $X$  is an abelian variety: the function field is then as large as the dimension permits, the torus is projective, and  $\mathcal{M}(X) = \mathbb{C}(X)$  is the field of rational functions of the projective variety. At the bottom,  $a(X) = 0$  means the only meromorphic functions on  $X$  are the constants, so  $X$  admits no nonconstant holomorphic map to any projective space and is as far from algebraic as a compact complex manifold can be. For a torus,  $a(X) = g$  is equivalent to the existence of a positive Hermitian form on  $V$  with imaginary part integral on  $\Lambda$ , which is the polarization condition of chapter 3; the intermediate values  $0 < a(X) < g$  occur and correspond to tori that are fibered over lower-dimensional abelian varieties. By proposition 1.4.2,  $g = 1$  forces  $a(X) = 1 = g$ , so the phenomenon of a deficient function field is invisible until  $g \geq 2$ .

The following example makes the failure concrete. It exhibits a two-dimensional torus, given by an explicit period matrix, that carries no nonconstant meromorphic functions, hence is not algebraic.

#### Example 1.4.3: A Generic Two-Dimensional Torus Is Not Algebraic

Fix  $V = \mathbb{C}^2$  and let  $\tau \in M_2(\mathbb{C})$  be a  $2 \times 2$  complex matrix, chosen generically in a sense made precise below. Take the lattice

$$\Lambda = \tau\mathbb{Z}^2 + \mathbb{Z}^2 \subset \mathbb{C}^2$$

spanned by the columns of the period matrix  $\Pi = (\tau \ I_2) \in M_{2 \times 4}(\mathbb{C})$ , and set  $X = \mathbb{C}^2/\Lambda$ . For  $\Lambda$  to be a full lattice, the columns of  $\Pi$  must be  $\mathbb{R}$ -linearly independent, which holds as long as  $\text{Im}\tau$  is invertible; assume this. The claim is that for  $\tau$  outside a countable union of proper analytic subsets of the space of such matrices,  $X$  admits no nonzero Hermitian form  $H$  on  $V$  whose imaginary part

$E = \text{Im}H$  is integral on  $\Lambda$ , and consequently  $a(X) = 0$ : the only meromorphic functions on  $X$  are constants, and  $X$  is not an abelian variety.

The key computation is a count of linear constraints. By the analysis of Hermitian forms integral on a lattice, which the reader will meet in proposition 2.3.2, a Hermitian  $H$  on  $V = \mathbb{C}^2$  with  $E = \text{Im}H$  integral on  $\Lambda$  is the same datum as an alternating integral form  $E: \Lambda \times \Lambda \rightarrow \mathbb{Z}$  satisfying the compatibility  $E(iv, iw) = E(v, w)$  with the complex structure. An alternating integral form on the rank-four lattice  $\Lambda$  is specified by an alternating matrix  $E \in M_4(\mathbb{Z})$ , a choice of  $\binom{4}{2} = 6$  integers. The compatibility condition  $E(iv, iw) = E(v, w)$ , written out in the period coordinates via  $\Pi$ , becomes the equation

$$\Pi E^{-1} \Pi^t = 0$$

on the entries of  $\tau$ . Because  $E$  is alternating,  $E^{-1}$  is again skew, so the  $2 \times 2$  complex matrix  $\Pi E^{-1} \Pi^t$  is itself skew-symmetric: its diagonal vanishes identically and its two off-diagonal entries are negatives of one another. The equation is therefore the vanishing of a single complex number, one  $\mathbb{C}$ -linear equation, hence two  $\mathbb{R}$ -linear equations, in the six integer entries of  $E$ . Written out, the constraint is the vanishing of the single complex-linear form

$$-a - d\tau_{11} + b\tau_{12} - e\tau_{21} + f(\tau_{12}\tau_{21} - \tau_{11}\tau_{22}) + c\tau_{22}$$

in which  $a, b, c, d, e, f$  range over the six independent entries of  $E$ ; its real and imaginary parts are two  $\mathbb{R}$ -linear equations whose coefficients are the real and imaginary parts of 1, the four entries of  $\tau$ , and  $\det \tau = \tau_{11}\tau_{22} - \tau_{12}\tau_{21}$ . A system of two real equations on six real unknowns cannot cut it down to the origin; its real solution space is four-dimensional. The point is that for generic  $\tau$  these coefficients are not  $\mathbb{Q}$ -linear combinations of one another, so the four-dimensional real kernel is not defined over  $\mathbb{Q}$  and meets the integer lattice  $\mathbb{Z}^6$  only at the origin, forcing  $E = 0$ . The two demands, that  $E$  be integral (which confines it to  $\mathbb{Z}^6$ ) and that it be compatible with the complex structure (which cuts out the  $\tau$ -dependent real kernel above), are thus jointly satisfiable by a nonzero  $E$  only when  $\tau$  lies in a countable union of proper analytic subsets; a generic  $\tau$  avoids all of them.

Granting that  $H = 0$  is the only integral Hermitian form, no positive such form exists, so  $X$  carries no polarization. Since  $X$  then admits no nonzero integral Hermitian form at all, its Néron-Severi group vanishes, and a complex torus whose line bundles are all topologically trivial has  $a(X) = 0$ . The full verification that the vanishing of the meromorphic function field follows from the absence of an integral Hermitian form, and the precise measure-theoretic statement of genericity, are carried out in Birkenhake and Lange, *Complex Abelian Varieties*; the mechanism recorded above is the linear-algebraic heart of that argument. A concrete instance is the non-symmetric matrix

$$\tau = \begin{pmatrix} 2^{1/3}i & \sqrt{2}i \\ \sqrt{3}i & 3i \end{pmatrix}$$

for which  $\text{Im}\tau$  is invertible, so  $\Lambda$  is a full lattice. Here the two real equations read  $-a + (3 \cdot 2^{1/3} - \sqrt{6})f = 0$  and  $\sqrt{2}b + 3c - 2^{1/3}d - \sqrt{3}e = 0$ . The first forces  $f = 0$  and hence  $a = 0$ , since  $3 \cdot 2^{1/3} - \sqrt{6}$  is irrational; the second forces  $b = c = d = e = 0$ , since  $1, \sqrt{2}, 2^{1/3}, \sqrt{3}$  are linearly independent over  $\mathbb{Q}$ . Thus  $E = 0$  is the only integral compatible form, so this particular  $X$  carries no polarization and has only constant meromorphic functions. That  $\tau$  is non-symmetric is what forces the vanishing: a symmetric  $\tau$  with  $\text{Im}\tau$  positive definite lies in the Siegel upper half-space and always furnishes a principal polarization, as section 3.3 makes precise.

Two lessons stand behind this example. The first is that non-algebraic complex tori are not exotic: they are the generic case in every dimension  $g \geq 2$ , since the algebraicity conditions cut out a countable union of proper analytic subsets in the space of lattices, whose complement is everything. The second is that the obstruction is entirely linear algebraic, a question about whether a certain integral alternating form compatible with the complex structure exists, with no reference to the transcendental machinery of meromorphic functions once the equivalence  $a(X) = g \iff$  polarization is granted. This is what makes the algebraicity question tractable: it reduces to the arithmetic of forms on  $\Lambda$ , which is exactly the reduction chapter 3 carries out.

**1.4.4 Foreshadowing: The Riemann Relations** The condition separating the algebraic tori from the rest was named repeatedly above without being stated: a complex torus  $X = V/\Lambda$  is an abelian variety precisely when  $V$  carries a positive definite Hermitian form  $H$  whose imaginary part  $E = \text{Im}H$  takes integer values on  $\Lambda$ . This is the content of the *Riemann relations*, proved as theorem 3.1.4 in chapter 3. The positivity of  $H$  is what produces enough sections of a line bundle to define a projective embedding, via the theta functions of chapter 4; the integrality of  $E = \text{Im}H$  on  $\Lambda$  is what makes the bundle descend to the quotient  $X$ . In dimension one every lattice admits such an  $H$ , which reproves proposition 1.4.2 from the general criterion; in dimension two or more the two demands, positivity and integrality, are jointly a strong constraint that example 1.4.3 shows a generic lattice fails. The organizing tension of Part I is exactly this gap: complex tori are effortless to construct and describe, yet only those satisfying the Riemann relations are algebraic, and the machinery of the next two chapters exists to characterize, and then to realize, that scarce algebraic subclass.

### Exercise 1.4.1

1. State precisely what it means, per definition 1.4.1, for a complex torus  $X = V/\Lambda$  to be an abelian variety. Then, for the map  $\varphi$  of proposition 1.4.2, verify directly from the Laurent expansions  $\wp(z) \sim z^{-2}$  and  $\wp'(z) \sim -2z^{-3}$  near  $z = 0$  that  $\varphi(z) \rightarrow [0 : 1 : 0]$  as  $z \rightarrow 0$ , so that the value  $\varphi(0) = [0 : 1 : 0]$  is the only one extending  $\varphi$  continuously across the lattice point.
2. Let  $X$  be a compact complex manifold of dimension  $g$  with algebraic dimension  $a(X) = \text{trdeg}_{\mathbb{C}} \mathcal{M}(X)$ . Show that  $a(X) = 0$  forces every holomorphic map  $X \rightarrow \mathbb{P}^N$  to be constant, and conclude that such an  $X$  is not an abelian variety. Where does the argument use that  $X$  is compact?
3. In the setting of example 1.4.3, the compatibility condition on an integral alternating form  $E$  becomes the vanishing of the single complex-linear form

$$-a - d\tau_{11} + b\tau_{12} - e\tau_{21} + f(\tau_{12}\tau_{21} - \tau_{11}\tau_{22}) + c\tau_{22}$$

in the six integer entries  $a, b, c, d, e, f$  of  $E$ . For the concrete matrix

$$\tau = \begin{pmatrix} 2^{1/3}i & \sqrt{2}i \\ \sqrt{3}i & 3i \end{pmatrix}$$

write out the real and imaginary parts of this form as two real-linear equations in  $a, \dots, f$ , and deduce that  $E = 0$  is the only integral solution, using the  $\mathbb{Q}$ -linear independence of  $1, \sqrt{2}, 2^{1/3}, \sqrt{3}$ .

4. The count in example 1.4.3 says that the compatibility condition  $E(iv, iw) = E(v, w)$  is two real-linear equations on the six real entries of  $E$ , leaving a four-dimensional real solution space.

Explain why this dimension count alone does not decide algebraicity, and identify precisely the extra input that lets a generic  $\tau$  force  $E = 0$  while a special  $\tau$  can admit a nonzero integral  $E$ . Illustrate the special case with a symmetric  $\tau$  in the Siegel upper half-space, exhibiting one nonzero integral compatible  $E$ .

5. Prove that every one-dimensional complex torus  $X = \mathbb{C}/\Lambda$  satisfies  $a(X) = 1$ , and hence is an abelian variety, by exhibiting a nonconstant meromorphic function on  $X$  and bounding  $a(X)$  above and below. You may use the properties of  $\wp$  recorded in proposition 1.4.2 together with the general bound  $0 \leq a(X) \leq \dim X$ . Explain how this recovers the conclusion of proposition 1.4.2 at the level of the function field.



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# *Line Bundles and the Appell-Humbert Theorem*

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Whether a complex torus embeds in projective space is a question about line bundles: an embedding into  $\mathbb{P}^N$  pulls back the hyperplane bundle to an ample line bundle on  $X$ , and conversely a positive enough line bundle manufactures an embedding. Before one can speak of positivity, one needs to know what the line bundles on a complex torus are. This chapter answers that completely. The Appell-Humbert theorem describes every line bundle on  $X = V/\Lambda$  by a startlingly small amount of data: a Hermitian form  $H$  on  $V$  whose imaginary part is integral on the lattice, together with a unitary semicharacter  $\chi$  recording how the bundle twists around the lattice directions.

The route there runs through the exponential sequence, which splits the Picard group into a continuous part (the bundles that are topologically trivial, parametrized by characters of the lattice) and a discrete part (the Néron-Severi group, measuring the topological type through the first Chern class). Reading the first Chern class as an alternating integer-valued form on  $\Lambda$  turns the analytic classification into a problem in the arithmetic of bilinear forms, and it is that translation which, in chapter 3, makes the algebraicity criterion computable. The chapter ends by realizing sections of a line bundle as classical theta functions, the holomorphic functions on  $V$  with prescribed quasi-periodicity that chapter 4 studies in earnest.

## 2.1 The Picard Group and the Exponential Sequence

A complex torus  $X = V/\Lambda$  is a compact complex Lie group, and section 1.3 exhausted the linear algebra of its cohomology: the integral cohomology is the exterior algebra  $\bigwedge^\bullet \Lambda^\vee$ , and the Hodge structure on  $H^1$  is the explicit decomposition  $V^* \oplus \bar{V}^*$ . What that analysis did not touch is the geometry that makes a torus into a variety in the first place, namely its holomorphic line bundles.

A line bundle is what allows a compact complex manifold to be mapped to projective space: a choice of line bundle  $L$  together with a basis of its global sections  $s_0, \dots, s_N$  produces the map  $x \mapsto [s_0(x) : \dots : s_N(x)]$  into  $\mathbb{P}^N$ , and every holomorphic map to projective space arises this way. The question of which complex tori are algebraic, raised in section 1.4 and answered in chapter 3, is at bottom the question of which tori carry enough line bundles to separate their points. This chapter classifies all of them.

This section sets up the bookkeeping. The holomorphic line bundles on any complex manifold form a group, the Picard group, and on a complex manifold there is one exact sequence of sheaves that controls it completely: the exponential sequence. Feeding it through the long exact sequence in cohomology splits the Picard group into a continuous part and a discrete part, and identifies each with a piece of the linear algebra already in hand. The continuous part turns out to be a complex torus in its own right, the discrete part a lattice of alternating forms. The next two sections make each part fully explicit; here the exponential sequence does the separating.

**Definition 2.1.1: Picard Group**

Let  $X$  be a complex manifold and let  $\mathcal{O}_X^*$  denote the sheaf of nowhere-vanishing holomorphic functions on  $X$ , a sheaf of abelian groups under multiplication. The *Picard group* of  $X$  is

$$\mathrm{Pic}(X) = H^1(X, \mathcal{O}_X^*)$$

the first sheaf cohomology of  $X$  with coefficients in  $\mathcal{O}_X^*$ . Its elements are the isomorphism classes of holomorphic line bundles on  $X$ , the group operation is tensor product  $L \otimes L'$ , the identity is the trivial bundle  $\mathcal{O}_X$ , and the inverse of  $L$  is the dual bundle  $L^{-1} = \mathrm{Hom}(L, \mathcal{O}_X)$ .

The identification of  $H^1(X, \mathcal{O}_X^*)$  with isomorphism classes of line bundles is the standard Čech description: a line bundle is glued from trivial pieces  $U_i \times \mathbb{C}$  by transition functions  $g_{ij} \in \mathcal{O}_X^*(U_i \cap U_j)$  satisfying the cocycle condition  $g_{ij}g_{jk} = g_{ik}$ , and two systems of transition functions define isomorphic bundles precisely when they differ by a coboundary. This is exactly the datum of a class in  $H^1(X, \mathcal{O}_X^*)$ , and tensor product of line bundles corresponds to multiplication of Čech cocycles, so the bijection is a group isomorphism. The full verification belongs to the general theory of line bundles on complex manifolds; see [16]. What matters here is that the Picard group is a cohomology group, because that is what makes it computable.

The reason line bundles govern maps to projective space is worth stating precisely, since it is the motivation for the entire chapter. On  $\mathbb{P}^N$  there is the hyperplane bundle  $\mathcal{O}(1)$ , the Serre twisting sheaf, whose global sections are exactly the linear forms  $X_0, \dots, X_N$ . Pulling back along a holomorphic map  $\varphi: X \rightarrow \mathbb{P}^N$  produces a line bundle  $L = \varphi^*\mathcal{O}(1)$  on  $X$  together with the  $N+1$  sections  $s_i = \varphi^*X_i$ , which have no common zero (the point  $[X_0 : \dots : X_N]$  is always defined). Conversely a line bundle  $L$  and a base-point-free system of sections  $s_0, \dots, s_N$  reconstruct  $\varphi$ . Thus classifying line bundles is the first step toward classifying the maps of  $X$  to projective space, and understanding  $\mathrm{Pic}(X)$  is prerequisite to understanding when  $X$  is projective at all.

**Example 2.1.2: The Picard Group of an Elliptic Curve**

For the elliptic curve  $E = \mathbb{C}/\Lambda$  with  $g = 1$ , the Picard group is understood through the degree homomorphism. A line bundle on a compact Riemann surface has a well-defined degree, the sum of the multiplicities of the zeros minus the poles of any meromorphic section, and this gives a surjective homomorphism  $\deg: \mathrm{Pic}(E) \rightarrow \mathbb{Z}$ . The kernel consists of the degree-zero line bundles,



and the Abel-Jacobi theorem for  $E$  ([4]) identifies it with  $E$  itself,

$$\mathrm{Pic}^0(E) \cong E \quad L \mapsto [\text{the point } p \text{ with } L \cong \mathcal{O}_E(p - 0)]$$

where  $0 \in E$  is the origin. The degree sequence  $0 \rightarrow \mathrm{Pic}^0(E) \rightarrow \mathrm{Pic}(E) \xrightarrow{\deg} \mathbb{Z} \rightarrow 0$  splits, since the class of  $\mathcal{O}_E(0)$  maps to  $1 \in \mathbb{Z}$ , giving

$$\mathrm{Pic}(E) \cong \mathbb{Z} \oplus E$$

The discrete factor  $\mathbb{Z}$  records the degree; the continuous factor  $E$  is the group of degree-zero bundles, itself an elliptic curve. This chapter recovers both factors for a general torus: this section identifies the continuous factor  $E$  as the dual torus, and section 2.3 replaces the discrete factor  $\mathbb{Z}$  by the group of integral alternating forms.

The example exhibits, in the case  $g = 1$ , exactly the two-part structure this section extracts in general: a continuous group of “degree-zero” line bundles, which is again a torus, and a discrete group of numerical invariants. The mechanism that separates the two parts for an arbitrary complex torus, and indeed for any compact complex manifold, is the exponential sequence.

The exponential sequence relates the additive sheaf  $\mathcal{O}_X$  to the multiplicative sheaf  $\mathcal{O}_X^*$  through the exponential map  $f \mapsto e^{2\pi i f}$ . The factor  $2\pi i$  is chosen so that the kernel is exactly the constant integer functions:  $e^{2\pi i f} = 1$  holds precisely when  $f$  is an integer. This is the one short exact sequence of sheaves available on every complex manifold, and its long exact cohomology sequence is the engine of the whole classification.

### Theorem 2.1.3: The Exponential Sheaf Sequence

Let  $X$  be a complex manifold. Write  $\underline{\mathbb{Z}}$  for the constant sheaf  $\mathbb{Z}$  on  $X$ . The sequence of sheaves of abelian groups

$$0 \rightarrow \underline{\mathbb{Z}} \rightarrow \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^* \rightarrow 0 \quad (2.1)$$

is exact, where the first map is the inclusion of locally constant integer functions and  $\exp(f) = e^{2\pi i f}$ . It induces a long exact sequence in cohomology

$$\cdots \rightarrow H^1(X, \underline{\mathbb{Z}}) \rightarrow H^1(X, \mathcal{O}_X) \rightarrow H^1(X, \mathcal{O}_X^*) \xrightarrow{\delta} H^2(X, \underline{\mathbb{Z}}) \rightarrow H^2(X, \mathcal{O}_X) \rightarrow \cdots \quad (2.2)$$

in which the middle term  $H^1(X, \mathcal{O}_X^*) = \mathrm{Pic}(X)$  is the Picard group and  $\delta$  is the connecting homomorphism.

#### Proof:

*Step 1: Exactness of the sheaf sequence (2.1).* Exactness of a sequence of sheaves is checked on stalks, so fix a point  $x \in X$  and work with germs of holomorphic functions at  $x$ . The inclusion  $\underline{\mathbb{Z}} \rightarrow \mathcal{O}_X$  sends an integer  $n$  to the germ of the constant function  $n$ ; it is injective because a nonzero constant is a nonzero germ. The map  $\exp$  sends a germ  $f$  to  $e^{2\pi i f}$ , which is a germ of a nowhere-vanishing holomorphic function, since  $e^{2\pi i f}$  never takes the value zero. Composing,  $\exp$  of an integer constant is  $e^{2\pi i n} = 1$ , so the image of  $\underline{\mathbb{Z}}$  lies in the kernel of  $\exp$ .

Conversely, suppose  $e^{2\pi i f} = 1$  as germs at  $x$ , with  $f$  holomorphic near  $x$ . Then  $2\pi i f$  takes values in  $2\pi i \mathbb{Z}$ ; since  $f$  is continuous and  $\mathbb{Z}$  is discrete,  $f$  is locally constant near  $x$  with integer value, hence a germ from  $\underline{\mathbb{Z}}$ . Thus the kernel of  $\exp$  is exactly the image of  $\underline{\mathbb{Z}}$ , which is exactness at  $\mathcal{O}_X$ .

*Step 2: Surjectivity of  $\exp$  on stalks.* Surjectivity of a map of sheaves is also a stalk condition. Let  $u$  be the germ at  $x$  of a nowhere-vanishing holomorphic function. On a small enough simply connected neighborhood  $U$  of  $x$ , the function  $u$  has no zeros, so a holomorphic logarithm  $\log u$  exists on  $U$ : since  $u: U \rightarrow \mathbb{C}^*$  and  $U$  is simply connected, the holomorphic 1-form  $\frac{du}{u}$  has a holomorphic primitive  $\ell$  on  $U$  with  $e^\ell = u$  after adjusting an additive constant. Setting  $f = \frac{1}{2\pi i} \ell$  gives a holomorphic germ with  $\exp(f) = e^{2\pi i f} = e^\ell = u$ . Hence every germ of  $\mathcal{O}_X^*$  lifts, and  $\exp$  is surjective as a map of sheaves. This existence of a local holomorphic logarithm is the one nontrivial input, and it is a theorem of one complex variable; see [16]. This establishes the exactness of (2.1).

*Step 3: The long exact sequence.* A short exact sequence of sheaves of abelian groups on any topological space induces a long exact sequence in sheaf cohomology, with connecting homomorphisms  $\delta: H^k(X, \mathcal{O}_X^*) \rightarrow H^{k+1}(X, \underline{\mathbb{Z}})$ . This is the derived-functor long exact sequence of the cohomology functor  $H^\bullet(X, -)$ , proved in [8] for the cohomology of any left-exact functor applied to a short exact sequence. Applied to (2.1), it reads

$$\cdots \rightarrow H^k(X, \underline{\mathbb{Z}}) \rightarrow H^k(X, \mathcal{O}_X) \rightarrow H^k(X, \mathcal{O}_X^*) \xrightarrow{\delta} H^{k+1}(X, \underline{\mathbb{Z}}) \rightarrow \cdots$$

and the segment in degrees 1 and 2 is exactly (2.2), with  $H^1(X, \mathcal{O}_X^*) = \text{Pic}(X)$  by definition 2.1.1, completing the proof.

The sheaf-cohomology group  $H^k(X, \underline{\mathbb{Z}})$  of the constant sheaf agrees with the singular cohomology  $H^k(X, \mathbb{Z})$  for a manifold, so the outer terms of (2.2) are the topological invariants computed in section 1.3, while the terms involving  $\mathcal{O}_X$  are the holomorphic ones. The exponential sequence is thus the precise mechanism that couples topology to holomorphic geometry: a line bundle is a class in the middle, and reading it leftward recovers its holomorphic content while reading it rightward through  $\delta$  recovers its topological content. Everything that follows is the extraction of these two readings.

The rightward reading is the more immediate one, because  $\delta$  carries a line bundle to a class in  $H^2(X, \mathbb{Z})$  that measures its topological type. This class is the first Chern class, and it is the primary discrete invariant of a line bundle.

#### Definition 2.1.4: First Chern Class

The *first Chern class* is the connecting homomorphism of the exponential sequence,

$$c_1: \text{Pic}(X) = H^1(X, \mathcal{O}_X^*) \xrightarrow{\delta} H^2(X, \mathbb{Z})$$

For a line bundle  $L$ , the class  $c_1(L) \in H^2(X, \mathbb{Z})$  is a homomorphism of abelian groups:  $c_1(L \otimes L') = c_1(L) + c_1(L')$ , since  $\delta$  is a homomorphism of cohomology groups.

The first Chern class is the topological shadow of a line bundle. Two line bundles with the same  $c_1$

are topologically isomorphic as complex line bundles, differing only in their holomorphic structure; conversely  $c_1$  records exactly the underlying topological line bundle, which on a manifold is classified by  $H^2(X, \mathbb{Z})$ . The image of  $c_1$  inside  $H^2(X, \mathbb{Z})$  is the group of topological types that can be realized holomorphically, and identifying it is the subject of section 2.3, where it emerges as the group of alternating integral forms  $E = \text{Im}H$ . For the moment the object of interest is the kernel.

**Proposition 2.1.5: The Degree-Zero Picard Group as a Complex Torus**

Let  $X = V/\Lambda$  be a complex torus of dimension  $g$ , and let  $\text{Pic}^0(X) = \ker(c_1) \subseteq \text{Pic}(X)$  be the group of line bundles with trivial first Chern class. Then the exponential sequence induces an isomorphism

$$\text{Pic}^0(X) \cong H^1(X, \mathcal{O}_X) / \text{im}(H^1(X, \mathbb{Z}) \rightarrow H^1(X, \mathcal{O}_X)) \quad (2.3)$$

Moreover  $H^1(X, \mathcal{O}_X) = H^{0,1} = \overline{V}^*$ , the space of antilinear functionals on  $V$ , and the image of  $H^1(X, \mathbb{Z}) \cong \Lambda^\vee$  is a full lattice in it. Consequently  $\text{Pic}^0(X)$  is a complex torus of dimension  $g$ .

**Proof:**

*Step 1: The kernel of  $c_1$  as a cokernel.* The relevant segment of the long exact sequence (2.2) is

$$H^1(X, \mathbb{Z}) \xrightarrow{\iota} H^1(X, \mathcal{O}_X) \xrightarrow{\exp_*} \text{Pic}(X) \xrightarrow{c_1} H^2(X, \mathbb{Z})$$

By exactness at  $\text{Pic}(X)$ , the kernel of  $c_1$  equals the image of  $\exp_*$ . By exactness at  $H^1(X, \mathcal{O}_X)$ , the kernel of  $\exp_*$  equals the image of  $\iota$ . Hence  $\exp_*$  induces an isomorphism from the cokernel of  $\iota$  onto its image  $\text{Pic}^0(X) = \ker(c_1)$ ,

$$\text{Pic}^0(X) \cong H^1(X, \mathcal{O}_X) / \text{im}(\iota)$$

which is (2.3).

*Step 2: Identifying  $H^1(X, \mathcal{O}_X)$ .* The Dolbeault isomorphism identifies  $H^1(X, \mathcal{O}_X)$  with the Hodge piece  $H^{0,1}(X)$  of  $H^1(X, \mathbb{C})$ . For a complex torus this piece was computed explicitly in proposition 1.3.4: the holomorphic cotangent space is  $H^{1,0} = V^*$ , spanned by  $dz_1, \dots, dz_g$ , and its conjugate is

$$H^1(X, \mathcal{O}_X) = H^{0,1} = \overline{V}^*$$

the space of  $\mathbb{C}$ -antilinear functionals on  $V$ , spanned by  $d\bar{z}_1, \dots, d\bar{z}_g$ . This has complex dimension  $g$ .

*Step 3: The image of  $H^1(X, \mathbb{Z})$  is a full lattice.* By theorem 1.3.2 the integral cohomology is  $H^1(X, \mathbb{Z}) \cong \Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$ , a free abelian group of rank  $2g$ . The map  $\iota$  is the composite of the change of coefficients  $H^1(X, \mathbb{Z}) \rightarrow H^1(X, \mathbb{C})$  with the projection  $H^1(X, \mathbb{C}) = H^{1,0} \oplus H^{0,1} \rightarrow H^{0,1}$  onto the second summand. The change of coefficients is injective with image the real form  $H^1(X, \mathbb{Z}) \otimes \mathbb{R} = H^1(X, \mathbb{R})$ , a real subspace of  $H^1(X, \mathbb{C})$  of real dimension  $2g$  that meets neither  $H^{1,0}$  nor  $H^{0,1}$  except in 0: indeed  $H^1(X, \mathbb{R})$  is the fixed space of complex conjugation, which interchanges  $H^{1,0}$  and  $H^{0,1}$ , so a real class lying in  $H^{0,1}$  would be conjugation-fixed and hence also in  $H^{1,0}$ , forcing it to be zero. It follows that the projection  $H^1(X, \mathbb{R}) \rightarrow H^{0,1}$  is a real-linear isomorphism onto the underlying real space of  $\overline{V}^*$ , of real dimension  $2g$ . The image  $\text{im}(\iota) = \iota(\Lambda^\vee)$  is therefore the image of the full lattice  $\Lambda^\vee$  under a real-linear isomorphism, hence a full lattice in  $H^{0,1} = \overline{V}^*$ .

*Step 4: The quotient is a complex torus.* Combining the three steps,

$$\mathrm{Pic}^0(X) \cong \overline{V^*} / \iota(\Lambda^\vee)$$

the quotient of the  $g$ -dimensional complex vector space  $\overline{V^*}$  by a full lattice. This is a complex torus of dimension  $g$  in the sense of definition 1.1.2, as we sought to show.

The proposition says that the degree-zero Picard group is not just a group but a complex torus of the same dimension as  $X$ , built from the antilinear dual  $\overline{V^*}$  and the lattice  $\Lambda^\vee$ . This torus is the *dual torus*  $\hat{X}$ , the central construction of chapter 5, where its geometry is developed in full and the pairing between  $X$  and  $\hat{X}$  made explicit. The description as  $\overline{V^*}/\Lambda^\vee$  is the analytic incarnation of the dual; the same object reappears in Part II as the identity component  $\mathrm{Pic}_{X/k}^0$  of the Picard scheme. For now the point is that the continuous part of  $\mathrm{Pic}(X)$  carries its own complex structure inherited from that of  $X$ .

There is a second description of  $\mathrm{Pic}^0(X)$  that the Appell-Humbert theorem of section 2.2 will make precise. A degree-zero line bundle is, by (2.3), a class in  $H^1(X, \mathcal{O}_X)$  modulo the integral lattice, and such a class is equivalent to a unitary character of the lattice  $\Lambda$ . Concretely, the isomorphism  $\mathrm{Pic}^0(X) \cong \mathrm{Hom}(\Lambda, U(1))$  presents a degree-zero bundle as a homomorphism  $\chi: \Lambda \rightarrow U(1)$ , the group of characters of  $\Lambda$ . This is the semicharacter datum with  $H = 0$  that theorem 2.2.8 attaches to a line bundle, and it is the concrete face of the abstract quotient (2.3).

#### Example 2.1.6: Degree-Zero Bundles on an Elliptic Curve

For  $g = 1$  the proposition reads  $\mathrm{Pic}^0(E) \cong \overline{V^*}/\Lambda^\vee$  with  $V = \mathbb{C}$  one-dimensional, so  $\overline{V^*} \cong \mathbb{C}$  and  $\Lambda^\vee$  is a rank-2 lattice in it. The quotient is a one-dimensional complex torus, and it is isomorphic to  $E$  itself, recovering the identification  $\mathrm{Pic}^0(E) \cong E$  of example 2.1.2. Under the character description, a degree-zero line bundle on  $E = \mathbb{C}/\Lambda$  corresponds to a character  $\chi: \Lambda \rightarrow U(1)$ ; writing  $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$ , such a character is determined by the two unit-modulus values  $\chi(1)$  and  $\chi(\tau)$ , so

$$\mathrm{Pic}^0(E) \cong \mathrm{Hom}(\Lambda, U(1)) \cong U(1) \times U(1)$$

as real Lie groups, a real 2-torus carrying the complex structure that makes it isomorphic to  $E$ . The trivial character  $\chi = 1$  is the trivial bundle  $\mathcal{O}_E$ , the origin of  $\mathrm{Pic}^0(E)$ .

The two examples bracket the general theory. The degree homomorphism of example 2.1.2 and the character description of example 2.1.6 are the  $g = 1$  shadows of the two exact-sequence readings: the rightward reading through  $c_1$  produces the discrete degree, and the leftward reading through  $\exp_*$  produces the continuous group of characters. For general  $g$  the same split holds, packaged in the short exact sequence extracted from the exponential sequence.

The exponential sequence has done its structural work. By definition 2.1.4 the map  $c_1$  is defined on all of  $\mathrm{Pic}(X)$ , and by proposition 2.1.5 its kernel is the complex torus  $\mathrm{Pic}^0(X)$ . The image of  $c_1$  is a subgroup of  $H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee$ , and section 2.3 will identify it as the group of alternating integral forms arising from Hermitian data, the Néron-Severi group  $\mathrm{NS}(X)$ . The result is the exact sequence

$$0 \longrightarrow \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}(X) \xrightarrow{c_1} \mathrm{NS}(X) \longrightarrow 0 \quad (2.4)$$

which separates every line bundle into its continuous invariant, a point of the dual torus  $\mathrm{Pic}^0(X)$ , and its discrete invariant, an alternating integral form  $c_1(L) \in \mathrm{NS}(X)$ . The continuous invariant is the content of a character of  $\Lambda$ ; the discrete invariant is the topological type. This is the clean

separation the exponential sequence provides, and the two invariants are exactly what the next two sections compute: section 2.2 realizes both parts at once through the Appell-Humbert data  $(H, \chi)$ , and section 2.3 pins down  $\text{NS}(X)$  as Hermitian forms with integral imaginary part.

### Exercise 2.1.1

1. Using definition 2.1.1, verify the group axioms claimed there directly from the description of line bundles by Čech transition functions. Show that under the correspondence between  $\text{Pic}(X)$  and systems of transition functions, the tensor product  $L \otimes L'$  corresponds to the product  $g_{ij}g'_{ij}$  of cocycles, that the trivial bundle  $\mathcal{O}_X$  corresponds to the constant cocycle  $g_{ij} = 1$ , and that the dual  $L^{-1}$  corresponds to  $g_{ij}^{-1}$ . Conclude that these operations make  $\text{Pic}(X)$  an abelian group.
2. In the exponential sheaf sequence (2.1) the first map  $\exp$  is  $f \mapsto e^{2\pi i f}$ . Explain precisely why the factor  $2\pi i$  is present by computing, for each complex constant  $c$ , the kernel of the map  $\mathbb{C} \rightarrow \mathbb{C}^*$  sending  $f \mapsto e^{cf}$  on locally constant functions, and determine for which  $c$  this kernel equals exactly the constant integer functions  $\mathbb{Z}$ . What sequence of sheaves results if  $c$  is replaced by 1 (that is, if one uses the ordinary exponential  $f \mapsto e^f$ )?
3. Let  $X = V/\Lambda$  be a complex torus of dimension  $g$ . Using proposition 2.1.5, compute the real dimension of  $\text{Pic}^0(X)$  as a real Lie group, and identify  $\text{Pic}^0(X)$  with a quotient of the form  $W/\Gamma$  for an explicit complex vector space  $W$  and lattice  $\Gamma$ . Then, for the elliptic curve  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$ , use the character description of example 2.1.6 to write down the two unit-modulus parameters determining a degree-zero line bundle, and identify which parameters give the trivial bundle  $\mathcal{O}_E$ .
4. example 2.1.2 shows  $\text{Pic}(E) \cong \mathbb{Z} \oplus E$  for an elliptic curve, splitting off the degree. Explain how this decomposition is the  $g = 1$  instance of the exact sequence (2.4), identifying the discrete quotient  $\text{NS}(E)$  with  $\mathbb{Z}$  and the kernel  $\text{Pic}^0(E)$  with  $E$ . Show first that (2.4) always splits as a sequence of abelian groups, since  $\text{NS}(X) \subseteq H^2(X, \mathbb{Z})$  is a finitely generated free abelian group and hence  $\mathbb{Z}$ -projective. Then explain that although a splitting always exists, in general there is no *canonical* one: a group-theoretic splitting  $s: \text{NS}(X) \rightarrow \text{Pic}(X)$  amounts to a lift of each generator of  $\text{NS}(X)$ , and for a general complex torus  $X$  of dimension  $g$  the group  $\text{NS}(X)$  has rank  $> 1$  with no distinguished basis (it records the alternating integral forms arising from Hermitian data of section 2.3), so no splitting is preferred over its many competitors. For  $E$ , by contrast,  $\text{NS}(E) \cong \mathbb{Z}$  has the distinguished positive generator  $\mathcal{O}_E(0)$  (the class of a point, an ample divisor of degree 1), and lifting that canonical generator gives a canonical splitting  $\text{Pic}(E) \cong \mathbb{Z} \oplus E$ .
5. Prove the assertion made in Step 3 of the proof of proposition 2.1.5: the real subspace  $H^1(X, \mathbb{R}) \subseteq H^1(X, \mathbb{C})$  meets each of  $H^{1,0}$  and  $H^{0,1}$  only in 0, and the projection  $H^1(X, \mathbb{R}) \rightarrow H^{0,1}$  is a real-linear isomorphism onto the underlying real space of  $\overline{V^*}$ . Use only that  $H^1(X, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$  with  $\overline{H^{1,0}} = H^{0,1}$ , that  $H^1(X, \mathbb{R})$  is the fixed space of complex conjugation, and a dimension count.
6. Prove that the connecting homomorphism produces the short exact sequence (2.4). Precisely: starting from the long exact sequence (2.2) and definition 2.1.4, show that  $\text{Pic}^0(X) = \ker(c_1)$  is the image of  $\exp_*$ , that the sequence

$$0 \longrightarrow \text{Pic}^0(X) \longrightarrow \text{Pic}(X) \xrightarrow{c_1} \text{im}(c_1) \longrightarrow 0$$

is exact, and that  $c_1$  is a homomorphism onto  $\text{im}(c_1)$ . Then explain why identifying  $\text{im}(c_1)$  with  $\text{NS}(X)$  (the content of section 2.3) gives (2.4).

## 2.2 Factors of Automorphy and the Appell-Humbert Theorem

The exponential sequence of section 2.1 organizes  $\text{Pic}(X)$  into a discrete part (the image of the first Chern class in  $H^2(X, \mathbb{Z})$ ) and a continuous part ( $\text{Pic}^0(X)$ ), but it does not yet exhibit a single line bundle concretely. To compute with a bundle one wants a formula: a rule for gluing the trivial bundle on  $V$  back into a bundle on the quotient  $X = V/\Lambda$ . Because  $V$  is contractible, every holomorphic line bundle on  $X$  pulls back to the trivial bundle on  $V$ , and the entire content of the bundle is packaged in how the two trivializations differ across the deck transformations by  $\Lambda$ . That packaging is a factor of automorphy. The main result of this section, the Appell-Humbert theorem, says that a factor of automorphy can always be normalized to a canonical form determined by a pair  $(H, \chi)$ : a Hermitian form on  $V$  and a unitary semicharacter of  $\Lambda$ . This turns the classification of line bundles into a problem in linear algebra over the lattice, and it is the computational engine for everything that follows.

A holomorphic line bundle on  $X$  pulled back to  $V$  becomes  $\pi^*L \cong V \times \mathbb{C}$ , since  $V$  is contractible and Stein. The lattice  $\Lambda$  acts on  $V \times \mathbb{C}$  by deck transformations covering its action on  $V$ , and any such action lifting translation must scale the fiber  $\mathbb{C}$  linearly and holomorphically. This is exactly the data of a factor of automorphy.

### Definition 2.2.1: Factor of Automorphy

A *factor of automorphy* for  $\Lambda$  acting on  $V$  is a holomorphic map

$$a: \Lambda \times V \rightarrow \mathbb{C}^*$$

satisfying the *cocycle condition*

$$a(\lambda + \mu, v) = a(\lambda, \mu + v) a(\mu, v) \quad \text{for all } \lambda, \mu \in \Lambda, v \in V$$

Given such an  $a$ , the group  $\Lambda$  acts on the trivial bundle  $V \times \mathbb{C}$  by

$$\lambda \cdot (v, t) = (v + \lambda, a(\lambda, v)t)$$

and the quotient

$$L(a) := (V \times \mathbb{C})/\Lambda$$

is a holomorphic line bundle on  $X = V/\Lambda$ .

The cocycle condition is precisely what makes the formula  $\lambda \cdot (v, t) = (v + \lambda, a(\lambda, v)t)$  a group action rather than a mere assignment: applying first  $\mu$  and then  $\lambda$  sends  $(v, t)$  to  $(v + \mu + \lambda, a(\lambda, \mu + v)a(\mu, v)t)$ , and this must agree with applying  $\lambda + \mu$  in one step, which sends  $(v, t)$  to  $(v + \lambda + \mu, a(\lambda + \mu, v)t)$ . Equating the fiber factors gives exactly  $a(\lambda + \mu, v) = a(\lambda, \mu + v)a(\mu, v)$ . Since  $\Lambda$  acts freely and properly discontinuously on  $V$  (hence on  $V \times \mathbb{C}$ ), the quotient  $L(a)$  is a manifold, and the projection  $V \times \mathbb{C} \rightarrow V$  descends to a holomorphic line bundle  $L(a) \rightarrow X$  whose fiber over  $v + \Lambda$  is the  $\mathbb{C}$  carried along  $v$ .

Two factors of automorphy can produce isomorphic bundles. An isomorphism of the trivialized bundles over  $V$  is a nowhere-vanishing holomorphic function  $h: V \rightarrow \mathbb{C}^*$ , and it intertwines the two  $\Lambda$ -actions exactly when the factors differ by the corresponding coboundary. This is the equivalence one must quotient by.

**Definition 2.2.2: Equivalence of Factors and Coboundaries**

A factor of automorphy  $a$  is a *coboundary* if there is a nowhere-vanishing holomorphic function  $h: V \rightarrow \mathbb{C}^*$  with

$$a(\lambda, v) = \frac{h(v + \lambda)}{h(v)} \quad \text{for all } \lambda \in \Lambda, v \in V$$

Two factors  $a, a'$  are *equivalent* if  $a'/a$  is a coboundary. The factors of automorphy form an abelian group under pointwise multiplication (the cocycles  $Z^1$ ), the coboundaries form a subgroup  $B^1$ , and the quotient  $Z^1/B^1$  is a group under multiplication.

The factor of automorphy is a group cocycle for  $\Lambda$  with coefficients in the multiplicative group  $\mathcal{O}_V^*$  of nowhere-vanishing holomorphic functions on  $V$ , where  $\Lambda$  acts on  $\mathcal{O}_V^*$  by translation. The cocycle condition of definition 2.2.1 is the 1-cocycle identity, and the coboundary relation of definition 2.2.2 is the image of the differential  $\mathcal{O}_V^* \rightarrow Z^1$ ,  $h \mapsto (\lambda \mapsto (v \mapsto h(v + \lambda)/h(v)))$ . Thus  $Z^1/B^1 = H^1(\Lambda, \mathcal{O}_V^*)$  in the sense of group cohomology ([8], group cohomology of a discrete group acting on a module). The construction  $a \mapsto L(a)$  realizes this cohomology group as line bundles.

**Proposition 2.2.3: Line Bundles as Factors of Automorphy**

The assignment  $a \mapsto L(a)$  induces an isomorphism of groups

$$Z^1/B^1 \xrightarrow{\sim} \text{Pic}(X), \quad [a] \mapsto [L(a)]$$

Every holomorphic line bundle on  $X$  arises from a factor of automorphy, and  $L(a) \cong L(a')$  as line bundles if and only if  $a$  and  $a'$  are equivalent.

**Proof:**

*Step 1: Every bundle arises from a factor.* Let  $L$  be a holomorphic line bundle on  $X$  and  $\pi: V \rightarrow X$  the projection. The pullback  $\pi^*L$  is a holomorphic line bundle on  $V$ . Since  $V = \mathbb{C}^g$  is contractible and Stein, every holomorphic line bundle on  $V$  is trivial:  $H^1(V, \mathcal{O}_V^*) = 0$ , because the exponential sequence on  $V$  gives  $H^1(V, \mathcal{O}_V) \rightarrow H^1(V, \mathcal{O}_V^*) \rightarrow H^2(V, \mathbb{Z})$  with both outer groups zero ([16], triviality of holomorphic line bundles on a contractible Stein manifold). Fix a trivialization  $\varphi: \pi^*L \xrightarrow{\sim} V \times \mathbb{C}$ .

The deck group  $\Lambda$  acts on  $\pi^*L$  (covering its translation action on  $V$ ) with quotient  $L$ , since  $\pi^*L = V \times_X L$ . Transporting this action through  $\varphi$  gives an action on  $V \times \mathbb{C}$  of the form  $\lambda \cdot (v, t) = (v + \lambda, a(\lambda, v)t)$  for some holomorphic  $a(\lambda, -): V \rightarrow \mathbb{C}^*$ ; that the action is by bundle automorphisms forces linearity in the fiber, and nowhere-vanishing since each  $\lambda$  acts invertibly. The associativity of the action is the cocycle condition. By construction  $L(a) = (V \times \mathbb{C})/\Lambda \cong \pi^*L/\Lambda = L$ .

*Step 2: Equivalent factors give isomorphic bundles.* Suppose  $a' = a \cdot \delta h$  with  $\delta h(\lambda, v) =$



$h(v + \lambda)/h(v)$  for a nowhere-vanishing holomorphic  $h: V \rightarrow \mathbb{C}^*$ . The map  $V \times \mathbb{C} \rightarrow V \times \mathbb{C}$ ,  $(v, t) \mapsto (v, h(v)t)$ , is a holomorphic bundle automorphism over  $V$ , and it intertwines the  $a$ -action with the  $a'$ -action:

$$(v, t) \xrightarrow{\lambda} (v + \lambda, a(\lambda, v)t) \xrightarrow{h} (v + \lambda, h(v + \lambda)a(\lambda, v)t)$$

while

$$(v, t) \xrightarrow{h} (v, h(v)t) \xrightarrow{\lambda} (v + \lambda, a'(\lambda, v)h(v)t)$$

and  $a'(\lambda, v)h(v) = a(\lambda, v)h(v + \lambda)$  by the coboundary relation. The two composites agree, so the automorphism descends to an isomorphism  $L(a) \cong L(a')$ .

*Step 3: Isomorphic bundles give equivalent factors.* Conversely, an isomorphism  $L(a) \cong L(a')$  pulls back to a  $\Lambda$ -equivariant bundle isomorphism  $V \times \mathbb{C} \rightarrow V \times \mathbb{C}$  over the identity of  $V$ , hence is multiplication by some nowhere-vanishing holomorphic  $h: V \rightarrow \mathbb{C}^*$  in each fiber. Equivariance is precisely  $a'(\lambda, v)h(v) = a(\lambda, v)h(v + \lambda)$ , that is,  $a'/a = \delta h$  is a coboundary. Steps 2 and 3 show  $L(a) \cong L(a')$  iff  $[a] = [a']$ .

*Step 4: The map is a homomorphism.* If  $a, a'$  are factors, the tensor product  $L(a) \otimes L(a')$  is  $(V \times \mathbb{C})/\Lambda$  for the fiberwise-product action, whose factor is  $a \cdot a'$ ; thus  $L(aa') \cong L(a) \otimes L(a')$ . The trivial bundle corresponds to the constant factor  $a \equiv 1$ . Therefore  $[a] \mapsto [L(a)]$  is a group homomorphism, and by Steps 1 and 3 it is a bijection, completing the proof.

This proposition reduces the classification of line bundles to the classification of factors of automorphy up to coboundary. The identity  $Z^1/B^1 \cong \text{Pic}(X)$  is not yet a computation: the space of holomorphic factors is enormous. The Appell-Humbert theorem accomplishes the reduction by showing that each equivalence class contains one, and only one, factor of a rigidly prescribed canonical shape. The shape involves a Hermitian form and a unitary semicharacter, which are introduced next.

The semicharacter is the discrete piece of the data. It records how the bundle winds around the lattice directions, and its defect from being an ordinary homomorphism is governed by the alternating form  $E = \text{Im}H$  evaluated on lattice vectors. Recall from section 1.3 the conventions on Hermitian forms fixed for this book:  $H: V \times V \rightarrow \mathbb{C}$  is linear in the first variable and conjugate-linear in the second, with  $H(w, v) = \overline{H(v, w)}$ , and  $E := \text{Im}H$  is an alternating  $\mathbb{R}$ -bilinear form satisfying  $E(iv, iw) = E(v, w)$ .

#### Definition 2.2.4: Semicharacter

Let  $H$  be a Hermitian form on  $V$  with  $E = \text{Im}H$  integral on  $\Lambda$ , that is,  $E(\Lambda, \Lambda) \subseteq \mathbb{Z}$ . A *semicharacter* for  $E$  is a map

$$\chi: \Lambda \rightarrow U(1) = \{z \in \mathbb{C} \mid |z| = 1\}$$

satisfying

$$\chi(\lambda + \mu) = \chi(\lambda) \chi(\mu) e^{i\pi E(\lambda, \mu)} \quad \text{for all } \lambda, \mu \in \Lambda$$

Since  $E(\lambda, \mu) \in \mathbb{Z}$ , the twisting factor is  $e^{i\pi E(\lambda, \mu)} = (-1)^{E(\lambda, \mu)}$ , so equivalently  $\chi(\lambda + \mu) = \chi(\lambda)\chi(\mu)(-1)^{E(\lambda, \mu)}$ . A pair  $(H, \chi)$  with  $H$  Hermitian,  $E = \text{Im}H$  integral on  $\Lambda$ , and  $\chi$  a semicharacter for  $E$  is called *Appell-Humbert data*.



A semicharacter is a homomorphism twisted by the sign cocycle  $(\lambda, \mu) \mapsto (-1)^{E(\lambda, \mu)}$ . When  $E$  vanishes on the lattice (in particular when  $H = 0$ ) the defect disappears and  $\chi$  is an ordinary group homomorphism  $\Lambda \rightarrow U(1)$ . In general the identity  $\chi(\lambda + \mu)/[\chi(\lambda)\chi(\mu)] = (-1)^{E(\lambda, \mu)}$  measures how far  $\chi$  is from being a character, and the discrepancy is symmetric in the sense that swapping  $\lambda, \mu$  multiplies by  $(-1)^{E(\mu, \lambda) - E(\lambda, \mu)} = (-1)^{-2E(\lambda, \mu)} = 1$ , consistent with the commutativity of  $\Lambda$ . That  $E$  is integral on  $\Lambda$  is exactly the condition needed for the sign to be well-defined; it is the integrality that section 2.3 identifies with the first Chern class lying in  $H^2(X, \mathbb{Z})$ .

Given Appell-Humbert data  $(H, \chi)$ , one now writes down the canonical factor and checks that it is a factor of automorphy. The Hermitian form supplies the continuous, curvature-carrying part of the factor, and the semicharacter supplies the discrete twisting; the two are locked together precisely by the semicharacter identity.

#### Definition 2.2.5: Appell-Humbert Bundle

Let  $(H, \chi)$  be Appell-Humbert data. The *canonical factor of automorphy* associated to  $(H, \chi)$  is

$$a_{(H, \chi)}(\lambda, v) = \chi(\lambda) \exp\left(\pi H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda)\right)$$

and the *Appell-Humbert bundle* is the line bundle

$$L(H, \chi) := L(a_{(H, \chi)}) = (V \times \mathbb{C})/\Lambda, \quad \lambda \cdot (v, t) = (v + \lambda, a_{(H, \chi)}(\lambda, v) t)$$

where  $H(v, \lambda)$  is linear in  $v$  and  $H(\lambda, \lambda) \in \mathbb{R}$  because  $H$  is Hermitian.

#### Proposition 2.2.6: The Canonical Factor is a Factor of Automorphy

For any Appell-Humbert data  $(H, \chi)$ , the map  $a_{(H, \chi)}$  is a factor of automorphy. Consequently  $L(H, \chi)$  is a well-defined holomorphic line bundle on  $X$ .

#### Proof:

Write  $a = a_{(H, \chi)}$  and  $f(\lambda, v) = \exp(\pi H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda))$ , so  $a(\lambda, v) = \chi(\lambda) f(\lambda, v)$ . For each fixed  $\lambda$ , the function  $v \mapsto a(\lambda, v)$  is holomorphic and nowhere-vanishing:  $H(v, \lambda)$  is  $\mathbb{C}$ -linear in  $v$ , so  $f(\lambda, -)$  is the exponential of a holomorphic function, and  $\chi(\lambda) \in U(1)$  is a nonzero constant. It remains to verify the cocycle condition

$$a(\lambda + \mu, v) = a(\lambda, \mu + v) a(\mu, v)$$

*Step 1: Expand the exponential part.* Using bilinearity of  $H$  (linear in the first slot, conjugate-linear in the second) compute the exponent of the right-hand product  $f(\lambda, \mu + v) f(\mu, v)$ :

$$\pi H(\mu + v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda) + \pi H(v, \mu) + \frac{\pi}{2} H(\mu, \mu)$$

Since  $H(\mu + v, \lambda) = H(\mu, \lambda) + H(v, \lambda)$ , this exponent equals

$$\pi H(v, \lambda) + \pi H(v, \mu) + \pi H(\mu, \lambda) + \frac{\pi}{2} (H(\lambda, \lambda) + H(\mu, \mu))$$

Now the exponent of the left-hand factor  $f(\lambda + \mu, v)$  is

$$\pi H(v, \lambda + \mu) + \frac{\pi}{2} H(\lambda + \mu, \lambda + \mu) = \pi H(v, \lambda) + \pi H(v, \mu) + \frac{\pi}{2} (H(\lambda, \lambda) + H(\lambda, \mu) + H(\mu, \lambda) + H(\mu, \mu))$$

using  $H(v, \lambda + \mu) = H(v, \lambda) + H(v, \mu)$  (additivity in the second slot, which holds for a conjugate-linear argument since conjugation affects only scalar multiplication, not addition) and the expansion of  $H(\lambda + \mu, \lambda + \mu)$ .

*Step 2: Compare the two exponents.* Subtracting the exponent of  $f(\lambda + \mu, v)$  from that of  $f(\lambda, \mu + v)f(\mu, v)$ , the terms  $\pi H(v, \lambda) + \pi H(v, \mu)$  and  $\frac{\pi}{2} (H(\lambda, \lambda) + H(\mu, \mu))$  cancel, leaving

$$\pi H(\mu, \lambda) - \frac{\pi}{2} (H(\lambda, \mu) + H(\mu, \lambda)) = \frac{\pi}{2} (H(\mu, \lambda) - H(\lambda, \mu))$$

By the Hermitian symmetry  $H(\lambda, \mu) = \overline{H(\mu, \lambda)}$ , the difference  $H(\mu, \lambda) - H(\lambda, \mu) = H(\mu, \lambda) - \overline{H(\mu, \lambda)} = 2i \operatorname{Im} H(\mu, \lambda) = 2i E(\mu, \lambda)$ . Therefore

$$\frac{f(\lambda, \mu + v) f(\mu, v)}{f(\lambda + \mu, v)} = \exp(i\pi E(\mu, \lambda))$$

*Step 3: Absorb the sign into the semicharacter.* Multiply through by the semicharacter values. The right-hand side of the cocycle is

$$a(\lambda, \mu + v) a(\mu, v) = \chi(\lambda) \chi(\mu) f(\lambda, \mu + v) f(\mu, v) = \chi(\lambda) \chi(\mu) e^{i\pi E(\mu, \lambda)} f(\lambda + \mu, v)$$

Because  $E$  is integral on  $\Lambda$  and alternating,  $E(\mu, \lambda) = -E(\lambda, \mu)$  is an integer, so  $e^{i\pi E(\mu, \lambda)} = (-1)^{E(\mu, \lambda)} = (-1)^{E(\lambda, \mu)} = e^{i\pi E(\lambda, \mu)}$ . The semicharacter identity of definition 2.2.4 then gives

$$\chi(\lambda) \chi(\mu) e^{i\pi E(\lambda, \mu)} = \chi(\lambda + \mu)$$

so that  $a(\lambda, \mu + v) a(\mu, v) = \chi(\lambda + \mu) f(\lambda + \mu, v) = a(\lambda + \mu, v)$ , which is the cocycle condition. Hence  $a$  is a factor of automorphy, and  $L(H, \chi)$  is a holomorphic line bundle on  $X$ , as we sought to show.

The proof shows exactly where the semicharacter condition comes from: it is forced. The exponential part  $f$  satisfies the cocycle condition only up to the sign  $e^{i\pi E(\mu, \lambda)}$ , and the twisting law  $\chi(\lambda + \mu) = \chi(\lambda) \chi(\mu) (-1)^{E(\lambda, \mu)}$  is designed to cancel precisely that sign. This is why one cannot use an ordinary character: the Hermitian part alone fails the cocycle identity by a sign, and only a semicharacter repairs it. The integrality  $E(\Lambda, \Lambda) \subseteq \mathbb{Z}$  is what makes  $(-1)^{E(\lambda, \mu)}$  meaningful and makes the two signs  $e^{\pm i\pi E(\lambda, \mu)}$  coincide on the lattice.

Before the general theorem, the canonical factor is worth seeing on the elliptic curve, where every quantity is a single complex number.

### Example 2.2.7: The Canonical Factor for an Elliptic Curve

Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ . Take the Hermitian form

$$H(v, w) = \frac{v \bar{w}}{\operatorname{Im} \tau}$$

which is positive definite, with imaginary part  $E(v, w) = \operatorname{Im}(v \bar{w} / \operatorname{Im} \tau)$ . On the ordered basis  $(1, \tau)$

of  $\Lambda$  one computes  $E(1, \tau) = \text{Im}(\bar{\tau}/\text{Im}\tau) = -1$  and  $E(\tau, 1) = +1$ , while  $E(1, 1) = E(\tau, \tau) = 0$ . Thus  $E$  takes integer values on  $\Lambda$  (it is  $\pm 1$  on the basis and extends  $\mathbb{Z}$ -bilinearly), so  $H$  is integral. A semicharacter  $\chi$  for  $E$  is determined by its values  $\chi(1), \chi(\tau) \in U(1)$  on the basis together with the twisting rule extending it to all of  $\Lambda$ . Taking  $\chi(1) = \chi(\tau) = 1$  and extending forces  $\chi(m+n\tau) = (-1)^{mn}$ , which satisfies the semicharacter identity: for  $\lambda = m_1 + n_1\tau$  and  $\mu = m_2 + n_2\tau$  one has  $E(\lambda, \mu) = m_2n_1 - m_1n_2$ , and  $\chi(\lambda + \mu)/[\chi(\lambda)\chi(\mu)] = (-1)^{(m_1+m_2)(n_1+n_2)-m_1n_1-m_2n_2} = (-1)^{m_1n_2+m_2n_1} = (-1)^{E(\lambda, \mu)}$ , the last equality because  $m_1n_2 + m_2n_1$  and  $m_2n_1 - m_1n_2$  differ by  $2m_1n_2$ . The canonical factor becomes, for  $\lambda \in \Lambda$  and  $v \in \mathbb{C}$ ,

$$a_{(H, \chi)}(\lambda, v) = \chi(\lambda) \exp\left(\frac{\pi v \bar{\lambda}}{\text{Im}\tau} + \frac{\pi}{2} \frac{|\lambda|^2}{\text{Im}\tau}\right)$$

This is the automorphy factor whose sections are the classical Jacobi theta functions, recovered in section 2.4; the exponential quadratic in  $\lambda$  is the source of the Gaussian factor in the theta series.

Everything is now in place for the central theorem of the chapter. It asserts that the map  $(H, \chi) \mapsto L(H, \chi)$  is a bijection onto the full Picard group: every line bundle is canonically one of these, and no two distinct data give the same bundle. The proof is a normalization argument. Given an arbitrary factor of automorphy, one first adjusts it by a coboundary so that its class is represented by a factor whose logarithm is affine in  $v$ , then reads off  $(H, \chi)$  from the coefficients. The technical heart is showing that every holomorphic factor is equivalent to one of this affine type, which rests on the vanishing  $H^1(V, \mathcal{O}_V) = 0$  from the Stein property of  $V$  together with the Hodge-theoretic description of  $\text{Pic}^0(X)$  established in section 2.1.

### Theorem 2.2.8: Appell-Humbert Theorem

Let  $X = V/\Lambda$  be a complex torus. The assignment

$$(H, \chi) \mapsto L(H, \chi)$$

is a bijection from the set of Appell-Humbert data (pairs  $(H, \chi)$  with  $H$  Hermitian,  $E = \text{Im}H$  integral on  $\Lambda$ , and  $\chi$  a semicharacter for  $E$ ) onto  $\text{Pic}(X)$ . Every holomorphic line bundle on  $X$  is isomorphic to a unique  $L(H, \chi)$ .

#### Proof:

By proposition 2.2.3,  $\text{Pic}(X) = Z^1/B^1$ , and by proposition 2.2.6 each  $L(H, \chi)$  is a line bundle. The proof establishes that every class in  $Z^1/B^1$  contains exactly one canonical factor  $a_{(H, \chi)}$ .

*Step 1: Passing to the logarithm.* Let  $a$  be any factor of automorphy. Because each  $a(\lambda, -): V \rightarrow \mathbb{C}^*$  is holomorphic on the simply connected  $V$ , it has a holomorphic logarithm: there exist holomorphic  $g_\lambda: V \rightarrow \mathbb{C}$  with  $a(\lambda, v) = \exp(2\pi i g_\lambda(v))$ , unique up to an additive integer constant. The cocycle condition  $a(\lambda + \mu, v) = a(\lambda, \mu + v)a(\mu, v)$  becomes, in the logarithm,

$$g_{\lambda+\mu}(v) \equiv g_\lambda(\mu + v) + g_\mu(v) \pmod{\mathbb{Z}}$$

This is a 1-cocycle for  $\Lambda$  with values in the holomorphic functions on  $V$ , twisted by  $\mathbb{Z}$ .

*Step 2: Normalizing to an affine cocycle.* The goal is to replace  $a$  by an equivalent factor whose

logarithm  $g_\lambda(v)$  is *affine* in  $v$ , that is, of the form  $g_\lambda(v) = \ell(\lambda, v) + c(\lambda)$  with  $\ell(\lambda, -): V \rightarrow \mathbb{C}$  a  $\mathbb{C}$ -linear functional depending on  $\lambda$  and  $c(\lambda)$  a constant. Modifying  $a$  by a coboundary  $\delta h$  with  $h = \exp(2\pi i k)$ ,  $k: V \rightarrow \mathbb{C}$  holomorphic, changes each  $g_\lambda$  by  $k(v + \lambda) - k(v)$ ; since  $V$  is Stein and simply connected, every holomorphic  $k: V \rightarrow \mathbb{C}$  is admissible.

The reduction is carried out on the second-order derivatives of  $g_\lambda$ . Each  $g_\lambda$  is holomorphic on  $V = \mathbb{C}^g$ , so its holomorphic Hessian

$$B_\lambda(v) := \left( \frac{\partial^2 g_\lambda}{\partial v_j \partial v_k}(v) \right)_{1 \leq j, k \leq g}$$

is a holomorphic map  $V \rightarrow \text{Sym}^2(\mathbb{C}^g)^\vee$ , symmetric in  $(j, k)$ . Differentiating the Step 1 relation twice in  $v$  (the additive  $\mathbb{Z}$ -constant and the first-order translation terms drop) gives

$$B_{\lambda+\mu}(v) = B_\lambda(v + \mu) + B_\mu(v)$$

so  $\lambda \mapsto B_\lambda$  is a 1-cocycle for  $\Lambda$  valued in the sheaf  $\mathcal{H} := \text{Hess}(\mathcal{O}_V)$  of holomorphic symmetric fields that arise as Hessians (equivalently the integrable ones,  $\partial_l \gamma_{jk} = \partial_j \gamma_{lk}$ ). Two vanishings on the Stein contractible  $V$  combine to make this cocycle a coboundary. First, the operator Hess is surjective on global sections: it sits in the exact sheaf sequence  $0 \rightarrow \mathcal{A} \rightarrow \mathcal{O}_V \xrightarrow{\text{Hess}} \mathcal{H} \rightarrow 0$ , where  $\mathcal{A}$  is the finite-dimensional space of affine functions (the kernel of Hess), and since  $H^1(V, \mathcal{A}) = 0$  ( $\mathcal{A}$  is a trivial finite-rank bundle and  $V$  is contractible) the global-section sequence  $\mathcal{O}_V(V) \rightarrow \mathcal{H}(V) \rightarrow H^1(V, \mathcal{A}) = 0$  shows every global integrable field, in particular each  $B_\lambda$ , is the Hessian of a global holomorphic function. Second, the  $\Lambda$ -cocycle  $\lambda \mapsto B_\lambda$  is a  $\Lambda$ -coboundary: because  $V = \mathbb{C}^g$  is holomorphically contractible with  $H^1(V, \mathcal{O}_V) = 0$  ([16]), the group cohomology  $H^1(\Lambda, \mathcal{O}_V(V)) = 0$  for the translation action ([8]), and its image under the Hess-surjection kills the class of  $B_\lambda$ ; concretely, solving the coboundary equation in  $\mathcal{O}_V(V)$  and applying Hess produces a global holomorphic  $k: V \rightarrow \mathbb{C}$  with  $\beta := \text{Hess}(k)$  satisfying  $B_\lambda(v) = \beta(v + \lambda) - \beta(v)$  for every  $\lambda$ . Replacing  $g_\lambda$  by  $g_\lambda - (k(\cdot + \lambda) - k)$  leaves a factor equivalent to  $a$  whose logarithm has Hessian  $B_\lambda - (\beta(\cdot + \lambda) - \beta) = 0$  for every  $\lambda$ . A holomorphic function on  $V$  with identically vanishing Hessian is affine, so after this coboundary adjustment the representative has

$$g_\lambda(v) = \ell(\lambda, v) + c(\lambda)$$

with  $c(\lambda) \in \mathbb{C}$  and  $\ell(\lambda, -) = dg_\lambda$  a functional. Because  $g_\lambda$  is holomorphic, its differential  $\ell(\lambda, -)$  is  $\mathbb{C}$ -linear, so  $\ell(\lambda, -) \in V^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$ . This  $\mathbb{C}$ -linearity is used decisively in Step 4.

*Step 3: The cocycle condition in affine form.* Substituting  $g_\lambda(v) = \ell(\lambda, v) + c(\lambda)$  into the Step 1 relation and using linearity of  $\ell(\lambda, -)$ ,

$$\ell(\lambda + \mu, v) + c(\lambda + \mu) \equiv \ell(\lambda, v) + \ell(\lambda, \mu) + c(\lambda) + \ell(\mu, v) + c(\mu) \pmod{\mathbb{Z}}$$

Comparing the parts that are linear in  $v$  gives

$$\ell(\lambda + \mu, v) = \ell(\lambda, v) + \ell(\mu, v)$$

so  $\lambda \mapsto \ell(\lambda, -)$  is additive in  $\lambda$ : it is a homomorphism  $\Lambda \rightarrow \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$ . Comparing the  $v$ -independent parts gives

$$c(\lambda + \mu) \equiv c(\lambda) + c(\mu) + \ell(\lambda, \mu) \pmod{\mathbb{Z}} \quad (*)$$

*Step 4: Extracting the Hermitian form.* Extend  $\lambda \mapsto \ell(\lambda, -)$  from  $\Lambda$  to all of  $V = \Lambda \otimes_{\mathbb{Z}} \mathbb{R}$  by  $\mathbb{R}$ -linearity in the first slot, writing  $\ell'(u, w)$  for the value of the extended functional with first argument  $u \in V$  evaluated at  $w \in V$ ; thus  $\ell'(u, -)$  is  $\mathbb{C}$ -linear in  $w$  for each  $u$ , and  $\ell'$  is  $\mathbb{R}$ -bilinear. Define an  $\mathbb{R}$ -bilinear alternating form on  $V$  by

$$E(v, w) := \ell'(w, v) - \ell'(v, w)$$

Antisymmetrizing Step 3's relation  $(*)$  (writing  $(*)$  with  $\lambda, \mu$  swapped and subtracting) eliminates  $c$ , yielding  $\ell(\mu, \lambda) - \ell(\lambda, \mu) \equiv 0 \pmod{\mathbb{Z}}$ ; since  $\ell'$  extends  $\ell$ , this reads

$$E(\lambda, \mu) \in \mathbb{Z} \quad \text{for } \lambda, \mu \in \Lambda$$

so  $E$  is an integer-valued alternating form on  $\Lambda$ , hence  $\mathbb{R}$ -valued on all of  $V$  by  $\mathbb{R}$ -bilinear extension.

It remains to see  $E$  is the imaginary part of a Hermitian form. Set

$$H(v, w) := E(iv, w) + i E(v, w)$$

following the correspondence fixed in section 1.3 ([7] for the general Hodge-structure statement). By the criterion recalled there,  $H$  is Hermitian, that is,  $\mathbb{C}$ -linear in  $v$  with  $H(w, v) = \overline{H(v, w)}$ , if and only if  $E$  satisfies the compatibility  $E(iv, iw) = E(v, w)$ . This compatibility is where the holomorphicity of  $a$  enters, through the  $\mathbb{C}$ -linearity of  $\ell(\lambda, -)$  established in Step 2. Compute the defect  $D(v, w) := E(iv, iw) - E(v, w)$  directly from the definition of  $E$  and the  $\mathbb{C}$ -linearity of  $\ell'$  in its second slot ( $\ell'(u, iw) = i \ell'(u, w)$ ):

$$D(v, w) = [\ell'(iw, iv) - \ell'(iv, iw)] - [\ell'(w, v) - \ell'(v, w)]$$

Using  $\ell'(iv, iw) = i \ell'(iv, w)$  and  $\ell'(iw, iv) = i \ell'(iw, v)$ , the bracketed difference  $\ell'(iw, iv) - \ell'(iv, iw) = i(\ell'(iw, v) - \ell'(iv, w))$ , so  $D$  is  $\mathbb{C}$ -bilinear in the complexified arguments, that is, a holomorphic alternating 2-form:  $D \in \bigwedge^2 V^*$ , of pure Hodge type  $(2, 0)$ . On the other hand  $E$  is  $\mathbb{R}$ -valued, so  $D = E(i \cdot, i \cdot) - E$  is  $\mathbb{R}$ -valued as well. A form that is simultaneously  $\mathbb{C}$ -bilinear (type  $(2, 0)$ ) and  $\mathbb{R}$ -valued must vanish: its complex conjugate is both equal to itself (reality) and of type  $(0, 2)$  (conjugating a  $(2, 0)$ -form), and  $(2, 0) \cap (0, 2) = 0$ . Hence  $D \equiv 0$ , giving  $E(iv, iw) = E(v, w)$ . Thus  $H$  is a Hermitian form with  $\text{Im} H = E$ : indeed  $\text{Im} H(v, w) = E(v, w)$  directly from the definition, and  $E$  is integral on  $\Lambda$ .

*Step 5: Matching the linear part to  $H$ .* The canonical factor  $a_{(H, 1)}$  (semicharacter  $\equiv 1$ ) has logarithm whose  $v$ -linear part is  $\frac{1}{2\pi i} \pi H(v, \lambda) = \frac{1}{2i} H(v, \lambda)$ , a  $\mathbb{C}$ -linear functional of  $v$ ; write  $L(\lambda, v) := \frac{1}{2i} H(v, \lambda)$ . Compare it with  $\ell(\lambda, v)$ , the  $\mathbb{C}$ -linear part produced in Step 2. Both are  $\mathbb{C}$ -linear in  $v$ , and both have the *same* alternating part under interchange of  $\lambda$  and  $v$ : indeed

$$L(\lambda, v) - L(v, \lambda) = \frac{1}{2i} (H(v, \lambda) - H(\lambda, v)) = \frac{1}{2i} (H(v, \lambda) - \overline{H(v, \lambda)}) = \text{Im} H(v, \lambda) = E(v, \lambda)$$

while  $\ell(\lambda, v) - \ell(v, \lambda) = \ell'(\lambda, v) - \ell'(v, \lambda) = E(v, \lambda)$  by the definition of  $E$  in Step 4. Their difference  $\ell(\lambda, v) - L(\lambda, v)$  is therefore symmetric in  $(\lambda, v)$  and  $\mathbb{C}$ -linear in each, hence the restriction to  $v \in \Lambda$  of a symmetric  $\mathbb{C}$ -bilinear form  $Q(u, v)$  on  $V$ . Setting  $k(v) := \frac{1}{2} Q(v, v)$ , a holomorphic quadratic, the coboundary  $\delta k$  has  $v$ -linear part  $k(v + \lambda) - k(v) = Q(\lambda, v) + \frac{1}{2} Q(\lambda, \lambda)$ , whose linear-in- $v$  term is exactly  $\ell(\lambda, v) - L(\lambda, v)$ . Subtracting this coboundary replaces  $\ell$  by  $L$ :

after one further coboundary adjustment the normalized logarithm has  $v$ -linear part  $\frac{1}{2i}H(v, \lambda)$ , so  $a$  and  $a_{(H,1)}$  have identical  $v$ -dependence up to a coboundary, and their ratio is a factor of automorphy  $\lambda \mapsto \chi(\lambda)$  with  $\chi(\lambda) \in \mathbb{C}^*$  independent of  $v$ .

*Step 6: Extracting the semicharacter.* The residual  $\chi = a/a_{(H,1)}$  inherits its cocycle law from  $(*)$ . Its cocycle condition reads  $\chi(\lambda + \mu) = \chi(\lambda)\chi(\mu)e^{2\pi i \rho(\lambda, \mu)}$ , where  $\rho(\lambda, \mu)$  is the discrepancy between the constant part of the logarithm of  $a$  and that of  $a_{(H,1)}$ . The canonical factor is a genuine cocycle (proposition 2.2.6), so its constant part is exact; the constant part of  $a$  obeys  $(*)$ , that is  $c(\lambda + \mu) - c(\lambda) - c(\mu) \equiv \ell(\lambda, \mu) \pmod{\mathbb{Z}}$ . Only the alternating part of  $\ell(\lambda, \mu)$  survives in  $\rho$  (the symmetric part was absorbed into  $H$  and  $a_{(H,1)}$  in Step 5), and that alternating part is

$$\frac{1}{2}(\ell(\lambda, \mu) - \ell(\mu, \lambda)) = \frac{1}{2}(\ell'(\lambda, \mu) - \ell'(\mu, \lambda)) = -\frac{1}{2}E(\lambda, \mu)$$

using  $E(\lambda, \mu) = \ell'(\mu, \lambda) - \ell'(\lambda, \mu)$  from Step 4. Since  $E(\lambda, \mu) \in \mathbb{Z}$ , one has  $e^{2\pi i (-\frac{1}{2}E(\lambda, \mu))} = e^{-i\pi E(\lambda, \mu)} = (-1)^{-E(\lambda, \mu)} = (-1)^{E(\lambda, \mu)} = e^{i\pi E(\lambda, \mu)}$ , so

$$\chi(\lambda + \mu) = \chi(\lambda)\chi(\mu)e^{i\pi E(\lambda, \mu)}$$

and  $\chi$  is a semicharacter for  $E$ . Finally,  $|\chi(\lambda)| = 1$ : writing  $\chi(\lambda) = |\chi(\lambda)|e^{i \arg \chi(\lambda)}$ , the modulus  $\lambda \mapsto \log |\chi(\lambda)|$  is an ordinary  $\mathbb{R}$ -valued homomorphism on  $\Lambda$  (the sign twist  $e^{i\pi E}$  is unimodular, so it drops out of the moduli), hence extends to an  $\mathbb{R}$ -linear functional  $u: V \rightarrow \mathbb{R}$ . To remove the modulus by a coboundary one must use a *holomorphic* gauge, as definition 2.2.2 demands: let  $\ell \in V^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$  be the unique  $\mathbb{C}$ -linear functional with  $\text{Re} \ell = u$ , which exists because  $l \mapsto \text{Re}(l(\cdot))$  is an  $\mathbb{R}$ -linear isomorphism  $V^* \xrightarrow{\sim} \text{Hom}_{\mathbb{R}}(V, \mathbb{R})$ . Then  $h(v) = e^{\ell(v)}$  is a nowhere-vanishing holomorphic function on  $V$ , its coboundary is  $\delta h(\lambda, v) = h(v + \lambda)/h(v) = e^{\ell(\lambda)}$ , and its modulus is  $|e^{\ell(\lambda)}| = e^{\text{Re} \ell(\lambda)} = e^{u(\lambda)} = |\chi(\lambda)|$ . Dividing the factor by this coboundary replaces  $\chi$  by  $\chi/|\chi|$  without changing the bundle and leaves  $\chi: \Lambda \rightarrow U(1)$ . Thus  $a$  is equivalent to  $a_{(H, \chi)}$ , proving *existence*: every line bundle is some  $L(H, \chi)$ .

*Step 7: Uniqueness.* Suppose  $L(H, \chi) \cong L(H', \chi')$ . By proposition 2.2.3 the factors  $a_{(H, \chi)}$  and  $a_{(H', \chi')}$  differ by a coboundary  $\delta h$ ,  $h = e^{2\pi i k}$ . Taking logarithms,  $\pi H(v, \lambda) + \frac{\pi}{2}H(\lambda, \lambda) - \pi H'(v, \lambda) - \frac{\pi}{2}H'(\lambda, \lambda) + \log \frac{\chi(\lambda)}{\chi'(\lambda)} \equiv 2\pi i(k(v + \lambda) - k(v)) \pmod{2\pi i\mathbb{Z}}$ . The left side is affine in  $v$  (linear part  $\pi(H - H')(v, \lambda)$ ), so  $k(v + \lambda) - k(v)$  is affine in  $v$  for every  $\lambda$ . Applying the holomorphic Hessian  $k'' := (\partial^2 k / \partial v_j \partial v_k)$ , affineness of the lattice-difference means  $k''(v + \lambda) = k''(v)$  for all  $\lambda \in \Lambda$ , so  $k''$  is  $\Lambda$ -periodic and descends to a holomorphic  $\text{Sym}^2 V^*$ -valued function on the compact torus  $X$ ; a holomorphic function on the compact connected  $X$  is constant, so  $k''$  is a constant symmetric form and  $k$  is a polynomial of degree  $\leq 2$ . The  $v$ -linear part of  $2\pi i \delta k$  is then a  $\mathbb{C}$ -bilinear form  $2\pi i Q(\lambda, v)$  *symmetric* in  $(\lambda, v)$ , whereas the  $v$ -linear part  $\pi(H - H')(v, \lambda)$  of the left side is  $\mathbb{C}$ -linear in  $v$  but conjugate-linear in  $\lambda$ . A  $\mathbb{C}$ -bilinear form and a form conjugate-linear in  $\lambda$  agree for all  $\lambda$  only if both vanish, so  $H - H' = 0$ , that is,  $H = H'$ . With  $H = H'$  the linear and quadratic parts cancel entirely, leaving  $\chi(\lambda)/\chi'(\lambda) = e^{2\pi i(k(v + \lambda) - k(v))}$  independent of  $v$  and of modulus 1. Since  $H = H'$ , the difference  $k(v + \lambda) - k(v)$  is  $v$ -independent, so  $k$  is affine, say  $k(v) = \psi(v) + c$  with  $\psi \in V^*$ ; then  $\chi(\lambda)/\chi'(\lambda) = e^{2\pi i \psi(\lambda)}$ . Both  $\chi$  and  $\chi'$  are semicharacters for the same  $E$ , so their ratio  $\lambda \mapsto e^{2\pi i \psi(\lambda)}$  is a genuine character  $\Lambda \rightarrow U(1)$ ; but a character valued in  $U(1)$  of the form  $e^{2\pi i \psi(\lambda)}$  with  $\psi$  a  $\mathbb{C}$ -linear functional forces  $\psi(\lambda) \in \mathbb{R}$  for all  $\lambda \in \Lambda$ , and a  $\mathbb{C}$ -linear functional real-valued on the full lattice  $\Lambda$  (hence on  $V = \Lambda \otimes_{\mathbb{Z}} \mathbb{R}$ ) vanishes identically. Thus the ratio is trivial and  $\chi = \chi'$ . Hence the data  $(H, \chi)$  is unique, completing the proof.

The Appell-Humbert theorem is the structural pillar of the analytic theory. It says that the infinite-dimensional-looking group  $\text{Pic}(X)$  is parametrized by a pair of finite pieces of linear-algebraic data: a Hermitian form, which is a matrix of complex numbers, and a semicharacter, which is a finite amount of  $U(1)$ -data on lattice generators. Every question about line bundles on  $X$  can be translated into this language: positivity of  $L$  becomes positivity of  $H$  (the polarization condition of section 3.1), the first Chern class becomes  $\text{Im}H$  (section 2.3), and the sections of  $L$  become theta functions with the quasi-periodicity dictated by  $a_{(H,\chi)}$  (section 2.4). The Hermitian form is the “continuous” invariant carrying the topological type, and the semicharacter is the “arithmetic” choice of a specific bundle within a fixed topological type.

Two immediate consequences pin down the pieces of  $\text{Pic}(X)$  predicted by the exponential sequence. When  $H = 0$ , the semicharacter condition degenerates to the character condition  $\chi(\lambda + \mu) = \chi(\lambda)\chi(\mu)$ , so  $\chi \in \text{Hom}(\Lambda, U(1))$ ; these are exactly the topologically trivial bundles. When  $H \neq 0$ , the class  $\text{Im}H$  is the first Chern class, and it ranges over the integral Hermitian forms. The theorem thus splits  $\text{Pic}(X)$  into a continuous torus of characters and a discrete lattice of Hermitian forms.

### Corollary 2.2.9: The Two Pieces of the Picard Group

Under the Appell-Humbert parametrization:

1. The bundles  $L(0, \chi)$  with  $\chi \in \text{Hom}(\Lambda, U(1))$  are exactly the elements of  $\text{Pic}^0(X)$ , and

$$\text{Pic}^0(X) \cong \text{Hom}(\Lambda, U(1))$$

2. There is an exact sequence

$$0 \rightarrow \text{Pic}^0(X) \rightarrow \text{Pic}(X) \xrightarrow{c_1} \text{NS}(X) \rightarrow 0, \quad c_1(L(H, \chi)) = H$$

where the quotient  $\text{NS}(X)$  is the group of Hermitian forms  $H$  with  $\text{Im}H$  integral on  $\Lambda$  that arise as first Chern classes, given its formal treatment as the Néron-Severi group in section 2.3.

### Proof:

1. When  $H = 0$  the factor  $a_{(0,\chi)}(\lambda, v) = \chi(\lambda)$  is a constant unitary factor, whose bundle  $L(0, \chi)$  is flat, hence topologically trivial and in  $\text{Pic}^0(X)$ . Conversely, by proposition 2.1.5 the group  $\text{Pic}^0(X)$  is the image of  $H^1(X, \mathcal{O}_X)$  in  $\text{Pic}(X)$ ; by exactness of the exponential sequence theorem 2.1.3 this image is the kernel of  $c_1$ , so  $\text{Pic}^0(X)$  consists exactly of the classes with trivial first Chern class. By the theorem these are the  $L(H, \chi)$  with  $\text{Im}H = 0$ , and a Hermitian form with  $\text{Im}H = 0$  is 0. The map  $\chi \mapsto L(0, \chi)$  is a homomorphism  $\text{Hom}(\Lambda, U(1)) \rightarrow \text{Pic}^0(X)$  by proposition 2.2.10, injective and surjective by the theorem’s bijectivity restricted to  $H = 0$ .
2. The map  $c_1$  sends  $L(H, \chi)$  to  $H$ ; it is well-defined and surjective onto  $\text{NS}(X)$  because every integral Hermitian  $H$  admits at least one semicharacter: fixing a  $\mathbb{Z}$ -basis of  $\Lambda$  and any integral bilinear  $B$  with  $B - B^t = E$  (for instance the strictly-upper-triangular part of  $E$  in that basis), the formula  $\chi(\lambda) = (-1)^{B(\lambda, \lambda)}$  satisfies the semicharacter identity, since  $B(\lambda + \mu, \lambda + \mu) - B(\lambda, \lambda) - B(\mu, \mu) = B(\lambda, \mu) + B(\mu, \lambda) \equiv B(\lambda, \mu) - B(\mu, \lambda) = E(\lambda, \mu) \pmod{2}$ . Its kernel is the set of data with  $H = 0$ , which is  $\text{Pic}^0(X)$  by part (1). Exactness

follows, completing the proof.

The group structure on Appell-Humbert data is the last piece needed to make the parametrization an isomorphism of groups rather than of sets. Tensoring line bundles multiplies factors of automorphy, and the canonical factors multiply by adding Hermitian forms and multiplying semicharacters, so the bijection of the theorem is a group isomorphism.

**Proposition 2.2.10: Group Law on Appell-Humbert Data**

The set of Appell-Humbert data is a group under

$$(H_1, \chi_1) \cdot (H_2, \chi_2) = (H_1 + H_2, \chi_1 \chi_2)$$

and  $(H, \chi) \mapsto L(H, \chi)$  is an isomorphism of this group onto  $\text{Pic}(X)$  (definition 2.1.1). Under it, tensor product of bundles corresponds to the sum  $(H_1 + H_2, \chi_1 \chi_2)$ , the trivial bundle to  $(0, 1)$ , and the inverse  $L(H, \chi)^{-1}$  to  $(-H, \chi^{-1})$ .

**Proof:**

First,  $(H_1 + H_2, \chi_1 \chi_2)$  is again valid data:  $H_1 + H_2$  is Hermitian with  $\text{Im}(H_1 + H_2) = E_1 + E_2$  integral on  $\Lambda$ , and  $\chi_1 \chi_2$  is a semicharacter for  $E_1 + E_2$  because

$$(\chi_1 \chi_2)(\lambda + \mu) = \chi_1(\lambda) \chi_1(\mu) e^{i\pi E_1(\lambda, \mu)} \chi_2(\lambda) \chi_2(\mu) e^{i\pi E_2(\lambda, \mu)} = (\chi_1 \chi_2)(\lambda) (\chi_1 \chi_2)(\mu) e^{i\pi (E_1 + E_2)(\lambda, \mu)}$$

The canonical factors multiply directly:

$$a_{(H_1, \chi_1)}(\lambda, v) a_{(H_2, \chi_2)}(\lambda, v) = \chi_1(\lambda) \chi_2(\lambda) \exp\left(\pi(H_1 + H_2)(v, \lambda) + \frac{\pi}{2}(H_1 + H_2)(\lambda, \lambda)\right) = a_{(H_1 + H_2, \chi_1 \chi_2)}(\lambda, v)$$

Since tensor product of line bundles multiplies factors of automorphy (proposition 2.2.3, Step 4),  $L(H_1, \chi_1) \otimes L(H_2, \chi_2) \cong L(H_1 + H_2, \chi_1 \chi_2)$ . The identity element is the trivial data  $(0, 1)$ , whose factor is  $\equiv 1$  (the trivial bundle  $\mathcal{O}_X$ ), and inverses are  $(-H, \chi^{-1})$  by the same multiplication. The map is therefore a group homomorphism, and it is a bijection by theorem 2.2.8, hence a group isomorphism onto  $\text{Pic}(X)$ , as we sought to show.

The isomorphism identifies the flat bundles  $L(0, \chi)$  with  $\text{Hom}(\Lambda, U(1))$  as a subgroup. Here  $\chi: \Lambda \rightarrow U(1)$  is a homomorphism of *abstract groups* from the lattice to the circle, not a homomorphism of tori; the group  $\text{Hom}(\Lambda, U(1))$  of all such characters is itself a real torus of dimension  $2g$ , the dual real torus of  $\Lambda$ . The lattice-level maps recorded by  $\text{Hom}(\Lambda, U(1))$  are the same kind of  $\Lambda$ -datum that underlies a homomorphism of complex tori (definition 1.1.6), where a holomorphic homomorphism  $X \rightarrow X'$  is pinned down by a  $\mathbb{Z}$ -linear map  $\Lambda \rightarrow \Lambda'$ ; the exponential sequence theorem 2.1.3 exhibited  $\text{Pic}^0(X)$  as  $H^1(X, \mathcal{O}_X)/H^1(X, \mathbb{Z})$ , and the present description  $\text{Pic}^0(X) \cong \text{Hom}(\Lambda, U(1))$  is the same group read through characters.

The worked example that closes the section returns to the elliptic curve and recovers, in the  $(H, \chi)$  language, the line bundles the reader already knows from the theory of divisors on a genus-one curve. On  $E = \mathbb{C}/\Lambda$ , line bundles are classified by degree and by a point of  $E$  (the divisor class of degree zero), and this classical picture matches the Appell-Humbert data exactly.



**Example 2.2.11: Line Bundles on an Elliptic Curve**

Let  $E = \mathbb{C}/\Lambda$ ,  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ . Here  $g = 1$ , and a Hermitian form on  $V = \mathbb{C}$  is  $H(v, w) = \alpha v\bar{w}$  for a real scalar  $\alpha \in \mathbb{R}$ , with  $E = \text{Im}H$  given by  $E(v, w) = \alpha \text{Im}(v\bar{w})$ .

1. *The topologically trivial bundles.* Taking  $H = 0$  forces  $\chi \in \text{Hom}(\Lambda, U(1))$ , a genuine character determined freely by  $\chi(1), \chi(\tau) \in U(1)$ . Thus

$$\text{Pic}^0(E) \cong \text{Hom}(\Lambda, U(1)) \cong U(1) \times U(1)$$

a real 2-torus. This matches the classical isomorphism  $\text{Pic}^0(E) \cong E$ : the degree-zero divisor classes on  $E$  form a group isomorphic to  $E$  itself, and  $E = \mathbb{C}/\Lambda$  is precisely the 2-torus  $U(1) \times U(1)$  as a real Lie group. A character  $\chi$  with  $\chi(1) = e^{2\pi i s}$ ,  $\chi(\tau) = e^{2\pi i t}$  corresponds to the point  $s + t\tau \in E$  (up to the normalization fixing the isomorphism), so the flat bundle  $L(0, \chi)$  is the degree-zero bundle  $\mathcal{O}_E(P - 0)$  associated to that point  $P$ .

2. *Integrality and the degree.* For  $H \neq 0$ , integrality of  $E = \text{Im}H$  on  $\Lambda$  requires  $E(1, \tau) \in \mathbb{Z}$ . With  $H(v, w) = \alpha v\bar{w}$  one has  $E(1, \tau) = \alpha \text{Im}\bar{\tau} = -\alpha \text{Im}\tau$ , so integrality forces  $\alpha \text{Im}\tau \in \mathbb{Z}$ , that is,  $\alpha = d/\text{Im}\tau$  for the integer  $d = E(\tau, 1) = \alpha \text{Im}\tau$ . This integer  $d$  is the degree of the line bundle: the first Chern class  $c_1(L(H, \chi)) = E$  evaluated on the fundamental class of  $E$  is  $E(\tau, 1) = d$  (section 2.3 makes this identification precise). Positivity  $H > 0$  (equivalently  $\alpha > 0$ ) corresponds to  $d > 0$ , an ample bundle.
3. *The degree-one theta bundle.* The choice  $d = 1$ , that is,

$$H(v, w) = \frac{v\bar{w}}{\text{Im}\tau}, \quad E(\tau, 1) = 1$$

with any semicharacter  $\chi$  for this  $E$  (for instance  $\chi(m + n\tau) = (-1)^{mn}$ , as in example 2.2.7), gives the positive line bundle of degree one. This is the principal polarization: as section 3.4 will show,  $\dim H^0(E, L) = 1$ , so it has a unique section up to scalar, whose zero divisor is a single point. That section is the classical Riemann theta function, and the bundle  $L(H, \chi)$  is  $\mathcal{O}_E(P)$  for the base point  $P$  determined by  $\chi$ . Every positive line bundle on  $E$  is a translate-and-power of this one:  $L(dH, \chi') \cong \mathcal{O}_E(D)$  for a divisor  $D$  of degree  $d$ , recovering the classical statement that  $\text{Pic}^d(E) \cong E$  for each  $d$ .

The full picture is the exact sequence  $0 \rightarrow \text{Pic}^0(E) \rightarrow \text{Pic}(E) \xrightarrow{\deg} \mathbb{Z} \rightarrow 0$  of corollary 2.2.9, with  $\text{Pic}^0(E) \cong E$  the connected component and  $\deg = c_1$  the topological degree, exactly reproducing the classification of line bundles on a genus-one curve by degree together with a point.

**Exercise 2.2.1**

1. Fix Appell-Humbert data  $(H, \chi)$  and write  $a = a_{(H, \chi)}$  for the canonical factor of definition 2.2.5. Compute  $a(0, v)$  for the zero lattice vector directly from the formula, and verify that the value you obtain is forced by the cocycle condition of definition 2.2.1 (that is, show any factor of automorphy satisfies  $a(0, v) = 1$  for all  $v$ ).
2. Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ , and take the positive definite Hermitian form  $H(v, w) = v\bar{w}/\text{Im}\tau$  of example 2.2.7. Consider the two candidate maps  $\chi_1(m + n\tau) = 1$  and  $\chi_2(m + n\tau) = (-1)^{mn}$  from  $\Lambda$  to  $U(1)$ . Using  $E(m_1 + n_1\tau, m_2 + n_2\tau) = m_2n_1 - m_1n_2$ , decide

which of  $\chi_1, \chi_2$  is a semicharacter for  $E = \text{Im}H$  in the sense of definition 2.2.4, and which is not.

3. Let  $H$  be Hermitian with  $E = \text{Im}H$  integral on  $\Lambda$ , and suppose  $\chi, \chi'$  are both semicharacters for the same  $E$ . Prove that the ratio  $\chi/\chi': \Lambda \rightarrow U(1)$  is an ordinary group homomorphism (a character), and conversely that if  $\chi$  is a semicharacter for  $E$  and  $\psi: \Lambda \rightarrow U(1)$  is a character then  $\chi\psi$  is again a semicharacter for  $E$ . Deduce that the set of semicharacters for a fixed integral  $E$  is a coset of  $\text{Hom}(\Lambda, U(1))$ , hence nonempty as soon as one semicharacter exists.
4. Let  $a$  be a factor of automorphy for  $\Lambda$  on  $V$  and let  $h: V \rightarrow \mathbb{C}^*$  be nowhere-vanishing holomorphic, with coboundary  $(\delta h)(\lambda, v) = h(v + \lambda)/h(v)$ . Prove directly from definition 2.2.1 that  $a \cdot \delta h$  is again a factor of automorphy, and that  $\delta h$  alone is a factor of automorphy. (This is the group-theoretic content behind definition 2.2.2: the coboundaries  $B^1$  form a subgroup of the cocycles  $Z^1$ .)
5. Work on the elliptic curve  $E = \mathbb{C}/\Lambda$  of example 2.2.11. Using the Appell-Humbert parametrization and the group law of proposition 2.2.10, prove that the tensor square  $L(H, \chi)^{\otimes 2}$  of the degree-one bundle  $H(v, w) = v\bar{w}/\text{Im}\tau$ ,  $\chi(m + n\tau) = (-1)^{mn}$  is the Appell-Humbert bundle  $L(2H, \chi^2)$ , and show its data has  $\chi^2 \equiv 1$ . Identify the degree of  $L(H, \chi)^{\otimes 2}$  via part (2) of example 2.2.11, and confirm the answer is consistent with  $c_1$  being a homomorphism as in corollary 2.2.9.
6. Prove that a flat unitary factor is a factor of automorphy and identify the bundle it produces. Precisely: let  $\chi: \Lambda \rightarrow U(1)$  be a group homomorphism and set  $a(\lambda, v) = \chi(\lambda)$  (independent of  $v$ ). Show  $a$  is a factor of automorphy, that it equals the canonical factor  $a_{(0, \chi)}$  of definition 2.2.5, and that  $L(a) = L(0, \chi)$  lies in  $\text{Pic}^0(X)$ , recovering the identification  $\text{Pic}^0(X) \cong \text{Hom}(\Lambda, U(1))$  of corollary 2.2.9. Explain where the semicharacter condition of definition 2.2.4 degenerates to the character condition in this case.

## 2.3 The Neron-Severi Group and the First Chern Class

The Appell-Humbert theorem describes a line bundle on  $X = V/\Lambda$  by a pair  $(H, \chi)$ , and the exponential sequence sorts these pairs into two registers: the semicharacter  $\chi$  varies continuously and records the topologically trivial part, while the Hermitian form  $H$  is the discrete invariant, the one that survives passage to cohomology. That discrete invariant is the subject of this section. The Hermitian forms  $H$  that occur, namely those whose imaginary part is integer-valued on the lattice, assemble into a finitely generated abelian group, the Neron-Severi group, and the map sending a bundle to its  $H$  is exactly the first Chern class. Everything that will distinguish an abelian variety from a general complex torus lives here: the question of projectivity, taken up in chapter 3, becomes the question of whether this group of forms contains one that is positive definite.

The technical heart of the section is a translation. A Hermitian form carries a complex structure inside it, while the first Chern class, living in  $H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee$ , is by nature a real alternating form on the lattice. These two kinds of object are the same data seen from two sides, and the dictionary between them is the correspondence  $H \leftrightarrow \text{Im}H$ . Making that dictionary precise turns the analytic classification of line bundles into a problem about integral alternating forms, which is arithmetic the reader can compute with. The elliptic curve shows the mechanism in miniature: on  $E = \mathbb{C}/\Lambda$  the Neron-Severi group is just  $\mathbb{Z}$ , and the integer attached to a line bundle is its degree.

The starting point is the observation, extracted from the Appell-Humbert theorem in section 2.2, that not every Hermitian form on  $V$  underlies a line bundle. The factor of automorphy is well defined on the lattice only when the semicharacter relation can be satisfied, and that relation forces  $\text{Im}H$  to take integer values on  $\Lambda$ . The forms meeting this constraint are precisely the ones that survive as invariants of bundles, and they deserve a name.

**Definition 2.3.1: Neron-Severi Group**

Let  $X = V/\Lambda$  be a complex torus. The *Neron-Severi group* of  $X$  is

$$\text{NS}(X) = \{ H : V \times V \rightarrow \mathbb{C} \text{ Hermitian} \mid \text{Im}H(\Lambda, \Lambda) \subseteq \mathbb{Z} \}$$

the group, under addition of forms, of Hermitian forms on  $V$  whose imaginary part is integer-valued on the lattice. Equivalently, by the Appell-Humbert theorem it is the image of the first Chern class

$$c_1 : \text{Pic}(X) \longrightarrow H^2(X, \mathbb{Z})$$

the group of line bundles modulo those that are topologically trivial.

Two descriptions of the same group are packaged here, and their agreement is the content of the section. The first is intrinsic linear algebra: a Hermitian form on  $V$  subject to one integrality constraint. The second is topological: the isomorphism classes of line bundles that remain once one forgets the continuous parameter  $\chi$ , recorded by their first Chern class. The Appell-Humbert theorem theorem 2.2.8 makes these coincide, since it identifies  $\text{Pic}(X)$  with the group of pairs  $(H, \chi)$  and identifies  $\text{Pic}^0(X)$ , the topologically trivial bundles, with the pairs  $(0, \chi)$ ; the quotient is exactly the group of admissible  $H$ . That  $\text{NS}(X)$  is a group under addition is immediate from the first description, and that it is free of finite rank will follow once it is realized inside  $\bigwedge^2 \Lambda^\vee$ , which is a free  $\mathbb{Z}$ -module of rank  $\binom{2g}{2}$ .

The name is borrowed from algebraic geometry, where for a smooth projective variety the Neron-Severi group is the group of divisors modulo algebraic equivalence, a finitely generated abelian group by the theorem of the base. For a complex torus the analytic definition above reproduces that group, and Part II will recover it a third time as  $\text{NS}(X) = \text{Pic}(X)/\text{Pic}^0(X)$  with  $\text{Pic}^0$  the connected component of the Picard scheme. The three faces of an abelian variety each supply their own construction of the same lattice of forms.

To compute with  $\text{NS}(X)$  one must resolve the tension already flagged: an element of  $\text{NS}(X)$  is a Hermitian form, an object built from the complex structure on  $V$ , whereas its first Chern class is an alternating form on the underlying real lattice, an object that has forgotten the complex structure. The bridge is the imaginary part. Passing from  $H$  to  $E = \text{Im}H$  discards no information provided one remembers a single compatibility that  $E$  inherits from the complex structure, and the whole of  $H$  can be rebuilt from  $E$  by an explicit formula. This is the central mechanism of the section.

**Proposition 2.3.2: Hermitian and Alternating Forms Correspond**

Let  $V$  be a complex vector space and  $\Lambda \subset V$  a full lattice. Sending a Hermitian form to its imaginary part,

$$H \mapsto E := \operatorname{Im} H$$

is a bijection from the set of Hermitian forms  $H$  on  $V$  whose imaginary part is integer-valued on  $\Lambda$  onto the set of alternating  $\mathbb{R}$ -bilinear forms  $E: V \times V \rightarrow \mathbb{R}$  that are integer-valued on  $\Lambda$  and satisfy the compatibility

$$E(iv, iw) = E(v, w) \quad \text{for all } v, w \in V \quad (2.5)$$

The inverse sends such an  $E$  to the Hermitian form

$$H(v, w) = E(iv, w) + iE(v, w) \quad (2.6)$$

Under the correspondence,  $H$  is positive definite if and only if the symmetric form  $(v, w) \mapsto E(iv, w)$  is positive definite.

**Proof:**

The argument has two directions, together with the verification of the reconstruction formula; the integrality condition rides along passively, since  $E = \operatorname{Im} H$  is integral on  $\Lambda$  exactly when the reconstructed  $H$  is admissible, so it suffices to establish the bijection between real forms and Hermitian forms and observe that it respects the integrality of  $E$ .

*Step 1: The imaginary part of a Hermitian form is alternating and complex-compatible.* Let  $H$  be Hermitian, so  $H$  is  $\mathbb{C}$ -linear in the first variable,  $\mathbb{C}$ -antilinear in the second, and  $H(w, v) = \overline{H(v, w)}$ . Write  $E = \operatorname{Im} H$  and  $S = \operatorname{Re} H$ , so  $H = S + iE$  with  $S, E$  real-valued. The Hermitian symmetry  $H(w, v) = \overline{H(v, w)}$  gives  $S(w, v) + iE(w, v) = S(v, w) - iE(v, w)$ , whence  $S$  is symmetric and  $E$  is alternating. For the compatibility, use that  $H$  is  $\mathbb{C}$ -linear in the first slot and  $\mathbb{C}$ -antilinear in the second:  $H(iv, iw) = i \cdot \bar{i} \cdot H(v, w) = i(-i)H(v, w) = H(v, w)$ . Taking imaginary parts of  $H(iv, iw) = H(v, w)$  yields (2.5). Since  $E = \operatorname{Im} H$  takes integer values on  $\Lambda$  by hypothesis,  $E$  lies in the target set.

*Step 2: The reconstruction formula recovers  $H$  from  $E$ .* Still with  $H = S + iE$  Hermitian, the real and imaginary parts are not independent, and  $S$  is expressible through  $E$  alone. Using  $\mathbb{C}$ -linearity in the first variable,  $H(iv, w) = iH(v, w) = i(S(v, w) + iE(v, w)) = -E(v, w) + iS(v, w)$ , while directly  $H(iv, w) = S(iv, w) + iE(iv, w)$ . Comparing imaginary parts of these two expressions gives

$$S(v, w) = E(iv, w) \quad (2.7)$$

Hence  $S$  is determined by  $E$ , and therefore so is  $H$ :

$$H(v, w) = S(v, w) + iE(v, w) = E(iv, w) + iE(v, w)$$

which is (2.6). So the map  $H \mapsto \operatorname{Im} H$  is injective.

*Step 3: Every complex-compatible alternating form arises this way.* Let  $E$  be a real alternating form with  $E(iv, iw) = E(v, w)$ , and define  $H$  by the formula (2.6),  $H(v, w) = E(iv, w) + iE(v, w)$ . It remains to check that  $H$  is Hermitian; then  $\operatorname{Im} H = E$  by construction and surjectivity follows.

First,  $H$  is  $\mathbb{R}$ -bilinear because  $E$  is. Since  $\mathbb{R}$ -linearity is already in hand,  $\mathbb{C}$ -linearity in the first

variable reduces to the single identity  $H(iv, w) = i H(v, w)$ . Expanding the left side by the definition and using  $E(i(iv), w) = E(-v, w) = -E(v, w)$ ,

$$H(iv, w) = E(i(iv), w) + i E(iv, w) = -E(v, w) + i E(iv, w)$$

while the right side is

$$i H(v, w) = i (E(iv, w) + i E(v, w)) = -E(v, w) + i E(iv, w)$$

and the two agree. For Hermitian symmetry,  $H(w, v) = \overline{H(v, w)}$  splits into the real-part identity  $E(iw, v) = E(iv, w)$  and the imaginary-part identity  $E(w, v) = -E(v, w)$ ; the latter is the alternating property of  $E$ . For the former, the compatibility (2.5) applied to the pair  $(iw, v)$  gives  $E(iw, v) = E(i \cdot iw, iv) = \overline{E(-w, iv)} = -E(w, iv) = E(iv, w)$ , the last step by the alternating property. So  $H(w, v) = \overline{H(v, w)}$ , and  $H$  is Hermitian. Antilinearity in the second variable follows from linearity in the first together with Hermitian symmetry.

*Step 4: Integrality and positivity.* If  $E$  is integer-valued on  $\Lambda$  then the reconstructed  $H$  has  $\text{Im}H = E$  integral on  $\Lambda$ , so  $H \in \text{NS}(X)$ ; conversely  $\text{Im}H$  integral is the defining condition on  $E$ . This matches the two sets slot for slot. Finally, by (2.7) the real part of  $H$  is the symmetric form  $S(v, w) = E(iv, w)$ , and  $H(v, v) = S(v, v) = E(iv, v)$  is real for every  $v$ , so  $H > 0$  precisely when this symmetric form is positive definite, as we sought to show.

The correspondence is the organizing principle of the entire analytic theory. On one side sits a Hermitian form, the natural home for positivity and hence for the notion of an ample line bundle. On the other sits an integral alternating form on  $\Lambda$ , the natural home for cohomology and hence for the first Chern class and the intersection theory of divisors. The single compatibility (2.5) is precisely the condition that an alternating form, a priori insensitive to the complex structure, be compatible with multiplication by  $i$ ; it is the analytic shadow of the Hodge-theoretic statement that the class lies in  $H^{1,1}$ . Chapter 3 will read the positivity clause as the Riemann relations, and the entire question of whether  $X$  is projective will reduce to whether the alternating form  $E$  can be chosen so that  $E(iv, w)$  is positive definite.

Before turning to the identification with the first Chern class, it is worth pinning down the compatibility condition on a concrete torus, where the abstract clause  $E(iv, iw) = E(v, w)$  becomes a linear equation on the entries of a matrix.

### Example 2.3.3: The Compatibility Condition on a Torus

Take  $g = 1$ ,  $V = \mathbb{C}$  with real basis  $\{1, i\}$ , and let  $E$  be a real alternating form on  $\mathbb{C}$ . An alternating form on a 2-dimensional real space is determined by a single number  $E(1, i) =: m$ , and every such form is automatically compatible with the complex structure:

$$E(i \cdot 1, i \cdot i) = E(i, -1) = -E(i, 1) = E(1, i)$$

so  $E(iv, iw) = E(v, w)$  holds for all  $v, w$  by bilinearity. Writing  $z = x + iy$  and  $w = u + iv$  and expanding  $E$  from  $E(1, i) = m$ ,  $E(1, 1) = E(i, i) = 0$  gives  $E(z, w) = m(xv - yu) = m \text{Im}(\bar{z}w)$ . The reconstructed Hermitian form  $H(z, w) = E(iz, w) + iE(z, w)$  then works out to

$$H(z, w) = -m \text{Re}(\bar{z}w) + im \text{Im}(\bar{z}w) = -m z \bar{w}$$

which is positive definite precisely when  $m < 0$ . So for  $g = 1$  every integral alternating form on  $\Lambda$  is complex-compatible, and the Neron-Severi group is all of  $\bigwedge^2 \Lambda^\vee \cong \mathbb{Z}$ , with the positive classes forming one of the two half-lines.

For  $g = 2$ , an alternating form on  $\mathbb{R}^4$  has  $\binom{4}{2} = 6$  parameters, while the compatibility (2.5) imposes linear relations cutting the space down. In the real basis  $\{e_1, ie_1, e_2, ie_2\}$ , the condition  $E(iv, iw) = E(v, w)$  leaves  $E(e_1, ie_1)$  and  $E(e_2, ie_2)$  free and imposes  $E(e_1, e_2) = E(ie_1, ie_2)$  together with  $E(e_1, ie_2) = -E(ie_1, e_2)$ , leaving a 4-dimensional space of compatible forms, matching the real dimension  $g^2 = 4$  of the space of Hermitian forms on  $\mathbb{C}^2$ .

The correspondence just established is stated for real vector spaces, but its content is arithmetic once restricted to the lattice: a complex compatible alternating form that is integer-valued on  $\Lambda$  is an element of  $\bigwedge^2 \Lambda^\vee$ , the group in which the first Chern class lives. What remains is to check that the alternating form the correspondence attaches to  $L(H, \chi)$  is its first Chern class, and not some other invariant that happens to resemble one. This is a direct computation with the connecting homomorphism of the exponential sequence.

**Proposition 2.3.4: The First Chern Class Is the Imaginary Part**

Let  $X = V/\Lambda$  and let  $L = L(H, \chi)$  be the line bundle attached by the Appell-Humbert theorem theorem 2.2.8 to the data  $(H, \chi)$ . Under the identification  $H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee$  of theorem 1.3.2, the first Chern class of  $L$  is

$$c_1(L) = E = \text{Im}H$$

the alternating integral form  $E(\lambda, \mu) = \text{Im}H(\lambda, \mu)$  on  $\Lambda$ . In particular  $c_1$  depends only on  $H$ , not on the semicharacter  $\chi$ , and the induced map  $\text{NS}(X) \rightarrow \bigwedge^2 \Lambda^\vee$ ,  $H \mapsto \text{Im}H$ , identifies  $\text{NS}(X)$  with the group of complex-compatible integral alternating forms on  $\Lambda$ .

**Proof:**

The first Chern class is the image of  $L$  under the connecting map  $\delta: H^1(X, \mathcal{O}_X^*) \rightarrow H^2(X, \mathbb{Z})$  of the long exact sequence attached to the exponential sequence  $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X^* \rightarrow 0$  (definition 2.1.4). Computing  $\delta$  amounts to lifting the cocycle that represents  $L$  from  $\mathcal{O}_X^*$  to  $\mathcal{O}_X$  and measuring the failure of the lift to be a cocycle, which lands in the constant sheaf  $\mathbb{Z}$ . The computation is cleanest in the description of  $L$  by a factor of automorphy, where the cocycle is given by the canonical factor from section 2.2.

*Step 1: From the factor of automorphy to a group 2-cocycle.* The bundle  $L(H, \chi)$  is  $V \times \mathbb{C}/\Lambda$  with  $\lambda \cdot (v, t) = (v + \lambda, a(\lambda, v)t)$ , where

$$a(\lambda, v) = \chi(\lambda) \exp\left(\pi H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda)\right)$$

is the canonical factor of automorphy. Since  $V$  is contractible, the cohomology of  $X = V/\Lambda$  with coefficients in a locally constant or holomorphic sheaf is computed by the group cohomology of  $\Lambda$  acting on functions on  $V$ ; this is the identification  $H^\bullet(X, \mathcal{O}_X) \cong H^\bullet(\Lambda, \mathcal{O}(V))$  underlying theorem 1.3.2. The line bundle  $L$  is represented in  $H^1(\Lambda, \mathcal{O}(V)^*)$  by the 1-cocycle  $\lambda \mapsto a(\lambda, -)$ , which satisfies the cocycle identity  $a(\lambda + \mu, v) = a(\lambda, \mu + v) a(\mu, v)$  dictated by the action. Choose a branch of the logarithm and lift  $a$  to a  $\mathbb{C}$ -valued cochain  $a(\lambda, v) = \exp(2\pi i g(\lambda, v))$  with

$$g(\lambda, v) = \frac{1}{2\pi i} \left( \pi H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda) \right) + c(\lambda) = \frac{1}{2i} H(v, \lambda) + \frac{1}{4i} H(\lambda, \lambda) + c(\lambda)$$

where  $c(\lambda) = \frac{1}{2\pi i} \log \chi(\lambda) \in \mathbb{R}$  is fixed modulo  $\mathbb{Z}$  by  $\chi(\lambda) = \exp(2\pi i c(\lambda))$ .

*Step 2: The connecting map is the coboundary of the lift.* The connecting homomorphism  $\delta$  sends the class of  $a$  to the class of the integer-valued 2-cocycle measuring the failure of the lift  $g$  to satisfy the same cocycle identity as  $a$ . Taking  $\frac{1}{2\pi i}$  log of the identity  $a(\lambda + \mu, v) = a(\lambda, \mu + v) a(\mu, v)$  turns products into sums up to an integer, and that integer is

$$F(\lambda, \mu) = g(\lambda + \mu, v) - g(\lambda, \mu + v) - g(\mu, v)$$

which lies in  $\mathbb{Z}$  and, as the computation now shows, is independent of  $v$ . Substitute the expression for  $g$  and collect by type of term. The terms linear in  $v$  contribute

$$\frac{1}{2i} (H(v, \lambda + \mu) - H(\mu + v, \lambda) - H(v, \mu)) = \frac{1}{2i} (-H(\mu, \lambda))$$

using additivity of  $H$  in each slot, so the  $v$ -dependence cancels. The quadratic terms  $\frac{1}{4i} H(-, -)$  on the diagonal contribute

$$\frac{1}{4i} (H(\lambda + \mu, \lambda + \mu) - H(\lambda, \lambda) - H(\mu, \mu)) = \frac{1}{4i} (H(\lambda, \mu) + H(\mu, \lambda))$$

and the semicharacter relation  $\chi(\lambda + \mu) = \chi(\lambda)\chi(\mu) e^{i\pi E(\lambda, \mu)}$  from definition 2.2.4 gives  $c(\lambda + \mu) - c(\lambda) - c(\mu) = \frac{1}{2} E(\lambda, \mu)$  modulo  $\mathbb{Z}$ .

*Step 3: Collecting the terms.* Adding the three contributions,

$$F(\lambda, \mu) = -\frac{1}{2i} H(\mu, \lambda) + \frac{1}{4i} (H(\lambda, \mu) + H(\mu, \lambda)) + \frac{1}{2} E(\lambda, \mu)$$

modulo  $\mathbb{Z}$ . The two  $H$ -terms combine to  $\frac{1}{4i} (H(\lambda, \mu) - H(\mu, \lambda)) = \frac{1}{4i} (H(\lambda, \mu) - \overline{H(\lambda, \mu)}) = \frac{1}{2} \text{Im} H(\lambda, \mu) = \frac{1}{2} E(\lambda, \mu)$ , using Hermitian symmetry  $H(\mu, \lambda) = \overline{H(\lambda, \mu)}$  and  $\frac{1}{2i} (z - \bar{z}) = \text{Im} z$ . Hence

$$F(\lambda, \mu) = \frac{1}{2} E(\lambda, \mu) + \frac{1}{2} E(\lambda, \mu) = E(\lambda, \mu)$$

which is integer-valued on  $\Lambda$  because  $E$  is integral there, so the congruence modulo  $\mathbb{Z}$  is an equality. The 2-cocycle  $F = E$  is alternating and  $\mathbb{Z}$ -bilinear, hence a well-defined element of  $\bigwedge^2 \Lambda^\vee \cong H^2(X, \mathbb{Z})$  (theorem 1.3.2). By construction it is  $\delta(L) = c_1(L)$ , so  $c_1(L) = E = \text{Im} H$ . The semicharacter entered only through the term  $\frac{1}{2} E(\lambda, \mu)$ , which combined with the Hermitian part to produce exactly  $E$ ; changing  $\chi$  within its coset alters  $c$  by a genuine character, whose coboundary is a symmetric integer-valued cochain and so does not change the alternating class  $F$ . Thus  $c_1(L)$  depends only on  $H$ , completing the proof.

The proposition closes the loop opened by the Appell-Humbert theorem. That theorem gave a bijection between line bundles and pairs  $(H, \chi)$ ; the exponential sequence organized  $\text{Pic}(X)$  into the extension  $0 \rightarrow \text{Pic}^0(X) \rightarrow \text{Pic}(X) \xrightarrow{c_1} \text{NS}(X) \rightarrow 0$ ; and now the first Chern class in that sequence is seen to be nothing other than the map  $H \mapsto \text{Im} H$  of proposition 2.3.2. The three descriptions of  $\text{NS}(X)$ , as admissible Hermitian forms, as the image of  $c_1$ , and as complex-compatible integral alternating forms on  $\Lambda$ , are now literally the same group, and one may compute in whichever is convenient. For questions of positivity one keeps  $H$ ; for questions of cohomology, intersection, and rank one passes to  $E$ .

Because  $E = \text{Im} H$  is an integral alternating form on the rank- $2g$  lattice  $\Lambda$ , the group  $\text{NS}(X)$  embeds in the free  $\mathbb{Z}$ -module  $\bigwedge^2 \Lambda^\vee$  of rank  $\binom{2g}{2}$ , and is therefore itself free of finite rank. Its rank is the invariant that measures how many independent line-bundle types a torus carries, and it earns a name.



**Definition 2.3.5: Picard Number**

The *Picard number* of a complex torus  $X = V/\Lambda$  is the rank of its Neron-Severi group,

$$\rho(X) = \text{rank}_{\mathbb{Z}} \text{NS}(X)$$

the number of independent classes of line bundles modulo topological triviality.

Two bounds frame the Picard number. On one side,  $\text{NS}(X)$  sits inside the real vector space of Hermitian forms on  $V = \mathbb{C}^g$ , which has real dimension  $g^2$ ; since  $\text{NS}(X)$  is a discrete subgroup of that space, its rank cannot exceed  $g^2$ , giving  $\rho(X) \leq g^2$ . On the other side,  $\rho(X) \geq 0$ , with equality exactly when the only admissible Hermitian form is  $H = 0$ , which is the generic situation for a complex torus of dimension at least two and the reason most complex tori are not projective. The gap between these bounds is where the arithmetic of the individual torus lives: a very special torus, such as one with complex multiplication or a product of isogenous factors, pushes  $\rho$  toward the upper bound  $g^2$ , while a very general torus sits at the floor  $\rho = 0$ . The elliptic curve, sitting at  $g = 1$ , has no room for either phenomenon: the bound reads  $0 \leq \rho \leq 1$ , and since an elliptic curve is projective it realizes  $\rho = 1$ .

**Example 2.3.6: The Neron-Severi Group of an Elliptic Curve**

Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ , as in example 1.1.4. Here  $g = 1$ , so  $V = \mathbb{C}$  is a real 2-dimensional space and every alternating form on  $V$  is automatically complex-compatible, as computed in example 2.3.3. An alternating integral form on  $\Lambda = \mathbb{Z}\langle 1, \tau \rangle$  is determined by its value on the basis pair, and every integer value occurs, so

$$\text{NS}(E) \cong \mathbb{Z}, \quad \rho(E) = 1$$

Under proposition 2.3.2 the positive generator is the class of the Hermitian form

$$H_1(z, w) = \frac{z\bar{w}}{\text{Im}\tau}$$

whose imaginary part  $E_1 = \text{Im}H_1$  is the alternating form with  $E_1(\tau, 1) = 1$ ; a general class is  $H_m = mH_1$  for  $m \in \mathbb{Z}$ , positive definite precisely when  $m > 0$ . The integer  $m$  is the degree of the corresponding line bundle:  $L(H_m, \chi)$  has  $\deg L = m$ , matching the classical fact from [4] that line bundles on an elliptic curve are classified up to  $\text{Pic}^0$  by their degree. The positive generator is the class of the origin as a divisor, the bundle  $\mathcal{O}_E(0)$  of degree 1, the principal polarization studied in chapter 3.

The elliptic curve exhausts the possibilities for  $g = 1$ , but as soon as  $g = 2$  the Picard number becomes a genuine invariant that separates one torus from another. A product of two elliptic curves already carries more line-bundle classes than either factor, because line bundles can be pulled back from each factor and, when the factors are isogenous, a further class appears from the graph of the isogeny. This is the first place where the higher-dimensional theory departs qualitatively from the elliptic-curve model.



**Example 2.3.7: The Neron-Severi Group of a Product  $E \times E$**

Let  $E = \mathbb{C}/\Lambda$  be an elliptic curve and  $X = E \times E$ , a complex torus of dimension  $g = 2$  with lattice  $\Lambda \oplus \Lambda \subset \mathbb{C}^2$  and upper bound  $\rho(X) \leq g^2 = 4$ . Three line-bundle classes are visible at once.

1. The pullback  $p_1^* \mathcal{O}_E(0)$  along the first projection, whose Chern class is the alternating form  $E_1$  supported on the first factor,  $E_1((\lambda, \lambda'), (\mu, \mu')) = E_0(\lambda, \mu)$  with  $E_0$  the generator of  $\text{NS}(E)$  from example 2.3.6.
2. The pullback  $p_2^* \mathcal{O}_E(0)$  along the second projection, with Chern class  $E_2((\lambda, \lambda'), (\mu, \mu')) = E_0(\lambda', \mu')$ .
3. The class of the diagonal  $\Delta \subset E \times E$ , the image of  $E$  under  $z \mapsto (z, z)$ ; its Chern class  $E_\Delta$  is not a combination of  $E_1$  and  $E_2$  because it pairs the two factors, taking the value  $E_0(\lambda, \mu')$  on the cross terms.

These three classes are  $\mathbb{Z}$ -linearly independent in  $\text{NS}(X)$ , so

$$\rho(E \times E) \geq 3$$

Whether the rank is exactly 3 or larger is decided by the arithmetic of  $E$ . If  $E$  has no complex multiplication, so  $\text{End}(E) = \mathbb{Z}$ , then the only homomorphisms  $E \rightarrow E$  are the integers  $[n]$ , and one checks that these three classes span:  $\rho(E \times E) = 3$ . If  $E$  has complex multiplication by an order in an imaginary quadratic field  $K$ , so  $\text{End}(E) \otimes \mathbb{Q} = K$  is 2-dimensional over  $\mathbb{Q}$ , then each endomorphism  $\alpha \in \text{End}(E)$  contributes the class of its graph  $\Gamma_\alpha = \{(z, \alpha z)\}$ , and these graphs supply a fourth independent class; the maximum  $\rho(E \times E) = 4 = g^2$  is attained. The Picard number thus reads off the endomorphism algebra of  $E$ , a first instance of the principle, developed fully in chapter 6, that the arithmetic of an abelian variety is encoded in its endomorphisms.

The two examples mark the two extremes the Picard number can occupy and show that the invariant is sensitive to arithmetic that is invisible at the level of the underlying real torus. What makes  $\text{NS}(X)$  the arithmetic heart of the classification is that every question about the algebraicity of  $X$  passes through it. A complex torus is a compact complex manifold regardless of its Neron-Severi group, but it is an abelian variety, embeddable in projective space, only when  $\text{NS}(X)$  contains a class that is not just nonzero but positive definite as a Hermitian form. The existence of such a class is a condition on the lattice  $\Lambda$  relative to the complex structure on  $V$ , and translating it into the language of the alternating form  $E$  produces the Riemann relations. That translation, and the definition of an abelian variety it makes possible, is the business of chapter 3, for which the correspondence  $H \leftrightarrow \text{Im}H$  of this section is the indispensable dictionary.

**Exercise 2.3.1**

1. Let  $V = \mathbb{C}$  and  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$  with  $\tau \in \mathbb{H}$ . Let  $E$  be the integral alternating form on  $\Lambda$  with  $E(1, \tau) = 3$ , extended  $\mathbb{R}$ -bilinearly to  $V$ . Verify that  $E$  satisfies the compatibility (2.5), use (2.6) to compute the associated Hermitian form  $H(z, w)$  explicitly, and determine for which sign of the coefficient  $H$  is positive definite.
2. Fix  $H \in \text{NS}(X)$  with imaginary part  $E = \text{Im}H$ . Show directly from the Hermitian symmetry  $H(w, v) = \overline{H(v, w)}$  that the real part  $S = \text{Re}H$  is a symmetric  $\mathbb{R}$ -bilinear form and that  $S(v, w) = E(iv, w)$ , without invoking proposition 2.3.2. Deduce that  $H$  is determined by  $E$

alone.

3. Let  $X = V/\Lambda$  be a complex torus of dimension  $g$ . Using proposition 2.3.4 and the description of  $\text{NS}(X)$  in definition 2.3.1, prove that  $c_1 : \text{Pic}(X) \rightarrow \bigwedge^2 \Lambda^\vee$  is a group homomorphism whose image is exactly the set of complex-compatible integral alternating forms on  $\Lambda$ , and that its kernel is  $\text{Pic}^0(X)$ . Conclude that  $\text{NS}(X)$  is free of finite rank and that the Picard number satisfies  $\rho(X) \leq g^2$ .
4. Let  $E$  be a complex-compatible integral alternating form on  $\Lambda$ , with reconstructed Hermitian form  $H$ . Prove that  $H$  is positive definite if and only if  $-H$  is negative definite and that the only  $H$  that is both positive semidefinite with a nonzero kernel and complex-compatible arises from an  $E$  that is degenerate as a form on  $V$ . Then, taking  $g = 1$  and  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$  as in example 2.3.6, show that no nonzero class in  $\text{NS}(E)$  is degenerate, so every nonzero class is either positive or negative definite.
5. Let  $E_0 = \mathbb{C}/\Lambda_0$  have no complex multiplication, and set  $X = E_0 \times E_0$ . Following example 2.3.7, write down the three alternating forms  $E_1, E_2, E_\Delta$  on  $\Lambda_0 \oplus \Lambda_0$  explicitly on a  $\mathbb{Z}$ -basis and prove they are  $\mathbb{Z}$ -linearly independent, so  $\rho(X) \geq 3$ . Then verify that each is complex-compatible, confirming that all three lie in  $\text{NS}(X)$ .

## 2.4 Sections and the Space $H^0(X, L)$

The Appell-Humbert theorem classifies every line bundle on  $X = V/\Lambda$  by a pair  $(H, \chi)$ , but a classification is not yet a geometry. What makes a line bundle useful is its space of global sections: a positive bundle carries enough sections to separate points and tangent vectors, and the map to projective space assembled from a basis of those sections is what realizes  $X$  as a projective variety. This section identifies the global sections of  $L(H, \chi)$  concretely. Because  $L(H, \chi)$  is built from the trivial bundle  $V \times \mathbb{C}$  by gluing along the factor of automorphy, a section is not an abstract cohomology class but a holomorphic function on  $V$  obeying a prescribed transformation law under the lattice. These functions are the theta functions, and the whole projective story of the next chapters rests on counting them.

Two facts organize the count. First, the transformation law is so rigid that a nonzero section can exist only when  $H$  is positive semidefinite: a Hermitian form that fails to be positive somewhere forces every candidate section to grow too slowly to survive on a compact torus, and Liouville's theorem kills it. Second, when  $H$  is strictly positive the sections form a finite-dimensional space whose dimension is exactly the Pfaffian of  $\text{Im}H$ , the product of the elementary divisors of the polarization. The first fact is proved here in full; the second is stated here and its proof, together with an explicit basis, is the business of section 3.4 and of chapter 4.

**Definition 2.4.1: Theta Function for a Line Bundle**

Let  $L = L(H, \chi)$  be the line bundle on  $X = V/\Lambda$  associated by the Appell-Humbert construction (definition 2.2.5) to a datum  $(H, \chi)$ , with canonical factor of automorphy

$$a_{(H, \chi)}(\lambda, v) = \chi(\lambda) \exp\left(\pi H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda)\right)$$

A *theta function* for  $(H, \chi)$  is a holomorphic function  $\theta: V \rightarrow \mathbb{C}$  satisfying the functional equation

$$\theta(v + \lambda) = a_{(H, \chi)}(\lambda, v) \theta(v) \quad \text{for all } v \in V, \lambda \in \Lambda \quad (2.8)$$

The complex vector space of all theta functions for  $(H, \chi)$  is denoted  $H^0(X, L)$ .

The identification of this space with the sheaf-cohomology group  $H^0(X, L)$  of global sections is a direct unwinding of the Appell-Humbert construction, and spelling it out fixes the dictionary used for the rest of the book. Recall from definition 2.2.5 that  $L(H, \chi)$  is the quotient of  $V \times \mathbb{C}$  by the  $\Lambda$ -action  $\lambda \cdot (v, t) = (v + \lambda, a_{(H, \chi)}(\lambda, v) t)$ . A global section of  $L$  is a holomorphic map  $s: X \rightarrow L$  splitting the projection  $L \rightarrow X$ ; pulling back along  $V \rightarrow X$ , it lifts to a holomorphic map  $\tilde{s}: V \rightarrow V \times \mathbb{C}$  of the form  $\tilde{s}(v) = (v, \theta(v))$  for a holomorphic  $\theta: V \rightarrow \mathbb{C}$ . Compatibility with the  $\Lambda$ -action,  $\tilde{s}(v + \lambda) = \lambda \cdot \tilde{s}(v)$ , reads  $(v + \lambda, \theta(v + \lambda)) = (v + \lambda, a_{(H, \chi)}(\lambda, v) \theta(v))$ , which is exactly (2.8). Thus a section is the same datum as a theta function, and the space of theta functions of definition 2.4.1 coincides with the sheaf-cohomology group  $H^0(X, L)$  of global sections of  $L$ . The functional equation is the reason these functions are called *quasi-periodic*:  $\theta$  is not  $\Lambda$ -periodic (that would make it a section of the trivial bundle), but it reproduces itself under lattice translation up to the explicit automorphy factor.

A theta function is a function on  $V$ , not on  $X$ , so only its zero set descends. The functional equation (2.8) shows  $a_{(H, \chi)}(\lambda, v)$  never vanishes, so the zero locus  $\{\theta = 0\} \subseteq V$  is  $\Lambda$ -invariant and descends to a well-defined divisor on  $X$ , the divisor of the section. When  $H > 0$  this divisor is the geometric object, a theta divisor, that chapter 4 studies in its own right. For now the point is structural: sections are quasi-periodic entire functions, and their count is controlled entirely by the growth that the factor of automorphy permits.

Before counting sections when they exist, the first question is when they exist at all. The functional equation constrains the modulus of  $\theta$  against the Hermitian form  $H$ , and on a compact torus modulus constraints become boundedness constraints. The following growth argument extracts from a single nonzero section the positivity of  $H$ .

**Proposition 2.4.2: Nonpositive Bundles Have No Sections**

Let  $L = L(H, \chi)$  be the line bundle associated to a datum  $(H, \chi)$  on  $X = V/\Lambda$ . If  $H$  is not positive semidefinite, then

$$H^0(X, L) = 0$$

Equivalently, if  $L$  admits a nonzero global section, then  $H \geq 0$ .

**Proof:**

Let  $\theta \in H^0(X, L)$  be a theta function for  $(H, \chi)$ , and consider the real-valued function

$$\varphi(v) = |\theta(v)| \exp\left(-\frac{\pi}{2} H(v, v)\right) \quad (2.9)$$

which is well-defined because  $H(v, v) \in \mathbb{R}$  for a Hermitian form.

*Step 1:  $\varphi$  descends to the compact torus.* The claim is that  $\varphi$  is  $\Lambda$ -periodic. Fix  $\lambda \in \Lambda$ . Since  $|\chi(\lambda)| = 1$  and  $H(\lambda, \lambda) \in \mathbb{R}$ , taking moduli in the functional equation (2.8) gives

$$|\theta(v + \lambda)| = |\theta(v)| \exp\left(\pi \operatorname{Re} H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda)\right)$$

On the other side, expanding  $H(v + \lambda, v + \lambda)$  by bilinearity and using  $H(\lambda, v) = \overline{H(v, \lambda)}$  yields

$$H(v + \lambda, v + \lambda) = H(v, v) + 2 \operatorname{Re} H(v, \lambda) + H(\lambda, \lambda)$$

Substituting both computations into (2.9),

$$\varphi(v + \lambda) = |\theta(v)| \exp\left(\pi \operatorname{Re} H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda)\right) \exp\left(-\frac{\pi}{2} H(v, v) - \pi \operatorname{Re} H(v, \lambda) - \frac{\pi}{2} H(\lambda, \lambda)\right)$$

The two terms involving  $\operatorname{Re} H(v, \lambda)$  cancel, as do the two terms involving  $H(\lambda, \lambda)$ , leaving  $\varphi(v + \lambda) = |\theta(v)| \exp(-\frac{\pi}{2} H(v, v)) = \varphi(v)$ . Thus  $\varphi$  is  $\Lambda$ -periodic, so it descends to a continuous function on the compact space  $X = V/\Lambda$  and is therefore bounded: there is a constant  $M \geq 0$  with

$$|\theta(v)| \leq M \exp\left(\frac{\pi}{2} H(v, v)\right) \quad \text{for all } v \in V \quad (2.10)$$

*Step 2: a negative direction forces  $\theta$  to vanish.* Suppose  $H$  is not positive semidefinite. Then there is a vector  $v_0 \in V$  with  $H(v_0, v_0) < 0$ . Fix an arbitrary  $w \in V$  and restrict  $\theta$  to the complex line through  $w$  in the direction  $v_0$ , setting

$$g(z) = \theta(w + zv_0) \quad (z \in \mathbb{C})$$

which is an entire function of  $z$ . Expanding the Hermitian form along this line, using linearity in the first slot and conjugate-linearity in the second,

$$H(w + zv_0, w + zv_0) = H(w, w) + 2 \operatorname{Re}(z H(v_0, w)) + |z|^2 H(v_0, v_0)$$

The bound (2.10) applied at  $v = w + zv_0$  therefore gives

$$|g(z)| \leq M \exp\left(\frac{\pi}{2} H(w, w) + \pi \operatorname{Re}(z H(v_0, w)) + \frac{\pi}{2} |z|^2 H(v_0, v_0)\right)$$

As  $|z| \rightarrow \infty$  the quadratic term  $\frac{\pi}{2} |z|^2 H(v_0, v_0)$  tends to  $-\infty$ , since  $H(v_0, v_0) < 0$ , while the linear term  $\pi \operatorname{Re}(z H(v_0, w))$  grows at most like  $|z|$ . The quadratic term dominates, so the exponent tends to  $-\infty$  and  $|g(z)| \rightarrow 0$  as  $|z| \rightarrow \infty$ . An entire function that tends to 0 at infinity is bounded, hence constant by Liouville's theorem, and its constant value must be 0. Thus  $g \equiv 0$ ; in particular  $g(0) = \theta(w) = 0$ .

Since  $w \in V$  was arbitrary,  $\theta$  vanishes identically. Therefore  $H^0(X, L) = 0$ , as we sought to show.

The mechanism deserves emphasis. The factor of automorphy is designed precisely so that the Gaussian-modulated modulus  $\varphi(v) = |\theta(v)| \exp(-\frac{\pi}{2} H(v, v))$  becomes periodic, converting the transcendental transformation law into a boundedness statement on a compact space. Once bounded, the section can grow no faster than  $\exp(\frac{\pi}{2} H(v, v))$ , and along any direction in which  $H$  is negative this is exponential decay, which Liouville cannot tolerate for a nonzero entire function. The positivity of

$H$  is thus not an extra hypothesis imposed by hand: it is forced by the mere existence of a section. This is the analytic shadow of a fact that will reappear algebraically in Part II, that an effective line bundle has nonnegative degree. Combined with the classification of theorem 2.2.8, it means that of all the line bundles on  $X$ , only those whose Hermitian form is positive semidefinite can contribute sections at all, and the interesting geometry lives among the strictly positive ones.

The converse question, how many sections a strictly positive bundle supports, has an exact answer in terms of the alternating form  $E = \text{Im}H$ . When  $H > 0$  the form  $E$  is nondegenerate and integral on  $\Lambda$ , and the elementary-divisor theorem produces a symplectic basis in which  $E$  has invariants  $d_1 | d_2 | \cdots | d_g$ , the type of the polarization. The dimension of the section space is the product of these invariants, equivalently the Pfaffian of  $E$ .

**Proposition 2.4.3: Dimension of the Space of Sections**

Let  $L = L(H, \chi)$  with  $H > 0$ , and let  $E = \text{Im}H$ , a nondegenerate alternating integral form on  $\Lambda$  with elementary divisors  $d_1 | d_2 | \cdots | d_g$ . Then  $H^0(X, L)$  is finite-dimensional, with

$$\dim_{\mathbb{C}} H^0(X, L) = \text{Pf}(E) = d_1 d_2 \cdots d_g = \sqrt{\det E}$$

In particular  $\dim_{\mathbb{C}} H^0(X, L) \geq 1$ , so every positive line bundle has a nonzero section.

The proof constructs an explicit basis of theta functions, indexed by a set of coset representatives whose cardinality is exactly  $d_1 \cdots d_g$ , and verifies that these functions span. That construction is the content of section 3.4, where the dimension formula is derived as the Riemann-Roch theorem for abelian varieties, and of chapter 4, where the basis is written down in classical coordinates as the Riemann theta functions with characteristics. The deferral is deliberate: the counting argument is cleanest once the reduction of  $E$  to symplectic normal form by the elementary-divisor theorem ([12]) is available in the polarization language of definition 3.2.3 in chapter 3, and the explicit Fourier-series form of the basis belongs with the classical theta function of chapter 4. Stated here, the formula records the payoff of the growth estimate above: positivity is not only necessary for sections but produces precisely  $\text{Pf}(E)$  of them, and this integer is the projective dimension budget that the Lefschetz embedding theorem will spend.

The dimension count is already transparent in the one-dimensional case, where it recovers a computation the reader has met analytically and where the sections are the classical theta functions of Jacobi.

**Example 2.4.4: Theta Functions on an Elliptic Curve**

Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z}\tau + \mathbb{Z}$  and  $\tau \in \mathbb{H}$ , the  $g = 1$  case. A Hermitian form on  $V = \mathbb{C}$  is  $H(v, w) = h v \bar{w}$  for a real scalar  $h$ , and  $H > 0$  means  $h > 0$ . The imaginary part  $E = \text{Im}H$  is the alternating form determined by its value on the basis  $(\tau, 1)$  of  $\Lambda$ ,

$$E(\tau, 1) = \text{Im}(h \tau \cdot \bar{1}) = h \text{Im}\tau$$

so  $E$  is integral on  $\Lambda$  precisely when  $h \text{Im}\tau \in \mathbb{Z}$ . Writing  $d = h \text{Im}\tau = E(\tau, 1)$ , this integer is the single elementary divisor, the type  $(d)$  of the polarization, and it is the degree of the corresponding line bundle. proposition 2.4.3 then gives

$$\dim_{\mathbb{C}} H^0(E, L) = \text{Pf}(E) = d$$

For the principal polarization  $d = 1$ , so  $h = 1/\text{Im}\tau$  and there is, up to scalar, a unique theta

function. It is the classical *Jacobi theta function*

$$\vartheta(z, \tau) = \sum_{n \in \mathbb{Z}} \exp(\pi i n^2 \tau + 2\pi i n z) \quad (2.11)$$

an entire function of  $z$  whose series converges because  $\text{Im} \tau > 0$  forces  $|\exp(\pi i n^2 \tau)| = \exp(-\pi n^2 \text{Im} \tau)$  to decay faster than any geometric rate. Its quasi-periodicity is the direct analogue of the functional equation (2.8): reindexing the series gives

$$\vartheta(z + 1, \tau) = \vartheta(z, \tau), \quad \vartheta(z + \tau, \tau) = \exp(-\pi i \tau - 2\pi i z) \vartheta(z, \tau)$$

The first identity is immediate since  $\exp(2\pi i n) = 1$ ; for the second, the substitution  $\pi i n^2 \tau + 2\pi i n \tau = \pi i \tau (n + 1)^2 - \pi i \tau$  followed by the reindexing  $m = n + 1$  produces the stated factor. These two transformation laws are exactly the values of a factor of automorphy on the basis  $\lambda = 1$  and  $\lambda = \tau$  of  $\Lambda$ ; up to the coboundary that passes between the classical normalization and the canonical Appell-Humbert factor,  $\vartheta$  is the section of the degree-one bundle predicted by proposition 2.4.3.

For a general positive type  $(d)$  with  $d \geq 1$ , the dimension is  $d$ , and a basis is furnished by the  $d$  theta functions with rational characteristics  $\vartheta \left[ \begin{smallmatrix} a/d \\ 0 \end{smallmatrix} \right] (z, \tau)$  for  $a = 0, 1, \dots, d-1$ , translates of (2.11) that chapter 4 introduces in general. The count  $\dim H^0(E, L) = d = \deg L$  is the analytic form of the Riemann-Roch statement for elliptic curves, recovering from the present transcendental viewpoint the dimension the reader computed algebraically in [4].

The elliptic case makes the general pattern legible. A section of a line bundle is a concrete holomorphic function on the universal cover with a completely explicit transformation law under the deck group, and once that law is fixed the sections form a finite-dimensional space whose dimension is a discrete invariant of the bundle, its degree in dimension one and its Pfaffian in general. Two features of this dimension are what power everything downstream. First, it is positive exactly when the bundle is positive, so the analytic condition  $H > 0$  that permits sections is the same condition that will permit a projective embedding. Second, it grows: replacing  $L$  by  $L^{\otimes n}$  multiplies each elementary divisor by  $n$  and hence multiplies  $\text{Pf}(E)$  by  $n^g$ , so high tensor powers of a positive bundle supply an abundance of sections.

### Exercise 2.4.1

1. Let  $L = L(H, \chi)$  on  $X = V/\Lambda$  and let  $\theta_1, \theta_2 \in H^0(X, L)$  be two nonzero theta functions for the same datum  $(H, \chi)$ , as in definition 2.4.1. Show that the quotient  $f = \theta_1/\theta_2$  is a meromorphic function on  $V$  that is  $\Lambda$ -periodic, hence descends to a meromorphic function on  $X$ . Explain why this makes the divisor of a section well-defined on  $X$  while the sections themselves are only functions on  $V$ .
2. Take the trivial datum  $H = 0$  and  $\chi = 1$ , so that  $a_{(H, \chi)}(\lambda, v) = 1$  for all  $\lambda, v$ . Describe  $H^0(X, L(0, 1))$  explicitly and compute its dimension. Then let  $\chi: \Lambda \rightarrow \mathbb{C}$  be a character with  $|\chi(\lambda)| = 1$  and  $\chi \neq 1$ , still with  $H = 0$ ; show that  $H^0(X, L(0, \chi)) = 0$ , and reconcile this with proposition 2.4.3.
3. In the proof of proposition 2.4.2 the function  $\varphi(v) = |\theta(v)| \exp(-\frac{\pi}{2} H(v, v))$  of (2.9) is shown to be  $\Lambda$ -periodic. Suppose instead  $H > 0$  and  $\theta \in H^0(X, L)$  is nonzero. Prove that  $\varphi$  attains a positive maximum on  $X$ , and deduce that  $\theta$  does not vanish identically but does have zeros, so its zero divisor is a nonempty effective divisor on  $X$ .

4. Consider the elliptic curve  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z}\tau + \mathbb{Z}$  of example 2.4.4, and the Jacobi theta function  $\vartheta(z, \tau)$  of (2.11). Using the quasi-periodicity relations recorded there, count the zeros of  $\vartheta(\cdot, \tau)$  in a fundamental parallelogram for  $\Lambda$  by integrating  $\frac{1}{2\pi i} \vartheta'/\vartheta$  around its boundary, and confirm that the answer agrees with  $\dim_{\mathbb{C}} H^0(E, L) = 1$  for the principal polarization.
5. Let  $L = L(H, \chi)$  with  $H > 0$  on  $X = V/\Lambda$ , and let  $E = \text{Im}H$  have elementary divisors  $d_1 | \cdots | d_g$ . For an integer  $n \geq 1$  the tensor power  $L^{\otimes n}$  is the bundle  $L(nH, \chi^n)$ . Using proposition 2.4.3, prove that

$$\dim_{\mathbb{C}} H^0(X, L^{\otimes n}) = n^g \dim_{\mathbb{C}} H^0(X, L)$$

and explain in one or two sentences why this growth is what makes high tensor powers useful for projective embeddings.

6. Prove that the space  $H^0(X, L)$  of definition 2.4.1 is a complex vector space, and that if  $L = L(H, \chi)$  with  $H$  positive definite then  $\dim_{\mathbb{C}} H^0(X, L) < \infty$  using only the bound (2.10) and a normal-families argument, without invoking the exact value in proposition 2.4.3.





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# *Polarizations and the Riemann Relations*

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The classification of chapter 2 reduces the algebraicity of a complex torus to a single question: among the Hermitian forms  $H$  with  $\text{Im}H$  integral on  $\Lambda$ , is there one that is positive definite? This chapter shows that the answer decides everything. A positive such form is a *polarization*, and its existence is exactly the Riemann relations, the classical bilinear conditions on the period matrix that separate the projective tori from the transcendental ones. A complex torus that carries a polarization is an *abelian variety*; one that does not is forever analytic, with too few meromorphic functions to cut out anything in projective space.

With the definition in hand, the chapter develops the two invariants a polarization carries. Its type, a tuple of elementary divisors  $(d_1, \dots, d_g)$ , records the arithmetic of the form on the lattice and specializes to the notion of a principal polarization when the type is trivial; the associated period matrices, normalized by the type, sweep out the Siegel upper half-space and give the first picture of a moduli space of abelian varieties. Finally, Riemann-Roch computes the number of independent theta functions a polarization supports,  $\dim H^0(X, L) = d_1 \cdots d_g$ , the count that chapter 4 will realize by writing the theta functions down explicitly and using them to embed the variety.

## 3.1 Positive Line Bundles and the Riemann Relations

The previous chapter classified every line bundle on a complex torus  $X = V/\Lambda$  by a pair  $(H, \chi)$ , where  $H$  is a Hermitian form on  $V$  whose imaginary part  $E = \text{Im}H$  is integral on the lattice, and  $\chi$  is a semicharacter for  $E$ . That classification is purely formal: it records which line bundles exist, but says nothing about which of them have sections, and therefore nothing about whether  $X$  can be embedded in projective space. The class  $c_1(L(H, \chi)) = H$  in  $\text{NS}(X)$  carries all the information relevant to this

question, and the answer turns entirely on one attribute of  $H$  that the Appell-Humbert data does not constrain, namely its *signature*. This section isolates the condition, positive definiteness of  $H$ , and proves that it is exactly the condition under which  $X$  is algebraic.

The statement that a positive Hermitian form with integral imaginary part exists is the Riemann relations. In the period-matrix language it becomes a pair of explicit conditions on the  $g \times 2g$  matrix  $\Pi$  of periods, one an algebraic identity and one a positivity, and these are the conditions Riemann wrote down for the periods of an abelian integral. The equivalence of the intrinsic Hermitian statement with the coordinate one is a linear-algebra computation carried out in full below. The consequence, developed across the rest of this chapter, is that the tori  $\mathbb{C}^g/\Lambda$  admitting such a form are precisely the abelian varieties, a vanishingly thin subset of all complex tori once  $g \geq 2$ .

**Definition 3.1.1: Positive Line Bundle**

Let  $X = V/\Lambda$  be a complex torus and  $L = L(H, \chi)$  a line bundle on  $X$ , with Hermitian form  $H$  and semicharacter  $\chi$  as in the Appell-Humbert classification. The bundle  $L$  is *positive* if its Hermitian form  $H$  is positive definite, written  $H > 0$ ; that is,  $H(v, v) > 0$  for every nonzero  $v \in V$ .

Positivity depends only on  $H$ , not on the semicharacter  $\chi$ , and  $H = c_1(L)$  is the image of  $L$  in  $\text{NS}(X)$  under the first Chern class. Positivity is therefore a property of the class of  $L$  in the Néron-Severi group  $\text{NS}(X)$  (definition 2.3.1), unchanged by tensoring  $L$  with any bundle from  $\text{Pic}^0(X)$ , which alters  $\chi$  but leaves  $H$  fixed. The definition is the analytic incarnation of ampleness: a line bundle on a projective variety is ample when a power of it embeds the variety in projective space, and the content of this chapter and the next is that  $H > 0$  is precisely the analytic signature of that geometric condition. The positive-definiteness requirement is restrictive. A Hermitian form on  $\mathbb{C}^g$  has a signature  $(p, q)$  with  $p + q = g$ , and only the extreme case  $(g, 0)$  is positive; the intermediate signatures give line bundles that are nontrivial in  $\text{NS}(X)$  but have no global sections, while  $(0, g)$  gives an anti-ample class.

The prototype is the elliptic curve, where the condition is automatically satisfiable and one sees the whole mechanism in a single complex variable.

**Example 3.1.2: Positive Bundles on an Elliptic Curve**

Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z}\tau + \mathbb{Z}$ ,  $\tau \in \mathbb{H}$ , so  $V = \mathbb{C}$  and  $g = 1$ . A Hermitian form on  $\mathbb{C}$  is  $H(v, w) = cv\bar{w}$  for a real scalar  $c$ , and  $H > 0$  means  $c > 0$ . Its imaginary part is  $E(v, w) = c\text{Im}(v\bar{w})$ , an alternating  $\mathbb{R}$ -form on  $\mathbb{C}$ . Evaluate  $E$  on the basis  $\lambda_1 = \tau$ ,  $\lambda_2 = 1$  of  $\Lambda$ :

$$E(\tau, 1) = c\text{Im}(\tau \cdot 1) = c\text{Im}(\tau)$$

while  $E(1, 1) = E(\tau, \tau) = 0$ . Integrality of  $E$  on  $\Lambda$  demands  $E(\tau, 1) \in \mathbb{Z}$ , so  $c\text{Im}\tau$  must be an integer. Taking the smallest positive value,  $c = 1/\text{Im}\tau > 0$ , gives a positive Hermitian form

$$H(v, w) = \frac{v\bar{w}}{\text{Im}\tau}$$

with  $E(\tau, 1) = 1$  and matrix  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  on the basis  $(\tau, 1)$ . Thus every elliptic curve carries a positive line bundle. This is the analytic counterpart of the ample bundle  $\mathcal{O}_E(O)$  attached to the origin in  $E$ , whose sections are the classical theta functions of  $\mathbb{C}/\Lambda$  ([4]); the constant  $1/\text{Im}\tau$  is the reciprocal of the covolume of  $\Lambda$ , which is why it is exactly this normalization that makes  $E$  integral.

The example shows the two constraints acting together: positivity of  $H$  and integrality of  $E = \operatorname{Im} H$  on  $\Lambda$ . A line bundle whose Hermitian form satisfies both is what makes  $X$  algebraic, and this pairing of conditions deserves a name.

**Definition 3.1.3: Polarization (Analytic)**

Let  $X = V/\Lambda$  be a complex torus. A *polarization* of  $X$  is a positive-definite Hermitian form  $H$  on  $V$  whose imaginary part  $E = \operatorname{Im} H$  takes integer values on  $\Lambda$ , i.e.  $E(\Lambda, \Lambda) \subseteq \mathbb{Z}$ . Equivalently, a polarization is a class in  $\operatorname{NS}(X)$  (definition 2.3.1) that is represented by a positive-definite Hermitian form; such a class is called a *positive class*. The torus  $X$  is *polarizable* if it admits a polarization.

A polarization is thus a point of  $\operatorname{NS}(X)$  lying in the open cone of positive-definite forms, subject to the integrality that membership in  $\operatorname{NS}(X)$  already imposes. By theorem 2.2.8, every such  $H$  is the Chern class of an actual line bundle  $L(H, \chi)$  on  $X$ , and different choices of semicharacter  $\chi$  produce line bundles differing by an element of  $\operatorname{Pic}^0(X)$ ; the polarization is the common class, an object slightly coarser than a line bundle. Recording a polarization as the Hermitian form  $H$  rather than as the bundle  $L$  is the convention throughout: what matters for algebraicity is the positive class, and the semicharacter is auxiliary data that the theory of theta functions in the next chapter pins down when an explicit bundle is needed. The word “polarization” is chosen to evoke a rigidification: a bare complex torus has no distinguished cohomology class, while a polarized torus comes equipped with one, and it is this extra datum that anchors the transcendental object to projective geometry.

The central result of the section states that polarizability is the precise dividing line for algebraicity, and gives the condition two forms, one intrinsic and one in coordinates.

**Theorem 3.1.4: Riemann Relations**

Let  $X = V/\Lambda$  be a complex torus of dimension  $g$ , with  $V = \mathbb{C}^g$ . The following are equivalent.

1.  $X$  admits a polarization: there is a positive-definite Hermitian form  $H$  on  $V$  whose imaginary part  $E = \operatorname{Im} H$  is integral on  $\Lambda$ .
2. (*Period-matrix form.*) For a period matrix  $\Pi \in M_{g \times 2g}(\mathbb{C})$  of  $X$ , there exists a nondegenerate alternating integral matrix  $E \in M_{2g}(\mathbb{Z})$  such that

$$\Pi E^{-1} \Pi^t = 0 \quad \text{and} \quad i \Pi E^{-1} \bar{\Pi}^t > 0 \quad (3.1)$$

where the second inequality asserts that the Hermitian matrix  $i \Pi E^{-1} \bar{\Pi}^t$  is positive definite.

When either condition holds,  $X$  is an abelian variety: it admits a holomorphic embedding into some projective space  $\mathbb{P}^N$ , and conversely every complex torus that so embeds satisfies these conditions.

Condition (1) is intrinsic to  $V$  and  $\Lambda$ ; condition (2) is its transcription into the periods, and the two Riemann relations (3.1) are the classical statement, an algebraic identity together with a positivity. The final sentence records the geometric payoff, which is the reason the conditions are studied at all: they are the exact criterion separating algebraic tori from transcendental ones. Before proving the theorem it is worth being clear about what each half of the equivalence contributes. The equivalence

of (1) and (2) is a self-contained linear-algebra computation, carried out below. The passage from (1) to a projective embedding is a genuine construction, requiring the sections of a positive line bundle to be produced explicitly; that construction is the theory of theta functions, and the resulting Lefschetz embedding theorem is proved in the next chapter. The reverse passage, from a projective embedding to a polarization, extracts the Hermitian form from the first Chern class of the very ample bundle furnished by the embedding.

**Proof:**

The proof has three parts: the equivalence of the two analytic conditions (1) and (2), and the two implications relating (1) to algebraicity. The core computation, the equivalence of (1) and (2), occupies the first two steps; the geometric implications are then addressed in turn.

*Step 1: From a polarization to the period relations.* Suppose  $H$  is a positive-definite Hermitian form on  $V = \mathbb{C}^g$  with  $E = \text{Im}H$  integral on  $\Lambda$ . Fix the standard  $\mathbb{C}$ -basis of  $V$ , and let  $\mathcal{H} \in M_g(\mathbb{C})$  be the matrix of  $H$ , so that  $H(v, w) = v^t \mathcal{H} \bar{w}$ ; the Hermitian symmetry  $H(w, v) = \overline{H(v, w)}$  is the condition  $\mathcal{H}^t = \mathcal{H}$ , and positive definiteness of  $H$  is positive definiteness of  $\mathcal{H}$ . Let  $\lambda_1, \dots, \lambda_{2g}$  be the  $\mathbb{Z}$ -basis of  $\Lambda$  that forms the columns of  $\Pi$ , and let  $E \in M_{2g}(\mathbb{Z})$  be the integer matrix with entries  $E_{jk} = E(\lambda_j, \lambda_k)$ , alternating because  $E$  is, and nondegenerate because  $H$  is nondegenerate on the real span  $V = \Lambda \otimes \mathbb{R}$ .

Assemble the periods into the invertible big period matrix

$$\hat{\Pi} = \begin{pmatrix} \Pi \\ \bar{\Pi} \end{pmatrix} \in M_{2g}(\mathbb{C})$$

whose invertibility records that  $\lambda_1, \dots, \lambda_{2g}$  is an  $\mathbb{R}$ -basis of  $V \cong \mathbb{R}^{2g}$ . Writing  $M = \Pi^t \mathcal{H} \bar{\Pi}$ , the  $(j, k)$  entry of  $M$  is  $\lambda_j^t \mathcal{H} \bar{\lambda}_k = H(\lambda_j, \lambda_k)$ , so  $E = \text{Im}M$  entrywise. Since  $\bar{M} = \bar{\Pi}^t \mathcal{H} \Pi = \bar{\Pi}^t \mathcal{H}^t \Pi$ , this reads

$$E = \frac{1}{2i} (M - \bar{M}) = \frac{1}{2i} (\Pi^t \mathcal{H} \bar{\Pi} - \bar{\Pi}^t \mathcal{H}^t \Pi) = \frac{1}{2i} \hat{\Pi}^t B \hat{\Pi}, \quad B = \begin{pmatrix} 0 & \mathcal{H} \\ -\bar{\mathcal{H}} & 0 \end{pmatrix} \quad (3.2)$$

the last equality being the block expansion  $\hat{\Pi}^t B \hat{\Pi} = \Pi^t \mathcal{H} \bar{\Pi} - \bar{\Pi}^t \mathcal{H} \Pi$ , valid because  $\mathcal{H}^t = \bar{\mathcal{H}}$ . The block matrix  $B$  is invertible, with

$$B^{-1} = \begin{pmatrix} 0 & -\bar{\mathcal{H}}^{-1} \\ \mathcal{H}^{-1} & 0 \end{pmatrix}$$

as one checks by multiplying out. Inverting (3.2) gives  $E^{-1} = 2i \hat{\Pi}^{-1} B^{-1} (\hat{\Pi}^t)^{-1}$ , hence

$$\hat{\Pi} E^{-1} \hat{\Pi}^t = 2i B^{-1} = 2i \begin{pmatrix} 0 & -\bar{\mathcal{H}}^{-1} \\ \mathcal{H}^{-1} & 0 \end{pmatrix} \quad (3.3)$$

Now read off the  $g \times g$  blocks of the left-hand side. Since  $\hat{\Pi} = \begin{pmatrix} \Pi \\ \bar{\Pi} \end{pmatrix}$ , the top-left block of  $\hat{\Pi} E^{-1} \hat{\Pi}^t$  is  $\Pi E^{-1} \Pi^t$  and the top-right block is  $\Pi E^{-1} \bar{\Pi}^t$ . Comparing with the right-hand side of (3.3), whose top-left block is 0 and whose top-right block is  $-2i \bar{\mathcal{H}}^{-1}$ , gives

$$\Pi E^{-1} \Pi^t = 0 \quad \text{and} \quad i \Pi E^{-1} \bar{\Pi}^t = i(-2i \bar{\mathcal{H}}^{-1}) = 2 \bar{\mathcal{H}}^{-1}$$

Since  $\mathcal{H}$  is positive definite, so is  $\mathcal{H}^{-1}$ , and so is its conjugate  $\bar{\mathcal{H}}^{-1}$ ; hence  $i \Pi E^{-1} \bar{\Pi}^t = 2 \bar{\mathcal{H}}^{-1} > 0$ . Both relations of (3.1) hold, establishing (1)  $\Rightarrow$  (2).

*Step 2: From the period relations to a polarization.* Conversely, suppose  $E \in M_{2g}(\mathbb{Z})$  is nondegenerate alternating integral and the two relations (3.1) hold for  $\Pi$ . Define an  $\mathbb{R}$ -bilinear alternating form on  $V = \Lambda \otimes \mathbb{R}$  by declaring  $E(\lambda_j, \lambda_k)$  to be the  $(j, k)$  entry of  $E$  and extending  $\mathbb{R}$ -bilinearly; it is integral on  $\Lambda$  by construction. Set

$$H(v, w) := E(iv, w) + iE(v, w)$$

Two things must be checked: that  $H$  is a Hermitian form (which requires the compatibility  $E(iv, iw) = E(v, w)$ ), and that it is positive definite. Both follow from (3.1).

For the compatibility, the relation  $\Pi E^{-1} \Pi^t = 0$  is exactly the vanishing of the top-left block of  $\hat{\Pi} E^{-1} \hat{\Pi}^t$ ; conjugating the whole matrix identity, the bottom-right block  $\bar{\Pi} E^{-1} \bar{\Pi}^t$  vanishes as well. Writing  $P = \Pi E^{-1} \bar{\Pi}^t$  for the top-right block and using that  $\hat{\Pi} E^{-1} \hat{\Pi}^t$  is alternating (its bottom-left block is therefore  $-P^t$ ), the big matrix is block-antidiagonal:

$$\hat{\Pi} E^{-1} \hat{\Pi}^t = \begin{pmatrix} 0 & P \\ -P^t & 0 \end{pmatrix} \quad (3.4)$$

This block-antidiagonality is equivalent to the compatibility  $E(iv, iw) = E(v, w)$ . Indeed, the computation of Step 1 is reversible: for any real alternating form  $E$  on  $V = \mathbb{R}^{2g}$ , let  $\mathcal{E}$  denote its matrix in the standard real coordinates and  $J$  the matrix of multiplication by  $i$ ; then a direct expansion of  $\hat{\Pi} E^{-1} \hat{\Pi}^t$  shows its top-left block vanishes if and only if  $J^t \mathcal{E} J = \mathcal{E}$ , which is the coordinate form of  $E(iv, iw) = E(v, w)$ . When  $E = \text{Im} H$  for a Hermitian  $H$  this compatibility holds automatically, giving (3.4) with  $P = -2i\bar{\mathcal{H}}^{-1}$  as in Step 1; conversely, (3.4) forces  $J^t \mathcal{E} J = \mathcal{E}$ , so the  $E$  furnished by the Riemann relations is compatible. Granting  $E(iv, iw) = E(v, w)$ , the form  $H(v, w) = E(iv, w) + iE(v, w)$  is Hermitian. It is  $\mathbb{R}$ -bilinear by construction, and  $\mathbb{C}$ -linear in the first variable because

$$H(iv, w) = E(-v, w) + iE(iv, w) = -E(v, w) + iE(iv, w) = i(E(iv, w) + iE(v, w)) = iH(v, w)$$

Conjugate symmetry  $H(w, v) = \overline{H(v, w)}$  splits into two pieces. On imaginary parts,  $\text{Im} H(w, v) = E(w, v) = -E(v, w) = -\text{Im} H(v, w)$  because  $E$  is alternating. On real parts,  $\text{Re} H(w, v) = E(iw, v)$  must equal  $\text{Re} H(v, w) = E(iv, w)$ , and this follows from the compatibility applied to the pair  $(iw, v)$ :  $E(iw, v) = E(i \cdot iw, iv) = E(-w, iv) = -E(w, iv) = E(iv, w)$ .

For positive definiteness, the second relation is decisive. The identity  $i \Pi E^{-1} \bar{\Pi}^t = 2\bar{\mathcal{H}}^{-1}$  derived in Step 1 holds as a formal consequence of (3.2) and (3.3) once  $H$  is Hermitian, where now  $\mathcal{H}$  is the matrix of the reconstructed  $H(v, w) = v^t \mathcal{H} \bar{w}$ ; reading it backwards, the hypothesis  $i \Pi E^{-1} \bar{\Pi}^t > 0$  forces  $\bar{\mathcal{H}}^{-1} > 0$ , hence  $\mathcal{H} > 0$ , hence  $H > 0$ . Thus  $H$  is a polarization, establishing (2)  $\Rightarrow$  (1) and completing the equivalence of the two analytic conditions.

*Step 3: A polarization yields a projective embedding.* Suppose  $X$  carries a polarization  $H$ . By theorem 2.2.8 there is a semicharacter  $\chi$  making  $L = L(H, \chi)$  a line bundle with  $c_1(L) = H$ , and since  $H > 0$  the bundle  $L$  is positive. The sections of a high tensor power of a positive line bundle are the Riemann theta functions, and they separate points and tangent directions on  $X$ . This is the Lefschetz embedding theorem, proved in chapter 4: for a positive line bundle  $L$  on a complex torus,  $L^{\otimes 3}$  is very ample, so the associated map  $X \rightarrow \mathbb{P}^N$  sending a point to the ratios of a basis of  $H^0(X, L^{\otimes 3})$  is a holomorphic embedding. The construction of the sections, and the verification that they separate points, is the content of the theta-function machinery

of that chapter, so the argument is deferred to it; what the present theorem contributes is the criterion, that a positive  $H$  exists, which is the input the embedding theorem consumes. Granting chapter 4, a polarizable torus embeds in projective space and is therefore an abelian variety.

*Step 4: A projective embedding yields a polarization.* Suppose conversely that  $X \hookrightarrow \mathbb{P}^N$  is a holomorphic embedding. The restriction to  $X$  of the hyperplane bundle  $\mathcal{O}_{\mathbb{P}^N}(1)$  is a very ample line bundle  $L$  on  $X$ , and its first Chern class  $c_1(L) \in H^2(X, \mathbb{Z})$  is the restriction of the Fubini-Study class, which is a positive  $(1, 1)$ -form on  $\mathbb{P}^N$ . Under the identification of  $\text{NS}(X)$  with Hermitian forms (definition 2.3.1 and proposition 2.3.2), a positive  $(1, 1)$ -class corresponds to a positive-definite Hermitian form  $H$ , whose imaginary part  $E = \text{Im}H = c_1(L)$  is the integral class; positivity of the Fubini-Study form restricts to positivity along the complex submanifold  $X$ , so  $H > 0$ , and  $E = \text{Im}H$  is integral because  $c_1(L)$  is an integral cohomology class. Hence  $H$  is a polarization, and  $X$  is polarizable. This closes the circle: polarizability and projective embeddability are the same condition, as we sought to show.

The theorem is the structural dividing line of the analytic theory. Every complex torus is a compact complex Lie group and a differentiable object of the utmost regularity, yet only the polarizable ones are algebraic, and (3.1) measures exactly the failure of the rest. The first relation  $\Pi E^{-1} \Pi^t = 0$  is a system of  $\binom{g}{2}$  complex equations on the entries of  $\Pi$ , cutting out a proper analytic subvariety of the space of all period matrices; the positivity is an open condition on the complement. For  $g = 1$  the first relation is vacuous, one antisymmetric equation among a single pair of indices contributes nothing, which is why every one-dimensional torus is an elliptic curve; for  $g \geq 2$  the relations are nontrivial constraints, and a torus with a generic period matrix satisfies none of them. The precise mechanism of this genericity is examined in the worked contrast at the end of the section.

The passage from the intrinsic Hermitian condition to the two period relations is worth isolating as a statement in its own right, because it is the form in which the Riemann relations are used in computation. The proposition below is the coordinate incarnation of theorem 3.1.4, extracted from the proof and recorded with the explicit dictionary between the Hermitian matrix and the periods.

**Proposition 3.1.5: The Period-Matrix Riemann Relations**

Let  $X = V/\Lambda$  have period matrix  $\Pi \in M_{g \times 2g}(\mathbb{C})$  with respect to a  $\mathbb{Z}$ -basis  $\lambda_1, \dots, \lambda_{2g}$  of  $\Lambda$ , and let  $E \in M_{2g}(\mathbb{Z})$  be a nondegenerate alternating integral matrix. Then  $E$  is the imaginary part on  $\Lambda$  of a Hermitian form  $H$  on  $V$  if and only if

$$\Pi E^{-1} \Pi^t = 0$$

and in that case  $H$  is positive definite if and only if

$$i \Pi E^{-1} \bar{\Pi}^t > 0$$

The Hermitian matrix is recovered from the periods by  $\mathcal{H} = 2(\overline{i \Pi E^{-1} \bar{\Pi}^t})^{-1}$ , where  $H(v, w) = v^t \mathcal{H} \bar{w}$ .

**Proof:**

The first relation is equivalent to the existence of a compatible  $H$ , and the second to its positivity; both equivalences were established in Steps 1 and 2 of theorem 3.1.4, and it remains only to record the reconstruction formula. From the identity  $i \Pi E^{-1} \bar{\Pi}^t = 2 \bar{\mathcal{H}}^{-1}$  obtained in (3.3), solve for  $\mathcal{H}$ : taking inverses gives  $\bar{\mathcal{H}} = 2(i \Pi E^{-1} \bar{\Pi}^t)^{-1}$ , and conjugating gives

$$\mathcal{H} = 2 \overline{(i \Pi E^{-1} \bar{\Pi}^t)^{-1}} = 2 (\overline{i \Pi E^{-1} \bar{\Pi}^t})^{-1}$$

which is the stated formula, completing the proof.

The utility of the proposition is that it turns the search for a polarization into linear algebra over  $\mathbb{Z}$  followed by a positivity check over  $\mathbb{R}$ . One writes down  $\Pi$ , chooses a candidate integer alternating matrix  $E$  (for instance the standard symplectic form on the basis at hand), tests the algebraic identity  $\Pi E^{-1} \bar{\Pi}^t = 0$ , and if it holds, forms the Hermitian matrix  $i \Pi E^{-1} \bar{\Pi}^t$  and checks that it is positive definite. The reconstruction formula shows that no information is lost in passing between  $H$  and the pair  $(\Pi, E)$ : the Hermitian form is a rational, indeed polynomial-in-the-inverse, function of the periods and the polarizing form. This is the calculation performed for the elliptic curve next, and it is the calculation that fails, provably, for the transcendental torus discussed afterward.

**Example 3.1.6: The Riemann Relations for an Elliptic Curve**

Take  $g = 1$ ,  $V = \mathbb{C}$ , and  $\Lambda = \mathbb{Z}\tau + \mathbb{Z}$  with  $\tau \in \mathbb{H}$ , so that the period matrix on the basis  $\lambda_1 = \tau$ ,  $\lambda_2 = 1$  is

$$\Pi = \begin{pmatrix} \tau & 1 \end{pmatrix} \in M_{1 \times 2}(\mathbb{C})$$

Choose the standard symplectic integer matrix

$$E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad E^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

nondegenerate and alternating over  $\mathbb{Z}$ . Test the first Riemann relation:

$$\Pi E^{-1} \bar{\Pi}^t = \begin{pmatrix} \tau & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \bar{\tau} \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & -\tau \end{pmatrix} \begin{pmatrix} \bar{\tau} \\ 1 \end{pmatrix} = \tau - \tau = 0$$

so the algebraic relation holds identically, for every  $\tau$ ; this is the reason there is no obstruction in dimension one. Now the positivity. Compute

$$i \Pi E^{-1} \bar{\Pi}^t = i \begin{pmatrix} 1 & -\tau \end{pmatrix} \begin{pmatrix} \bar{\tau} \\ 1 \end{pmatrix} = i(\bar{\tau} - \tau) = i(-2i \operatorname{Im} \tau) = 2 \operatorname{Im} \tau$$

which is a positive real number because  $\tau \in \mathbb{H}$ . Both Riemann relations hold, so  $E$  is the imaginary part of a positive Hermitian form, and  $X = \mathbb{C}/\Lambda$  is an abelian variety. The reconstruction formula gives  $\mathcal{H} = 2 \overline{(2 \operatorname{Im} \tau)}^{-1} = 1/\operatorname{Im} \tau$ , so

$$H(v, w) = \frac{v \bar{w}}{\operatorname{Im} \tau}$$

recovering the positive form found by hand in example 3.1.2. Every elliptic curve is therefore an abelian variety, and the polarization it carries is principal (its type is (1), since  $E$  has determinant 1). This is the  $g = 1$  instance of the general theory, and it succeeds for every  $\tau$  precisely because the single algebraic relation degenerates to the tautology  $\tau - \tau = 0$ .

The contrast with higher dimension is the point of the whole development. When  $g \geq 2$  the first Riemann relation is a nonempty system, and for a lattice with generic periods it has no integral

solution at all. This is not an abstract possibility but an explicit phenomenon, already exhibited in section 1.4: the two-dimensional torus of example 1.4.3, built from a period matrix whose entries are algebraically independent transcendentals, admits no nonzero integral Hermitian form, hence no polarization, hence no projective embedding, and in fact has no nonconstant meromorphic functions whatsoever. Set against example 3.1.6, the two examples bracket the theorem exactly. The elliptic curve is the case where the algebraic relation is vacuous and a polarization always exists; the generic torus of example 1.4.3 is the case where the relation is a genuine constraint and no polarization exists. The Riemann relations are the wall between them, the condition an integral alternating form must satisfy against the periods before it can be the shadow of a positive line bundle, and they are the reason that among all complex tori of dimension  $g \geq 2$ , the abelian varieties form a countable union of proper subvarieties of the moduli of tori rather than the whole space. Everything that follows in this book concerns the tori on the algebraic side of that wall.

### Exercise 3.1.1

1. Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z}\tau + \mathbb{Z}$  and  $\tau = 2i$ . Following example 3.1.2, write down the positive Hermitian form  $H$  on  $V = \mathbb{C}$  whose imaginary part is integral on  $\Lambda$  and takes the value 1 on the basis  $(\tau, 1)$ , and give the matrix of  $E = \text{Im}H$  on that basis.
2. A Hermitian form  $H$  on  $\mathbb{C}^g$  has signature  $(p, q)$  with  $p + q = g$ . Using definition 3.1.1 and the discussion following it, state exactly which signature makes  $L(H, \chi)$  positive, and explain why two line bundles with the same Hermitian form but different semicharacters are either both positive or both non-positive.
3. Let  $\Lambda = \mathbb{Z}\tau_1 + \mathbb{Z}\tau_2 + \mathbb{Z} + \mathbb{Z}$  inside  $V = \mathbb{C}^2$  give the period matrix

$$\Pi = \begin{pmatrix} \tau_1 & 0 & 1 & 0 \\ 0 & \tau_2 & 0 & 1 \end{pmatrix}, \quad \tau_1, \tau_2 \in \mathbb{H}$$

and take  $E$  to be the standard symplectic integer matrix  $E = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}$ . Verify the first Riemann relation  $\Pi E^{-1} \Pi^t = 0$  and compute  $i \Pi E^{-1} \bar{\Pi}^t$ ; conclude via proposition 3.1.5 that  $X = V/\Lambda$  is an abelian variety, and recover the Hermitian matrix  $\mathcal{H}$ .

4. With the same  $\Pi$  as in exercise 3.1.1 (3), replace the symplectic  $E$  by the nondegenerate alternating integral matrix  $E'' = \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$  with  $D = \text{diag}(2, 3)$ . Redo the computation to check both Riemann relations, recover the Hermitian matrix  $\mathcal{H}''$ , and compute  $\det E''$ . Using the criterion for principality applied to the elliptic case in example 3.1.6, decide whether this polarization is principal, and compare with the polarization from exercise 3.1.1 (3).
5. Prove the reconstruction formula of proposition 3.1.5: if  $E$  is nondegenerate alternating integral and both relations (3.1) hold for  $\Pi$ , then the Hermitian matrix of the polarization  $H$  satisfies

$$\mathcal{H} = 2 \left( i \Pi E^{-1} \bar{\Pi}^t \right)^{-1}$$

where  $H(v, w) = v^t \mathcal{H} \bar{w}$ . You may use the identity  $\hat{\Pi} E^{-1} \hat{\Pi}^t = 2i B^{-1}$  from (3.3).

## 3.2 The Definition of an Abelian Variety

The Riemann relations of section 3.1 single out, among all complex tori, exactly those that carry a positive Hermitian form integral on the lattice. This condition is the analytic gate to projective space:



a torus on the positive side of the gate embeds as a closed algebraic subvariety of some  $\mathbb{P}^N$ , while a torus on the wrong side supports too few meromorphic functions to embed anywhere. The term *abelian variety* names the objects on the good side. This section fixes that definition, records the two numerical invariants a polarization carries (its type and, in the extremal case, its principality), and states the equivalence with projectivity whose forward half is the Lefschetz embedding theorem of chapter 4.

A caution on terminology precedes the definition. Chapter 1 introduced a *provisional* notion (definition 1.4.1): a complex torus that happens to admit some holomorphic embedding into projective space. That definition is geometrically transparent but analytically opaque, since it quantifies over embeddings one cannot see. The polarization reformulates the same class of tori in terms of a single piece of linear-algebraic data on  $V$  and  $\Lambda$ , a positive Hermitian form, and it is this reformulation that makes the class computable. The two descriptions agree, and theorem 3.1.4 is precisely the bridge between them.

### Definition 3.2.1: Abelian Variety

An *abelian variety* (over  $\mathbb{C}$ ) is a complex torus  $X = V/\Lambda$  that admits a polarization, that is, a positive definite Hermitian form  $H > 0$  on  $V$  whose imaginary part  $E = \text{Im}H$  takes integer values on  $\Lambda$  (definition 3.1.3). A *polarized abelian variety* is a pair  $(X, H)$  consisting of an abelian variety together with a choice of such a form.

This refines definition 1.4.1. The provisional definition asked for a holomorphic embedding  $X \hookrightarrow \mathbb{P}^N$ ; definition 3.2.1 asks instead for a polarization. That the two requirements pick out the same tori is the content of the Riemann relations together with the embedding theorem: theorem 3.1.4 supplies the polarization from the embedding by pulling back the Fubini-Study form, and the Lefschetz theorem of chapter 4 supplies the embedding from the polarization by constructing enough theta functions. From now on “abelian variety” means definition 3.2.1, and the provisional notion is retired.

The definition is a condition on  $X$  alone, not on a chosen  $H$ : an abelian variety is a torus for which *at least one* polarization exists, and different polarizations on the same torus are distinct geometric structures that cut out different projective embeddings. This is why the polarized pair  $(X, H)$  is recorded as a separate object. The bare torus underlying an abelian variety may carry many inequivalent polarizations, and the moduli problem of section 3.3 is sensitive to the choice.

### Example 3.2.2: The Elliptic Curve as an Abelian Variety

Take  $g = 1$ , so  $X = E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$  and  $\tau \in \mathbb{H}$ . A Hermitian form on  $V = \mathbb{C}$  is  $H(v, w) = cv\bar{w}$  for a real scalar  $c$ , and  $H > 0$  means  $c > 0$ . Its imaginary part is  $E(v, w) = \text{Im}H(v, w) = c\text{Im}(v\bar{w})$ , an alternating  $\mathbb{R}$ -form on  $\mathbb{C}$ . Integrality on  $\Lambda$  requires  $E(1, \tau) \in \mathbb{Z}$  and  $E(\tau, 1) \in \mathbb{Z}$ ; since  $E(1, \tau) = c\text{Im}(\bar{\tau}) = -c\text{Im}\tau$  and  $E(\tau, 1) = c\text{Im}\tau$ , and the remaining pairings  $E(1, 1) = E(\tau, \tau) = 0$  vanish, integrality amounts to the single condition  $c\text{Im}\tau \in \mathbb{Z}$ . Choosing  $c = (\text{Im}\tau)^{-1}$  gives  $E(\tau, 1) = 1$ , a positive Hermitian form integral on  $\Lambda$ . Every one-dimensional complex torus therefore admits a polarization, so every elliptic curve is an abelian variety, matching proposition 1.4.2: in dimension one the gate is always open.

The one-dimensional case makes the structure visible in miniature: a polarization is a positive multiple of the natural symplectic form on  $\mathbb{C}$ , normalized so that the fundamental parallelogram has integer area against  $E$ . In higher dimensions the same picture holds pointwise on  $V$ , but the integrality of  $E$  on a rank- $2g$  lattice is a strong constraint that most tori fail, which is why the class

of abelian varieties is a strict subclass of all complex tori for  $g \geq 2$  (as the transcendental torus of example 1.4.3 showed).

A polarization carries more information than the mere fact of its existence. The alternating integral form  $E = \text{Im}H$  has, like any nondegenerate alternating form over  $\mathbb{Z}$ , a normal form under change of symplectic basis, and the diagonal of that normal form is an invariant of  $E$  that measures how far the polarization is from being “as tight as possible.” The elementary-divisor theorem supplies the normal form, and its diagonal is the type.

#### Definition 3.2.3: Type of a Polarization

Let  $H$  be a polarization of  $X = V/\Lambda$ , with  $E = \text{Im}H$  the associated nondegenerate alternating form, integral on  $\Lambda$ . By the elementary-divisor theorem ([12]) there is a *symplectic basis*  $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$  of  $\Lambda$  in which the Gram matrix of  $E$  has the block form

$$\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}, \quad D = \text{diag}(d_1, \dots, d_g)$$

with positive integers  $d_1 | d_2 | \dots | d_g$ . The tuple  $(d_1, \dots, d_g)$  is the *type* of the polarization; it is independent of the chosen basis.

The type reads off directly from the elementary divisors of the alternating form  $E$  on the free  $\mathbb{Z}$ -module  $\Lambda$ . Concretely,  $E$  is given by an antisymmetric integer matrix once a  $\mathbb{Z}$ -basis of  $\Lambda$  is fixed; Smith normal form for antisymmetric integer matrices, which is the symplectic refinement of the elementary-divisor theorem, brings this matrix to the block form above by an integral symplectic change of basis. The invariance of  $(d_1, \dots, d_g)$  is the invariance of elementary divisors:  $d_1 \cdots d_k$  is the greatest common divisor of the  $2k \times 2k$  Pfaffian minors of  $E$ , hence a function of  $E$  and  $\Lambda$  alone, not of the basis realizing the normal form. Because  $H > 0$  forces  $E$  to be nondegenerate, all  $d_i \geq 1$ , and  $\det E = (d_1 \cdots d_g)^2$ , so the Pfaffian  $\text{Pf}(E) = d_1 \cdots d_g$  is the product of the type, a number that will reappear in section 3.4 as the dimension of the space of theta functions.

The extremal case is the one where the type is as small as it can be.

#### Definition 3.2.4: Principal Polarization

A polarization  $H$  of  $X$  is *principal* if its type is  $(1, \dots, 1)$ , that is, if the alternating form  $E = \text{Im}H$  admits a symplectic basis of  $\Lambda$  in which its Gram matrix is  $\begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$ . A *principally polarized abelian variety* (a *ppav*) is a pair  $(X, H)$  with  $H$  a principal polarization.

A principal polarization is one for which  $E$  is *unimodular* on  $\Lambda$ , that is,  $\det E = 1$  and  $\text{Pf}(E) = 1$ . Equivalently,  $E$  induces an isomorphism  $\Lambda \xrightarrow{\sim} \Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$  rather than an injection with finite cokernel; the cokernel  $\Lambda^\vee/E(\Lambda)$  has order  $(d_1 \cdots d_g)^2 = \det E$  in general and is trivial exactly for a principal polarization. Principal polarizations are the ones that behave most simply under duality and are the source of the canonical examples: the theta divisor of a ppav is unique up to translation (section 3.4 gives  $\dim H^0(X, L) = 1$  in this case), and the Jacobian of a curve is canonically a ppav. Not every abelian variety admits a principal polarization, but every abelian variety is isogenous to one that does, a point that belongs to the later theory of isogenies.

**Example 3.2.5: Types on the Square Torus**

Let  $g = 1$  and  $X = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}i)$ , the square torus. The polarization  $H(v, w) = v\bar{w}$  has  $E(v, w) = \text{Im}(v\bar{w})$ . Take the symplectic basis  $\lambda_1 = i, \mu_1 = 1$ ; then  $E(\lambda_1, \mu_1) = E(i, 1) = \text{Im}(i \cdot 1) = \text{Im}(i) = 1$ , while  $E(\lambda_1, \lambda_1) = E(\mu_1, \mu_1) = 0$ , so the Gram matrix in the ordered basis  $(\lambda_1, \mu_1)$  is  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  and the type is (1): this is a principal polarization. Scaling the form to  $H'(v, w) = 3v\bar{w}$  multiplies  $E$  by 3, giving Gram matrix  $\begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}$  and type (3). The bundle  $L'$  with  $c_1(L') = E' = 3E$  is still positive and still integral, hence still a polarization, but a non-principal one; by the Riemann-Roch count of section 3.4 it will carry  $\text{Pf}(E') = 3$  independent theta functions rather than one. In dimension one the type is a single integer, the degree of the associated line bundle.

For  $g > 1$  the type records  $g$  separate divisibilities. A two-dimensional abelian variety can carry a polarization of type  $(1, 1)$  (principal), or of type  $(1, d)$  with  $d > 1$ , or of type  $(d_1, d_2)$  with  $1 < d_1 | d_2$ ; these are distinct geometric structures on the same or different tori, and the moduli of section 3.3 are organized type by type. The type is thus the first place where the higher-dimensional theory diverges from the elliptic-curve case, where a polarization is determined up to scale by a single degree.

With the definition and its invariants in place, the equivalence that motivates the whole chapter can be stated cleanly. It ties three conditions together: the analytic condition (a polarization exists), the Riemann-relation condition on the period matrix, and the geometric condition (the torus is projective).

**Theorem 3.2.6: Polarizable if and only if Projective**

Let  $X = V/\Lambda$  be a complex torus. The following are equivalent.

1.  $X$  is an abelian variety, that is,  $X$  admits a polarization  $H > 0$  with  $E = \text{Im}H$  integral on  $\Lambda$ .
2. The period matrix  $\Pi$  of  $X$  satisfies the Riemann relations: there is a nondegenerate alternating integral matrix  $E \in M_{2g}(\mathbb{Z})$  with  $\Pi E^{-1} \Pi^t = 0$  and  $i \Pi E^{-1} \bar{\Pi}^t > 0$ .
3.  $X$  admits a closed holomorphic embedding into some projective space  $\mathbb{P}^N$ , and its image is a smooth projective algebraic variety.

**Proof:**

The equivalence of (1) and (2) is theorem 3.1.4: statement (2) is the coordinate incarnation of the positive integral Hermitian form in (1), obtained by writing  $H$  against the period matrix  $\Pi$ , and proposition 3.1.5 records the translation in both directions. Nothing further is needed here.

*Projective implies polarizable, (3)  $\Rightarrow$  (1).* Suppose  $X \hookrightarrow \mathbb{P}^N$  is a closed holomorphic embedding. The Fubini-Study Kähler form  $\omega_{\text{FS}}$  on  $\mathbb{P}^N$  restricts to a closed positive  $(1, 1)$ -form on  $X$  whose cohomology class lies in  $H^2(X, \mathbb{Z})$ , being the restriction of the integral hyperplane class. Under the identification  $H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee$  of proposition 1.3.5, this class is an alternating integral form  $E$  on  $\Lambda$ ; positivity of  $\omega_{\text{FS}}$  makes the associated Hermitian form  $H(v, w) = E(iv, w) + iE(v, w)$  positive definite, and the compatibility  $E(iv, iw) = E(v, w)$  is automatic because  $\omega_{\text{FS}}$  is of type  $(1, 1)$  for the complex structure. Thus  $H$  is a polarization and  $X$  is an abelian variety.

*Polarizable implies projective, (1)  $\Rightarrow$  (3).* This is the Lefschetz embedding theorem, proved

in full in chapter 4. Given a polarization  $H$ , the associated positive line bundle  $L = L(H, \chi)$  (definition 3.1.1) has, for  $n \geq 3$ , enough global theta functions that the map  $X \rightarrow \mathbb{P}^N$  sending  $x$  to the ratios of a basis of  $H^0(X, L^{\otimes n})$  is a closed holomorphic embedding; that  $L^{\otimes 3}$  is already very ample is the precise statement. The construction of the theta functions and the verification that they separate points and tangent vectors is carried out in chapter 4, so the embedding is deferred to there rather than reproduced now. Chow's theorem then upgrades the analytic image to a projective algebraic variety. Combining the two implications with the equivalence of (1) and (2) completes the cycle, as we sought to show.

The theorem is the analytic definition of an abelian variety made precise and useful. Condition (1) is what one checks in practice, since it is a finite linear-algebra problem on  $V$  and  $\Lambda$ ; condition (2) is what one computes from an explicit period matrix; and condition (3) is what the name “variety” promises. The only piece not proved here is the passage from a polarization to an actual embedding, which requires the theta functions of chapter 4 and is the reason polarizations were worth isolating in the first place. Everything downstream, the moduli space of section 3.3 and the Riemann-Roch count of section 3.4, works with condition (1) and never needs to see the embedding directly.

The class of abelian varieties is closed under the operations one expects of a good class of geometric objects, and the simplest of these is the product. Products let one build higher-dimensional examples from lower ones and, in the principally polarized case, preserve the principality.

**Example 3.2.7: Products of Principally Polarized Abelian Varieties**

Let  $(X_1, H_1)$  and  $(X_2, H_2)$  be abelian varieties with  $X_j = V_j/\Lambda_j$  and polarizations  $H_j$ . The product torus is  $X_1 \times X_2 = (V_1 \oplus V_2)/(\Lambda_1 \oplus \Lambda_2)$ , and the form

$$H(v_1 + v_2, w_1 + w_2) = H_1(v_1, w_1) + H_2(v_2, w_2)$$

is Hermitian on  $V_1 \oplus V_2$ . It is positive definite because each  $H_j$  is, and a sum of positive forms on complementary summands is positive on the direct sum. Its imaginary part  $E = E_1 \oplus E_2$  is integral on  $\Lambda_1 \oplus \Lambda_2$  because each  $E_j$  is integral on  $\Lambda_j$  and the cross terms vanish. Thus  $H$  is a polarization and  $X_1 \times X_2$  is an abelian variety.

Suppose now that both  $H_1$  and  $H_2$  are principal. Choose symplectic bases of  $\Lambda_1$  and  $\Lambda_2$  in which  $E_1$  and  $E_2$  have Gram matrices  $\begin{pmatrix} 0 & I_{g_1} \\ -I_{g_1} & 0 \end{pmatrix}$  and  $\begin{pmatrix} 0 & I_{g_2} \\ -I_{g_2} & 0 \end{pmatrix}$ . Concatenating the two  $\lambda$ -blocks and the two  $\mu$ -blocks produces a symplectic basis of  $\Lambda_1 \oplus \Lambda_2$  in which  $E = E_1 \oplus E_2$  has Gram matrix  $\begin{pmatrix} 0 & I_{g_1+g_2} \\ -I_{g_1+g_2} & 0 \end{pmatrix}$ , so the type of  $H$  is  $(1, \dots, 1)$ . A product of principally polarized abelian varieties is therefore principally polarized. In particular  $E \times E$ , with the product of the standard principal polarization on each factor, is a principally polarized abelian surface.

The product construction shows that principal polarizations are stable under the most basic building operation, but it also shows that not every ppav is indecomposable:  $E \times E$  is a principally polarized surface that splits as a product. The indecomposable principally polarized abelian surfaces, the ones that do not split as a product of two elliptic curves, are the Jacobians of genus-two curves, and this is the first instance of the single most important source of principally polarized abelian varieties in the subject.

**Example 3.2.8: The Canonical Polarization of a Jacobian**

Let  $C$  be a smooth projective curve of genus  $g \geq 1$  over  $\mathbb{C}$ . Its Jacobian is the complex torus

$$\text{Jac}(C) = H^0(C, \Omega_C^1)^* / H_1(C, \mathbb{Z})$$

of dimension  $g$ , where  $H_1(C, \mathbb{Z})$  embeds in the dual of the space of holomorphic 1-forms by integration,  $\gamma \mapsto (\omega \mapsto \int_\gamma \omega)$ . The cup-product intersection pairing on  $H_1(C, \mathbb{Z})$  is a nondegenerate alternating integral form, and with respect to a symplectic basis of 1-cycles (an  $a_i, b_i$  basis with  $a_i \cdot b_j = \delta_{ij}$  and  $a_i \cdot a_j = b_i \cdot b_j = 0$ ) its Gram matrix is exactly  $\begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$ . This form is the imaginary part of a positive Hermitian form on  $H^0(C, \Omega_C^1)^*$ , the one coming from the positivity of the Hodge inner product  $\frac{i}{2} \int_C \omega \wedge \bar{\eta}$ ; the positivity is the Riemann bilinear relations for the curve  $C$ . The resulting polarization is principal by construction, since the intersection form is unimodular, and it depends on nothing but  $C$ . Every curve thus manufactures a principally polarized abelian variety of dimension  $g$ , canonically and with no choices.

The Jacobian construction is the reason principal polarizations are central rather than exceptional: they are the polarizations that arise from geometry without any auxiliary data, and the map sending a curve to its Jacobian is the bridge between the theory of curves and the theory of abelian varieties. That this map is close to injective, so that a curve can be recovered from its principally polarized Jacobian, is the Torelli theorem, and the full algebraic construction of the Jacobian together with its canonical polarization occupies chapter 12. The analytic version sketched here already shows the mechanism: the intersection form on  $H_1(C, \mathbb{Z})$  is unimodular for topological reasons, and the Hodge metric supplies the positivity, so the two ingredients of a principal polarization are both present the moment one writes down the Jacobian.

The word “abelian variety” now has a precise analytic meaning. It is a complex torus carrying a positive integral Hermitian form, equivalently a torus whose period matrix satisfies the Riemann relations, equivalently a projective complex torus. All three descriptions are of the same objects, and the equivalence is the analytic heart of Part I. The definition is, however, tied to  $\mathbb{C}$  at every step: it invokes a Hermitian form on a complex vector space, a period matrix, and holomorphic embeddings, none of which survive the passage to a general field. Part II gives the field-independent algebraic definition, an abelian variety as a complete connected group variety over an arbitrary field, in which projectivity is a theorem rather than a hypothesis and no metric or transcendental input appears. That the two definitions agree over  $\mathbb{C}$ , so that the analytic abelian varieties of this chapter are exactly the complex points of the algebraic ones, is the comparison theorem taken up in chapter 11, where the analytic period data of Part I is matched against the algebraic structure of Part II.

**Exercise 3.2.1**

1. Let  $X = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$  and  $\tau = 2i$ , and set  $H(v, w) = cv\bar{w}$  with  $c > 0$ . Following example 3.2.2, determine the smallest  $c > 0$  for which  $H$  is a polarization of  $X$ , and compute the type of the resulting polarization. Is it principal?
2. Consider  $X = \mathbb{C}^2/\Lambda$  with  $\Lambda = \mathbb{Z}^4$  under the standard inclusion  $\mathbb{Z}^4 \subset \mathbb{R}^4 = \mathbb{C}^2$ , and let  $E$  be the alternating form whose Gram matrix in the standard basis  $e_1, e_2, e_3, e_4$  of  $\Lambda$  is

$$\begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \\ -2 & 0 & 0 & 0 \\ 0 & -6 & 0 & 0 \end{pmatrix}$$

Assuming  $E$  is the imaginary part of a positive definite Hermitian form, compute the type

- $(d_1, d_2)$  of the polarization in the sense of definition 3.2.3, the Pfaffian  $\text{Pf}(E)$ , and  $\det E$ . Verify the relation  $\det E = (d_1 d_2)^2$  recorded after definition 3.2.3.
3. Let  $H$  be a polarization of  $X = V/\Lambda$  with type  $(d_1, \dots, d_g)$ , and let  $n \geq 1$  be an integer. Show that  $nH$  is again a polarization of  $X$  and that its type is  $(nd_1, \dots, nd_g)$ . Deduce that if  $g \geq 1$  and  $n \geq 2$ , then  $nH$  is never principal, and conclude that the square torus of example 3.2.5 carries polarizations of infinitely many distinct types.
  4. example 3.2.7 shows that a product of principally polarized abelian varieties is principally polarized. Generalize the type computation there: if  $(X_1, H_1)$  has type  $(a_1, \dots, a_{g_1})$  and  $(X_2, H_2)$  has type  $(b_1, \dots, b_{g_2})$ , the product polarization  $H = H_1 \oplus H_2$  on  $X_1 \times X_2$  has  $\text{Pf}(E) = \text{Pf}(E_1) \text{Pf}(E_2)$  where  $E = \text{Im}H$ . Show this, and give an explicit example of two type-(1) polarizations on elliptic curves whose product has type  $(1, 1)$ , together with an example of a type-(1) and a type-(2) polarization whose product has type  $(1, 2)$  rather than  $(2, 2)$ .
  5. Prove directly, without invoking the Lefschetz embedding theorem, that if a complex torus  $X = V/\Lambda$  admits a closed holomorphic embedding into some  $\mathbb{P}^N$  then it is an abelian variety in the sense of definition 3.2.1. This is the implication  $(3) \Rightarrow (1)$  of theorem 3.2.6; reconstruct its proof, stating precisely where positivity and integrality of the pulled-back class each come from.
  6. A principal polarization  $H$  of  $X = V/\Lambda$  is one for which  $E = \text{Im}H$  is unimodular on  $\Lambda$ , equivalently for which the induced map  $\Lambda \rightarrow \Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$ ,  $\lambda \mapsto E(\lambda, -)$ , is an isomorphism (definition 3.2.4). Prove that this map is always injective when  $H > 0$ , that its cokernel is finite of order  $\det E = (d_1 \cdots d_g)^2$ , and hence that  $H$  is principal if and only if this cokernel is trivial. Here  $(d_1, \dots, d_g)$  is the type of  $H$ .

### 3.3 The Period Matrix, the Siegel Upper Half-Space, and Moduli

The Riemann relations decide, for a single torus, whether it is an abelian variety. Once that question is settled, a second one takes its place: the enumeration and organization of all abelian varieties of a given dimension. For an elliptic curve the answer is already familiar from [4]. A lattice in  $\mathbb{C}$  normalizes to  $\mathbb{Z}\tau + \mathbb{Z}$  with  $\tau$  in the upper half-plane, two such lattices give isomorphic curves exactly when their parameters are related by  $\text{SL}_2(\mathbb{Z})$ , and the quotient  $\text{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$  is the moduli space of elliptic curves, uniformized by the  $j$ -invariant. This section carries out the same program in dimension  $g$ . The role of the upper half-plane is played by the Siegel upper half-space  $\mathfrak{H}_g$ , the role of  $\text{SL}_2(\mathbb{Z})$  by the symplectic group  $\text{Sp}(2g, \mathbb{Z})$ , and the quotient  $\mathfrak{H}_g / \text{Sp}(2g, \mathbb{Z})$  is the first picture of the moduli space of principally polarized abelian varieties of dimension  $g$ .

The mechanism is a normalization of the period matrix. A principal polarization supplies a symplectic basis of the lattice (definition 3.2.3), and in a complex basis adapted to it the period matrix collapses to the compact form  $(\tau \ I_g)$  for a single  $g \times g$  matrix  $\tau$ . The Riemann relations, which in a general basis are a pair of bilinear conditions on the full period matrix, become the two transparent conditions that  $\tau$  is symmetric and has positive definite imaginary part. That is the definition of  $\mathfrak{H}_g$ , so every principally polarized abelian variety is a point of  $\mathfrak{H}_g$ , and the ambiguity in the construction (the freedom in the choice of symplectic basis) is precisely the  $\text{Sp}(2g, \mathbb{Z})$ -action.

**Definition 3.3.1: Siegel Upper Half-Space**

The *Siegel upper half-space* of genus  $g$  is

$$\mathfrak{H}_g = \{ \tau \in M_g(\mathbb{C}) \mid \tau = \tau^t, \operatorname{Im} \tau > 0 \}$$

the set of symmetric complex  $g \times g$  matrices whose imaginary part  $\operatorname{Im} \tau = \frac{1}{2i}(\tau - \bar{\tau})$  is a positive definite real symmetric matrix. It is an open subset of the  $\mathbb{C}$ -vector space  $\operatorname{Sym}^2 \mathbb{C}^g$  of symmetric  $g \times g$  matrices, hence a complex manifold of dimension

$$\dim_{\mathbb{C}} \mathfrak{H}_g = \frac{g(g+1)}{2}$$

the number of entries on and above the diagonal of a symmetric matrix.

The condition  $\tau = \tau^t$  cuts the symmetric matrices out of all of  $M_g(\mathbb{C})$ , a linear condition leaving the  $g(g+1)/2$  independent entries. The condition  $\operatorname{Im} \tau > 0$  is open and nonempty, so it selects an open region rather than a lower-dimensional slice, and the dimension is unchanged. For  $g = 1$  a  $1 \times 1$  matrix is automatically symmetric and  $\operatorname{Im} \tau > 0$  is the single inequality defining the upper half-plane, so  $\mathfrak{H}_1 = \mathbb{H}$  and  $\dim_{\mathbb{C}} \mathfrak{H}_1 = 1$ ; the Siegel upper half-space is the exact multivariable analogue of the domain that uniformizes elliptic curves. The positivity of  $\operatorname{Im} \tau$  is the shadow of the positivity of the polarization: it is what will make the Hermitian form  $H$  come out positive definite, and it is what makes  $\mathfrak{H}_g$  a bounded symmetric domain rather than all of  $\operatorname{Sym}^2 \mathbb{C}^g$ .

**Example 3.3.2: The Siegel Upper Half-Space in Genus Two**

For  $g = 2$  a point of  $\mathfrak{H}_2$  is a symmetric matrix

$$\tau = \begin{pmatrix} \tau_{11} & \tau_{12} \\ \tau_{12} & \tau_{22} \end{pmatrix} \quad \tau_{ij} \in \mathbb{C}$$

with three free entries  $\tau_{11}, \tau_{12}, \tau_{22}$ , so  $\dim_{\mathbb{C}} \mathfrak{H}_2 = 3 = \frac{2 \cdot 3}{2}$ . Writing  $\tau_{ij} = x_{ij} + iy_{ij}$ , the positivity condition  $\operatorname{Im} \tau > 0$  reads

$$y_{11} > 0 \quad \text{and} \quad y_{11}y_{22} - y_{12}^2 > 0$$

the leading-principal-minor (Sylvester) criterion for the symmetric real matrix  $\operatorname{Im} \tau = \begin{pmatrix} y_{11} & y_{12} \\ y_{12} & y_{22} \end{pmatrix}$  to be positive definite. The diagonal points  $\tau = \operatorname{diag}(\tau_1, \tau_2)$  with  $\tau_1, \tau_2 \in \mathbb{H}$  form a copy of  $\mathbb{H} \times \mathbb{H}$  inside  $\mathfrak{H}_2$ ; these are the period matrices of the products  $E_1 \times E_2$  of two elliptic curves, and the off-diagonal entry  $\tau_{12}$  measures how far a principally polarized abelian surface is from splitting as such a product.

The example makes the parameter count concrete and displays the smallest case where the Siegel domain is more than a half-plane: a principally polarized abelian surface depends on three complex parameters, of which two record a pair of elliptic curves and the third records how they are glued. To turn a torus into such a point, one must first put its period matrix into normal form.

Recall the setup from section 3.1. A period matrix for  $X = V/\Lambda$  is a matrix  $\Pi \in M_{g \times 2g}(\mathbb{C})$  whose columns are the coordinates, in a fixed  $\mathbb{C}$ -basis of  $V$ , of a  $\mathbb{Z}$ -basis  $\lambda_1, \dots, \lambda_{2g}$  of  $\Lambda$ . A principal polarization (definition 3.2.4) is a positive Hermitian form  $H$  whose imaginary part  $E = \operatorname{Im} H$  is a unimodular integral alternating form on  $\Lambda$ , and by definition 3.2.3 (with type  $(1, \dots, 1)$ ) there is a symplectic basis  $\mu_1, \dots, \mu_g, \nu_1, \dots, \nu_g$  of  $\Lambda$  in which  $E$  has matrix  $\begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$ . The next result says



that using this basis, and one adapted change of complex coordinates, brings  $\Pi$  to the form  $(\tau I_g)$  with  $\tau \in \mathfrak{H}_g$ , and that the Riemann relations become exactly the two conditions defining  $\mathfrak{H}_g$ .

**Proposition 3.3.3: Normalized Period Matrix of a Principally Polarized Abelian Variety**

Let  $X = V/\Lambda$  be a principally polarized abelian variety of dimension  $g$ , with polarization  $H$  and  $E = \text{Im}H$ . Choose a symplectic basis  $\mu_1, \dots, \mu_g, \nu_1, \dots, \nu_g$  of  $\Lambda$  for  $E$ , so that

$$E(\mu_i, \mu_j) = 0 \quad E(\nu_i, \nu_j) = 0 \quad E(\mu_i, \nu_j) = \delta_{ij}$$

Then  $\nu_1, \dots, \nu_g$  is a  $\mathbb{C}$ -basis of  $V$ . Expressing  $\Lambda$  in this basis, the period matrix takes the form  $\Pi = (\tau I_g)$ , where  $\tau \in M_g(\mathbb{C})$  is the matrix whose  $j$ -th column is  $\mu_j$  written in the basis  $\nu_1, \dots, \nu_g$ . Moreover  $\tau \in \mathfrak{H}_g$ , that is

$$\tau = \tau^t \quad \text{and} \quad \text{Im} \tau > 0$$

and these two conditions are precisely the Riemann relations for the period matrix  $(\tau I_g)$  with respect to  $E$ . The lattice is  $\Lambda = \tau \mathbb{Z}^g + \mathbb{Z}^g$ .

**Proof:**

The argument has three steps: first that the  $\nu_j$  form a complex basis, then the normalization of the period matrix, and finally the translation of the two Riemann relations into symmetry and positivity of  $\tau$ .

*Step 1: The vectors  $\nu_1, \dots, \nu_g$  are a  $\mathbb{C}$ -basis of  $V$ .* Since  $\Lambda$  has rank  $2g$  over  $\mathbb{Z}$  and  $V = \Lambda \otimes_{\mathbb{Z}} \mathbb{R}$  has real dimension  $2g$ , the  $2g$  vectors  $\mu_i, \nu_j$  form an  $\mathbb{R}$ -basis of  $V$ . It suffices to show that  $\nu_1, \dots, \nu_g$  are linearly independent over  $\mathbb{C}$ , for then they span the  $g$ -dimensional complex space  $V$ . Suppose  $\sum_j c_j \nu_j = 0$  with  $c_j \in \mathbb{C}$ ; write  $c_j = a_j + ib_j$  with  $a_j, b_j \in \mathbb{R}$ . Then

$$\sum_j a_j \nu_j + \sum_j b_j (i\nu_j) = 0$$

The  $\nu_j$  are Lagrangian for  $E$ , so  $E(\nu_j, \nu_k) = 0$  for all  $j, k$ ; and by the identity  $\text{Re}H(v, w) = E(iv, w)$  of section 3.1, together with the symmetry  $\text{Re}H(v, w) = \text{Re}H(w, v)$ ,

$$E(i\nu_j, \nu_k) = \text{Re}H(\nu_j, \nu_k) = \text{Re}H(\nu_k, \nu_j) = E(i\nu_k, \nu_j)$$

The matrix  $S$  with entries  $S_{jk} = E(i\nu_j, \nu_k) = \text{Re}H(\nu_j, \nu_k)$  is therefore real symmetric, and it is positive definite because  $H$  is positive definite and  $\text{Re}H$  is its real part, whose restriction to the real span of the  $\nu_j$  is positive definite (a positive definite Hermitian form has positive definite real part on any real subspace on which it is defined). Pairing the displayed relation against each  $\nu_k$  under  $E$  gives  $\sum_j b_j S_{jk} = 0$  for all  $k$ , hence  $b = 0$  since  $S$  is invertible; the relation then reads  $\sum_j a_j \nu_j = 0$  with real  $a_j$ , and pairing against  $i\nu_k$  under  $E$  gives  $\sum_j a_j S_{kj} = 0$ , whence  $a = 0$ . Thus  $c = 0$  and the  $\nu_j$  are  $\mathbb{C}$ -linearly independent, as we sought to show.

*Step 2: The period matrix is  $(\tau I_g)$ .* Adopt  $\nu_1, \dots, \nu_g$  as the  $\mathbb{C}$ -basis of  $V$ . In these coordinates  $\nu_j$  is the  $j$ -th standard basis vector, so the block of the period matrix recording  $\nu_1, \dots, \nu_g$  is the identity  $I_g$ . Let  $\tau \in M_g(\mathbb{C})$  be the matrix whose  $j$ -th column is the coordinate vector of  $\mu_j$  in the basis  $\nu_1, \dots, \nu_g$ , so that  $\mu_j = \sum_k \tau_{kj} \nu_k$ . Ordering the lattice basis as  $\mu_1, \dots, \mu_g, \nu_1, \dots, \nu_g$ ,



the period matrix is

$$\Pi = (\tau \ I_g)$$

and the lattice is  $\Lambda = \tau\mathbb{Z}^g + \mathbb{Z}^g$ , the  $\mathbb{Z}$ -span of the columns of  $\tau$  together with the standard basis vectors. This is the claimed normal form.

*Step 3: The Riemann relations become  $\tau = \tau^t$  and  $\text{Im}\tau > 0$ .* In the symplectic basis  $E$  has matrix  $J = \begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$ , with inverse  $J^{-1} = \begin{pmatrix} 0 & -I_g \\ I_g & 0 \end{pmatrix}$ . The Riemann relations for the period matrix  $\Pi = (\tau \ I_g)$  with respect to  $E$  (proposition 3.1.5) are

$$\Pi J^{-1} \Pi^t = 0 \quad \text{and} \quad i \Pi J^{-1} \bar{\Pi}^t > 0$$

Compute the first product. Writing  $\Pi = (\tau \ I_g)$  in blocks,

$$\Pi J^{-1} = (\tau \ I_g) \begin{pmatrix} 0 & -I_g \\ I_g & 0 \end{pmatrix} = (I_g \ -\tau)$$

so that

$$\Pi J^{-1} \Pi^t = (I_g \ -\tau) \begin{pmatrix} \tau^t \\ I_g \end{pmatrix} = \tau^t - \tau$$

The first Riemann relation  $\Pi J^{-1} \Pi^t = 0$  is therefore equivalent to  $\tau = \tau^t$ : the matrix  $\tau$  is symmetric. For the second, using  $\bar{\Pi} = (\bar{\tau} \ I_g)$ ,

$$i \Pi J^{-1} \bar{\Pi}^t = i (I_g \ -\tau) \begin{pmatrix} \bar{\tau}^t \\ I_g \end{pmatrix} = i(\bar{\tau}^t - \tau)$$

Since  $\tau$  is symmetric,  $\bar{\tau}^t = \bar{\tau}$ , and

$$i(\bar{\tau} - \tau) = i(-2i \text{Im}\tau) = 2 \text{Im}\tau$$

The second Riemann relation  $i \Pi J^{-1} \bar{\Pi}^t > 0$  is thus equivalent to  $\text{Im}\tau > 0$ . Both relations hold because  $H$  is a polarization, so  $\tau$  is symmetric with positive definite imaginary part, that is  $\tau \in \mathfrak{H}_g$ , completing the proof.

The proposition is the structural heart of the section. It converts a principally polarized abelian variety, an object presented by a full  $g \times 2g$  complex period matrix subject to two matrix identities, into a single point  $\tau$  of the domain  $\mathfrak{H}_g$ , with the two identities absorbed into the definition of the domain itself. Nothing has been lost: from  $\tau$  one recovers the lattice  $\Lambda = \tau\mathbb{Z}^g + \mathbb{Z}^g$ , hence the torus  $X = \mathbb{C}^g/\Lambda$ , and the Hermitian form  $H(v, w) = v^t(\text{Im}\tau)^{-1}\bar{w}$  whose imaginary part is unimodular integral on  $\Lambda$  and which is positive definite exactly because  $\text{Im}\tau > 0$ ; this is the polarization back again. So the map from (symplectic basis)-marked principally polarized abelian varieties to  $\mathfrak{H}_g$  is a bijection. The freedom in the marking is what remains to be quotiented out.

#### Example 3.3.4: The Product of Two Elliptic Curves as a Period Matrix

Let  $E_1 = \mathbb{C}/(\mathbb{Z}\tau_1 + \mathbb{Z})$  and  $E_2 = \mathbb{C}/(\mathbb{Z}\tau_2 + \mathbb{Z})$  with  $\tau_1, \tau_2 \in \mathbb{H}$ , each carrying its principal polarization from [4]. The product  $X = E_1 \times E_2$  is the torus  $\mathbb{C}^2/\Lambda$  with lattice generated by the columns of

$$\Pi = \begin{pmatrix} \tau_1 & 0 & 1 & 0 \\ 0 & \tau_2 & 0 & 1 \end{pmatrix}$$

whose product polarization has  $E = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}$  in the evident symplectic basis  $\mu_1 = (\tau_1, 0)$ ,  $\mu_2 = (0, \tau_2)$ ,  $\nu_1 = (1, 0)$ ,  $\nu_2 = (0, 1)$ . The block of  $\nu$ -periods is already  $I_2$ , so no change of coordinates is needed, and the associated Siegel point is

$$\tau = \begin{pmatrix} \tau_1 & 0 \\ 0 & \tau_2 \end{pmatrix} \in \mathfrak{H}_2$$

a diagonal matrix. It is symmetric, and  $\text{Im}\tau = \text{diag}(\text{Im}\tau_1, \text{Im}\tau_2) > 0$  since  $\tau_1, \tau_2 \in \mathbb{H}$ ; both Riemann relations reduce to their  $g = 1$  instances on each factor. This is the diagonal locus  $\mathbb{H} \times \mathbb{H} \subset \mathfrak{H}_2$  of example 3.3.2: products of elliptic curves are exactly the principally polarized abelian surfaces whose Siegel point is diagonal.

The normal form is not unique, because the symplectic basis is not unique. Two symplectic bases of  $\Lambda$  for the same  $E$  differ by an integral matrix that preserves the form  $J$ , and such matrices are exactly the integral symplectic group. Changing the symplectic basis changes the normalized  $\tau$ , and the resulting equivalence relation on  $\mathfrak{H}_g$  is what identifies period matrices that describe isomorphic abelian varieties. The next definition records the group and its action.

**Definition 3.3.5: The Symplectic Group and its Action on  $\mathfrak{H}_g$**

The *integral symplectic group* is

$$\text{Sp}(2g, \mathbb{Z}) = \{M \in \text{GL}_{2g}(\mathbb{Z}) \mid M^t J M = J\} \quad J = \begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$$

the group of integer matrices preserving the standard alternating form  $J$ . Writing  $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$  in  $g \times g$  blocks, the group acts on the Siegel upper half-space  $\mathfrak{H}_g$  by

$$M \cdot \tau = (A\tau + B)(C\tau + D)^{-1}$$

the *Siegel modular transformation* associated to  $M$ .

The block condition  $M^t J M = J$  unpacks into  $A^t C = C^t A$ ,  $B^t D = D^t B$ , and  $A^t D - C^t B = I_g$ ; these are the defining relations of the symplectic group in block form, and for  $g = 1$  they collapse to the single relation  $ad - bc = 1$ , so  $\text{Sp}(2, \mathbb{Z}) = \text{SL}_2(\mathbb{Z})$ . That the formula  $M \cdot \tau = (A\tau + B)(C\tau + D)^{-1}$  lands back in  $\mathfrak{H}_g$ , that  $C\tau + D$  is invertible for  $\tau \in \mathfrak{H}_g$ , and that the assignment is a group action are the content of the following verification, which also identifies the action with change of symplectic basis.

**Proposition 3.3.6: The Siegel Action is Well-Defined and Realizes Change of Symplectic Basis**

For  $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \text{Sp}(2g, \mathbb{Z})$  and  $\tau \in \mathfrak{H}_g$ , the matrix  $C\tau + D$  is invertible, the point  $M \cdot \tau = (A\tau + B)(C\tau + D)^{-1}$  lies in  $\mathfrak{H}_g$ , and  $\tau \mapsto M \cdot \tau$  is an action of  $\text{Sp}(2g, \mathbb{Z})$  on  $\mathfrak{H}_g$ . Two points  $\tau, \tau' \in \mathfrak{H}_g$  arise from the same principally polarized abelian variety, by two choices of symplectic basis, if and only if  $\tau' = M \cdot \tau$  for some  $M \in \text{Sp}(2g, \mathbb{Z})$ .

**Proof:**

*Step 1:  $C\tau + D$  is invertible and  $M \cdot \tau \in \mathfrak{H}_g$ .* Write  $P = A\tau + B$  and  $Q = C\tau + D$ . The block relations  $A^t D - C^t B = I_g$ ,  $A^t C = C^t A$ , and  $B^t D = D^t B$  from definition 3.3.5, together with  $\tau = \tau^t$  and the reality of  $A, B, C, D$ , give the identity

$$\bar{Q}^t P - \bar{P}^t Q = \tau - \bar{\tau} = 2i \operatorname{Im} \tau \quad (3.5)$$

obtained by expanding both products and cancelling the terms  $\bar{\tau}(C^t A - A^t C)\tau = 0$  and  $D^t B - B^t D = 0$ , while the remaining terms contribute  $-\bar{\tau}(A^t D - C^t B) + (D^t A - B^t C)\tau = -\bar{\tau} + \tau$  using  $D^t A - B^t C = (A^t D - C^t B)^t = I_g$ . If  $Qx = 0$  for some  $x \in \mathbb{C}^g$ , then multiplying the identity on the left by  $\bar{x}^t$  and on the right by  $x$  makes the left-hand side vanish (both  $\bar{x}^t \bar{Q}^t = \overline{Qx}^t = 0$  and  $Qx = 0$ ), so  $2i \bar{x}^t (\operatorname{Im} \tau) x = 0$ ; since  $\operatorname{Im} \tau > 0$  this forces  $x = 0$ . Hence  $Q = C\tau + D$  is invertible, and  $M \cdot \tau = PQ^{-1}$  is defined.

It remains to check  $W := M \cdot \tau = PQ^{-1} \in \mathfrak{H}_g$ . For symmetry, the relations give  $P^t Q = Q^t P$ : expanding,  $P^t Q - Q^t P = \tau(A^t C - C^t A)\tau + \tau(A^t D - C^t B) - (D^t A - B^t C)\tau + (B^t D - D^t B) = \tau - \tau = 0$ , using  $A^t C = C^t A$ ,  $B^t D = D^t B$ , and  $A^t D - C^t B = (D^t A - B^t C)^t = I_g$  with  $\tau$  symmetric. Hence  $W^t = (Q^t)^{-1} P^t = (Q^t)^{-1} (P^t Q) Q^{-1} = (Q^t)^{-1} (Q^t P) Q^{-1} = PQ^{-1} = W$ , so  $W$  is symmetric. For positivity, symmetry of  $W$  gives  $\bar{W} = (\bar{Q}^t)^{-1} \bar{P}^t$ , and using (3.5),

$$\bar{Q}^t (W - \bar{W}) Q = \bar{Q}^t P - \bar{P}^t Q = 2i \operatorname{Im} \tau$$

so that  $\operatorname{Im} W = (\bar{Q}^t)^{-1} (\operatorname{Im} \tau) Q^{-1}$ . This is the congruence of the positive definite matrix  $\operatorname{Im} \tau$  by the invertible matrix  $Q^{-1}$ , hence positive definite. Therefore  $W = M \cdot \tau$  is symmetric with  $\operatorname{Im} W > 0$ , that is  $M \cdot \tau \in \mathfrak{H}_g$ .

*Step 2: The assignment is a group action.* The identity matrix  $I_{2g}$  has  $C = 0$ ,  $D = I_g$ , so  $I_{2g} \cdot \tau = \tau$ . For the composite of  $M_1 = \begin{pmatrix} A_1 & B_1 \\ C_1 & D_1 \end{pmatrix}$  and  $M_2 = \begin{pmatrix} A_2 & B_2 \\ C_2 & D_2 \end{pmatrix}$ , the automorphy factor satisfies the cocycle relation

$$C_{21}\tau + D_{21} = (C_2(M_1 \cdot \tau) + D_2)(C_1\tau + D_1)$$

where  $\begin{pmatrix} A_{21} & B_{21} \\ C_{21} & D_{21} \end{pmatrix} = M_2 M_1$ , obtained by multiplying out the bottom block row of  $M_2 M_1$  and substituting  $M_1 \cdot \tau = (A_1\tau + B_1)(C_1\tau + D_1)^{-1}$ . Dividing the top block row identity by this factor gives  $(M_2 M_1) \cdot \tau = M_2 \cdot (M_1 \cdot \tau)$ , so the assignment is a left action of  $\operatorname{Sp}(2g, \mathbb{Z})$  on  $\mathfrak{H}_g$ .

*Step 3: The action realizes change of symplectic basis.* Let  $\Pi = (\tau I_g)$  be a normalized period matrix as in proposition 3.3.3, with symplectic basis columns  $\mu_1, \dots, \mu_g, \nu_1, \dots, \nu_g$ . A second  $\mathbb{Z}$ -basis of  $\Lambda$  is  $\Pi' = \Pi N$  for a unique  $N \in \operatorname{GL}_{2g}(\mathbb{Z})$ , and  $\Pi'$  is again a symplectic basis for  $E$  if and only if  $N$  preserves the matrix  $J$  of  $E$ , that is  $N^t J N = J$ , so  $N \in \operatorname{Sp}(2g, \mathbb{Z})$ . The transformed period matrix  $\Pi N = (\tau I_g) N$  need not be in normal form; but both principally polarized abelian varieties it presents are the same  $(X, H)$ , since  $\Pi$  and  $\Pi N$  span the same lattice and carry the same  $E$ . Renormalizing  $\Pi N$  by the recipe of proposition 3.3.3 returns a point  $\tau' \in \mathfrak{H}_g$  that is, by construction, obtained from  $\tau$  by an integral symplectic change of the coordinates  $(\mu, \nu)$ ; the Siegel transformation of definition 3.3.5, applied to the symplectic matrix  $M = N^t \in \operatorname{Sp}(2g, \mathbb{Z})$ , produces exactly this  $\tau' = M \cdot \tau$  (both are the normalized period point of the same polarized torus in the new basis, and the normalization is unique by proposition 3.3.3). Conversely every  $M \in \operatorname{Sp}(2g, \mathbb{Z})$  arises from such a change of basis. Thus two points  $\tau, \tau' \in \mathfrak{H}_g$  present the same principally polarized abelian variety, through different symplectic bases, if and only if  $\tau' = M \cdot \tau$  for some  $M \in \operatorname{Sp}(2g, \mathbb{Z})$ , as we sought to show.

The proposition completes the dictionary. A principally polarized abelian variety, once a symplectic basis is chosen, is a point of  $\mathfrak{H}_g$ ; forgetting the choice of basis quotients by  $\mathrm{Sp}(2g, \mathbb{Z})$ ; and points in the same orbit describe isomorphic polarized varieties. Everything needed for the moduli statement is now in place. The one point deserving emphasis is what “isomorphic” means here: an isomorphism of principally polarized abelian varieties is a biholomorphism  $X \rightarrow X'$  carrying the polarization of  $X$  to that of  $X'$ , equivalently a  $\mathbb{C}$ -linear isomorphism  $V \rightarrow V'$  carrying  $\Lambda$  to  $\Lambda'$  and  $H$  to  $H'$ . Under normalization this is exactly a change of symplectic basis, so the orbit space is the set of isomorphism classes.

### Theorem 3.3.7: Moduli of Principally Polarized Abelian Varieties

The map sending a principally polarized abelian variety of dimension  $g$  to the  $\mathrm{Sp}(2g, \mathbb{Z})$ -orbit of its normalized Siegel point induces a bijection

$$\left\{ \begin{array}{c} \text{principally polarized abelian} \\ \text{varieties of dimension } g \end{array} \right\} / \cong \xrightarrow{\sim} \mathfrak{H}_g / \mathrm{Sp}(2g, \mathbb{Z})$$

between isomorphism classes of  $g$ -dimensional principally polarized abelian varieties and the orbit space of the Siegel upper half-space under the integral symplectic group. This orbit space is a complex analytic space of dimension

$$\dim_{\mathbb{C}} \mathfrak{H}_g / \mathrm{Sp}(2g, \mathbb{Z}) = \dim_{\mathbb{C}} \mathfrak{H}_g = \frac{g(g+1)}{2}$$

the dimension of the moduli space  $\mathcal{A}_g$  of principally polarized abelian varieties of dimension  $g$ .

#### **Proof:**

By proposition 3.3.3, every principally polarized abelian variety of dimension  $g$ , together with a choice of symplectic basis, determines a point  $\tau \in \mathfrak{H}_g$ , and conversely every  $\tau \in \mathfrak{H}_g$  determines such data: the torus  $X_\tau = \mathbb{C}^g / (\tau \mathbb{Z}^g + \mathbb{Z}^g)$  with the Hermitian form  $H_\tau(v, w) = v^t (\mathrm{Im} \tau)^{-1} \bar{w}$ . This form is positive definite because  $\mathrm{Im} \tau > 0$ , and its imaginary part  $E_\tau = \mathrm{Im} H_\tau$  has, in the basis given by the columns of  $\Pi = (\tau \ I_g)$ , the standard symplectic matrix  $J$  (this is the content of Step 3 of proposition 3.3.3 read backwards), so  $E_\tau$  is a unimodular integral alternating form and  $H_\tau$  is a principal polarization. Thus  $\tau \mapsto (X_\tau, H_\tau, \text{basis})$  is a two-sided inverse to normalization, giving a bijection between  $\mathfrak{H}_g$  and basis-marked principally polarized abelian varieties.

By proposition 3.3.6, two points of  $\mathfrak{H}_g$  give isomorphic principally polarized abelian varieties (forgetting the basis) if and only if they lie in the same  $\mathrm{Sp}(2g, \mathbb{Z})$ -orbit. Passing to orbits on the left and to isomorphism classes on the right, the bijection descends to the stated bijection between isomorphism classes and  $\mathfrak{H}_g / \mathrm{Sp}(2g, \mathbb{Z})$ .

For the dimension, the action of  $\mathrm{Sp}(2g, \mathbb{Z})$  on  $\mathfrak{H}_g$  is properly discontinuous (the group is discrete and acts by biholomorphisms of a bounded symmetric domain), so the quotient inherits the structure of a complex analytic space of the same dimension as  $\mathfrak{H}_g$  (a quotient of a complex manifold by a properly discontinuous action has the same dimension everywhere; the fixed loci are lower-dimensional suborbifolds, not lower-dimensional components). Hence  $\dim_{\mathbb{C}} \mathfrak{H}_g / \mathrm{Sp}(2g, \mathbb{Z}) = \dim_{\mathbb{C}} \mathfrak{H}_g = g(g+1)/2$  by definition 3.3.1, as we sought to show.

This is the first sight of the moduli space of abelian varieties, denoted  $\mathcal{A}_g$ : as a set, the isomorphism classes of  $g$ -dimensional principally polarized abelian varieties; as a space, the quotient  $\mathfrak{H}_g / \mathrm{Sp}(2g, \mathbb{Z})$ ,

of dimension  $g(g+1)/2$ . Two cautions fix the level of the claim. First, the theorem describes  $\mathcal{A}_g$  as an analytic quotient and as a set of isomorphism classes; it is a *coarse* description, in that points of the quotient are in bijection with isomorphism classes but the quotient does not by itself carry a universal family of abelian varieties over it. The obstruction is exactly the extra automorphisms that some polarized varieties possess (every abelian variety has the automorphism  $[-1]$ , and special ones have more), which is why  $\mathcal{A}_g$  carries the structure of an orbifold, or in the algebraic category a Deligne-Mumford stack, rather than a smooth manifold or variety. Developing  $\mathcal{A}_g$  as a genuine moduli *scheme*, and its compactifications and Siegel modular forms, is outside this book's scope; it is pointed to in chapter 14 and taken up in the companion volume `NateShimuraVarieties`. Second, the quotient here is constructed purely analytically, from period matrices and the symplectic action; the arithmetic of the space, its special points and their fields of definition, is developed later. The special points where  $X$  has complex multiplication are the CM points, treated in chapter 6, and they are the seeds from which the arithmetic theory of  $\mathcal{A}_g$  and of Shimura varieties grows, developed in the companion volume `NateShimuraVarieties`.

The genus-one case is the sanity check, and it recovers a construction the reader has already met in full.

### Example 3.3.8: Genus One: the Modular Curve and the $j$ -Line

Take  $g = 1$ . A  $1 \times 1$  matrix  $\tau = (\tau_{11})$  is automatically symmetric, and  $\text{Im}\tau > 0$  is the single condition  $\text{Im}\tau_{11} > 0$ , so

$$\mathfrak{H}_1 = \mathbb{H} = \{\tau \in \mathbb{C} \mid \text{Im}\tau > 0\}$$

is the upper half-plane, of dimension  $\frac{1 \cdot 2}{2} = 1$ . The symplectic group is  $\text{Sp}(2, \mathbb{Z}) = \text{SL}_2(\mathbb{Z})$ , since for a  $2 \times 2$  integer matrix the condition  $M^t J M = J$  with  $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  is the single equation  $\det M = 1$ . The Siegel action specializes, for  $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ , to

$$M \cdot \tau = \frac{a\tau + b}{c\tau + d}$$

the classical action of the modular group on  $\mathbb{H}$  by Möbius transformations. Theorem 3.3.7 then reads: principally polarized abelian varieties of dimension 1, up to isomorphism, correspond to  $\text{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ . A 1-dimensional abelian variety is an elliptic curve, and its unique principal polarization is the one furnished by a point (the class of a degree-one divisor); so this is exactly the statement, proved analytically in [4], that  $\text{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$  is the moduli space of elliptic curves, uniformized by the  $j$ -invariant  $j: \text{SL}_2(\mathbb{Z}) \backslash \mathbb{H} \xrightarrow{\sim} \mathbb{C}$ . Thus  $\mathcal{A}_1$  is the affine  $j$ -line, and  $\mathfrak{H}_g / \text{Sp}(2g, \mathbb{Z})$  is its higher-dimensional replacement, with the  $j$ -invariant giving way to Siegel modular forms. The two elliptic points  $\tau = i$  and  $\tau = e^{2\pi i/3}$ , where the curve has extra automorphisms, are the  $g = 1$  instances of the orbifold points that prevent  $\mathcal{A}_g$  from being a fine moduli space.

### Exercise 3.3.1

1. Verify directly from definition 3.3.1 that the matrix

$$\tau = \begin{pmatrix} i & 1 \\ 1 & 2i \end{pmatrix}$$

lies in  $\mathfrak{H}_2$ , and that

$$\tau' = \begin{pmatrix} i & 2i \\ 2i & i \end{pmatrix}$$

does not. In each case identify precisely which of the two conditions ( $\tau = \tau^t$ ,  $\text{Im}\tau > 0$ ) hold and which fail.

2. Let  $\tau_1, \tau_2 \in \mathbb{H}$  and let  $\tau = \text{diag}(\tau_1, \tau_2) \in \mathfrak{H}_2$  be the Siegel point of the product  $E_1 \times E_2$  from example 3.3.4. Compute the lattice  $\Lambda = \tau\mathbb{Z}^2 + \mathbb{Z}^2$  and the Hermitian form  $H(v, w) = v^t(\text{Im}\tau)^{-1}\bar{w}$  explicitly, and check that  $H$  splits as an orthogonal direct sum of the two one-dimensional Hermitian forms attached to  $E_1$  and  $E_2$ .
3. Take  $g = 1$ , so  $\mathfrak{H}_1 = \mathbb{H}$  and  $\text{Sp}(2, \mathbb{Z}) = \text{SL}_2(\mathbb{Z})$  by example 3.3.8. Using the identity for  $\text{Im}W$  established in the proof of proposition 3.3.6, show that for  $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$  and  $\tau \in \mathbb{H}$  one has

$$\text{Im}(M \cdot \tau) = \frac{\text{Im}\tau}{|c\tau + d|^2}$$

and deduce that  $M \cdot \tau \in \mathbb{H}$ .

4. Fix  $B \in M_g(\mathbb{Z})$  with  $B = B^t$  and set  $M = \begin{pmatrix} I_g & B \\ 0 & I_g \end{pmatrix}$ . Show that  $M \in \text{Sp}(2g, \mathbb{Z})$  and that  $M \cdot \tau = \tau + B$  for every  $\tau \in \mathfrak{H}_g$ . Conclude that  $\tau$  and  $\tau + B$  present the same principally polarized abelian variety, and interpret this as the higher-dimensional analogue of the identification  $\tau \sim \tau + 1$  on  $\mathbb{H}$ .
5. Prove that the block condition  $M^t J M = J$  of definition 3.3.5, for  $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$  in  $g \times g$  blocks, is equivalent to the three relations

$$A^t C = C^t A \quad B^t D = D^t B \quad A^t D - C^t B = I_g$$

Deduce that  $M^t \in \text{Sp}(2g, \mathbb{Z})$  whenever  $M \in \text{Sp}(2g, \mathbb{Z})$ , and that for  $g = 1$  the three relations collapse to  $\det M = 1$ .

6. Let  $\tau \in \mathfrak{H}_g$  and let  $M = \begin{pmatrix} 0 & -I_g \\ I_g & 0 \end{pmatrix}$ , the higher-dimensional analogue of the modular inversion  $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ . Prove that  $M \in \text{Sp}(2g, \mathbb{Z})$  and that  $M \cdot \tau = -\tau^{-1}$ , and show directly (without invoking proposition 3.3.6) that  $-\tau^{-1} \in \mathfrak{H}_g$ .

## 3.4 Riemann-Roch on an Abelian Variety

A polarization is a positive line bundle in the sense of definition 3.1.1, and a positive line bundle is supposed to have sections. Chapter 2 already isolated the space that carries them: for  $L = L(H, \chi)$  the space  $H^0(X, L)$  consists of the holomorphic functions  $\theta: V \rightarrow \mathbb{C}$  transforming by the canonical factor of automorphy, the *theta functions* of definition 2.4.1, and proposition 2.4.3 announced that this space is finite-dimensional with dimension the Pfaffian  $\text{Pf}(\text{Im}H)$  when  $H > 0$ . This section proves that count. It is the analytic Riemann-Roch theorem for an abelian variety: a single integer, read off the type of the polarization, controls how many independent theta functions the bundle supports, and hence how large a projective space the variety can embed into once those functions are written down.

The number that emerges is  $d_1 d_2 \cdots d_g$ , the product of the elementary divisors of the polarization from definition 3.2.3. Two features make this the correct generalization of the elliptic-curve count  $\dim H^0(E, L) = \deg L$ . First, all higher cohomology vanishes, so the dimension equals the Euler characteristic and the answer is a topological invariant of the bundle. Second, in the principal case  $(1, \dots, 1)$  the count collapses to  $\dim H^0(X, L) = 1$ : a principal polarization has, up to translation,

exactly one effective divisor, the theta divisor that will organize the entire geometry of Jacobians in Chapter 12.

The strategy for the dimension count is a Fourier decomposition. A theta function is quasiperiodic for the full lattice  $\Lambda$ ; but  $\Lambda$  contains a sublattice on which the factor of automorphy is trivial up to an ordinary character, and along that sublattice a theta function is periodic. Expanding in the resulting Fourier series turns the transformation law under the remaining lattice directions into a finite system of linear recursions on the Fourier coefficients, and counting the free coefficients gives  $d_1 \cdots d_g$ . The positivity  $H > 0$  enters exactly where it must: it is what forces the series to converge, so that every formal solution of the recursion is an actual holomorphic function.

**3.4.1 Convention: reduction to a normal form** Throughout this section  $L = L(H, \chi)$  with  $H > 0$  and  $E = \text{Im}H$  of type  $(d_1, \dots, d_g)$ ,  $d_1 \mid \cdots \mid d_g$ , as in definition 3.2.3. The dimension  $\dim H^0(X, L)$  is unchanged by an isomorphism of line bundles, so the argument may be run in any  $\mathbb{Z}$ -basis of  $\Lambda$  adapted to  $E$ ; the symplectic basis of definition 3.2.3 is the convenient one. Moreover the count below is carried out for a fixed but arbitrary semicharacter  $\chi$  and returns  $d_1 \cdots d_g$  no matter which  $\chi$  is chosen, so the answer in fact depends only on  $H$ ; that the underlying bundles  $L(H, \chi) \otimes L_0$  with  $L_0 \in \text{Pic}^0(X)$  are related by translation, which explains this  $\chi$ -independence conceptually, is established in proposition 3.4.5.

The reduction recorded above is what lets the proof choose coordinates freely. The dimension is an invariant of  $H$  alone, so the semicharacter  $\chi$  may be normalized and the lattice put in symplectic form without loss; only the positive Hermitian form  $H$ , through its elementary divisors, survives into the final count.

**Theorem 3.4.2: Riemann-Roch on an Abelian Variety**

Let  $X = V/\Lambda$  be a complex torus and let  $L = L(H, \chi)$  be a line bundle with  $H > 0$ . Let  $E = \text{Im}H$ , of type  $(d_1, \dots, d_g)$ . Then

$$\dim_{\mathbb{C}} H^0(X, L) = \text{Pf}(E) = d_1 d_2 \cdots d_g = \sqrt{\det E}$$

In particular  $H^0(X, L) \neq 0$ , so a polarization always has a nonzero global section.

**Proof:**

By box 3.4.1 the dimension depends only on  $H$ , and by definition 2.4.1 an element of  $H^0(X, L)$  is a holomorphic function  $\theta: V \rightarrow \mathbb{C}$  satisfying the quasiperiodicity

$$\theta(v + \lambda) = a_{(H, \chi)}(\lambda, v) \theta(v) = \chi(\lambda) \exp\left(\pi H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda)\right) \theta(v) \quad (3.6)$$

for every  $\lambda \in \Lambda$ . The proof decomposes  $\Lambda$  into a sublattice on which  $\theta$  is periodic and a complementary one that acts by a shift, expands  $\theta$  in a Fourier series along the first, and counts the coefficients that the second permits.

*Step 1: A symplectic splitting of the lattice.* By definition 3.2.3 choose a  $\mathbb{Z}$ -basis  $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$  of  $\Lambda$  in which  $E$  has the matrix  $\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$  with  $D = \text{diag}(d_1, \dots, d_g)$ ; that is,  $E(\lambda_j, \mu_k) = d_j \delta_{jk}$  and  $E(\lambda_j, \lambda_k) = E(\mu_j, \mu_k) = 0$ . Set

$$\Lambda_1 = \sum_j \mathbb{Z} \lambda_j, \quad \Lambda_2 = \sum_j \mathbb{Z} \mu_j$$



so that  $\Lambda = \Lambda_1 \oplus \Lambda_2$  and  $E$  vanishes on each of  $\Lambda_1, \Lambda_2$  separately. Because  $\Lambda$  is a full lattice,  $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$  is an  $\mathbb{R}$ -basis of  $V$ ; in particular  $\lambda_1, \dots, \lambda_g$  are  $\mathbb{R}$ -linearly independent. They are moreover  $\mathbb{C}$ -linearly independent. If not, the real subspace  $V_1 = \sum_j \mathbb{R}\lambda_j$  would contain a complex line, so there would be a nonzero  $w \in V_1$  with  $iw \in V_1$ ; but  $E$  vanishes on  $\Lambda_1$ , hence on its real span  $V_1$ , and then  $H(w, w) = E(iw, w) = 0$  with  $w \neq 0$ , contradicting  $H > 0$ . Thus  $\lambda_1, \dots, \lambda_g$  are  $\mathbb{C}$ -linearly independent, and by the dimension count they form a  $\mathbb{C}$ -basis of  $V$ . Use it to identify  $V \cong \mathbb{C}^g$  and write  $v = \sum_j z_j \lambda_j$  with coordinates  $z = (z_1, \dots, z_g)$ .

*Step 2: Periodicity along  $\Lambda_1$  and the Fourier series.* The idea is to strip the transformation law (3.6) down to its core content by multiplying  $\theta$  by a fixed nowhere-zero *holomorphic* exponential, the standard passage to a *classical theta* normalization. The prefactor must stay holomorphic, so it cannot be the non-holomorphic  $e^{-\pi H(v,v)/2}$ ; instead it is built from the  $\mathbb{C}$ -bilinear form that agrees with  $H$  along  $\Lambda_1$ . Because  $E = \text{Im}H$  vanishes on  $\Lambda_1$  (Step 1), the restriction of  $H$  to the real span  $V_1 = \sum_j \mathbb{R}\lambda_j$  is real symmetric, so the numbers  $H(\lambda_j, \lambda_k)$  are real and symmetric in  $j, k$ . Let  $B$  be the symmetric  $\mathbb{C}$ -bilinear form on  $V \cong \mathbb{C}^g$  with  $B(\lambda_j, \lambda_k) = H(\lambda_j, \lambda_k)$  on the  $\mathbb{C}$ -basis of Step 1, extended  $\mathbb{C}$ -bilinearly; in the coordinates  $z$  it is the holomorphic quadratic  $B(v, v) = \sum_{j,k} H(\lambda_j, \lambda_k) z_j z_k$ . Set

$$\theta'(v) = \exp\left(-\frac{\pi}{2} B(v, v)\right) \theta(v)$$

a *holomorphic* function, since both factors are. The prefactor is nowhere zero, so  $\theta \mapsto \theta'$  is a bijection between holomorphic  $\theta$  obeying (3.6) and holomorphic  $\theta'$  satisfying the transformed law, and it preserves linear independence, so no dimension is lost by working with  $\theta'$ . Substituting  $\theta = e^{\pi B(v,v)/2} \theta'$  into (3.6) and using  $B(v + \lambda, v + \lambda) = B(v, v) + 2B(v, \lambda) + B(\lambda, \lambda)$  (valid because  $B$  is  $\mathbb{C}$ -bilinear and symmetric) gives, for every  $\lambda \in \Lambda$ ,

$$\theta'(v + \lambda) = \chi(\lambda) \exp\left(\pi(H(v, \lambda) - B(v, \lambda)) + \frac{\pi}{2}(H(\lambda, \lambda) - B(\lambda, \lambda))\right) \theta'(v) \quad (3.7)$$

Now take  $\lambda = \lambda_j \in \Lambda_1$ . Here  $B$  was defined to agree with  $H$  on the  $\mathbb{C}$ -basis, and both  $v \mapsto H(v, \lambda_j)$  and  $v \mapsto B(v, \lambda_j)$  are the  $\mathbb{C}$ -linear functional with the same values  $H(\lambda_k, \lambda_j)$  on  $\lambda_k$ , so  $H(v, \lambda_j) = B(v, \lambda_j)$  identically, and likewise  $H(\lambda_j, \lambda_j) = B(\lambda_j, \lambda_j)$ . Every exponent in (3.7) therefore cancels, leaving

$$\theta'(v + \lambda_j) = \chi(\lambda_j) \theta'(v)$$

genuine periodicity up to the fixed *unitary constant*  $\chi(\lambda_j) \in U(1)$ , with  $\theta'$  still holomorphic. Writing  $\chi(\lambda_j) = e^{2\pi i \varphi_j}$  and setting  $\psi(v) = \exp(-2\pi i \sum_j \varphi_j z_j) \theta'(v)$ , still holomorphic since the new exponent is  $\mathbb{C}$ -linear in  $z$ , absorbs the constant: under  $z_j \mapsto z_j + 1$  the linear exponential contributes  $e^{-2\pi i \varphi_j}$ , which cancels  $\chi(\lambda_j)$ , so  $\psi(v + \lambda_j) = \psi(v)$  for every  $j$ . Thus  $\psi$  is holomorphic and  $\mathbb{Z}^g$ -periodic in  $z$ , and it admits a Fourier expansion

$$\psi(v) = \sum_{m \in \mathbb{Z}^g} c_m e^{2\pi i \langle m, z \rangle}, \quad \langle m, z \rangle = \sum_j m_j z_j \quad (3.8)$$

with coefficients  $c_m \in \mathbb{C}$ , absolutely and uniformly convergent on compact sets because  $\psi$  is holomorphic and 1-periodic in each coordinate  $z_j$ . The original theta is recovered as  $\theta(v) = \exp(\frac{\pi}{2} B(v, v) + 2\pi i \sum_j \varphi_j z_j) \psi(v)$  through this fixed holomorphic prefactor.

*Step 3: The recursion imposed by  $\Lambda_2$ .* It remains to impose (3.6) for  $\lambda = \mu_k \in \Lambda_2$ . Through the fixed holomorphic prefactor relating  $\theta$  and  $\psi$  recorded at the end of Step 2, that quasiperiodicity of  $\theta$  under  $\mu_k$  is equivalent to a transformation law for  $\psi$  of the same shape:  $\psi(v + \mu_k)$  equals



$\psi(v)$  times a nowhere-zero holomorphic exponential in  $z$ . In the  $\mathbb{C}$ -basis  $\lambda_1, \dots, \lambda_g$  of Step 1 write  $\mu_k = \sum_j \tau_{jk} \lambda_j$ , so that  $\tau = (\tau_{jk}) \in M_g(\mathbb{C})$  has  $\mu_k$  for its  $k$ -th column and the period matrix in the ordered basis  $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$  is  $\Pi = (I_g \ \tau)$ .

Two properties of  $\tau$  are needed, and both come directly from the Riemann relations of proposition 3.1.5 for this  $\Pi$  and this  $E$ , whose matrix in the present basis is  $\mathcal{E} = \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$  with inverse  $\mathcal{E}^{-1} = \begin{pmatrix} 0 & -D^{-1} \\ D^{-1} & 0 \end{pmatrix}$ . This is *not* the principal normalization of proposition 3.3.3: there the  $\mathbb{C}$ -basis is the second lattice half and  $D = I_g$ , whereas here the  $\mathbb{C}$ -basis is the first half  $\lambda_1, \dots, \lambda_g$  and  $D = \text{diag}(d_1, \dots, d_g)$ , so the symmetry and positivity of that proposition do not transfer and must be recomputed. Blockwise  $\Pi \mathcal{E}^{-1} = (\tau D^{-1} \ -D^{-1})$ , so the first Riemann relation reads

$$\Pi \mathcal{E}^{-1} \Pi^t = \tau D^{-1} - D^{-1} \tau^t = 0,$$

that is  $D\tau = (D\tau)^t$ : the matrix  $D\tau$  is symmetric, though  $\tau$  itself is symmetric only in the principal case  $d_1 = \dots = d_g$ . Using  $\tau D^{-1} = D^{-1} \tau^t$ , the second relation is

$$i \Pi \mathcal{E}^{-1} \bar{\Pi}^t = i(\tau D^{-1} - D^{-1} \bar{\tau}^t) = i D^{-1}(\tau - \bar{\tau})^t = -2 D^{-1}(\text{Im} \tau)^t = -2 \text{Im} \tau D^{-1},$$

the last equality because  $D \text{Im} \tau = \text{Im}(D\tau)$  is symmetric. Since  $H > 0$  this Hermitian matrix is positive definite, so the real symmetric matrix

$$Y := -D \text{Im} \tau = -\text{Im}(D\tau)$$

is positive definite; equivalently  $\text{Im}(D\tau) < 0$ . This is the correct form of the positivity in the present swapped, type- $(d_1, \dots, d_g)$  configuration, and  $Y > 0$  is the sole consequence of  $H > 0$  that the convergence argument will use.

Translation by  $\mu_k$  is the shift  $z_j \mapsto z_j + \tau_{jk}$ . Substituting (3.8) into the transformation law for  $\psi$  and matching the coefficient of  $e^{2\pi i \langle m, z \rangle}$  on both sides yields, for each  $k$ , a relation of the form

$$c_{m+d_k e_k} = w_k(m) c_m, \quad w_k(m) = C_k \exp\left(-2\pi i \sum_j (m_j + d_k \delta_{jk}) \tau_{jk}\right), \quad (3.9)$$

where  $e_k$  is the  $k$ -th standard basis vector,  $C_k \in \mathbb{C}^*$  is a constant of modulus  $|C_k| = \exp(2\pi \sum_j \varphi_j (\text{Im} \tau)_{jk} - \pi d_k (\text{Im} \tau)_{kk})$  built from  $\chi(\mu_k) = e^{2\pi i \varphi_k}$  and the diagonal shift, and the index moves by exactly  $d_k$  in the  $k$ -th slot because  $\mathcal{H} \text{Im} \tau = -D$  (the matrix form of the second relation, with  $\mathcal{H}_{pq} = H(\lambda_p, \lambda_q)$  the real symmetric Gram matrix of Step 2) forces the linear part of the multiplier to be  $e^{2\pi i d_k z_k}$ , while  $E(\lambda_k, \mu_k) = d_k$  governs the quasiperiod in that direction. Thus each coefficient  $c_{m+d_k e_k}$  is determined, through the nonzero multiplier  $w_k(m)$ , by  $c_m$ . Consistency of the  $g$  recursions among themselves is automatic: the  $\mu_k$  commute and  $E(\mu_j, \mu_k) = 0$ , so the shifts in different coordinates compose without contradiction.

*Step 4: Counting free coefficients.* The recursion (3.9) partitions the index set  $\mathbb{Z}^g$  into orbits under the translation lattice  $D\mathbb{Z}^g = \bigoplus_j d_j \mathbb{Z}$ . On each orbit every coefficient is determined by any one of them via the nonzero multipliers, so an orbit contributes at most one free parameter: the value of  $c_m$  at a single chosen representative. The number of orbits is the index

$$[\mathbb{Z}^g : D\mathbb{Z}^g] = d_1 d_2 \cdots d_g$$

Hence the space of *formal* solutions of the system (3.9) has dimension at most  $d_1 \cdots d_g$ : a solution is a choice of  $c_m$  on one representative per orbit, freely, with all other coefficients then forced.

*Step 5: Convergence, from positivity.* A formal solution defines an actual holomorphic  $\psi$ , and hence through the fixed holomorphic prefactor an actual theta function  $\theta$ , precisely when the series (3.8) converges. From the explicit multiplier (3.9) its modulus is

$$\log |w_k(m)| = 2\pi \sum_j (m_j + \varphi_j)(\operatorname{Im}\tau)_{jk} + \pi d_k(\operatorname{Im}\tau)_{kk},$$

linear in  $m$ ; telescoping the  $N$  steps that move  $m$  to  $m + Nd_k e_k$  sums  $\log |w_k(m + \ell d_k e_k)|$  over  $\ell = 0, \dots, N-1$ , and the  $\ell$ -dependence through  $(\operatorname{Im}\tau)_{kk}$  contributes the quadratic term  $2\pi(\operatorname{Im}\tau)_{kk}d_k \binom{N}{2} = \pi d_k(\operatorname{Im}\tau)_{kk}(N^2 - N)$ , giving

$$|c_{m+Nd_k e_k}| = |c_m| \exp\left(\pi d_k(\operatorname{Im}\tau)_{kk} N^2 + O(N)\right) = |c_m| \exp\left(-\pi Y_{kk} N^2 + O(N)\right),$$

where  $Y = -D \operatorname{Im}\tau$  is the positive-definite matrix of Step 3 and the coefficient of  $-N^2$  is  $\pi Y_{kk} = -\pi d_k(\operatorname{Im}\tau)_{kk} > 0$  exactly because  $Y > 0$  has positive diagonal, itself the consequence of  $H > 0$ . The same estimate holds in every coordinate direction; for a general shift  $m \mapsto m + Dn$ ,  $n \in \mathbb{Z}^g$ , the same telescoping produces  $\exp(-\pi n^t Y n + O(|n|))$ , a genuine Gaussian because  $Y > 0$ , with the off-diagonal entries of  $Y$  contributing the cross terms. Thus each coefficient decays faster than any geometric rate as its index leaves a bounded region, and the series converges absolutely and locally uniformly to a holomorphic function on all of  $V$ . This is where the hypothesis  $H > 0$  is used, and used essentially: without it  $Y$  would fail to be positive definite, the quadratic form in the exponent would change sign, the series would diverge, and the formal solutions would not be functions. Every one of the  $d_1 \cdots d_g$  free choices therefore produces a genuine element of  $H^0(X, L)$ , and distinct choices give linearly independent functions since they differ already in their Fourier coefficients.

Combining Steps 4 and 5, the space  $H^0(X, L)$  has dimension exactly  $d_1 d_2 \cdots d_g$ . Finally, for a nondegenerate alternating integral matrix in the symplectic normal form  $\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$  the Pfaffian is  $\operatorname{Pf}(E) = d_1 \cdots d_g$  and  $\det E = (d_1 \cdots d_g)^2$ , so  $\operatorname{Pf}(E) = \sqrt{\det E}$ , giving

$$\dim_{\mathbb{C}} H^0(X, L) = \operatorname{Pf}(E) = d_1 d_2 \cdots d_g = \sqrt{\det E}$$

as we sought to show.

The theorem is the analytic heart of the projective geometry of abelian varieties. It says that positivity is not a qualitative condition for algebraicity alone but a quantitative one: the polarization prescribes the exact dimension of the linear system of theta functions, and that dimension is a product of the elementary divisors, computable from the period matrix by a determinant. The proof also explains the mechanism. The lattice splits into a periodic half and a translating half; Fourier analysis handles the periodic half and turns the translating half into a finite orbit count, and positivity is precisely the input that promotes formal solutions into convergent, holomorphic ones. Nothing here yet writes a single theta function down. The count is an existence-and-dimension statement; chapter 4 supplies the explicit basis, the Riemann theta series indexed by the  $d_1 \cdots d_g$  orbit representatives, and uses it to construct the projective embedding whose target dimension this count determines.

One loose end remains before the count deserves the name Riemann-Roch. The classical theorem computes an Euler characteristic, an alternating sum of cohomology dimensions, and it is only because the higher terms vanish that the alternating sum reduces to  $\dim H^0$ . The next result records that vanishing.

**Proposition 3.4.3: Vanishing of Higher Cohomology**

Let  $X = V/\Lambda$  be a complex torus and  $L = L(H, \chi)$  with  $H > 0$ . Then

$$H^i(X, L) = 0 \quad \text{for all } i > 0$$

**Proof Key ideas:**

The vanishing is standard, and the general- $i$  statement rests on the theory of the cohomology of line bundles on a complex torus computed through the Dolbeault complex or through the theta-group action; the full computation of all  $H^i(X, L)$  for  $L(H, \chi)$  is carried out in Mumford, *Abelian Varieties*, and is cited here rather than reproduced because it repeats the harmonic-oscillator style Fourier analysis of theorem 3.4.2 once for each cohomological degree without adding a new idea. The mechanism is this. On a complex torus  $H^i(X, L) = H_{\bar{\partial}}^{0,i}(X, L)$  is computed by the  $\bar{\partial}$ -complex with coefficients in  $L$ . Diagonalizing the Hermitian form  $H$  and running the Fourier expansion of theorem 3.4.2 in each of the  $g$  complex directions separately, the  $\bar{\partial}$ -Laplacian acting on  $L$ -valued  $(0, i)$ -forms decouples into a sum of  $g$  one-dimensional operators. Because  $H > 0$ , each of these one-dimensional operators is strictly positive on the antiholomorphic directions, so a harmonic  $L$ -valued  $(0, i)$ -form with  $i > 0$  must vanish; only the  $(0, 0)$ -part supports harmonic representatives, and those are exactly the holomorphic sections counted in theorem 3.4.2. Positivity is again the load-bearing hypothesis: it is what makes every higher  $\bar{\partial}$ -cohomology group vanish, completing the argument.

The vanishing is the second half of what makes the dimension count a true Riemann-Roch theorem. With no higher cohomology to correct for, the Euler characteristic of the bundle reduces to the single number computed above, and that number is therefore a topological invariant, insensitive to the analytic choices in the construction of  $L$ .

**Corollary 3.4.4: Euler Characteristic of a Positive Bundle**

Let  $L$  be a line bundle with first Chern class  $H > 0$  of type  $(d_1, \dots, d_g)$ , and set  $E = \text{Im}H$ . Then the Euler characteristic of  $L$ , written with the standard symbol  $\chi(-)$  for an alternating sum of cohomology dimensions, is

$$\chi(L) = \sum_{i \geq 0} (-1)^i \dim H^i(X, L) = \text{Pf}(E) = d_1 d_2 \cdots d_g$$

**Proof:**

By proposition 3.4.3 every term with  $i > 0$  vanishes, so  $\chi(L) = \dim H^0(X, L)$ , which equals  $\text{Pf}(E) = d_1 \cdots d_g$  by theorem 3.4.2, completing the proof.

That the Euler characteristic equals  $\text{Pf}(E) = \frac{1}{g!} \int_X c_1(L)^g$  is the abelian-variety incarnation of Hirzebruch-Riemann-Roch: the topological expression  $\int_X e^{c_1(L)} \text{td}(X)$  collapses, because the tangent bundle of a torus is trivial and so its Todd class is 1, to the top self-intersection  $\frac{1}{g!} \int_X c_1(L)^g$ , and that integral is exactly the Pfaffian of  $E = c_1(L)$ . The analytic count of theorem 3.4.2 and the topological count of Hirzebruch thus agree, each confirming the other.

The most consequential special case is the smallest one. When the type is trivial the product of elementary divisors is 1, so the polarization supports a one-dimensional space of theta functions, a single section up to scale, and hence a single effective divisor. This is the defining feature of a principal polarization and the reason such polarizations dominate the theory.

**Proposition 3.4.5: Principal Polarizations Have a Unique Theta Divisor**

Let  $L = L(H, \chi)$  define a principal polarization on  $X$ , so that  $H > 0$  has type  $(1, \dots, 1)$  in the sense of definition 3.2.4. Then

$$\dim_{\mathbb{C}} H^0(X, L) = 1$$

Consequently  $L$  has, up to scaling, a unique nonzero global section  $\theta$ , and the associated effective divisor  $\Theta = \{\theta = 0\}$  is the unique effective divisor in the linear equivalence class of  $L$ . A different choice of the semicharacter  $\chi$ , or of a bundle in  $\text{Pic}^0(X)$  to twist by, replaces  $\Theta$  by a translate; so a principal polarization determines a divisor  $\Theta$  uniquely up to translation.

**Proof:**

By definition 3.2.4 a principal polarization has type  $(1, \dots, 1)$ , so  $d_1 = \dots = d_g = 1$  and theorem 3.4.2 gives  $\dim_{\mathbb{C}} H^0(X, L) = 1 \cdot 1 \cdots 1 = 1$ . A one-dimensional space of sections has a nonzero element  $\theta$  unique up to a scalar; its divisor  $\Theta = \{\theta = 0\}$  is therefore well defined independently of the scalar. Any effective divisor  $D$  linearly equivalent to  $L$  is the zero divisor of some global section of  $L$ , hence of a scalar multiple of  $\theta$ , so  $D = \Theta$ ; thus  $\Theta$  is the only effective divisor in the class.

For the last statement, the polarization is the class  $H \in \text{NS}(X)$ , and by the Appell-Humbert classification the line bundles with first Chern class  $H$  are the twists  $L(H, \chi) \otimes L_0$  with  $L_0 \in \text{Pic}^0(X)$ ; each is again positive of the same type, so each has a one-dimensional space of sections and a single theta divisor. Two such bundles differ by an element of  $\text{Pic}^0(X) = \{L(0, \chi') : \chi' \in \text{Hom}(\Lambda, U(1))\}$ , and it suffices to realize the passage from  $L$  to  $L \otimes L(0, \chi')$  as a translation. This is a direct computation with the canonical factor. For  $a \in V$  write  $t_a : X \rightarrow X$  for translation by the image of  $a$ , lifted to  $v \mapsto v + a$  on  $V$ ; then  $t_a^* L(H, \chi)$  has factor of automorphy

$$a_{(H, \chi)}(\lambda, v + a) = a_{(H, \chi)}(\lambda, v) e^{\pi H(a, \lambda)}$$

since  $H$  is linear in its first slot. The extra factor  $e^{\pi H(a, \lambda)}$  differs by a coboundary from the constant  $U(1)$ -valued character  $\lambda \mapsto e^{2\pi i E(a, \lambda)}$ . Indeed the  $\mathbb{R}$ -linear functional  $\ell(v) = \pi E(ia, v) - \pi i E(a, v)$  is in fact  $\mathbb{C}$ -linear, since  $E(iu, iw) = E(u, w)$  gives  $\ell(iv) = i \ell(v)$ ; so  $h(v) = e^{\ell(v)}$  is a nowhere-zero holomorphic function on  $V$ , and its coboundary  $h(v + \lambda)/h(v) = e^{\ell(\lambda)}$  satisfies

$$e^{\pi H(a, \lambda)} = e^{2\pi i E(a, \lambda)} e^{\ell(\lambda)}$$

using  $H(a, \lambda) = E(ia, \lambda) + iE(a, \lambda)$ . Multiplying the factor of  $t_a^* L(H, \chi)$  by this coboundary replaces  $e^{\pi H(a, \lambda)}$  by the constant character  $e^{2\pi i E(a, \lambda)}$  without changing the isomorphism class, so, in  $\text{Pic}(X)$ ,

$$t_a^* L(H, \chi) \cong L(H, \chi) \otimes L(0, \chi_a), \quad \chi_a(\lambda) = e^{2\pi i E(a, \lambda)}$$

Now  $H > 0$  makes  $E = \text{Im} H$  nondegenerate on  $V$ , so the  $\mathbb{R}$ -linear map  $V \rightarrow \text{Hom}(\Lambda, \mathbb{R})$ ,  $a \mapsto E(a, -)$ , is an isomorphism; composing with the projection  $\text{Hom}(\Lambda, \mathbb{R}) \twoheadrightarrow \text{Hom}(\Lambda, \mathbb{R})/\text{Hom}(\Lambda, \mathbb{Z}) = \text{Hom}(\Lambda, U(1))$  shows that  $a \mapsto \chi_a$  maps  $V$  onto  $\text{Hom}(\Lambda, U(1))$ . Thus every  $\chi' \in \text{Hom}(\Lambda, U(1))$

equals  $\chi_a$  for some  $a$ , and  $L \otimes L(0, \chi') \cong t_a^* L$ : tensoring by the degree-zero bundle really is pulling back along a translation, and it therefore carries the theta divisor  $\Theta$  of  $L$  to its translate  $t_a^{-1}(\Theta)$ . Hence the theta divisors of the various bundles carrying the principal polarization are translates of one another, and  $\Theta$  is determined by  $H$  up to translation, completing the proof.

The uniqueness is what gives a principally polarized abelian variety its distinguished geometry. An ample class in general has a positive-dimensional linear system, a whole family of divisors moving inside it; the principal case has a single divisor, rigid up to translation, and that divisor carries the polarization as its own class. The theta divisor  $\Theta$  is thus an intrinsic feature of the polarized variety rather than an auxiliary choice, and it is the object through which Jacobians of curves acquire their canonical structure. A curve of genus  $g$  produces a principally polarized abelian variety of dimension  $g$ , its Jacobian, and the theta divisor is the image of the symmetric product of the curve; the Torelli theorem of chapter 12 recovers the curve from the pair  $(\text{Jac } C, \Theta)$ , which is possible only because  $\Theta$  is canonical, which is possible only because  $\dim H^0 = 1$ .

Before turning to consequences, it is worth grounding the whole apparatus in the one dimension where every quantity is already familiar. In genus one the type is a single integer, the Pfaffian is that integer, and the count reproduces the Riemann-Roch dimension that elliptic-curve theory supplies by divisors.

#### Example 3.4.6: Riemann-Roch for Elliptic Curves

Let  $g = 1$ , so  $E = \mathbb{C}/\Lambda$  is an elliptic curve and  $V = \mathbb{C}$ . A Hermitian form on  $\mathbb{C}$  is  $H(v, w) = h v \bar{w}$  for a real scalar  $h$ , and  $H > 0$  means  $h > 0$ . Its imaginary part  $E = \text{Im} H$  is a single alternating integral form on the rank-two lattice  $\Lambda$ , and its type is the single elementary divisor  $d_1 = E(\lambda_1, \lambda_2)$  for an oriented  $\mathbb{Z}$ -basis  $\lambda_1, \lambda_2$  of  $\Lambda$ ; this integer is positive because  $H > 0$ . The Pfaffian of a  $2 \times 2$  alternating matrix  $\begin{pmatrix} 0 & d_1 \\ -d_1 & 0 \end{pmatrix}$  is  $d_1$ , so theorem 3.4.2 reads

$$\dim_{\mathbb{C}} H^0(E, L) = \text{Pf}(E) = d_1$$

This matches the classical elliptic-curve Riemann-Roch. A line bundle  $L$  on  $E$  of degree  $\deg L = n \geq 1$  has  $\dim H^0(E, L) = n$  by the Riemann-Roch theorem for curves of genus one ([4], where the divisor form  $\ell(D) = \deg D$  for  $\deg D \geq 1$  is proved). The bridge between the two counts is that for a line bundle on an elliptic curve the degree equals the single elementary divisor of its first Chern class,

$$\deg L = \int_E c_1(L) = E(\lambda_1, \lambda_2) = d_1$$

the middle equality being the evaluation of the fundamental class against  $c_1(L) = \text{Im} H$  under  $H^2(E, \mathbb{Z}) \cong \mathbb{Z}$ . So the analytic formula  $\dim H^0 = d_1$  and the classical formula  $\dim H^0 = \deg L$  are the same statement, and the higher-dimensional Pfaffian  $d_1 \cdots d_g$  is the natural replacement for  $\deg L$  once  $g > 1$ , where a line bundle no longer has a single degree but a whole tuple of elementary divisors.

The elliptic-curve case fixes the reading of the general theorem. For  $g = 1$  there is one elementary divisor, it is the degree, and  $\dim H^0$  is that degree; a principal polarization is a degree-one bundle, with a one-dimensional space of sections and the single point of its divisor serving as the origin. For  $g > 1$  the polarization no longer reduces to a number but to the tuple  $(d_1, \dots, d_g)$ , and the theorem asserts that their product, not their sum or their maximum, counts the theta functions. That product is exactly the input chapter 4 requires: it is the number of coordinates in the projective embedding a polarization furnishes, the  $d_1 \cdots d_g$  Riemann theta functions written out explicitly and used to map  $X$  into  $\mathbb{P}^{d_1 \cdots d_g - 1}$  when the polarization is base-point free. Among all polarizations the principal ones

sit at the base of this count, with  $\dim H^0 = 1$  and a single canonical theta divisor, and they are the case the theory returns to again and again because Jacobians of curves realize them: every curve manufactures a principally polarized abelian variety, and the rigidity of its lone theta divisor is what lets the geometry of the curve be read off from it.

### Exercise 3.4.1

1. Let  $X = V/\Lambda$  be an abelian variety of dimension  $g = 3$  and let  $L = L(H, \chi)$  with  $H > 0$  of type  $(2, 2, 6)$ . Compute  $\dim_{\mathbb{C}} H^0(X, L)$ ,  $\text{Pf}(E)$ , and  $\det E$  for  $E = \text{Im}H$ . What is the smallest projective space  $\mathbb{P}^N$  that the linear system  $|L|$  could conceivably map  $X$  into, and why does this number depend only on  $H$  and not on  $\chi$ ?
2. Let  $L = L(H, \chi)$  with  $H > 0$  of type  $(d_1, \dots, d_g)$  and let  $n \geq 1$  be an integer. Recall that  $L^{\otimes n} = L(nH, \chi^n)$ . Show that  $\dim_{\mathbb{C}} H^0(X, L^{\otimes n}) = n^g d_1 \cdots d_g$ , and deduce that the Euler characteristic  $\chi(L^{\otimes n})$  is a polynomial in  $n$  of degree  $g$  with leading coefficient  $d_1 \cdots d_g$ .
3. In Step 2 of the proof of theorem 3.4.2 the passage from  $\theta$  to  $\theta' = \exp(-\frac{\pi}{2}B(v, v))\theta$  removes the Hermitian part of the factor of automorphy. Explain precisely why the correcting exponent uses the  $\mathbb{C}$ -bilinear form  $B$  rather than the Hermitian form  $H$  itself, that is, why  $\exp(-\frac{\pi}{2}H(v, v))\theta$  would not serve. Then verify by direct substitution that with  $B$  as defined there, the reduced law (3.7) becomes genuine periodicity  $\theta'(v + \lambda_j) = \chi(\lambda_j)\theta'(v)$  for  $\lambda_j \in \Lambda_1$ .
4. Consider a complex torus  $X = V/\Lambda$  of dimension  $g$  carrying a line bundle  $L = L(H, \chi)$  whose Hermitian form  $H$  is nondegenerate but has signature  $(p, q)$  with  $q > 0$ , so  $H$  is not positive definite. Locate the exact point in the proof of theorem 3.4.2 where positivity is used essentially, and explain what fails: does the count of formal solutions in Step 4 change, and does the space  $H^0(X, L)$  still have dimension  $d_1 \cdots d_g$ ? Reconcile your answer with the elliptic-curve picture of example 3.4.6, where  $H < 0$  corresponds to a line bundle of negative degree.
5. Let  $(X, L)$  be a principally polarized abelian variety, so  $L = L(H, \chi)$  with  $H > 0$  of type  $(1, \dots, 1)$ , and let  $\Theta$  be the theta divisor of proposition 3.4.5. Prove that the polarized variety  $(X, \Theta)$  has no nontrivial automorphism-free translation ambiguity in the following precise sense: for  $a \in V$  the pullback  $t_a^*L$  is isomorphic to  $L$  as a line bundle if and only if the image of  $a$  in  $X$  lies in the finite subgroup  $K(L) = \{x \in X \mid t_x^*L \cong L\}$ , and for a principal polarization this subgroup is trivial. Use the computation of  $t_a^*L(H, \chi)$  recorded in the proof of proposition 3.4.5 and identify  $K(L)$  with a quotient of lattices.
6. Combine the Riemann-Roch count with the higher-cohomology vanishing to prove the following divisor-theoretic statement. Let  $L = L(H, \chi)$  with  $H > 0$  of type  $(d_1, \dots, d_g)$ . Show that  $L$  admits a nonzero global section, hence an effective divisor  $D$  with  $\mathcal{O}_X(D) \cong L$ , and that the number of linearly independent such divisors is exactly  $d_1 \cdots d_g$ . Then show that for a principal polarization the effective divisor is unique, and explain how this recovers the last assertion of proposition 3.4.5 that  $\Theta$  is determined by  $H$  up to translation.

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## *Theta Functions*

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Chapter 3 counted the sections of a positive line bundle without writing a single one down: the Riemann-Roch theorem gave  $\dim H^0(X, L) = d_1 \cdots d_g$  by a Fourier argument that produced the dimension and nothing more. This chapter supplies the functions themselves. They are the theta functions, holomorphic functions on  $V$  with the prescribed quasi-periodicity of the Appell-Humbert factor, and once they are in hand the projective geometry of an abelian variety becomes explicit: a basis of theta functions is exactly a system of homogeneous coordinates, and the map it defines is the embedding whose existence Chapter 3 could only assert.

Three threads run through the chapter. The first is the theta group, the finite Heisenberg group  $\mathcal{G}(L)$  of symmetries of  $L$ , whose unique irreducible representation is the section space  $H^0(X, L)$ ; this recovers the dimension count of chapter 3 as a fact about the Heisenberg group and fixes the natural basis. The second is the classical Riemann theta function and its translates with characteristics, the concrete series that realize that basis and converge precisely because the period matrix lies in the Siegel upper half-space. The third is the payoff: the Lefschetz theorem, that the third power of a polarization is very ample, so every polarizable torus embeds in projective space and the provisional definition of an abelian variety from chapter 1 is at last discharged. The chapter closes with the theta divisor, the zero locus of the single section of a principal polarization, the geometric object through which Jacobians will later carry their structure.

### 4.1 Canonical Factors and the Theta Group

The dimension count of theorem 3.4.2 answered the question “how many sections does a positive line bundle have?” with the number  $d_1 \cdots d_g$ , the product of the elementary divisors of the polarization. That count was extracted by a direct convergence estimate on quasi-periodic functions, and it left the space  $H^0(X, L)$  looking like an accident of Fourier analysis: a vector space of the right dimension, with no structure singling out one basis over another. This section supplies the missing



structure. Attached to every line bundle  $L$  on  $X = V/\Lambda$  is a finite abelian group  $K(L)$ , and above it a nonabelian group  $\mathcal{G}(L)$  of symmetries of  $L$ , the *theta group*. The group  $\mathcal{G}(L)$  acts on  $H^0(X, L)$ , and when  $L$  is a polarization this action is irreducible; the dimension of the unique such irreducible representation recovers  $d_1 \cdots d_g$  without a single convergence estimate. The theta group is the intrinsic symmetry of a polarized abelian variety, and its commutator is the germ of the Weil pairing that organizes chapter 5.

Everything rests on a lattice larger than  $\Lambda$ , on which the alternating form  $E = \text{Im}H$  is still integral. Recall from definition 3.1.3 that a line bundle  $L = L(H, \chi)$  carries the Hermitian form  $H$  and its imaginary part  $E$ , an alternating  $\mathbb{R}$ -form that is integral on  $\Lambda$ :  $E(\Lambda, \Lambda) \subseteq \mathbb{Z}$ . The next definition asks for the largest sublattice of  $V$  on which  $E$  remains integral against all of  $\Lambda$ .

**Definition 4.1.1: The Lattice  $\Lambda(L)$**

Let  $X = V/\Lambda$  and let  $L = L(H, \chi)$  be a line bundle with  $E = \text{Im}H$  integral on  $\Lambda$ . Define

$$\Lambda(L) := \{ v \in V : E(v, \lambda) \in \mathbb{Z} \text{ for all } \lambda \in \Lambda \}$$

the set of vectors on which  $E$  pairs integrally with the whole lattice.

Because  $E(\Lambda, \Lambda) \subseteq \mathbb{Z}$ , every lattice vector lies in  $\Lambda(L)$ , so  $\Lambda \subseteq \Lambda(L)$ . The set  $\Lambda(L)$  is a subgroup of  $V$ : it is the preimage of the integral lattice  $\Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$  under the  $\mathbb{R}$ -linear map  $v \mapsto E(v, -)|_\Lambda$  from  $V$  to  $\text{Hom}_\mathbb{R}(\Lambda, \mathbb{R}) = \text{Hom}_\mathbb{R}(V, \mathbb{R})$ . When  $E$  is nondegenerate this map is an isomorphism of  $\mathbb{R}$ -vector spaces, so  $\Lambda(L)$  is the preimage of a lattice under a linear isomorphism, hence itself a full lattice in  $V$  containing  $\Lambda$  with finite index. The quotient measures how far  $E$  is from being unimodular on  $\Lambda$ .

**Definition 4.1.2: The Group  $K(L)$**

With notation as in definition 4.1.1, define the finite abelian group

$$K(L) := \Lambda(L)/\Lambda$$

the cokernel of the inclusion  $\Lambda \hookrightarrow \Lambda(L)$ .

The group  $K(L)$  is the analytic incarnation of the kernel of the polarization isogeny  $\varphi_L$  that chapter 5 will construct: a point  $\bar{v} \in K(L)$  is exactly a translation of  $X$  under which  $L$  is preserved up to isomorphism. For now it is a finite abelian group whose order is dictated entirely by the type of the polarization, as the following computation shows. The proof is a direct application of the symplectic normal form for  $E$  recorded in definition 3.2.3.

The order of  $K(L)$  is the payload of the next result, and it will be reproved representation-theoretically at the end of the section. The motivating point is that  $|K(L)|$  is a perfect square,  $(d_1 \cdots d_g)^2$ , whose square root is the dimension of  $H^0(X, L)$ ; the theta group will explain why.



**Proposition 4.1.3: The Order of  $K(L)$** 

Let  $L = L(H, \chi)$  with  $H > 0$  a polarization of type  $(d_1, \dots, d_g)$ , so  $E = \text{Im} H$  is nondegenerate integral alternating on  $\Lambda$ . Then

$$K(L) \cong \left( \bigoplus_{i=1}^g \mathbb{Z}/d_i \mathbb{Z} \right)^2$$

and in particular

$$|K(L)| = (d_1 d_2 \cdots d_g)^2 = \det E$$

**Proof:**

By definition 3.2.3 there is a symplectic basis  $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$  of  $\Lambda$  in which  $E$  has matrix

$$\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}, \quad D = \text{diag}(d_1, \dots, d_g), \quad d_1 | d_2 | \cdots | d_g$$

that is,  $E(\lambda_i, \mu_j) = d_i \delta_{ij}$  and  $E(\lambda_i, \lambda_j) = E(\mu_i, \mu_j) = 0$ . Since  $\Lambda$  has real rank  $2g$  and spans  $V$  over  $\mathbb{R}$ , the vectors  $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$  form an  $\mathbb{R}$ -basis of  $V$ , so every  $v \in V$  has unique real coordinates against them.

*Step 1: Coordinatize  $\Lambda(L)$ .* Write a general  $v \in V$  in the symplectic basis as  $v = \sum_i x_i \lambda_i + \sum_i y_i \mu_i$  with  $x_i, y_i \in \mathbb{R}$ . From the matrix above,  $E(\lambda_i, \lambda_j) = 0$  and  $E(\mu_i, \lambda_j) = -E(\lambda_j, \mu_i) = -d_i \delta_{ij}$ , while  $E(\lambda_i, \mu_j) = d_i \delta_{ij}$  and  $E(\mu_i, \mu_j) = 0$ . Pairing  $v$  against the basis vectors of  $\Lambda$ ,

$$E(v, \lambda_j) = \sum_i y_i E(\mu_i, \lambda_j) = -d_j y_j, \quad E(v, \mu_j) = \sum_i x_i E(\lambda_i, \mu_j) = d_j x_j$$

Thus  $v \in \Lambda(L)$  if and only if  $d_j x_j \in \mathbb{Z}$  and  $d_j y_j \in \mathbb{Z}$  for every  $j$ , that is, if and only if  $x_j \in \frac{1}{d_j} \mathbb{Z}$  and  $y_j \in \frac{1}{d_j} \mathbb{Z}$  for all  $j$ . Hence

$$\Lambda(L) = \sum_{i=1}^g \frac{1}{d_i} \mathbb{Z} \lambda_i + \sum_{i=1}^g \frac{1}{d_i} \mathbb{Z} \mu_i$$

*Step 2: Take the quotient.* The lattice  $\Lambda = \sum_i \mathbb{Z} \lambda_i + \sum_i \mathbb{Z} \mu_i$  sits inside  $\Lambda(L)$  coordinatewise, and the quotient splits as a direct sum over the  $2g$  coordinate directions:

$$K(L) = \Lambda(L)/\Lambda \cong \bigoplus_{i=1}^g \left( \frac{1}{d_i} \mathbb{Z}/\mathbb{Z} \right) \oplus \bigoplus_{i=1}^g \left( \frac{1}{d_i} \mathbb{Z}/\mathbb{Z} \right)$$

Each factor  $\frac{1}{d_i} \mathbb{Z}/\mathbb{Z}$  is cyclic of order  $d_i$ , isomorphic to  $\mathbb{Z}/d_i \mathbb{Z}$  via multiplication by  $d_i$ . Grouping the two copies gives  $K(L) \cong \left( \bigoplus_i \mathbb{Z}/d_i \mathbb{Z} \right)^2$ , as claimed.

*Step 3: Compute the order.* The order is the product of the orders of the cyclic factors:

$$|K(L)| = \prod_{i=1}^g d_i \cdot \prod_{i=1}^g d_i = (d_1 \cdots d_g)^2$$

Finally, in the symplectic basis  $\det E = \det \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix} = \det(D)^2 = (d_1 \cdots d_g)^2$ , and the determinant of the Gram matrix of an alternating form is a basis-independent invariant up to the square of the change-of-basis determinant, which is 1 for a change between  $\mathbb{Z}$ -bases of  $\Lambda$ . Hence  $|K(L)| = \det E$ , as we sought to show.

The proof exhibits  $K(L)$  as  $2g$  cyclic pieces, one  $\mathbb{Z}/d_i$  for each “ $\lambda$ -direction” and one for each “ $\mu$ -direction” of the symplectic basis. The square in  $(d_1 \cdots d_g)^2$  is not cosmetic: it is the statement that  $K(L)$  pairs with itself. Half of  $K(L)$  comes from the  $\lambda$ -directions and half from the  $\mu$ -directions, and the alternating form  $E$  pairs the two halves nondegenerately, exactly as the standard symplectic form pairs the two Lagrangian halves of a symplectic vector space. This self-pairing is the commutator of the theta group, developed below; the appearance of  $\sqrt{|K(L)|} = d_1 \cdots d_g$  as  $\dim H^0(X, L)$  is then the statement that  $H^0(X, L)$  is a Lagrangian-sized representation of  $K(L)$ .

#### Example 4.1.4: $K(L)$ for a Principal Polarization

A principal polarization has type  $(1, \dots, 1)$ , so  $E = \text{Im} H$  is *unimodular* on  $\Lambda$ : its Gram matrix is the standard symplectic matrix  $\begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$ , with  $\det E = 1$ . Following definition 4.1.1, the condition  $E(v, \lambda) \in \mathbb{Z}$  for all  $\lambda \in \Lambda$  then already forces  $v \in \Lambda$ , because the coordinate map  $v \mapsto E(v, -)|_\Lambda$  carries  $\Lambda$  isomorphically onto  $\Lambda^\vee$ . Thus  $\Lambda(L) = \Lambda$  and  $K(L) = 0$  is trivial, with  $|K(L)| = 1 = 1^2$ . By theorem 3.4.2 the corresponding space of sections is one-dimensional: a principally polarized abelian variety has, up to scalar, a single theta function, whose divisor is the theta divisor. The higher-type case  $|K(L)| = (d_1 \cdots d_g)^2 > 1$  measures precisely the failure of this uniqueness.

With  $K(L)$  in hand, the nonabelian group sitting above it comes into view. The line bundle  $L$  has automorphisms, and it also has “partial symmetries”: for a translation  $t_x: X \rightarrow X$  by a point  $x = \bar{v} \in X$ , one can ask whether the pulled-back bundle  $t_x^* L$  is isomorphic to  $L$ . The set of  $x$  for which this holds is exactly  $K(L)$  (this is the analytic content of definition 4.1.2, and will be identified with  $\ker \varphi_L$  in chapter 5). The theta group collects, for each such  $x$ , the isomorphisms  $t_x^* L \xrightarrow{\sim} L$  themselves, not just the fact that one exists. Because the automorphisms of a line bundle over a compact connected complex manifold are the nonzero scalars  $\mathbb{C}^*$ , choosing such an isomorphism is a  $\mathbb{C}^*$ -torsor of choices, and the resulting group is a central extension of  $K(L)$  by  $\mathbb{C}^*$ .

#### Definition 4.1.5: The Theta Group

Let  $L$  be a line bundle on  $X = V/\Lambda$ . The *theta group* (or *Heisenberg group*) of  $L$  is

$$\mathcal{G}(L) := \{ (x, \psi) : x \in K(L), \psi: t_x^* L \xrightarrow{\sim} L \text{ an isomorphism of line bundles} \}$$

with multiplication

$$(x, \psi) \cdot (x', \psi') = (x + x', \psi \circ t_x^* \psi')$$

where  $t_x^* \psi': t_{x+x'}^* L = t_x^*(t_{x'}^* L) \xrightarrow{\sim} t_x^* L$  is the pullback of  $\psi'$  along  $t_x$ , so that the composite lands in  $L$ .

Equivalently, and this is the description used in the sequel,  $\mathcal{G}(L)$  is the group of pairs  $(x, \psi)$  acting on the total space of  $L$  by a bundle map covering the translation  $t_x$  of the base. The composition law is exactly the composition of such covering maps: to translate by  $x + x'$  and identify the result with  $L$ , first translate by  $x'$  and identify (via  $\psi'$ ), then translate by  $x$  and identify (via  $\psi$ ), pulling  $\psi'$  back along  $t_x$  so the two identifications compose. This makes  $\mathcal{G}(L)$  literally the symmetry group of

the pair  $(X, L)$  over the translations that fix the isomorphism class of  $L$ .

**Proposition 4.1.6: The Theta Group is a Central Extension**

Let  $L$  be a line bundle on  $X$ . The map sending a nonzero scalar  $c \in \mathbb{C}^*$  to the automorphism  $(0, c) \in \mathcal{G}(L)$  (translation by 0, identified with  $L$  via multiplication by  $c$ ) and the map  $(x, \psi) \mapsto x$  fit into a short exact sequence of groups

$$1 \longrightarrow \mathbb{C}^* \longrightarrow \mathcal{G}(L) \longrightarrow K(L) \longrightarrow 0 \quad (4.1)$$

in which the image of  $\mathbb{C}^*$  is central. Equivalently,  $\mathcal{G}(L)$  acts  $\mathbb{C}$ -linearly on  $H^0(X, L)$ , the central  $\mathbb{C}^*$  acting by scalar multiplication.

**Proof:**

*Step 1: Exactness.* The second map  $\pi: (x, \psi) \mapsto x$  is a homomorphism by the multiplication law, and it is surjective because every  $x \in K(L)$  admits an isomorphism  $t_x^* L \cong L$  by the defining property of  $K(L)$  (definition 4.1.2). Its kernel consists of pairs  $(0, \psi)$  with  $\psi: L \xrightarrow{\sim} L$  an automorphism of  $L$ . The torus  $X$  is compact and connected (proposition 1.1.3), so by the maximum principle its global holomorphic functions are the constants; hence a line-bundle automorphism is multiplication by a nowhere-zero global function, that is, by a nonzero constant  $c \in \mathbb{C}^*$ . This identifies  $\ker \pi$  with  $\mathbb{C}^*$  and gives (4.1).

*Step 2: Centrality.* An element  $(0, c)$  with  $c \in \mathbb{C}^*$  acts on the fibres of  $L$  by the scalar  $c$  and covers the identity translation. For any  $(x, \psi)$ ,

$$(0, c) \cdot (x, \psi) = (x, c \cdot \psi), \quad (x, \psi) \cdot (0, c) = (x, \psi \circ t_x^*(c)) = (x, c \cdot \psi)$$

because pulling back the scalar  $c$  along any translation returns the same scalar  $c$  (scalars are global constants), and scalar multiplication commutes with the linear isomorphism  $\psi$ . Hence the two products agree and  $\mathbb{C}^*$  is central.

*Step 3: The action on sections.* A pair  $(x, \psi) \in \mathcal{G}(L)$  acts on  $s \in H^0(X, L)$  by

$$(x, \psi) \cdot s := \psi \circ t_x^* s \circ t_{-x} = \psi(s \circ t_{-x})$$

translating the section by  $-x$  and reidentifying via  $\psi$ ; this is again a global section of  $L$ , and the composition law of definition 4.1.5 makes it a left action. The central  $(0, c)$  sends  $s$  to  $c s$ , scalar multiplication. This is the promised linear representation, completing the proof.

The exact sequence (4.1) is the structural heart of the section. It says that  $\mathcal{G}(L)$  is not much larger than the finite group  $K(L)$ : it is  $K(L)$  with a copy of  $\mathbb{C}^*$  threaded through its center, exactly the pattern of the Heisenberg group  $\begin{pmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{pmatrix}$ , whose center is the scalars along the top-right entry and whose quotient by the center is the abelian  $(*, *)$ . The nonabelianness of  $\mathcal{G}(L)$  is concentrated in a single invariant, the commutator, which because  $\mathbb{C}^*$  is central and  $K(L)$  is abelian descends to a pairing on  $K(L)$  itself.

To see this, notice that for  $(x, \psi), (y, \varphi) \in \mathcal{G}(L)$  the commutator  $(x, \psi)(y, \varphi)(x, \psi)^{-1}(y, \varphi)^{-1}$  projects under  $\pi$  to  $x + y - x - y = 0$  in the abelian group  $K(L)$ , hence lies in the central  $\mathbb{C}^*$ . It is therefore a scalar, and because  $\mathbb{C}^*$  is central the scalar is unchanged under replacing  $(x, \psi)$  by  $(x, c\psi)$  or  $(y, \varphi)$  by  $(y, c'\varphi)$ : the commutator depends only on the images  $x, y \in K(L)$ . This produces a well-defined

pairing  $K(L) \times K(L) \rightarrow \mathbb{C}^*$ , whose value the next result computes explicitly from the alternating form  $E$ .

**Proposition 4.1.7: The Commutator Pairing**

Let  $L = L(H, \chi)$  with  $E = \text{Im}H$ . The commutator in  $\mathcal{G}(L)$  induces a map

$$e^L : K(L) \times K(L) \longrightarrow \mathbb{C}^*, \quad e^L(\bar{v}, \bar{w}) = (w, \varphi)(v, \psi)(w, \varphi)^{-1}(v, \psi)^{-1}$$

where  $(v, \psi), (w, \varphi) \in \mathcal{G}(L)$  are any lifts of  $\bar{v}, \bar{w} \in K(L)$ ; the value depends only on  $\bar{v}, \bar{w}$ , and it is given by the formula

$$e^L(\bar{v}, \bar{w}) = \exp(2\pi i E(v, w)) \quad (4.2)$$

for any representatives  $v, w \in \Lambda(L)$  of  $\bar{v}, \bar{w} \in K(L) = \Lambda(L)/\Lambda$ . The pairing  $e^L$  is alternating and nondegenerate when  $H > 0$ .

**Proof:**

*Step 1: Well-definedness of the formula on  $K(L)$ .* The right side of (4.2) must not depend on the choice of representatives  $v \in \bar{v}$  and  $w \in \bar{w}$  modulo  $\Lambda$ . Replace  $v$  by  $v + \lambda$  with  $\lambda \in \Lambda$ ; then

$$E(v + \lambda, w) = E(v, w) + E(\lambda, w)$$

Now  $w \in \Lambda(L)$ , so by definition 4.1.1  $E(w, \lambda) \in \mathbb{Z}$  for all  $\lambda \in \Lambda$ , and  $E(\lambda, w) = -E(w, \lambda) \in \mathbb{Z}$  since  $E$  is alternating. Thus  $E(v + \lambda, w) \equiv E(v, w) \pmod{\mathbb{Z}}$  and  $\exp(2\pi i E(v + \lambda, w)) = \exp(2\pi i E(v, w))$ . Symmetrically the value is unchanged when  $w$  is altered by a lattice vector. This is precisely the point where the enlargement to  $\Lambda(L)$  is used: the inclusion  $E(\Lambda(L), \Lambda) \subseteq \mathbb{Z}$  is what makes  $e^L$  well-defined on the quotient  $K(L)$ .

*Step 2: The commutator equals the formula.* Realize  $\mathcal{G}(L)$  through the canonical factor of automorphy of definition 2.2.5. A section of  $L$  is a holomorphic  $\theta: V \rightarrow \mathbb{C}$  with  $\theta(u + \lambda) = a_{(H, \chi)}(\lambda, u) \theta(u)$ , where  $a_{(H, \chi)}(\lambda, u) = \chi(\lambda) \exp(\pi H(u, \lambda) + \frac{\pi}{2} H(\lambda, \lambda))$ . For  $v \in \Lambda(L)$ , the translation operator on functions

$$(U_v \theta)(u) := \exp\left(\pi H(u, v) + \frac{\pi}{2} H(v, v)\right)^{-1} \theta(u + v)$$

carries sections of  $L$  to sections of  $L$ ; the prefactor is the value at  $v$  of the canonical automorphy factor, chosen so that  $U_v \theta$  again satisfies the transformation law for  $\Lambda$ . To verify this, compute  $(U_v \theta)(u + \lambda)$  from the definitions: pushing  $\theta(u + v + \lambda) = a_{(H, \chi)}(\lambda, u + v) \theta(u + v)$  through the prefactor, the ratio  $(U_v \theta)(u + \lambda) / (a_{(H, \chi)}(\lambda, u) (U_v \theta)(u))$  collapses, after the  $H(\cdot, \lambda)$  and  $H(\cdot, v)$  terms cancel in the exponent, to  $\exp(\pi(H(v, \lambda) - H(\lambda, v))) = \exp(2\pi i E(v, \lambda))$ . This equals 1 for every  $\lambda \in \Lambda$  precisely when  $E(v, \lambda) \in \mathbb{Z}$  for all  $\lambda$ , that is, exactly when  $v \in \Lambda(L)$  (definition 4.1.1); this is why the enlargement to  $\Lambda(L)$  is the exact condition for  $U_v$  to preserve the transformation law. Indeed  $U_v$  is a concrete lift of translation by  $\bar{v}$  to a symmetry of  $L$ , hence an element of  $\mathcal{G}(L)$  over  $\bar{v} \in K(L)$ . Composing,

$$(U_v U_w \theta)(u) = \exp\left(-\pi H(u, v) - \frac{\pi}{2} H(v, v) - \pi H(u + v, w) - \frac{\pi}{2} H(w, w)\right) \theta(u + v + w)$$

and the same expression with  $v, w$  interchanged for  $(U_w U_v \theta)(u)$ . Subtracting the two exponents, the terms  $-\frac{\pi}{2} H(v, v)$  and  $-\frac{\pi}{2} H(w, w)$  cancel, and expanding  $H(u + v, w) = H(u, w) + H(v, w)$  and

$H(u+w, v) = H(u, v) + H(w, v)$  shows all terms linear in  $u$  cancel as well, leaving only the constant  $\pi(H(w, v) - H(v, w))$ . Using  $H(v, w) - H(w, v) = H(v, w) - \overline{H(v, w)} = 2i \operatorname{Im} H(v, w) = 2i E(v, w)$ , this constant is  $-2\pi i E(v, w)$ , so  $(U_v U_w \theta)(u) = e^{-2\pi i E(v, w)} (U_w U_v \theta)(u)$ , that is,

$$U_w U_v = \exp(2\pi i E(v, w)) U_v U_w$$

Hence the group commutator  $U_w U_v U_w^{-1} U_v^{-1}$  is the scalar  $\exp(2\pi i E(v, w))$ , and this is the value of  $e^L(\bar{v}, \bar{w})$  in the ordering convention of (4.2). (Reversing the order replaces  $E(v, w)$  by  $E(w, v) = -E(v, w)$ , consistent with the alternating property.)

*Step 3: Alternating and nondegenerate.* The form is alternating because  $E$  is:  $e^L(\bar{v}, \bar{v}) = \exp(2\pi i E(v, v)) = \exp(0) = 1$ . For nondegeneracy, suppose  $\bar{w} \in K(L)$  satisfies  $e^L(\bar{v}, \bar{w}) = 1$  for all  $\bar{v} \in K(L)$ , that is,  $E(v, w) \in \mathbb{Z}$  for all  $v \in \Lambda(L)$ . Work in the symplectic coordinates of proposition 4.1.3, where  $\Lambda(L) = \sum_i \frac{1}{d_i} \mathbb{Z} \lambda_i + \sum_i \frac{1}{d_i} \mathbb{Z} \mu_i$  and  $w = \sum_i a_i \lambda_i + \sum_i b_i \mu_i$  with  $a_i, b_i \in \frac{1}{d_i} \mathbb{Z}$ . Testing against  $v = \frac{1}{d_j} \lambda_j \in \Lambda(L)$  gives  $E(v, w) = b_j \in \mathbb{Z}$ , and testing against  $v = \frac{1}{d_j} \mu_j$  gives  $E(v, w) = -a_j \in \mathbb{Z}$ . Thus every  $a_j, b_j \in \mathbb{Z}$ , so  $w \in \Lambda$  and  $\bar{w} = 0$ . Hence  $e^L$  is nondegenerate, completing the proof.

The pairing  $e^L$  is the arithmetic shadow of the entire theta-group construction. It is a nondegenerate alternating pairing on a finite abelian group, and such a pairing on  $K(L) \cong (\bigoplus \mathbb{Z}/d_i)^2$  realizes the group as an orthogonal sum of hyperbolic planes  $\mathbb{Z}/d_i \oplus \mathbb{Z}/d_i$ , the two summands of proposition 4.1.3 pairing against each other. Restricted to the  $n$ -torsion this pairing is the *Weil pairing*  $e_n: X[n] \times X[n] \rightarrow \mu_n$  that chapter 5 constructs analytically and Part II reconstructs over an arbitrary field as a duality of finite group schemes; the commutator of the theta group is where it is born. The nondegeneracy just proved is what forces  $H^0(X, L)$  to be irreducible under  $\mathcal{G}(L)$ , by the following classical fact about Heisenberg groups.

A finite Heisenberg group, a central extension of a finite abelian group  $K$  by  $\mathbb{C}^*$  whose commutator pairing is nondegenerate, has a unique irreducible representation, up to isomorphism, on which the center acts by scalars. This is the Stone-von Neumann theorem in its finite form. The uniqueness is a general fact about the Heisenberg group and its proof, a counting argument via the characters of a maximal isotropic subgroup, is standard; the representation-theoretic details are given in Mumford, *Abelian Varieties*, and are cited here rather than reproved (the mechanism is the same as for the infinite Heisenberg group in quantum mechanics). What matters for this book is the payoff: the space of sections of a polarization is that unique representation.

#### Theorem 4.1.8: Stone-von Neumann for the Theta Group

Let  $L = L(H, \chi)$  with  $H > 0$  a polarization of type  $(d_1, \dots, d_g)$ . Then  $\mathcal{G}(L)$  has, up to isomorphism, a unique irreducible representation on which the central subgroup  $\mathbb{C}^*$  acts by the identity character (scalar multiplication). Its dimension is

$$\sqrt{|K(L)|} = d_1 d_2 \cdots d_g$$

and the representation of  $\mathcal{G}(L)$  on  $H^0(X, L)$  from proposition 4.1.6 is isomorphic to it. In particular

$$\dim_{\mathbb{C}} H^0(X, L) = d_1 d_2 \cdots d_g$$

reproving theorem 3.4.2 representation-theoretically.

**Proof:**

*Step 1: Uniqueness and dimension of the abstract representation.* By proposition 4.1.7 the extension (4.1) has nondegenerate commutator pairing  $e^L$  on  $K(L)$ , so  $\mathcal{G}(L)$  is a finite Heisenberg group over  $K(L)$ . The Stone-von Neumann theorem for such a group provides a unique irreducible representation  $W$  with  $\mathbb{C}^*$  acting by the identity character; its dimension is  $\sqrt{|K(L)|}$ . Concretely, choose a maximal isotropic subgroup  $M \subseteq K(L)$  for  $e^L$ ; nondegeneracy gives  $|M|^2 = |K(L)|$ . The preimage  $\tilde{M} := \pi^{-1}(M) \subseteq \mathcal{G}(L)$  is abelian (isotropic subgroups lift to abelian subgroups, since the commutator of two lifts is  $e^L$  evaluated on their images, which is 1 on  $M$ ), so a choice of character of  $\tilde{M}$  extending the identity character of  $\mathbb{C}^*$  induces a representation  $W = \text{Ind}_{\tilde{M}}^{\mathcal{G}(L)}$  of dimension  $[K(L) : M] = |K(L)|/|M| = |M| = \sqrt{|K(L)|}$ . By proposition 4.1.3,  $\sqrt{|K(L)|} = \sqrt{(d_1 \cdots d_g)^2} = d_1 \cdots d_g$ . The uniqueness is the content cited to Mumford above.

*Step 2:  $H^0(X, L)$  carries this representation.* By proposition 4.1.6,  $\mathcal{G}(L)$  acts on  $V_0 := H^0(X, L)$  with  $\mathbb{C}^*$  acting by scalars, that is,  $V_0$  is a representation of  $\mathcal{G}(L)$  on which the center acts by the identity character. Any such representation is a direct sum of copies of the unique irreducible  $W$  (all irreducibles with a fixed nontrivial central character are isomorphic to  $W$ , again by Stone-von Neumann). To conclude  $V_0 \cong W$  it suffices to show  $V_0 \neq 0$  and that the multiplicity is one.

*Step 3: Nonvanishing and multiplicity one.* That  $V_0 = H^0(X, L) \neq 0$  for  $H > 0$  is proposition 2.4.3: a positive line bundle has nonzero sections. Write  $V_0 \cong W^{\oplus m}$  for some  $m \geq 1$ . To pin down  $m$ , restrict to the maximal abelian  $\tilde{M} = \pi^{-1}(M)$  of Step 1, realized by the commuting operators  $U_v$  of proposition 4.1.7 for  $v$  in the lattice  $\Lambda(M)$  (with  $\Lambda \subseteq \Lambda(M) \subseteq \Lambda(L)$  the sublattice corresponding to  $M$ ). The joint eigenspaces of this commuting family on  $V_0$  each have dimension  $m$ , since on  $W$  each eigencharacter of  $\tilde{M}$  occurs exactly once. But a joint eigenfunction is a holomorphic  $\theta: V \rightarrow \mathbb{C}$  satisfying a full system of quasi-periodicity relations for the lattice  $\Lambda(M)$ , whose Fourier coefficients are then determined up to a single overall scalar; hence every joint eigenspace is at most one-dimensional. Comparing gives  $m = 1$ . (This last step is the abstract shadow of the explicit basis constructed in the next section, where the joint eigenfunctions are exhibited as the theta functions with characteristics.) Hence  $V_0 \cong W$  and

$$\dim_{\mathbb{C}} H^0(X, L) = \dim_{\mathbb{C}} W = d_1 \cdots d_g$$

This recovers the value of theorem 3.4.2 with no convergence estimate, completing the proof.

The theorem reverses the logic of chapter 3. There the dimension  $d_1 \cdots d_g$  was computed by counting lattice points and estimating a theta series; here it is forced by a symmetry principle. The space  $H^0(X, L)$  is the unique irreducible representation of  $\mathcal{G}(L)$  with the given central character, and its dimension is pinned by the order of  $K(L)$  alone. This is why the basis of theta functions constructed in the next section is not arbitrary: the theta group acts on  $H^0(X, L)$ , its maximal isotropic subgroups give simultaneous eigenbases, and the functions with characteristics  $a \in \bigoplus_i (\frac{1}{d_i} \mathbb{Z})/\mathbb{Z}$  are exactly the eigenvectors of one such isotropic system. The theta group is the intrinsic symmetry that makes the natural basis of theta functions *canonical*, and its commutator  $e^L$  is the germ of the Weil pairing constructed analytically in chapter 5.

The whole construction is transparent in dimension one, where it reduces to the finite Heisenberg groups underlying the classical Jacobi theta functions.

**Example 4.1.9: The Theta Group of an Elliptic Curve**

Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ , and let  $L$  be a line bundle of degree  $d \geq 1$ , so  $E = \text{Im}H$  is the alternating form on  $\Lambda$  with  $E(1, \tau) = d$  (the polarization of type  $(d)$ , matching [4]). There is a single elementary divisor  $d_1 = d$ , so  $g = 1$  and the type is  $(d)$ .

1. *Principal case*  $d = 1$ . Here  $E(1, \tau) = 1$ , so  $E$  is unimodular and  $\Lambda(L) = \Lambda$ , giving  $K(L) = 0$  (example 4.1.4). The theta group is just its center  $\mathbb{C}^*$ , and  $H^0(E, L)$  is one-dimensional: the unique classical theta function  $\vartheta(z, \tau)$  up to scalar. This is the analytic source of the single-point theta divisor on an elliptic curve.
2. *Degree*  $d$ . By definition 4.1.1,  $v \in \Lambda(L)$  iff  $E(v, 1) \in \mathbb{Z}$  and  $E(v, \tau) \in \mathbb{Z}$ . Writing  $v = x + y\tau$  with  $x, y \in \mathbb{R}$  and using  $E(1, \tau) = d$ , these conditions read  $-dy \in \mathbb{Z}$  and  $dx \in \mathbb{Z}$ , so

$$\Lambda(L) = \frac{1}{d}\mathbb{Z} + \frac{1}{d}\mathbb{Z}\tau, \quad K(L) = \Lambda(L)/\Lambda \cong \frac{1}{d}\mathbb{Z}/\mathbb{Z} \oplus \frac{1}{d}\mathbb{Z}/\mathbb{Z} \cong (\mathbb{Z}/d)^2$$

consistent with proposition 4.1.3:  $|K(L)| = d^2 = \det E$ . The commutator pairing (4.2) on the generators  $\frac{1}{d}$  and  $\frac{\tau}{d}$  is

$$e^L\left(\frac{1}{d}, \frac{\tau}{d}\right) = \exp\left(2\pi i E\left(\frac{1}{d}, \frac{\tau}{d}\right)\right) = \exp\left(2\pi i \cdot \frac{1}{d^2} E(1, \tau)\right) = \exp\left(\frac{2\pi i}{d}\right)$$

a primitive  $d$ -th root of unity. Thus  $\mathcal{G}(L)$  is a central extension of  $(\mathbb{Z}/d)^2$  by  $\mathbb{C}^*$ , generated over the center by the two lifted translation operators  $S := U_{1/d}$  and  $T := U_{\tau/d}$  of proposition 4.1.7, translation by  $\frac{1}{d}$  and by  $\frac{\tau}{d}$  together with their canonical automorphy prefactors. Their commutator is the root of unity just computed,

$$TST^{-1}S^{-1} = e^L\left(\frac{1}{d}, \frac{\tau}{d}\right) = e^{2\pi i/d} =: \zeta_d$$

so  $S$  and  $T$  satisfy  $TS = \zeta_d ST$ . These are the operators under which the  $d$  classical Jacobi theta functions with characteristics transform among themselves;  $S$  is the shift by a  $d$ -torsion translation and  $T$  the quasi-periodic “modulation” encoded by the automorphy factor.

3. *Stone-von Neumann*. By theorem 4.1.8 the unique irreducible representation of  $\mathcal{G}(L)$  with the standard central character has dimension  $\sqrt{d^2} = d$ , and  $H^0(E, L)$  is that representation. Hence  $\dim H^0(E, L) = d = \deg L$ , recovering the Riemann-Roch count example 3.4.6 for the elliptic curve, now as the dimension of the standard representation of the finite Heisenberg group  $H(\mathbb{Z}/d)$ .

The pairing  $\zeta_d = e^{2\pi i/d}$  read off in part (2) is the Weil pairing  $e_d$  on  $E[d]$ , and the operators  $S, T$  are the theta-group symmetries that permute the  $d$  Jacobi theta functions. This is the  $g = 1$  prototype of the entire section: the theta group is a finite Heisenberg group, its commutator is the Weil pairing, and the sections of  $L$  form its Schrödinger representation.

**Exercise 4.1.1**

1. Let  $X = V/\Lambda$  carry a polarization  $L = L(H, \chi)$  of type  $(1, 2, 6)$ , so  $g = 3$ . Using proposition 4.1.3, write down the group  $K(L)$  explicitly as a product of cyclic groups, compute  $|K(L)|$ , and state  $\dim_{\mathbb{C}} H^0(X, L)$ .



2. Work with the elliptic curve  $E = \mathbb{C}/\Lambda$ ,  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ , and a line bundle of degree  $d = 3$ , as in example 4.1.9. Compute  $\Lambda(L)$  and  $K(L)$  from definitions 4.1.1 and 4.1.2, and evaluate the commutator pairing  $e^L(\frac{1}{d}, \frac{\tau}{d})$  from (4.2). Confirm that the two lifted operators  $S = U_{1/d}$  and  $T = U_{\tau/d}$  satisfy  $TS = \zeta_3 ST$  with  $\zeta_3 = e^{2\pi i/3}$ .
3. Let  $\bar{v} \in K(L)$  be fixed. Prove directly from proposition 4.1.7 that the map  $\bar{w} \mapsto e^L(\bar{v}, \bar{w})$  is a group homomorphism  $K(L) \rightarrow \mathbb{C}^*$ , and that when  $H > 0$  the assignment  $\bar{v} \mapsto e^L(\bar{v}, -)$  is an isomorphism of  $K(L)$  onto its character group  $\text{Hom}(K(L), \mathbb{C}^*)$ .
4. A subgroup  $M \subseteq K(L)$  is *isotropic* for  $e^L$  if  $e^L(\bar{v}, \bar{w}) = 1$  for all  $\bar{v}, \bar{w} \in M$ . Prove that an isotropic subgroup satisfies  $|M|^2 \leq |K(L)|$ , with equality precisely when  $M$  is maximal isotropic. (Use the nondegeneracy of proposition 4.1.7 and exercise 4.1.1 (3).) Then exhibit a maximal isotropic subgroup in the elliptic-curve case  $d = 3$  of example 4.1.9 and verify  $|M| = 3 = \sqrt{|K(L)|}$ .
5. Let  $\tilde{M} := \pi^{-1}(M) \subseteq \mathcal{G}(L)$  be the preimage of an isotropic subgroup  $M$  under the projection  $\pi$  of (4.1). Prove that  $\tilde{M}$  is abelian, and that if  $M$  is maximal isotropic then  $\tilde{M}$  is a maximal abelian subgroup of  $\mathcal{G}(L)$ . Conclude, using theorem 4.1.8, that  $\dim_{\mathbb{C}} H^0(X, L) = [K(L) : M]$  for any maximal isotropic  $M$ .

## 4.2 The Classical Riemann Theta Function

The previous section identified the section space  $H^0(X, L)$  of a polarization abstractly, as the unique irreducible representation of the theta group, and read off its dimension from the order of  $K(L)$ . What that argument does not supply is a single function. This section writes one down. For a principally polarized abelian variety in normalized coordinates the period matrix is  $(\tau \ I_g)$  with  $\tau$  in the Siegel upper half-space (proposition 3.3.3), and there is a completely explicit holomorphic function on  $\mathbb{C}^g$ , given by a Fourier series indexed by the integer lattice, whose quasi-periodicity under the lattice  $\Lambda = \tau\mathbb{Z}^g + \mathbb{Z}^g$  is exactly the transformation law of a section of the principal polarization. It is the Riemann theta function, and it is the direct generalization to  $g$  variables of the Jacobi theta series the reader met in the elliptic case (example 2.4.4).

The construction begins from the coordinates furnished by the normalization. A principally polarized abelian variety of dimension  $g$  is, by proposition 3.3.3, the torus  $X = \mathbb{C}^g/\Lambda$  with  $\Lambda = \tau\mathbb{Z}^g + \mathbb{Z}^g$  for a matrix  $\tau \in \mathfrak{H}_g$  (definition 3.3.1); the lattice is spanned by the columns of  $\tau$  together with the standard basis vectors of  $\mathbb{Z}^g$ . A section of the principal polarization is a holomorphic function on  $\mathbb{C}^g$  reproducing itself, up to the canonical factor of automorphy, under translation by each of these  $2g$  generators. Riemann's insight is that the two families of generators play asymmetric roles: translation by the integer vectors  $\mathbb{Z}^g$  should leave the function unchanged, which forces a Fourier expansion in  $e^{2\pi i n^t z}$ , and translation by the columns of  $\tau$  then dictates a Gaussian weight  $e^{\pi i n^t \tau n}$  on the Fourier coefficients. The result is a single series.



**Definition 4.2.1: The Riemann Theta Function**

Let  $\tau \in \mathfrak{H}_g$  be a point of the Siegel upper half-space (definition 3.3.1). The *Riemann theta function* associated to  $\tau$  is the function  $\theta(-, \tau): \mathbb{C}^g \rightarrow \mathbb{C}$  defined by the series

$$\theta(z, \tau) = \sum_{n \in \mathbb{Z}^g} \exp(\pi i n^t \tau n + 2\pi i n^t z) \quad (4.3)$$

where  $z \in \mathbb{C}^g$  is written as a column vector,  $n$  ranges over the integer lattice  $\mathbb{Z}^g$ , and  $n^t \tau n$  and  $n^t z$  are the usual matrix products (so  $n^t \tau n = \sum_{j,k} n_j \tau_{jk} n_k$  is a scalar). The convergence of the series is established in proposition 4.2.2.

The series (4.3) is the canonical section of the principal polarization on the torus with period matrix  $(\tau \ I_g)$ , the explicit holomorphic function whose existence proposition 2.4.3 and the Stone-von Neumann argument of theorem 4.1.8 could guarantee but not exhibit. Two structural features are already visible in the formula. The summation over  $\mathbb{Z}^g$  builds in periodicity under the integer part of the lattice: replacing  $z$  by  $z + m$  for  $m \in \mathbb{Z}^g$  multiplies each term by  $e^{2\pi i n^t m} = 1$ , so  $\theta$  is literally  $\mathbb{Z}^g$ -periodic, and this is why  $\theta$  is a Fourier series in the first place. The Gaussian factor  $e^{\pi i n^t \tau n}$ , which decays because  $\text{Im} \tau > 0$ , is what makes the series converge and what governs its behaviour under the other half of the lattice, the columns of  $\tau$ . The quasi-periodicity computed below shows that these two behaviours are exactly the values of the canonical factor of automorphy on a symplectic basis, so  $\theta$  descends to a section of the line bundle it names.

Before any of that, the series must be shown to define a function at all. The decay of the Gaussian weight is the entire content of the convergence, and it is worth extracting the estimate in full, because the same estimate reappears every time a theta series is differentiated, translated, or specialized.

**Proposition 4.2.2: Absolute and Uniform Convergence of the Theta Series**

For each fixed  $\tau \in \mathfrak{H}_g$  the series (4.3) converges absolutely, and uniformly on every compact subset of  $\mathbb{C}^g$ . Consequently  $\theta(-, \tau)$  is a holomorphic function on  $\mathbb{C}^g$ . Moreover the convergence is uniform in  $\tau$  as well, on compact subsets of  $\mathfrak{H}_g \times \mathbb{C}^g$ , so  $\theta$  is holomorphic jointly in  $(z, \tau)$  on  $\mathfrak{H}_g \times \mathbb{C}^g$ .

**Proof:**

The strategy is to bound the modulus of the general term of (4.3) by a Gaussian in  $n$  and apply the Weierstrass  $M$ -test.

*Step 1: The modulus of a term.* Fix  $n \in \mathbb{Z}^g$ . Since the term is  $\exp(\pi i n^t \tau n + 2\pi i n^t z)$ , its modulus is the exponential of the real part of the exponent. Writing  $\text{Im}$  for the imaginary part of a complex number,

$$|\exp(\pi i n^t \tau n + 2\pi i n^t z)| = \exp\left(\text{Re}(\pi i n^t \tau n) + \text{Re}(2\pi i n^t z)\right) = \exp\left(-\pi n^t (\text{Im} \tau) n - 2\pi n^t \text{Im} z\right)$$

where the first term uses  $\text{Re}(i w) = -\text{Im} w$  applied to the complex number  $w = n^t \tau n$ , giving  $\text{Re}(\pi i n^t \tau n) = -\pi \text{Im}(n^t \tau n) = -\pi n^t (\text{Im} \tau) n$  since  $n$  is a real vector; and the second uses  $\text{Re}(2\pi i n^t z) = -2\pi \text{Im}(n^t z) = -2\pi n^t \text{Im} z$ .

*Step 2: A Gaussian majorant on a compact set.* Let  $K \subset \mathbb{C}^g$  be compact. Since  $\text{Im} \tau$  is a positive definite real symmetric matrix (definition 3.3.1), its smallest eigenvalue  $c > 0$  gives the bound

$n^t(\text{Im}\tau)n \geq c|n|^2$  for every  $n \in \mathbb{R}^g$ , where  $|n|$  is the Euclidean norm. And since  $K$  is compact there is a radius  $R \geq 0$  with  $|\text{Im}z| \leq R$  for all  $z \in K$ , so by Cauchy-Schwarz  $|n^t \text{Im}z| \leq |n|R$ . Combining these with Step 1,

$$|\exp(\pi i n^t \tau n + 2\pi i n^t z)| \leq \exp(-\pi c |n|^2 + 2\pi R |n|) \quad \text{for all } z \in K \quad (4.4)$$

The right-hand side is independent of  $z$ , and completing the square in the exponent,  $-\pi c |n|^2 + 2\pi R |n| = -\pi c (|n| - R/c)^2 + \pi R^2/c$ , shows it is bounded above by  $\exp(\pi R^2/c) \exp(-\frac{\pi c}{4} |n|^2)$  once  $|n| \geq 2R/c$ , since then  $(|n| - R/c)^2 \geq |n|^2/4$ . Thus, discarding the finitely many  $n$  with  $|n| < 2R/c$  from the tail, the general term is dominated by a constant multiple of  $\exp(-\frac{\pi c}{4} |n|^2)$ .

*Step 3: Summability of the majorant.* The series  $\sum_{n \in \mathbb{Z}^g} \exp(-\frac{\pi c}{4} |n|^2)$  converges: it factors as

$$\sum_{n \in \mathbb{Z}^g} \exp\left(-\frac{\pi c}{4} |n|^2\right) = \prod_{j=1}^g \left( \sum_{n_j \in \mathbb{Z}} \exp\left(-\frac{\pi c}{4} n_j^2\right) \right) = \left( \sum_{m \in \mathbb{Z}} e^{-\pi c m^2/4} \right)^g$$

and each one-dimensional factor is a convergent series, being dominated by the convergent geometric series  $\sum_m e^{-(\pi c/4)|m|}$  for  $|m| \geq 1$  (as  $m^2 \geq |m|$ ). By the Weierstrass  $M$ -test the majorant from (4.4) makes the series (4.3) converge absolutely and uniformly on  $K$ . A uniform limit of holomorphic functions is holomorphic, and each partial sum is holomorphic in  $z$ , so  $\theta(-, \tau)$  is holomorphic on  $\mathbb{C}^g$ .

*Step 4: Joint holomorphy.* Let  $K' \subset \mathfrak{H}_g$  be compact. On  $K'$  the smallest eigenvalue of  $\text{Im}\tau$  attains a positive minimum  $c' > 0$  by continuity and compactness, so the constant  $c$  in Step 2 may be taken to be  $c'$  uniformly over  $\tau \in K'$ ; the bound (4.4) then holds with  $c'$  in place of  $c$  for all  $(\tau, z) \in K' \times K$ , and Step 3 gives a single summable majorant independent of  $(\tau, z)$ . Hence the series converges uniformly on  $K' \times K$ . Each term is holomorphic in  $(z, \tau)$  jointly, so the sum is holomorphic on  $\mathfrak{H}_g \times \mathbb{C}^g$ , completing the proof.

The estimate (4.4) isolates the mechanism: positivity of  $\text{Im}\tau$  turns the oscillatory factor  $e^{\pi i n^t \tau n}$  into a genuine Gaussian  $e^{-\pi n^t (\text{Im}\tau) n}$ , whose super-exponential decay in  $|n|$  overwhelms the at-most-exponential growth  $e^{2\pi R |n|}$  coming from the compact set of  $z$ -values. This is the analytic reason the Siegel condition  $\text{Im}\tau > 0$  is not optional: outside  $\mathfrak{H}_g$  the weight fails to decay in some direction and the series diverges, so the theta function exists precisely on the domain that parametrizes principally polarized abelian varieties. The joint holomorphy in  $(z, \tau)$  is what allows  $\theta$  to be viewed as a single holomorphic object over the whole family, the germ of a section of a line bundle on the universal abelian variety over  $\mathfrak{H}_g$ ; that global viewpoint belongs to the theory of Siegel modular forms, but the convergence estimate is already its foundation.

With  $\theta$  established as a holomorphic function, its defining property is the transformation law under  $\Lambda = \tau \mathbb{Z}^g + \mathbb{Z}^g$ . The two halves of the lattice behave differently, and the computation is a reindexing of the series in each case.

**Proposition 4.2.3: Quasi-Periodicity of the Riemann Theta Function**

Let  $\tau \in \mathfrak{H}_g$  and let  $\theta = \theta(-, \tau)$  be the Riemann theta function of definition 4.2.1. For every  $m \in \mathbb{Z}^g$  and every  $z \in \mathbb{C}^g$ , the periodicity relation

$$\theta(z + m, \tau) = \theta(z, \tau) \quad (4.5)$$

holds under the integer part of the lattice, and the quasi-periodicity relation

$$\theta(z + \tau m, \tau) = \exp(-\pi i m^t \tau m - 2\pi i m^t z) \theta(z, \tau) \quad (4.6)$$

holds under the columns of  $\tau$ .

**Proof:**

Both identities are proved by manipulating the absolutely convergent series (4.3), which may be reindexed freely.

*Step 1: Periodicity under  $\mathbb{Z}^g$ .* For  $m \in \mathbb{Z}^g$ , substituting  $z + m$  for  $z$  in (4.3) multiplies the  $n$ -th term by  $\exp(2\pi i n^t m)$ . Since  $n, m \in \mathbb{Z}^g$ , the inner product  $n^t m$  is an integer, so  $\exp(2\pi i n^t m) = 1$  and every term is unchanged. Hence

$$\theta(z + m, \tau) = \sum_{n \in \mathbb{Z}^g} \exp(\pi i n^t \tau n + 2\pi i n^t (z + m)) = \sum_{n \in \mathbb{Z}^g} \exp(\pi i n^t \tau n + 2\pi i n^t z) = \theta(z, \tau)$$

which is (4.5).

*Step 2: The exponent under a  $\tau$ -translation.* For  $m \in \mathbb{Z}^g$ , substitute  $z + \tau m$  for  $z$  in (4.3). The  $n$ -th exponent becomes

$$\pi i n^t \tau n + 2\pi i n^t (z + \tau m) = \pi i n^t \tau n + 2\pi i n^t z + 2\pi i n^t \tau m$$

The idea is to absorb the cross term  $2\pi i n^t \tau m$  into the quadratic form by completing the square in  $n$ . Because  $\tau$  is symmetric (definition 3.3.1),  $n^t \tau m = m^t \tau n$ , and

$$\pi i n^t \tau n + 2\pi i n^t \tau m = \pi i (n + m)^t \tau (n + m) - \pi i m^t \tau m$$

as one checks by expanding  $(n + m)^t \tau (n + m) = n^t \tau n + 2n^t \tau m + m^t \tau m$  using the symmetry of  $\tau$ . Substituting, the  $n$ -th exponent equals

$$\pi i (n + m)^t \tau (n + m) + 2\pi i n^t z - \pi i m^t \tau m$$

*Step 3: Reindexing.* Set  $k = n + m$ , a bijection of  $\mathbb{Z}^g$  with itself. Then  $n = k - m$ , so  $2\pi i n^t z = 2\pi i k^t z - 2\pi i m^t z$ , and the  $n$ -th exponent becomes

$$\pi i k^t \tau k + 2\pi i k^t z - \pi i m^t \tau m - 2\pi i m^t z$$

The last two terms are independent of the summation index  $k$  and factor out of the sum. Therefore

$$\theta(z + \tau m, \tau) = \exp(-\pi i m^t \tau m - 2\pi i m^t z) \sum_{k \in \mathbb{Z}^g} \exp(\pi i k^t \tau k + 2\pi i k^t z) = \exp(-\pi i m^t \tau m - 2\pi i m^t z) \theta(z, \tau)$$

which is (4.6), completing the proof.

The two identities are precisely the values of the canonical Appell-Humbert factor of automorphy on the symplectic basis of  $\Lambda = \tau\mathbb{Z}^g + \mathbb{Z}^g$ . The columns of  $I_g$ , the integer generators, act trivially by (4.5), reflecting the vanishing of the factor's exponent on the isotropic sublattice  $\mathbb{Z}^g$ ; the columns of  $\tau$  act through the nowhere-vanishing multiplier  $\exp(-\pi i m^t \tau m - 2\pi i m^t z)$  of (4.6). Up to the holomorphic coboundary passing between this classical normalization and the canonical factor  $a_{(H,\chi)}(\lambda, v)$  of definition 2.2.5, that multiplier is the value of the factor of automorphy of the principal polarization, whose Hermitian form is  $H(v, w) = v^t (\text{Im}\tau)^{-1} \bar{w}$  in the normalized coordinates, on the lattice vector  $\lambda = \tau m$ . Thus  $\theta$  satisfies a functional equation of the shape (2.8) defining a section of  $L$ , and it is nonzero, so it spans the one-dimensional space  $H^0(X, L)$  predicted by proposition 3.4.5 for a principal polarization. The existence that chapter 3 could only infer from a dimension count is now witnessed by an explicit series.

For polarizations of higher type, and for organizing the whole space of sections, one needs not a single theta function but a finite family of them, obtained by shifting the summation lattice and the argument by constant vectors. These shifts are the characteristics, and they produce a two-parameter family of theta functions whose members transform by the same law up to explicit unimodular constants.

**Definition 4.2.4: Theta Function with Characteristics**

Let  $\tau \in \mathfrak{H}_g$  and let  $a, b \in \mathbb{R}^g$ . The *theta function with characteristic*  $\begin{bmatrix} a \\ b \end{bmatrix}$  is

$$\theta \begin{bmatrix} a \\ b \end{bmatrix}(z, \tau) = \sum_{n \in \mathbb{Z}^g} \exp\left(\pi i (n+a)^t \tau (n+a) + 2\pi i (n+a)^t (z+b)\right) \quad (4.7)$$

for  $z \in \mathbb{C}^g$ . The vectors  $a$  and  $b$  are the *characteristics*;  $a$  shifts the summation lattice and  $b$  shifts the argument. The plain Riemann theta function of definition 4.2.1 is the case  $a = b = 0$ , that is  $\theta(z, \tau) = \theta \begin{bmatrix} 0 \\ 0 \end{bmatrix}(z, \tau)$ .

The characteristic version is not a new kind of object but a translate of the old one, and recording the exact relation makes both its convergence and its transformation law immediate. Expanding the exponent in (4.7) and comparing with (4.3) yields the identity

$$\theta \begin{bmatrix} a \\ b \end{bmatrix}(z, \tau) = \exp\left(\pi i a^t \tau a + 2\pi i a^t (z+b)\right) \theta(z + \tau a + b, \tau) \quad (4.8)$$

valid for all  $a, b \in \mathbb{R}^g$  and  $z \in \mathbb{C}^g$ : substituting  $z + \tau a + b$  into (4.3) and using the symmetry of  $\tau$  to write  $\pi i n^t \tau n + 2\pi i n^t \tau a = \pi i (n+a)^t \tau (n+a) - \pi i a^t \tau a$  reproduces the summand of (4.7) after multiplying by the displayed prefactor. Since  $\theta$  is holomorphic and converges absolutely and uniformly on compacta (proposition 4.2.2), so does  $\theta \begin{bmatrix} a \\ b \end{bmatrix}$ , being  $\theta$  evaluated at the translated argument  $z + \tau a + b$  times a holomorphic prefactor. The characteristics thus produce a whole family of holomorphic sections without any new convergence analysis, and their transformation law follows from that of  $\theta$  by tracking the two constant shifts.

**Proposition 4.2.5: Quasi-Periodicity of Theta with Characteristics**

Let  $\tau \in \mathfrak{H}_g$  and  $a, b \in \mathbb{R}^g$ . For every  $m \in \mathbb{Z}^g$  and  $z \in \mathbb{C}^g$ , translation by the integer part of the lattice gives

$$\theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right](z + m, \tau) = \exp(2\pi i a^t m) \theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right](z, \tau) \quad (4.9)$$

and translation by the columns of  $\tau$  gives

$$\theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right](z + \tau m, \tau) = \exp(-2\pi i b^t m) \exp(-\pi i m^t \tau m - 2\pi i m^t z) \theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right](z, \tau) \quad (4.10)$$

**Proof:**

*Step 1: Translation by an integer vector.* Substituting  $z + m$  for  $z$  in (4.7) multiplies the  $n$ -th term by  $\exp(2\pi i (n + a)^t m)$ . Splitting  $(n + a)^t m = n^t m + a^t m$  and using  $n^t m \in \mathbb{Z}$ , this factor equals  $\exp(2\pi i a^t m)$ , which is independent of  $n$  and pulls out of the sum. This gives (4.9).

*Step 2: Translation by a  $\tau$ -column, via the translate identity.* Apply (4.8) at the argument  $z + \tau m$ :

$$\theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right](z + \tau m, \tau) = \exp\left(\pi i a^t \tau a + 2\pi i a^t (z + \tau m + b)\right) \theta(z + \tau m + \tau a + b, \tau)$$

The plain theta on the right is quasi-periodic under  $\tau m$ : writing  $w = z + \tau a + b$  and applying (4.6) from proposition 4.2.3 at  $w$ ,

$$\theta(w + \tau m, \tau) = \exp(-\pi i m^t \tau m - 2\pi i m^t w) \theta(w, \tau)$$

with  $m^t w = m^t z + m^t \tau a + m^t b$ . Substituting and collecting exponents, the total prefactor multiplying  $\theta(z + \tau a + b, \tau)$  is the exponential of

$$\pi i a^t \tau a + 2\pi i a^t z + 2\pi i a^t \tau m + 2\pi i a^t b - \pi i m^t \tau m - 2\pi i m^t z - 2\pi i m^t \tau a - 2\pi i m^t b$$

The cross terms  $2\pi i a^t \tau m$  and  $-2\pi i m^t \tau a$  cancel by the symmetry  $a^t \tau m = m^t \tau a$ .

*Step 3: Recognizing the characteristic theta.* By (4.8) again,  $\theta(z + \tau a + b, \tau) = \exp(-\pi i a^t \tau a - 2\pi i a^t (z + b)) \theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right](z, \tau)$ . Multiplying this into the prefactor from Step 2, the terms  $\pi i a^t \tau a$ ,  $2\pi i a^t z$ , and  $2\pi i a^t b$  cancel against their negatives, leaving the exponential of

$$-\pi i m^t \tau m - 2\pi i m^t z - 2\pi i m^t b$$

Since  $m^t b = b^t m$ , this is exactly  $\exp(-2\pi i b^t m) \exp(-\pi i m^t \tau m - 2\pi i m^t z)$ , and the whole expression equals the right-hand side of (4.10) times  $\theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right](z, \tau)$ , as we sought to show.

The two constants  $e^{2\pi i a^t m}$  and  $e^{-2\pi i b^t m}$  are unimodular multipliers that alter the factor of automorphy only by a character of the lattice, so every  $\theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right]$  is a section of a line bundle differing from the principal one by a degree-zero twist, that is by an element of  $\text{Pic}^0(X)$ . When  $a$  and  $b$  range over the finite set of rational representatives  $a \in \bigoplus_i (\frac{1}{d_i} \mathbb{Z}) / \mathbb{Z}$  and  $b = 0$ , the multiplier  $e^{2\pi i a^t m}$  tracks exactly the character by which the theta group permutes the sections, and the  $d_1 \cdots d_g$  resulting functions transform under a common factor of automorphy, so they span a single space  $H^0(X, L)$ ; that they form a basis is the subject of section 4.3. For the principal polarization the only relevant characteristics are the half-integer ones  $a, b \in \frac{1}{2} \mathbb{Z}^g$ , which produce the sixteen classical theta constants in genus two and, in every genus, the even and odd theta functions whose zeros locate the theta

divisor of section 4.5.

The construction closes the circle with the elliptic case, where  $g = 1$  and the whole apparatus collapses to a single classical series the reader has already computed.

**Example 4.2.6: Recovering the Jacobi Theta Function**

Take  $g = 1$ . The Siegel upper half-space is  $\mathfrak{H}_1 = \mathbb{H}$  (example 3.3.8), so  $\tau \in \mathbb{H}$  is a single complex number with  $\text{Im}\tau > 0$ , and  $z \in \mathbb{C}$ . A column vector  $n \in \mathbb{Z}^1$  is an integer, and the matrix products degenerate to  $n^t \tau n = n^2 \tau$  and  $n^t z = nz$ . The defining series (4.3) becomes

$$\theta(z, \tau) = \sum_{n \in \mathbb{Z}} \exp(\pi i n^2 \tau + 2\pi i n z)$$

which is exactly the Jacobi theta function  $\vartheta(z, \tau)$  of (2.11) met in example 2.4.4. The Riemann theta function is thus the  $g$ -variable generalization of  $\vartheta$ , and the two notations agree in genus one.

The quasi-periodicity specializes to match as well. For  $m \in \mathbb{Z}$ , (4.5) reads  $\vartheta(z + m, \tau) = \vartheta(z, \tau)$ , and for  $m = 1$ , (4.6) reads

$$\vartheta(z + \tau, \tau) = \exp(-\pi i \tau - 2\pi i z) \vartheta(z, \tau)$$

These are precisely the two transformation laws recorded in example 2.4.4 for the section of the degree-one line bundle on  $E = \mathbb{C}/(\mathbb{Z}\tau + \mathbb{Z})$ . There the laws were derived by an ad hoc reindexing of the one-variable series and identified as the values of a factor of automorphy on the basis  $\lambda = 1$  and  $\lambda = \tau$  of  $\Lambda$ ; here they fall out as the  $g = 1$  instances of proposition 4.2.3. For a positive polarization of type  $(d)$  with  $d \geq 1$ , the  $d$  sections promised in example 2.4.4 are the characteristic thetas  $\theta \left[ \begin{smallmatrix} a/d \\ 0 \end{smallmatrix} \right] (z, \tau)$  for  $a = 0, 1, \dots, d-1$  of definition 4.2.4, and (4.9) shows they are permuted up to the root-of-unity multipliers  $e^{2\pi i am/d}$ , the genus-one shadow of the theta-group action of section 4.1.

The genus-one specialization confirms what the general estimate already implied: the Riemann theta function is not a new invention but the resolution of an existence claim. Chapter 3 proved, by counting orbits and summing a Gaussian, that a principal polarization has a one-dimensional space of sections, without producing a section; the elliptic case then exhibited the single section as Jacobi's series. The Riemann theta function does the same in every dimension at once, turning the abstract guarantee of proposition 2.4.3 and theorem 4.1.8 into a concrete holomorphic function on  $\mathbb{C}^g$ , computable term by term, whose zeros and whose translates carry all the projective geometry that the remainder of the chapter extracts.

**Exercise 4.2.1**

1. Take  $g = 1$  and  $\tau = i$ , so that  $\mathfrak{H}_1 = \mathbb{H}$  (example 3.3.8) and the Gaussian weight of (4.3) is real. Write down the first three distinct terms of the series  $\theta(0, i)$  (that is, the contributions of  $n = 0$  and  $n = \pm 1$ ), giving each as an explicit real number of the form  $e^{-\pi^k}$ , and identify the term of largest modulus.
2. Fix  $\tau \in \mathfrak{H}_g$  with smallest eigenvalue  $c > 0$  of  $\text{Im}\tau$ . Using the modulus computation in Step 1 of the proof of proposition 4.2.2, show that at  $z = 0$  every term of (4.3) has modulus at most 1, and determine exactly which indices  $n \in \mathbb{Z}^g$  attain the value 1.
3. Use the two transformation laws of proposition 4.2.3 to compute the multiplier relating

$\theta(z + \tau m + m', \tau)$  to  $\theta(z, \tau)$  for arbitrary  $m, m' \in \mathbb{Z}^g$ . Then verify that applying the  $\tau m$  shift is consistent whether the integer shift  $m'$  is performed first or second, and explain in one sentence why this consistency is forced.

4. Specialize the translate identity (4.8) to a half-integer characteristic. For  $g = 1$ ,  $a = b = \frac{1}{2}$ , express  $\theta\left[\begin{smallmatrix} 1/2 \\ 1/2 \end{smallmatrix}\right](z, \tau)$  as an explicit exponential prefactor times a translate of the plain theta  $\theta(-, \tau)$ , and use (4.5) together with the parity of the series to show  $\theta\left[\begin{smallmatrix} 1/2 \\ 1/2 \end{smallmatrix}\right](0, \tau) = 0$ .
5. Prove that for every  $a, b \in \mathbb{R}^g$  the identity (4.8) holds:

$$\theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right](z, \tau) = \exp\left(\pi i a^t \tau a + 2\pi i a^t(z + b)\right) \theta(z + \tau a + b, \tau)$$

by expanding the exponent of  $\theta(z + \tau a + b, \tau)$  from (4.3) and completing the square in the summation index, exactly as in the proof of proposition 4.2.3. Deduce that  $\theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right]$  is holomorphic on  $\mathbb{C}^g$  and converges absolutely and uniformly on compacta.

6. Combine (4.6) with (4.3) to prove that the plain Riemann theta function satisfies  $\theta(-z, \tau) = \theta(z, \tau)$  for all  $z \in \mathbb{C}^g$ , and then show that this evenness is compatible with the quasi-periodicity in the sense that the multiplier in (4.6) at  $z$  is the reciprocal of the multiplier at  $-z - \tau m$ . State what this says about the zeros of  $\theta$  under  $z \mapsto -z$ .

### 4.3 The Space of Theta Functions and Its Dimension

Two threads have been left deliberately open. In section 2.4 the dimension formula  $\dim_{\mathbb{C}} H^0(X, L) = \text{Pf}(\text{Im}H)$  was stated as proposition 2.4.3 and its proof postponed; in section 3.4 the Riemann-Roch theorem theorem 3.4.2 counted the sections of a positive line bundle abstractly, as  $d_1 \cdots d_g$ , without producing a single one of them. This section closes both threads at once, and it does so in the strongest possible form: not by recounting the dimension, but by exhibiting an explicit basis. The theta functions with characteristics from section 4.2, indexed by a finite set of rational shift vectors, turn out to be exactly  $d_1 \cdots d_g$  in number and linearly independent, and they span. Once a basis is in hand the sections of  $L$  become concrete coordinate functions, and the ratio of any two of them is a genuine meromorphic function on  $X$ . That observation is the germ of the projective embedding developed in section 4.4: a basis of  $H^0(X, L)$  assembles into a map  $X \rightarrow \mathbb{P}^{d_1 \cdots d_g - 1}$ , and the whole enterprise of realizing an abelian variety inside projective space reduces to understanding this one map.

Throughout,  $X = V/\Lambda$  with  $V = \mathbb{C}^g$ , and  $L = L(H, \chi)$  is a positive line bundle,  $H > 0$ , with  $E = \text{Im}H$  integral on  $\Lambda$  of type  $(d_1, \dots, d_g)$ . By choosing a symplectic basis realizing this type (definition 3.2.3), the lattice is normalized so that  $\Lambda = \tau\mathbb{Z}^g + D\mathbb{Z}^g$  with  $\tau \in \mathfrak{H}_g$  and  $D = \text{diag}(d_1, \dots, d_g)$ , and the sections of  $L$  are the holomorphic  $\theta: V \rightarrow \mathbb{C}$  obeying the quasi-periodicity attached to the canonical factor of automorphy of  $L(H, \chi)$ . The classical theta function  $\theta(z, \tau)$  of definition 4.2.1 and its variants with characteristics  $\theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right]$  of definition 4.2.4 are exactly the holomorphic functions with this behavior, and the transformation law of proposition 4.2.5 is precisely the quasi-periodicity a section must satisfy.

The counting begins with the shift index set. A theta function with characteristic  $a$  transforms under the sublattice  $D\mathbb{Z}^g$  by the automorphy factor of  $L$ , but its behavior under the finer lattice  $\mathbb{Z}^g$  picks out which coset of  $a$  modulo  $\mathbb{Z}^g$  one has chosen. The distinct sections are therefore indexed by the

shift vectors  $a$  modulo  $\mathbb{Z}^g$  that are compatible with the type, and these form a finite abelian group of order  $d_1 \cdots d_g$ .

**Definition 4.3.1: The Characteristic Group**

For a positive line bundle  $L$  of type  $(d_1, \dots, d_g)$  on  $X = V/\Lambda$ , the *characteristic group* is the finite abelian group

$$\mathfrak{a}(L) := \left( \bigoplus_{i=1}^g \frac{1}{d_i} \mathbb{Z} \right) / \mathbb{Z}^g \cong \bigoplus_{i=1}^g \mathbb{Z}/d_i \mathbb{Z}$$

of order  $d_1 \cdots d_g$ . For  $a \in \mathfrak{a}(L)$ , represented by a vector  $a \in \bigoplus_i \frac{1}{d_i} \mathbb{Z}$ , write  $\theta_a := \theta \begin{bmatrix} a \\ 0 \end{bmatrix}$  for the associated theta function with characteristic of definition 4.2.4.

The group  $\mathfrak{a}(L)$  is a set of representatives for one of the two Lagrangian pieces of  $K(L)$ . Recall from proposition 4.1.3 that the finite group  $K(L) = \Lambda(L)/\Lambda$  has order  $(d_1 \cdots d_g)^2$  and, in the symplectic normalization, decomposes as a product of two isotropic subgroups each isomorphic to  $\bigoplus_i \mathbb{Z}/d_i \mathbb{Z}$ . The characteristics in  $\mathfrak{a}(L)$  label the translates  $t_a^*$  under one such subgroup, and the square root  $\sqrt{|K(L)|} = d_1 \cdots d_g$  is exactly the order of  $\mathfrak{a}(L)$ . This is the first sign that the theta functions  $\theta_a$  will match the Stone-von Neumann count of theorem 4.1.8.

The main result asserts that these  $d_1 \cdots d_g$  functions are a basis. Linear independence is the transparent half and follows by comparing Fourier expansions; spanning is the substantive half, and there are two routes to it. One matches an arbitrary section's Fourier coefficients against the  $\theta_a$  directly; the other invokes the representation theory of the theta group, where the  $\theta_a$  are visibly a basis because they diagonalize the action of a maximal abelian subgroup. Both are given, the Fourier argument in full as the primary proof and the representation-theoretic viewpoint as the conceptual companion.

**Theorem 4.3.2: Basis of Theta Functions**

Let  $X = V/\Lambda$  and let  $L = L(H, \chi)$  be a positive line bundle with  $H > 0$  of type  $(d_1, \dots, d_g)$ , in the symplectic normalization  $\Lambda = \tau \mathbb{Z}^g + D \mathbb{Z}^g$  with  $\tau \in \mathfrak{H}_g$  and  $D = \text{diag}(d_1, \dots, d_g)$ . Then the  $d_1 \cdots d_g$  theta functions

$$\{ \theta_a \mid a \in \mathfrak{a}(L) \}, \quad \theta_a = \theta \begin{bmatrix} a \\ 0 \end{bmatrix}(\cdot, \tau)$$

form a  $\mathbb{C}$ -basis of  $H^0(X, L)$ . In particular  $\dim_{\mathbb{C}} H^0(X, L) = d_1 \cdots d_g$ , confirming proposition 2.4.3 and theorem 3.4.2 with an explicit basis.

**Proof:**

After the symplectic normalization the sections of  $L$  are, by proposition 4.2.5, precisely the holomorphic functions  $\theta: \mathbb{C}^g \rightarrow \mathbb{C}$  satisfying the two quasi-periodicity laws

$$\theta(z + Dm) = \theta(z), \quad \theta(z + \tau m) = \exp(-\pi i m^t \tau m - 2\pi i m^t z) \theta(z) \quad (m \in \mathbb{Z}^g) \quad (4.11)$$

where the first line uses that  $D\mathbb{Z}^g \subseteq \Lambda$  carries the trivial part of the automorphy factor and the second reproduces the transformation of proposition 4.2.3 on the  $\tau$ -periods. Denote by  $\mathcal{T}$  the  $\mathbb{C}$ -vector space of all such  $\theta$ ; the claim is that  $\{\theta_a\}_{a \in \mathfrak{a}(L)}$  is a basis of  $\mathcal{T} = H^0(X, L)$ .



*Step 1: Fourier expansion along the first period.* The first law in (4.11) makes  $\theta$  periodic under  $D\mathbb{Z}^g$ , so it admits a Fourier expansion in the variable  $z$  against the dual lattice of  $D\mathbb{Z}^g$ . Writing  $D^{-1} = \text{diag}(d_1^{-1}, \dots, d_g^{-1})$ , the characters of  $\mathbb{C}^g/D\mathbb{Z}^g$  are  $z \mapsto \exp(2\pi i r^t D^{-1} z)$  for  $r \in \mathbb{Z}^g$ , so

$$\theta(z) = \sum_{r \in \mathbb{Z}^g} c_r \exp(2\pi i r^t D^{-1} z)$$

for uniquely determined coefficients  $c_r \in \mathbb{C}$ . The task is to see how the second law in (4.11) constrains the  $c_r$ .

*Step 2: The recursion on Fourier coefficients.* Substitute the expansion into the second law. On the one hand,

$$\theta(z + \tau m) = \sum_{r \in \mathbb{Z}^g} c_r \exp(2\pi i r^t D^{-1} z) \exp(2\pi i r^t D^{-1} \tau m)$$

On the other hand the right-hand side of the law multiplies  $\theta(z)$  by  $\exp(-\pi i m^t \tau m - 2\pi i m^t z)$ . Matching the two forces, for every frequency index, an identity between the coefficient of a given exponential  $\exp(2\pi i r^t D^{-1} z)$  on each side. On the right the factor  $\exp(-2\pi i m^t z) = \exp(-2\pi i (Dm)^t D^{-1} z)$  shifts the frequency index by  $Dm$ ; comparing the coefficient of  $\exp(2\pi i r^t D^{-1} z)$  gives

$$c_r \exp(2\pi i r^t D^{-1} \tau m) = c_{r+Dm} \exp(-\pi i m^t \tau m) \quad (4.12)$$

The content of (4.12) is that  $c_r$  and  $c_{r+Dm}$  determine one another for every  $m \in \mathbb{Z}^g$ : the coefficients are governed by the coset of  $r$  in  $\mathbb{Z}^g/D\mathbb{Z}^g$ .

*Step 3: Each coset has one free coefficient.* The cosets  $\mathbb{Z}^g/D\mathbb{Z}^g$  are in bijection with  $\mathfrak{a}(L)$  via  $r \mapsto D^{-1}r \bmod \mathbb{Z}^g$ , since  $D^{-1}\mathbb{Z}^g/\mathbb{Z}^g = \bigoplus_i (\frac{1}{d_i}\mathbb{Z})/\mathbb{Z}$ . Fix a coset and a representative  $r_0$ ; (4.12) then expresses every  $c_{r_0+Dm}$  as  $c_{r_0}$  times an explicit exponential factor, so the entire family of coefficients over the coset is determined by the single value  $c_{r_0}$ . Consistency of the recursion (that traversing the coset by different paths gives the same answer) is exactly the cocycle identity built into the automorphy factor, and holds because  $\tau$  is symmetric: the accumulated phase for the shift  $m + m'$  agrees with composing the shifts by  $m$  and  $m'$ , using  $m^t \tau m' = m'^t \tau m$ . Consequently  $\dim_{\mathbb{C}} \mathcal{T}$  equals the number of cosets, namely  $|\mathbb{Z}^g/D\mathbb{Z}^g| = d_1 \cdots d_g$ .

*Step 4: The  $\theta_a$  realize the free coefficients.* It remains to identify a section supported on each coset with a theta function with characteristic. Expanding the definition of definition 4.2.4 in the normalization at hand,

$$\theta_a(z) = \theta\left[\begin{smallmatrix} a \\ 0 \end{smallmatrix}\right](z, \tau) = \sum_{n \in \mathbb{Z}^g} \exp(\pi i (n+a)^t \tau (n+a) + 2\pi i (n+a)^t z)$$

which, upon collecting the coefficient of  $\exp(2\pi i (n+a)^t z)$ , is a Fourier series whose nonzero frequencies  $D^{-1}r = n+a$  run exactly over the coset  $a + \mathbb{Z}^g$ , that is, over the single coset of  $\mathbb{Z}^g/D\mathbb{Z}^g$  corresponding to  $a \in \mathfrak{a}(L)$ . Thus  $\theta_a$  is a nonzero element of  $\mathcal{T}$  whose Fourier support meets only the coset labelled by  $a$ . Since distinct  $a \in \mathfrak{a}(L)$  give disjoint Fourier supports, the

$\theta_a$  are linearly independent: a vanishing combination  $\sum_a \lambda_a \theta_a = 0$  has, on each coset, only the term  $\lambda_a \theta_a$  contributing, and each  $\theta_a$  has a nonzero coefficient there (the  $n = 0$  term  $\exp(\pi i a^t \tau a)$  is nonzero), forcing every  $\lambda_a = 0$ .

*Step 5: Independence plus the dimension count gives a basis.* By Step 3,  $\dim_{\mathbb{C}} \mathcal{T} = d_1 \cdots d_g = |\mathfrak{a}(L)|$ ; by Step 4 the  $|\mathfrak{a}(L)|$  functions  $\theta_a$  are linearly independent elements of  $\mathcal{T}$ . A linearly independent set whose size equals the dimension of the space spans it, so  $\{\theta_a\}_{a \in \mathfrak{a}(L)}$  is a basis of  $\mathcal{T} = H^0(X, L)$ . The equality  $\dim_{\mathbb{C}} H^0(X, L) = d_1 \cdots d_g$  recovers proposition 2.4.3 and theorem 3.4.2, now with the basis displayed explicitly, completing the proof.

The argument does more than reprove the dimension formula: it pins down which functions the abstract count was counting. Every section of  $L$  is a superposition  $\sum_a \lambda_a \theta_a$ , and the recursion (4.12) shows that a section is nothing but a choice of one Fourier coefficient per coset of  $\mathbb{Z}^g / D\mathbb{Z}^g$ , with all remaining coefficients propagated by the quasi-periodicity. The theta functions with characteristics are the sections that concentrate on a single coset, and the transparency of the answer, one basis vector per element of  $\mathfrak{a}(L)$ , is what makes theta functions the computational engine of the whole theory. For a principal polarization  $D = I_g$ , the group  $\mathfrak{a}(L)$  is trivial and the basis is a single function, recovering  $\dim H^0(X, L) = 1$  from proposition 3.4.5: there is one theta function, unique up to scale, and its zero locus is the theta divisor.

The Fourier computation is self-contained, but the structural reason the count comes out to  $d_1 \cdots d_g$  rather than to  $(d_1 \cdots d_g)^2 = |K(L)|$  is the theta group. The theta group  $\mathcal{G}(L)$  of theorem 4.1.8 acts on  $H^0(X, L)$  as its unique irreducible representation with the central  $\mathbb{C}^*$  acting by scalars, and that representation has dimension  $\sqrt{|K(L)|} = d_1 \cdots d_g$ . The characteristics  $\mathfrak{a}(L)$  are a maximal abelian subgroup of  $K(L)$ , lifting to a maximal abelian subgroup of  $\mathcal{G}(L)$ , and the  $\theta_a$  are precisely the simultaneous eigenvectors for its action.

**4.3.3 Remark: The Theta Basis as a Weight Basis** The representation-theoretic reading of theorem 4.3.2 is that  $\{\theta_a\}$  is a weight basis. Under the commutator pairing  $e^L$  of proposition 4.1.7, the finite group  $K(L)$  splits as a product  $K_1 \times K_2$  of two maximal isotropic subgroups, dual to one another under  $e^L$ ; here  $K_1$  is the image of the characteristics  $\mathfrak{a}(L)$  and  $K_2$  its complement. The subgroup  $K_1$  lifts to a maximal abelian subgroup  $A \subset \mathcal{G}(L)$ , and on the irreducible representation  $H^0(X, L)$  the group  $A$  acts by a direct sum of  $|K_1| = d_1 \cdots d_g$  distinct characters, one for each element of the dual group  $K_2 \cong \widehat{K_1}$ . The  $\theta_a$  are exactly the corresponding character eigenlines:  $\theta_a$  is the eigenvector on which  $A$  acts through the character indexed by  $a$ . This is why the count is  $\sqrt{|K(L)|}$  and not  $|K(L)|$ : Stone-von Neumann forces the irreducible to have dimension equal to the square root of the order of the finite group, and the Fourier computation above is that theorem made explicit for the Heisenberg group of a polarized torus.

The two proofs are complementary. The Fourier argument is elementary and produces the basis by hand; the Stone-von Neumann argument explains, without computation, why a basis of that size had to exist and why the natural indexing set is a group. Both will be used downstream: the explicit functions when computing embeddings and theta relations, the group-theoretic picture when the theta group acts on families and its action must be tracked functorially.

With a basis fixed, the sections of  $L$  become homogeneous coordinates. Choosing an ordering

$a_0, a_1, \dots, a_N$  of  $\mathfrak{a}(L)$  with  $N = d_1 \cdots d_g - 1$ , the tuple  $(\theta_{a_0}, \dots, \theta_{a_N})$  sends a point of  $X$  to a point of projective space, provided the sections do not all vanish there. This is the theta map, the concrete device that turns the abstract line bundle  $L$  into a map to  $\mathbb{P}^N$ .

**Proposition 4.3.4: The Theta Map**

Let  $L = L(H, \chi)$  be positive of type  $(d_1, \dots, d_g)$ , with basis  $\{\theta_a\}_{a \in \mathfrak{a}(L)}$  of  $H^0(X, L)$  from theorem 4.3.2, and fix an ordering  $a_0, \dots, a_N$  of  $\mathfrak{a}(L)$ ,  $N = d_1 \cdots d_g - 1$ . On the open set

$$U := \{x \in X \mid \theta_{a_j}(x) \neq 0 \text{ for some } j\}$$

where the sections have no common zero, the assignment

$$\varphi_{|L|}: U \rightarrow \mathbb{P}^N, \quad x \mapsto [\theta_{a_0}(x) : \theta_{a_1}(x) : \cdots : \theta_{a_N}(x)]$$

is a well-defined holomorphic map, independent of the choice of basis up to the induced projective linear automorphism of  $\mathbb{P}^N$ .

**Proof:**

The functions  $\theta_a$  are not functions on  $X$  but sections of  $L$ : each satisfies the quasi-periodicity (4.11), so under  $z \mapsto z + \lambda$  every  $\theta_a$  is multiplied by the *same* automorphy factor  $a_{(H, \chi)}(\lambda, z)$ , which is independent of  $a$ . Consequently the ratio  $\theta_{a_j}(z + \lambda)/\theta_{a_k}(z + \lambda) = \theta_{a_j}(z)/\theta_{a_k}(z)$  is  $\Lambda$ -invariant, so the homogeneous tuple  $[\theta_{a_0}(z) : \cdots : \theta_{a_N}(z)]$  depends only on the class of  $z$  in  $X$ . This is the precise sense in which a common automorphy factor produces a well-defined point of projective space: rescaling all coordinates by a common nonzero factor is the identity in  $\mathbb{P}^N$ .

On  $U$  at least one coordinate is nonzero, so the tuple defines an actual point of  $\mathbb{P}^N$ , and in a neighborhood where  $\theta_{a_k} \neq 0$  the map is given in the affine chart by the holomorphic functions  $\theta_{a_j}/\theta_{a_k}$ ; hence  $\varphi_{|L|}$  is holomorphic on  $U$ . Finally, a different basis  $\{\theta'_b\}$  of the same space  $H^0(X, L)$  is related to  $\{\theta_a\}$  by an invertible matrix  $M \in \mathrm{GL}_{N+1}(\mathbb{C})$ , and replacing the coordinates by  $M$  applied to them changes  $\varphi_{|L|}$  by the projective linear automorphism of  $\mathbb{P}^N$  induced by  $M$ , as we sought to show.

The map  $\varphi_{|L|}$  is the object the next section studies in depth. Two features are already visible. First, whether  $\varphi_{|L|}$  extends to all of  $X$ , and whether it is injective and an immersion, depends on the base locus and on separation properties of the sections, which is why  $L$  itself need not embed but a suitable power does. Second, the target dimension  $N = d_1 \cdots d_g - 1$  is dictated entirely by the type: a principal polarization gives  $N = 0$  and a constant map, a type  $(1, \dots, 1, 2)$  polarization gives a map to  $\mathbb{P}^1$ , and only once the type is large enough do the sections separate points. The systematic way to guarantee enough sections is to pass to a power  $L^{\otimes n}$ , whose type is  $(nd_1, \dots, nd_g)$  and whose section space has dimension  $n^g d_1 \cdots d_g$ ; this is the mechanism behind the Lefschetz embedding theorem of section 4.4.

The collection of hypersurfaces cut out by the sections of  $L$ , as the section ranges over  $H^0(X, L)$ , is a classical object worth naming: the complete linear system of  $L$ . It is the intrinsic, basis-free version of the theta map, and it is the language in which the classification of maps to projective space is usually phrased.

**Definition 4.3.5: The Complete Linear System of a Line Bundle**

Let  $L$  be a line bundle on  $X$  with  $H^0(X, L) \neq 0$ . The *complete linear system* of  $L$  is the projective space

$$|L| := \mathbb{P}(H^0(X, L)) = (H^0(X, L) \setminus \{0\})/\mathbb{C}^*$$

of nonzero sections up to scalar. A point of  $|L|$  is the divisor  $\text{div}(s)$  of a nonzero section  $s$ , so  $|L|$  is identified with the set of effective divisors  $D \geq 0$  on  $X$  with  $\mathcal{O}_X(D) \cong L$ . The *base locus* of  $|L|$  is the common zero set  $\text{Bl}(L) := \bigcap_{s \in H^0(X, L)} \text{div}(s)$ ; when  $L$  is positive of type  $(d_1, \dots, d_g)$ , the theta functions  $\{\theta_a\}_{a \in \mathfrak{a}(L)}$  of theorem 4.3.2 furnish a set of projective coordinates on  $|L| \cong \mathbb{P}^{d_1 \cdots d_g - 1}$ .

The complete linear system reframes the theta map coordinate-free. A choice of basis of  $H^0(X, L)$  identifies  $|L|$  with  $\mathbb{P}^N$  and, dually, sends a point  $x \in X$  off the base locus to the hyperplane of sections vanishing at  $x$ ; this is the map  $\varphi_{|L|}$  of proposition 4.3.4 read through the isomorphism  $|L| \cong \mathbb{P}^N$ . Points of the base locus are exactly the points where  $\varphi_{|L|}$  is undefined, so  $\varphi_{|L|}$  is a morphism on all of  $X$  precisely when  $|L|$  is base-point-free. For a positive line bundle of type  $(d_1, \dots, d_g)$  the dimension of  $|L|$  is  $d_1 \cdots d_g - 1$ , and the effective divisors it parametrizes are the translates and combinations of the theta divisor; the principal case,  $\dim |L| = 0$ , is the statement that the theta divisor is rigid, a single effective divisor with no linear freedom.

The genus-two case is the smallest one where the type varies nontrivially. In dimension one the type is a single integer  $d_1 = \deg L$  and the section count is that integer; in dimension two the type is a pair  $(d_1, d_2)$  with  $d_1 | d_2$ , and the section count  $d_1 d_2$  already exhibits the full phenomenon that distinct polarizations of the same underlying torus give projective spaces of different dimensions.

**Example 4.3.6: Theta Bases in Genus Two**

Let  $X = V/\Lambda$  be a two-dimensional complex torus,  $g = 2$ , carrying a positive line bundle  $L = L(H, \chi)$  with  $E = \text{Im} H$  of type  $(d_1, d_2)$ ,  $d_1 | d_2$ . theorem 4.3.2 gives

$$\dim_{\mathbb{C}} H^0(X, L) = d_1 d_2, \quad \varphi_{|L|}: X \dashrightarrow \mathbb{P}^{d_1 d_2 - 1}$$

and the three smallest types illustrate the range of behavior.

1. *Principal, type (1, 1).* Here  $\mathfrak{a}(L) = (\frac{1}{1}\mathbb{Z} \oplus \frac{1}{1}\mathbb{Z})/\mathbb{Z}^2$  is trivial, so  $\dim H^0(X, L) = 1$ : there is a single theta function  $\theta = \theta \begin{bmatrix} 0 \\ 0 \end{bmatrix} (z, \tau)$ , unique up to scale, and  $\varphi_{|L|}$  maps  $X$  to  $\mathbb{P}^0$ , a point. The linear system  $|L|$  is a single point, and its unique effective divisor is the theta divisor  $\Theta = \text{div}(\theta)$ , a curve inside  $X$ . This is the case relevant to Jacobians of genus-two curves: the Jacobian is principally polarized, and when  $X$  is such a Jacobian  $\Theta$  is a smooth genus-two curve, the image of the curve under Abel-Jacobi.
2. *Type (1, 2).* Now  $\mathfrak{a}(L) = (\mathbb{Z} \oplus \frac{1}{2}\mathbb{Z})/\mathbb{Z}^2 \cong \mathbb{Z}/2$ , of order 2, so  $\dim H^0(X, L) = 2$ . A basis is  $\{\theta_{(0,0)}, \theta_{(0,1/2)}\}$ , the two theta functions with characteristics  $a = (0, 0)$  and  $a = (0, \frac{1}{2})$ , and  $\varphi_{|L|}: X \dashrightarrow \mathbb{P}^1$  is a map to the projective line. The two sections have a common zero locus in general, so  $\varphi_{|L|}$  is only a rational map, undefined on the base locus.
3. *Type (2, 2).* Here  $\mathfrak{a}(L) = (\frac{1}{2}\mathbb{Z} \oplus \frac{1}{2}\mathbb{Z})/\mathbb{Z}^2 \cong (\mathbb{Z}/2)^2$ , of order 4, so  $\dim H^0(X, L) = 4$  with basis

$$\{ \theta_{(0,0)}, \theta_{(1/2,0)}, \theta_{(0,1/2)}, \theta_{(1/2,1/2)} \}$$

the four theta functions with half-integer characteristics, and  $\varphi_{|L|}: X \dashrightarrow \mathbb{P}^3$ . This type is realized by  $L_0^{\otimes 2}$  for a principal  $L_0$ , and the four sections are the classical second-order theta functions; the resulting map factors through  $X/\{\pm 1\}$  and realizes the Kummer surface inside  $\mathbb{P}^3$  as a quartic.

The pattern is uniform: the characteristic group  $\mathfrak{a}(L) \cong \mathbb{Z}/d_1 \oplus \mathbb{Z}/d_2$  indexes the basis, its order  $d_1 d_2$  is the dimension of  $H^0(X, L)$ , and the target of the theta map is  $\mathbb{P}^{d_1 d_2 - 1}$ . Passing from a principal polarization to its square multiplies each  $d_i$  by 2 and takes the section count from 1 to 4, which is exactly the jump from a point to  $\mathbb{P}^3$  that makes the theta map informative.

The genus-two computation makes the abstract dimension of theorem 3.4.2 tangible: the number  $\text{Pf}(E) = d_1 \cdots d_g$  is not a bookkeeping figure but the count of coordinate functions available for mapping  $X$  to projective space, and the theta functions with characteristics are those coordinates written out. The principal case gives too few sections to embed, which is why a Jacobian is not cut out by its own theta divisor but by a power of it; the type  $(2, 2)$  case gives just enough sections to separate the generic point from its negative, which is the Kummer construction. The general principle, that enough sections force an embedding, is the content of the Lefschetz theorem, and theorem 4.3.2 supplies the explicit sections on which its proof runs.

### Exercise 4.3.1

1. Let  $X = V/\Lambda$  carry a positive line bundle  $L$  of type  $(2, 4)$ . Using definition 4.3.1, identify the characteristic group  $\mathfrak{a}(L)$  as an abstract abelian group, list a set of representatives for it inside  $\bigoplus_i \frac{1}{d_i} \mathbb{Z}$ , and state  $\dim_{\mathbb{C}} H^0(X, L)$  and the target  $\mathbb{P}^N$  of the theta map. Which of the elementary divisors  $d_1, d_2$  contributes each cyclic factor?
2. Fix  $g = 1$ , so  $\Lambda = \tau\mathbb{Z} + d\mathbb{Z}$  with  $\tau \in \mathfrak{H}_1$  and  $D = (d)$ , and let  $L$  have type  $(d)$ . Write out the recursion (4.12) explicitly in this case, and verify directly that the Fourier coefficients  $c_r$  of a section are determined by their residues  $r \bmod d$ . Confirm that the number of free coefficients is  $d$ , matching theorem 4.3.2.
3. In the setting of theorem 4.3.2, verify the claim used in Step 4 of its proof that the Fourier supports of distinct  $\theta_a$  are disjoint. That is, show that the set of frequency indices  $r \in \mathbb{Z}^g$  appearing with a possibly nonzero coefficient in  $\theta_a$  is exactly the coset of  $\mathbb{Z}^g/D\mathbb{Z}^g$  corresponding to  $a \in \mathfrak{a}(L)$ , and that different  $a$  give different cosets.
4. Let  $L_0$  be a principal polarization, so  $L_0$  has type  $(1, \dots, 1)$ , and consider  $L = L_0^{\otimes n}$ , whose type is  $(n, \dots, n)$ . Using theorem 4.3.2 and definition 4.3.1, compute  $\dim_{\mathbb{C}} H^0(X, L)$  and  $\mathfrak{a}(L)$  as an abstract group. Compare with the type  $(2, 2)$  Kummer case of example 4.3.6 when  $g = 2$  and  $n = 2$ .
5. Prove that the theta map  $\varphi_{|L|}$  of proposition 4.3.4 is invariant under a common rescaling of the sections, and deduce that the map  $|L| \cong \mathbb{P}^N$  of definition 4.3.5 does not depend on the choice of theta basis  $\{\theta_a\}_{a \in \mathfrak{a}(L)}$  beyond a projective linear automorphism of  $\mathbb{P}^N$ . Use only the fact that every  $\theta_a$  satisfies the same quasi-periodicity (4.11).
6. Prove that for a positive line bundle  $L$  of type  $(d_1, \dots, d_g)$ , the theta map  $\varphi_{|L|}$  of proposition 4.3.4 is a constant map to a point if and only if  $L$  is principally polarized, that is,  $d_1 = \dots = d_g = 1$ . Interpret the base locus  $\text{Bl}(L)$  of definition 4.3.5 in the principal case.

## 4.4 Projective Embeddings: The Lefschetz Theorem

Every thread of Part I has been converging on a single promise. The Riemann relations (theorem 3.1.4) identified the complex tori that ought to be projective as exactly the polarizable ones; the definition of an abelian variety (definition 3.2.1) took the existence of a polarization as the working hypothesis; and theorem 3.2.6 then asserted the equivalence “polarizable if and only if projective” with one implication, the passage from a polarization to an actual embedding, left as a debt to be paid in this chapter. The theta functions built in the previous three sections are the currency. This section pays the debt: given a polarization  $L$ , the sections of  $L^{\otimes 3}$  are numerous enough, and separate points and tangent directions well enough, that the map they define embeds  $X$  as a closed submanifold of projective space. The result is due to Lefschetz, and it is the analytic keystone of the entire theory, since it turns the linear-algebraic condition of positivity into a projective variety.

The strategy is classical and rests on a single geometric device: the translates of the theta divisor. A polarization  $L$  has, by Riemann-Roch (theorem 3.4.2), a computable and positive number of global sections, and each nonzero section vanishes on a divisor. Translating that divisor around the torus produces a family of divisors that, taken together, cover  $X$  and separate its points, because two distinct points of an abelian variety can always be pried apart by a translate of a positive divisor. Passing from  $L$  to  $L^{\otimes 2}$  turns “covered” into “base-point-free”, and passing to  $L^{\otimes 3}$  turns “separated” into “embedded”. The exponents 2 and 3 are not incidental: they are the minimal powers for which the translation argument produces, respectively, a section avoiding a given point and a pair of sections separating any two points together with any tangent direction.

Throughout,  $L = L(H, \chi)$  is a polarization of  $X = V/\Lambda$ , so  $H > 0$  and  $E = \text{Im}H$  is integral on  $\Lambda$ ; write  $(d_1, \dots, d_g)$  for its type (definition 3.2.3) and  $d := d_1 \cdots d_g = \text{Pf}(E)$  for the product, which by theorem 3.4.2 equals  $\dim_{\mathbb{C}} H^0(X, L)$ . For a point  $a \in X$ , let  $t_a: X \rightarrow X$  denote translation by  $a$ , and recall that  $t_a^*L$  is again a line bundle with the same first Chern class  $E$  as  $L$ , so that  $t_a^*L$  and  $L$  define the same polarization and differ only by an element of  $\text{Pic}^0(X)$ . A section  $s \in H^0(X, L)$  pulled back along  $t_a$  is a section of  $t_a^*L$ ; its divisor is the translate  $D_s - a$  of the divisor  $D_s$  of  $s$ . This interplay between translation and the theta divisor is the mechanism the whole argument exploits.

The first step is to remove base points. A *base point* of a line bundle  $M$  is a point at which every global section vanishes; the bundle is *base-point-free* when it has none, equivalently when the sections define an everywhere-regular map  $\varphi_{|M|}: X \rightarrow \mathbb{P}(H^0(X, M)^*)$ . A single polarization  $L$  need not be base-point-free, for a principal polarization has  $\dim H^0(X, L) = 1$  (proposition 3.4.5) and its one section vanishes on the theta divisor, which is then a common (indeed the only possible) base locus. The remedy is to square. The sections of  $L^{\otimes 2}$  include all products  $s_a \cdot s_b$  of translated sections, and these products have enough freedom to avoid any prescribed point.

### Proposition 4.4.1: $L^{\otimes 2}$ Is Base-Point-Free

Let  $L = L(H, \chi)$  be a polarization of the abelian variety  $X = V/\Lambda$ . Then the line bundle  $L^{\otimes 2}$  is base-point-free: for every point  $x_0 \in X$  there is a global section  $s \in H^0(X, L^{\otimes 2})$  with  $s(x_0) \neq 0$ . Equivalently, the theta functions of  $L^{\otimes 2}$  have no common zero.

#### **Proof:**

*Step 1: Translated sections of  $L$  multiply into sections of  $L^{\otimes 2}$ .* Fix a nonzero section  $s_0 \in H^0(X, L)$ ; such a section exists because  $\dim H^0(X, L) = d \geq 1$  by theorem 3.4.2, the type having

all  $d_i \geq 1$ . Let  $D_0 = \text{div}(s_0)$  be its divisor, an effective divisor on  $X$  with  $\mathcal{O}_X(D_0) \cong L$ . For  $a \in X$ , represented by  $\alpha \in V$ , the translate  $t_a^*L$  is computed from the Appell-Humbert data (theorem 2.2.8, definition 2.2.5): translating the canonical factor of automorphy by  $\alpha$  multiplies the semicharacter  $\chi$  by the character  $\lambda \mapsto e^{2\pi i E(\alpha, \lambda)}$ , so

$$t_a^*L \cong L(H, \chi_\alpha), \quad \chi_\alpha(\lambda) = \chi(\lambda) e^{2\pi i E(\alpha, \lambda)}$$

with the same Hermitian form  $H$ , hence the same Chern class  $E$ ; in particular  $t_a^*L$  is again a polarization of the same type as  $L$ , with  $\dim H^0(X, t_a^*L) = d$ . Applying this with  $\alpha$  and with  $-\alpha$  and multiplying the data (proposition 2.2.10),

$$t_a^*L \otimes t_{-a}^*L \cong L(2H, \chi_\alpha \chi_{-\alpha}) = L(2H, \chi^2) = L^{\otimes 2}$$

where the middle equality holds because  $\chi_\alpha(\lambda) \chi_{-\alpha}(\lambda) = \chi(\lambda)^2 e^{2\pi i E(\alpha, \lambda)} e^{-2\pi i E(\alpha, \lambda)} = \chi(\lambda)^2$ , the two exponential factors cancelling. Taking the specific pulled-back sections  $s' := t_a^*s_0 \in H^0(X, t_a^*L)$  and  $s'' := t_{-a}^*s_0 \in H^0(X, t_{-a}^*L)$ , both nonzero, their product  $s' \cdot s''$  is a nonzero section of  $L^{\otimes 2}$  whose divisor is  $(D_0 - a) + (D_0 + a)$ , the sum of two translates of  $D_0$ . (A generic nonzero section of  $t_a^*L$  need not have divisor  $D_0 - a$  when  $d > 1$ ; the pullbacks  $t_a^*s_0, t_{-a}^*s_0$  are chosen precisely to realize these two translates.)

*Step 2: Choose the translates to miss a prescribed point.* Fix  $x_0 \in X$ ; the goal is a section of  $L^{\otimes 2}$  nonvanishing at  $x_0$ , equivalently a pair of translates of  $D_0$  neither of which passes through  $x_0$ . A translate  $D_0 + a$  contains  $x_0$  if and only if  $x_0 \in \text{supp}(D_0 + a)$ , that is, if and only if  $a \in x_0 - \text{supp}(D_0)$ . Since  $D_0$  is an effective divisor on the  $g$ -dimensional torus  $X$ , its support is a hypersurface, an analytic subset of complex codimension one, so  $x_0 - \text{supp}(D_0)$  is a proper analytic subset of  $X$ ; its complement is nonempty and dense. Pick any  $a$  in that complement. Then  $x_0 \notin D_0 + a$ , and also  $x_0 \notin D_0 - a$  unless  $a$  lies in the reflected set  $\text{supp}(D_0) - x_0$ , likewise a proper analytic subset. Because  $X$  is not the union of two proper analytic subsets, some  $a$  avoids both, and for that  $a$  the section  $s' \cdot s''$  from Step 1 vanishes at neither of the two components of its divisor at  $x_0$ , hence

$$(s' \cdot s'')(x_0) \neq 0$$

exhibiting a section of  $L^{\otimes 2}$  nonzero at  $x_0$ .

Since  $x_0 \in X$  was arbitrary, no point is a base point of  $L^{\otimes 2}$ , so  $L^{\otimes 2}$  is base-point-free, completing the proof.

The proof isolates the one property of abelian varieties that makes the whole embedding theory work: an effective divisor can be translated, and its translates sweep out the torus. On a general projective variety a positive line bundle carries no such homogeneity, and base-point-freeness of powers is a subtler matter; on an abelian variety the group law supplies exactly the room needed. The section  $s' \cdot s''$  produced above is, in the language of the previous section, a product of two theta functions with shifted arguments, and this is the concrete form the translates take once a Riemann theta function (definition 4.2.1) is chosen: the addition formula for theta functions expresses such products in terms of the theta basis of  $H^0(X, L^{\otimes 2})$  (theorem 4.3.2), so that Step 1 is an identity among explicit holomorphic functions on  $V$ .

With base points gone after squaring, the map  $\varphi_{|L^{\otimes 2}|}$  is everywhere regular, but it need not yet be injective, nor an immersion. To force injectivity and the immersion property one more tensor power



is needed, and the extra room it provides is what separates points and tangent vectors.

The heart of the theorem is the separation argument. A regular map  $\varphi_{|M|}: X \rightarrow \mathbb{P}^N$  defined by a base-point-free bundle  $M$  is a closed embedding precisely when it is injective and its differential is injective at every point, that is, when the sections of  $M$  *separate points* (for distinct  $x \neq y$  there is a section vanishing at  $x$  but not at  $y$ ) and *separate tangent vectors* (for each  $x$  and each nonzero tangent vector  $v \in T_x X$  there is a section vanishing at  $x$  whose differential is nonzero on  $v$ ). The next result records that  $L^{\otimes 3}$ , and more generally every  $L^{\otimes n}$  with  $n \geq 3$ , achieves both.

#### Theorem 4.4.2: Lefschetz Embedding Theorem

Let  $L = L(H, \chi)$  be a polarization of the abelian variety  $X = V/\Lambda$ , of type  $(d_1, \dots, d_g)$  and with  $d = d_1 \cdots d_g$ . For every integer  $n \geq 3$  the line bundle  $L^{\otimes n}$  is very ample: the map

$$\varphi_{|L^{\otimes n}|}: X \longrightarrow \mathbb{P}(H^0(X, L^{\otimes n})^*) = \mathbb{P}^N, \quad N = n^g d - 1$$

associated to a basis of theta functions of  $L^{\otimes n}$  is a closed holomorphic embedding of  $X$  onto a smooth projective subvariety. In particular  $\varphi_{|L^{\otimes 3}|}: X \hookrightarrow \mathbb{P}^N$  with  $N = 3^g d - 1$  is a closed holomorphic embedding.

#### Proof:

The dimension count is immediate:  $L^{\otimes n}$  has Chern class  $nE$ , hence is a polarization of type  $(nd_1, \dots, nd_g)$ , so by theorem 3.4.2 its space of sections has dimension  $\dim H^0(X, L^{\otimes n}) = \text{Pf}(nE) = (nd_1)(nd_2) \cdots (nd_g) = n^g d$ , and the target is  $\mathbb{P}^N$  with  $N = n^g d - 1$ . It remains to show, for  $n \geq 3$ , that  $L^{\otimes n}$  is base-point-free, separates points, and separates tangent vectors. The cases  $n = 3$  and  $n = 4$  are treated directly below; the case  $n \geq 5$  then follows because  $L^{\otimes n} = L^{\otimes 3} \otimes L^{\otimes(n-3)}$  and tensoring a very ample bundle with a base-point-free bundle keeps it very ample. Here  $L^{\otimes(n-3)}$  is base-point-free for  $n \geq 5$ : any power  $L^{\otimes m}$  with  $m \geq 2$  is a tensor product of copies of  $L^{\otimes 2}$  and  $L^{\otimes 3}$ , hence base-point-free by proposition 4.4.1 and Step 1 below, and  $n-3 \geq 2$  precisely when  $n \geq 5$ . This reduction does *not* apply at  $n = 4$ , where  $L^{\otimes(n-3)} = L$  has a base point in the principal case, so  $n = 4$  must be, and is, handled on its own. So fix first  $n = 3$ ; the modifications for  $n = 4$  are indicated at the end of each step.

*Step 1: Base-point-freeness.* The naive factorization  $L^{\otimes 3} = L^{\otimes 2} \otimes L$  does not help, because  $L$  itself may have a base point: in the principal case  $\dim H^0(X, L) = 1$  (proposition 3.4.5) and its single section vanishes on the theta divisor, so no section of  $L$  avoids a point  $x_0$  lying there. The correct device is again to spread the theta divisor by translation, but now using *three* translates so that the  $\text{Pic}^0(X)$ -twists cancel. Fix a nonzero section  $s_0 \in H^0(X, L)$  with effective divisor  $D_0$ . For  $a, b, c \in X$  with  $a + b + c = 0$  the Appell-Humbert computation of Step 1 of proposition 4.4.1, applied three times, gives

$$t_a^* L \otimes t_b^* L \otimes t_c^* L \cong L^{\otimes 3}$$

with the three character twists  $e^{2\pi i E(\alpha, \cdot)}$ ,  $e^{2\pi i E(\beta, \cdot)}$ ,  $e^{2\pi i E(\gamma, \cdot)}$  multiplying to  $e^{2\pi i E(\alpha + \beta + \gamma, \cdot)} = 1$  on  $\Lambda$  since  $\alpha + \beta + \gamma \in \Lambda$  and  $E$  is integral there (the same identity is spelled out in Step 2 below). Following the convention fixed above, the divisor of  $t_a^* s_0$  is the translate  $D_0 - a$ , so the product of the three pulled-back sections  $t_a^* s_0 \cdot t_b^* s_0 \cdot t_c^* s_0$  is a section of  $L^{\otimes 3}$  with divisor  $(D_0 - a) + (D_0 - b) + (D_0 - c)$ . Given  $x_0$ , choose  $a$  in the complement of the proper analytic subset  $\text{supp}(D_0) - x_0$ , then  $b$  likewise avoiding  $\text{supp}(D_0) - x_0$ , and set  $c = -a - b$ , adjusting



$b$  within its open dense set of admissible values so that  $c$  too avoids  $\text{supp}(D_0) - x_0$ ; this is possible because each constraint excludes only a proper analytic subset. Then none of the three translates passes through  $x_0$ , so the section is nonzero at  $x_0$ . Thus  $L^{\otimes 3}$  is base-point-free and  $\varphi|_{L^{\otimes 3}}$  is a well-defined holomorphic map on all of  $X$ . For  $n = 4$  the same reasoning applies with  $L^{\otimes 4} = L^{\otimes 2} \otimes L^{\otimes 2}$ : a product of two sections of  $L^{\otimes 2}$  each nonvanishing at  $x_0$  (proposition 4.4.1) is a section of  $L^{\otimes 4}$  nonzero at  $x_0$ , so  $L^{\otimes 4}$  is base-point-free as well.

*Step 2: Separation of points.* Let  $x \neq y$  be distinct points of  $X$ ; the goal is a section of  $L^{\otimes 3}$  vanishing at  $x$  but not at  $y$ . As in proposition 4.4.1, fix a nonzero section  $s_0 \in H^0(X, L)$  with effective divisor  $D_0$ , and consider divisors of the form

$$D_a := (D_0 - a) + (D_0 - b) + (D_0 - c)$$

with  $a + b + c = 0$  in  $X$ , which is the divisor of a section of  $L^{\otimes 3}$ : indeed the three-fold analogue of the identity used above,

$$t_a^* L \otimes t_b^* L \otimes t_c^* L \cong L^{\otimes 3} \quad \text{whenever } a + b + c = 0,$$

holds by the same Appell-Humbert computation as in Step 1 of proposition 4.4.1. Representing  $a, b, c$  by  $\alpha, \beta, \gamma \in V$  with  $\mu := \alpha + \beta + \gamma \in \Lambda$ , the left side is  $L(3H, \chi_\alpha \chi_\beta \chi_\gamma)$ , and  $\chi_\alpha \chi_\beta \chi_\gamma(\lambda) = \chi(\lambda)^3 e^{2\pi i E(\mu, \lambda)} = \chi(\lambda)^3$ , since  $E(\mu, \lambda) \in \mathbb{Z}$  by integrality of  $E$  on  $\Lambda$ ; thus the product is  $L(3H, \chi^3) = L^{\otimes 3}$ . A product of nonzero sections of the three translated bundles is then a section  $s_{a,b,c}$  of  $L^{\otimes 3}$  with  $\text{div}(s_{a,b,c}) = D_a$ . Now  $x$  lies on  $D_a$  if and only if  $x$  lies on one of the three translates, that is, if and only if one of  $a, b, c$  lies in  $\text{supp}(D_0) - x$ ; and  $y$  avoids  $D_a$  if and only if none of  $a, b, c$  lies in  $\text{supp}(D_0) - y$ . The task is to choose  $a \in \text{supp}(D_0) - x$  with  $a \notin \text{supp}(D_0) - y$ , then  $b$  and  $c = -a - b$  generic avoiding  $\text{supp}(D_0) - y$ ; the resulting section vanishes at  $x$  but not at  $y$ . Both are proper analytic subsets of  $X$ , so once a valid  $a$  is fixed the genericity of  $b$  (hence of  $c = -a - b$ ) is again the codimension-one argument of Step 1. The one point that a single divisor  $D_0$  does not settle is whether the hypersurface  $\text{supp}(D_0) - x$  is contained in  $\text{supp}(D_0) - y$ : this fails exactly when  $\text{supp}(D_0)$  is invariant under the translation  $x - y$ , and a positive divisor can carry a nonzero finite (torsion) translation stabilizer, so for such a pair no translate of  $D_0$  separates  $x$  from  $y$ . That the sections of  $L^{\otimes 3}$  nonetheless separate every pair  $x \neq y$  is the transversality input supplied by the theta-group argument deferred to Mumford below, which produces enough sections beyond the products of translates of a single  $D_0$  to cover these exceptional pairs. For  $n = 4$  the same reduction runs with four translates:  $D_a := (D_0 - a) + (D_0 - b) + (D_0 - c) + (D_0 - e)$  with  $a + b + c + e = 0$  is the divisor of a section of  $L^{\otimes 4}$  by the same Appell-Humbert computation, and the choice of  $a \in \text{supp}(D_0) - x$  with  $a \notin \text{supp}(D_0) - y$  and the remaining  $b, c$  generic with  $e = -a - b - c$  all avoiding  $\text{supp}(D_0) - y$  is governed by the same deferred input.

*Step 3: Separation of tangent vectors.* Fix  $x \in X$  and a nonzero tangent vector  $v \in T_x X$ ; the goal is a section  $s$  of  $L^{\otimes 3}$  with  $s(x) = 0$  but  $ds_x(v) \neq 0$ , so that  $\varphi|_{L^{\otimes 3}}$  is an immersion at  $x$ . Work again with  $D_a = (D_0 - a) + (D_0 - b) + (D_0 - c)$ ,  $a + b + c = 0$ . Choose  $a$  so that  $D_0 - a$  passes through  $x$  as a *smooth* point of that single translate and with tangent hyperplane not annihilating  $v$ : concretely,  $x \in \text{supp}(D_0 - a)$  means  $a \in \text{supp}(D_0) - x$ , and as  $a$  ranges over this hypersurface the point of  $D_0$  through which the translate meets  $x$  ranges over all of  $\text{supp}(D_0)$ , in particular over its smooth locus, where the tangent hyperplane to  $D_0 - a$  at  $x$  is a translate of the tangent hyperplane to  $D_0$  and varies. Here  $D_0$  is a hypersurface and not a union of translation-invariant subtori, since a positive divisor cannot be invariant under a positive-dimensional subgroup, as

its self-intersection would then vanish, whereas  $(D_0^g) = g!d > 0$ . This last self-intersection is computed on an abelian variety, where  $\mathrm{td}(X) = 1$ , from  $(D_0^g) = \int_X c_1(L)^g$  together with the degree- $g$  term of  $\int_X e^{c_1(L)}$ , namely  $\int_X \frac{1}{g!} c_1(L)^g = \chi(L) = \mathrm{Pf}(E) = d$ ; hence  $\int_X c_1(L)^g = g!d$ , the factor  $\mathrm{Pf}(E) = d$  being the section count of theorem 3.4.2. Consequently the Gauss map of  $D_0$  is nonconstant on its smooth locus, so these tangent hyperplanes are not all equal, and the fact that some such  $a$  places a smooth branch through  $x$  with tangent hyperplane missing the prescribed  $v$  is the transversality input deferred to Mumford below. Fix such an  $a$ , so that  $D_0 - a$  vanishes to order exactly one at  $x$  with  $d(\text{local equation})_x(v) \neq 0$ . Then choose  $b$  and  $c = -a - b$  generic so that neither  $D_0 - b$  nor  $D_0 - c$  passes through  $x$ , which is possible by the codimension-one argument of Step 1. The section  $s_{a,b,c}$  with divisor  $D_a$  then vanishes at  $x$  only along the single smooth branch  $D_0 - a$ , so near  $x$  it is a unit multiple of the local equation of  $D_0 - a$ , whence

$$s_{a,b,c}(x) = 0, \quad d(s_{a,b,c})_x(v) \neq 0$$

separating the tangent vector  $v$ . The requirement  $n \geq 3$  enters exactly here: three translates give one degree of freedom to place a smooth branch through  $x$  tangent to a chosen hyperplane and two more to push the remaining branches off  $x$ , whereas two translates suffice only for point separation and one for neither. For  $n = 4$  one uses four translates  $D_a = (D_0 - a) + (D_0 - b) + (D_0 - c) + (D_0 - e)$  with  $a + b + c + e = 0$ : place the single smooth branch  $D_0 - a$  through  $x$  with tangent hyperplane not annihilating  $v$  exactly as above, and push the three remaining branches  $D_0 - b$ ,  $D_0 - c$ ,  $D_0 - e = D_0 + a + b + c$  off  $x$ , which the extra degree of freedom only makes easier.

By Steps 1 to 3 the sections of  $L^{\otimes 3}$  are base-point-free and separate both points and tangent vectors, so  $\varphi_{|L^{\otimes 3}|}: X \rightarrow \mathbb{P}^N$  is an injective holomorphic immersion. Since  $X$  is compact, an injective immersion is a homeomorphism onto its image and hence a closed holomorphic embedding, and by Chow's theorem ([16]) the image is a smooth projective algebraic variety. The step-by-step modifications establish the same conclusion verbatim for  $n = 4$ , and the reduction at the start extends it from  $n = 3, 4$  to all  $n \geq 5$ , proving very ampleness for every  $n \geq 3$ , as we sought to show.

The separation arguments above are self-contained modulo one standard input used in Steps 2 and 3: that the sections of  $L^{\otimes 3}$  have enough independent behavior to separate the exceptional point pairs left over when a single theta divisor has a torsion translation stabilizer, and to isolate any tangent direction. A fully detailed verification, tracking the theta functions explicitly and confirming that  $\varphi_{|L^{\otimes 3}|}$  is injective with everywhere-injective differential, is carried out in Mumford, *Abelian Varieties*; the argument there is representation-theoretic, using the theta group of section 4.1 to show that the sections of  $L^{\otimes n}$  for  $n \geq 3$  generate enough jets at every point, and it is included there because this transversality claim is the one geometric fact whose full proof is longer than the linear-algebraic core of the theorem. The mechanism recorded above, translating the theta divisor to place branches and tangent hyperplanes at will, is the geometric heart of that argument.

The theorem is the analytic realization of an abelian variety as a projective variety, and it does more than assert projectivity: it names an explicit embedding. The coordinates of  $\varphi_{|L^{\otimes 3}|}$  are the  $3^g d$  theta functions of theorem 4.3.2 for the bundle  $L^{\otimes 3}$ , so the embedded torus is cut out by explicit theta relations, the quadratic and higher identities among these functions. For a principal polarization,  $d = 1$  and the embedding lands in  $\mathbb{P}^{3^g - 1}$ ; for  $g = 1$  this is  $\mathbb{P}^2$ , recovering the plane cubic, as the example below makes precise. The bound  $n \geq 3$  is sharp in general:  $L$  itself and  $L^{\otimes 2}$  can fail to be very ample, the former having only a single section, hence a base point, when  $d = 1$  and the latter failing to separate certain points on some abelian varieties, so cubing is the universal guarantee.

The theorem discharges the outstanding debt of section 3.2 in one stroke.

**Corollary 4.4.3: Every Polarizable Complex Torus Is Projective**

Let  $X = V/\Lambda$  be a complex torus admitting a polarization, that is, an abelian variety in the sense of definition 3.2.1. Then  $X$  admits a closed holomorphic embedding into projective space, and its image is a smooth projective algebraic variety. This supplies the missing implication of theorem 3.2.6, whose equivalence is thereby complete: a complex torus is polarizable if and only if it is projective.

**Proof:**

Let  $H > 0$  be a polarization of  $X$  with  $E = \text{Im}H$  integral on  $\Lambda$ , and let  $L = L(H, \chi)$  be an associated positive line bundle (definition 3.1.1), which exists by the Appell-Humbert theorem for any semicharacter  $\chi$  of  $E$ . By theorem 4.4.2 the bundle  $L^{\otimes 3}$  is very ample, so  $\varphi_{|L^{\otimes 3}|} : X \hookrightarrow \mathbb{P}^N$  is a closed holomorphic embedding onto a smooth projective variety. This establishes the forward implication “polarizable  $\Rightarrow$  projective” of theorem 3.2.6, whose converse was proved there. Polarizability and projectivity are therefore equivalent, completing the proof.

This corollary is the payoff of Part I. Chapter 3 proposed the polarization as the algebraicity criterion and proved everything except that a polarization forces projectivity; that gap, flagged in definition 3.2.1 and left open in theorem 3.2.6, is now closed. The word “abelian variety” has, over  $\mathbb{C}$ , a fully justified meaning: a polarizable complex torus is a projective algebraic group, and the passage from the transcendental quotient  $\mathbb{C}^g/\Lambda$  to a system of homogeneous polynomial equations in  $\mathbb{P}^N$  is effected concretely by theta functions. Nothing in Part I now rests on an unproved forward reference; the analytic theory is complete, and Part II will recover the same class of objects over an arbitrary field by algebraic means.

The embedding furnished by the theorem is a construction worth naming, both because it is canonical up to the choice of a basis of sections and because it is the object that later chapters manipulate.

**Definition 4.4.4: Theta Embedding**

Let  $L = L(H, \chi)$  be a polarization of the abelian variety  $X$  of type  $(d_1, \dots, d_g)$ , and let  $\theta_0, \dots, \theta_N$  with  $N = 3^g d_1 \cdots d_g - 1$  be a basis of the space  $H^0(X, L^{\otimes 3})$  of theta functions of  $L^{\otimes 3}$  (theorem 4.3.2). The *theta embedding* associated to  $L$  is the closed holomorphic embedding

$$\varphi_{|L^{\otimes 3}|} : X \longrightarrow \mathbb{P}^N, \quad x \longmapsto [\theta_0(x) : \cdots : \theta_N(x)]$$

of theorem 4.4.2. Its image is the abelian variety  $X$  presented as a smooth projective subvariety of  $\mathbb{P}^N$ , cut out by the theta relations among  $\theta_0, \dots, \theta_N$ .

The theta embedding depends on the polarization  $L$  and, through the choice of basis, on a coordinate system in  $H^0(X, L^{\otimes 3})$ , but two choices of basis differ by a projective linear change of coordinates, so the embedded variety is well defined up to the action of  $\text{PGL}_{N+1}$ . Different polarizations on the same torus give distinct embeddings, into projective spaces of different dimensions when the types differ, which is why the polarization, and not the bare torus, is the datum that a projective realization remembers. For a principally polarized abelian variety the theta functions with characteristics of theorem 4.3.2 give the coordinates explicitly, and the embedding is the one classically written in terms of the Riemann theta function  $\theta\left[\begin{smallmatrix} a \\ 0 \end{smallmatrix}\right]$  with characteristics  $a \in (\frac{1}{3}\mathbb{Z}^g)/\mathbb{Z}^g$ .

The definition becomes concrete in dimension one, where the theta embedding is an object the reader has already met from an entirely different route.

**Example 4.4.5: The Plane Cubic as a Theta Embedding**

Take  $g = 1$  and  $X = E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ . The principal polarization has  $H(v, w) = (\operatorname{Im}\tau)^{-1}v\bar{w}$  and type (1), so  $d = d_1 = 1$  and the associated line bundle  $L = L(H, \chi)$  has  $\dim H^0(E, L) = \operatorname{Pf}(E) = 1$  by theorem 3.4.2: a single theta function, the classical Jacobi theta  $\theta(z, \tau)$  of theorem 3.1.4 and theorem 4.3.2, spanning the sections of the degree-one bundle  $L$ . This one section vanishes to order one at a single point, so  $L$  itself is far from very ample, and the theorem's insistence on cubing is visible already here:  $\varphi|_L$  is not even defined, having a base point at the zero of  $\theta$ .

Cube the polarization. The bundle  $L^{\otimes 3}$  has Chern class  $3E$ , type (3), and by theorem 3.4.2

$$\dim_{\mathbb{C}} H^0(E, L^{\otimes 3}) = \operatorname{Pf}(3E) = 3$$

so  $L^{\otimes 3}$  has degree three and three independent theta functions. By theorem 4.3.2 a basis is furnished by the three theta functions with characteristics

$$\theta_j(z) = \theta \left[ \begin{smallmatrix} j/3 \\ 0 \end{smallmatrix} \right] (z, \tau) = \sum_{n \in \mathbb{Z}} \exp \left( \pi i \left( n + \frac{j}{3} \right)^2 \tau + 2\pi i \left( n + \frac{j}{3} \right) z \right), \quad j = 0, 1, 2$$

and the theta embedding is

$$\varphi|_{L^{\otimes 3}}: E \longrightarrow \mathbb{P}^2, \quad z \longmapsto [\theta_0(z) : \theta_1(z) : \theta_2(z)]$$

a map of the elliptic curve into  $\mathbb{P}^2 = \mathbb{P}^{3^1 \cdot 1 - 1}$ . By theorem 4.4.2 this is a closed holomorphic embedding, and its image is a smooth plane curve of degree three, since a degree-three line bundle on a genus-one curve embeds it as a cubic: the image meets a generic line in  $\deg L^{\otimes 3} = 3$  points.

This recovers, from the theta viewpoint, the Weierstrass embedding of proposition 1.4.2. There the embedding  $E \hookrightarrow \mathbb{P}^2$  was built from the pair  $(\wp, \wp')$ , realizing  $E$  as the cubic  $Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3$ ; here it is built from the triple  $(\theta_0, \theta_1, \theta_2)$  of theta functions with characteristics. The two embeddings target the same  $\mathbb{P}^2$  and their images are projectively equivalent smooth plane cubics, because both are the complete linear system of a degree-three line bundle on  $E$  and any two such systems differ by a projective linear change of coordinates. Concretely, the Weierstrass coordinates are recovered from the theta functions through the classical identity  $\wp(z) = -\frac{d^2}{dz^2} \log \theta(z, \tau) + c$  for a constant  $c$  (and its  $z$ -derivative for  $\wp'$ ), so the point  $[\wp(z) : \wp'(z) : 1]$  and the point  $[\theta_0(z) : \theta_1(z) : \theta_2(z)]$  are carried to one another by a fixed element of  $\operatorname{PGL}_3(\mathbb{C})$ . The classical fact that an elliptic curve is a plane cubic and the general Lefschetz theorem for abelian varieties are, in dimension one, the same statement seen through two lenses, the Weierstrass functions and the theta functions, and proposition 1.4.2 is the  $g = 1$  instance of theorem 4.4.2. For  $g = 1$  the analytic theory of Part I thus reduces, at its final step, to the theory of [4].

### Exercise 4.4.1

1. Let  $X = V/\Lambda$  be an abelian variety of dimension  $g = 3$  carrying a polarization  $L = L(H, \chi)$  of

type  $(1, 2, 4)$ . Using theorem 4.4.2, compute  $\dim_{\mathbb{C}} H^0(X, L^{\otimes n})$  and the dimension  $N$  of the projective space  $\mathbb{P}^N$  into which  $\varphi_{|L^{\otimes n}|}$  maps for  $n = 2, 3, 4$ . Which of these three bundles does theorem 4.4.2 guarantee to be very ample?

2. In the setting of example 4.4.5, take  $g = 1$  with the principal polarization, so  $d = 1$ .
  1. Using theorem 3.4.2, compute  $\dim_{\mathbb{C}} H^0(E, L^{\otimes n})$  for every  $n \geq 1$ , and state for which  $n$  the map  $\varphi_{|L^{\otimes n}|}$  is a well-defined morphism on all of  $E$ .
  2. Explain, from the section, why  $L$  itself is not base-point-free and why the image of  $\varphi_{|L^{\otimes 3}|}$  meets a generic line of  $\mathbb{P}^2$  in exactly three points.
3. Fix a polarization  $L = L(H, \chi)$  and points  $a, b, c \in X$  with  $a + b + c = 0$ , represented by  $\alpha, \beta, \gamma \in V$  with  $\mu := \alpha + \beta + \gamma \in \Lambda$ . Following the Appell-Humbert computation of proposition 4.4.1, verify from the twist formula

$$t_a^* L \cong L(H, \chi_\alpha), \quad \chi_\alpha(\lambda) = \chi(\lambda) e^{2\pi i E(\alpha, \lambda)}$$

that  $t_a^* L \otimes t_b^* L \otimes t_c^* L \cong L^{\otimes 3}$ , identifying precisely where the integrality of  $E$  on  $\Lambda$  is used. Show by an example that the isomorphism fails, in general, when  $a + b + c \neq 0$ : exhibit for which line bundle in the family  $\{L(3H, \psi)\}$  the triple product is then an isomorphism.

4. Prove that if  $M_1$  and  $M_2$  are base-point-free line bundles on  $X$ , then  $M_1 \otimes M_2$  is base-point-free. Deduce, using proposition 4.4.1, that  $L^{\otimes 2m}$  is base-point-free for every  $m \geq 1$ , and that  $L^{\otimes 4}$  in particular is base-point-free without appeal to the three-translate argument of theorem 4.4.2.
5. Let  $L = L(H, \chi)$  be a polarization of type  $(d_1, \dots, d_g)$  with  $d = d_1 \cdots d_g$ , and let  $D_0$  be the effective divisor of a nonzero section of  $L$ .
  1. Show that the top self-intersection number is  $(D_0^g) = g! d$ , using  $\text{td}(X) = 1$  on an abelian variety together with theorem 3.4.2.
  2. Deduce that  $\text{supp}(D_0)$  is not invariant under any positive-dimensional subtorus of  $X$ ; that is, no positive divisor on  $X$  has a positive-dimensional translation stabilizer.

Identify where this fact is used in the proof of theorem 4.4.2.

6. The proof of theorem 4.4.2 treats  $n = 3$  and  $n = 4$  directly and then reduces every  $n \geq 5$  to these. State the two general facts about line bundles that the reduction rests on, and explain in detail why the reduction gives very ampleness of  $L^{\otimes n}$  for all  $n \geq 5$  but cannot be applied at  $n = 4$ . In particular, say exactly which power  $L^{\otimes(n-3)}$  appears and why  $n - 3 \geq 2$  is the decisive inequality.

## 4.5 The Theta Divisor

The preceding sections turned a polarization into projective geometry: a positive line bundle  $L$  has a space of global sections built from theta functions, and a high tensor power of  $L$  embeds the abelian variety into projective space (theorem 4.3.2, theorem 4.4.2). When the polarization is principal, that space of sections collapses to a single line. A nonzero section then has a zero locus that is canonical up to scaling, and this zero locus, the theta divisor, is the geometric object in which a principally polarized abelian variety records its polarization. It is a divisor rather than a cohomology class, a

subvariety one can point at, and it will turn out to carry enough information to reconstruct a curve from its Jacobian. This section defines the theta divisor, identifies its class and its self-intersection, and pins down the sense in which it is unique.

The starting point is the dimension count from Riemann-Roch. For a principal polarization the space of sections is one-dimensional (proposition 3.4.5), so up to a scalar there is exactly one theta function, and its zero locus is a well-defined effective divisor.

#### Definition 4.5.1: Theta Divisor

Let  $X = V/\Lambda$  be an abelian variety with a principal polarization, and let  $L = L(H, \chi)$  be a line bundle with  $H > 0$  of type  $(1, \dots, 1)$  representing it. By proposition 3.4.5,  $\dim_{\mathbb{C}} H^0(X, L) = 1$ ; let  $\theta$  be a nonzero section, unique up to a nonzero scalar. The *theta divisor* of the polarization is the effective divisor

$$\Theta := \operatorname{div}(\theta)$$

on  $X$ , the zero locus of  $\theta$  counted with multiplicity. Since  $\theta$  is unique up to a scalar and scaling does not change its zero locus,  $\Theta$  is independent of the choice of  $\theta$ .

The definition is legitimate precisely because the polarization is principal. For a polarization of type  $(d_1, \dots, d_g)$  with  $d_1 \cdots d_g > 1$  there is a whole projective space  $\mathbb{P}(H^0(X, L))$  of sections, each with its own zero locus, and no single divisor is singled out. Principality is what forces  $\dim H^0(X, L) = 1$ , and hence what makes “the” theta divisor a meaningful phrase. The section  $\theta$  is a theta function in the sense of definition 2.4.1: a holomorphic  $\theta: V \rightarrow \mathbb{C}$  transforming by the canonical factor of automorphy, so its zero set is a  $\Lambda$ -invariant analytic hypersurface in  $V$  descending to the codimension-one subvariety  $\Theta \subset X$ .

The theta divisor is not a bookkeeping device attached to  $L$ ; it recovers  $L$  completely. The next result says that the line bundle  $\mathcal{O}_X(\Theta)$  associated to the divisor is  $L$  itself, so  $\Theta$  carries the full polarization, and it computes the one numerical invariant a principal polarization has: the self-intersection number  $(\Theta^g) = g!$ .

#### Proposition 4.5.2: The Theta Divisor Defines the Polarization

With notation as in definition 4.5.1, the divisor  $\Theta$  satisfies the following.

1. There is a canonical isomorphism  $\mathcal{O}_X(\Theta) \cong L$ ; in particular  $c_1(\mathcal{O}_X(\Theta)) = c_1(L) = H$ , so  $\mathcal{O}_X(\Theta)$  represents the given principal polarization.
2.  $\mathcal{O}_X(\Theta)$  is ample, and  $\Theta$  is an ample divisor.
3. The self-intersection number is

$$(\Theta^g) = g!$$

#### Proof:

We treat the three assertions in turn.

1. *The line bundle of the divisor.* By construction  $\theta \in H^0(X, L)$  is a global section of  $L$  whose divisor of zeros is  $\Theta = \operatorname{div}(\theta)$ . On any complex manifold, a nonzero global section  $s$  of a holomorphic line bundle  $M$  gives a canonical isomorphism  $\mathcal{O}_X(\operatorname{div}(s)) \cong M$ : locally  $s$  is a holomorphic function whose divisor is  $\operatorname{div}(s)$ , and comparing the transition data of

$M$  with those of the divisor line bundle of  $\text{div}(s)$  shows they agree (this is the standard correspondence between effective divisors with a chosen defining section and line bundles with a section, see [16]). Applying this with  $M = L$  and  $s = \theta$  gives  $\mathcal{O}_X(\Theta) \cong L$ . Taking first Chern classes and invoking  $c_1(L(H, \chi)) = H$  (proposition 2.3.4) yields  $c_1(\mathcal{O}_X(\Theta)) = H$ , so  $\mathcal{O}_X(\Theta)$  represents the principal polarization.

2. *Ampleness.* Since  $H > 0$ , the class  $L$  is a polarization (definition 3.1.3), and by the Lefschetz embedding theorem (theorem 4.4.2) some power  $L^{\otimes n}$  is very ample; hence  $L$  is ample. By part (1),  $\mathcal{O}_X(\Theta) \cong L$  is ample, and an ample line bundle admitting the effective section  $\theta$  has ample associated divisor  $\Theta$ .
3. *The self-intersection number.* The self-intersection  $(\Theta^g)$  is the top power of the first Chern class,

$$(\Theta^g) = \int_X c_1(\mathcal{O}_X(\Theta))^g = \int_X c_1(L)^g$$

Under the identification  $H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee$  of theorem 1.3.2,  $c_1(L) = E = \text{Im}H$  is the alternating form of the polarization (proposition 2.3.4), and the top self-intersection of a line bundle on a  $g$ -dimensional complex torus is computed by the Pfaffian of  $E$  times  $g!$ . Concretely, choose a symplectic basis  $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$  of  $\Lambda$  in which  $E$  has matrix  $\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$  with  $D = \text{diag}(d_1, \dots, d_g)$  (definition 3.2.3), and let  $x_1, \dots, x_g, y_1, \dots, y_g$  be the dual real coordinates on  $V/\Lambda$ . Then

$$E = \sum_{i=1}^g d_i dx_i \wedge dy_i$$

as an element of  $\bigwedge^2 \Lambda^\vee$ , and its  $g$ -th exterior power is

$$E^{\wedge g} = g! (d_1 d_2 \cdots d_g) dx_1 \wedge dy_1 \wedge \cdots \wedge dx_g \wedge dy_g$$

The fundamental class  $[X]$  pairs with the volume form  $dx_1 \wedge dy_1 \wedge \cdots \wedge dx_g \wedge dy_g$  to give 1 (this monomial is a generator of  $H^{2g}(X, \mathbb{Z}) \cong \bigwedge^{2g} \Lambda^\vee \cong \mathbb{Z}$ ), so

$$(\Theta^g) = \int_X E^{\wedge g} = g! (d_1 d_2 \cdots d_g) = g! \text{Pf}(E)$$

For the principal polarization all  $d_i = 1$ , so  $\text{Pf}(E) = 1$  and  $(\Theta^g) = g!$ , as we sought to show.

The three statements fit together as one picture. The class of  $\Theta$  is  $H$ , so no information is lost in passing from the polarization  $L$  to the geometric divisor  $\Theta$ ; the ampleness realizes  $\Theta$  as a projective hypersurface, cut out inside the embedding of theorem 4.4.2; and the number  $g!$  is the degree of  $X$  under the map defined by a suitable power of  $\mathcal{O}_X(\Theta)$ , the sole discrete invariant a principal polarization carries. The count  $(\Theta^g) = g! \cdot \text{Pf}(E)$  for a general type is exactly  $g!$  times the Riemann-Roch dimension  $\text{Pf}(E) = d_1 \cdots d_g$  of theorem 3.4.2, which for the principal case is the statement  $\dim H^0 = 1$  made numerical. That the self-intersection is  $g!$  rather than 1 is the familiar phenomenon that a hypersurface of “degree one” in the polarization still meets  $g$  generic translates of itself in  $g!$  points, the number of ways to order the  $g$  transverse sheets.

Nothing so far has fixed  $\Theta$  as a subset of  $X$ , only its class and its numerical invariants. The freedom



in the construction is the freedom in the line bundle representing the polarization: the semicharacter  $\chi$  can be altered, or  $L$  tensored by a bundle in  $\text{Pic}^0(X)$ , without changing the class  $H$ . Each such change moves  $\Theta$  by a translation. The next result makes this precise, so that “the” theta divisor is well-defined up to translation and no more.

**Proposition 4.5.3: Uniqueness up to Translation**

Let  $X$  be a principally polarized abelian variety with polarization class  $H > 0$  of type  $(1, \dots, 1)$ . The theta divisor of definition 4.5.1 is unique up to translation by a point of  $X$ : if  $\Theta$  and  $\Theta'$  are theta divisors attached to two line bundles  $L(H, \chi)$  and  $L(H, \chi')$  representing the same polarization  $H$ , then  $\Theta' = t_a^* \Theta$  for some  $a \in X$ , where  $t_a: X \rightarrow X$  is translation by  $a$ . Conversely, every translate  $t_a^* \Theta$  is the theta divisor of a line bundle representing  $H$ .

**Proof:**

*Step 1: Two representatives differ by a bundle in  $\text{Pic}^0(X)$ .* Two line bundles  $L = L(H, \chi)$  and  $L' = L(H, \chi')$  with the same Hermitian form  $H$  have the same first Chern class  $c_1 = H$  (proposition 2.3.4), so  $L' \otimes L^{-1}$  lies in the kernel of  $c_1: \text{Pic}(X) \rightarrow \text{NS}(X)$ , which is  $\text{Pic}^0(X)$  (theorem 2.2.8). Thus  $L' = L \otimes P$  for a unique  $P = L(0, \chi' \chi^{-1}) \in \text{Pic}^0(X)$ .

*Step 2: Tensoring by  $\text{Pic}^0(X)$  is translation of the bundle.* The polarization isogeny  $\varphi_L: X \rightarrow \text{Pic}^0(X)$  sending  $x \mapsto t_x^* L \otimes L^{-1}$  is, for a principal polarization, an isomorphism (this is the analytic statement that  $H > 0$  of type  $(1, \dots, 1)$  makes  $\varphi_L$  bijective; the map  $\varphi_L$  is developed in chapter 5, but its bijectivity in the principal case is the content of  $\dim H^0 = 1$  we already have). Hence there is a unique  $a \in X$  with  $P = \varphi_L(a) = t_a^* L \otimes L^{-1}$ , so

$$L' = L \otimes P = t_a^* L$$

as line bundles.

*Step 3: Translation of bundles is translation of divisors.* A section  $\theta$  of  $L$  pulls back under  $t_a$  to a section  $t_a^* \theta$  of  $t_a^* L = L'$ , and pullback of a section pulls back its divisor:  $\text{div}(t_a^* \theta) = t_a^* \text{div}(\theta) = t_a^* \Theta$ , which is the translate of  $\Theta$  by  $-a$ . Since  $\dim H^0(X, L') = 1$ , the section  $t_a^* \theta$  is, up to scalar, the unique defining section of  $L'$ , so  $\Theta' = \text{div}(t_a^* \theta) = t_a^* \Theta$ . This proves  $\Theta$  and  $\Theta'$  are translates.

*Step 4: Converse.* Conversely, given any  $a \in X$ , the bundle  $t_a^* L \cong L \otimes \varphi_L(a)$  has the same class  $H$  and its section has divisor  $t_a^* \Theta$ ; setting  $P = \varphi_L(a)$  and  $\chi'$  the semicharacter of  $L \otimes P$  exhibits  $t_a^* \Theta$  as a theta divisor for  $H$ . Every translate thus arises, completing the proof.

The translation ambiguity is not a defect to be removed but the actual state of affairs: a principal polarization determines a divisor class, and a divisor within that class only after a choice of origin-compatible line bundle. What the proposition guarantees is that the ambiguity is exactly a torsor under  $X$  acting by translation, so any geometric statement about  $\Theta$  that is translation-invariant, its singularities, its intersection numbers, the pair  $(X, \Theta)$  up to isomorphism, is well-defined independently of the choice. This is the sense in which one speaks of the theta divisor of a principally polarized abelian variety without further qualification. There is, however, a canonical way to reduce the ambiguity from a full translate to a sign, by choosing the representative that is symmetric under  $x \mapsto -x$ .



**Definition 4.5.4: Symmetric Theta Divisor**

Let  $[-1]: X \rightarrow X$  denote the inversion  $x \mapsto -x$  of the group  $X$ . A theta divisor  $\Theta$  is *symmetric* if

$$[-1]^*\Theta = \Theta$$

as divisors on  $X$ . Equivalently,  $\Theta$  is symmetric when the semicharacter  $\chi$  of the representing bundle  $L(H, \chi)$  is normalized so that  $[-1]^*L \cong L$  and the section  $\theta$  is even or odd,  $\theta(-v) = \pm \theta(v)$ .

A symmetric theta divisor exists because  $[-1]^*L$  again represents  $H$  (inversion is a group automorphism, so it preserves the Hermitian form  $H$ ), so by proposition 4.5.3 the divisor  $[-1]^*\Theta$  is a translate of  $\Theta$ ; translating by a suitable half-period makes the two coincide. Among the translates of a theta divisor there are exactly  $|X[2]| = 2^{2g}$  symmetric ones, the orbit of 2-torsion, so symmetry cuts the translation ambiguity down to a finite set fixed by inversion rather than eliminating it entirely. The value of the normalization is that it fixes a canonical point structure: a symmetric  $\Theta$  is invariant under  $[-1]$ , so it interacts cleanly with the 2-torsion  $X[2]$  and with the even/odd theta functions that furnish the classical theta constants, and it is the representative one uses when the geometry of  $\Theta$  near the origin, its multiplicity at 2-torsion points, matters. The role of  $\Theta$  as the geometric heart of the polarization is sharpest in the symmetric normalization, where the divisor is genuinely a canonical feature of the pointed group  $(X, 0)$  and not only of  $X$  up to translation.

The whole construction is transparent in dimension one, where the theta divisor is a single point of an elliptic curve and the count  $(\Theta^g) = g!$  reads  $(\Theta^1) = 1 = 1!$ .

**Example 4.5.5: The Theta Divisor of an Elliptic Curve**

Let  $X = E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ , a one-dimensional complex torus. Here  $g = 1$  and any polarization is automatically principal: the alternating form  $E$  has a single elementary divisor  $d_1$ , and rescaling  $H$  by  $1/d_1$  makes it of type (1), so the type is (1). Take  $L = L(H, \chi)$  with  $H$  the principal polarization normalized so that  $E = \text{Im}H$  is the standard alternating form on  $\Lambda$  with  $E(1, \tau) = 1$ .

By proposition 3.4.5,  $\dim_{\mathbb{C}} H^0(E, L) = 1$ , and the unique section (up to scalar) is the classical Jacobi theta function

$$\theta(z, \tau) = \sum_{n \in \mathbb{Z}} \exp(\pi i n^2 \tau + 2\pi i n z)$$

of example 4.2.6. Its quasi-periodicity,  $\theta(z + 1) = \theta(z)$  and  $\theta(z + \tau) = \exp(-\pi i \tau - 2\pi i z)\theta(z)$ , shows that the zero set is  $\Lambda$ -invariant, so it descends to a divisor on  $E$ . The number of zeros of  $\theta$  in a fundamental parallelogram is computed by the argument principle: integrating  $\frac{1}{2\pi i} d \log \theta$  around the boundary and using the quasi-periodicity, the four sides contribute a net winding of 1, so  $\theta$  has exactly one zero (counted with multiplicity) modulo  $\Lambda$ . That zero is the half-period  $z_0 = \frac{1+\tau}{2}$ , the unique odd 2-torsion point where  $\theta$  vanishes, and

$$\Theta = [z_0]$$

a single point of  $E$  with multiplicity one, an effective divisor of degree 1.

The class of  $\Theta$  is  $c_1(\mathcal{O}_E(\Theta)) = H$ , matching  $\dim H^0(E, \mathcal{O}_E(\Theta)) = 1$  from Riemann-Roch on the elliptic curve (example 3.4.6). The self-intersection number in dimension one is just the degree of the divisor,

$$(\Theta^1) = \deg \Theta = 1 = 1!$$

in agreement with proposition 4.5.2. The symmetric normalization (definition 4.5.4) is the choice of  $z_0 \in E[2]$ : since  $\theta$  with characteristics can be shifted so that its zero sits at any of the four 2-torsion points, the symmetric theta divisors on  $E$  are exactly the four points of  $E[2] = \frac{1}{2}\Lambda/\Lambda$ , on which  $[-1]$  acts trivially.

The elliptic case is the whole subject in miniature: a principally polarized abelian variety of dimension one is a pair  $(E, [z_0])$ , a curve together with a marked point, and the theta divisor is that point. This is why an elliptic curve is so often presented as a curve with a chosen origin. In higher dimensions the theta divisor stops being a point and becomes a genuine hypersurface of dimension  $g - 1$ , and its geometry, in particular its singular locus, becomes an object of study in its own right, but the principle is the same: the divisor is where the abelian variety stores its polarization in visible geometric form. The definition of a principal polarization (definition 3.2.4) is exactly the condition  $\dim H^0 = 1$  that makes this storage possible, and it is the class of principally polarized abelian varieties, those carrying a theta divisor, that the moduli space  $\mathfrak{H}_g/\mathrm{Sp}(2g, \mathbb{Z})$  of theorem 3.3.7 parametrizes.

The theta divisor is the point at which this book's two later strands converge. Chapter 12 realizes it concretely for the Jacobian  $\mathrm{Jac}(C)$  of a smooth projective curve  $C$  of genus  $g$ : the image of the  $(g - 1)$ -fold symmetric product of  $C$  under the Abel-Jacobi map is a theta divisor for the canonical principal polarization, so every curve manufactures not just a principally polarized abelian variety but an explicit theta divisor inside it. The reason this matters is the Torelli theorem, also taken up in chapter 12: the pair  $(\mathrm{Jac}(C), \Theta)$  determines the curve  $C$  up to isomorphism. The polarized Jacobian remembers its curve, and it remembers it through the geometry of the theta divisor, so that a hypersurface in an abelian variety encodes the entire projective curve that gave rise to it.

### Exercise 4.5.1

1. Let  $X$  be a principally polarized abelian variety of dimension  $g$ , with theta divisor  $\Theta$  as in definition 4.5.1. State the self-intersection number  $(\Theta^g)$ , and compute it explicitly for  $g = 1, 2, 3$ . Explain in one sentence why the answer is not 1 even though the polarization is “degree one” in the sense that  $\dim_{\mathbb{C}} H^0(X, L) = 1$ .
2. Let  $X = V/\Lambda$  carry a polarization of type  $(d_1, \dots, d_g)$  with associated alternating form  $E$ , not assumed principal. Using the computation in the proof of proposition 4.5.2, show that for any effective divisor  $D$  with  $c_1(\mathcal{O}_X(D)) = E$  one has  $(D^g) = g! d_1 d_2 \cdots d_g$ , and deduce that a polarization admitting a theta divisor in the sense of definition 4.5.1 is forced to have  $(D^g) = g!$ .
3. Fix a principally polarized abelian variety  $X$  with theta divisor  $\Theta$  attached to  $L = L(H, \chi)$ . For  $a \in X$ , identify the line bundle  $\mathcal{O}_X(t_a^* \Theta)$  in terms of  $L$  and the polarization isogeny  $\varphi_L: X \rightarrow \mathrm{Pic}^0(X)$ , and prove that the map  $a \mapsto \mathcal{O}_X(t_a^* \Theta)$  is a bijection from  $X$  onto the set of line bundles representing the polarization  $H$ . (You may use the results of proposition 4.5.3.)
4. Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$  be the elliptic curve of example 4.5.5, and let  $\theta(z, \tau)$  be the Jacobi theta function. Verify directly, using the quasi-periodicity relations  $\theta(z + 1) = \theta(z)$  and  $\theta(z + \tau) = \exp(-\pi i \tau - 2\pi i z) \theta(z)$  and the argument principle, that  $\theta$  has exactly one zero modulo  $\Lambda$  in a fundamental parallelogram. Then confirm that  $z_0 = \frac{1+\tau}{2}$  is a zero by checking that  $\theta(z_0, \tau)$  is odd about  $z_0$ .
5. Let  $\Theta$  be a symmetric theta divisor on a principally polarized abelian variety  $X$  of dimension  $g$  (definition 4.5.4). Prove that the number of symmetric theta divisors representing the same polarization is exactly  $2^{2g}$ . Then, in the elliptic case  $g = 1$  of example 4.5.5, identify these

four symmetric theta divisors as subsets of  $E$ .



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# *The Dual Torus, Isogenies, and Poincaré Reducibility*

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The first four chapters studied a single abelian variety and its line bundles. This chapter turns to the maps between abelian varieties and to the dual object that organizes them. The degree-zero line bundles on  $X$ , which chapter 2 exhibited as a complex torus in their own right, are assembled here into the dual abelian variety  $\hat{X}$ : the same  $\overline{V}^*/\Lambda^\vee$  met before, now regarded as a variety that varies functorially with  $X$  and satisfies  $\hat{\hat{X}} = X$ . A polarization, previously a positive Hermitian form, reappears as an actual homomorphism  $\varphi_L: X \rightarrow \hat{X}$ , and the type of the polarization becomes the degree of that isogeny.

Isogenies are the right notion of equivalence for the arithmetic of abelian varieties: a surjection with finite kernel, invertible up to that kernel. The torsion subgroups  $X[n]$ , the Tate-module shadows that Part II will exploit, and the Weil pairing that a polarization places on them are all introduced here in their transparent analytic form. The chapter ends with Poincaré complete reducibility, the statement that a polarization splits any abelian subvariety off up to isogeny, so that every abelian variety is isogenous to a product of simple ones. This is what reduces the study of endomorphism algebras, the subject of chapter 6, to the case of a simple variety, whose endomorphism algebra is a division algebra.

## 5.1 The Dual Complex Torus

The degree-zero Picard group  $\text{Pic}^0(X)$  of a complex torus was already computed in chapter 2: the exponential sequence exhibited it as the quotient  $\overline{V}^*/\iota(\Lambda^\vee)$  of the antilinear dual by an integral lattice, hence as a complex torus of the same dimension  $g$  as  $X$  (proposition 2.1.5). At that point  $\text{Pic}^0(X)$  was a group of isomorphism classes of line bundles that happened to carry a complex

structure. This chapter promotes it to a geometric object in its own right, the *dual torus*  $\hat{X}$ , and organizes the constructions that make  $X \mapsto \hat{X}$  a duality: a contravariant functor on complex tori, biduality  $\hat{\hat{X}} = X$ , and (in the next sections) the isogeny  $X \rightarrow \hat{X}$  attached to a polarization. The dual torus is the target into which a polarization maps  $X$ , so pinning it down precisely is the prerequisite for everything about polarizations that follows.

The definition below builds  $\hat{X}$  directly from the linear algebra of  $V$  and  $\Lambda$ , without reference to line bundles, and then proposition 5.1.3 reconciles this construction with the Picard-group description. Working from the linear algebra first is what makes the functoriality and biduality transparent: both are statements about antilinear duals of vector spaces and their integral lattices, and neither is obvious from the definition of  $\text{Pic}^0(X)$  as a set of bundle classes.

The right ambient space is the space of antilinear functionals on  $V$ . Recall from the notational conventions of chapter 1 that  $\bar{V}^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$  denotes the  $\mathbb{C}$ -antilinear functionals  $\ell: V \rightarrow \mathbb{C}$ , those satisfying  $\ell(cv) = \bar{c}\ell(v)$  for  $c \in \mathbb{C}$ ; this is a  $g$ -dimensional complex vector space, the conjugate of the ordinary dual  $V^*$ .

**Definition 5.1.1: Dual Complex Torus**

Let  $X = V/\Lambda$  be a complex torus of dimension  $g$ . Set

$$\hat{V} := \bar{V}^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$$

the space of  $\mathbb{C}$ -antilinear functionals on  $V$ , a complex vector space of dimension  $g$ , and let

$$\hat{\Lambda} := \{ \ell \in \hat{V} : \text{Im} \ell(\lambda) \in \mathbb{Z} \text{ for all } \lambda \in \Lambda \}$$

be the subgroup of antilinear functionals whose imaginary part is integer-valued on  $\Lambda$ . Then  $\hat{\Lambda}$  is a full lattice in  $\hat{V}$ , and the *dual complex torus* of  $X$  is the quotient

$$\hat{X} := \hat{V} / \hat{\Lambda}$$

a complex torus of dimension  $g$ .

The definition packages the parameter space of degree-zero line bundles as a variety. A degree-zero bundle on  $X$  is, by proposition 2.1.5 and the character description that followed it, a unitary character  $\chi: \Lambda \rightarrow U(1)$ ; writing  $\chi(\lambda) = e^{2\pi i \varphi(\lambda)}$  presents such a character by a real-valued phase  $\varphi$  on  $\Lambda$  modulo  $\mathbb{Z}$ , and the antilinear functional  $\ell$  with  $\text{Im} \ell = \varphi$  (up to the lattice) is exactly the data  $\hat{X}$  records. The condition  $\text{Im} \ell(\lambda) \in \mathbb{Z}$  is the condition that the phase  $\ell$  be trivial modulo the integral lattice, so that  $\ell$  and  $\ell'$  define the same character precisely when  $\ell - \ell' \in \hat{\Lambda}$ . Thus  $\hat{X}$  is the moduli space of degree-zero line bundles, made into a complex torus by giving its universal cover the antilinear-dual complex structure. That  $\hat{\Lambda}$  is a genuine lattice, and  $\hat{X}$  a genuine torus, is the content of the following verification, carried out before the reconciliation with  $\text{Pic}^0(X)$  so that the definition stands on the linear algebra alone.

**Proof of the lattice claim in definition 5.1.1:**

Only the assertion that  $\hat{\Lambda}$  is a full lattice in  $\hat{V}$  requires proof; granting it,  $\hat{X} = \hat{V} / \hat{\Lambda}$  is a complex torus of dimension  $g$  by definition 1.1.2.

*Step 1: The imaginary-part pairing.* Sending an antilinear functional to the imaginary part of its restriction to  $V$  gives a map

$$\Phi: \hat{V} \longrightarrow \text{Hom}_{\mathbb{R}}(V, \mathbb{R}), \quad \ell \longmapsto \text{Im}\ell$$

This map is  $\mathbb{R}$ -linear. It is a bijection: an antilinear functional  $\ell$  is determined by  $\text{Im}\ell$ , because  $\text{Re}\ell(v) = -\text{Im}\ell(iv)$ . Indeed  $\ell(iv) = \bar{i}\ell(v) = -i\ell(v)$  by antilinearity, so  $\text{Im}\ell(iv) = \text{Im}(-i\ell(v)) = -\text{Re}\ell(v)$ , which recovers the real part from the imaginary part. Conversely, given any  $\mathbb{R}$ -linear  $u: V \rightarrow \mathbb{R}$ , the formula  $\ell(v) := -u(iv) + iu(v)$  defines an antilinear functional with  $\text{Im}\ell = u$ : additivity is clear, and  $\ell(iv) = -u(-v) + iu(iv) = u(v) + iu(iv) = -i(-u(iv) + iu(v)) = -i\ell(v)$ , so  $\ell$  is antilinear. Hence  $\Phi$  is an  $\mathbb{R}$ -linear isomorphism from the  $2g$ -dimensional real space  $\hat{V}$  onto the  $2g$ -dimensional real space  $\text{Hom}_{\mathbb{R}}(V, \mathbb{R})$ .

*Step 2: Transporting the integral lattice.* Under  $\Phi$  the defining condition of  $\hat{\Lambda}$  reads  $\Phi(\ell)(\lambda) = \text{Im}\ell(\lambda) \in \mathbb{Z}$  for all  $\lambda \in \Lambda$ , so

$$\Phi(\hat{\Lambda}) = \{ u \in \text{Hom}_{\mathbb{R}}(V, \mathbb{R}) : u(\Lambda) \subseteq \mathbb{Z} \} = \text{Hom}(\Lambda, \mathbb{Z}) = \Lambda^{\vee}$$

the dual lattice, where the middle equality uses that an  $\mathbb{R}$ -linear functional integral on  $\Lambda$  is determined by its values on a  $\mathbb{Z}$ -basis of  $\Lambda$ , which is an  $\mathbb{R}$ -basis of  $V$  (the lattice is full). Now  $\Lambda^{\vee}$  is a full lattice in  $\text{Hom}_{\mathbb{R}}(V, \mathbb{R})$ : it is the free abelian group on the basis dual to a  $\mathbb{Z}$ -basis of  $\Lambda$ , and that basis is an  $\mathbb{R}$ -basis of  $\text{Hom}_{\mathbb{R}}(V, \mathbb{R})$ . Since  $\Phi$  is an  $\mathbb{R}$ -linear isomorphism,  $\hat{\Lambda} = \Phi^{-1}(\Lambda^{\vee})$  is a full lattice in  $\hat{V}$ , as we sought to show.

The proof isolates the mechanism that makes the dual work: the  $\mathbb{R}$ -linear isomorphism  $\ell \mapsto \text{Im}\ell$  between the antilinear dual  $\hat{V}$  and the ordinary real dual  $\text{Hom}_{\mathbb{R}}(V, \mathbb{R})$ , under which the integral lattice  $\hat{\Lambda}$  corresponds to the dual lattice  $\Lambda^{\vee}$ . This is the same identification that appeared inside proposition 2.1.5, where  $\iota(\Lambda^{\vee})$  was the image of  $\Lambda^{\vee}$  in  $H^{0,1} = \overline{V}^*$ ; the present formulation makes the lattice intrinsic to the antilinear dual rather than an image under a cohomological projection. The two descriptions of  $\hat{\Lambda}$ , the imaginary-part condition and the dual lattice  $\Lambda^{\vee}$ , are the two faces of the dual torus that the next proposition matches.

The first concrete instance is  $g = 1$ , worked out here before the general reconciliation so that the abstract definition has an example within reach.

### Example 5.1.2: The Dual of an Elliptic Curve

Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ , as in example 1.1.4. Here  $V = \mathbb{C}$ , and an antilinear functional  $\ell: \mathbb{C} \rightarrow \mathbb{C}$  has the form  $\ell(v) = \bar{v}w$  for a unique  $w \in \mathbb{C}$  (antilinearity forces  $\ell$  to be a conjugate-linear multiple), so  $\hat{V} = \overline{\mathbb{C}}^* \cong \mathbb{C}$  via  $\ell \leftrightarrow w$ . The dual lattice is

$$\hat{\Lambda} = \{ \ell_w : \text{Im}(\bar{\lambda}w) \in \mathbb{Z} \text{ for } \lambda \in \{1, \tau\} \}$$

Writing the two integrality conditions  $\text{Im}(w) \in \mathbb{Z}$  and  $\text{Im}(\bar{\tau}w) \in \mathbb{Z}$ , these cut out a rank-2 lattice in the  $w$ -plane, real-linearly isomorphic to  $\Lambda^{\vee}$  under  $w \mapsto (\text{Im}w, \text{Im}\bar{\tau}w)$ . The quotient  $\hat{E} = \hat{V}/\hat{\Lambda}$  is therefore a one-dimensional complex torus.

Every one-dimensional complex torus is an elliptic curve (proposition 1.4.2), so  $\hat{E}$  is an elliptic curve. It is in fact isomorphic to  $E$ : by proposition 5.1.3 below,  $\hat{E} \cong \text{Pic}^0(E)$ , and example 2.1.2 identified  $\text{Pic}^0(E) \cong E$  through the Abel-Jacobi theorem, sending a degree-zero bundle  $\mathcal{O}_E(p - 0)$

to the point  $p$ . Thus

$$\hat{E} \cong \text{Pic}^0(E) \cong E$$

and the elliptic curve is *self-dual*. This self-duality is the  $g = 1$  shadow of the principal polarization: the isomorphism  $E \xrightarrow{\sim} \hat{E}$  is the polarization isogeny  $\varphi_L$  of section 5.3 for  $L = \mathcal{O}_E(0)$ , which is an isomorphism exactly because the natural polarization of an elliptic curve is principal. For general  $g$  an abelian variety need not be self-dual; the elliptic curve is the special case where the canonical map  $X \rightarrow \hat{X}$  happens to be an isomorphism.

The construction of  $\hat{X}$  from the antilinear dual and the construction of  $\text{Pic}^0(X)$  from the exponential sequence produce the same complex torus. Establishing this canonically, and not up to a non-specified isomorphism, is what turns  $\hat{X}$  from a piece of linear algebra into the geometric parameter space of degree-zero bundles.

**Proposition 5.1.3: The Dual Torus Is the Degree-Zero Picard Group**

Let  $X = V/\Lambda$  be a complex torus. There is a canonical isomorphism of complex tori

$$\hat{X} \xrightarrow{\sim} \text{Pic}^0(X)$$

identifying  $\hat{X} = \hat{V}/\hat{\Lambda}$  with the quotient  $\overline{V^*}/\iota(\Lambda^\vee)$  of proposition 2.1.5. Explicitly, the class of an antilinear functional  $\ell \in \hat{V}$  maps to the degree-zero line bundle  $L(0, \chi_\ell)$  with unitary semicharacter

$$\chi_\ell(\lambda) = e^{2\pi i \text{Im}(\ell(\lambda))} = e^{-\pi i (\ell(\lambda) - \overline{\ell(\lambda)})}$$

for  $\lambda \in \Lambda$ .

**Proof:**

*Step 1: The two universal covers agree.* The antilinear dual  $\hat{V} = \overline{V^*}$  is by definition the space  $\overline{V^*}$  of  $\mathbb{C}$ -antilinear functionals appearing in proposition 2.1.5, where it was identified with  $H^{0,1} = H^1(X, \mathcal{O}_X)$ . So the universal covers of  $\hat{X}$  and of the torus  $\overline{V^*}/\iota(\Lambda^\vee)$  representing  $\text{Pic}^0(X)$  are literally the same complex vector space; there is nothing to choose in identifying them.

*Step 2: The two lattices agree.* It remains to check  $\hat{\Lambda} = \iota(\Lambda^\vee)$  inside  $\hat{V} = \overline{V^*}$ . In proposition 2.1.5 the lattice  $\iota(\Lambda^\vee)$  was the image of  $H^1(X, \mathbb{Z}) \cong \Lambda^\vee$  under the projection  $H^1(X, \mathbb{R}) \rightarrow H^{0,1}$ , which is a real-linear isomorphism onto  $\overline{V^*}$ . Concretely, a class in  $H^1(X, \mathbb{Z}) = \text{Hom}(\Lambda, \mathbb{Z})$  is an integral  $\mathbb{R}$ -linear functional  $u: V \rightarrow \mathbb{R}$  with  $u(\Lambda) \subseteq \mathbb{Z}$ , and its image in  $H^{0,1} = \overline{V^*}$  is the antilinear functional  $\ell$  with  $\text{Im} \ell = u$ , by the computation of the projection in the proof of proposition 2.1.5. Thus  $\iota(\Lambda^\vee)$  is exactly the set of  $\ell \in \hat{V}$  with  $\text{Im} \ell$  integer-valued on  $\Lambda$ , which is  $\hat{\Lambda}$  by definition 5.1.1. The two lattices coincide, so the identity map on  $\hat{V}$  descends to an isomorphism of complex tori  $\hat{X} = \hat{V}/\hat{\Lambda} \xrightarrow{\sim} \overline{V^*}/\iota(\Lambda^\vee) = \text{Pic}^0(X)$ .

*Step 3: The semicharacter description.* Under the character description following proposition 2.1.5, the class of  $\ell \in H^1(X, \mathcal{O}_X) = \overline{V^*}$  modulo  $\iota(\Lambda^\vee)$  corresponds to the unitary character  $\chi_\ell: \Lambda \rightarrow U(1)$  obtained by exponentiating: a class in  $H^1(X, \mathcal{O}_X)/\iota(\Lambda^\vee)$  is sent by  $\exp_*$  to the



line bundle whose semicharacter records the phase  $\text{Im} \ell$  on  $\Lambda$ , that is,

$$\chi_\ell(\lambda) = e^{2\pi i \text{Im} \ell(\lambda)}$$

Since  $\text{Im} \ell(\lambda) = \frac{1}{2i}(\ell(\lambda) - \overline{\ell(\lambda)})$ , this equals  $e^{-\pi i (\ell(\lambda) - \overline{\ell(\lambda)})}$ . Because  $H = 0$ , the semicharacter identity of definition 2.2.4 degenerates to  $\chi_\ell(\lambda + \mu) = \chi_\ell(\lambda)\chi_\ell(\mu)$ :  $\chi_\ell$  is an ordinary character, matching the description  $\text{Pic}^0(X) \cong \text{Hom}(\Lambda, U(1))$ . The bundle is  $L(0, \chi_\ell)$  in the Appell-Humbert notation of theorem 2.2.8, and  $\ell \mapsto L(0, \chi_\ell)$  is the canonical isomorphism, completing the proof.

The proposition is the identification that gives the dual torus its meaning. On the left,  $\hat{X}$  is a complex torus manufactured from antilinear functionals, an object of linear algebra; on the right,  $\text{Pic}^0(X)$  is the group of degree-zero line bundles on  $X$ , an object of geometry. That they are canonically the same complex torus means every degree-zero line bundle on  $X$  is a point of  $\hat{X}$ , and conversely, in a way compatible with the group law and the complex structure. The reconciliation of the two descriptions of  $\hat{\Lambda}$ , the imaginary-part-integrality condition of definition 5.1.1 and the image  $\iota(\Lambda^\vee)$  of the dual lattice from proposition 2.1.5, is the concrete content: both describe the same rank- $2g$  subgroup of  $\hat{V}$ , and the semicharacter  $\chi_\ell$  makes the correspondence explicit at the level of line bundles. From here on  $\hat{X}$  and  $\text{Pic}^0(X)$  are used interchangeably, and the Poincaré bundle of the algebraic theory (Part II) will make the point of  $\hat{X}$  tautologically a bundle on  $X$ .

A duality is not a construction but a functor, and the value of  $\hat{X}$  is that  $X \mapsto \hat{X}$  reverses arrows. A homomorphism  $f: X \rightarrow Y$  pulls back line bundles on  $Y$  to line bundles on  $X$ , and this pullback carries degree-zero bundles to degree-zero bundles, hence induces a homomorphism  $\hat{Y} \rightarrow \hat{X}$  in the opposite direction. The linear-algebraic incarnation of this pullback is the antilinear transpose, which is what makes the induced map manifestly a homomorphism of complex tori.

#### Proposition 5.1.4: Functoriality of the Dual

Let  $f: X \rightarrow Y$  be a homomorphism of complex tori, induced (proposition 1.1.7) by the  $\mathbb{C}$ -linear map  $F: V \rightarrow W$  with  $F(\Lambda_X) \subseteq \Lambda_Y$ , where  $X = V/\Lambda_X$  and  $Y = W/\Lambda_Y$ . Then:

1. The antilinear transpose  $\hat{F}: \hat{W} \rightarrow \hat{V}$ ,  $\ell \mapsto \ell \circ F$ , is  $\mathbb{C}$ -linear and satisfies  $\hat{F}(\hat{\Lambda}_Y) \subseteq \hat{\Lambda}_X$ , so it descends to a homomorphism of complex tori

$$\hat{f}: \hat{Y} \longrightarrow \hat{X}$$

called the *dual homomorphism*. Under proposition 5.1.3,  $\hat{f}$  is the pullback of line bundles  $f^*: \text{Pic}^0(Y) \rightarrow \text{Pic}^0(X)$ .

2. The assignment  $f \mapsto \hat{f}$  is contravariant and functorial:  $\widehat{\text{id}_X} = \text{id}_{\hat{X}}$ , and for a second homomorphism  $h: Y \rightarrow Z$  one has  $\widehat{h \circ f} = \hat{f} \circ \hat{h}$ .

#### Proof:

1. *The transpose and its lattice condition.* For  $\ell \in \hat{W} = \text{Hom}_{\overline{\mathbb{C}}}(W, \mathbb{C})$ , the composite  $\ell \circ F: V \rightarrow \mathbb{C}$  is antilinear, being the composite of the  $\mathbb{C}$ -linear  $F$  with the antilinear  $\ell$ , so  $\hat{F}(\ell) = \ell \circ F \in \hat{V}$ . The map  $\hat{F}$  is  $\mathbb{C}$ -linear in  $\ell$ : scalar multiplication and addition of functionals are computed pointwise, and  $(c\ell) \circ F = c(\ell \circ F)$  for  $c \in \mathbb{C}$  because  $c\ell$  scales

values, so  $\hat{F}(c\ell) = c\hat{F}(\ell)$ . To check the lattice condition, let  $\ell \in \hat{\Lambda}_Y$ , so  $\text{Im}\ell(\mu) \in \mathbb{Z}$  for all  $\mu \in \Lambda_Y$ . For  $\lambda \in \Lambda_X$  one has  $F(\lambda) \in \Lambda_Y$  by hypothesis, hence

$$\text{Im}(\hat{F}(\ell)(\lambda)) = \text{Im}(\ell(F(\lambda))) \in \mathbb{Z}$$

so  $\hat{F}(\ell) \in \hat{\Lambda}_X$ . Thus  $\hat{F}(\hat{\Lambda}_Y) \subseteq \hat{\Lambda}_X$ , and by proposition 1.1.7 the  $\mathbb{C}$ -linear  $\hat{F}$  carrying  $\hat{\Lambda}_Y$  into  $\hat{\Lambda}_X$  descends to a homomorphism  $\hat{f}: \hat{Y} \rightarrow \hat{X}$  of complex tori.

That  $\hat{f}$  is the pullback  $f^*$  is a computation with semicharacters. By proposition 5.1.3 a point  $\ell \in \hat{W}$  represents the bundle  $L(0, \chi_\ell)$  on  $Y$  with  $\chi_\ell(\mu) = e^{2\pi i \text{Im}\ell(\mu)}$ . Its pullback  $f^*L(0, \chi_\ell)$  is the bundle on  $X$  obtained by pulling back the factor of automorphy along  $F$ , whose semicharacter is  $\lambda \mapsto \chi_\ell(F(\lambda)) = e^{2\pi i \text{Im}\ell(F(\lambda))} = e^{2\pi i \text{Im}(\hat{F}(\ell))(\lambda)} = \chi_{\hat{F}(\ell)}(\lambda)$ . This is the semicharacter attached to  $\hat{F}(\ell)$ , so  $f^*L(0, \chi_\ell) = L(0, \chi_{\hat{F}(\ell)})$ , which is exactly the bundle represented by  $\hat{f}(\ell)$ . Hence  $\hat{f} = f^*$  under the identification of proposition 5.1.3.

2. *Contravariance and functoriality.* The identity  $\text{id}_X$  is induced by  $\text{id}_V$ , whose antilinear transpose is  $\text{id}_{\hat{V}}$ , so  $\widehat{\text{id}_X} = \text{id}_{\hat{X}}$ . If  $h: Y \rightarrow Z$  is induced by the  $\mathbb{C}$ -linear  $G: W \rightarrow U$  with  $G(\Lambda_Y) \subseteq \Lambda_Z$ , then  $h \circ f$  is induced by  $G \circ F$ , and for  $\ell \in \hat{U}$ ,

$$\widehat{G \circ F}(\ell) = \ell \circ (G \circ F) = (\ell \circ G) \circ F = \hat{F}(\hat{G}(\ell))$$

so  $\widehat{h \circ f} = \hat{f} \circ \hat{h}$  on universal covers, and hence on the tori. The functor  $X \mapsto \hat{X}$ ,  $f \mapsto \hat{f}$  is therefore contravariant, as we sought to show.

The proposition records that duality is a genuine contravariant functor on the category of complex tori, computed on universal covers by the antilinear transpose  $F \mapsto \hat{F} = (-) \circ F$ . Two descriptions of  $\hat{f}$  have been matched: the linear-algebraic one,  $\ell \mapsto \ell \circ F$ , which is manifestly a homomorphism of tori, and the geometric one, pullback of degree-zero line bundles along  $f$ , which is manifestly functorial. The equality of the two is what allows the dual to be computed either way, a convenience used repeatedly in the sequel. Contravariance is the reason a polarization, which will be a self-map up to the identification  $X \cong \hat{X}$ , produces a symmetric object: dualizing reverses the direction of  $\varphi_L: X \rightarrow \hat{X}$ , and the Rosati involution of chapter 6 is built from exactly this reversal.

Applying the dual twice should return to the original torus, and it does, canonically. Biduality is the statement that promotes  $X \mapsto \hat{X}$  from a functor to a duality: not only is  $\hat{\hat{X}}$  isomorphic to  $X$ , the isomorphism is natural, given by a single formula independent of any choice of basis or lattice.

#### Theorem 5.1.5: Biduality

Let  $X = V/\Lambda$  be a complex torus. The map

$$\kappa_X: X \longrightarrow \hat{\hat{X}}, \quad v + \Lambda \longmapsto (\ell \mapsto \overline{\ell(v)})$$

is a canonical isomorphism of complex tori. It is natural in  $X$ : for every homomorphism  $f: X \rightarrow Y$  one has  $\hat{\hat{f}} \circ \kappa_X = \kappa_Y \circ f$ . Consequently  $\hat{\hat{X}} = X$  canonically.

**Proof:**

*Step 1: The formula defines a bidual element.* Write  $\hat{V} = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$  and  $\hat{\hat{V}} = \text{Hom}_{\mathbb{C}}(\hat{V}, \mathbb{C})$ . For fixed  $v \in V$ , consider the assignment  $K(v): \ell \mapsto \overline{\ell(v)}$  on  $\hat{V}$ . It is a functional on  $\hat{V}$ , and it is antilinear in  $\ell$ : for  $c \in \mathbb{C}$ ,  $(c\ell)(v) = c\ell(v)$ , so  $K(v)(c\ell) = \overline{c\ell(v)} = \bar{c}\overline{\ell(v)} = \bar{c}K(v)(\ell)$ . Hence  $K(v) \in \hat{\hat{V}}$ . Moreover  $v \mapsto K(v)$  is  $\mathbb{C}$ -linear:  $\ell$  is antilinear, so  $\ell(cv) = \bar{c}\ell(v)$  and  $K(cv)(\ell) = \overline{\ell(cv)} = \overline{\bar{c}\ell(v)} = c\overline{\ell(v)} = cK(v)(\ell)$ , and additivity in  $v$  is immediate. Thus  $K: V \rightarrow \hat{\hat{V}}$  is a  $\mathbb{C}$ -linear map.

*Step 2:  $K$  is an isomorphism of vector spaces.* Both  $V$  and  $\hat{\hat{V}}$  have complex dimension  $g$ , so it suffices to check that  $K$  is injective. If  $K(v) = 0$ , then  $\overline{\ell(v)} = 0$ , hence  $\ell(v) = 0$ , for every antilinear functional  $\ell$  on  $V$ . But the antilinear functionals separate points of  $V$ : if  $v \neq 0$ , extend  $v$  to a  $\mathbb{C}$ -basis and take the antilinear functional dual to  $v$  in that basis, which does not vanish at  $v$ . Therefore  $v = 0$ , so  $K$  is injective, hence a  $\mathbb{C}$ -linear isomorphism  $V \xrightarrow{\sim} \hat{\hat{V}}$ .

*Step 3:  $K$  matches the lattices.* It remains to show  $K(\Lambda) = \hat{\hat{\Lambda}}$ , where  $\hat{\hat{\Lambda}} = \{\Xi \in \hat{\hat{V}} : \text{Im}\Xi(\ell) \in \mathbb{Z} \text{ for all } \ell \in \hat{\Lambda}\}$  is the lattice defining  $\hat{\hat{X}}$  by definition 5.1.1 applied to  $\hat{X}$ . First,  $K(\Lambda) \subseteq \hat{\hat{\Lambda}}$ : for  $\lambda \in \Lambda$  and  $\ell \in \hat{\Lambda}$ ,

$$\text{Im}(K(\lambda)(\ell)) = \text{Im}(\overline{\ell(\lambda)}) = -\text{Im}\ell(\lambda) \in \mathbb{Z}$$

since  $\text{Im}\ell(\lambda) \in \mathbb{Z}$  by definition of  $\hat{\Lambda}$ . Thus  $K(\lambda) \in \hat{\hat{\Lambda}}$ . For the reverse inclusion, both  $K(\Lambda)$  and  $\hat{\hat{\Lambda}}$  are full lattices in  $\hat{\hat{V}}$  (the first because  $K$  is an  $\mathbb{R}$ -linear isomorphism carrying the full lattice  $\Lambda$ , the second by definition 5.1.1), and  $K(\Lambda) \subseteq \hat{\hat{\Lambda}}$ ; a containment of full lattices is an equality once the index is 1. The index is 1 because dualizing preserves lattice covolume up to the fixed pairing: taking the imaginary-part pairing  $\Phi$  of Step 1 of the proof of definition 5.1.1,  $\hat{\Lambda}$  is  $\mathbb{R}$ -linearly identified with  $\Lambda^\vee$ , and  $\hat{\hat{\Lambda}}$  with  $(\Lambda^\vee)^\vee$ , and the canonical identification  $(\Lambda^\vee)^\vee = \Lambda$  of the double  $\mathbb{Z}$ -dual of a lattice sends  $\lambda$  to evaluation at  $\lambda$ , which is exactly  $K(\lambda)$  under  $\Phi$ . Hence  $K(\Lambda) = \hat{\hat{\Lambda}}$ , and  $K$  descends to an isomorphism of complex tori  $\kappa_X: X \xrightarrow{\sim} \hat{\hat{X}}$ .

*Step 4: Naturality.* Let  $f: X \rightarrow Y$  be induced by  $F: V \rightarrow W$ . Then  $\hat{f}$  is induced by  $\hat{F} = (-) \circ F$  (proposition 5.1.4), and  $\hat{\hat{f}}$  by  $\hat{\hat{F}} = (-) \circ \hat{F}$ . For  $v \in V$  and  $\ell \in \hat{W}$ ,

$$(\hat{\hat{F}}(K_X(v)))(\ell) = K_X(v)(\hat{F}(\ell)) = K_X(v)(\ell \circ F) = \overline{(\ell \circ F)(v)} = \overline{\ell(F(v))} = K_Y(F(v))(\ell)$$

so  $\hat{\hat{F}} \circ K_X = K_Y \circ F$  on universal covers, hence  $\hat{\hat{f}} \circ \kappa_X = \kappa_Y \circ f$  on the tori. The isomorphism  $\kappa_X$  is therefore natural in  $X$ , completing the proof.

Biduality closes the construction: the passage to the dual torus is an involution on complex tori, canonically returning  $X$  from  $\hat{\hat{X}}$ . The conjugation in the formula  $v \mapsto (\ell \mapsto \overline{\ell(v)})$  is forced by the two layers of antilinearity, each dual being built from conjugate-linear functionals, and it is what makes  $\kappa_X$  a  $\mathbb{C}$ -linear rather than  $\mathbb{R}$ -linear isomorphism. Naturality is the substantive part: it means the identification  $X = \hat{\hat{X}}$  is compatible with all homomorphisms simultaneously, so one may treat a complex torus and its bidual as the same object without ever tracking the isomorphism. This is what

licenses the standard abuses of the sequel, writing  $\hat{\hat{X}} = X$ , regarding the dual of  $\hat{f}: \hat{Y} \rightarrow \hat{X}$  as  $f$  again, and treating the polarization isogeny  $\varphi_L: X \rightarrow \hat{X}$  as a map whose own dual  $\hat{\varphi}_L: \hat{\hat{X}} = X \rightarrow \hat{X}$  can be compared with it (proposition 5.3.6).

The dual torus is the second complex torus canonically attached to  $X$ , after  $X$  itself, and it is the object toward which the theory now turns. The functor  $X \mapsto \hat{X}$  converts the degree-zero Picard group, a group of line bundles, into a variety that varies functorially and contravariantly with  $X$ , and biduality makes that variety carry  $X$  back. A polarization is the datum of a homomorphism  $X \rightarrow \hat{X}$  with a positivity property, so with the dual torus in hand the polarization can finally be read as a map, the isogeny  $\varphi_L$  constructed in section 5.3; the intervening section 5.2 develops the isogenies and torsion in which  $\varphi_L$  takes its values.

### Exercise 5.1.1

1. Let  $V = \mathbb{C}$  and let  $\ell: \mathbb{C} \rightarrow \mathbb{C}$  be the antilinear functional  $\ell(v) = \bar{v}$ , so  $\ell = \ell_w$  with  $w = 1$  in the notation of example 5.1.2. For  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$  and  $\tau = i$ , decide whether  $\ell \in \hat{\Lambda}$  by checking the two integrality conditions of definition 5.1.1. Do the same for  $\tau = \frac{1}{2} + i$ .
2. The semicharacter attached to  $\ell$  in proposition 5.1.3 is  $\chi_\ell(\lambda) = e^{2\pi i \operatorname{Im} \ell(\lambda)}$ . Starting from the identity  $\operatorname{Im} z = \frac{1}{2i}(z - \bar{z})$ , rewrite this in terms of the difference  $\ell(\lambda) - \overline{\ell(\lambda)}$ , and show that  $\chi_\ell(\lambda)$  has modulus 1. Then show that the assignment  $\ell \mapsto \chi_\ell$  depends only on the class of  $\ell$  modulo  $\hat{\Lambda}$ , so that it is well defined on  $\hat{X} = \hat{V}/\hat{\Lambda}$ .
3. Let  $\Phi: \hat{V} \rightarrow \operatorname{Hom}_{\mathbb{R}}(V, \mathbb{R})$ ,  $\ell \mapsto \operatorname{Im} \ell$ , be the imaginary-part map from the proof of the lattice claim in definition 5.1.1. Show that  $\Phi$  is an isomorphism of complex tori  $\hat{X} \xrightarrow{\sim} \operatorname{Hom}_{\mathbb{R}}(V, \mathbb{R})/\Lambda^\vee$  of *real* Lie groups, and give the complex structure that  $\operatorname{Hom}_{\mathbb{R}}(V, \mathbb{R})$  must carry for  $\Phi$  to be an isomorphism of *complex* tori. Explain why this complex structure is not the one induced from an identification of  $\operatorname{Hom}_{\mathbb{R}}(V, \mathbb{R})$  with the real dual of the underlying real vector space of  $V$  in the naive way.
4. Let  $X = V/\Lambda$  and let  $n \geq 1$ . The multiplication-by- $n$  map  $n_X: X \rightarrow X$  is induced by  $n \cdot \operatorname{id}_V$ . Compute its dual  $\widehat{n_X}: \hat{X} \rightarrow \hat{X}$  using proposition 5.1.4, and conclude that  $\widehat{n_X} = n_{\hat{X}}$ . Using theorem 5.1.5, deduce that the natural isomorphism  $\kappa_X: X \rightarrow \hat{\hat{X}}$  intertwines  $n_X$  with  $n_{\hat{\hat{X}}}$ .
5. Let  $f: X \rightarrow Y$  be a surjective homomorphism of complex tori, induced by  $F: V \rightarrow W$ . Prove that the antilinear transpose  $\hat{F}: \hat{W} \rightarrow \hat{V}$  of proposition 5.1.4 is injective, and conclude that the dual homomorphism  $\hat{f}: \hat{Y} \rightarrow \hat{X}$  has finite kernel. State, and prove using biduality (theorem 5.1.5), the reversed statement: if  $g: Z \rightarrow X$  is a homomorphism with injective inducing map  $G: V_Z \rightarrow V$ , then the dual  $\hat{g}: \hat{X} \rightarrow \hat{Z}$  is surjective.
6. Let  $f: X \rightarrow Y$  be an isogeny of complex tori, that is, a surjective homomorphism with finite kernel; write  $F: V \rightarrow W$  for the inducing  $\mathbb{C}$ -linear map. Prove that the dual  $\hat{f}: \hat{Y} \rightarrow \hat{X}$  is again an isogeny, and that  $\ker \hat{f}$  and  $\ker f$  have the same order. Deduce from theorem 5.1.5 that  $f$  and  $\hat{\hat{f}}$  have kernels of the same order, consistently with  $\hat{\hat{f}}$  being identified with  $f$ .

## 5.2 Isogenies, Torsion, and the Weil Pairing

The dual torus of section 5.1 records how line bundles on  $X$  vary; the present section records how abelian varieties map to one another. The morphisms that matter for the classification of abelian varieties are not the isomorphisms, of which there are few, but the isogenies: surjective homomorphisms with finite kernel. Two abelian varieties that are isogenous share almost everything of arithmetic interest (their endomorphism algebras, their  $\ell$ -adic representations, their  $L$ -functions), differing only by a finite amount of torsion data. This is why isogeny, not isomorphism, is the working equivalence relation of the subject, and it is the analytic shadow of the isogeny theory that Part II will rebuild scheme-theoretically over an arbitrary field.

Two constructions organize this section. The first is the torsion subgroup  $X[n]$ , the kernel of multiplication by  $n$ , which turns out to be a free  $\mathbb{Z}/n$ -module of rank  $2g$ ; running over all  $n$  (or all powers of a fixed prime) it is the analytic precursor of the Tate module of Part II. The second is the Weil pairing, an alternating pairing on torsion that a polarization manufactures out of the commutator of the theta group of section 4.1. Over  $\mathbb{C}$  both constructions are transparent: torsion is  $\frac{1}{n}\Lambda/\Lambda$ , and the pairing is  $e^{2\pi i E}$  read off from the alternating form  $E = \text{Im}H$ .

### Definition 5.2.1: Isogeny

An *isogeny*  $f: X \rightarrow Y$  between complex tori is a surjective homomorphism with finite kernel. Equivalently, it is a homomorphism  $f: X \rightarrow Y$  with  $\dim X = \dim Y$  and  $\ker f$  finite. The complex tori  $X$  and  $Y$  are *isogenous*, written  $X \sim Y$ , if there exists an isogeny  $X \rightarrow Y$ .

The equivalence of the two formulations is worth spelling out, since it is the reason isogenies can be detected dimensionally. Recall from proposition 1.1.7 that a homomorphism  $f: X \rightarrow Y$  of complex tori  $X = V/\Lambda$  and  $Y = V'/\Lambda'$  is induced by a  $\mathbb{C}$ -linear map  $F: V \rightarrow V'$  with  $F(\Lambda) \subseteq \Lambda'$ . If  $f$  is surjective with finite kernel, then  $F$  is surjective (its image is a complex subtorus of full measure, hence all of  $Y$ ) and injective (its kernel is a subspace of  $V$  contained in the preimage of the finite set  $\ker f$ , hence zero), so  $F$  is an isomorphism and  $\dim X = \dim V = \dim V' = \dim Y$ . Conversely, if  $\dim X = \dim Y$  and  $\ker f$  is finite, then  $F: V \rightarrow V'$  is a  $\mathbb{C}$ -linear map between spaces of equal dimension whose kernel is contained in the finite set  $F^{-1}(\Lambda') \cap (\text{preimage of } \ker f)$ , forcing  $F$  injective and hence bijective; surjectivity of  $f$  follows because  $F(V) = V'$  meets every coset of  $\Lambda'$ .

The finiteness of the kernel is what makes an isogeny quasi-invertible, and the order of that kernel is its degree, the numerical invariant that governs it.

**Proposition 5.2.2: The Degree of an Isogeny**

Let  $f: X \rightarrow Y$  be an isogeny of complex tori, induced by the  $\mathbb{C}$ -linear isomorphism  $F: V \xrightarrow{\sim} V'$  with  $F(\Lambda) \subseteq \Lambda'$ . Then  $F(\Lambda)$  is a full sublattice of  $\Lambda'$  of finite index, and

$$\ker f \cong F^{-1}(\Lambda')/\Lambda \cong \Lambda'/F(\Lambda)$$

is a finite abelian group. Its order

$$\deg f := |\ker f| = [\Lambda' : F(\Lambda)]$$

is the *degree* of  $f$ . It is multiplicative: if  $g: Y \rightarrow Z$  is a further isogeny, then  $\deg(g \circ f) = (\deg g)(\deg f)$ .

**Proof:**

Since  $F$  is a  $\mathbb{C}$ -linear isomorphism, it carries the full lattice  $\Lambda$  (of rank  $2g$ ) to the full lattice  $F(\Lambda) \subseteq \Lambda'$ , also of rank  $2g$ ; two full lattices in the same space are commensurable, so  $[\Lambda' : F(\Lambda)]$  is finite.

*Step 1: Identifying the kernel.* A point  $v + \Lambda \in X = V/\Lambda$  lies in  $\ker f$  precisely when  $F(v) \in \Lambda'$ , that is, when  $v \in F^{-1}(\Lambda')$ . Thus

$$\ker f = F^{-1}(\Lambda')/\Lambda$$

as subgroups of  $V/\Lambda$ . Because  $F$  is injective, it maps the sublattice  $\Lambda \subseteq F^{-1}(\Lambda')$  isomorphically onto  $F(\Lambda) \subseteq \Lambda'$  and maps  $F^{-1}(\Lambda')$  isomorphically onto  $\Lambda'$ . Passing to quotients,

$$F^{-1}(\Lambda')/\Lambda \xrightarrow{\sim} \Lambda'/F(\Lambda)$$

is an isomorphism of abelian groups, so  $\ker f \cong \Lambda'/F(\Lambda)$ , a finite group of order  $[\Lambda' : F(\Lambda)]$ .

*Step 2: Multiplicativity.* Let  $g: Y \rightarrow Z = V''/\Lambda''$  be induced by  $G: V' \xrightarrow{\sim} V''$  with  $G(\Lambda') \subseteq \Lambda''$ . Then  $g \circ f$  is induced by  $G \circ F$ , and the index of  $(G \circ F)(\Lambda)$  in  $\Lambda''$  factors through the intermediate lattice  $G(\Lambda')$ :

$$[\Lambda'' : (G \circ F)(\Lambda)] = [\Lambda'' : G(\Lambda')] \cdot [G(\Lambda') : G(F(\Lambda))]$$

Since  $G$  is injective,  $[G(\Lambda') : G(F(\Lambda))] = [\Lambda' : F(\Lambda)]$ , and the right-hand side is  $(\deg g)(\deg f)$ , completing the proof.

The identification  $\ker f \cong \Lambda'/F(\Lambda)$  makes the degree computable directly from lattices: an isogeny is, on lattices, an inclusion of one full lattice into another, and its degree is the index of that inclusion. This is the analytic content of the statement that an isogeny is a finite-to-one covering of complex tori of the same dimension, the covering being  $\deg f$  sheeted.

Isogeny is genuinely an equivalence relation, and the nontrivial point is symmetry: an isogeny in one direction produces one in the reverse direction. The mechanism is that any isogeny can be inverted up to multiplication by an integer.

**Proposition 5.2.3: Isogeny is an Equivalence Relation**

The relation  $X \sim Y$  of definition 5.2.1 is reflexive, symmetric, and transitive. Concretely, if  $f: X \rightarrow Y$  is an isogeny of degree  $d$ , then there is an isogeny  $g: Y \rightarrow X$  with

$$g \circ f = [d]_X \quad \text{and} \quad f \circ g = [d]_Y$$

where  $[d]$  denotes multiplication by  $d$ ; in particular  $\deg g = d^{2g-1}$ .

**Proof:**

Reflexivity is the identity, and transitivity follows from the multiplicativity of degree in proposition 5.2.2, since a composite of surjections with finite kernels is again one. For symmetry, let  $f$  be induced by  $F: V \xrightarrow{\sim} V'$  with  $F(\Lambda) \subseteq \Lambda'$  of index  $d = \deg f$ .

The kernel  $\ker f = F^{-1}(\Lambda')/\Lambda$  is a finite abelian group of order  $d$ , so it is annihilated by  $d$ : every element has order dividing  $|\ker f| = d$  by Lagrange. This means  $d \cdot F^{-1}(\Lambda') \subseteq \Lambda$ , that is,  $F^{-1}(d\Lambda') \subseteq \Lambda$ , hence  $d\Lambda' \subseteq F(\Lambda)$ . Consider the  $\mathbb{C}$ -linear isomorphism  $G := dF^{-1}: V' \xrightarrow{\sim} V$ . Then

$$G(\Lambda') = dF^{-1}(\Lambda') \subseteq \Lambda$$

so  $G$  induces a homomorphism  $g: Y \rightarrow X$ . By construction  $g \circ f$  is induced by  $G \circ F = dF^{-1}F = d \cdot \text{id}_V$ , which induces  $[d]_X$ ; and  $f \circ g$  is induced by  $F \circ G = F(dF^{-1}) = d \cdot \text{id}_{V'}$ , which induces  $[d]_Y$ . Both composites are isogenies (degree  $d^{2g}$  by the torsion count below), so  $g$  is surjective with finite kernel, hence an isogeny. Finally  $\deg(g \circ f) = \deg[d]_X = d^{2g}$  by proposition 5.2.4 below, and  $\deg(g \circ f) = (\deg g)(\deg f) = (\deg g)d$ , so  $\deg g = d^{2g}/d = d^{2g-1}$ , as we sought to show.

The isogeny  $g$  of proposition 5.2.3 is a *quasi-inverse*: it inverts  $f$  after clearing the denominator  $d$ . This is exactly the sense in which isogenous abelian varieties are “the same up to a finite amount of data”, and it is why one inverts isogenies only after tensoring the endomorphism ring with  $\mathbb{Q}$ , a passage taken up in section 6.1. The proof leaned on the degree of multiplication by  $n$ , which is the next order of business.

Among isogenies the multiplication maps  $[n]$  recur most often. Understanding the kernel of  $[n]$  amounts to understanding the torsion of  $X$ , and this is a computation the lattice description settles at once.

**Proposition 5.2.4: The Structure of Torsion**

Let  $X = V/\Lambda$  be a complex torus of dimension  $g$ , and let  $n \geq 1$  be an integer. The multiplication map  $[n]: X \rightarrow X$  is an isogeny, and its kernel, the  *$n$ -torsion subgroup*, is

$$X[n] := \ker[n] = \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$$

a free  $\mathbb{Z}/n\mathbb{Z}$ -module of rank  $2g$ , of order  $n^{2g}$ . Consequently

$$\deg[n] = |X[n]| = n^{2g}$$

**Proof:**

Multiplication by  $n$  on  $X = V/\Lambda$  is induced by the  $\mathbb{C}$ -linear isomorphism  $n \cdot \text{id}_V : V \rightarrow V$ , which certainly carries  $\Lambda$  into itself, so  $[n]$  is a homomorphism induced by an invertible linear map, hence dimensions agree.

A point  $v + \Lambda$  lies in  $\ker[n]$  precisely when  $nv \in \Lambda$ , that is, when  $v \in \frac{1}{n}\Lambda$ . Therefore

$$X[n] = \frac{1}{n}\Lambda/\Lambda$$

Multiplication by  $n$  carries  $\frac{1}{n}\Lambda$  isomorphically onto  $\Lambda$  and carries  $\Lambda$  onto  $n\Lambda$ , so it induces an isomorphism

$$\frac{1}{n}\Lambda/\Lambda \xrightarrow{\sim} \Lambda/n\Lambda$$

Now  $\Lambda \cong \mathbb{Z}^{2g}$  is a free abelian group of rank  $2g$  (it is a full lattice in  $V = \mathbb{C}^g \cong \mathbb{R}^{2g}$ ), so

$$\Lambda/n\Lambda \cong \mathbb{Z}^{2g}/n\mathbb{Z}^{2g} \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$$

a free  $\mathbb{Z}/n\mathbb{Z}$ -module of rank  $2g$ . Its order is  $n^{2g}$ , which is finite, so  $[n]$  has finite kernel and is an isogeny with  $\deg[n] = |X[n]| = n^{2g}$ , completing the proof.

The formula  $\deg[n] = n^{2g}$  is the analytic prototype of a statement that in Part II requires the theorem of the cube for its proof: over an arbitrary field there is no lattice to count, and the degree of  $[n]$  must be extracted from the behavior of line bundles under  $[n]$  (in the algebraic development of Part II). Here the lattice makes it immediate. The identification  $X[n] = \frac{1}{n}\Lambda/\Lambda$  also exhibits the torsion as the reduction of the lattice: choosing a  $\mathbb{Z}$ -basis  $\lambda_1, \dots, \lambda_{2g}$  of  $\Lambda$ , the  $n^{2g}$  points  $\frac{1}{n}(a_1\lambda_1 + \dots + a_{2g}\lambda_{2g})$  with  $a_i \in \{0, 1, \dots, n-1\}$  are exactly the  $n$ -torsion points. Assembling the groups  $X[n]$  over the powers of a fixed prime  $\ell$ , along the maps  $[\ell]: X[\ell^{m+1}] \rightarrow X[\ell^m]$ , produces the analytic Tate module  $\varprojlim_m X[\ell^m] \cong (\mathbb{Z}_\ell)^{2g}$ , whose algebraic incarnation as a Galois representation is one of the central objects of Part II.

**Example 5.2.5: Torsion on an Elliptic Curve**

For  $g = 1$ , take  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ . The proposition gives

$$E[n] = \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^2$$

generated by the images of  $\frac{1}{n}$  and  $\frac{\tau}{n}$ , and  $\deg[n] = n^2$ . This matches the classical count for elliptic curves: the multiplication-by- $n$  map on an elliptic curve has degree  $n^2$ , and its kernel  $E[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$  when  $n$  is invertible in the ground field (see *Elliptic Curves* [4] for the algebraic proof via the division polynomials, which recovers the same group without reference to a lattice). The two generators  $\frac{1}{n}, \frac{\tau}{n}$  are the analytic avatars of the two “directions” in the  $n$ -torsion that the Weil pairing will detect below.

Every isogeny has a dual, and the duality is contravariant and degree-preserving. The construction is pullback of line bundles: an isogeny  $f: X \rightarrow Y$  pulls back a degree-zero bundle on  $Y$  to one on  $X$ , and under the identification  $\hat{X} \cong \text{Pic}^0(X)$  of proposition 5.1.3 this is a homomorphism  $\hat{f}: \hat{Y} \rightarrow \hat{X}$ . Analytically it is transparent.



**Proposition 5.2.6: The Dual Isogeny**

Let  $f: X \rightarrow Y$  be an isogeny of complex tori. Then the dual homomorphism  $\hat{f}: \hat{Y} \rightarrow \hat{X}$  of proposition 5.1.4 is again an isogeny, and

$$\deg \hat{f} = \deg f$$

Moreover there is an isogeny  $g: Y \rightarrow X$  with  $g \circ f = [\deg f]_X$  and  $f \circ g = [\deg f]_Y$ , so that  $\hat{f}$  and  $f$  fit into the duality of theorem 5.1.5: under  $\hat{\hat{X}} = X$ , the double dual  $\hat{\hat{f}}$  is  $f$  itself.

**Proof:**

Write  $f$  as induced by  $F: V \xrightarrow{\sim} V'$  with  $F(\Lambda) \subseteq \Lambda'$  of index  $d = \deg f$ . Recall from definition 5.1.1 that  $\hat{X} = \hat{V}/\hat{\Lambda}$  with  $\hat{V} = \overline{V}^*$  the space of  $\mathbb{C}$ -antilinear functionals and  $\hat{\Lambda} = \{\ell \in \hat{V} : \text{Im} \ell(\lambda) \in \mathbb{Z} \text{ for all } \lambda \in \Lambda\}$ , and from proposition 5.1.4 that  $\hat{f}$  is induced by the transpose  $\hat{F}: \hat{V}' \rightarrow \hat{V}$ ,  $\hat{F}(\ell) = \ell \circ F$ .

*Step 1:  $\hat{F}$  is an isomorphism.* Since  $F: V \rightarrow V'$  is a  $\mathbb{C}$ -linear isomorphism, precomposition  $\ell \mapsto \ell \circ F$  is a  $\mathbb{C}$ -linear isomorphism  $\overline{V}'^* \rightarrow \overline{V}^*$ , that is,  $\hat{F}: \hat{V}' \xrightarrow{\sim} \hat{V}$ . It carries  $\hat{\Lambda}'$  into  $\hat{\Lambda}$ : if  $\ell \in \hat{\Lambda}'$ , then for every  $\lambda \in \Lambda$  one has  $\text{Im}(\ell \circ F)(\lambda) = \text{Im} \ell(F\lambda) \in \mathbb{Z}$  because  $F\lambda \in F(\Lambda) \subseteq \Lambda'$ . Thus  $\hat{F}(\hat{\Lambda}') \subseteq \hat{\Lambda}$  and  $\hat{f}$  is a homomorphism induced by a linear isomorphism, hence has  $\dim \hat{Y} = \dim \hat{X}$  and finite kernel, that is, an isogeny.

*Step 2: The degree is preserved.* By proposition 5.2.2,  $\deg \hat{f} = [\hat{\Lambda} : \hat{F}(\hat{\Lambda}')] = [\hat{\Lambda} : \hat{F}(\hat{\Lambda}')] = [\Lambda^\vee : \hat{F}((\Lambda')^\vee)] = [\Lambda' : F(\Lambda)] = \deg f$ . The pairing  $\langle \ell, v \rangle = \text{Im} \ell(v)$  identifies  $\hat{\Lambda}$  with the dual lattice  $\Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$  and  $\hat{\Lambda}'$  with  $(\Lambda')^\vee$ , under which  $\hat{F}$  becomes the transpose of the inclusion  $F(\Lambda) \hookrightarrow \Lambda'$ . For a finite-index inclusion  $M \hookrightarrow N$  of free  $\mathbb{Z}$ -modules of the same rank, the transpose  $N^\vee \hookrightarrow M^\vee$  has the same index, since the elementary divisors  $d_1, \dots, d_{2g}$  of the inclusion (with  $\prod d_i = [N : M]$ ) are unchanged under transposition. Hence

$$\deg \hat{f} = [\Lambda^\vee : \hat{F}((\Lambda')^\vee)] = [\Lambda' : F(\Lambda)] = \deg f$$

*Step 3: The quasi-inverse and biduality.* The isogeny  $g$  with  $g \circ f = [d]_X$  and  $f \circ g = [d]_Y$  is the one produced in proposition 5.2.3. Dualizing  $g \circ f = [d]_X$  and using contravariant functoriality (proposition 5.1.4) together with  $\widehat{[d]} = [d]$ , one gets  $\hat{f} \circ \hat{g} = [d]_{\hat{X}}$ , confirming  $\hat{f}$  is an isogeny with quasi-inverse  $\hat{g}$ . Under the canonical isomorphism  $\hat{\hat{X}} = X$  of theorem 5.1.5, the double transpose  $\hat{\hat{F}} = F$ , so  $\hat{\hat{f}} = f$ , completing the proof.

The dual isogeny is the analytic model for the operation that in Part II is defined by the universal property of the Picard scheme, and the equality  $\deg \hat{f} = \deg f$  is the first hint of a duality that a polarization will make into an honest self-duality: the polarization isogeny  $\varphi_L$  of section 5.3 satisfies  $\hat{\varphi}_L = \varphi_L$ , so an abelian variety with a polarization is nearly self-dual, and exactly self-dual when the polarization is principal.

The dual isogeny puts  $\hat{Y}[n]$  in duality with  $X[n]$  once a polarization is present, and this duality is the

Weil pairing. Its cleanest source is the theta group of section 4.1: the commutator of the Heisenberg extension is an alternating pairing on  $K(L)$ , and specializing  $L$  to  $L^n$  realizes the  $n$ -torsion inside  $K(L^n)$  and reads off a pairing valued in the  $n$ -th roots of unity. The following definition records the pairing between  $X[n]$  and the  $n$ -torsion of the dual; the polarization then folds it into a pairing on  $X[n]$  alone.

**Definition 5.2.7: The Weil Pairing**

Let  $X = V/\Lambda$  be a complex torus of dimension  $g$  and  $n \geq 1$ . Write  $\mu_n = \{z \in \mathbb{C}^* : z^n = 1\}$  for the group of  $n$ -th roots of unity. The *Weil pairing* is the map

$$e_n : X[n] \times \hat{X}[n] \longrightarrow \mu_n$$

defined as follows. Identify  $\hat{X} = \hat{V}/\hat{\Lambda}$  with the antilinear-functional torus of definition 5.1.1. A point of  $X[n]$  is  $\bar{v} = v + \Lambda$  with  $nv \in \Lambda$ , and a point of  $\hat{X}[n]$  is  $\bar{\ell} = \ell + \hat{\Lambda}$  with  $n\ell \in \hat{\Lambda}$ . Set

$$e_n(\bar{v}, \bar{\ell}) := \exp(2\pi i n \operatorname{Im} \ell(v))$$

This is well defined and  $\mu_n$ -valued. Replacing  $v$  by  $v + \lambda$  with  $\lambda \in \Lambda$  changes the argument by  $2\pi i n \operatorname{Im} \ell(\lambda) = 2\pi i \operatorname{Im}(n\ell)(\lambda) \in 2\pi i \mathbb{Z}$ , since  $n\ell \in \hat{\Lambda}$ ; replacing  $\ell$  by  $\ell + \mu$  with  $\mu \in \hat{\Lambda}$  changes it by  $2\pi i \operatorname{Im} \mu(nv) \in 2\pi i \mathbb{Z}$ , since  $nv \in \Lambda$ . And  $(e_n(\bar{v}, \bar{\ell}))^n = \exp(2\pi i \operatorname{Im}((n\ell)(nv))) = 1$ , because  $n\ell \in \hat{\Lambda}$  and  $nv \in \Lambda$  give  $\operatorname{Im}((n\ell)(nv)) \in \mathbb{Z}$ .

The definition matches the theta-group construction of section 4.1. Recall from proposition 4.1.7 that a positive line bundle  $L = L(H, \chi)$  has theta group  $\mathcal{G}(L)$ , a central extension of  $K(L)$  by  $\mathbb{C}^*$ , whose commutator is the nondegenerate alternating pairing  $e^L(\bar{v}, \bar{w}) = e^{2\pi i E(v, w)}$  on  $K(L)$ . Applying this to  $L^n$  and using  $\ker \varphi_L = K(L)$  (which section 5.3 will establish) identifies  $\hat{X}[n]$  with a subgroup pairing against  $X[n]$ , and the pairing  $e_n$  above is the resulting duality pairing  $X[n] \times \hat{X}[n] \rightarrow \mu_n$ . The two routes, the antilinear-functional formula and the theta-group commutator, give the same  $e_n$ ; the formula above is the more explicit, and the theta-group description is the one that survives into the algebraic theory of Part II.

Interpreting  $e_n$  requires knowing it is a perfect duality, which is the content of the next result.

**Proposition 5.2.8: Nondegeneracy of the Weil Pairing**

The Weil pairing  $e_n : X[n] \times \hat{X}[n] \rightarrow \mu_n$  of definition 5.2.7 is bilinear and nondegenerate: it is a homomorphism in each variable, and the induced maps

$$X[n] \longrightarrow \operatorname{Hom}(\hat{X}[n], \mu_n), \quad \hat{X}[n] \longrightarrow \operatorname{Hom}(X[n], \mu_n)$$

are isomorphisms. It is also compatible with dual isogenies: for an isogeny  $f : X \rightarrow Y$  and  $n$ -torsion points  $\bar{v} \in X[n]$ ,  $\bar{\ell} \in \hat{Y}[n]$ ,

$$e_n^Y(f(\bar{v}), \bar{\ell}) = e_n^X(\bar{v}, \hat{f}(\bar{\ell}))$$

**Proof:**

*Step 1: Bilinearity.* The formula  $e_n(\bar{v}, \bar{\ell}) = \exp(2\pi i n \operatorname{Im}\ell(v))$  is additive in  $v$  because  $\ell$  is  $\mathbb{R}$ -linear, so  $\operatorname{Im}\ell(v_1 + v_2) = \operatorname{Im}\ell(v_1) + \operatorname{Im}\ell(v_2)$ ; and additive in  $\ell$  because  $\operatorname{Im}(\ell_1 + \ell_2)(v) = \operatorname{Im}\ell_1(v) + \operatorname{Im}\ell_2(v)$ . The exponential converts these sums to products, so  $e_n$  is a homomorphism in each variable.

*Step 2: Nondegeneracy.* Both  $X[n]$  and  $\hat{X}[n]$  have order  $n^{2g}$  by proposition 5.2.4 (applied to  $X$  and to  $\hat{X}$ , both of dimension  $g$ ), and  $\operatorname{Hom}(\hat{X}[n], \mu_n)$  has order  $n^{2g}$  as well since  $\mu_n \cong \mathbb{Z}/n\mathbb{Z}$ . So it suffices to show the left kernel is trivial: if  $\bar{v} \in X[n]$  pairs trivially with every  $\bar{\ell} \in \hat{X}[n]$ , then  $\bar{v} = 0$ . Suppose  $nv \in \Lambda$  and  $n \operatorname{Im}\ell(v) \in \mathbb{Z}$  for all  $\ell$  with  $n\ell \in \hat{\Lambda}$ . Writing  $w := nv \in \Lambda$ , the condition  $n \operatorname{Im}\ell(v) \in \mathbb{Z}$  becomes  $\operatorname{Im}\ell(w) \in \mathbb{Z}$  for all  $\ell \in \frac{1}{n}\hat{\Lambda}$ , since  $n \operatorname{Im}\ell(v) = \operatorname{Im}\ell(nv) = \operatorname{Im}\ell(w)$ . Recall  $\hat{\Lambda} = \{\ell : \operatorname{Im}\ell(\Lambda) \subseteq \mathbb{Z}\}$  is the dual lattice of  $\Lambda$  under the pairing  $(\ell, u) \mapsto \operatorname{Im}\ell(u)$ , which is a perfect  $\mathbb{R}$ -bilinear pairing  $\hat{V} \times V \rightarrow \mathbb{R}$  restricting to a perfect  $\mathbb{Z}$ -pairing  $\hat{\Lambda} \times \Lambda \rightarrow \mathbb{Z}$ . Under this perfect pairing,  $\operatorname{Im}\ell(w) \in \mathbb{Z}$  for all  $\ell \in \frac{1}{n}\hat{\Lambda}$  forces  $w \in n\Lambda$ . Hence  $v \in \Lambda$  and  $\bar{v} = 0$  in  $X[n]$ . The left-hand map is thus injective between groups of the same order, an isomorphism; by symmetry so is the right-hand map.

*Step 3: Compatibility with dual isogenies.* Let  $f$  be induced by  $F: V \rightarrow V'$ , so  $\hat{f}$  is induced by  $\hat{F}(\ell) = \ell \circ F$  (proposition 5.1.4). For  $\bar{v} \in X[n]$  and  $\bar{\ell} \in \hat{Y}[n]$ ,

$$e_n^Y(f\bar{v}, \bar{\ell}) = \exp(2\pi i n \operatorname{Im}\ell(Fv)) = \exp(2\pi i n \operatorname{Im}(\ell \circ F)(v)) = e_n^X(\bar{v}, \hat{f}\bar{\ell})$$

which is the stated adjunction, completing the proof.

Nondegeneracy is what makes the Weil pairing an instrument rather than a curiosity: it identifies the dual torsion  $\hat{X}[n]$  with the Pontryagin dual  $\operatorname{Hom}(X[n], \mu_n)$ , so that any structure carried by  $X[n]$  is reflected in  $\hat{X}[n]$  and conversely. The adjunction  $e_n^Y(f\bar{v}, \bar{\ell}) = e_n^X(\bar{v}, \hat{f}\bar{\ell})$  says the Weil pairing turns an isogeny into its dual, which is the analytic origin of the compatibility that governs the algebraic Weil pairing of Part II. When a polarization  $\varphi_L: X \rightarrow \hat{X}$  is present, composing  $e_n$  with  $\varphi_L$  in the second slot produces an alternating pairing  $X[n] \times X[n] \rightarrow \mu_n$ ,  $(\bar{v}, \bar{w}) \mapsto e_n(\bar{v}, \varphi_L(\bar{w}))$ ; by definition 5.2.7 this equals  $\exp(2\pi i n E(w, v))$  with  $E = \operatorname{Im}H$ , recovering the alternating incarnation of the polarization on torsion. The pairing a polarization thereby places on torsion is the bilinear structure that all subsequent classification (the Rosati involution, Albert's types) will exploit. For  $g = 1$  this is the classical Weil pairing  $e_n: E[n] \times E[n] \rightarrow \mu_n$  of *Elliptic Curves* [4], alternating and nondegenerate, whose values encode the Galois action on torsion.

The example that ties the section together is multiplication by  $n$  itself, the isogeny whose degree, kernel, and self-duality have been computed piecemeal above.

**Example 5.2.9: Multiplication by  $n$  as an Isogeny**

Fix  $X = V/\Lambda$  of dimension  $g$  and  $n \geq 1$ . Multiplication by  $n$ ,

$$[n]: X \longrightarrow X$$

is an isogeny with the following features, drawn together from the section.

1. *Degree.* By proposition 5.2.4,  $\deg[n] = |X[n]| = n^{2g}$ . For  $g = 1$  this is  $n^2$ , matching example 5.2.5 and the classical elliptic-curve degree.
2. *Kernel.*  $\ker[n] = X[n] = \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$ , the full  $n$ -torsion.

3. *Self-duality.* The dual of  $[n]: X \rightarrow X$  is  $\widehat{[n]} = [n]: \hat{X} \rightarrow \hat{X}$ : indeed  $[n]$  is induced by  $n \cdot \text{id}_V$ , whose transpose on  $\hat{V} = \overline{V}^*$  is again  $n \cdot \text{id}_{\hat{V}}$ , so  $\widehat{[n]} = [n]$  and  $\deg \widehat{[n]} = n^{2g} = \deg [n]$ , consistent with proposition 5.2.6.
4. *Quasi-inverse.* The quasi-inverse of  $[n]$  from proposition 5.2.3 is  $[n]$  itself:  $[n] \circ [n] = [n^2] = [\deg [n]/n^{2g-1}]$  is not the identity, but  $[n]$  becomes invertible after inverting  $n$ , that is, in  $\text{End}(X) \otimes \mathbb{Q}$ , where  $[n]^{-1} = [1/n]$ . This is the concrete reason isogenies are inverted only rationally.

The multiplication maps  $[n]$  are the source of all the torsion  $X[n]$ , hence of the Tate module and of the Weil pairing: every  $n$ -torsion point is a point of  $\ker [n]$ , and the pairing  $e_n$  of definition 5.2.7 lives on this kernel and its dual. The endomorphism  $[n]$  commutes with every homomorphism of tori, so its kernels  $X[n]$  carry a canonical action of  $\text{End}(X)$ , and this is the module structure through which the endomorphism algebra of section 6.1 will act on torsion.

Isogeny is the working equivalence of abelian varieties: it collapses the isomorphism classes into a coarser, more tractable classification in which every abelian variety is a product of simple ones (section 5.4), and in which the endomorphism ring becomes a semisimple algebra over  $\mathbb{Q}$ . The Weil pairing supplies the other half of the structure. Torsion alone is a bare  $(\mathbb{Z}/n)^{2g}$ ; a polarization, through  $\varphi_L$  and the pairing  $e_n$ , endows it with a nondegenerate alternating form, and it is this pairing, not the torsion group itself, that a polarization contributes. The dual torus  $\hat{X}$  of definition 5.1.1 is the object that makes the pairing a duality:  $e_n$  pairs  $X[n]$  against  $\hat{X}[n]$  perfectly, and a polarization is what turns that duality into a self-duality of  $X$ .

### Exercise 5.2.1

1. Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ , and let  $F = 2 \cdot \text{id}_{\mathbb{C}}$ , which carries  $\Lambda$  into  $\Lambda$  and induces the isogeny  $[2]: E \rightarrow E$ . Compute the degree  $[2]$  directly as the lattice index  $[\Lambda : 2\Lambda]$  of proposition 5.2.2, exhibit the kernel  $E[2] = \frac{1}{2}\Lambda/\Lambda$  as an explicit set of four coset representatives, and confirm the answer against example 5.2.5.
2. Consider the isogeny  $f: E \rightarrow E'$  of elliptic curves induced by  $F = \text{id}_{\mathbb{C}}: \mathbb{C} \rightarrow \mathbb{C}$  carrying the lattice  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$  into the lattice  $\Lambda' = \frac{1}{3}\mathbb{Z} + \mathbb{Z}\tau$ . Compute  $\deg f$  and identify  $\ker f$  as an abstract group using proposition 5.2.2, then produce the quasi-inverse  $g: E' \rightarrow E$  of proposition 5.2.3 explicitly as a linear map and verify  $g \circ f = [\deg f]_E$ .
3. Let  $X = V/\Lambda$  have dimension  $g$ , and let  $m, n \geq 1$  be coprime. Prove that  $X[mn] \cong X[m] \times X[n]$  as abelian groups, and deduce that  $X[m] \cap X[n] = 0$  inside  $X$ . Use the description of torsion from proposition 5.2.4.
4. Take  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$  and  $n = 2$ . The dual torus is  $\hat{E} = \hat{V}/\hat{\Lambda}$  with  $\hat{V} = \overline{V}^*$ , and  $\hat{E}[2] \cong (\mathbb{Z}/2\mathbb{Z})^2$ . Using the formula of definition 5.2.7, show that the Weil pairing  $e_2: E[2] \times \hat{E}[2] \rightarrow \mu_2$  is nondegenerate by computing its values on a basis. Concretely, let  $\bar{v}_1, \bar{v}_2 \in E[2]$  be the classes of  $\frac{1}{2}$  and  $\frac{\tau}{2}$ , and let  $\bar{\ell}_1, \bar{\ell}_2 \in \hat{E}[2]$  be antilinear functionals with  $\text{Im} \ell_i(\frac{1}{2}), \text{Im} \ell_i(\frac{\tau}{2})$  chosen so that  $(\ell_i)$  is dual to  $(v_j)$ ; write out the  $2 \times 2$  matrix of values  $e_2(\bar{v}_j, \bar{\ell}_i)$  and check it has trivial kernel.
5. Let  $\varphi: X \rightarrow \hat{X}$  be a homomorphism of complex tori that is self-dual in the sense  $\hat{\varphi} = \varphi$  under the biduality  $\hat{\hat{X}} = X$ . Define the pairing  $E_n: X[n] \times X[n] \rightarrow \mu_n$  by  $E_n(\bar{v}, \bar{w}) = e_n(\bar{v}, \varphi(\bar{w}))$ ,

where  $e_n$  is the Weil pairing of definition 5.2.7. Prove that  $E_n$  is skew-symmetric, that is,  $E_n(\bar{w}, \bar{v}) = E_n(\bar{v}, \bar{w})^{-1}$ , using the compatibility of  $e_n$  with dual isogenies from proposition 5.2.8 and the self-duality  $\hat{\varphi} = \varphi$ .

### 5.3 The Polarization Isogeny and Its Dual

A polarization has so far worn two faces. In definition 3.1.3 it was a positive Hermitian form  $H > 0$  on  $V$  whose imaginary part is integral on  $\Lambda$ ; equivalently, by the correspondence between Hermitian and alternating forms, a positive class in the Néron-Severi group. Both descriptions live on the linear-algebra side of the torus. The dual torus  $\hat{X}$  of definition 5.1.1 now supplies a third: a polarization is a homomorphism from  $X$  to  $\hat{X}$ , built from the line bundle it comes from. This section constructs that homomorphism, computes its kernel and degree, and identifies the one property that separates the maps to  $\hat{X}$  arising from polarizations from arbitrary homomorphisms  $X \rightarrow \hat{X}$ , namely that they are symmetric under biduality.

The construction is the geometric shadow of the map  $x \mapsto t_x^* L \otimes L^{-1}$  that already appeared implicitly when the group  $\Lambda(L)$  was defined. Translating a line bundle  $L$  by a point  $x$  and comparing the result to  $L$  produces a line bundle whose first Chern class vanishes, hence a point of  $\text{Pic}^0(X) = \hat{X}$ . Letting  $x$  vary gives a homomorphism  $X \rightarrow \hat{X}$ , and over  $\mathbb{C}$  this homomorphism is an isogeny precisely when  $L$  is nondegenerate, an isomorphism precisely when the polarization is principal.

#### Definition 5.3.1: The polarization isogeny

Let  $L = L(H, \chi)$  be a line bundle on  $X = V/\Lambda$ . For  $x \in X$  let  $t_x: X \rightarrow X$  denote translation by  $x$ . The *polarization homomorphism* associated to  $L$  is

$$\varphi_L: X \rightarrow \hat{X} = \text{Pic}^0(X), \quad x \mapsto t_x^* L \otimes L^{-1}$$

When  $L$  is a polarization (that is,  $H > 0$  with  $E = \text{Im} H$  integral on  $\Lambda$ ),  $\varphi_L$  is called the *polarization isogeny*.

The map is well defined: for every  $x$  the class  $c_1(t_x^* L \otimes L^{-1})$  vanishes, because translation acts trivially on cohomology (a translation is homotopic to the identity on the torus), so  $t_x^* L$  and  $L$  have the same first Chern class and their difference lands in  $\text{Pic}^0(X) = \hat{X}$ . That  $\varphi_L$  is a homomorphism, rather than a map of sets, is the theorem of the square in its analytic form; it will be transparent from the linear-algebra description in proposition 5.3.3, where  $\varphi_L$  is induced by an honest  $\mathbb{R}$ -linear map of vector spaces.

**5.3.2 Convention: two maps named  $\varphi$**  The subscripted symbol  $\varphi_L$  of definition 5.3.1 is a homomorphism from  $X$  to the *dual torus*  $\hat{X}$ . It should not be confused with the map  $\varphi_{|L^{\otimes n}|}: X \rightarrow \mathbb{P}^N$  of chapter 4, the morphism to *projective space* defined by the linear system of a suitable power of  $L$  (the Lefschetz embedding, theorem 4.4.2). Both are written with the letter  $\varphi$  by long tradition, but their targets differ:  $\varphi_L$  lands in  $\hat{X}$ , while  $\varphi_{|L^{\otimes n}|}$  lands in a projective space. The context, and the presence or absence of a linear system in the subscript, always distinguishes them.

To compute with  $\varphi_L$  one needs its description on the universal cover, where translation and the

factor of automorphy become explicit. The result is that  $\varphi_L$  is linear, induced by the map sending  $v$  to the antilinear functional  $H(v, -)$ . This is where the Hermitian form of the polarization enters the picture concretely, and it is the identity from which every subsequent computation of kernel and degree flows.

**Proposition 5.3.3: Analytic form of the polarization homomorphism**

Let  $L = L(H, \chi)$  on  $X = V/\Lambda$ . The homomorphism  $\varphi_L: X \rightarrow \hat{X} = \hat{V}/\hat{\Lambda}$  is induced by the  $\mathbb{R}$ -linear map

$$\varphi_H: V \rightarrow \hat{V} = \overline{V}^*, \quad v \mapsto H(v, -)$$

Here  $H(v, -)$  is  $\mathbb{C}$ -antilinear in its argument, so it lies in  $\hat{V} = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$ , and  $\varphi_H$  carries  $\Lambda$  into  $\hat{\Lambda}$ , because  $\text{Im}H(v, -) = E(v, -)$  is integral on  $\Lambda$ .

**Proof:**

*Step 1: The target is  $\hat{V}$ .* Fix  $v \in V$ . The assignment  $w \mapsto H(v, w)$  is a function  $V \rightarrow \mathbb{C}$ . Since  $H$  is conjugate-linear in its second variable (definition 3.1.3 and the convention that a Hermitian form is linear in the first and conjugate-linear in the second),  $H(v, \alpha w) = \bar{\alpha} H(v, w)$  for  $\alpha \in \mathbb{C}$ , so  $H(v, -)$  is a  $\mathbb{C}$ -antilinear functional, an element of  $\hat{V} = \overline{V}^*$ . The map  $\varphi_H: v \mapsto H(v, -)$  is  $\mathbb{C}$ -linear in  $v$  because  $H$  is linear in its first variable, and in particular  $\mathbb{R}$ -linear.

*Step 2:  $\varphi_H$  carries  $\Lambda$  into  $\hat{\Lambda}$ .* Recall from definition 5.1.1 that

$$\hat{\Lambda} = \{ \ell \in \hat{V} : \text{Im} \ell(\lambda) \in \mathbb{Z} \text{ for all } \lambda \in \Lambda \}$$

Let  $\lambda \in \Lambda$ . For every  $\mu \in \Lambda$ ,

$$\text{Im}(\varphi_H(\lambda)(\mu)) = \text{Im}H(\lambda, \mu) = E(\lambda, \mu)$$

and  $E(\lambda, \mu) \in \mathbb{Z}$  because  $E = \text{Im}H$  is integral on  $\Lambda$  (this integrality is exactly the condition that  $H$  defines a class in  $\text{NS}(X)$ , definition 3.1.3). Hence  $\varphi_H(\lambda) \in \hat{\Lambda}$ , so  $\varphi_H(\Lambda) \subseteq \hat{\Lambda}$  and  $\varphi_H$  descends to a homomorphism of tori  $V/\Lambda \rightarrow \hat{V}/\hat{\Lambda}$ .

*Step 3: The descended map is  $\varphi_L$ .* It remains to identify the homomorphism induced by  $\varphi_H$  with  $x \mapsto t_x^* L \otimes L^{-1}$ . Fix  $x \in X$  with a lift  $u \in V$ . The bundle  $t_x^* L$  has factor of automorphy  $a_{(H, \chi)}(\lambda, v + u)$ , so  $t_x^* L \otimes L^{-1}$  has factor

$$\frac{a_{(H, \chi)}(\lambda, v + u)}{a_{(H, \chi)}(\lambda, v)} = \exp(\pi H(u, \lambda))$$

independent of  $v$ : the  $\chi(\lambda)$  and  $\frac{\pi}{2}H(\lambda, \lambda)$  factors cancel, and  $H(v + u, \lambda) - H(v, \lambda) = H(u, \lambda)$  by linearity in the first slot. This factor has trivial Hermitian part, so  $t_x^* L \otimes L^{-1} \in \text{Pic}^0(X)$ , and it is equivalent, modulo the holomorphic coboundary used in the modulus reduction of proposition 3.4.5, to its unitary part, the character

$$\lambda \mapsto \exp(2\pi i \text{Im}H(u, \lambda))$$

on  $\Lambda$  (using  $H(u, \lambda) - \overline{H(u, \lambda)} = 2i \text{Im}H(u, \lambda)$ ), which is  $\exp(2\pi i \text{Im}\varphi_H(u)(\lambda))$ . Under the identification  $\text{Pic}^0(X) \cong \hat{V}/\hat{\Lambda}$  of proposition 5.1.3, sending  $\ell \in \hat{V}$  to the character  $\lambda \mapsto \exp(2\pi i \text{Im}\ell(\lambda))$ , this is precisely the image of  $\varphi_H(u) \bmod \hat{\Lambda}$ . Thus  $\varphi_L(x) = \varphi_H(u) \bmod \hat{\Lambda}$ , as we sought to show.

The proposition turns a map defined by translating line bundles into a map of vector spaces given by a single formula,  $v \mapsto H(v, -)$ . Everything about  $\varphi_L$  that one wants to know, whether it is an isogeny, what its kernel is, how it interacts with the dual, is now a question about the linear map  $\varphi_H$  and the Hermitian form  $H$ . In particular  $\varphi_L$  is injective (as a map of tori) exactly when  $\varphi_H$  is injective on  $V$ , which holds iff  $H$  is nondegenerate as a Hermitian form; this is the analytic root of the equivalence between nondegeneracy of  $H$  and  $\varphi_L$  being an isogeny, made precise next.

The kernel of  $\varphi_L$  is now within reach. A point  $x$  lies in  $\ker \varphi_L$  iff a lift  $u$  of  $x$  satisfies  $\varphi_H(u) \in \hat{\Lambda}$ , and unwinding what membership in  $\hat{\Lambda}$  means recovers exactly the group  $\Lambda(L)$  from chapter 4. The kernel is therefore the finite group  $K(L)$ , and its order is the determinant of  $E$ , which is the square of the Pfaffian.

**Proposition 5.3.4: Kernel and degree of the polarization isogeny**

Let  $L = L(H, \chi)$  with  $H > 0$  of type  $(d_1, \dots, d_g)$  (definition 3.2.3). Then

$$\ker \varphi_L = \Lambda(L)/\Lambda = K(L)$$

(with  $\Lambda(L)$  and  $K(L)$  as in definition 4.1.2). Consequently  $\varphi_L$  is an isogeny iff  $H$  is nondegenerate, and in that case

$$\deg \varphi_L = |K(L)| = \det E = (d_1 d_2 \cdots d_g)^2$$

**Proof:**

By proposition 5.3.3,  $\varphi_L$  is induced by the  $\mathbb{R}$ -linear map  $\varphi_H: V \rightarrow \hat{V}$ ,  $v \mapsto H(v, -)$ . A point  $x = u \bmod \Lambda$  lies in  $\ker \varphi_L$  iff  $\varphi_H(u) \in \hat{\Lambda}$ , that is, iff

$$\operatorname{Im} H(u, \lambda) = E(u, \lambda) \in \mathbb{Z} \quad \text{for all } \lambda \in \Lambda$$

This is precisely the defining condition of  $\Lambda(L) = \{v \in V : E(v, \lambda) \in \mathbb{Z} \text{ for all } \lambda \in \Lambda\}$  from definition 4.1.2. Hence the set of lifts  $u$  with  $x \in \ker \varphi_L$  is  $\Lambda(L)$ , and

$$\ker \varphi_L = \Lambda(L)/\Lambda = K(L)$$

Since  $H$  is positive definite it is in particular nondegenerate, so  $E = \operatorname{Im} H$  is a nondegenerate alternating form; then  $\Lambda(L)$  is a full lattice containing  $\Lambda$  with finite index, and  $K(L)$  is finite. A homomorphism of tori of equal dimension with finite kernel is an isogeny (definition 5.2.1), and  $\varphi_L$  is such because  $\varphi_H$  is an  $\mathbb{R}$ -linear isomorphism  $V \rightarrow \hat{V}$  (both have real dimension  $2g$  and  $\varphi_H$  is injective by nondegeneracy of  $H$ ), hence surjective on tori. If instead  $H$  were degenerate,  $\varphi_H$  would have nonzero kernel in  $V$  and  $\varphi_L$  would have positive-dimensional kernel, hence fail to be an isogeny; this proves the stated equivalence.

For the degree,  $\deg \varphi_L = |\ker \varphi_L| = |K(L)|$  by proposition 5.2.2. The order of  $K(L)$  was computed in proposition 4.1.3: choosing a symplectic basis in which  $E$  has matrix  $\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$  with  $D = \operatorname{diag}(d_1, \dots, d_g)$  (definition 3.2.3), one finds  $K(L) \cong (\bigoplus_i \mathbb{Z}/d_i)^2$ , so

$$|K(L)| = (d_1 d_2 \cdots d_g)^2 = \det \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix} = \det E$$

completing the proof.



The kernel of the polarization isogeny is the same finite group  $K(L)$  that carried the commutator pairing of the theta group in chapter 4. The two appearances are not a coincidence: the pairing  $e^L$  on  $K(L)$  is, up to sign, the Weil pairing pulled through  $\varphi_L$ , and the nondegeneracy of that pairing is the nondegeneracy of  $H$  seen inside the finite group. The degree formula  $\deg \varphi_L = (d_1 \cdots d_g)^2 = (\text{Pf } E)^2$  makes the point that the polarization isogeny is never injective unless the polarization is principal: the failure of injectivity is measured by the type, and it disappears exactly when the type is  $(1, \dots, 1)$ . That last observation is important enough to record as the isogeny-theoretic definition of a polarization. A polarization, presented so far as a Hermitian form or a Néron-Severi class, can equally be given as the homomorphism  $\varphi_L$  to the dual, and the principal polarizations are exactly those for which  $\varphi_L$  is an isomorphism.

**Definition 5.3.5: Polarization as an isogeny to the dual**

A *polarization* on  $X$  may be presented as the isogeny  $\varphi_L: X \rightarrow \hat{X}$  attached (via definition 5.3.1) to an ample line bundle  $L$ ; its *degree* is  $\deg \varphi_L = (d_1 \cdots d_g)^2$ , where  $(d_1, \dots, d_g)$  is the type of the polarization (definition 3.2.3). The polarization is *principal* precisely when  $\varphi_L: X \rightarrow \hat{X}$  is an isomorphism, equivalently when its type is  $(1, \dots, 1)$ .

The equivalence of “principal” with “ $\varphi_L$  an isomorphism” is immediate from proposition 5.3.4:  $\varphi_L$  is an isomorphism iff  $\deg \varphi_L = 1$  iff  $d_1 \cdots d_g = 1$  iff every  $d_i = 1$ , since the  $d_i$  are positive integers. A principal polarization thus identifies  $X$  with its own dual, canonically once the polarization is fixed. This is the analytic prototype of the autoduality that Jacobians will exhibit in chapter 12: the theta divisor of a curve furnishes a principal polarization, hence an isomorphism  $\text{Jac}(C) \cong \widehat{\text{Jac}(C)}$ .

What distinguishes the homomorphisms  $X \rightarrow \hat{X}$  coming from line bundles among all homomorphisms to the dual is a symmetry condition. Biduality (theorem 5.1.5) identifies  $\hat{\hat{X}}$  canonically with  $X$ ; under this identification the dual of a homomorphism  $f: X \rightarrow \hat{X}$  is again a homomorphism  $\hat{f}: X \rightarrow \hat{X}$ , and one may ask whether  $\hat{f} = f$ . For  $f = \varphi_L$  the answer is yes: the polarization isogeny is symmetric. This is the property that makes  $\varphi_L$  a polarization rather than a general isogeny to the dual.

**Proposition 5.3.6: The polarization isogeny is symmetric**

Under the canonical identification  $\hat{\hat{X}} = X$  of theorem 5.1.5, the dual homomorphism  $\hat{\varphi}_L: \hat{X} = X \rightarrow \hat{X}$  equals  $\varphi_L$ . That is,  $\varphi_L$  is symmetric.

**Proof:**

Work with the analytic descriptions. By proposition 5.3.3,  $\varphi_L$  is induced by  $\varphi_H: V \rightarrow \hat{V}$ ,  $v \mapsto H(v, -)$ . The dual homomorphism  $\hat{\varphi}_L$  is, on universal covers, the transpose  $\varphi_H^*: \hat{V} \rightarrow \hat{V}$ , followed by the biduality identification  $\hat{\hat{V}} \cong V$ ; recall from proposition 5.1.4 that dualizing a homomorphism of tori corresponds analytically to taking the  $\mathbb{C}$ -antilinear transpose of the inducing map.

Unwind the transpose explicitly. The bidual space  $\hat{\hat{V}} = \overline{(\overline{V^*})}^*$  is canonically  $V$  via the evaluation pairing  $v \leftrightarrow (\ell \mapsto \overline{\ell(v)})$ , which is the analytic content of theorem 5.1.5. For  $v, w \in V$ , the



transpose  $\varphi_H^*$  acts by

$$(\varphi_H^*(w))(v) = \overline{\varphi_H(v)(w)} = \overline{H(v, w)} = H(w, v)$$

where the last equality is the Hermitian symmetry  $H(w, v) = \overline{H(v, w)}$ . Thus  $\varphi_H^*(w) = H(w, -) = \varphi_H(w)$  for every  $w \in V$ , so  $\varphi_H^* = \varphi_H$  under the biduality identification. Passing to the induced maps of tori,  $\hat{\varphi}_L = \varphi_L$ , as we sought to show.

The symmetry  $\hat{\varphi}_L = \varphi_L$  is the exact analytic image of the symmetry of the Hermitian form:  $H(w, v) = \overline{H(v, w)}$  says that  $H$  is Hermitian, and it is this identity, and no more, that forces  $\varphi_H^* = \varphi_H$ . A general homomorphism  $X \rightarrow \hat{X}$  corresponds to an  $\mathbb{R}$ -bilinear form on  $V$  with no symmetry constraint and need not be symmetric; the symmetric ones are exactly those coming from Hermitian forms, that is, from classes in  $\text{NS}(X)$ , and the positive symmetric ones are the polarizations. Symmetry is therefore the algebraic feature that will single out polarizations in Part II, where  $\varphi_L$  is redefined without reference to  $\mathbb{C}$  and the Hermitian form is no longer available; the condition  $\hat{\varphi}_L = \varphi_L$  survives verbatim and becomes the definition of a symmetric homomorphism to the dual.

The elliptic-curve case closes the circle with the earlier chapters, where a polarization first appeared as a positive real number times the standard form and then as a positive integer generating  $\text{NS}(E)$ . On an elliptic curve every principal polarization gives a degree-one map, hence an isomorphism, so the polarization isogeny recovers the classical fact that an elliptic curve is its own dual.

### Example 5.3.7: The polarization isogeny of an elliptic curve

Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ , and take the principal polarization: the Hermitian form

$$H(v, w) = \frac{v \bar{w}}{\text{Im}\tau}$$

with  $E_0 := \text{Im}H$  the alternating form having  $E_0(1, \tau) = 1$  on the basis  $\{1, \tau\}$  of  $\Lambda$  (the generator of  $\text{NS}(E) \cong \mathbb{Z}$ , of type (1)). Since  $g = 1$ , the type is the single integer  $d_1 = 1$ , so by proposition 5.3.4,

$$\deg \varphi_L = (d_1)^2 = 1$$

and  $\varphi_L: E \rightarrow \hat{E}$  is an isomorphism. Concretely,  $\varphi_H: \mathbb{C} \rightarrow \hat{\mathbb{C}} = \overline{\mathbb{C}}^*$  sends  $v$  to the antilinear functional  $w \mapsto v\bar{w}/\text{Im}\tau$ , which is a bijection because  $H$  is nondegenerate; and  $\ker \varphi_L = \Lambda(L)/\Lambda$  is trivial because  $E_0$  already pairs  $\Lambda$  to  $\mathbb{Z}$  nondegenerately (so  $\Lambda(L) = \Lambda$ ). This reproduces the identification  $\hat{E} \cong E$  of example 5.1.2 and matches the  $g = 1$  statement of example 5.2.9 that  $E$  carries a principal polarization. The degree is the square of the Pfaffian  $\text{Pf}(E_0) = d_1 = 1$ , consistent with the general formula  $\deg \varphi_L = (\text{Pf } E)^2$ .

Across the three appearances the same object has been redescribed without change of content. A polarization is a positive Hermitian form  $H$  on  $V$  with  $\text{Im}H$  integral on  $\Lambda$  (definition 3.1.3); equivalently, a positive class in  $\text{NS}(X)$ ; and now, equivalently again, a symmetric isogeny  $\varphi_L: X \rightarrow \hat{X}$  to the dual torus, self-dual under biduality, whose degree is the square of the Pfaffian  $\text{Pf}(E) = d_1 \cdots d_g$  (definition 3.2.3). The isogeny description is the one that transports to arbitrary fields: it uses only the dual torus, translation of line bundles, and the notion of a symmetric homomorphism, none of which require the complex numbers, and it is the form in which polarizations reappear in Part II when the Hermitian form is no longer available.

**Exercise 5.3.1**

1. Let  $L = L(H, \chi)$  be a polarization of type  $(1, 2, 3)$  on an abelian variety  $X$  of dimension  $g = 3$ . Using proposition 5.3.4, compute  $\deg \varphi_L$  and describe the group structure of  $\ker \varphi_L = K(L)$ . Is this polarization principal?
2. Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ , and let  $L$  be the principal polarization of example 5.3.7. Fix a nonzero integer  $n$  and consider the line bundle  $L^{\otimes n}$ , with Hermitian form  $nH$ . Compute  $\varphi_{L^{\otimes n}}$  in terms of  $\varphi_L$ , determine its degree, and identify  $\ker \varphi_{L^{\otimes n}}$  as a subgroup of  $E$ .
3. Show directly from the analytic description  $\varphi_H: v \mapsto H(v, -)$  of proposition 5.3.3 that the assignment  $L \mapsto \varphi_L$  is additive, that is,  $\varphi_{L_1 \otimes L_2} = \varphi_{L_1} + \varphi_{L_2}$  for line bundles  $L_1 = L(H_1, \chi_1)$  and  $L_2 = L(H_2, \chi_2)$  on  $X$ . Deduce that  $\varphi_{L^{-1}} = -\varphi_L$ .
4. A homomorphism  $f: X \rightarrow \hat{X}$  is called *symmetric* if  $\hat{f} = f$  under the biduality identification  $\hat{\hat{X}} = X$  of theorem 5.1.5. proposition 5.3.6 shows every  $\varphi_L$  is symmetric. Prove the converse at the level of inducing maps: if  $\psi: V \rightarrow \hat{V} = \bar{V}^*$  is an  $\mathbb{R}$ -linear map inducing a homomorphism  $X \rightarrow \hat{X}$  and satisfying  $\psi^* = \psi$  (with  $\psi^*$  the antilinear transpose of proposition 5.3.6), then the pairing

$$H_\psi(v, w) := \psi(v)(w)$$

is a Hermitian form on  $V$ . Conclude that the symmetric homomorphisms  $X \rightarrow \hat{X}$  are exactly those induced by classes in  $\text{NS}(X)$ .

5. Let  $L$  be a polarization on  $X$  and let  $f: Y \rightarrow X$  be an isogeny of abelian varieties, so that  $M := f^*L$  is a line bundle on  $Y$ . Working analytically, with  $f$  induced by a  $\mathbb{C}$ -linear isomorphism  $F: V_Y \rightarrow V_X$  carrying  $\Lambda_Y$  into  $\Lambda_X$  with finite index, show that the Hermitian form of  $M$  is  $H_M(v, w) = H_L(Fv, Fw)$ , and prove that

$$\varphi_M = \hat{f} \circ \varphi_L \circ f$$

where  $\hat{f}: \hat{X} \rightarrow \hat{Y}$  is the dual homomorphism. Use this to compute  $\deg \varphi_M$  in terms of  $\deg \varphi_L$  and  $\deg f$ .

**5.4 Poincaré Complete Reducibility**

The polarization isogeny  $\varphi_L$  of definition 5.3.1 is more than a bridge from an abelian variety to its dual: it is a rigidifying structure that lets one split an abelian variety into indivisible pieces. The guiding analogy is linear algebra. A finite-dimensional vector space carrying a positive-definite inner product decomposes as an orthogonal direct sum of any subspace and its orthogonal complement; there is nothing to obstruct the splitting because the inner product is nondegenerate everywhere it is restricted. An abelian variety with a polarization behaves the same way, with one adjustment forced by the geometry: the pieces meet not in the zero subspace but in a finite group, so the decomposition is a direct sum only up to isogeny. This is Poincaré's complete reducibility theorem, and it reduces the classification of abelian varieties up to isogeny to the classification of the indivisible ones, the simple abelian varieties. The endomorphism algebra of a simple abelian variety turns out to be a division algebra, and chapter 6 will read off from Albert's classification exactly which division algebras can occur.

The objects to be split off are the abelian subvarieties. A subtorus of a complex torus is a connected complex subgroup; when the ambient torus is an abelian variety, the interesting subtori are those that are themselves abelian varieties.

**Definition 5.4.1: Abelian Subvariety**

Let  $X = V/\Lambda$  be an abelian variety. An *abelian subvariety* of  $X$  is a complex subtorus  $Y \subseteq X$  that is itself an abelian variety. Concretely,  $Y = W/(\Lambda \cap W)$  for a complex subspace  $W \subseteq V$  with  $\Lambda \cap W$  a full lattice in  $W$ , and  $Y$  is required to be polarizable.

Two facts make abelian subvarieties easy to produce and to polarize. First, an abelian subvariety is exactly what arises as the image or the connected component of the kernel of a homomorphism: if  $f: X \rightarrow X'$  is a homomorphism of abelian varieties, then  $f(X)$  is an abelian subvariety of  $X'$ , and the identity component  $(\ker f)^0$  is an abelian subvariety of  $X$ . This is immediate on the level of the linear map inducing  $f$  (proposition 1.1.7): the image and kernel of a  $\mathbb{C}$ -linear  $F: V \rightarrow V'$  carrying  $\Lambda$  into  $\Lambda'$  are complex subspaces meeting the lattices in full lattices. Second, a subtorus of an abelian variety inherits a polarization by restriction. If  $H$  is the positive Hermitian form of a polarization on  $X$  and  $W \subseteq V$  is the subspace defining  $Y$ , then  $H|_{W \times W}$  is again positive definite, and  $\text{Im}(H|_{W \times W}) = (\text{Im} H)|_{W \times W}$  is integral on the full lattice  $\Lambda \cap W$ . Thus  $H|_W$  is a polarization on  $Y$ , and every abelian subvariety of an abelian variety is automatically polarizable: the requirement of polarizability in definition 5.4.1 is a consequence of being a subtorus of a polarized torus, not an extra hypothesis to be checked.

**Example 5.4.2: Abelian Subvarieties of a Product**

Let  $A$  and  $B$  be abelian varieties of dimensions  $a$  and  $b$ , and consider  $X = A \times B$ , an abelian variety of dimension  $a + b$  (the product of two polarizations is a polarization, coordinatewise). The two factors  $A \times \{0\}$  and  $\{0\} \times B$  are abelian subvarieties: they are the images of the inclusions and the kernels of the two projections. So is the graph of any homomorphism  $\psi: A \rightarrow B$ , namely  $\Gamma_\psi = \{(a, \psi(a)) : a \in A\}$ , which is the image of the homomorphism  $a \mapsto (a, \psi(a))$  and is isomorphic to  $A$ . When  $A = B = E$  is an elliptic curve, the diagonal  $\Delta = \{(x, x)\} \subset E \times E$  and the antidiagonal  $\{(x, -x)\}$  are abelian subvarieties, each a copy of  $E$ , and they meet in the finite group  $\{(x, x) : x \in E[2]\}$  of order 4. This finiteness of the intersection is the phenomenon the complete reducibility theorem will exploit.

The construction of the complementary subvariety is the mechanism of the theorem; isolating it before the statement makes the proof transparent. Given an abelian subvariety  $Y \subseteq X$  and a polarization  $L$  on  $X$ , the inclusion  $\iota: Y \hookrightarrow X$  has a dual  $\hat{\iota}: \hat{X} \rightarrow \hat{Y}$  (proposition 5.1.4), which is surjective because  $\iota$  is injective on a subtorus. Composing the polarization isogeny with this restriction gives a homomorphism

$$\pi = \hat{\iota} \circ \varphi_L: X \xrightarrow{\varphi_L} \hat{X} \xrightarrow{\hat{\iota}} \hat{Y}$$

whose analytic description makes the orthogonality transparent. Recall from proposition 5.3.3 that  $\varphi_L$  is induced by the  $\mathbb{R}$ -linear map  $v \mapsto H(v, -)$  from  $V$  to  $\hat{V} = \bar{V}^*$ . Under the surjection  $\hat{V} \rightarrow \hat{W}$  dual to  $W \hookrightarrow V$ , the functional  $H(v, -)$  restricts to  $H(v, -)|_W$ . Hence  $\pi$  is induced by

$$v \mapsto H(v, -)|_W \in \bar{W}^*$$

and a vector  $v \in V$  lies in the analytic kernel of  $\pi$  precisely when  $H(v, w) = 0$  for all  $w \in W$ , that is, when  $v$  is  $H$ -orthogonal to the subspace  $W$  defining  $Y$ . The complement will be the subtorus cut out by exactly this orthogonality condition.

**Theorem 5.4.3: Poincaré Complete Reducibility**

Let  $X$  be an abelian variety with a polarization given by the positive Hermitian form  $H$ , and let  $Y \subseteq X$  be an abelian subvariety. Set

$$Z = (\ker(\hat{\iota} \circ \varphi_L))^0,$$

the identity component of the kernel of  $\hat{\iota} \circ \varphi_L: X \rightarrow \hat{Y}$ , where  $\iota: Y \hookrightarrow X$  is the inclusion. Then  $Z$  is an abelian subvariety of  $X$ , called the *complementary abelian subvariety* of  $Y$  with respect to the polarization, and:

1.  $Y + Z = X$ ;
2.  $Y \cap Z$  is finite.

Consequently the addition map  $Y \times Z \rightarrow X$ ,  $(y, z) \mapsto y + z$ , is an isogeny, and  $X$  is isogenous to  $Y \times Z$ .

**Proof:**

Write  $X = V/\Lambda$  and  $Y = W/(\Lambda \cap W)$  for a complex subspace  $W \subseteq V$ . Let  $H$  be the positive Hermitian form of the polarization and  $E = \text{Im}H$  its alternating part.

*Step 1: Identify  $Z$  with the  $H$ -orthogonal complement of  $W$ .* By the analytic computation preceding the theorem, the homomorphism  $\pi = \hat{\iota} \circ \varphi_L$  is induced by the  $\mathbb{R}$ -linear map  $V \rightarrow \bar{W}^*$ ,  $v \mapsto H(v, -)|_W$ . Its kernel as a real subspace is

$$W^\perp = \{v \in V : H(v, w) = 0 \text{ for all } w \in W\}$$

the orthogonal complement of  $W$  under the positive-definite Hermitian form  $H$ . Because  $H$  is  $\mathbb{C}$ -linear in its first argument,  $W^\perp$  is a complex subspace of  $V$ ; because  $H$  is positive definite,  $V = W \oplus W^\perp$  as complex vector spaces, and  $W \cap W^\perp = 0$ . The identity component of  $\ker \pi$  is the subtorus of  $X$  with tangent space  $W^\perp$ , so

$$Z = W^\perp/(\Lambda \cap W^\perp)$$

provided  $\Lambda \cap W^\perp$  is a full lattice in  $W^\perp$ ; this is automatic for the identity component of the kernel of a homomorphism, since  $\ker \pi$  is a closed subgroup of the compact torus  $X$  and its identity component is a subtorus. Being a subtorus of the polarized  $X$ ,  $Z$  is an abelian subvariety, with polarization  $H|_{W^\perp}$  (positive definite, integral imaginary part on  $\Lambda \cap W^\perp$ ).

*Step 2:  $Y \cap Z$  is finite.* On the level of tangent spaces, the intersection  $Y \cap Z$  has tangent space  $W \cap W^\perp = 0$ , since  $V = W \oplus W^\perp$ . A subtorus with zero tangent space is a point, so the identity component of  $Y \cap Z$  is trivial. But  $Y \cap Z$  is a closed subgroup of the compact torus  $X$  with trivial identity component, hence finite, establishing (2).

*Step 3:  $Y + Z = X$ .* The addition map  $\mu: Y \times Z \rightarrow X$  is a homomorphism of complex tori, induced on tangent spaces by the addition map  $W \oplus W^\perp \rightarrow V$ ,  $(w, w') \mapsto w + w'$ . This is an isomorphism of complex vector spaces, precisely because  $V = W \oplus W^\perp$ . A homomorphism of complex tori whose induced map on tangent spaces is an isomorphism is surjective with finite kernel (its kernel is discrete, hence finite by compactness of the source). Surjectivity gives  $Y + Z = \mu(Y \times Z) = X$ , establishing (1).

*Step 4: The addition map is an isogeny.* By Step 3 the map  $\mu: Y \times Z \rightarrow X$  induces an isomorphism on tangent spaces, so it is a surjective homomorphism with finite kernel between abelian varieties of the same dimension  $\dim Y + \dim Z = \dim_{\mathbb{C}} W + \dim_{\mathbb{C}} W^{\perp} = \dim_{\mathbb{C}} V = \dim X$ . This is exactly the definition of an isogeny (definition 5.2.1). Its kernel is  $\{(t, -t) : t \in Y \cap Z\}$ , of order  $|Y \cap Z|$ , which is finite by Step 2. Therefore  $X$  is isogenous to  $Y \times Z$ , completing the proof.

The theorem says an abelian subvariety is, up to isogeny, a direct factor. This is false for a general commutative group without a polarization: a subgroup need not have a complement (the subgroup  $\mathbb{Z} \subset \mathbb{Z}$  of index 2, or a nonsplit extension of one elliptic curve by another, admits no complementary subgroup). What forces the splitting is the positive-definite form  $H$ , which supplies an orthogonal complement in the tangent space; the geometry of the torus then converts the tangent-space splitting  $V = W \oplus W^{\perp}$  into a splitting of  $X$  up to the finite ambiguity  $Y \cap Z$ . The passage from an honest direct-sum decomposition of vector spaces to a mere isogeny of tori is the price of the lattice:  $\Lambda$  need not respect the decomposition  $V = W \oplus W^{\perp}$ , so  $(\Lambda \cap W) \oplus (\Lambda \cap W^{\perp})$  is in general a finite-index sublattice of  $\Lambda$ , and the index is  $|Y \cap Z|$ .

Two features of the complement deserve emphasis. It depends on the polarization: a different polarization on  $X$  can produce a different complement  $Z$  for the same  $Y$ , since the orthogonality is measured against  $H$ . And the construction is symmetric, since  $(W^{\perp})^{\perp} = W$  for a nondegenerate Hermitian form; the complement of  $Z$  is again  $Y$ , so the roles of  $Y$  and  $Z$  are interchangeable.

**Example 5.4.4: Complement of the Diagonal in  $E \times E$**

Let  $E = \mathbb{C}/\Lambda$  be an elliptic curve with its principal polarization  $H_E$  (the positive Hermitian form with  $\text{Im}H_E$  generating the alternating forms on  $\Lambda$ ), and give  $X = E \times E$  the product polarization  $H = H_E \oplus H_E$  on  $V = \mathbb{C}^2$ . Take  $Y = \Delta$ , the diagonal, with tangent space  $W = \{(z, z) : z \in \mathbb{C}\}$ . The Hermitian form is  $H((v_1, v_2), (w_1, w_2)) = H_E(v_1, w_1) + H_E(v_2, w_2)$ . A vector  $(u_1, u_2)$  is  $H$ -orthogonal to all  $(z, z)$  precisely when  $H_E(u_1, z) + H_E(u_2, z) = 0$  for all  $z$ , that is,  $H_E(u_1 + u_2, z) = 0$  for all  $z$ ; since  $H_E$  is nondegenerate, this forces  $u_1 + u_2 = 0$ . Hence

$$W^{\perp} = \{(u, -u) : u \in \mathbb{C}\}$$

is the antidiagonal, and the complement of the diagonal is the antidiagonal subvariety  $Z = \{(x, -x)\} \cong E$ . Their intersection  $\Delta \cap Z$  consists of the points  $(x, x)$  with  $x = -x$ , that is,  $x \in E[2]$ , a group of order 4; and indeed the addition map  $\Delta \times Z \rightarrow E \times E$  is an isogeny of degree 4. This recovers the linear-algebra picture in which the diagonal and antidiagonal are orthogonal lines whose lattice sublattice has index 4 in  $\Lambda \oplus \Lambda$ .

Iterating the theorem splits an abelian variety into indivisible pieces. The pieces that admit no further splitting are the simple abelian varieties.

**Definition 5.4.5: Simple Abelian Variety**

An abelian variety  $X$  of positive dimension is *simple* if it has no abelian subvariety other than 0 and  $X$  itself.

The word “simple” is used exactly as in the theory of modules or groups: an object with no nontrivial subobjects of the same kind. An elliptic curve is simple, since a nonzero proper abelian subvariety would have dimension strictly between 0 and 1, which is impossible. In higher dimension simplicity is a genuine restriction: a product  $E_1 \times E_2$  of elliptic curves is never simple, and more subtly, an

abelian variety can fail to be simple even when it is not visibly a product, if it is isogenous to one. Simple abelian varieties are the atoms out of which every abelian variety is assembled up to isogeny.

**Corollary 5.4.6: Isotypic Decomposition up to Isogeny**

Every abelian variety  $X$  of positive dimension is isogenous to a product

$$X \sim X_1^{n_1} \times X_2^{n_2} \times \cdots \times X_r^{n_r}$$

of powers of pairwise non-isogenous simple abelian varieties  $X_1, \dots, X_r$ . The simple factors  $X_i$  (up to isogeny) and their multiplicities  $n_i$  are uniquely determined by  $X$ , up to permutation of the factors.

**Proof:**

*Step 1: Existence.* Argue by induction on  $\dim X$ . If  $X$  is simple, there is nothing to prove ( $r = 1$ ,  $n_1 = 1$ ). Otherwise  $X$  has an abelian subvariety  $Y$  with  $0 \neq Y \neq X$ ; choosing  $Y$  of minimal positive dimension makes  $Y$  simple, since any nonzero proper abelian subvariety of  $Y$  would be a smaller abelian subvariety of  $X$ . Fix a polarization on  $X$  and let  $Z$  be the complementary abelian subvariety of theorem 5.4.3, so that  $X \sim Y \times Z$  with  $\dim Z = \dim X - \dim Y < \dim X$ . By induction  $Z$  is isogenous to a product of powers of simple abelian varieties; multiplying by the simple factor  $Y$  and collecting isogenous factors into common powers gives the asserted decomposition of  $X$ .

*Step 2: Uniqueness.* Suppose  $X \sim \prod_i X_i^{n_i} \sim \prod_j Y_j^{m_j}$  are two such decompositions into powers of pairwise non-isogenous simple abelian varieties. For a simple abelian variety  $S$  and any abelian variety  $A$  with simple isogeny factors  $A_k$ , one has  $\mathrm{Hom}(S, A) \otimes \mathbb{Q} = \bigoplus_k (\mathrm{Hom}(S, A_k) \otimes \mathbb{Q})$ ; since a homomorphism from the simple  $S$  to a simple  $T$  is 0 unless  $S \sim T$  (proposition 5.4.7 below), only the factors isogenous to  $S$  contribute, and viewed as a left vector space over the division algebra  $\mathrm{End}^0(S)$ ,

$$\dim_{\mathrm{End}^0(S)} (\mathrm{Hom}(S, A) \otimes \mathbb{Q})$$

counts precisely the multiplicity of the isogeny class of  $S$  in  $A$ . Isogenous abelian varieties have isomorphic such Hom-spaces, because an isogeny  $A \rightarrow A'$  induces an isomorphism  $\mathrm{Hom}(S, A) \otimes \mathbb{Q} \rightarrow \mathrm{Hom}(S, A') \otimes \mathbb{Q}$  (compose with the isogeny and with a quasi-inverse in  $\mathrm{Hom}(A', A) \otimes \mathbb{Q}$ ). Applying this with  $S = X_i$  to both decompositions shows that each  $X_i$  occurs among the  $Y_j$  with the same multiplicity  $n_i = m_j$ , and conversely; hence the two decompositions agree up to permutation, as we sought to show.

The corollary reduces the classification of abelian varieties up to isogeny to two separate problems: the classification of the simple abelian varieties themselves, and the bookkeeping of multiplicities. Up to isogeny an abelian variety is thus a formal “monomial”  $\prod X_i^{n_i}$  in simple factors, and the entire isogeny category of abelian varieties is a semisimple category whose simple objects are the simple abelian varieties. This is the exact analogue, for abelian varieties, of the decomposition of a semisimple module into isotypic components, and it is the reason the endomorphism algebra will assemble into a product of matrix algebras over division rings.

That last point rests on a single fact about simple abelian varieties: a nonzero endomorphism of a simple abelian variety is invertible, once one is allowed to invert isogenies. This is the input chapter 6 needs.

**Proposition 5.4.7: Endomorphism Algebra of a Simple Abelian Variety**

Let  $X$  be a simple abelian variety. Then every nonzero homomorphism  $f: X \rightarrow Y$  from  $X$  to a simple abelian variety  $Y$  is an isogeny; in particular every nonzero endomorphism of  $X$  is an isogeny. Consequently the endomorphism algebra

$$\text{End}^0(X) = \text{End}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$$

is a division algebra over  $\mathbb{Q}$ .

**Proof:**

*Step 1: A nonzero endomorphism of a simple abelian variety is an isogeny.* Let  $f \in \text{End}(X)$  be nonzero. The image  $f(X)$  is an abelian subvariety of  $X$ , and it is nonzero because  $f \neq 0$ ; since  $X$  is simple,  $f(X) = X$ , so  $f$  is surjective. The identity component  $(\ker f)^0$  is an abelian subvariety of  $X$ ; it is not all of  $X$ , again because  $f \neq 0$ , so by simplicity  $(\ker f)^0 = 0$ . A homomorphism with finite kernel (the kernel has trivial identity component, hence is finite by compactness) that is surjective between varieties of equal dimension is an isogeny (definition 5.2.1). Thus every nonzero  $f \in \text{End}(X)$  is an isogeny.

*Step 2: Isogenies are invertible in  $\text{End}^0(X)$ .* An isogeny  $f: X \rightarrow X$  of degree  $d = |\ker f|$  has a quasi-inverse: since  $\ker f \subseteq X[d]$  (every element of the finite group  $\ker f$  has order dividing  $d$ ), the multiplication map  $[d]$  factors as  $[d] = g \circ f$  for a unique homomorphism  $g \in \text{End}(X)$ . In  $\text{End}^0(X)$  this reads  $g \circ f = [d] = d \cdot \text{id}$ , an invertible scalar, so

$$f^{-1} = \frac{1}{d} g \in \text{End}^0(X)$$

is a two-sided inverse (the same argument on the other side, or the observation that  $\text{End}^0(X)$  is finite-dimensional over  $\mathbb{Q}$  so a left inverse is a two-sided inverse). Hence every nonzero element of  $\text{End}^0(X)$  is invertible.

*Step 3:  $\text{End}^0(X)$  is a division algebra.*  $\text{End}^0(X)$  is an associative  $\mathbb{Q}$ -algebra with unit  $\text{id}$ , and it is finite-dimensional over  $\mathbb{Q}$ : the rational representation  $\rho_r$  embeds it into  $\text{End}(\Lambda \otimes \mathbb{Q}) \cong M_{2g}(\mathbb{Q})$  (proposition 1.2.6), so  $\dim_{\mathbb{Q}} \text{End}^0(X) \leq (2g)^2$ . By Steps 1 and 2 every nonzero element is invertible. An associative unital algebra in which every nonzero element is invertible is a division algebra, as we sought to show. The statement for a homomorphism  $f: X \rightarrow Y$  between simple abelian varieties follows verbatim:  $f(X) = Y$  by simplicity of  $Y$  and  $(\ker f)^0 = 0$  by simplicity of  $X$ , so  $f$  is an isogeny.

This is the fact that turns the isotypic decomposition of corollary 5.4.6 into a structure theorem for the endomorphism algebra. For pairwise non-isogenous simple abelian varieties  $X_i$  there are no nonzero homomorphisms between distinct factors, by proposition 5.4.7, so  $\text{End}^0(\prod_i X_i^{n_i})$  decomposes as a product  $\prod_i \text{End}^0(X_i^{n_i})$ , and endomorphisms of the power  $X_i^{n_i}$  are  $n_i \times n_i$  matrices with entries in  $D_i = \text{End}^0(X_i)$ . Since each  $D_i$  is a division algebra, this yields

$$\text{End}^0(X) \cong \prod_i M_{n_i}(D_i),$$

a product of matrix algebras over division algebras, which is the Wedderburn form of a semisimple algebra. The one input not supplied by pure algebra is that each  $D_i$  is a division algebra, and that is exactly proposition 5.4.7. Chapter 6 takes this as its starting point: it introduces the Rosati



involution on  $\text{End}^0(X)$ , shows it is positive, and invokes Albert's classification to determine precisely which division algebras with a positive involution can arise, and hence which endomorphism algebras a simple abelian variety can possess.

**Example 5.4.8: Endomorphism Algebra of  $E \times E$**

Let  $E = \mathbb{C}/\Lambda$  be an elliptic curve without complex multiplication, so that  $\text{End}^0(E) = \mathbb{Q}$ . The product  $X = E \times E$  is isogenous to  $E^2$ , a power of the single simple abelian variety  $E$ , so it is not simple; its isotypic decomposition (corollary 5.4.6) is  $X \sim E^2$  with one simple factor of multiplicity 2. An endomorphism of  $E \times E$  is a homomorphism  $E^2 \rightarrow E^2$ , and by proposition 5.4.7 such a homomorphism is recorded by a matrix of homomorphisms  $E \rightarrow E$ :

$$\text{End}^0(E \times E) \cong M_2(\text{End}^0(E)) = M_2(\mathbb{Q})$$

Concretely, an endomorphism sends  $(x, y) \mapsto (ax + by, cx + dy)$  for  $a, b, c, d \in \text{End}^0(E) = \mathbb{Q}$ , and composition of endomorphisms corresponds to matrix multiplication. This is a semisimple  $\mathbb{Q}$ -algebra of dimension  $4 = n^2[D : \mathbb{Q}] = 2^2 \cdot 1$ , consistent with the bound  $\dim_{\mathbb{Q}} \text{End}^0(X) \leq (2g)^2 = 16$  for  $g = 2$ ; the algebra is not a division algebra (the matrix  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  is a nonzero non-invertible element), which is the algebraic shadow of the fact that  $E \times E$  is not simple. Had  $E$  instead had complex multiplication by an imaginary quadratic field  $K$ , so  $\text{End}^0(E) = K$ , the same computation would give  $\text{End}^0(E \times E) \cong M_2(K)$ , a matrix algebra over the division algebra  $K$ .

The two computations in example 5.4.8 display the general pattern in miniature. Splitting an abelian variety by a polarization, via the complementary subvariety of theorem 5.4.3, reduces every abelian variety up to isogeny to a product of powers of simple ones; and the endomorphism algebra of each simple factor is a division algebra by proposition 5.4.7. The classification problem for abelian varieties up to isogeny thereby becomes the classification of simple abelian varieties, and with it the classification of the division algebras that can serve as their endomorphism algebras, the question that chapter 6 answers through the Rosati involution and Albert's theorem.

### Exercise 5.4.1

1. Let  $X = V/\Lambda$  be an abelian variety with polarization given by the positive Hermitian form  $H$ , and let  $Y = W/(\Lambda \cap W)$  be an abelian subvariety with complementary subvariety  $Z = W^\perp/(\Lambda \cap W^\perp)$  as in theorem 5.4.3. Show that  $\dim Y + \dim Z = \dim X$ , and that the complement of  $Z$  (with respect to the same polarization) is again  $Y$ .
2. Fix an elliptic curve  $E = \mathbb{C}/\Lambda$  with principal polarization  $H_E$ , and give  $X = E \times E$  the product polarization  $H = H_E \oplus H_E$ . Following example 5.4.4, let  $Y_k = \{(z, kz) : z \in \mathbb{C}\}/\Lambda_k$  be the abelian subvariety cut out by the tangent line  $W_k = \{(z, kz) : z \in \mathbb{C}\}$  for a fixed integer  $k \neq 0$ . Compute the tangent space  $W_k^\perp$  of the complementary subvariety  $Z_k$ , and identify the order of  $Y_k \cap Z_k$ .
3. Let  $X$  be an abelian variety and  $f \in \text{End}(X)$  an idempotent up to isogeny, meaning  $f^2 = f$  in  $\text{End}^0(X)$ . Show that  $X$  is isogenous to  $\text{im}(f) \times \ker(f)^0$ , where  $\text{im}(f)$  and  $\ker(f)^0$  are the image and identity component of the kernel of any integer multiple  $nf \in \text{End}(X)$ . (Do not assume  $X$  is simple; use theorem 5.4.3 on a suitable subvariety.)
4. Let  $X$  be a simple abelian variety and let  $D = \text{End}^0(X)$ . Using proposition 5.4.7, prove that  $D$  has no zero divisors and that every element of  $D$  satisfies a monic polynomial with rational coefficients. Deduce that  $D$  contains a number field  $\mathbb{Q}(\alpha)$  for every  $\alpha \in D \setminus \mathbb{Q}$ .



5. Let  $X$  be an abelian variety with isotypic decomposition  $X \sim X_1^{n_1} \times \cdots \times X_r^{n_r}$  as in corollary 5.4.6, with  $D_i = \text{End}^0(X_i)$ . Prove that  $X$  is simple if and only if  $r = 1$  and  $n_1 = 1$ , and that  $X$  is isogenous to a power of a single simple abelian variety if and only if  $\text{End}^0(X)$  is a simple  $\mathbb{Q}$ -algebra (has no nontrivial two-sided ideals).



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# *Endomorphism Algebras, the Rosati Involution, and Complex Multiplication*

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The endomorphisms of an abelian variety carry its arithmetic. Chapter 1 already saw that they form a ring of finite rank, constrained by the integrality of the rational representation; Poincaré reducibility then showed that the endomorphism algebra of a simple abelian variety is a division algebra. This chapter determines which division algebras can occur. The instrument is the Rosati involution, an anti-involution of  $\text{End}^0(X)$  manufactured from a polarization, whose defining property is positivity: the trace form  $(f, g) \mapsto \text{Tr}(fg')$  is positive definite. Positivity is a severe constraint, and Albert's classification extracts its consequences completely, sorting the possible endomorphism algebras into four types of division algebra with involution.

The chapter closes with the fourth and most structured type, complex multiplication. An abelian variety of dimension  $g$  has complex multiplication when its endomorphism algebra contains a CM field of degree  $2g$ , the largest commutative symmetry it can support; such varieties are built directly from the arithmetic of the field through a CM type, and they are the points at which the transcendental theory of Part I meets the arithmetic of Part II most sharply. Complex multiplication is also the gateway to Shimura varieties: the moduli space of chapter 3 acquires distinguished CM points defined over number fields, and the reflex field that governs them is where the theory of this book hands off to *Shimura Varieties* [11].

## 6.1 The Endomorphism Algebra

The endomorphism ring  $\text{End}(X)$  records every self-map of an abelian variety that respects the group law. It carries the arithmetic of  $X$ : for the elliptic curve it distinguishes the generic case from complex multiplication, and in general it governs which isogeny decompositions  $X$  admits. As a ring, though,  $\text{End}(X)$  is awkward. An isogeny  $f: X \rightarrow X$  has finite kernel but no inverse in  $\text{End}(X)$ , so the ring has no division and its structure resists description. The remedy is to tensor with  $\mathbb{Q}$ . Poincaré reducibility (theorem 5.4.3) has already resolved  $X$  up to isogeny into a product of simple pieces; passing to  $\mathbb{Q}$  makes that decomposition visible at the level of the endomorphism ring, converting it into a semisimple algebra whose Wedderburn factors read off the simple constituents of  $X$ .

### Definition 6.1.1: The Endomorphism Algebra

Let  $X$  be an abelian variety. Its *endomorphism algebra* is the  $\mathbb{Q}$ -algebra

$$\text{End}^0(X) := \text{End}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$$

obtained by extending scalars from  $\mathbb{Z}$  to  $\mathbb{Q}$ . Its elements are formal fractions  $f/n$  with  $f \in \text{End}(X)$  and  $n$  a positive integer, and multiplication is composition. An element  $f \in \text{End}(X)$  is identified with  $f \otimes 1$ , so  $\text{End}(X)$  embeds in  $\text{End}^0(X)$ .

The passage to  $\mathbb{Q}$  does one thing: it inverts isogenies. An isogeny  $f: X \rightarrow X$  has, by the isogeny relation of proposition 5.2.6, a complementary isogeny  $g$  with  $g \circ f = [n]$  and  $f \circ g = [n]$  for some  $n \geq 1$ ; in  $\text{End}^0(X)$  the element  $[n]$  is the invertible scalar  $n$ , so  $f^{-1} = g/n$ . An element of  $\text{End}^0(X)$  is thus a unit precisely when it is (a rational multiple of) an isogeny. Endomorphisms that were only isogenies onto their image in  $\text{End}(X)$  become genuine units in  $\text{End}^0(X)$ , and the ring acquires the division needed for the structure theory to apply. The trade is that  $\text{End}^0(X)$  forgets the lattice  $\text{End}(X)$  inside it, retaining only the isogeny class of  $X$ ; this is precisely the information the endomorphism algebra is meant to carry.

For the elliptic curve  $E = \mathbb{C}/\Lambda$  the two levels are already familiar from *Elliptic Curves* [4]. Generically  $\text{End}(E) = \mathbb{Z}$  and  $\text{End}^0(E) = \mathbb{Q}$ ; when  $E$  has complex multiplication by an order in an imaginary quadratic field  $K$ ,  $\text{End}(E)$  is that order and  $\text{End}^0(E) = K$ . Tensoring with  $\mathbb{Q}$  discards the choice of order and keeps the field  $K$ , which is what determines the isogeny class. The worked cases appear in example 6.1.5 below.

### Theorem 6.1.2: Semisimplicity of the Endomorphism Algebra

Let  $X$  be an abelian variety isogenous to  $\prod_i X_i^{n_i}$ , where the  $X_i$  are pairwise non-isogenous simple abelian varieties (corollary 5.4.6). Then

$$\text{End}^0(X) \cong \prod_i M_{n_i}(D_i), \quad D_i := \text{End}^0(X_i)$$

where each  $D_i$  is a division algebra over  $\mathbb{Q}$  (proposition 5.4.7). In particular  $\text{End}^0(X)$  is a semisimple  $\mathbb{Q}$ -algebra, and the pair of data  $(D_i, n_i)$  is an isogeny invariant of  $X$ .

**Proof:**

Isogenous abelian varieties have isomorphic endomorphism algebras: an isogeny  $\alpha: X \rightarrow X'$  induces  $f \mapsto \alpha f \alpha^{-1}$ , an isomorphism  $\text{End}^0(X) \rightarrow \text{End}^0(X')$  of  $\mathbb{Q}$ -algebras, since  $\alpha^{-1}$  exists in  $\text{End}^0$  by proposition 5.2.6. Thus  $\text{End}^0(X)$  depends only on the isogeny class, and it suffices to compute it for  $X = \prod_i X_i^{n_i}$ .

*Step 1: No maps between distinct isotypic blocks.* Let  $S$  and  $T$  be simple abelian varieties with  $S$  not isogenous to  $T$ . A homomorphism  $h: S \rightarrow T$  has image an abelian subvariety of  $T$ ; simplicity of  $T$  forces the image to be 0 or all of  $T$ . If the image is all of  $T$  then  $\ker h$  is a proper abelian subvariety of the simple  $S$ , hence finite, so  $h$  is an isogeny, contradicting  $S \not\sim T$ . Therefore  $h = 0$ , and  $\text{Hom}(S, T) = 0$ ; tensoring with  $\mathbb{Q}$ ,  $\text{Hom}^0(S, T) = 0$  as well. Consequently a homomorphism between the blocks  $X_i^{n_i}$  and  $X_j^{n_j}$  vanishes for  $i \neq j$ , and  $\text{End}^0(X)$  splits as the product of the endomorphism algebras of the individual blocks:

$$\text{End}^0\left(\prod_i X_i^{n_i}\right) \cong \prod_i \text{End}^0(X_i^{n_i})$$

*Step 2: A single block is a matrix algebra.* Fix one simple factor  $S = X_i$  and write  $D = \text{End}^0(S)$ , a division algebra by proposition 5.4.7. An endomorphism of  $S^n$  is, by distributing over the  $n$  summands, an  $n \times n$  array of homomorphisms  $S \rightarrow S$ : the component  $h_{k\ell} \in \text{Hom}^0(S, S) = D$  is the composite  $S \xrightarrow{\iota_\ell} S^n \xrightarrow{f} S^n \xrightarrow{\pi_k} S$ , where  $\iota_\ell$  and  $\pi_k$  are the inclusion of and projection onto the  $\ell$ -th and  $k$ -th factors. The assignment  $f \mapsto (h_{k\ell})$  is additive, and composition of endomorphisms corresponds to matrix multiplication because  $\pi_k \iota_\ell = \delta_{k\ell} \text{id}_S$ . It is bijective, with inverse sending a matrix  $(h_{k\ell})$  to the endomorphism  $f = \sum_{k,\ell} \iota_k h_{k\ell} \pi_\ell$ . Hence

$$\text{End}^0(S^n) \cong M_n(D)$$

*Step 3: Assembly and semisimplicity.* Combining the two steps,

$$\text{End}^0(X) \cong \prod_i \text{End}^0(X_i^{n_i}) \cong \prod_i M_{n_i}(D_i)$$

Each  $M_{n_i}(D_i)$  is a simple algebra with the division algebra  $D_i$  as its underlying skew field, so the product is semisimple; this is exactly the form delivered by the Wedderburn-Artin theorem of *Core Representation Theory* [3], run here in reverse from the geometric decomposition. The simple factors of a semisimple algebra, together with their multiplicities and their skew fields, are determined up to isomorphism by the algebra; since the isotypic decomposition  $\prod_i X_i^{n_i}$  is unique up to isogeny and permutation (corollary 5.4.6), the data  $(D_i, n_i)$  is an isogeny invariant of  $X$ , completing the proof.

Ordinarily one is handed an abstract semisimple algebra and decomposes it into matrix algebras over division rings; here the geometry supplies the decomposition, and the endomorphism algebra inherits it. The division algebras  $D_i$  are the arithmetic content: the integers  $n_i$  record how many copies of each simple factor occur, while the  $D_i$  encode the internal symmetry of the simple pieces themselves. Classifying endomorphism algebras of abelian varieties therefore reduces to two tasks: describing which division algebras arise as  $\text{End}^0$  of a simple abelian variety, and bounding their size. The Rosati

involution (section 6.2) and Albert's theorem (section 6.3) carry out the first; the dimension bound below begins the second.

The reduction also explains why simplicity is the atomic notion. A simple abelian variety is one that cannot be broken up further by isogeny, and its endomorphism algebra is a bare division algebra with no matrix structure at all. Everything else is built from these atoms by taking powers (which introduces the matrix algebras  $M_{n_i}$ ) and products (which introduces the direct product). The next proposition constrains how large the whole assembly can be.

Before stating it, recall the source of the constraint. Every endomorphism acts on the rational homology  $\Lambda \otimes \mathbb{Q}$ , and this action is faithful and  $\mathbb{Q}$ -linear on a space of dimension  $2g$ . The endomorphism algebra therefore sits inside a  $(2g)^2$ -dimensional matrix algebra, which caps its own dimension.

**Proposition 6.1.3: Finite Dimensionality of  $\text{End}^0(X)$**

Let  $X$  be an abelian variety of dimension  $g$ . Then  $\text{End}^0(X)$  is a finite-dimensional  $\mathbb{Q}$ -algebra, and

$$\dim_{\mathbb{Q}} \text{End}^0(X) \leq (2g)^2$$

**Proof:**

The rational representation of definition 1.2.3 attaches to each endomorphism its induced action on rational homology,

$$\rho_r: \text{End}^0(X) \longrightarrow \text{End}(\Lambda \otimes_{\mathbb{Z}} \mathbb{Q})$$

a homomorphism of  $\mathbb{Q}$ -algebras. Since  $\Lambda$  has rank  $2g$ , the target is  $\text{End}(\Lambda \otimes \mathbb{Q}) \cong M_{2g}(\mathbb{Q})$ , of  $\mathbb{Q}$ -dimension  $(2g)^2$ .

The representation  $\rho_r$  is faithful. An endomorphism  $f \in \text{End}(X)$  is determined by its analytic action  $\rho_a(f)$  on  $V$ , which is in turn determined by the restriction of  $\rho_r(f)$  to  $\Lambda$  under  $V = \Lambda \otimes \mathbb{R}$ ; concretely, if  $\rho_r(f) = 0$  then  $f$  kills the lattice, so  $f = 0$  as a map of tori. The same holds after tensoring with  $\mathbb{Q}$ : if  $\rho_r(\xi) = 0$  for  $\xi \in \text{End}^0(X)$ , write  $\xi = f/n$  with  $f \in \text{End}(X)$ , whence  $\rho_r(f) = n\rho_r(\xi) = 0$  and  $f = 0$ , so  $\xi = 0$ . That  $\rho_r$  is well defined and integral valued is the content of proposition 1.2.6, whose characteristic polynomial computation shows  $\rho_r(f)$  has integer characteristic polynomial of degree  $2g$ .

A faithful homomorphism embeds  $\text{End}^0(X)$  as a  $\mathbb{Q}$ -subalgebra of  $M_{2g}(\mathbb{Q})$ , so

$$\dim_{\mathbb{Q}} \text{End}^0(X) \leq \dim_{\mathbb{Q}} M_{2g}(\mathbb{Q}) = (2g)^2$$

which is finite, as we sought to show.

Finite dimensionality is what makes the Wedderburn structure of theorem 6.1.2 meaningful: a semisimple algebra of infinite dimension would have infinitely many or infinite-dimensional factors, and no classification could follow. The bound  $(2g)^2$  is the size of the ambient matrix algebra  $M_{2g}(\mathbb{Q})$ , and it is rarely attained; for a simple abelian variety with  $\text{End}^0(X) = \mathbb{Q}$  the dimension is 1, at the opposite extreme. The bound does, however, force  $[D_i : \mathbb{Q}]$  and the matrix sizes  $n_i$  to satisfy  $\sum_i n_i^2 [D_i : \mathbb{Q}] \leq (2g)^2$ , a numerical constraint that Albert's classification sharpens into an exact table.

With semisimplicity and the dimension bound in hand, the structural vocabulary can be fixed. The organizing dichotomy is whether the algebra is a division algebra, equivalently whether  $X$  is simple.

**Definition 6.1.4: Simplicity and the Endomorphism Structure**

An abelian variety  $X$  is simple (definition 5.4.5) if and only if  $\text{End}^0(X)$  is a division algebra. In general  $\text{End}^0(X)$  is a semisimple  $\mathbb{Q}$ -algebra, and its *endomorphism structure* is the collection of Wedderburn data  $(D_i, n_i)$  of theorem 6.1.2: the division algebras  $D_i = \text{End}^0(X_i)$  over  $\mathbb{Q}$  attached to the simple isogeny factors  $X_i$ , together with their multiplicities  $n_i$ . This structure is an isogeny invariant of  $X$ .

The equivalence packaged into the definition unpacks theorem 6.1.2:  $X$  is simple exactly when the isotypic decomposition has a single factor  $X = X_1$  with  $n_1 = 1$ , so  $\text{End}^0(X) = D_1$  is a bare division algebra with no matrix part. A nonsimple  $X$  shows its reducibility in the endomorphism algebra as either a nontrivial matrix factor  $M_{n_i}(D_i)$  with  $n_i \geq 2$  (a repeated simple factor) or a nontrivial product (distinct simple factors), or both. The endomorphism structure is thus a faithful algebraic shadow of the geometric decomposition, and because it is an isogeny invariant it can be computed in whichever model of the isogeny class is most convenient.

**Example 6.1.5: Endomorphism Algebras of Elliptic Curves**

For an elliptic curve  $E$  ( $g = 1$ ) the dimension bound of proposition 6.1.3 reads  $\dim_{\mathbb{Q}} \text{End}^0(E) \leq 4$ , and every elliptic curve is simple, so  $\text{End}^0(E)$  is a division algebra of  $\mathbb{Q}$ -dimension 1, 2, or 4. The three cases are exactly the possibilities catalogued in *Elliptic Curves* [4].

1. *The generic case.*  $\text{End}^0(E) = \mathbb{Q}$ , of dimension 1. Here  $\text{End}(E) = \mathbb{Z}$ : the only endomorphisms are the multiplication-by- $n$  maps. Over  $\mathbb{C}$  this is the situation for a lattice  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$  with  $\tau$  not lying in an imaginary quadratic field, which is the case for all but a countable set of  $\tau$ .
2. *Complex multiplication.*  $\text{End}^0(E) = K$ , an imaginary quadratic field, of dimension 2. This occurs when  $\tau$  generates an imaginary quadratic field  $K = \mathbb{Q}(\tau)$ ; then  $\text{End}(E)$  is an order in  $K$ , and  $\text{End}^0(E) = K$ . The curve  $E$  has *complex multiplication* by  $K$ , the  $g = 1$  instance of the CM abelian varieties of section 6.4.
3. *The quaternionic case.*  $\text{End}^0(E)$  a quaternion algebra over  $\mathbb{Q}$ , of dimension 4. This does not arise over  $\mathbb{C}$ : a 1-dimensional complex torus has commutative endomorphism algebra. It occurs only for supersingular elliptic curves over a field of characteristic  $p$ , where  $\text{End}^0(E)$  is the quaternion algebra over  $\mathbb{Q}$  ramified exactly at  $p$  and  $\infty$ . The phenomenon belongs to the algebraic theory of Part II and is taken up there.

The reducible case  $E \times E$  illustrates the matrix structure. Its isotypic decomposition is a single simple factor  $E$  with multiplicity 2 (example 5.4.8), so by theorem 6.1.2

$$\text{End}^0(E \times E) \cong M_2(\text{End}^0(E))$$

In the generic case this is  $M_2(\mathbb{Q})$ , of dimension 4, well below the bound of proposition 6.1.3 for  $g = \dim(E \times E) = 2$ , which is  $(2 \cdot 2)^2 = 16$ ; when  $E$  has CM by  $K$  it is  $M_2(K)$ , of dimension 8. More generally, for a product  $\prod_i E_i^{n_i}$  of elliptic curves with the  $E_i$  pairwise non-isogenous, the endomorphism algebra is  $\prod_i M_{n_i}(\text{End}^0(E_i))$ , a product of matrix algebras over fields (in the CM and generic cases) that makes the isogeny decomposition legible at a glance.

The three elliptic cases are the smallest instance of Albert's four-type classification, previewed here so the general statement has a concrete anchor. In the language of section 6.3, the generic  $\text{End}^0(E) = \mathbb{Q}$  is a totally real field of Type I, the CM field is a Type IV center, and the supersingular quaternion algebra is of Type III (totally definite). The general lesson is already visible:  $\text{End}^0(X)$  is a semisimple  $\mathbb{Q}$ -algebra, and Wedderburn reduces its classification to that of the division algebras  $D_i$ . Which division algebras can occur is not arbitrary; the Rosati involution equips each with a positive anti-involution, and Albert's theorem (theorem 6.3.4) determines the resulting list.

### Exercise 6.1.1

1. Let  $E$  be an elliptic curve with complex multiplication by an imaginary quadratic field  $K$ . Using definition 6.1.1 and the elliptic cases of example 6.1.5, identify  $\text{End}^0(E)$ , compute  $\dim_{\mathbb{Q}} \text{End}^0(E)$ , and verify that the value obeys the bound of proposition 6.1.3 for  $g = 1$ .
2. Let  $X$  be an abelian variety of dimension  $g = 5$  isogenous to  $A^2 \times B$ , where  $A$  is a simple abelian surface with  $\text{End}^0(A) = \mathbb{Q}$  and  $B$  is a simple elliptic curve, not isogenous to  $A$ , with complex multiplication by an imaginary quadratic field  $K$ . Write down  $\text{End}^0(X)$  as a product of matrix algebras using theorem 6.1.2, compute its  $\mathbb{Q}$ -dimension, and check the inequality  $\sum_i n_i^2 [D_i : \mathbb{Q}] \leq (2g)^2$  against proposition 6.1.3.
3. An element  $\xi \in \text{End}^0(X)$  is a unit precisely when it is a rational multiple of an isogeny of  $X$ . Prove this, using the complementary-isogeny relation of proposition 5.2.6 for the forward direction and the definition of a unit for the converse.
4. Let  $S$  be a simple abelian variety and  $n \geq 1$ . Following the argument of theorem 6.1.2, prove directly that  $\text{End}^0(S^n) \cong M_n(D)$  with  $D = \text{End}^0(S)$ , giving the maps in both directions and checking that composition corresponds to matrix multiplication.
5. Deduce the equivalence packaged in definition 6.1.4: an abelian variety  $X$  is simple if and only if  $\text{End}^0(X)$  is a division algebra. Then decide, for each of the following, whether the algebra can be the endomorphism algebra of a simple abelian variety of dimension  $g$ , citing proposition 6.1.3:
  1.  $M_2(\mathbb{Q})$  for  $g = 2$ ;
  2. a quartic field  $K$  with  $[K : \mathbb{Q}] = 4$ , for  $g = 2$ .
6. Let  $X$  and  $X'$  be isogenous abelian varieties. Prove that  $\text{End}^0(X) \cong \text{End}^0(X')$  as  $\mathbb{Q}$ -algebras, and conclude that the Wedderburn data  $(D_i, n_i)$  of theorem 6.1.2 is an isogeny invariant. Explain in one or two sentences why the same statement fails for the ring  $\text{End}(X)$ .

## 6.2 The Rosati Involution and Its Positivity

The endomorphism algebra  $\text{End}^0(X)$  of the previous section is a semisimple  $\mathbb{Q}$ -algebra, but so far it carries no extra structure tied to the geometry of  $X$ . A polarization supplies that structure. The polarization isogeny  $\varphi_L : X \rightarrow \hat{X}$  of definition 5.3.1 lets one transport an endomorphism  $f$  of  $X$  across to the dual and back, producing a second endomorphism  $f'$ . The resulting operation  $f \mapsto f'$  is an anti-involution of  $\text{End}^0(X)$ , and its associated bilinear form  $(f, g) \mapsto \text{Tr}(fg')$  is positive definite. This single positivity property is the one strong constraint a polarization places on  $\text{End}^0(X)$ , and



Albert's classification (section 6.3) is exactly the extraction of its consequences.

Throughout,  $L$  is a fixed polarization on the abelian variety  $X$ , so that  $\varphi_L: X \rightarrow \hat{X}$  is an isogeny (definition 5.3.5). In the isogeny category  $\varphi_L$  is invertible: it has finite kernel and equal source and target dimension, so it becomes an isomorphism after tensoring  $\text{Hom}(X, \hat{X})$  with  $\mathbb{Q}$ , and  $\varphi_L^{-1} \in \text{Hom}^0(\hat{X}, X) := \text{Hom}(\hat{X}, X) \otimes_{\mathbb{Z}} \mathbb{Q}$  is well defined. The dual construction  $f \mapsto \hat{f}$  of proposition 5.1.4 extends  $\mathbb{Q}$ -linearly to isogeny classes as well.

#### Definition 6.2.1: Rosati Involution

Let  $L$  be a polarization on  $X$  with polarization isogeny  $\varphi_L: X \rightarrow \hat{X}$  (definition 5.3.1). The *Rosati involution* associated to  $L$  is the map

$$': \text{End}^0(X) \rightarrow \text{End}^0(X), \quad f \mapsto f' := \varphi_L^{-1} \circ \hat{f} \circ \varphi_L$$

where  $\hat{f}: \hat{X} \rightarrow \hat{X}$  is the dual of  $f$  (proposition 5.1.4) and  $\varphi_L^{-1} \in \text{Hom}^0(\hat{X}, X)$  is the inverse of  $\varphi_L$  in the isogeny category.

The composite  $\varphi_L^{-1} \circ \hat{f} \circ \varphi_L$  reads off the diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & X \\ \downarrow \varphi_L & & \downarrow \varphi_L \\ \hat{X} & \xrightarrow{\hat{f}} & \hat{X} \end{array}$$

run backwards along the right edge: apply  $\varphi_L$  to land in  $\hat{X}$ , apply  $\hat{f}$ , then return along  $\varphi_L^{-1}$ . When  $\varphi_L$  is an honest isomorphism (the principal case)  $f'$  is again an honest endomorphism; in general  $f'$  lives in  $\text{End}^0(X)$  and need not preserve the integral lattice  $\text{End}(X)$ . The construction depends on  $L$  only through  $\varphi_L$ , and different polarizations generally give different involutions. What follows shows that  $f \mapsto f'$  reverses products and squares to the identity, so that  $\text{End}^0(X)$  is an algebra with involution, and then that the involution is positive.

#### Proposition 6.2.2: The Rosati Involution Is an Anti-Involution

The Rosati involution  $f \mapsto f'$  of definition 6.2.1 is a  $\mathbb{Q}$ -linear anti-involution of  $\text{End}^0(X)$ :

1.  $(f + g)' = f' + g'$  and  $(cf)' = cf'$  for  $c \in \mathbb{Q}$ ;
2.  $(fg)' = g'f'$ ;
3.  $(f')' = f$ ;
4.  $[n]' = [n]$  for every  $n \in \mathbb{Z}$ , and more generally  $c' = c$  for every  $c \in \mathbb{Q} \subseteq \text{End}^0(X)$ .

The motivating point is that all four properties are formal consequences of one fact about dualization: it is a contravariant  $\mathbb{Q}$ -linear functor (proposition 5.1.4), so it fixes scalars, reverses composites, and squares to the identity under biduality. The Rosati involution conjugates dualization by  $\varphi_L$ , and conjugation preserves each of these features. The self-duality  $\hat{\varphi}_L = \varphi_L$  of proposition 5.3.6 is what makes the squaring property  $(f')' = f$  come out clean.

**Proof:**

We verify the four properties in turn, using that  $g \mapsto \hat{g}$  is  $\mathbb{Q}$ -linear, contravariant, and compatible with biduality (proposition 5.1.4), and that under the canonical identification  $\hat{\hat{X}} = X$  (theorem 5.1.5) one has  $\hat{\hat{f}} = f$ .

(1) *Linearity.* Dualization is additive and  $\mathbb{Q}$ -linear, so  $\widehat{f+g} = \hat{f} + \hat{g}$  and  $\widehat{cf} = c\hat{f}$ . Conjugating by  $\varphi_L$  preserves sums and  $\mathbb{Q}$ -scalars because composition in  $\text{End}^0(X)$  is  $\mathbb{Q}$ -bilinear:

$$(f+g)' = \varphi_L^{-1}(\hat{f} + \hat{g})\varphi_L = \varphi_L^{-1}\hat{f}\varphi_L + \varphi_L^{-1}\hat{g}\varphi_L = f' + g'$$

and likewise  $(cf)' = cf'$ .

(2) *Anti-multiplicativity.* Contravariance of dualization gives  $\widehat{fg} = \hat{g}\hat{f}$ . Hence

$$(fg)' = \varphi_L^{-1}\widehat{fg}\varphi_L = \varphi_L^{-1}\hat{g}\hat{f}\varphi_L = (\varphi_L^{-1}\hat{g}\varphi_L)(\varphi_L^{-1}\hat{f}\varphi_L) = g'f'$$

where the inserted  $\varphi_L\varphi_L^{-1} = \text{id}$  between  $\hat{g}$  and  $\hat{f}$  splits the conjugate of the product into the product of the conjugates in reversed order.

(3) *Involutivity.* By definition  $f' = \varphi_L^{-1}\hat{f}\varphi_L$ , so applying the involution again,

$$(f')' = \varphi_L^{-1}(\widehat{f'})\varphi_L = \varphi_L^{-1}\widehat{\varphi_L^{-1}\hat{f}\varphi_L}\varphi_L$$

Dualizing the inner composite reverses it and applies hats:  $\widehat{\varphi_L^{-1}\hat{f}\varphi_L} = \hat{\varphi}_L\hat{\hat{f}}\hat{\varphi}_L^{-1} = \hat{\varphi}_L f (\hat{\varphi}_L)^{-1}$ , using  $\hat{\hat{f}} = f$  under biduality and  $\widehat{\varphi_L^{-1}} = (\hat{\varphi}_L)^{-1}$ . Now  $\hat{\varphi}_L = \varphi_L$  by proposition 5.3.6 (here one identifies  $\hat{\hat{X}} = X$ , so that  $\hat{\varphi}_L: X \rightarrow \hat{X}$  is compared with  $\varphi_L: X \rightarrow \hat{X}$ ). Substituting,

$$(f')' = \varphi_L^{-1}(\varphi_L f \varphi_L^{-1})\varphi_L = f$$

the two conjugating pairs cancelling.

(4) *Scalars are fixed.* For  $c \in \mathbb{Q}$  the endomorphism  $c$  commutes with every isogeny and satisfies  $\hat{c} = c$  (dualization is  $\mathbb{Q}$ -linear and fixes the identity, so it fixes each rational multiple of it). Hence

$$c' = \varphi_L^{-1}\hat{c}\varphi_L = \varphi_L^{-1}c\varphi_L = c$$

since  $c$  commutes past  $\varphi_L$ . In particular  $[n]' = [n]$ , completing the proof.

An algebra equipped with such a map is an algebra with anti-involution, and  $\text{End}^0(X)$  is one for every choice of polarization. Property (2) is the substantive one: the Rosati involution is genuinely an *anti*-automorphism, not an automorphism, so it interacts with the multiplicative structure the way transposition interacts with matrix multiplication. That analogy is exact: for  $X = E \times E$  a product of elliptic curves without complex multiplication, carrying the product polarization,  $\text{End}^0(X) \cong M_2(\mathbb{Q})$  and the Rosati involution is the transpose, since the  $H$ -adjoint of Step 1 of theorem 6.2.3 computes to the conjugate transpose and  $H$  is real here. Property (4) records that the central copy of  $\mathbb{Q}$  is fixed pointwise, which will let one restrict the involution to the center of each simple factor in Albert's classification.

The reason the Rosati involution deserves a name of its own, rather than being one anti-involution among many, is the positivity of its associated trace form. Recall from proposition 1.2.6 that every

$f \in \text{End}^0(X)$  has a rational representation  $\rho_r(f)$  acting on  $\Lambda \otimes \mathbb{Q}$ , whose characteristic polynomial has degree  $2g$  and rational (in fact, for  $f \in \text{End}(X)$ , integer) coefficients; write  $\text{Tr}(f)$  for the trace of  $\rho_r(f)$ , a  $\mathbb{Q}$ -valued linear functional on  $\text{End}^0(X)$ . Pairing  $f$  against  $g'$  through this trace produces a symmetric bilinear form, and the content of the next theorem is that this form is definite.

### Theorem 6.2.3: Positivity of the Rosati Involution

Let  $X = V/\Lambda$  be an abelian variety with polarization  $L$  and associated Rosati involution  $f \mapsto f'$ . The bilinear form

$$\langle f, g \rangle := \text{Tr}(fg'), \quad f, g \in \text{End}^0(X)$$

is a symmetric, positive-definite bilinear form on the  $\mathbb{Q}$ -vector space  $\text{End}^0(X)$ . Equivalently,

$$\text{Tr}(ff') > 0 \quad \text{for every } f \in \text{End}^0(X), f \neq 0$$

Positivity is proved analytically. The rational trace  $\text{Tr}$  is computed most transparently from the analytic representation  $\rho_a$  of definition 1.2.2, and the Rosati involution has an equally concrete analytic description as the adjoint with respect to the positive Hermitian form  $H$  underlying the polarization. Once both are in analytic terms,  $\text{Tr}(ff')$  becomes twice the real part of the trace of  $\rho_a(f)$  times its  $H$ -adjoint, which is a weighted sum of squared absolute values of matrix entries, hence positive.

#### Proof:

Write  $F := \rho_a(f) \in \text{End}_{\mathbb{C}}(V)$  for the analytic representation of  $f$  (definition 1.2.2), the  $\mathbb{C}$ -linear map on  $V = \mathbb{C}^g$  induced by  $f$ . Let  $H$  be the positive-definite Hermitian form on  $V$  defining the polarization  $L$  (definition 3.1.3), linear in the first variable and conjugate-linear in the second.

*Step 1: The Rosati involution is the  $H$ -adjoint.* Denote by  $F^*$  the adjoint of  $F$  with respect to  $H$ , that is, the unique  $\mathbb{C}$ -linear endomorphism of  $V$  with

$$H(Fv, w) = H(v, F^*w) \quad \text{for all } v, w \in V$$

which exists and is unique because  $H$  is nondegenerate. The claim is

$$\rho_a(f') = F^*$$

so that the Rosati involution is analytically the  $H$ -adjoint. To see this, recall from proposition 5.3.3 that  $\varphi_L$  is induced by the  $\mathbb{R}$ -linear map  $\varphi_H: V \rightarrow \hat{V} = \overline{V}^*$ ,  $v \mapsto H(v, -)$ , and that the dual  $\hat{f}$  is induced on  $\hat{V}$  by the transpose (pullback) of  $F$ . Unwinding  $f' = \varphi_L^{-1} \hat{f} \varphi_L$  on the level of analytic representations, an endomorphism  $G := \rho_a(f')$  is characterized by  $\varphi_H \circ G = ({}^t F) \circ \varphi_H$ , i.e.  $H(Gv, w) = H(v, Fw)$  for all  $v, w$ . Comparing with the defining relation of the adjoint gives  $G = F^*$ , which is the claim. In particular  $F^*$  is again  $\mathbb{C}$ -linear, consistent with  $f' \in \text{End}^0(X)$ .

*Step 2: The rational trace as an analytic trace.* By proposition 1.2.5, the rational representation and the analytic representation are related by  $\rho_r \otimes_{\mathbb{Q}} \mathbb{R} \cong \rho_a \oplus \overline{\rho_a}$ . Taking traces of  $\rho_r(h)$  for any  $h \in \text{End}^0(X)$ ,

$$\text{Tr}(h) = \text{tr}_{\mathbb{C}} \rho_a(h) + \overline{\text{tr}_{\mathbb{C}} \rho_a(h)} = 2 \text{Re } \text{tr}_{\mathbb{C}} \rho_a(h)$$

where  $\text{tr}_{\mathbb{C}}$  is the ordinary trace of a  $\mathbb{C}$ -linear endomorphism of  $V$ . Applying this to  $h = ff'$  and using Step 1,

$$\text{Tr}(ff') = 2 \text{Re } \text{tr}_{\mathbb{C}}(\rho_a(f)\rho_a(f')) = 2 \text{Re } \text{tr}_{\mathbb{C}}(FF^*)$$

*Step 3: Positivity via a Hermitian trace.* Choose an  $H$ -orthonormal  $\mathbb{C}$ -basis  $e_1, \dots, e_g$  of  $V$ , so that  $H(e_j, e_k) = \delta_{jk}$ ; such a basis exists because  $H > 0$ . Write  $F = (F_{jk})$  for the matrix of  $F$  in this basis,  $Fe_k = \sum_j F_{jk}e_j$ . With respect to an  $H$ -orthonormal basis the  $H$ -adjoint  $F^*$  is the conjugate transpose:  $(F^*)_{jk} = \overline{F_{kj}}$ . Therefore  $FF^*$  has diagonal entries

$$(FF^*)_{jj} = \sum_k F_{jk}(F^*)_{kj} = \sum_k F_{jk}\overline{F_{jk}} = \sum_k |F_{jk}|^2$$

which are real and nonnegative, so

$$\mathrm{tr}_{\mathbb{C}}(FF^*) = \sum_{j,k} |F_{jk}|^2 \geq 0$$

is a real number. Consequently

$$\mathrm{Tr}(ff') = 2 \operatorname{Re} \mathrm{tr}_{\mathbb{C}}(FF^*) = 2 \sum_{j,k} |F_{jk}|^2 \geq 0$$

with equality if and only if every entry  $F_{jk}$  vanishes, i.e.  $F = \rho_a(f) = 0$ . Since  $\rho_a$  is injective on  $\mathrm{End}^0(X)$  (proposition 1.1.7: an endomorphism is determined by its analytic lift),  $F = 0$  forces  $f = 0$ . Hence  $\mathrm{Tr}(ff') > 0$  whenever  $f \neq 0$ .

*Step 4: Symmetry.* First,  $\mathrm{Tr}(h) = \mathrm{Tr}(h')$  for every  $h \in \mathrm{End}^0(X)$ : by Step 2,  $\mathrm{Tr}(h) = 2 \operatorname{Re} \mathrm{tr}_{\mathbb{C}} \rho_a(h)$  and  $\mathrm{Tr}(h') = 2 \operatorname{Re} \mathrm{tr}_{\mathbb{C}} (\rho_a(h))^*$  (using Step 1), and the trace of the  $H$ -adjoint is the complex conjugate of the trace, so the two real parts agree. Second,  $(gf')' = f''g' = fg'$  by proposition 6.2.2. Combining the two,

$$\langle g, f \rangle = \mathrm{Tr}(gf') = \mathrm{Tr}((gf')') = \mathrm{Tr}(fg') = \langle f, g \rangle$$

so the form is symmetric. Combined with Step 3, it is symmetric and positive definite, as we sought to show.

Positivity is the one nontrivial constraint the geometry of a polarized abelian variety imposes on its endomorphism algebra. The bare algebraic facts, that  $\mathrm{End}^0(X)$  is semisimple and admits an anti-involution fixing the center of each simple factor, are shared by many algebras. What is special is that this particular anti-involution is *positive*: the trace form it defines is definite. This rules out, for a simple abelian variety, all but a short list of division algebras with involution, and it fixes the type of involution each can carry. Extracting that list is precisely Albert's classification (section 6.3), which reads off from positivity alone the four possible types of  $\mathrm{End}^0(X)$ .

The positive-definite form of theorem 6.2.3 organizes  $\mathrm{End}^0(X)$  into a Euclidean space, and it is convenient to name it.

**Definition 6.2.4: Rosati Form**

The *Rosati form* associated to a polarization  $L$  on  $X$  is the positive-definite symmetric bilinear form

$$\langle f, g \rangle := \text{Tr}(fg')$$

on the  $\mathbb{Q}$ -vector space  $\text{End}^0(X)$ , where  $f \mapsto f'$  is the Rosati involution of  $L$  (definition 6.2.1) and  $\text{Tr}$  is the trace of the rational representation  $\rho_r$  (proposition 1.2.6). It makes  $\text{End}^0(X)$  a finite-dimensional  $\mathbb{Q}$ -vector space with a positive-definite inner product, with respect to which the Rosati involution is self-adjoint:  $\langle fh, g \rangle = \langle h, f'g \rangle$ .

The self-adjointness  $\langle fh, g \rangle = \langle h, f'g \rangle$  is immediate from the trace identity  $\text{Tr}(fhg') = \text{Tr}(h(f'g'))$ , using  $(f'g)' = g'f'' = g'f$  and the cyclic invariance of the trace; it says the Rosati involution is the adjoint operation for the Rosati inner product, matching Step 1 of the proof at the level of  $\text{End}^0(X)$  rather than of  $V$ . This inner-product structure is what makes positivity usable: a positive-definite form has no isotropic vectors, so any element  $f$  with  $f' = f$  and  $\text{Tr}(f^2) = 0$  must vanish, and the self-adjoint elements form a formally real subspace. These are the levers Albert's classification pulls.

The elliptic-curve case grounds the whole discussion, and there the Rosati involution is a familiar operation in disguise.

**Example 6.2.5: The Rosati Involution on an Elliptic Curve**

Let  $E = \mathbb{C}/\Lambda$  be an elliptic curve with its principal polarization, so  $g = 1$  and  $\varphi_L: E \rightarrow \hat{E}$  is an isomorphism (example 5.3.7). By example 6.1.5 the endomorphism algebra  $\text{End}^0(E)$  is one of two kinds:  $\mathbb{Q}$ , or an imaginary quadratic field  $K = \mathbb{Q}(\sqrt{-d})$  (the CM case). The analytic representation identifies  $\text{End}^0(E)$  with a subalgebra of  $\text{End}_{\mathbb{C}}(V) = \mathbb{C}$  (definition 1.2.2): an endomorphism induced by multiplication by  $\alpha \in \mathbb{C}$  has  $\rho_a = \alpha$ .

By Step 1 of the proof of theorem 6.2.3, the Rosati involution is the  $H$ -adjoint on  $V = \mathbb{C}$ . Here  $H$  is a positive real multiple of the standard Hermitian form  $H(v, w) = cv\bar{w}$  with  $c > 0$ , and the adjoint of multiplication by  $\alpha$  with respect to  $H(v, w) = cv\bar{w}$  is multiplication by  $\bar{\alpha}$ , since

$$H(\alpha v, w) = c\alpha v\bar{w} = c v\overline{\alpha w} = H(v, \bar{\alpha} w)$$

Thus  $\rho_a(f') = \overline{\rho_a(f)}$ , and the two cases read off at once.

1. If  $\text{End}^0(E) = \mathbb{Q}$ , every  $f = \rho_a^{-1}(q)$  with  $q \in \mathbb{Q} \subseteq \mathbb{R}$  satisfies  $\bar{q} = q$ , so  $f' = f$ : the Rosati involution is the *identity*. Positivity says  $\text{Tr}(f^2) > 0$  for  $f \neq 0$ ; since  $\text{Tr}(f) = 2\text{Re}(q) = 2q$  (Step 2 of the proof, with  $g = 1$ ) and  $\text{Tr}(f^2) = 2q^2$ , this is the elementary inequality  $2q^2 > 0$  for  $q \neq 0$ .
2. If  $\text{End}^0(E) = K = \mathbb{Q}(\sqrt{-d})$  is imaginary quadratic, an element  $f$  with  $\rho_a(f) = \alpha \in K \subset \mathbb{C}$  has  $f' = \rho_a^{-1}(\bar{\alpha})$ , and  $\bar{\alpha}$  is the image of  $\alpha$  under the nontrivial automorphism of  $K$ : the Rosati involution is *complex conjugation* on  $K$ . The Rosati form is then

$$\text{Tr}(ff') = 2\text{Re}(\alpha\bar{\alpha}) = 2|\alpha|^2 = 2N_{K/\mathbb{Q}}(\alpha)$$

twice the field norm  $N_{K/\mathbb{Q}}(\alpha) = \alpha\bar{\alpha}$  of  $K$ . Positivity  $\text{Tr}(ff') > 0$  for  $\alpha \neq 0$  is exactly the positivity of the norm form of an imaginary quadratic field, which is positive definite because  $K$  has no real embedding.

In both cases the Rosati involution is the unique positive anti-involution the field carries: the identity on the totally real  $\mathbb{Q}$ , complex conjugation on the CM field  $K$ . This is the  $g = 1$  shadow of Albert's dichotomy, with  $\mathbb{Q}$  a Type I example and  $K$  a Type IV example (theorem 6.1.2 organizes the general simple case into which these fit).

The elliptic-curve computation exhibits in miniature the mechanism the general theory rests on. The Rosati involution restricts to complex conjugation on the CM field, and its positivity is the positivity of the norm form; for a totally real endomorphism field the involution is trivial and positivity is the positivity of the sum of squares of the real embeddings. In the general case a simple abelian variety has  $\text{End}^0(X)$  a division algebra (proposition 5.4.7), and the Rosati involution restricts to a positive involution on it whose fixed field is totally real. Positivity forces the fixed field to be totally real and forces the ambient division algebra into one of four shapes; Albert's classification (section 6.3) records which four.

### Exercise 6.2.1

1. Let  $c \in \mathbb{Q} \subseteq \text{End}^0(X)$  be a rational scalar and let  $f \in \text{End}^0(X)$  be arbitrary. Using only the properties recorded in proposition 6.2.2, compute  $(cf)'$  and  $(fc)'$  in terms of  $c$  and  $f'$ , and confirm that the two agree.
2. An element  $f \in \text{End}^0(X)$  is called *symmetric* if  $f' = f$  and *anti-symmetric* if  $f' = -f$ . Show that every  $f \in \text{End}^0(X)$  has a unique decomposition  $f = s + a$  with  $s$  symmetric and  $a$  anti-symmetric, and identify  $s$  and  $a$  explicitly. (You may use that 2 is invertible in  $\mathbb{Q}$ .)
3. Work in the setting of theorem 6.2.3, and write  $F = \rho_a(f)$ ,  $F^*$  for its  $H$ -adjoint as in Step 1 of the proof.
  1. Show that  $\text{Tr}(f'f) = \text{Tr}(ff')$  directly from the analytic formula of Steps 2 and 3, without invoking the symmetry of the Rosati form.
  2. Deduce that a symmetric endomorphism  $f = f'$  with  $\text{Tr}(f^2) = 0$  must be zero.
4. Let  $E$  be an elliptic curve without complex multiplication, so  $\text{End}^0(E) = \mathbb{Q}$  by example 6.2.5, and let  $X = E \times E$  carry the product polarization, so that  $\text{End}^0(X) \cong M_2(\mathbb{Q})$  and the Rosati involution is the transpose  $A \mapsto A^t$  (as noted after proposition 6.2.2).
  1. Verify on  $M_2(\mathbb{Q})$  that  $A \mapsto A^t$  satisfies each of the four properties of proposition 6.2.2.
  2. For  $A = (a_{ij}) \in M_2(\mathbb{Q})$  compute the Rosati form  $\langle A, A \rangle = \text{Tr}(AA^t)$  under the reduced-trace normalization  $\text{Tr} = \text{tr}$  on  $M_2(\mathbb{Q})$ , and check directly that it is positive definite.
5. Prove that the Rosati form is invariant under the Rosati involution in the following sense: for all  $f, g \in \text{End}^0(X)$ ,

$$\langle f', g' \rangle = \langle g, f \rangle$$

where  $\langle \cdot, \cdot \rangle$  is the form of definition 6.2.4. Use proposition 6.2.2 and the trace identity  $\text{Tr}(h) = \text{Tr}(h')$  established in Step 4 of the proof of theorem 6.2.3.

6. Let  $f \in \text{End}^0(X)$  be symmetric ( $f' = f$ ) and nonzero. Using the analytic description of the Rosati involution (Step 1 of the proof of theorem 6.2.3), show that  $F = \rho_a(f)$  is a self-adjoint operator on  $(V, H)$ , and conclude that  $\text{Tr}(f^2) = 2 \sum_j \mu_j^2$ , where  $\mu_1, \dots, \mu_g$  are the (real) eigenvalues of  $F$ . Deduce the positivity  $\text{Tr}(f^2) > 0$  for such  $f$  from this formula, and explain why the eigenvalues are real.

### 6.3 Albert's Classification

Among all finite-dimensional division algebras over  $\mathbb{Q}$ , only a handful admit a positive involution at all, and Albert enumerated them completely. The result is that the endomorphism algebra of a simple abelian variety, which by proposition 5.4.7 is a division algebra and by theorem 6.2.3 carries a positive involution, cannot be an arbitrary division algebra: it belongs to one of exactly four families, sorted by its center and by the totally real field the involution fixes. This section states and proves the classification, deferring only the exhaustive numerical constraint table to the standard references. The right object to classify is not the division algebra alone but the pair consisting of the algebra and its involution.

#### Definition 6.3.1: Division Algebra with Positive Involution

A *division algebra with positive involution* over  $\mathbb{Q}$  is a pair  $(D, ')$  where  $D$  is a finite-dimensional division algebra over  $\mathbb{Q}$  whose center  $K$  is a field, and  $': D \rightarrow D$  is a  $\mathbb{Q}$ -linear anti-involution (so  $(a + b)' = a' + b'$ ,  $(ab)' = b'a'$ , and  $a'' = a$ ) that is *positive*: the symmetric bilinear form  $(a, b) \mapsto \text{Tr}_{D/\mathbb{Q}}(ab')$  on  $D$  is positive definite, where  $\text{Tr}_{D/\mathbb{Q}}$  is the trace of left multiplication on the  $\mathbb{Q}$ -vector space  $D$  (equivalently  $\text{Tr}_{K/\mathbb{Q}}$  composed with  $d$  times the reduced trace  $\text{Trd}_{D/K}$ , since  $\text{Tr}_{D/K} = d \text{Trd}_{D/K}$  for  $d = [D: K]^{1/2}$ ). Write  $K_0 = \{a \in K \mid a' = a\}$  for the fixed field of the involution inside the center.

The definition abstracts exactly the structure the Rosati involution supplies. For a simple abelian variety  $X$ , the algebra  $D = \text{End}^0(X)$  is a division algebra by proposition 5.4.7, and the Rosati involution of definition 6.2.1 is a positive  $\mathbb{Q}$ -linear anti-involution by theorem 6.2.3, with the positive form there being  $(f, g) \mapsto \text{Tr}(fg')$ . So  $(\text{End}^0(X), ')$  is a division algebra with positive involution in the sense above, and classifying these pairs classifies the algebras that can occur.

The first structural fact is that the fixed field  $K_0$  is forced to be totally real. Positivity is the whole of the input.

#### Proposition 6.3.2: The Fixed Field Is Totally Real

Let  $(D, ')$  be a division algebra with positive involution over  $\mathbb{Q}$ , with center  $K$  and fixed field  $K_0 = K'$ . Then  $K_0$  is a totally real number field: every embedding  $K_0 \hookrightarrow \mathbb{C}$  has image in  $\mathbb{R}$ . Moreover the center  $K$  is either equal to  $K_0$  (the involution fixes the center) or is a totally imaginary quadratic extension of  $K_0$  (a CM field, on which  $'$  restricts to complex conjugation).

#### Proof:

*Step 1: Positivity on the center.* The restriction of  $'$  to  $K$  is a  $\mathbb{Q}$ -linear involution of the field  $K$ , so its fixed field  $K_0$  is a subfield with  $[K: K_0] \in \{1, 2\}$ . Because  $D$  is a division algebra over  $\mathbb{Q}$ , the trace form  $(a, b) \mapsto \text{Tr}_{D/\mathbb{Q}}(ab')$  restricts to a positive definite form on  $K_0$ : for  $a \in K_0$  nonzero,  $a' = a$ , so  $\text{Tr}_{D/\mathbb{Q}}(aa') = \text{Tr}_{D/\mathbb{Q}}(a^2) > 0$  by positivity.

*Step 2: Every embedding of  $K_0$  is real.* Fix an embedding  $\sigma: K_0 \hookrightarrow \mathbb{C}$  and suppose it were not real, i.e.  $\sigma(K_0) \not\subseteq \mathbb{R}$ . The trace  $\text{Tr}_{K_0/\mathbb{Q}}(a^2)$  is a positive multiple of  $\sum_{\sigma} \sigma(a)^2$ , the sum running over the  $[K_0: \mathbb{Q}]$  embeddings of  $K_0$  into  $\mathbb{C}$ , each counted once (the reduced trace on  $K_0$  is the field trace, and the  $D/\mathbb{Q}$  trace is a positive integer multiple of it). If some  $\sigma$  were complex, its



conjugate  $\bar{\sigma}$  is a distinct embedding, and one can choose  $a \in K_0$  with  $\sigma(a)$  a nonzero purely imaginary number to make the pair's contribution  $\sigma(a)^2 + \bar{\sigma}(a)^2 = 2\operatorname{Re}(\sigma(a)^2)$  negative while the real embeddings contribute arbitrarily little. Concretely, approximating the idempotent-like element that isolates the  $\sigma$ -component (which exists since  $K_0 \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^r \times \mathbb{C}^s$  as an  $\mathbb{R}$ -algebra, with  $s$  the number of complex places) produces  $a \in K_0$  with  $\operatorname{Tr}_{K_0/\mathbb{Q}}(a^2) < 0$ , contradicting Step 1. Hence  $s = 0$ : every embedding is real and  $K_0$  is totally real.

*Step 3: The center over  $K_0$ .* If  $[K : K_0] = 1$  then  $K = K_0$  is totally real and  $'$  fixes the center. If  $[K : K_0] = 2$ , then  $K = K_0(\alpha)$  with  $\alpha' = -\alpha$  (choose  $\alpha$  in the  $'$ -anti-invariant line), and  $\beta := \alpha^2 \in K_0$  is fixed. The same positivity argument, applied place by place at each real embedding  $\sigma$  of  $K_0$  (as in Step 2, isolating the  $\sigma$ -component), forces  $\sigma(\beta) = \sigma(\alpha)^2 < 0$  at every real place: were  $\sigma(\beta) > 0$  for some  $\sigma$ , one could produce an anti-invariant  $a \in K$  with  $\operatorname{Tr}_{K/\mathbb{Q}}(aa') < 0$ , contradicting positivity. Thus  $\beta = \alpha^2 \in K_0$  is totally negative, so  $K = K_0(\sqrt{\beta})$  is totally imaginary over the totally real  $K_0$ . Thus  $K$  is a CM field and  $'$ , sending  $\alpha \mapsto -\alpha$ , is complex conjugation at every archimedean place, completing the proof.

The proposition already carves the possibilities into two worlds. Either the center is the totally real field  $K_0$ , and the involution is of the *first kind* (trivial on the center), or the center is a CM field with the involution of the *second kind* (complex conjugation on the center). Albert's theorem refines the first kind into three sub-cases according to whether  $D$  is commutative or a quaternion algebra and, in the quaternion case, whether the involution is orthogonal or symplectic. The result is four types.

#### Definition 6.3.3: The Four Albert Types

Let  $(D, ')$  be a division algebra with positive involution over  $\mathbb{Q}$ , with center  $K$  and totally real fixed field  $K_0$ . The pair is of one of the following four types.

1. *Type I.*  $D = K = K_0$  is a totally real field and  $'$  is the identity.
2. *Type II.*  $K = K_0$  is totally real,  $D$  is a quaternion division algebra over  $K_0$  that is *totally indefinite* (split at every real place,  $D \otimes_{K_0, \sigma} \mathbb{R} \cong M_2(\mathbb{R})$  for every embedding  $\sigma$ ), and  $'$  is of *orthogonal type*.
3. *Type III.*  $K = K_0$  is totally real,  $D$  is a quaternion division algebra over  $K_0$  that is *totally definite* (ramified at every real place,  $D \otimes_{K_0, \sigma} \mathbb{R}$  is the Hamilton quaternions for every embedding  $\sigma$ ), and  $'$  is of *symplectic type*.
4. *Type IV.*  $K$  is a CM field (a totally imaginary quadratic extension of the totally real  $K_0$ ),  $D$  is a division algebra with center  $K$ , and  $'$  is of the *second kind*, restricting to complex conjugation on  $K$ .

Here the Hamilton quaternions are the unique division algebra of dimension 4 over  $\mathbb{R}$ , and the involution kind (orthogonal or symplectic) refers to the type of the symmetric bilinear form fixed by  $'$  on the reduced  $2 \times 2$  representation of the split quaternion algebra.

The distinction between orthogonal and symplectic involutions in Types II and III is the involution's behavior on a quaternion algebra  $D$  over  $K_0$ . Every  $K_0$ -linear involution of a quaternion algebra is either the standard (conjugation) involution or a twist of it; the standard involution is symplectic, and a nonstandard one is orthogonal. Positivity selects: a totally indefinite quaternion algebra admits a positive involution only of orthogonal type, and a totally definite one only of symplectic



type, which is why the two quaternionic cases split cleanly into Types II and III.

#### Theorem 6.3.4: Albert's Classification

Every division algebra with positive involution  $(D, ')$  over  $\mathbb{Q}$  is of exactly one of the four types of definition 6.3.3. The type is determined by the pair  $(K, K_0)$  and the kind of the involution:

- if  $D$  is commutative, then  $D = K$ ; it is Type I when  $K = K_0$  and the maximal Type IV case (a field) when  $K$  is CM;
- if  $D$  is noncommutative with  $K = K_0$ , then  $D$  is a quaternion algebra over  $K_0$  and the type is II or III according as the positive involution is orthogonal or symplectic, equivalently according as  $D$  is totally indefinite or totally definite;
- if  $D$  is noncommutative with  $K$  a CM field, the type is IV.

Consequently the endomorphism algebra  $\text{End}^0(X)$  of a simple abelian variety  $X$ , together with its Rosati involution, is of one of these four types.

#### Proof:

Let  $(D, ')$  be a division algebra with positive involution, center  $K$ , fixed field  $K_0$ . By proposition 6.3.2,  $K_0$  is totally real, and either  $K = K_0$  or  $K$  is a CM field with  $'$  complex conjugation on it. These two cases are the involutions of the first and second kind. The proof is a case analysis on  $[K : K_0]$  and on the dimension  $[D : K]$ , which for a division algebra with center a field is a perfect square  $d^2$  (Wedderburn theory of central simple algebras, *Core Representation Theory* [3]).

*Step 1: Second kind gives Type IV.* If  $K$  is a CM field, then by definition the involution restricts to complex conjugation on the center, and  $(D, ')$  is of Type IV whatever the value of  $d$ . When  $d = 1$  the algebra  $D = K$  is the field itself (the maximal commutative Type IV case, realized by CM abelian varieties); when  $d > 1$  it is a genuine division algebra over the CM field  $K$ . No further constraint on the kind is needed, since complex conjugation on a CM field is the unique nontrivial involution fixing  $K_0$ .

*Step 2: First kind, commutative, is Type I.* Suppose  $K = K_0$  and  $D$  is commutative, so  $D = K = K_0$  is a totally real field. The involution is a  $\mathbb{Q}$ -linear involution of a field fixing  $K_0 = K$  pointwise, hence trivial. Positivity is automatic and the trace form on a totally real field is positive definite. This is Type I.

*Step 3: First kind, noncommutative, is quaternionic.* Suppose  $K = K_0$  and  $D$  is noncommutative of degree  $d = [D : K]^{1/2} > 1$ . The existence of an involution of the first kind already forces  $D$  to have order dividing 2 in the Brauer group of  $K$ : the involution gives an isomorphism  $D \cong D^{\text{op}}$ , and  $D^{\text{op}}$  is the Brauer inverse of  $D$ , so  $[D] = [D]^{-1}$  and  $2[D] = 0$  in  $\text{Br}(K)$  (*Core Representation Theory* [3]). This alone does not bound the degree: exponent 2 only forces  $d$  to be a power of 2, and there exist central division algebras of exponent 2 and degree 4 over a number field (Albert's biquaternion algebras). What rules those out here is not the involution alone but its *positivity*. This is a theorem of Albert: a division algebra over a totally real number field that carries a positive involution of the first kind has degree  $d \leq 2$ . The obstruction is global. At

each real place  $\sigma$  the completion  $D \otimes_{K_0, \sigma} \mathbb{R}$  is a real central simple algebra (a matrix algebra over  $\mathbb{R}$  or over the Hamilton quaternions), and a first-kind involution positive at  $\sigma$  exists on such a completion in every degree; positivity at a single place does not cap the degree, exactly as the transpose is a positive first-kind involution on  $M_n(\mathbb{R})$  for every  $n$ . The constraint is that a *single*  $\mathbb{Q}$ -algebra involution must be positive at *all* real places of  $K_0$  simultaneously while  $D$  remains a division algebra globally, and Albert's analysis shows this is impossible once  $d > 2$ ; the argument is a real-place analysis of positive first-kind involutions on the division algebra, carried out in Albert's *Structure of Algebras* and reproduced in Mumford's *Abelian Varieties*. Combined with  $d > 1$  this gives  $d = 2$ , so  $D$  is a quaternion algebra over  $K_0$ .

*Step 4: Orthogonal versus symplectic.* An involution of the first kind on a quaternion algebra  $D$  is either the standard involution  $x \mapsto \bar{x}$  (whose fixed space is the center  $K_0$ , of dimension 1 over  $K_0$ ) or a nonstandard one (whose fixed space of  $'$ -symmetric elements has dimension 3). The standard involution is *symplectic*; a nonstandard involution is *orthogonal*. At each real place  $\sigma$  of  $K_0$ , positivity of the involution forces a compatibility with the local invariant of  $D$ : the split quaternion algebra  $M_2(\mathbb{R})$  carries a positive involution only of orthogonal type, while the ramified quaternion algebra (the Hamilton quaternions) carries a positive involution only of symplectic type (the standard conjugation  $x \mapsto \bar{x}$  is positive because the reduced trace satisfies  $\text{Trd}(x\bar{x}) = 2 \text{Nrd}(x) > 0$ ). Since a positive involution over  $\mathbb{Q}$  is positive at every real place simultaneously, either  $D$  is split at every real place and  $'$  is everywhere orthogonal (Type II, totally indefinite), or  $D$  is ramified at every real place and  $'$  is everywhere symplectic (Type III, totally definite). The two cannot mix, because the involution's kind is a global datum: it is orthogonal or symplectic as an involution of the  $\mathbb{Q}$ -algebra  $D$ , and this single choice must be positive at every real place at once.

The four cases exhaust the possibilities of proposition 6.3.2 and are mutually exclusive by their invariants  $(K, K_0, \text{kind})$ , so every  $(D, ')$  is of exactly one type. Applying this to  $D = \text{End}^0(X)$  with its Rosati involution, which is a division algebra with positive involution when  $X$  is simple, gives the final assertion, completing the proof.

Albert's theorem is the exact answer to the question that opened this chapter: which division algebras arise as  $\text{End}^0$  of a simple abelian variety? The space of finite-dimensional division algebras over  $\mathbb{Q}$  is a continuum, indexed by centers and by local invariants at every place; positivity cuts it down to four families, each governed by a totally real field  $K_0$  and a small amount of quaternionic or CM data. The refinement of this coarse classification into precise numerical bounds relates three integers: the degree  $[K_0: \mathbb{Q}]$ , the degree  $d = [D: K]^{1/2}$ , and the dimension  $g$  of the abelian variety. The next proposition records these constraints, citing the exhaustive table to the standard sources.

**Proposition 6.3.5: Structural Constraints on the Type**

Let  $X$  be a simple abelian variety of dimension  $g$  over an algebraically closed field,  $D = \text{End}^0(X)$  its endomorphism algebra,  $K$  the center of  $D$ ,  $K_0$  the totally real fixed field of the Rosati involution,  $e_0 = [K_0 : \mathbb{Q}]$ , and  $d = [D : K]^{1/2}$  the degree of  $D$  over its center. Then  $K_0$  is totally real,  $(D, ')$  is of one of the four types of theorem 6.3.4, and the following divisibility constraints relate the invariants to  $g$ :

- Type I:  $D = K_0$ ,  $e_0 | g$ ;
- Type II:  $d = 2$ ,  $K = K_0$ ,  $2e_0 | g$ ;
- Type III:  $d = 2$ ,  $K = K_0$ ,  $2e_0 | g$ ;
- Type IV:  $[K : K_0] = 2$ , and  $e_0 d^2 | g$ .

In particular  $[D : \mathbb{Q}]$  divides  $2g$  in Types I, II, III (where  $[D : \mathbb{Q}] = e_0 d^2$  with  $d \in \{1, 2\}$ ) and  $\text{End}^0(X)$  has  $\mathbb{Q}$ -dimension at most  $(2g)^2$  in all cases (proposition 6.1.3). The precise upper bounds in each type, and the finer constraints in positive characteristic, are as in Mumford's *Abelian Varieties* and Birkenhake-Lange's *Complex Abelian Varieties*.

**Proof:**

That  $K_0$  is totally real and that  $(D, ')$  is of one of the four types is proposition 6.3.2 and theorem 6.3.4. The divisibility constraints come from the action of  $D$  on the rational homology  $H_1(X, \mathbb{Q})$ , a  $\mathbb{Q}$ -vector space of dimension  $2g$  which by proposition 1.2.6 carries the faithful rational representation  $\rho_r$  of  $\text{End}^0(X)$ . Because  $X$  is simple,  $H_1(X, \mathbb{Q})$  is a module over the division algebra  $D$ , hence a free  $D$ -module, so  $[D : \mathbb{Q}] = e_0 d^2 [K : K_0]$  divides  $\dim_{\mathbb{Q}} H_1(X, \mathbb{Q}) = 2g$ . Reading off the value  $[K : K_0] \in \{1, 2\}$  and  $d \in \{1, 2, \dots\}$  in each type yields the stated divisibilities. In Types II and III,  $[D : \mathbb{Q}] = e_0 d^2 [K : K_0] = e_0 \cdot 4 \cdot 1 = 4e_0$ , so  $4e_0 | 2g$ , which is exactly  $2e_0 | g$ ; no analytic refinement is needed. In Type IV,  $[D : \mathbb{Q}] = e_0 d^2 \cdot 2$ , so  $2e_0 d^2 | 2g$ , i.e.  $e_0 d^2 | g$  directly. Only Type I needs the refinement: there  $[D : \mathbb{Q}] = e_0$  gives  $e_0 | 2g$ , and  $e_0 | g$  follows from the fact that the analytic representation is a form of the rational one of half the dimension.

The Type I refinement from  $e_0 | 2g$  to  $e_0 | g$  requires tracking how the analytic representation  $\rho_a$  of definition 1.2.2 sits inside  $\rho_r \cong \rho_a \oplus \bar{\rho}_a$  (proposition 1.2.5): the tangent space  $V = H^{1,0}$  is a  $D$ -module of dimension  $g$  over  $\mathbb{C}$ , and the compatibility of a positive involution with a CM or totally real structure constrains the multiplicities of the simple  $D \otimes \mathbb{R}$ -summands in  $V$ . The full bookkeeping producing the exhaustive table of admissible  $(e_0, d, g)$  in each type, together with the reduction data  $[K : K_0]$  and the multiplicity restrictions on  $V$ , is a finite but intricate case analysis; it is carried out in full in Mumford's *Abelian Varieties* and Birkenhake-Lange's *Complex Abelian Varieties*, and is cited here rather than reproduced, per the batch contract's sanctioned deferral. What is proved in full above is the qualitative content: the four types, the totally real fixed field, and the divisibility of  $[D : \mathbb{Q}]$  into  $2g$ , which is what the applications in this book require, completing the proof.

The constraints already rule out many possibilities. For an elliptic curve,  $g = 1$ , so  $[D : \mathbb{Q}] | 2$ : the endomorphism algebra is  $\mathbb{Q}$  (Type I with  $e_0 = 1$ ) or an imaginary quadratic field (Type IV with  $e_0 = 1$ ,  $d = 1$ ), and no quaternion algebra can occur over  $\mathbb{C}$ , matching example 6.1.5. Every type is

realized somewhere in the range  $g \geq 1$ , and the following example exhibits a witness for each.

**Example 6.3.6: Realizations of the Four Types**

Each Albert type occurs as the endomorphism algebra of a simple abelian variety.

1. *Type I.* The generic elliptic curve  $E = \mathbb{C}/\Lambda$  with  $\text{End}^0(E) = \mathbb{Q}$  is Type I with  $K_0 = \mathbb{Q}$ ,  $e_0 = 1$ , and  $g = 1$ . A higher-dimensional Type I example is a *real multiplication* abelian surface: a simple abelian surface  $X$  with  $g = 2$  whose endomorphism algebra is a real quadratic field  $K_0 = \mathbb{Q}(\sqrt{m})$ ,  $m > 0$ , so  $e_0 = 2 = g$ , saturating the constraint  $e_0 | g$ . Such surfaces populate the Hilbert modular surfaces and are the first place beyond elliptic curves where a nontrivial totally real endomorphism field appears.
2. *Type IV.* A CM elliptic curve, for instance  $E: y^2 = x^3 - x$  with  $\text{End}^0(E) = \mathbb{Q}(i)$ , is Type IV with the CM field  $K = \mathbb{Q}(i)$ ,  $K_0 = \mathbb{Q}$ ,  $d = 1$ , and  $g = 1$ ; the Rosati involution is complex conjugation on  $\mathbb{Q}(i)$ . The general Type IV case with  $D = K$  a field is a CM abelian variety  $X_\Phi = \mathbb{C}^\Phi / \Phi(\mathcal{O}_K)$  for a CM field  $K$  of degree  $2g$  and a CM type  $\Phi$ , the construction taken up in section 6.4; there  $K$  is a maximal commutative semisimple subalgebra of  $\text{End}^0(X)$  of dimension  $2g$ , the maximal Type IV situation. Noncommutative Type IV, with  $D$  a genuine division algebra over a CM field, first occurs at  $g \geq 4$ , since the constraint  $e_0 d^2 | g$  of proposition 6.3.5 already forces  $4 | g$  in the minimal case  $e_0 = 1$ ,  $d = 2$ .
3. *Type II.* A simple abelian surface with indefinite quaternionic multiplication realizes Type II:  $g = 2$ ,  $K_0 = \mathbb{Q}$ ,  $e_0 = 1$ , and  $D$  an indefinite ( $D \otimes \mathbb{R} \cong M_2(\mathbb{R})$ ) quaternion division algebra over  $\mathbb{Q}$  with  $2e_0 = 2 = g$ . These are the fake elliptic curves, abelian surfaces with multiplication by an indefinite quaternion order, parametrized by Shimura curves; they are the smallest-dimensional quaternionic case.
4. *Type III.* The divisibility  $2e_0 | g$  permits  $g = 2$  for a definite quaternion algebra ( $D \otimes \mathbb{R}$  the Hamilton quaternions), but this necessary constraint is not sufficient: over  $\mathbb{C}$  a definite quaternion algebra over  $\mathbb{Q}$  does not occur as  $\text{End}^0$  of a simple abelian *surface*. Indeed the simple abelian surfaces over  $\mathbb{C}$  have endomorphism algebra  $\mathbb{Q}$ , a real quadratic field (Type I), an indefinite quaternion algebra (Type II), or a quartic CM field (Type IV); the definite quaternion case is excluded by the multiplicity restrictions on the tangent space (Shimura; Birkenhake-Lange, *Complex Abelian Varieties*, Ch. 9). A genuine Type III witness first appears in higher dimension: with  $K_0 = \mathbb{Q}$  and  $d = 2$  the smallest simple abelian variety with  $\text{End}^0$  a definite quaternion algebra over  $\mathbb{Q}$  has  $g = 4$ , the finer bound being part of the exhaustive table deferred to those references.

**Exercise 6.3.1**

1. Classify the following pairs  $(D, ')$  by their Albert type, using definition 6.3.3, and in each case name the center  $K$ , the fixed field  $K_0$ , the degree  $d = [D : K]^{1/2}$ , and the kind of the involution.
  1.  $D = \mathbb{Q}(\sqrt{2})$  with  $'$  the identity.
  2.  $D = \mathbb{Q}(i)$  with  $'$  complex conjugation.
  3.  $D$  the Hamilton quaternion division algebra over  $\mathbb{Q}$  (ramified at the real place) with  $'$  the standard conjugation  $x \mapsto \bar{x}$ .

4.  $D$  an indefinite quaternion division algebra over  $\mathbb{Q}$  (so  $D \otimes \mathbb{R} \cong M_2(\mathbb{R})$ ) with a positive involution.
2. Let  $D = \mathbb{Q}(i)$ , the Gaussian rationals, viewed as a division algebra over  $\mathbb{Q}$  with center  $K = D$ . Let  $'$  be complex conjugation, so  $K_0 = \mathbb{Q}$ . Compute the trace form  $(a, b) \mapsto \text{Tr}_{D/\mathbb{Q}}(ab')$  explicitly on the basis  $\{1, i\}$ , exhibit its Gram matrix, and verify directly that it is positive definite, confirming that  $(\mathbb{Q}(i), ')$  is a division algebra with positive involution in the sense of definition 6.3.1.
3. Show that the standard conjugation  $x \mapsto \bar{x}$  on the Hamilton quaternion algebra  $\mathbb{H} = \mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$  is a positive involution over  $\mathbb{R}$ , by computing  $\text{Trd}(x\bar{x})$  for  $x = x_0 + x_1i + x_2j + x_3k$  and relating it to the reduced norm. This is the local computation underlying the symplectic case of Step 4 of the proof of theorem 6.3.4.
4. Use the divisibility constraints of proposition 6.3.5 to determine, for each Albert type, the smallest dimension  $g$  of a simple abelian variety  $X$  over  $\mathbb{C}$  whose endomorphism algebra  $\text{End}^0(X)$  realizes that type with the smallest admissible invariants  $(e_0, d)$ . Compare your answers with the witnesses recorded in example 6.3.6, and explain why the divisibility bound alone does not always guarantee a realization.
5. Prove that if  $(D, ')$  is a division algebra with positive involution over  $\mathbb{Q}$  whose center  $K$  equals the fixed field  $K_0$  (an involution of the first kind), then  $K_0$  is totally real and the trace form restricts to a positive definite form on  $K_0$ . Deduce that no CM field can be the fixed field of a positive involution. (This is the content of proposition 6.3.2 for the first-kind case; give the argument in full.)
6. Let  $X$  be a simple abelian variety over  $\mathbb{C}$  of dimension  $g = 3$ . Using proposition 6.3.5, enumerate the Albert types and invariants  $(K, K_0, d)$  that  $\text{End}^0(X)$  can have, and in each case give  $[\text{End}^0(X) : \mathbb{Q}]$ . Conclude that no quaternionic case (Type II or III) can occur when  $g = 3$ .

## 6.4 Complex Multiplication

Albert's classification (theorem 6.3.4) fixes the shape of the algebras that can occur as  $\text{End}^0(X)$ , but it says nothing about which abelian variety realizes a given algebra, nor how large the algebra can grow before the abelian variety it acts on is pinned down. The extremal case is the one where the commutative part of  $\text{End}^0(X)$  is as large as the rational representation permits: a field of degree  $2g$  acting on a  $g$ -dimensional abelian variety. This is complex multiplication. Its construction is purely a matter of embedding a number field into  $\mathbb{C}^g$ , yet the resulting tori turn out to be defined over number fields and to generate abelian extensions of those fields, exactly as the  $j$ -invariant of a CM elliptic curve generates the Hilbert class field of an imaginary quadratic field.

This section builds the CM abelian varieties from a field and a combinatorial datum, the CM type, and identifies them inside Albert's Type IV. The arithmetic main theorem, which governs their fields of definition and the reciprocity law relating the two, is the subject of the downstream volume and is only stated here.

The fields that can act are constrained by the positivity of the Rosati involution: an involution of the second kind must fix a totally real field and conjugate on the field itself. The class of fields with

this property has a name.

**Definition 6.4.1: CM Field**

A *CM field* is a totally imaginary quadratic extension  $K$  of a totally real number field  $K_0$ . Thus  $K_0$  is a number field all of whose embeddings  $K_0 \hookrightarrow \mathbb{C}$  land in  $\mathbb{R}$ , and  $K = K_0(\sqrt{-\delta})$  for some totally positive  $\delta \in K_0$ , so that  $[K : \mathbb{Q}] = 2[K_0 : \mathbb{Q}]$  and no embedding of  $K$  into  $\mathbb{C}$  has image in  $\mathbb{R}$ . Complex conjugation on  $\mathbb{C}$  induces, via any embedding  $K \hookrightarrow \mathbb{C}$ , a single automorphism of  $K$  independent of the embedding, the *complex conjugation* of  $K$ , written  $a \mapsto \bar{a}$ ; its fixed field is  $K_0$ .

The independence of the conjugation from the chosen embedding is what makes a CM field behave like  $\mathbb{C}$  over  $\mathbb{R}$ : there is an intrinsic “real part”  $K_0$  and an intrinsic conjugation, and every place at infinity is complex. For  $K_0 = \mathbb{Q}$  the definition recovers the imaginary quadratic fields, the fields that govern the  $g = 1$  theory (*Elliptic Curves* [4]). See *Algebraic Number Theory* [1] for totally real and CM fields, their rings of integers, and the fact that the conjugation is well defined.

**Example 6.4.2: The Cyclotomic Field  $\mathbb{Q}(\zeta_n)$**

For  $n \geq 3$  the cyclotomic field  $K = \mathbb{Q}(\zeta_n)$  is a CM field. Its maximal real subfield is  $K_0 = \mathbb{Q}(\zeta_n + \zeta_n^{-1})$ , which is totally real of degree  $\varphi(n)/2$ , and  $K = K_0(\zeta_n)$  is a totally imaginary quadratic extension, since  $\zeta_n$  satisfies  $x^2 - (\zeta_n + \zeta_n^{-1})x + 1 = 0$  with discriminant  $(\zeta_n + \zeta_n^{-1})^2 - 4 = (\zeta_n - \zeta_n^{-1})^2 < 0$  under every real embedding of  $K_0$ . Complex conjugation is  $\zeta_n \mapsto \zeta_n^{-1} = \bar{\zeta}_n$ . The smallest case  $n = 3$  gives  $K = \mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ , imaginary quadratic, with  $K_0 = \mathbb{Q}$ : this is the field of complex multiplication of the curve  $y^2 = x^3 + 1$ .

To turn a CM field into an abelian variety we must embed it into a complex vector space, and a  $\mathbb{C}$ -algebra homomorphism  $K \rightarrow \mathbb{C}^g$  is a choice of  $g$  of the  $2g$  embeddings of  $K$ . The embeddings come in conjugate pairs, and choosing one from each pair is precisely the datum that makes the embedded lattice compatible with a complex structure.

**Definition 6.4.3: CM Type**

Let  $K$  be a CM field with  $[K : \mathbb{Q}] = 2g$ , so that  $|\text{Hom}(K, \mathbb{C})| = 2g$ . A *CM type* for  $K$  is a subset  $\Phi \subset \text{Hom}(K, \mathbb{C})$  such that

$$\Phi \sqcup \bar{\Phi} = \text{Hom}(K, \mathbb{C})$$

where  $\bar{\Phi} = \{\bar{\varphi} : \varphi \in \Phi\}$  and  $\bar{\varphi}(a) := \overline{\varphi(a)}$ . Equivalently,  $\Phi$  contains exactly one embedding from each conjugate pair  $\{\varphi, \bar{\varphi}\}$ . In particular  $|\Phi| = g = [K_0 : \mathbb{Q}]$ . The pair  $(K, \Phi)$  is a *CM pair* of degree  $2g$ .

A CM type is a system of representatives for the complex conjugation acting on the  $2g$  embeddings. Since conjugation is fixed-point-free on  $\text{Hom}(K, \mathbb{C})$  (no embedding is real, by total imaginarity), the  $2g$  embeddings split into  $g$  conjugate pairs, and a CM type picks one from each: there are exactly  $2^g$  of them. The choice matters. Different CM types on the same field produce non-isomorphic, in general non-isogenous, abelian varieties, and the type is the combinatorial shadow of the analytic structure.

**Example 6.4.4: CM Types on an Imaginary Quadratic Field**

Let  $K$  be imaginary quadratic, so  $[K : \mathbb{Q}] = 2$  and  $g = 1$ . There are two embeddings  $\varphi, \bar{\varphi} : K \rightarrow \mathbb{C}$ , forming a single conjugate pair, so there are exactly  $2^1 = 2$  CM types,  $\Phi = \{\varphi\}$  and  $\Phi' = \{\bar{\varphi}\}$ . The two are interchanged by conjugation and give complex-conjugate, hence isomorphic, elliptic curves. For a genuinely higher-dimensional example take  $K = \mathbb{Q}(\zeta_5)$ , a CM field of degree 4 with  $g = 2$  (example 6.4.2). The four embeddings send  $\zeta_5$  to the four primitive fifth roots of unity  $\zeta, \zeta^2, \zeta^3, \zeta^4$ , paired by conjugation as  $\{\zeta, \zeta^4\}$  and  $\{\zeta^2, \zeta^3\}$  since  $\zeta^4 = \bar{\zeta}$  and  $\zeta^3 = \bar{\zeta}^2$ . Choosing  $\Phi = \{\zeta \mapsto \zeta, \zeta \mapsto \zeta^2\}$  gives one of the  $2^2 = 4$  CM types.

The action of  $K$  on the associated abelian variety is what deserves the name “complex multiplication”: the field multiplies the tangent space in  $g$  independent ways at once. Before constructing the variety, the defining property is worth isolating, so that it can be checked against Albert’s list.

**Definition 6.4.5: Abelian Variety with Complex Multiplication**

An abelian variety  $X$  of dimension  $g$  has *complex multiplication* if its endomorphism algebra  $\text{End}^0(X)$  contains a commutative semisimple  $\mathbb{Q}$ -subalgebra of dimension  $2g$  over  $\mathbb{Q}$ ; when that subalgebra is a field  $K$ , one says  $X$  has *complex multiplication by  $K$* , and  $2g = [K : \mathbb{Q}]$  is forced. Since the rational representation  $\rho_r$  is faithful of degree  $2g$  (proposition 1.2.6), a commutative subalgebra of  $\text{End}^0(X)$  has  $\mathbb{Q}$ -dimension at most  $2g$ , so complex multiplication is the maximal commutative case.

The bound  $\dim_{\mathbb{Q}} \leq 2g$  comes directly from the faithfulness of  $\rho_r$ : a commutative semisimple algebra acting faithfully on the  $2g$ -dimensional  $\mathbb{Q}$ -vector space  $\Lambda \otimes \mathbb{Q}$  has dimension at most  $2g$ , because a maximal commutative subalgebra of  $\text{End}_{\mathbb{Q}}(\Lambda \otimes \mathbb{Q}) \cong M_{2g}(\mathbb{Q})$  that is a product of fields is a maximal étale subalgebra, of dimension equal to the size  $2g$  of a simultaneous eigenbasis after extending scalars (*Core Algebra* [12]). Complex multiplication saturates this bound. When  $X$  is simple the relevant subalgebra is a field, and the definition asks for a maximal subfield of degree  $2g$ ; the general definition allows a product of CM fields, matching the reducible case. As will be seen in proposition 6.4.10, positivity of the Rosati form forces such a maximal field to be a CM field, which is why the name attaches to that class.

We now construct, from a CM pair, an abelian variety carrying the action. The construction is the direct generalization of  $E = \mathbb{C}/\mathcal{O}_K$  from the imaginary quadratic case.



**Theorem 6.4.6: Construction of CM Abelian Varieties**

Let  $(K, \Phi)$  be a CM pair with  $[K : \mathbb{Q}] = 2g$  and  $|\Phi| = g$ , and let  $\mathcal{O}_K$  be the ring of integers of  $K$ . Embed  $K$  into  $\mathbb{C}^\Phi \cong \mathbb{C}^g$  by

$$\Phi: K \longrightarrow \mathbb{C}^\Phi, \quad a \longmapsto (\varphi(a))_{\varphi \in \Phi}$$

Then:

1.  $\Phi(\mathcal{O}_K)$  is a full lattice in  $\mathbb{C}^\Phi$ , so  $X_\Phi := \mathbb{C}^\Phi / \Phi(\mathcal{O}_K)$  is a complex torus of dimension  $g$ .
2.  $K$  acts on  $X_\Phi$  by  $\mathbb{Q}$ -algebra endomorphisms: for  $k \in K$ , multiplication by the diagonal matrix  $\text{diag}(\varphi(k))_{\varphi \in \Phi}$  on  $\mathbb{C}^\Phi$  preserves  $\Phi(K)$  and induces  $K \hookrightarrow \text{End}^0(X_\Phi)$ . In particular  $X_\Phi$  has complex multiplication by  $K$ .
3.  $X_\Phi$  is an abelian variety.

**Proof:**

*Step 1: The lattice is full.* The map  $\Phi: K \rightarrow \mathbb{C}^\Phi$  is  $\mathbb{Q}$ -linear, so it suffices to show that  $\Phi(\mathcal{O}_K)$  spans  $\mathbb{C}^\Phi$  over  $\mathbb{R}$ , equivalently that  $\Phi$  extends to an  $\mathbb{R}$ -linear isomorphism  $K \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow{\sim} \mathbb{C}^\Phi$ . Now  $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$  decomposes according to the archimedean places of  $K$ ; since  $K$  is totally imaginary of degree  $2g$ , it has  $r_1 = 0$  real places and  $r_2 = g$  complex places, so  $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{C}^g$  as an  $\mathbb{R}$ -algebra, of real dimension  $2g$ . The  $g$  complex places correspond bijectively to the conjugate pairs  $\{\varphi, \bar{\varphi}\}$ , and a CM type  $\Phi$  selects one embedding from each pair; the map  $a \otimes 1 \mapsto (\varphi(a))_{\varphi \in \Phi}$  is exactly the resulting isomorphism  $K \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow{\sim} \mathbb{C}^\Phi$ , since each complex place contributes the pair  $(\varphi, \bar{\varphi})$  and retaining  $\varphi$  records the full complex coordinate. Hence  $\Phi(K \otimes \mathbb{R}) = \mathbb{C}^\Phi$ , and  $\Phi(\mathcal{O}_K)$ , the image of a full  $\mathbb{Z}$ -lattice  $\mathcal{O}_K \subset K$  of rank  $2g$ , is a full lattice of rank  $2g$  in  $\mathbb{C}^\Phi$ .

*Step 2:  $K$  acts by endomorphisms.* Fix  $k \in K$  and let  $M_k := \text{diag}(\varphi(k))_{\varphi \in \Phi} \in \text{End}_{\mathbb{C}}(\mathbb{C}^\Phi)$ . For  $a \in \mathcal{O}_K$ ,

$$M_k \cdot \Phi(a) = (\varphi(k)\varphi(a))_{\varphi \in \Phi} = (\varphi(ka))_{\varphi \in \Phi} = \Phi(ka)$$

so  $M_k$  carries  $\Phi(\mathcal{O}_K)$  into  $\Phi(k\mathcal{O}_K) \subseteq \Phi(K)$ . If  $k \in \mathcal{O}_K$  this lands in  $\Phi(\mathcal{O}_K)$ , so  $M_k$  preserves the lattice and, being  $\mathbb{C}$ -linear, descends to a holomorphic endomorphism of  $X_\Phi$  (proposition 1.1.7). This gives a ring homomorphism  $\mathcal{O}_K \rightarrow \text{End}(X_\Phi)$ ,  $k \mapsto M_k$ , injective because  $M_k = 0$  forces  $\varphi(k) = 0$ , hence  $k = 0$ . Tensoring with  $\mathbb{Q}$  yields an injection of  $\mathbb{Q}$ -algebras  $K = \mathcal{O}_K \otimes \mathbb{Q} \hookrightarrow \text{End}^0(X_\Phi)$ : for general  $k \in K$  some integer multiple  $nk$  lies in  $\mathcal{O}_K$ , and  $M_k = \frac{1}{n}M_{nk}$  is a genuine element of  $\text{End}^0(X_\Phi)$ . Since  $\dim_{\mathbb{Q}} K = 2g = 2 \dim X_\Phi$  and  $K$  is a field,  $X_\Phi$  has complex multiplication by  $K$  in the sense of definition 6.4.5.

*Step 3:  $X_\Phi$  is polarizable.* It remains to exhibit a positive Hermitian form  $H$  on  $\mathbb{C}^\Phi$  whose imaginary part is integral on  $\Phi(\mathcal{O}_K)$ , i.e. a polarization, so that  $X_\Phi$  is an abelian variety (definition 3.2.1). Choose  $\xi \in K$  purely imaginary for the conjugation,  $\bar{\xi} = -\xi$ , with  $\text{Im}\varphi(\xi) < 0$  for every  $\varphi \in \Phi$ ; such a  $\xi$  exists because  $K = K_0(\xi_0)$  with  $\bar{\xi}_0 = -\xi_0$ , and multiplying  $\xi_0$  by a suitable totally positive element of  $K_0$  arranges all  $g$  imaginary parts to have the same sign, which we take negative by replacing  $\xi$  with  $-\xi$  if needed (an approximation available since  $K_0$  is



dense in  $K_0 \otimes \mathbb{R} = \mathbb{R}^g$  and the totally positive elements form an open cone). Define, for  $a, b \in K$ ,

$$E(\Phi(a), \Phi(b)) := \text{Tr}_{K/\mathbb{Q}}(\xi a \bar{b})$$

extended  $\mathbb{R}$ -bilinearly to  $\mathbb{C}^\Phi = \Phi(K) \otimes \mathbb{R}$ . Then  $E$  takes integer values on  $\Phi(\mathcal{O}_K) \times \Phi(\mathfrak{d}^{-1}\mathcal{O}_K)$  for the inverse different  $\mathfrak{d}^{-1}$ , and after rescaling  $\xi$  by an integer  $E(\Phi(\mathcal{O}_K), \Phi(\mathcal{O}_K)) \subseteq \mathbb{Z}$ . It is alternating:  $E(\Phi(b), \Phi(a)) = \text{Tr}(\xi b \bar{a}) = \text{Tr}(\bar{\xi} \bar{b} \bar{a}) = \text{Tr}(\bar{\xi} \bar{b} a) = -\text{Tr}(\xi a \bar{b}) = -E(\Phi(a), \Phi(b))$ , using  $\text{Tr}(\bar{z}) = \text{Tr}(z)$  and  $\bar{\xi} = -\xi$ . In the coordinates of  $\mathbb{C}^\Phi$  the trace runs over the pairs  $\{\varphi, \bar{\varphi}\}$ , contributing  $\varphi(\xi a \bar{b})$  and its conjugate, so  $E(\Phi(a), \Phi(b)) = \sum_{\varphi \in \Phi} (\varphi(\xi) \varphi(a) \overline{\varphi(b)} + \overline{\varphi(\xi) \varphi(a) \overline{\varphi(b)}}) = \sum_{\varphi} 2\text{Re}(\varphi(\xi) \varphi(a) \overline{\varphi(b)})$ . Writing  $\varphi(\xi) = it_\varphi$  with  $t_\varphi = \text{Im}\varphi(\xi) < 0$ , each summand is  $2\text{Re}(it_\varphi \varphi(a) \overline{\varphi(b)}) = -2t_\varphi \text{Im}(\varphi(a) \overline{\varphi(b)})$ , so  $E(\Phi(a), \Phi(b)) = \sum_{\varphi} (-2t_\varphi \text{Im}(\varphi(a) \overline{\varphi(b)}))$ , which is the imaginary part of the Hermitian form

$$H(u, w) = \sum_{\varphi \in \Phi} (-2t_\varphi) u_\varphi \overline{w_\varphi}$$

on  $\mathbb{C}^\Phi$ ; here  $H$  is diagonal with entries  $-2t_\varphi = -2\text{Im}\varphi(\xi) > 0$ , so  $H > 0$ . Thus  $E = \text{Im}H$  is a polarization and  $X_\Phi$  is an abelian variety, completing the proof.

The construction is entirely explicit: a CM field, a choice of half of its embeddings, and the lattice of algebraic integers embedded through them. Every step is linear algebra over  $K$ , yet the output is an abelian variety with an action of  $K$  so large that  $K$  is a maximal commutative subalgebra of its endomorphism algebra. The positivity in Step 3 is where the CM hypothesis earns its keep: a totally imaginary  $\xi$  with all  $g$  embeddings on the same side of the real axis exists precisely because  $K$  is totally imaginary over the totally real  $K_0$ , and this is the analytic incarnation of the Rosati positivity that will reappear in proposition 6.4.10. For  $g = 1$  this is nothing but  $E = \mathbb{C}/\mathcal{O}_K$  with  $K$  imaginary quadratic, the CM elliptic curve of *Elliptic Curves* [4], recorded below in example 6.4.11.

#### Example 6.4.7: The CM Abelian Surface from $\mathbb{Q}(\zeta_5)$

Take  $K = \mathbb{Q}(\zeta_5)$ ,  $g = 2$ , with the CM type  $\Phi = \{\varphi_1, \varphi_2\}$ ,  $\varphi_1(\zeta_5) = \zeta$ ,  $\varphi_2(\zeta_5) = \zeta^2$  (where  $\zeta = e^{2\pi i/5}$ ), from example 6.4.4. The ring of integers is  $\mathcal{O}_K = \mathbb{Z}[\zeta_5]$ , with  $\mathbb{Z}$ -basis  $1, \zeta_5, \zeta_5^2, \zeta_5^3$ . Embedding through  $\Phi$  sends this basis to the columns

$$\begin{pmatrix} 1 & \zeta & \zeta^2 & \zeta^3 \\ 1 & \zeta^2 & \zeta^4 & \zeta^6 \end{pmatrix} = \begin{pmatrix} 1 & \zeta & \zeta^2 & \zeta^3 \\ 1 & \zeta^2 & \zeta^4 & \zeta \end{pmatrix}$$

(using  $\zeta^6 = \zeta$ ), a  $2 \times 4$  period matrix whose four columns span a full lattice  $\Lambda = \Phi(\mathbb{Z}[\zeta_5])$  in  $\mathbb{C}^2$ . The surface  $X_\Phi = \mathbb{C}^2/\Lambda$  carries an action of  $\mathbb{Z}[\zeta_5]$ : multiplication by  $\zeta_5$  acts as  $\text{diag}(\zeta, \zeta^2)$  on  $\mathbb{C}^2$ , cyclically permuting the basis columns since  $\zeta_5 \cdot \zeta_5^j = \zeta_5^{j+1}$ . Thus  $\mathbb{Q}(\zeta_5) \hookrightarrow \text{End}^0(X_\Phi)$ , a field of degree  $4 = 2 \dim X_\Phi$ , so  $X_\Phi$  is a simple abelian surface with complex multiplication by  $\mathbb{Q}(\zeta_5)$ . It is isogenous to the Jacobian of the genus-two quotient curve  $y^2 = x^5 - 1$ , on which  $\zeta_5$  acts by  $x \mapsto \zeta_5 x$ , one of the oldest examples of a higher-dimensional CM abelian variety.

The construction depends on a second field, hidden in the CM type, that controls the arithmetic: the field generated by the traces of  $K$  under  $\Phi$ . This is the reflex field, and it is the field over which the CM abelian variety and its moduli point are naturally defined.

**Definition 6.4.8: Reflex Field and Reflex Type**

Let  $(K, \Phi)$  be a CM pair. Extend each  $\varphi \in \Phi$  to an embedding of the Galois closure  $\tilde{K}$  of  $K$  into  $\mathbb{C}$ , and let  $G = \text{Gal}(\tilde{K}/\mathbb{Q})$  act on  $\text{Hom}(K, \mathbb{C})$ . The *reflex field*  $K^*$  of  $(K, \Phi)$  is the subfield of  $\mathbb{C}$  fixed by

$$\{ \sigma \in G : \sigma\Phi = \Phi \}$$

equivalently,  $K^* = \mathbb{Q}(\{ \text{Tr}_\Phi(a) := \sum_{\varphi \in \Phi} \varphi(a) : a \in K \})$ , the field generated over  $\mathbb{Q}$  by the “ $\Phi$ -traces”. The reflex field is itself a CM field. Regarding a CM type as the subset  $\tilde{\Phi} = \{ \sigma \in G : \sigma|_K \in \Phi \}$  of  $G$  that induces it, the associated *reflex type*  $\Phi^*$  on  $K^*$  is the set of restrictions to  $K^*$  of the inverses,

$$\Phi^* = \{ \sigma^{-1}|_{K^*} : \sigma \in G, \sigma|_K \in \Phi \}$$

a set of  $[K_0^* : \mathbb{Q}]$  embeddings  $K^* \hookrightarrow \mathbb{C}$  (the restriction  $\sigma^{-1}|_{K^*}$  depending only on the coset, so the count is correct), making  $(K^*, \Phi^*)$  a CM pair, the *reflex* of  $(K, \Phi)$ .

When  $K$  is abelian over  $\mathbb{Q}$ , as for the cyclotomic examples, the Galois action is transparent and  $K^*$  is a subfield of  $K$  itself; in general  $K^*$  may differ from  $K$ , and the passage  $(K, \Phi) \mapsto (K^*, \Phi^*)$  is an involution up to isogeny. The reflex field is where the arithmetic lives: the CM abelian variety  $X_\Phi$  has a model over a number field, and the reflex norm  $N_{\Phi^*} : K^* \rightarrow K$  transports idele classes of  $K^*$  to the Galois action on the torsion of  $X_\Phi$ . This is the content of the CM main theorem, stated below but not proved here.

**6.4.9 Key Ideas: The CM Main Theorem** A CM abelian variety  $X_\Phi$  with CM by  $(K, \Phi)$  is defined over a number field, and its field of moduli is a specific abelian extension of the reflex field  $K^*$ . The Galois action on the torsion points  $X_\Phi[\mathfrak{m}]$  is described by the *reflex norm* through the reciprocity map of class field theory: an idele class of  $K^*$  acts on  $X_\Phi$  and its torsion via its reflex norm to  $K$ , generalizing the statement that the  $j$ -invariant of a CM elliptic curve generates the Hilbert class field of the imaginary quadratic field. The proof requires the full apparatus of class field theory and the theory of canonical models, developed in *Class Field Theory* [15] and *Shimura Varieties* [11]; it lies beyond the scope of this book, whose contribution is the construction  $X_\Phi$  and its endomorphisms, established in full in theorem 6.4.6.

It remains to place CM abelian varieties inside Albert’s classification and confirm that a maximal commutative subfield must be a CM field, which is what makes definition 6.4.1 the right notion.

**Proposition 6.4.10: CM Abelian Varieties Have Endomorphism Algebra of Type IV**

Let  $X$  be a simple abelian variety of dimension  $g$  with complex multiplication by a field  $K$  of degree  $2g$ , so  $K = \text{End}^0(X)$  is its full endomorphism algebra. Then  $K$  is a CM field, and  $\text{End}^0(X)$  is of Albert Type IV with  $K$  a maximal subfield equal to the center. Conversely, every CM field of degree  $2g$  arises as  $\text{End}^0(X)$  for a simple CM abelian variety of dimension  $g$ .

**Proof:**

Since  $X$  is simple,  $\text{End}^0(X)$  is a division algebra (proposition 5.4.7); as it contains the field  $K$  of maximal degree  $2g$  and  $\dim_{\mathbb{Q}} \text{End}^0(X) \leq 2g$  (proposition 1.2.6), the containment is an equality  $\text{End}^0(X) = K$ . Thus  $\text{End}^0(X)$  is a field, so its center is  $K$  itself and it falls under Albert Type IV, the only type with commutative endomorphism algebra of degree  $2g$  (theorem 6.3.4).

It remains to show  $K$  is a CM field. The Rosati involution  $f \mapsto f'$  attached to any polarization of  $X$  (theorem 6.2.3) restricts to an involution of the field  $K = \text{End}^0(X)$ , and its fixed field  $K_0 := \{k \in K : k' = k\}$  is a subfield. Two cases arise. If the Rosati involution is trivial on  $K$ , then  $K = K_0$  and the positivity  $\text{Tr}(kk') = \text{Tr}(k^2) > 0$  for all  $k \neq 0$  forces every  $\mathbb{Q}$ -embedding of  $K$  to send  $k^2$  to a positive real for every  $k$ , so  $K$  is totally real; but then  $K$  would be a totally real field of degree  $2g$  acting on the simple  $X$  as an Albert Type I algebra, whose numerical constraint requires  $[K : \mathbb{Q}] = [K_0 : \mathbb{Q}]$  to divide  $g$  (proposition 6.3.5), impossible for  $2g \nmid g$  with  $g \geq 1$ . Hence the involution is nontrivial,  $[K : K_0] = 2$ , and  $K_0$  is its fixed field. Positivity gives, for  $k \in K_0 \setminus \{0\}$ ,  $\text{Tr}(kk') = \text{Tr}(k^2) > 0$ , so  $K_0$  is totally real; and for  $k \in K$  with  $k' = \bar{k}$  the second-kind involution restricts to complex conjugation on  $K$  under every embedding, so  $K$  is totally imaginary over  $K_0$ . Therefore  $K$  is a CM field, its conjugation the Rosati involution.

The converse is theorem 6.4.6: given a CM field  $K$  of degree  $2g$  and any CM type  $\Phi$ , the abelian variety  $X_{\Phi}$  has  $K \hookrightarrow \text{End}^0(X_{\Phi})$ , and choosing  $\Phi$  so that  $X_{\Phi}$  is simple (which holds when  $\Phi$  is a *primitive* type, one not induced from a proper CM subfield) yields  $\text{End}^0(X_{\Phi}) = K$  exactly. This completes the proof.

The proposition closes the classification loop: the maximal commutative case of Type IV is exactly complex multiplication by a CM field, and definition 6.4.1 is not an arbitrary choice but the class of fields the Rosati involution allows to act maximally. The nontriviality of the Rosati involution on  $K$  is the algebraic counterpart of the totally imaginary  $\xi$  in Step 3 of theorem 6.4.6: both express that a CM field carries an intrinsic conjugation fixing a totally real subfield.

**Example 6.4.11: CM Elliptic Curves**

The case  $g = 1$  is the origin of the whole story. Let  $K$  be an imaginary quadratic field,  $K = \mathbb{Q}(\sqrt{-d})$  with  $d > 0$  squarefree, and  $\mathcal{O}_K$  its ring of integers. There is a unique conjugate pair of embeddings  $\{\varphi, \bar{\varphi}\} = \text{Hom}(K, \mathbb{C})$ , so a CM type is a single embedding  $\Phi = \{\varphi\}$ , and the construction of theorem 6.4.6 produces

$$E = X_{\Phi} = \mathbb{C}/\varphi(\mathcal{O}_K) = \mathbb{C}/\mathcal{O}_K$$

an elliptic curve. Its endomorphism algebra is  $\text{End}^0(E) = K$ , of degree  $2 = 2 \dim E$ , so  $E$  has complex multiplication by  $K$  and, by proposition 6.4.10,  $\text{End}^0(E)$  is of Type IV with  $K$  its own center: the Rosati involution is complex conjugation on  $K$ , and  $K_0 = \mathbb{Q}$ . This is exactly the CM theory of *Elliptic Curves* [4], where the deeper arithmetic is proved: the  $j$ -invariant  $j(E)$  is an algebraic integer, generating the Hilbert class field of  $K$  over  $K$ , and the torsion of  $E$  generates the ray class fields, the imaginary quadratic instance of the CM main theorem. For  $d = 1$ ,  $K = \mathbb{Q}(i)$  and  $E = \mathbb{C}/\mathbb{Z}[i]$  is the curve  $y^2 = x^3 - x$ ; for  $d = 3$ ,  $K = \mathbb{Q}(\zeta_3)$  and  $E = \mathbb{C}/\mathbb{Z}[\zeta_3]$  is  $y^2 = x^3 + 1$ .



## Part II

# Abelian Varieties over Arbitrary Fields



Part I built the theory of abelian varieties on the complex numbers, where a lattice  $\Lambda \subset V = \mathbb{C}^g$  makes every construction explicit: an abelian variety was a polarizable complex torus, and its line bundles, its dual, and its endomorphisms were computations with the pair  $(V, \Lambda)$ . None of that scaffolding survives over a general field. There is no universal cover, no lattice to count, and no Hermitian form; an abelian variety must be built and studied as an algebraic object in its own right.

This Part rebuilds the theory over an arbitrary field  $k$ , defining an abelian variety to be a complete connected group variety and extracting its structure from the geometry of that group alone. The device that replaces the lattice is the theorem of the cube, from which the rigidity of the group law, the behavior of line bundles under translation, and the degree  $n^{2g}$  of multiplication by  $n$  all follow. Chapter 7 lays the groundwork: it fixes the notion of an abelian variety as a group scheme, proves the rigidity lemma that forces the group law to be commutative and every morphism to be a translate of a homomorphism, and sets up the line bundles and the Picard functor on which the theorem of the cube will act. Over  $\mathbb{C}$  the algebraic and analytic theories describe the same objects, a comparison this Part makes precise once the dual abelian variety and its polarizations are available algebraically.

### Exercise 6.4.1

1. Let  $K = \mathbb{Q}(\zeta_7)$ , the seventh cyclotomic field. Verify from definition 6.4.1 that  $K$  is a CM field: identify its maximal totally real subfield  $K_0$ , compute  $[K : \mathbb{Q}]$  and  $[K_0 : \mathbb{Q}]$ , and describe complex conjugation on  $K$ . Then count the number of distinct CM types on  $K$  and state the value of  $g$  for the resulting CM abelian varieties.
2. Let  $(K, \Phi)$  be a CM pair with  $[K : \mathbb{Q}] = 2g$ , and let  $\bar{\Phi} = \{\bar{\varphi} : \varphi \in \Phi\}$  be the conjugate type. Show directly from theorem 6.4.6 that the tori  $X_\Phi$  and  $X_{\bar{\Phi}}$  are isomorphic as complex tori. (Compare example 6.4.4, where this is the observation that the two CM types on an imaginary quadratic field give isomorphic elliptic curves.)
3. Fix  $K = \mathbb{Q}(\zeta_5)$  and the CM type  $\Phi = \{\varphi_1, \varphi_2\}$  with  $\varphi_1(\zeta_5) = \zeta$ ,  $\varphi_2(\zeta_5) = \zeta^2$ , as in example 6.4.7. Following Step 3 of theorem 6.4.6, produce an explicit purely imaginary  $\xi \in K$  with  $\bar{\xi} = -\xi$ , and determine the sign of  $\text{Im}\varphi_j(\xi)$  for  $j = 1, 2$ . Explain what further adjustment of  $\xi$  the construction requires so that the Hermitian form  $H$  built from it is positive definite, and carry it out.
4. Let  $X$  be a simple abelian variety of dimension  $g$  whose endomorphism algebra  $\text{End}^0(X)$  contains a totally real field  $F$  with  $[F : \mathbb{Q}] = 2g$ . Prove that no such  $X$  exists. (This is the case ruled out inside the proof of proposition 6.4.10; reconstruct that step, stating precisely which numerical constraint on Albert Type I is used.)
5. Prove the trace-form computation underlying Step 3 of theorem 6.4.6: with  $\xi \in K$  satisfying  $\bar{\xi} = -\xi$  and  $E(\Phi(a), \Phi(b)) := \text{Tr}_{K/\mathbb{Q}}(\xi a \bar{b})$  for  $a, b \in K$ , show that  $E$  is alternating, and that in the coordinates of  $\mathbb{C}^\Phi$  it equals the imaginary part of the Hermitian form  $H(u, w) = \sum_{\varphi \in \Phi} (-2\text{Im}\varphi(\xi)) u_\varphi \overline{w_\varphi}$ . Deduce the exact condition on  $\xi$  under which  $H > 0$ .





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# *Group Varieties and Rigidity*

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Over  $\mathbb{C}$  an abelian variety could be handed over as a quotient  $\mathbb{C}^g/\Lambda$ ; over a general field it must be recognized by a property. The defining property is that of a *proper connected group variety*: a group object in the category of varieties over  $k$  whose underlying variety is geometrically connected and proper. Properness is the algebraic surrogate for compactness, and it is the source of the theory's rigidity. A morphism from a proper connected variety to a group variety that collapses one fibre collapses a whole family of them, and from this single constraint the commutativity of the group law and the shape of every morphism follow, with no explicit equation in sight.

Chapter 8 will turn the same properness into the theorem of the cube, the engine of the algebraic theory. That engine runs on line bundles, so this chapter closes by assembling them: the Picard functor of an abelian variety and the seesaw principle, in the form the theorem of the cube consumes. The scheme-theoretic vocabulary of group varieties, properness, and the Picard functor is imported throughout from *Algebraic Geometry* [13] and specialized here to the proper connected case, where properness converts general facts about morphisms into rigidity.

## 7.1 Group Varieties and Group Schemes

This section fixes the language in which an abelian variety is defined over an arbitrary field: a *group variety* is a variety carrying multiplication, identity, and inversion morphisms subject to the group axioms, an *abelian variety* is a group variety that is in addition proper and geometrically connected, and the functor-of-points reformulation packages the same data as a representable functor into groups. The scheme theory these definitions rest on (varieties and schemes over  $k$ , properness, smoothness, representable functors) is developed in *Algebraic Geometry* [13] and cited here rather than rebuilt.

A group is a set with three operations (multiplication, an identity, and inversion) subject to associativity, the unit law, and the inverse law. Each axiom is an equation between functions, and each

such equation transcribes into a commutative diagram. Once phrased diagrammatically the notion of a group makes sense in any category with finite products, and applying it to varieties over  $k$  yields the object this Part studies.

### Definition 7.1.1: Group Variety

Let  $k$  be a field. A *group variety* over  $k$  is a variety  $G$  over  $k$  equipped with three morphisms

$$m: G \times_k G \rightarrow G, \quad e: \operatorname{Spec} k \rightarrow G, \quad \iota: G \rightarrow G$$

(multiplication, the identity section, and inversion) such that the following diagrams commute.

1. *Associativity*:

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{m \times \operatorname{id}} & G \times G \\ \operatorname{id} \times m \downarrow & & \downarrow m \\ G \times G & \xrightarrow{m} & G \end{array}$$

2. *Unit law*: writing  $e$  also for the composite  $G \rightarrow \operatorname{Spec} k \xrightarrow{e} G$ , both composites

$$G \xrightarrow{(e, \operatorname{id})} G \times G \xrightarrow{m} G, \quad G \xrightarrow{(\operatorname{id}, e)} G \times G \xrightarrow{m} G$$

equal  $\operatorname{id}_G$ .

3. *Inverse law*: both composites

$$G \xrightarrow{(\iota, \operatorname{id})} G \times G \xrightarrow{m} G, \quad G \xrightarrow{(\operatorname{id}, \iota)} G \times G \xrightarrow{m} G$$

equal the constant morphism  $G \rightarrow \operatorname{Spec} k \xrightarrow{e} G$ .

A *homomorphism* of group varieties  $f: G \rightarrow G'$  is a morphism of varieties compatible with the multiplications,  $f \circ m = m' \circ (f \times f)$ .

The definition says only that the group operations are algebraic: they are given by polynomial (more precisely, by morphisms of varieties) formulas, not by abstract set maps. For each field extension  $K/k$  the  $K$ -points  $G(K)$  then form an ordinary group, with multiplication induced by  $m$  and identity  $e$ , and a homomorphism induces a group homomorphism  $G(K) \rightarrow G'(K)$  for every  $K$ . A group variety is thus an algebraic family of groups, one for each extension, varying in a way controlled by the geometry of  $G$ . The commutativity of the multiplication is deliberately *not* imposed: it is a theorem for the proper case, proved in section 7.2 from properness alone, and it fails for the affine examples below.

Two conventions are worth recording at once. First, the identity is a  $k$ -point  $e \in G(k)$ , so the group law has a rational basepoint; homogeneity below trades on it. Second, “variety” here means a geometrically integral separated scheme of finite type over  $k$ ; the products are fibre products over  $k$ , and all morphisms are  $k$ -morphisms. When these hypotheses are relaxed, allowing the underlying scheme to be non-reduced, one arrives at the group-scheme notion taken up at the end of this section, which is indispensable in characteristic  $p$ .

Among group varieties the abelian varieties are singled out by a topological condition on the underlying space: completeness. Over  $\mathbb{C}$  this is compactness, and Part I built compactness into the torus  $V/\Lambda$

from the start. Over a general field the right notion is properness, and it is the hypothesis on which the theory of section 7.2 rests.

**Definition 7.1.2: Abelian Variety (algebraic)**

An *abelian variety* over a field  $k$  is a group variety  $A$  over  $k$  that is *proper* (equivalently, complete) and *geometrically connected*:  $A \times_k \bar{k}$  is connected.

This is the algebraic counterpart of the analytic definition 3.2.1 of Part I, where an abelian variety was a complex torus admitting a polarization. The two definitions look unrelated (one demands a positive Hermitian form, the other demands properness of a scheme), yet over  $\mathbb{C}$  they pick out the same objects: a complex torus is projective, hence a proper group variety, precisely when it is polarizable, and conversely the analytification of a proper group variety over  $\mathbb{C}$  is a compact complex Lie group, so a complex torus. That the two routes agree is proved in chapter 11, where the analytic and algebraic theories are compared over  $\mathbb{C}$ ; until then the algebraic development is self-contained and makes no appeal to the analytic construction.

Properness is doing all the work in definition 7.1.2. A group variety need not be proper (the affine examples  $\mathbb{G}_m$  and  $\mathbb{G}_a$  below are group varieties that are not), and it is exactly the completeness of the underlying variety that forces the group to be commutative, forces morphisms to be translates of homomorphisms, and ultimately forces projectivity. The mechanism is the rigidity lemma of section 7.2, which turns “the image of a complete variety is closed” into strong constraints on maps out of  $A$ . The geometric-connectedness clause removes a triviality: a disjoint union of two copies of an abelian variety, glued by translation, would satisfy every group axiom while properly speaking being an extension of a finite group by a connected group; insisting on connectedness (geometrically, to survive base change to  $\bar{k}$ ) pins down the connected object.

Before producing examples, two structural facts should be settled, both consequences of the group law rather than of properness. A group variety over a field is automatically smooth, and any two of its points are interchangeable by an automorphism of the underlying variety. Together they say that a group variety is as regular and as homogeneous as its identity point, and that no point is geometrically distinguished.

**Proposition 7.1.3: Translations are automorphisms**

Let  $G$  be a group variety over  $k$  and  $a \in G(k)$ . The *translation* morphism

$$t_a: G \rightarrow G, \quad t_a = m \circ (a, \text{id})$$

(that is,  $t_a(x) = ax$  on points) is an automorphism of the variety  $G$ , with inverse  $t_{a^{-1}}$ . More generally, for  $a \in G(K)$  with  $K/k$  any extension,  $t_a$  is an automorphism of  $G_K = G \times_k K$ .

**Proof:**

The morphism  $t_a$  is the composite

$$G \xrightarrow{(a, \text{id})} G \times_k G \xrightarrow{m} G,$$

where  $(a, \text{id})$  sends  $x$  to  $(a, x)$  (using the  $k$ -point  $a: \text{Spec } k \rightarrow G$  in the first factor), so  $t_a$  is a morphism of varieties. Applying the unit and associativity axioms of definition 7.1.1 on points,

for every extension  $K$  and every  $x \in G(K)$  one has

$$(t_{a^{-1}} \circ t_a)(x) = a^{-1}(ax) = (a^{-1}a)x = ex = x, \quad (t_a \circ t_{a^{-1}})(x) = a(a^{-1}x) = x,$$

so  $t_{a^{-1}} \circ t_a$  and  $t_a \circ t_{a^{-1}}$  agree with  $\text{id}_G$  on  $K$ -points for all  $K$ . A morphism of  $k$ -varieties is determined by its effect on points valued in all extensions (equivalently, by Yoneda applied to the functor of points, [13]), so  $t_{a^{-1}} \circ t_a = \text{id} = t_a \circ t_{a^{-1}}$ . Hence  $t_a$  is an isomorphism with inverse  $t_{a^{-1}}$ . The same computation over  $K$  proves the statement for  $a \in G(K)$ , completing the proof.

Translations make a group variety *homogeneous*: given any two  $\bar{k}$ -points  $x, y$  of  $G$ , the translation  $t_{yx^{-1}}$  is an automorphism of  $G_{\bar{k}}$  carrying  $x$  to  $y$ . Every point of  $G$  therefore looks like every other: there is no intrinsic way to distinguish the identity from any other point except by having chosen it. This homogeneity is the lever for the next result: a local property that holds at even one point of a group variety must hold at every point.

**Proposition 7.1.4: A group variety is smooth**

Every group variety  $G$  over a field  $k$  is smooth over  $k$ .

**Proof:**

Smoothness may be checked after base change to the algebraic closure, so assume  $k = \bar{k}$ . A variety of finite type over a perfect field is smooth on a dense open subset: this is generic smoothness, proved in *Algebraic Geometry* [13] (the smooth locus of a reduced finite-type  $k$ -scheme is dense open, since the function field is separably generated over  $\bar{k}$ ). Let  $U \subseteq G$  be that dense open smooth locus, and fix a smooth point  $u \in U(\bar{k})$ ; such a point exists because  $U$  is nonempty and  $\bar{k}$  is algebraically closed.

Now let  $x \in G(\bar{k})$  be arbitrary. The translation  $t_{xu^{-1}}$  of proposition 7.1.3 is an automorphism of  $G$  carrying  $u$  to  $x$ . Smoothness of a variety at a point is a local property preserved by automorphisms, so  $G$  is smooth at  $x = t_{xu^{-1}}(u)$  because it is smooth at  $u$ . As  $x$  was arbitrary,  $G$  is smooth at every closed point, hence smooth, completing the proof.

The proof is the prototype of every homogeneity argument in this Part: establish a property somewhere by a general theorem (here generic smoothness), then translate it everywhere by proposition 7.1.3. That a group variety is automatically smooth is the reason the word “smooth” rarely appears in the hypotheses of later results: it is free. In characteristic  $p$  this is exactly what separates a group variety from a general group scheme: the kernel of the Frobenius homomorphism on an abelian variety is a group scheme that is not smooth (it is non-reduced), so it is not a group variety, and the finer group-scheme language becomes necessary. That refinement is recorded next.

The group axioms of definition 7.1.1 were stated as commutative diagrams precisely so that they make sense with “variety” replaced by “scheme”. Dropping the reduced-and-integral hypothesis on the underlying space gives the notion of a group scheme, and the natural way to phrase it is through the functor of points, which is also the most useful form in practice.

**Definition 7.1.5: Group Scheme**

A *group scheme* over  $k$  is a scheme  $G$  over  $k$  together with morphisms  $m: G \times_k G \rightarrow G$ ,  $e: \operatorname{Spec} k \rightarrow G$ ,  $\iota: G \rightarrow G$  satisfying the associativity, unit, and inverse diagrams of definition 7.1.1. Equivalently, by Yoneda, a group scheme is a scheme  $G$  over  $k$  whose functor of points

$$h_G: \operatorname{Sch}/k \rightarrow \operatorname{Set}, \quad T \mapsto G(T) = \operatorname{Hom}_k(T, G)$$

factors through the forgetful functor  $\operatorname{Grp} \rightarrow \operatorname{Set}$ : the set  $G(T)$  carries a group structure, functorially in  $T$ . A group scheme is precisely a representable functor  $\operatorname{Sch}/k \rightarrow \operatorname{Grp}$ .

The equivalence of the two descriptions is the Yoneda lemma applied to the category of schemes over  $k$ , for which see *Algebraic Geometry* [13]: giving the morphisms  $m, e, \iota$  subject to the diagrams is the same as giving a natural group structure on each  $G(T)$ , because a morphism  $G \times G \rightarrow G$  is the same as a natural transformation  $h_G \times h_G \rightarrow h_G$ , and the group axioms as diagrams correspond exactly to the group axioms on each set  $G(T)$ . The functorial description is the one to keep in mind: a group scheme is a rule assigning a group to every  $k$ -scheme, compatibly with base change, and representable by an actual scheme  $G$ .

Group varieties are the reduced, smooth group schemes over a field. Indeed, a group variety is a group scheme whose underlying scheme is a variety, hence reduced and (by proposition 7.1.4) smooth; conversely a smooth group scheme of finite type with geometrically integral identity component is a group variety on each component. Over a field of characteristic zero every group scheme of finite type is smooth (Cartier's theorem, [13]), so the distinction is invisible there; it is genuine only in characteristic  $p$ , where non-reduced group schemes such as  $\mu_p = \ker(\mathbb{G}_m \xrightarrow{p} \mathbb{G}_m)$  and  $\alpha_p = \ker(\mathbb{G}_a \xrightarrow{F} \mathbb{G}_a)$  arise as kernels of isogenies and must be handled scheme-theoretically. Those finite group schemes are the subject of chapter 9; for the geometry of an abelian variety itself the group-variety language of definition 7.1.2 suffices, and the two are used interchangeably wherever smoothness makes them coincide.

The definitions call for two contrasting examples: one abelian variety, and one pair of group varieties that are not abelian varieties. The elliptic curve supplies the first, in dimension one; the multiplicative and additive groups supply the second, and the contrast isolates properness as the decisive hypothesis.

**Example 7.1.6: The Elliptic Curve as an Abelian Variety**

Let  $E/k$  be an elliptic curve: a smooth projective geometrically connected curve of genus 1 equipped with a rational point  $O \in E(k)$ . As developed in *Elliptic Curves* [4], the chord-and-tangent construction defines a group law on  $E$  with identity  $O$ : for points  $P, Q \in E$ , the line through them meets the cubic in a third point  $R$ , and  $P + Q$  is the reflection of  $R$  through  $O$ . What matters here is that this group law is *algebraic*. The addition map  $m: E \times E \rightarrow E$ , the identity  $O: \operatorname{Spec} k \rightarrow E$ , and the inversion  $\iota: E \rightarrow E$  are morphisms of varieties: in Weierstrass coordinates the sum  $P + Q$  is given by rational functions of the coordinates of  $P$  and  $Q$  (with the collinearity and reflection formulas), and these rational functions extend to morphisms because  $E$  is smooth and complete, so no denominator can vanish identically. The associativity, unit, and inverse axioms of definition 7.1.1 hold as identities of these rational functions, verified in [4].

Thus  $(E, m, O, \iota)$  is a group variety over  $k$ . It is proper, being a projective curve, and geometrically connected, being a geometrically integral curve; by definition 7.1.2 it is an abelian variety, of dimension 1. This is the  $g = 1$  case, the running model for every construction in this Part. The group law is commutative (as it must be, by the theorem of section 7.2), which is visible directly

from the symmetry of the chord construction in  $P$  and  $Q$ .

An elliptic curve is exactly a one-dimensional abelian variety, and conversely: a one-dimensional abelian variety over  $k$ , being a smooth proper geometrically connected group variety of dimension one with a rational identity, is a genus-one curve with a marked point, hence an elliptic curve. The theory of this Part is the higher-dimensional generalization of [4], and the reader is invited to specialize each statement to  $g = 1$  as a sanity check.

The next example shows that dropping properness admits genuinely different objects.

### Example 7.1.7: The Multiplicative and Additive Groups

Two affine group varieties over  $k$  show that a group variety need not be an abelian variety.

1. *The multiplicative group  $\mathbb{G}_m$ .* As a variety,  $\mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$ , the complement of the origin in the affine line; its  $K$ -points are  $\mathbb{G}_m(K) = K^\times$  for every extension  $K/k$ . Multiplication  $m(x, y) = xy$ , identity  $e = 1$ , and inversion  $\iota(x) = x^{-1}$  are morphisms (the coordinate ring map  $t \mapsto t \otimes t$  defines  $m$ , and  $t \mapsto t^{-1}$  defines  $\iota$ , which is a morphism because  $t$  is invertible on  $\mathbb{G}_m$ ). So  $\mathbb{G}_m$  is a group variety of dimension 1.
2. *The additive group  $\mathbb{G}_a$ .* As a variety,  $\mathbb{G}_a = \operatorname{Spec} k[t] = \mathbb{A}^1$ , the affine line; its  $K$ -points are  $\mathbb{G}_a(K) = (K, +)$ . Addition  $m(x, y) = x + y$  (the coordinate map  $t \mapsto t \otimes 1 + 1 \otimes t$ ), identity  $e = 0$ , and inversion  $\iota(x) = -x$  are morphisms. So  $\mathbb{G}_a$  is a group variety of dimension 1.

Neither is an abelian variety. Each is affine:  $\mathbb{G}_m$  and  $\mathbb{G}_a$  are  $\operatorname{Spec}$  of a nonzero finitely generated  $k$ -algebra, and an affine variety of positive dimension is not proper. Concretely, a proper variety over  $k$  has only constant global regular functions, whereas the coordinate  $t$  is a nonconstant regular function on either group; so neither is complete. Both are smooth connected group varieties, consistent with proposition 7.1.4, yet the failure of properness excludes them from definition 7.1.2.

The contrast is exact. All three of  $E$ ,  $\mathbb{G}_m$ ,  $\mathbb{G}_a$  are smooth connected one-dimensional group varieties; the elliptic curve is an abelian variety and the two affine groups are not, and the sole distinction is properness. Everything special about abelian varieties (commutativity of the group law, the rigidity of morphisms, projectivity) is a consequence of that one hypothesis, and none of it holds for  $\mathbb{G}_m$  or  $\mathbb{G}_a$ . Both affine groups are themselves commutative, so their exclusion is not a matter of any group axiom failing; it is the non-properness alone that places them outside the theory.

The lesson of the section is methodological. Over  $\mathbb{C}$  an abelian variety could be handed to the reader as a quotient  $V/\Lambda$ , with every property manifest in the construction. Over an arbitrary field there is no such quotient, and the object is instead specified by three intrinsic demands (a group law, properness, and geometric connectedness), each of which is checkable on the variety  $A$  without reference to any transcendental model. Properness is the demand that does the work: it is the source of the theory's rigidity, and the mechanism by which the rigidity arises is the lemma developed in section 7.2.

### Exercise 7.1.1

1. Fix the multiplicative group  $\mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$  of example 7.1.7. Write down the  $k$ -algebra homomorphism  $k[t, t^{-1}] \rightarrow k[t, t^{-1}] \otimes_k k[t, t^{-1}]$  that defines multiplication  $m$ , the map defining the identity  $e: \operatorname{Spec} k \rightarrow \mathbb{G}_m$ , and the map defining inversion  $\iota$ . Verify on the coordinate ring that the composite  $\mathbb{G}_m \xrightarrow{(\iota, \operatorname{id})} \mathbb{G}_m \times_k \mathbb{G}_m \xrightarrow{m} \mathbb{G}_m$  equals the constant morphism through  $e$ ,

that is, check the inverse law of definition 7.1.1.

2. Let  $a \in \mathbb{G}_a(k) = k$  be a  $k$ -point of the additive group. Using the definition of the translation  $t_a$  in proposition 7.1.3, describe the  $k$ -algebra endomorphism of  $k[t]$  induced by  $t_a: \mathbb{G}_a \rightarrow \mathbb{G}_a$ , and confirm directly that it is an automorphism with inverse the endomorphism induced by  $t_{-a}$ .
3. Show that the coordinate function  $t$  on  $\mathbb{G}_m$  is a nonconstant global regular function, and deduce, using the criterion recorded in example 7.1.7 (a proper variety over  $k$  has only constant global regular functions), that  $\mathbb{G}_m$  is not proper. Explain in one sentence why this excludes  $\mathbb{G}_m$  from definition 7.1.2 even though  $\mathbb{G}_m$  is a smooth connected commutative group variety.
4. Let  $f: G \rightarrow G'$  be a homomorphism of group varieties over  $k$  in the sense of definition 7.1.1, so  $f \circ m = m' \circ (f \times f)$ . Prove that  $f$  carries the identity to the identity and commutes with inversion: on  $K$ -points,  $f(e_G) = e_{G'}$  and  $f(x^{-1}) = f(x)^{-1}$  for every extension  $K/k$  and every  $x \in G(K)$ .
5. Prove the following strengthening of the homogeneity remark following proposition 7.1.3: a group variety  $G$  over  $k$  acts transitively on its  $\bar{k}$ -points through translations. That is, for any two points  $x, y \in G(\bar{k})$  there is an automorphism of  $G_{\bar{k}}$  carrying  $x$  to  $y$ . Then use this together with proposition 7.1.4 to give a one-line proof that the local ring  $\mathcal{O}_{G,x}$  at any point  $x \in G(\bar{k})$  is regular.
6. Let  $G$  be a group scheme over a field  $k$  of characteristic  $p > 0$ , in the sense of definition 7.1.5, and let  $T = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$  be the dual numbers. Using the functorial description of definition 7.1.5, explain what structure the set  $G(T)$  carries and why it need not agree with the underlying reduced scheme's points; illustrate with  $\mu_p = \operatorname{Spec} k[t]/(t^p - 1)$ , computing  $\mu_p(\bar{k})$  and comparing it with the fact that  $\mu_p$  is non-reduced. Explain briefly why  $\mu_p$  is therefore a group scheme but not a group variety.

## 7.2 The Rigidity Lemma

The rigidity lemma is the algebraic fact that replaces the lattice of Part I: a complete variety admits no nonconstant morphism to an affine variety, so a family of maps out of it, once pinned on a single slice, cannot flex on any other. This section proves the lemma and extracts the two structural consequences that make an abelian variety a well-behaved group: every morphism is a translate of a homomorphism, and the group law is commutative.

The lemma concerns a morphism  $f: X \times Y \rightarrow Z$  that is constant on one slice  $X \times \{y_0\}$ . The claim is that it is then constant on *every* slice, so that  $f$  depends only on the  $Y$ -coordinate. The hypothesis that carries the argument is completeness of  $X$ : it is what makes the projection  $\operatorname{pr}_Y$  a closed map, and closedness is what lets a local collapse near  $y_0$  spread to all of  $Y$ .

**Theorem 7.2.1: Rigidity Lemma**

Let  $X$  be a complete variety over  $k$  that is geometrically connected and possesses a  $k$ -point, let  $Y$  and  $Z$  be varieties over  $k$ , and let  $f: X \times Y \rightarrow Z$  be a morphism. Suppose there is a point  $y_0 \in Y(k)$  for which

$$f(X \times \{y_0\}) = \{z_0\}$$

is a single point  $z_0 \in Z$ . Then  $f$  factors through the projection  $\text{pr}_Y: X \times Y \rightarrow Y$ : there is a morphism  $g: Y \rightarrow Z$  with  $f = g \circ \text{pr}_Y$ . Equivalently,  $f(x, y)$  is independent of  $x$ , and  $g(y) = f(x, y)$  for any  $x$ .

**Proof:**

The argument has three movements: choose an affine neighborhood of  $z_0$ ; show that for  $y$  close to  $y_0$  the whole slice  $f(X \times \{y\})$  collapses into that neighborhood, hence to a point; then spread the collapse to all of  $Y$  using that  $X$  has a  $k$ -point.

*Step 1: An affine neighborhood and the closed complement.* Pick an affine open neighborhood  $U \subseteq Z$  of  $z_0$ . Its complement  $Z \setminus U$  is closed in  $Z$ . Consider the preimage

$$W := f^{-1}(Z \setminus U) \subseteq X \times Y$$

a closed subset. Because  $X$  is complete, the projection  $\text{pr}_Y: X \times Y \rightarrow Y$  is a closed morphism: it carries closed sets to closed sets (this is the defining property of properness, and completeness of  $X$  is properness of  $X \rightarrow \text{Spec } k$ , stable under the base change to  $Y$ ; see *Algebraic Geometry* [13]). Hence

$$Y_0 := \text{pr}_Y(W) = \{y \in Y : f(x, y) \notin U \text{ for some } x \in X\}$$

is closed in  $Y$ . By hypothesis the slice over  $y_0$  maps entirely to  $z_0 \in U$ , so no  $x$  has  $f(x, y_0) \notin U$ ; thus  $y_0 \notin Y_0$ . The complement  $Y_1 := Y \setminus Y_0$  is therefore a nonempty open subset of  $Y$  containing  $y_0$ , and it is the locus of points  $y$  over which the entire slice lands in  $U$ :

$$y \in Y_1 \iff f(X \times \{y\}) \subseteq U$$

*Step 2: Each slice over  $Y_1$  is a single point.* Fix a point  $y \in Y_1$ . The slice  $X \times \{y\}$  is isomorphic to  $X$ , hence complete and geometrically connected, and its image  $f(X \times \{y\})$  lies in the affine variety  $U$  by the previous step. A complete variety has no nonconstant morphism to an affine variety: the image of a complete variety is complete and closed, and a complete subvariety of an affine variety is finite, being both affine and complete, hence of dimension zero (*Algebraic Geometry* [13]). Since  $X \times \{y\}$  is connected, its image is a connected finite set, that is, a single point. Thus for every  $y \in Y_1$ ,

$$f(X \times \{y\}) = \{g_1(y)\}$$

for a unique point  $g_1(y) \in U$ , and the slice over  $y_0$  recovers  $g_1(y_0) = z_0$ . The map  $f$  is therefore constant on each slice over the open set  $Y_1$ .

*Step 3: Spreading to all of  $Y$ .* Let  $x_0 \in X(k)$  be a  $k$ -point, which exists by hypothesis, and define

$$g: Y \rightarrow Z, \quad g := f \circ (x_0 \times \text{id}_Y),$$



that is,  $g(y) = f(x_0, y)$ , the restriction of  $f$  to the section  $\{x_0\} \times Y \cong Y$ . This is a morphism, defined on all of  $Y$ . It remains to show  $f = g \circ \text{pr}_Y$  as morphisms  $X \times Y \rightarrow Z$ , that is,  $f(x, y) = f(x_0, y)$  for all  $(x, y)$ .

Both  $f$  and  $g \circ \text{pr}_Y$  are morphisms  $X \times Y \rightarrow Z$ . On the open dense subset  $X \times Y_1$  they agree: for  $y \in Y_1$  the slice  $f(X \times \{y\})$  is the single point  $g_1(y)$ , so in particular  $f(x, y) = f(x_0, y) = g(y) = (g \circ \text{pr}_Y)(x, y)$ . Since  $Y_1$  is a nonempty open subset of the irreducible variety  $Y$ , the product  $X \times Y_1$  is dense in  $X \times Y$ . Two morphisms from a variety to a separated scheme that agree on a dense subset agree everywhere (*Algebraic Geometry* [13]): their equalizer is closed and contains a dense set. Hence  $f = g \circ \text{pr}_Y$  on all of  $X \times Y$ , which is the asserted factorization, as we sought to show.

The name “rigidity” records the phenomenon: once a family of maps out of a complete variety is pinned down on a single slice, it cannot flex. The maps have no room to vary, because varying would produce a nonconstant morphism from a complete variety to an affine chart, and there are none. The completeness of  $X$  is doing all the work, and doing it twice. It makes  $\text{pr}_Y$  closed, which locates an open set  $Y_1$  over which the slices collapse; and it forces each such slice, mapping into an affine chart, to be a point. The  $k$ -point  $x_0$  then supplies a single section along which the whole map is measured, propagating the collapse from the open set  $Y_1$  to all of  $Y$  by density. Every hypothesis earns its place: without completeness the projection need not be closed and the affine chart could receive a positive-dimensional image; without a  $k$ -point there is no section to define  $g$ ; without connectedness a slice could map to a finite set of more than one point.

The first payoff identifies the morphisms of abelian varieties that respect the identity. In the analytic theory this was automatic from the  $\mathbb{C}$ -linear lift; here it is a consequence of rigidity applied to the map measuring the failure of  $f$  to be additive.

### Corollary 7.2.2: Identity-Fixing Morphisms Are Homomorphisms

Let  $A$  and  $B$  be abelian varieties over  $k$  (definition 7.1.2), and let  $f: A \rightarrow B$  be a morphism of varieties with  $f(e_A) = e_B$ , where  $e_A$  and  $e_B$  are the identity elements. Then  $f$  is a homomorphism of group varieties.

#### Proof:

Write the group laws of  $A$  and  $B$  additively. Consider the morphism

$$\varphi: A \times A \rightarrow B, \quad \varphi(a, a') = f(a + a') - f(a) - f(a')$$

which is a morphism because it is built from  $f$ , the group law of  $A$ , and the group law and inversion of  $B$  by composition. Proving  $f$  a homomorphism amounts to showing  $\varphi$  is constantly  $e_B$ .

The variety  $A$  is complete, geometrically connected, and has the  $k$ -point  $e_A$ , so theorem 7.2.1 applies to  $\varphi$  with  $X = Y = A$ . On the slice  $A \times \{e_A\}$ ,

$$\varphi(a, e_A) = f(a + e_A) - f(a) - f(e_A) = f(a) - f(a) - e_B = e_B$$

using  $f(e_A) = e_B$ ; so  $\varphi$  collapses the slice  $A \times \{e_A\}$  to the single point  $e_B$ . By rigidity  $\varphi$  factors through  $\text{pr}_2: A \times A \rightarrow A$ , that is,  $\varphi(a, a')$  is independent of  $a$ . Evaluating at  $a = e_A$  gives

$\varphi(a, a') = \varphi(e_A, a') = e_B$  for all  $a'$  by the identical computation on the other slice. Hence  $\varphi \equiv e_B$ , that is,  $f(a + a') = f(a) + f(a')$  for all  $a, a'$ , and  $f$  is a homomorphism, completing the proof.

The corollary is the algebraic replacement for the analytic statement that a holomorphic map of tori fixing the origin is a homomorphism (part (2) of proposition 1.1.7). Both say the group structure of an abelian variety is not extra data attached to the variety but is determined by it once an identity is chosen: a map of the underlying varieties that sends identity to identity has no choice but to be a group homomorphism. The mechanism differs entirely. Analytically the additivity defect  $F(v + w) - F(v) - F(w)$  was a continuous map into a discrete lattice, hence constant; algebraically the defect  $\varphi$  is a morphism out of a complete variety, hence constant along the complete factor by rigidity. The lattice and the completeness play the same structural role, and this is the first instance of a substitution repeated throughout Part II.

Removing the hypothesis  $f(e_A) = e_B$  costs only a translation, and this is the general shape of a morphism of abelian varieties.

### Corollary 7.2.3: Morphisms Are Translates of Homomorphisms

Let  $A$  and  $B$  be abelian varieties over  $k$ , and let  $f: A \rightarrow B$  be any morphism of varieties. Then there is a unique homomorphism  $h: A \rightarrow B$  and a unique point  $c = f(e_A) \in B(k)$  with

$$f = t_c \circ h,$$

where  $t_c: B \rightarrow B$  is translation by  $c$ . In particular  $f$  is a homomorphism if and only if  $f(e_A) = e_B$ .

#### Proof:

Set  $c := f(e_A)$  and  $h := t_{-c} \circ f: A \rightarrow B$ , the composite of  $f$  with translation by  $-c$ ; translation by a point is an automorphism of the variety  $B$ , so  $h$  is a morphism. It fixes the identity,

$$h(e_A) = t_{-c}(f(e_A)) = f(e_A) - c = c - c = e_B,$$

so by corollary 7.2.2 the morphism  $h$  is a homomorphism. Composing back,  $t_c \circ h = t_c \circ t_{-c} \circ f = f$ , which is the asserted decomposition. For uniqueness, evaluating  $f = t_c \circ h$  at  $e_A$  forces  $c = t_c(h(e_A)) = t_c(e_B) = f(e_A)$ , pinning down  $c$ , and then  $h = t_{-c} \circ f$  is determined. The final statement is immediate:  $f = t_c \circ h$  is itself a homomorphism exactly when  $c = e_B$ , that is, when  $f(e_A) = e_B$ , as we sought to show.

The decomposition mirrors part (2) of proposition 1.1.7 verbatim at the level of statement: every morphism of the underlying varieties is a homomorphism followed by a translation, and the translation is measured by the image of the identity. It has a useful reading for endomorphisms. The morphisms  $A \rightarrow A$  of the variety are the translates  $t_c \circ h$  with  $h \in \text{End}(A)$  and  $c \in A(k)$ ; the endomorphisms of the group are exactly those fixing the origin, and the translations account for the rest. This is why the endomorphism *ring* is defined using identity-fixing maps, and why choosing a different origin on the same variety changes the group law only by conjugation with a translation.

The second structural payoff of rigidity is that the group law of an abelian variety, imposed only as a group-variety structure, is forced to be commutative. No commutativity was assumed; it is a theorem, and the proof is a one-line application of the previous corollary to inversion.

**Corollary 7.2.4: The Group Law Is Commutative**

The group law of an abelian variety  $A$  over  $k$  is commutative.

**Proof:**

Let  $\iota: A \rightarrow A$  be inversion,  $\iota(a) = -a$ . It is a morphism of the variety  $A$  fixing the identity,  $\iota(e_A) = e_A$ , so by corollary 7.2.2 it is a homomorphism of group varieties:

$$-(a + a') = \iota(a + a') = \iota(a) + \iota(a') = (-a) + (-a')$$

for all  $a, a' \in A$ . On the other hand, in any group inversion reverses order,  $-(a + a') = (-a') + (-a)$ . Comparing the two expressions for  $-(a + a')$  gives  $(-a) + (-a') = (-a') + (-a)$ , and replacing  $a$  by  $-a$  and  $a'$  by  $-a'$  (permissible since  $\iota$  is a bijection) yields  $a + a' = a' + a$  for all  $a, a' \in A$ , completing the proof.

The commutativity of the group law is the reason for the name “abelian” variety, and it is not a definitional convention but a consequence of completeness. A complete connected group variety cannot carry a noncommutative group law, because rigidity forces its inversion to be a homomorphism, and a group whose inversion is a homomorphism is abelian. The contrast with affine group varieties is exactly here:  $\mathrm{GL}_n$  and its noncommutative subgroups are group varieties, but they are affine, not complete, so rigidity does not apply and their group laws are free to be noncommutative. Completeness is the dividing line, and abelian varieties sit on the commutative side of it.

The elliptic-curve case shows that rigidity recovers facts already established analytically and by hand for  $g = 1$ , now as instances of a uniform algebraic principle.

**Example 7.2.5: Rigidity for an Elliptic Curve**

Let  $E$  be an elliptic curve over  $k$ , that is, a smooth projective curve of genus one with a marked  $k$ -rational point  $O$ , taken as the identity. As a projective curve  $E$  is complete, and it is geometrically connected with the  $k$ -point  $O$ , so it is an abelian variety of dimension one and theorem 7.2.1 applies to it. Two consequences that *Elliptic Curves* [4] obtained by direct computation with the chord-and-tangent law reappear as corollaries.

First, the group law on  $E$  is commutative (corollary 7.2.4), a fact visible geometrically in the chord construction (the line through  $P$  and  $Q$  is the line through  $Q$  and  $P$ ) and now seen to be forced by completeness alone, independently of any equation for  $E$ . Second, the choice of origin is the only choice made: a different rational point  $O'$  as identity produces the same underlying curve with a group law differing from the first by the translation  $t_{O'-O}$ , and every morphism  $E \rightarrow E'$  of the underlying curves is a translate of a homomorphism (corollary 7.2.3). In particular the isogenies  $E \rightarrow E'$ , the surjective homomorphisms with finite kernel, are exactly the nonconstant morphisms carrying  $O$  to  $O'$ : a nonconstant morphism fixing the marked points is a homomorphism by corollary 7.2.2, and it is an isogeny because a nonconstant morphism of curves is surjective with finite fibers. This matches the characterization of isogenies in *Elliptic Curves* [4], where an isogeny is defined as a nonconstant morphism preserving the identity, and shows that the definition loses nothing by requiring only that the origin be fixed: the homomorphism property comes for free from rigidity.

The three corollaries together say that an abelian variety is a group in essentially only one way. Its

group law is commutative, so the category of abelian varieties over  $k$  is a category of commutative group varieties; its morphisms are translates of homomorphisms, so the group structure is transported by any morphism up to a shift; and a morphism fixing the identity is automatically a homomorphism, so the identity is the only piece of group-theoretic data one must specify. All three are consequences of completeness through the single rigidity lemma. This is the algebraic substitute for the linearization of proposition 1.1.7 in Part I: where the analytic theory lifted a map to the universal cover and used the vanishing of the antiholomorphic derivative to make the lift  $\mathbb{C}$ -linear, the algebraic theory has no cover and no derivative, and uses instead that a complete variety admits no nonconstant map to an affine chart. The complex structure and the completeness play interchangeable roles, each rigidifying the maps between abelian varieties enough to pin their group structure to the choice of an origin, and it is this interchange that lets the computations of Part I and the constructions of Part II describe the same objects over  $\mathbb{C}$ .

### Exercise 7.2.1

1. State the three hypotheses on  $X$  in the Rigidity Lemma (theorem 7.2.1) and match each to the exact place in the proof where it is used. For each hypothesis, describe in one sentence what goes wrong if it is dropped.
2. Let  $A$  be an abelian variety over  $k$  and let  $c \in A(k)$  be a point. Using corollary 7.2.3, decide whether the translation  $t_c: A \rightarrow A$ ,  $t_c(a) = a + c$ , is a homomorphism of group varieties, and exhibit its expression as  $t_{c'} \circ h$  with  $h$  a homomorphism and  $c' \in A(k)$ . For which  $c$  is  $t_c$  a homomorphism?
3. Let  $A$  and  $B$  be abelian varieties over  $k$  and let  $f, g: A \rightarrow B$  be two homomorphisms of group varieties. Show that  $f + g: A \rightarrow B$ , defined by  $(f + g)(a) = f(a) + g(a)$ , is again a homomorphism. Where does the commutativity of  $B$  (corollary 7.2.4) enter?
4. Let  $A, B, C$  be abelian varieties over  $k$ , and let  $F: A \times B \rightarrow C$  be a morphism of varieties (not assumed to respect any group structure) with  $F(e_A, e_B) = e_C$ . Prove that

$$F(a, b) = F(a, e_B) + F(e_A, b)$$

for all  $a \in A, b \in B$ ; that is,  $F$  splits as the sum of its two “axis” restrictions. (Hint: apply corollary 7.2.2 to  $F$  viewed as a morphism on  $A \times B$ , whose group law is the product group law.)

5. Let  $G$  be a group variety over  $k$  that is complete, geometrically connected, and possesses a  $k$ -point taken as identity  $e$  (so  $G$  satisfies the standing hypotheses of theorem 7.2.1 with  $X = G$ ). Prove directly from the Rigidity Lemma that  $G$  is commutative, without first passing through corollary 7.2.2. (Hint: apply the lemma to the commutator morphism.)
6. Let  $E$  and  $E'$  be elliptic curves over  $k$  with marked points  $O$  and  $O'$ , and let  $f: E \rightarrow E'$  be a nonconstant morphism of the underlying curves. Following example 7.2.5, prove that  $f$  is an isogeny if and only if  $f(O) = O'$ , and that in general  $f$  is the translate  $t_{f(O)-O'} \circ h$  of an isogeny  $h$  when  $f(O) \neq O'$  but  $f$  is still nonconstant. State precisely which facts about curves, beyond the results of this section, the argument uses.

## 7.3 Abelian Varieties Are Projective

Properness makes an abelian variety a complete variety, but completeness is not enough. To do algebraic geometry on  $X$  one needs coordinates: a line bundle with sections enough to embed  $X$  into a projective space, so that  $X$  becomes a system of homogeneous equations rather than an abstract complete group. This section supplies exactly that. Every abelian variety over any field carries an ample line bundle, and an ample line bundle on a complete variety produces a projective embedding. Projectivity is what the theorem of the cube (chapter 8) manipulates and what the dual abelian variety (chapter 10) is built from, and it is a theorem, not a hypothesis: nothing in the definition of an abelian variety mentions  $\mathbb{P}^N$ .

The route is to manufacture an ample bundle by hand. On a curve any divisor of positive degree is ample, but in higher dimension ampleness is a subtler positivity, and the group law is what makes it accessible. The plan is to take an effective divisor  $D$  on  $X$  that is not preserved by translation by any nontrivial subgroup, form the line bundle  $L = \mathcal{O}_X(D)$ , and show that a suitable multiple of  $L$  is ample. The mechanism that converts the no-stabilizer condition into ampleness is the theorem of the square, proved in chapter 8: it computes how  $L$  transforms under translation and thereby controls the sections of  $L^{\otimes 2}$  and  $L^{\otimes 3}$ , exactly as translation of the theta divisor did in the analytic argument of section 4.4.

Fix an abelian variety  $X$  of dimension  $g$  over a field  $k$ , in the sense of definition 7.1.2: a proper, geometrically connected, smooth group variety over  $k$ . It suffices to prove projectivity after the base change to an algebraic closure  $\bar{k}$ , since properness and the existence of an ample bundle both descend from  $X_{\bar{k}}$  to  $X$  (a  $k$ -scheme is projective when its base change to  $\bar{k}$  is; see *Algebraic Geometry* [13]). Assume therefore that  $k$  is algebraically closed and work with closed points.

Two facts about line bundles under translation drive the construction; the second is the theorem of the square, whose proof is the business of chapter 8. State them first, so the ampleness argument reads as a single line.

The first is a rigidity statement in the same family as section 7.2. For an abelian variety  $X$  and a line bundle  $L$ , the translate  $t_x^*L$  measures how  $L$  moves as its argument slides by  $x$ , and the classes  $t_x^*L \otimes L^{-1}$  trace out a subgroup of  $\text{Pic}(X)$  as  $x$  ranges over  $X$ . That subgroup, and the locus of  $x$  fixing  $L$ , is the object the whole argument turns on.

**7.3.1 Preview: The Theorem of the Square** Let  $X$  be an abelian variety over  $k$  and  $L$  a line bundle on  $X$ . For closed points  $x, y \in X$ , with  $t_x: X \rightarrow X$  translation by  $x$ ,

$$t_{x+y}^*L \otimes L \cong t_x^*L \otimes t_y^*L$$

in  $\text{Pic}(X)$ . Equivalently, the map

$$\varphi_L: X \longrightarrow \text{Pic}(X), \quad x \longmapsto t_x^*L \otimes L^{-1}$$

is a group homomorphism. Its kernel

$$K(L) = \{x \in X : t_x^*L \cong L\}$$

is the subgroup of translations preserving the class of  $L$ . This is proved in chapter 8 as a corollary of the theorem of the cube; it is invoked here only to construct an ample bundle.

The theorem of the square says that translation acts on the class of  $L$  through a homomorphism, so the failure of  $L$  to be translation-invariant is governed by the single subgroup  $K(L)$ . When  $K(L)$  is finite,  $L$  is positive in the sense that matters: no positive-dimensional family of translations fixes it, which is precisely the algebraic form of the condition that in chapter 3 distinguished a polarization from a degenerate Hermitian form. The construction below produces an  $L$  with  $K(L)$  finite, and then multiples of such an  $L$  are ample.

The second ingredient is an existence statement, and it costs nothing: a complete variety of positive dimension always carries a nonconstant effective divisor, because it carries nonconstant rational functions after passing to an affine chart, or more directly because its function field has transcendence degree  $g \geq 1$  over  $k$ . What must be arranged is that this divisor have no translation stabilizer, and this is where a genericity choice enters.

**Proposition 7.3.2: An Abelian Variety Carries an Ample Line Bundle**

Let  $X$  be an abelian variety of dimension  $g \geq 1$  over an algebraically closed field  $k$ . Then there exists an ample line bundle on  $X$ . More precisely, there is an effective divisor  $D$  on  $X$ , not passing through the identity, such that the subgroup

$$K(\mathcal{O}_X(D)) = \{x \in X : t_x^* \mathcal{O}_X(D) \cong \mathcal{O}_X(D)\}$$

is finite, and for any such  $D$  the line bundle  $\mathcal{O}_X(D)^{\otimes 3}$  is ample.

**Proof:**

*Step 1: Produce an irreducible effective divisor.* Since  $g \geq 1$ , the function field  $k(X)$  has transcendence degree  $g \geq 1$  over  $k$ , so  $X$  admits a nonconstant rational function and hence a nonzero effective divisor. Choose an irreducible effective divisor  $D_0$  on  $X$  whose support does not contain the identity  $e$ ; such a  $D_0$  exists because a general effective divisor avoids a fixed point and one of its irreducible components may be taken for  $D_0$ . Nothing so far controls the translations fixing the class of  $\mathcal{O}_X(D_0)$ ; that is the work of the next step.

*Step 2: Spread the divisor by translation to shrink the stabilizer.* By the theorem of the square (box 7.3.1), for any line bundle  $M$  the locus

$$K(M) = \{x \in X : t_x^* M \cong M\}$$

is the kernel of the homomorphism  $\varphi_M : X \rightarrow \text{Pic}(X)$ , hence a closed subgroup of  $X$ . Take

$$D = D_0 + (D_0 - a) + (D_0 - b)$$

a sum of three generic translates of the irreducible  $D_0$ , with  $a, b \in X$  generic closed points chosen so that the three components are distinct and none passes through  $e$ , and set  $L = \mathcal{O}_X(D)$ . If  $x \in K(L)$ , then  $t_x^* L \cong L$ , so  $D - x$  and  $D$  are linearly equivalent effective divisors of the same three-component shape; for generic  $a, b$  no linear equivalence identifies  $D - x$  with  $D$  unless the translation  $t_x$  already carries the support of  $D_0$  to itself, because the three parallel translates of the single irreducible hypersurface  $\text{supp}(D_0)$  pin down  $x$  up to the stabilizer of  $\text{supp}(D_0)$ . That stabilizer is finite: an irreducible hypersurface invariant under a positive-dimensional connected subgroup  $T \subseteq X$  is a union of cosets of  $T$ , so its dimension exceeds  $g - 1$  unless  $\dim T = 0$ , contradicting that it is a proper hypersurface. Hence  $K(L)$  is contained in a finite group and is

finite. The passage from “ $D - x \sim D$ ” to “ $t_x$  stabilizes  $\text{supp}(D_0)$ ” for generic translates is the one point where the theorem of the square does real work, and its complete justification is part of the seesaw and square machinery of chapter 8.

*Step 3: Finite  $K(L)$  makes a multiple ample.* Write  $L = \mathcal{O}_X(D)$  with  $K(L)$  finite,  $D$  effective and not through  $e$ . The theorem of the square gives, for closed points  $x, y$ ,

$$t_x^* L \otimes t_y^* L \cong t_{x+y}^* L \otimes L$$

and, applied to  $L^{\otimes 3}$ , expresses each translate of a section of  $L^{\otimes 3}$  through translates of the effective divisor  $D$ . The claim is that  $L^{\otimes 3}$  is very ample, hence ample. The mechanism is the algebraic form of the translation argument of section 4.4: for closed points with  $u + v + w = 0$ , the divisor

$$D_{u,v,w} = (D + u) + (D + v) + (D + w)$$

is the divisor of a section of  $L^{\otimes 3}$  by the theorem of the square, and since  $K(L)$  is finite the translates of  $\text{supp}(D)$  are numerous enough to avoid any prescribed point and to separate any two distinct points and any tangent direction. This point-and-tangent separation is the same transversality input isolated in theorem 4.4.2 over  $\mathbb{C}$ ; its complete verification is carried out in Mumford, *Abelian Varieties*, and rests on the theorem of the cube (chapter 8), which is why the argument is completed there and only its construction is given here. The construction of  $D$  with  $K(L)$  finite is the one part special to abelian varieties; granting the separation,  $L^{\otimes 3}$  is very ample and in particular ample, completing the proof.

The construction is the algebraic shadow of the analytic one. Over  $\mathbb{C}$  a polarization was a positive Hermitian form  $H$ , its positivity recorded in the finiteness of the group  $K(L) = \Lambda(L)/\Lambda$  (section 5.3); here, with no Hermitian form available, positivity is defined directly as the finiteness of the analogous group  $K(L)$ , and it is arranged by a divisor with no translation stabilizer. The theorem of the square plays the role that the Appell-Humbert computation of  $t_a^* L$  played in section 4.4: it says how the class of  $L$  changes under translation, and the whole ampleness argument is choosing translates so that the changes cancel. The long transversality verification is deferred, as it was in the analytic proof of theorem 4.4.2, to Mumford and to the theorem of the cube; the construction of an ample bundle, the one step that uses the group law, is given in full.

With an ample bundle in hand, projectivity is a general fact about complete varieties, imported from *Algebraic Geometry* [13]: an ample line bundle embeds a complete variety as a closed subscheme of a projective space. Nothing further about the group structure is needed for this last step.

### Theorem 7.3.3: Abelian Varieties Are Projective

Every abelian variety  $X$  over a field  $k$  is projective. Concretely, if  $L$  is an ample line bundle on  $X$  (which exists by proposition 7.3.2), then for  $n$  sufficiently large the complete linear system  $|L^{\otimes n}|$  defines a closed immersion

$$\varphi_{|L^{\otimes n}|}: X \hookrightarrow \mathbb{P}_k^N$$

realizing  $X$  as a smooth closed subvariety of a projective space over  $k$ . In particular  $X$  is a projective group variety.



**Proof:**

By proposition 7.3.2 the abelian variety  $X_{\bar{k}}$  over an algebraic closure carries an ample line bundle; ampleness descends, so  $X$  carries an ample line bundle  $L$  over  $k$  as well (*Algebraic Geometry* [13], descent of ampleness along  $\bar{k}/k$ ). For an ample  $L$  on the complete (proper) variety  $X$ , some power  $L^{\otimes n}$  is very ample, and a very ample line bundle on a proper  $k$ -scheme defines a closed immersion into  $\mathbb{P}_k^N$  for  $N = \dim_k H^0(X, L^{\otimes n}) - 1$ ; this is the defining property of ampleness together with the properness of  $X$  (*Algebraic Geometry* [13]). The image is closed in  $\mathbb{P}_k^N$  because  $X$  is proper and the immersion is proper, so  $X$  is a projective variety; it is smooth because  $X$  is (definition 7.1.2), and it carries its group-variety structure unchanged. Hence  $X$  is a projective group variety, as we sought to show.

The theorem completes the recognition of an abelian variety begun in section 7.1. The definition asked only for a proper connected group variety; the theorem upgrades “proper” to “projective”, so an abelian variety is an object with honest homogeneous coordinates, and the apparatus of projective algebraic geometry, ample and very ample bundles, intersection numbers, the Picard scheme, becomes available. The word “variety” in “abelian variety” is therefore no accident: these are projective varieties, and the projectivity is forced, not assumed, by properness and the group law together. That every abelian variety is projective is what allows chapter 10 to construct the dual as the Picard scheme  $\text{Pic}^0$  of a projective variety, whose representability (chapter 10) rests on projectivity.

The analytic story ran the other way. Over  $\mathbb{C}$ , projectivity was a genuine restriction on complex tori: the Riemann relations (theorem 3.1.4) singled out the polarizable tori, and a generic complex torus of dimension  $g \geq 2$  is not projective at all, carrying no nonconstant meromorphic functions (example 1.4.3). There “abelian variety” named a proper subclass of complex tori, cut out by the existence of a positive line bundle. Here the situation is reversed: an abelian variety is defined at the outset as a proper group variety, and *every* such object is projective, over every field. The gap between complex tori and abelian varieties has no algebraic counterpart, because there is no algebraic object playing the role of a non-algebraic complex torus. What over  $\mathbb{C}$  was a criterion to be verified is over a general field a theorem that holds automatically, and the reason is that a proper group variety cannot be anything but projective.

The dimension-one case is the familiar one, and it needs no new construction.

**Example 7.3.4: An Elliptic Curve Is Projective**

An elliptic curve  $E$  over a field  $k$  is, by definition, a smooth projective geometrically connected curve of genus one with a marked  $k$ -point  $O$ , together with its group law (*Elliptic Curves* [4]). Projectivity is part of the definition there, but it is also produced by the construction above, and seeing how is the  $g = 1$  shadow of proposition 7.3.2. Take the effective divisor

$$D = 3(O)$$

the point  $O$  with multiplicity three. On a genus-one curve a divisor of degree  $d \geq 1$  has  $\dim_k H^0(E, \mathcal{O}_E(D)) = d$  by Riemann-Roch, so  $\mathcal{O}_E(3(O))$  has three sections; a basis is  $\{1, x, y\}$  where  $x, y$  are the Weierstrass coordinates, with  $x$  having a double pole at  $O$  and  $y$  a triple pole. The associated morphism

$$\varphi_{|3(O)|}: E \longrightarrow \mathbb{P}_k^2, \quad P \longmapsto [1 : x(P) : y(P)]$$

is a closed immersion realizing  $E$  as the plane cubic

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$



the Weierstrass model. Here  $K(\mathcal{O}_E(3(O)))$  is finite, indeed  $K(\mathcal{O}_E(3(O)))$  is the 3-torsion subgroup  $E[3]$ , because on an elliptic curve  $t_x^* \mathcal{O}_E((O))$  has the same degree as  $\mathcal{O}_E((O))$  and differs from it by an element of  $\text{Pic}^0(E) = E$ , so  $t_x^* \mathcal{O}_E(n(O)) \cong \mathcal{O}_E(n(O))$  exactly when  $nx = O$ ; with  $n = 3$  this is  $x \in E[3]$ , a group of order 9 (in characteristic away from 3). The degree of the ample bundle is 3, matching  $|K| = 9 = 3^2 = (\deg)^2$  for a principal polarization taken to its cube, the  $g = 1$  instance of the count of proposition 5.3.4.

Over  $\mathbb{C}$  this same embedding is the degree-three theta embedding of example 4.4.5: the Weierstrass coordinates  $[1 : \wp : \wp']$  and the theta coordinates  $[\theta_0 : \theta_1 : \theta_2]$  are two bases of the same three-dimensional space  $H^0(E, \mathcal{O}_E(3(O)))$ , carried to one another by an element of  $\text{PGL}_3$ , and both present  $E$  as a plane cubic. The algebraic construction over an arbitrary  $k$  and the analytic construction over  $\mathbb{C}$  (chapter 4) are, in dimension one, the same closed immersion into  $\mathbb{P}^2$  seen from two sides.

### Exercise 7.3.1

1. State the map  $\varphi_L: X \rightarrow \text{Pic}(X)$  of box 7.3.1 and its kernel  $K(L)$ . Using the theorem of the square, verify directly that  $\varphi_L$  is a group homomorphism and that  $K(L)$  is a subgroup of  $X$ .
2. In example 7.3.4 the kernel  $K(\mathcal{O}_E(3(O)))$  is identified with the 3-torsion  $E[3]$ . Carry out the analogous computation for  $L = \mathcal{O}_E(n(O))$  on an elliptic curve  $E$ : show that  $t_x^* L \cong L$  if and only if  $nx = O$ , so that  $K(\mathcal{O}_E(n(O))) = E[n]$ . State the resulting order of  $K$  when  $n$  is invertible in  $k$ , and check that it matches  $(\deg L)^2$ .
3. proposition 7.3.2 produces an ample bundle by taking a cube  $\mathcal{O}_X(D)^{\otimes 3}$ . Explain why the proof forms the three-component divisor  $D = D_0 + (D_0 - a) + (D_0 - b)$  rather than working with the single irreducible  $D_0$ : identify precisely which property of  $\mathcal{O}_X(D)$  is not guaranteed for  $\mathcal{O}_X(D_0)$ , and where in the argument that property is used to conclude ampleness.
4. Let  $H \subseteq X$  be an irreducible hypersurface on an abelian variety  $X$  of dimension  $g$ , and let  $T = \text{Stab}(H)^0$  be the identity component of its translation stabilizer  $\text{Stab}(H) = \{x \in X : H + x = H\}$ . Prove that  $H$  is a union of cosets of  $T$ , and deduce that if  $T$  is nontrivial then  $H$  is the preimage  $\pi^{-1}(\bar{H})$  of an irreducible hypersurface  $\bar{H}$  under the quotient  $\pi: X \rightarrow X/T$ . Conclude that the geometric input to Step 2 of proposition 7.3.2 is the statement that a hypersurface with no such cylinder structure, i.e. one for which  $T$  is trivial, has finite  $\text{Stab}(H)$ , and explain why this is what the generic choice of  $D_0$  arranges.
5. theorem 7.3.3 asserts that every abelian variety over any field  $k$  is projective, whereas over  $\mathbb{C}$  a generic complex torus of dimension  $g \geq 2$  carries no ample bundle and is not projective (example 1.4.3). Reconcile these two statements: explain why there is no contradiction, identifying the property in the algebraic definition (definition 7.1.2) that a generic complex torus fails to satisfy, and hence why “every abelian variety is projective” is not the same claim as “every complex torus is projective”.
6. Let  $X$  be an abelian variety over an algebraically closed field  $k$  and  $L$  an ample line bundle on  $X$ . Prove that  $K(L)$  is finite. (Hint:  $K(L)$  is a closed subgroup by box 7.3.1; suppose its identity component  $K(L)^0$  has positive dimension and derive a contradiction with ampleness by restricting  $L$  to  $K(L)^0$  and using that translation by  $x \in K(L)^0$  fixes the class of  $L$ .)

## 7.4 Line Bundles and the Picard Functor

This section does two things. It organizes the line bundles of an abelian variety into the Picard functor  $\mathrm{Pic}_{X/k}$  and its identity component  $\mathrm{Pic}_{X/k}^0$ , the object that will become the dual abelian variety, whose representability is imported from *Algebraic Geometry* [13]. And it proves in full the seesaw principle, the criterion for when a line bundle on a product  $X \times T$  is pulled back from the second factor, contributing nothing along  $X$ . That criterion is elementary and is what the induction of the theorem of the cube (chapter 8) consumes at each of its factor-peeling steps.

Over a fixed field the group of line bundles on  $X$  is  $\mathrm{Pic}(X)$ , but a single group cannot see how line bundles vary in families, and it is the variation that carries geometric content. A line bundle on  $X \times T$  is a family of line bundles on  $X$  parametrized by the points of  $T$ , and the right object to classify such families is not a group but a functor: to each  $T$  it assigns the group of families over  $T$ , and to each morphism  $T' \rightarrow T$  the operation of pulling a family back. The subtlety is that a line bundle on  $X \times T$  is slightly more data than a  $T$ -family of bundles on  $X$ : it also carries a bundle pulled back from  $T$ , which is invisible along each fibre  $X \times \{t\}$ . Quotienting that out is what the sheafification below accomplishes.

### Definition 7.4.1: The Picard Functor

Let  $X$  be a variety over a field  $k$ . The *relative Picard functor*  $\mathrm{Pic}_{X/k}$  is the functor

$$\mathrm{Pic}_{X/k}: (\mathrm{Sch}/k)^{\mathrm{op}} \longrightarrow \mathrm{Ab}$$

obtained by sheafifying, in the fppf topology on  $\mathrm{Sch}/k$ , the presheaf

$$T \longmapsto \mathrm{Pic}(X \times_k T) / \mathrm{pr}_T^* \mathrm{Pic}(T)$$

where  $\mathrm{pr}_T: X \times_k T \rightarrow T$  is the projection and  $\mathrm{pr}_T^* \mathrm{Pic}(T)$  is the subgroup of line bundles pulled back from the base. Its value on  $T = \mathrm{Spec} k$  is  $\mathrm{Pic}(X)$  when  $X$  has a  $k$ -rational point.

The presheaf  $T \mapsto \mathrm{Pic}(X \times_k T) / \mathrm{pr}_T^* \mathrm{Pic}(T)$  records, for each base  $T$ , the families of line bundles on  $X$  up to the ambiguity of a bundle from the base, and it is the naive candidate for the functor of families. It fails to be a sheaf: a family may exist locally on  $T$ , agreeing on overlaps up to isomorphism, without gluing to a global line bundle on  $X \times_k T$ , because the isomorphisms on overlaps need not themselves be compatible. Sheafifying repairs exactly this, and the fppf topology is the correct one because line bundles descend along faithfully flat morphisms. The construction, the choice of topology, and the comparison with the naive presheaf are developed in *Algebraic Geometry* [13]; here the functor is the object to be represented, and its representability is what makes the dual abelian variety a variety rather than a formless group.

### Theorem 7.4.2: Representability of the Picard Functor

Let  $X$  be a projective variety over a field  $k$  that is geometrically integral and has a  $k$ -rational point. Then the Picard functor  $\mathrm{Pic}_{X/k}$  is representable by a  $k$ -scheme, locally of finite type over  $k$ , called the *Picard scheme* of  $X$ .

**Proof Key ideas:**

Representability of the Picard functor is a theorem of Grothendieck's *Fondements de la géométrie algébrique* (FGA), proved by faithfully flat descent together with the representability of the relative Hilbert and Quot functors: a line bundle on  $X \times_k T$  is encoded, via its graph or its associated divisor after twisting by a fixed ample bundle, as a point of a Quot scheme, and the Picard scheme is cut out as a quotient. The full descent argument is scheme-theoretic and belongs to the general theory; it is given in *Algebraic Geometry* [13]. It is cited rather than reproduced because it is the representability machinery of a prerequisite, not a fact about abelian varieties, and reproving it here would duplicate that book to no local benefit.

The Picard scheme is locally of finite type but need not be connected or even of finite type: its group of connected components can be infinite, and these components record discrete invariants of line bundles, the algebraic analogue of the degree of a bundle on a curve (example 7.4.4). The component of the identity is the piece that varies continuously, and it is where a  $g$ -dimensional abelian variety hides inside the Picard scheme.

**Definition 7.4.3: The Identity Component of the Picard Scheme**

Let  $X$  be a projective geometrically integral variety over  $k$  with a  $k$ -rational point, and let  $\text{Pic}_{X/k}$  be its Picard scheme. The *identity component*  $\text{Pic}_{X/k}^0$  is the connected component of  $\text{Pic}_{X/k}$  containing the identity element (the class of the trivial bundle). A line bundle whose class lies in  $\text{Pic}_{X/k}^0$  is called *algebraically trivial*: its class is connected to the trivial class by a connected family of line bundles.

Algebraic triviality is the algebraic surrogate for the analytic notion of a degree-zero, topologically trivial line bundle. Over  $\mathbb{C}$  the identity component  $\text{Pic}_{X/k}^0$  recovers exactly the dual torus  $\hat{X}$  of definition 5.1.1 in Part I, the group of line bundles with vanishing first Chern class, presented there as  $\text{Pic}^0(X) = \overline{V}^*/\hat{\Lambda}$ . The comparison is the seam between the two Parts: the analytic  $\text{Pic}^0(X)$  and the algebraic  $\text{Pic}_{X/k}^0$  are the same object viewed through two constructions. When  $X$  is an abelian variety,  $\text{Pic}_{X/k}^0$  is again an abelian variety of the same dimension, the *dual abelian variety*, whose construction and biduality are the business of chapter 10. For now it is named, not built.

**Example 7.4.4: The Degree-Zero Picard Scheme of an Elliptic Curve**

Let  $E$  be an elliptic curve over  $k$ , a projective group variety of dimension one (definition 7.1.2). A line bundle on  $E$  has a well-defined degree, and the degree map  $\deg: \text{Pic}(E) \rightarrow \mathbb{Z}$  is a surjective homomorphism whose kernel is the group of degree-zero bundles. Over a curve, degree distinguishes the connected components of the Picard scheme: two line bundles lie in the same component precisely when they have the same degree, so

$$\text{Pic}_{E/k} = \coprod_{d \in \mathbb{Z}} \text{Pic}_{E/k}^d$$

with  $\text{Pic}_{E/k}^0 = \text{Pic}_{E/k}^0$  the degree-zero component. The theory of the group law on  $E$  (*Elliptic Curves* [4]) identifies this component with  $E$  itself: sending a point  $P \in E$  to the class of the degree-zero bundle  $\mathcal{O}_E(P - O)$ , where  $O$  is the origin, is an isomorphism of group varieties

$$E \xrightarrow{\sim} \text{Pic}_{E/k}^0, \quad P \longmapsto [\mathcal{O}_E(P - O)]$$

This is the algebraic form of the analytic identification  $\text{Pic}^0(E) \cong E$  from example 2.1.2: there it came from the Abel-Jacobi theorem for the complex torus  $\mathbb{C}/\Lambda$ ; here it comes from the group

law on the curve, with no reference to  $\mathbb{C}$ . That the dual of an elliptic curve is the curve itself is the first instance of the autoduality that a principal polarization produces, developed in general in chapter 10.

The example fixes the pattern the dual abelian variety generalizes: the identity component of the Picard scheme of an abelian variety is again an abelian variety, and for  $E$  it is  $E$  again. To construct that dual in general, and to run the theorem of the cube that feeds it, requires a way to recognize when a line bundle on a product is trivial along the first factor. That is the seesaw principle.

The name records the mechanism. A line bundle  $L$  on a product  $X \times T$  can be tested one factor at a time. Restricting to a fibre  $X \times \{t\}$  tests it along  $X$ ; restricting to a fibre  $\{x\} \times T$  tests it along  $T$ . The seesaw principle says that if  $L$  is trivial on enough of the first kind of fibre, then  $L$  owes nothing to  $X$  and is pulled back from  $T$ . The hypothesis that makes this work is the completeness of  $X$ : on a complete variety the only global functions are constants, and it is this rigidity that forces a fibrewise-trivial bundle to be constant along  $X$ .

#### Theorem 7.4.5: The Seesaw Principle

Let  $X$  be a complete variety over an algebraically closed field  $k$ , let  $T$  be a variety over  $k$ , and let  $L$  be a line bundle on  $X \times T$ . Suppose there is a dense subset  $S \subseteq T$  such that for every  $t \in S$  the restriction  $L|_{X \times \{t\}}$  is trivial. Then there is a line bundle  $M$  on  $T$  such that

$$L \cong \text{pr}_T^* M$$

where  $\text{pr}_T: X \times T \rightarrow T$  is the projection. If moreover  $L|_{X \times \{t\}}$  is trivial for *every*  $t \in T$ , then  $M$  may be taken to be  $M = \text{pr}_{T*} L$ , the pushforward of  $L$  along  $\text{pr}_T$ .

#### Proof:

Write  $p = \text{pr}_T: X \times T \rightarrow T$  for the projection, and for  $t \in T$  write  $X_t = X \times \{t\}$  and  $L_t = L|_{X_t}$ , the restriction of  $L$  to the fibre over  $t$ . The argument is a base change computation controlled by the completeness of  $X$ : on a complete variety the only global functions are constants, and this rigidity lets triviality of a fibre bundle be detected by two cohomological inequalities, one for the bundle and one for its inverse.

*Step 1: Completeness detects triviality of a fibre bundle by sections of the bundle and of its inverse.* For each  $t \in T$  the fibre  $X_t$  is isomorphic to  $X$ , hence complete and geometrically connected, so its only global functions are constants:  $H^0(X_t, \mathcal{O}_{X_t}) = k$ . Let  $N$  be a line bundle on  $X_t$ . If  $N$  is trivial then both  $N$  and  $N^{-1}$  are trivial, so each has a nonzero global section (a nonzero constant); thus  $\dim_k H^0(X_t, N) \geq 1$  and  $\dim_k H^0(X_t, N^{-1}) \geq 1$ . Conversely, suppose  $N$  and  $N^{-1}$  both carry a nonzero global section, say  $s \in H^0(X_t, N)$  and  $s' \in H^0(X_t, N^{-1})$ . Their tensor product  $s \otimes s'$  is a global section of  $N \otimes N^{-1} = \mathcal{O}_{X_t}$ , that is, a global function on  $X_t$ , hence a constant. This constant is nonzero: were it zero, then  $s \otimes s'$  would vanish identically, forcing  $s$  or  $s'$  to vanish on a dense open set of the irreducible variety  $X_t$  and hence everywhere, contrary to  $s, s' \neq 0$ . So  $s \otimes s'$  is a nonzero constant, which means  $s$  vanishes nowhere; a line bundle with a nowhere-vanishing global section is trivial. Therefore

$$N \text{ is trivial} \iff \dim_k H^0(X_t, N) \geq 1 \text{ and } \dim_k H^0(X_t, N^{-1}) \geq 1$$

Applied to  $N = L_t$ , the hypothesis that  $L_t$  is trivial for  $t \in S$  says that both  $\dim_k H^0(X_t, L_t) \geq 1$

and  $\dim_k H^0(X_t, L_t^{-1}) \geq 1$  hold for every  $t$  in the dense set  $S$ .

*Step 2: Semicontinuity propagates triviality from  $S$  to all of  $T$ .* The projection  $p$  is proper (its fibres are the complete variety  $X$ ), and  $L$  is a line bundle on  $X \times T$ , so it is locally free and in particular flat over  $T$ : indeed  $X \times T \rightarrow T$  is flat because it is the base change to  $T$  of the structure morphism  $X \rightarrow \operatorname{Spec} k$ , and a locally free sheaf on a scheme flat over  $T$  is flat over  $T$ . The semicontinuity theorem for cohomology and base change therefore applies to both  $L$  and its inverse  $L^{-1}$ : each of the functions

$$t \mapsto \dim_k H^0(X_t, L_t), \quad t \mapsto \dim_k H^0(X_t, L_t^{-1})$$

is upper semicontinuous on  $T$ , so for each integer  $n$  the loci  $\{t : \dim_k H^0(X_t, L_t) \geq n\}$  and  $\{t : \dim_k H^0(X_t, L_t^{-1}) \geq n\}$  are closed. The base-change theory is that of *Algebraic Geometry* [13]. By Step 1 both  $\dim_k H^0(X_t, L_t) \geq 1$  and  $\dim_k H^0(X_t, L_t^{-1}) \geq 1$  hold on the dense set  $S$ , so the intersection

$$\{t : \dim_k H^0(X_t, L_t) \geq 1\} \cap \{t : \dim_k H^0(X_t, L_t^{-1}) \geq 1\}$$

is closed and contains  $S$ , hence is all of  $T$ . By the criterion of Step 1,  $L_t$  is therefore trivial for every  $t \in T$ , not merely for  $t \in S$ ; in particular  $\dim_k H^0(X_t, L_t) = 1$  for every  $t \in T$ , since a trivial bundle on the complete geometrically connected  $X_t$  has exactly the constants as its global sections.

*Step 3: The pushforward is a line bundle, and its pullback maps to  $L$ .* Set  $M := p_* L$ . Because  $\dim_k H^0(X_t, L_t) = 1$  is constant in  $t$ , the theorem on cohomology and base change [13] gives that  $p_* L$  is locally free of rank one, that is, a line bundle on  $T$ , and that its formation commutes with base change: for each  $t$  the natural map  $(p_* L) \otimes k(t) \rightarrow H^0(X_t, L_t)$  is an isomorphism. The adjunction between  $p^*$  and  $p_*$  supplies the counit

$$\varphi: p^* p_* L = p^* M \longrightarrow L$$

a morphism of line bundles on  $X \times T$ . It remains to see that  $\varphi$  is an isomorphism.

*Step 4: Fibrewise nonvanishing forces  $\varphi$  to be an isomorphism.* A morphism of line bundles is an isomorphism precisely when it vanishes nowhere, and this may be checked on fibres. Fix  $t \in T$ . Restricting  $\varphi$  to  $X_t$  and using that the formation of  $M = p_* L$  commutes with base change, the restricted map is the evaluation

$$\varphi_t: H^0(X_t, L_t) \otimes_k \mathcal{O}_{X_t} \longrightarrow L_t$$

sending a global section and a point to the value of the section there. Since  $L_t$  is trivial (Step 2) with  $\dim_k H^0(X_t, L_t) = 1$ , a generator  $s_t$  of  $H^0(X_t, L_t)$  is a nowhere-vanishing section, so  $\varphi_t$  is an isomorphism at every point of  $X_t$ . Thus  $\varphi$  is an isomorphism on every fibre of  $p$ , hence an isomorphism, giving  $L \cong p^* M$  with  $M = p_* L$ , as we sought to show.

The seesaw principle reduces a question about a line bundle on a product to a question fibre by fibre: if the bundle is trivial along each copy of  $X$ , it is determined entirely by the base  $T$ . Two features carry the argument. The completeness of  $X$  makes triviality of a fibre bundle a closed condition, since it is detected by two semicontinuous quantities, a section of  $L_t$  and a section of

its inverse, and their product is a nonzero constant exactly when  $L_t$  is trivial. Semicontinuity then propagates the triviality from a dense subset to the whole base, and cohomology-and-base-change upgrades the fibrewise statement to a global isomorphism  $L \cong p^*M$ . The hypothesis of completeness is not cosmetic: on an affine  $X$  a nontrivial bundle can still be trivialized by a global function, the closedness of the trivial locus fails, and the conclusion fails with it. This is one of the places where properness of an abelian variety, established in theorem 7.3.3, does load-bearing work.

The seesaw comes in a symmetric two-variable form that is the shape the theorem of the cube invokes. Testing along the other factor as well pins a line bundle down completely, and the first application below is exactly this rigidity: a line bundle trivial on both families of fibres is trivial.

**Proposition 7.4.6: Bundles Trivial on Coordinate Fibres are Trivial**

Let  $X$  and  $Y$  be complete geometrically connected varieties over an algebraically closed field  $k$ , each with a  $k$ -rational point,  $x_0 \in X$  and  $y_0 \in Y$ . Let  $L$  be a line bundle on  $X \times Y$  whose restrictions to the coordinate fibres through  $(x_0, y_0)$  are trivial:

$$L|_{X \times \{y\}} \text{ is trivial for all } y \in Y, \quad L|_{\{x_0\} \times Y} \text{ is trivial}$$

Then  $L$  is trivial on  $X \times Y$ .

**Proof:**

Regard  $Y$  as the base  $T$  and  $X$  as the complete fibre. The hypothesis that  $L|_{X \times \{y\}}$  is trivial for every  $y \in Y$  is the full-strength hypothesis of theorem 7.4.5, so there is a line bundle  $M$  on  $Y$  with

$$L \cong \text{pr}_Y^* M$$

where  $\text{pr}_Y: X \times Y \rightarrow Y$  is the projection. It remains to identify  $M$ . Restrict this isomorphism to the fibre  $\{x_0\} \times Y$ , which the projection  $\text{pr}_Y$  maps isomorphically onto  $Y$ . On that fibre  $\text{pr}_Y^* M$  restricts to  $M$  itself (via the isomorphism  $\{x_0\} \times Y \cong Y$ ), while  $L$  restricts to  $L|_{\{x_0\} \times Y}$ , which is trivial by hypothesis. Hence  $M$  is trivial on  $Y$ , and therefore  $L \cong \text{pr}_Y^* M$  is trivial on  $X \times Y$ , completing the proof.

The proposition is the seesaw stated as a rigidity: a line bundle on a product of complete varieties that is trivial on every horizontal fibre and on one vertical fibre is trivial outright. In this form it is the base case of the induction proving the theorem of the cube, where  $L$  lives on a triple product  $X \times Y \times Z$  and the argument trivializes it one factor at a time by restricting to slices and invoking exactly this two-variable criterion. The completeness of all three factors, which for abelian varieties is their properness (definition 7.1.2), is what licenses each restriction.

**Exercise 7.4.1**

1. Let  $X$  be a variety over  $k$ . The relative Picard functor  $\text{Pic}_{X/k}$  of definition 7.4.1 sheafifies the presheaf  $T \mapsto \text{Pic}(X \times_k T) / \text{pr}_T^* \text{Pic}(T)$ . Compute the value of this presheaf on  $T = \text{Spec } k$  and show it equals  $\text{Pic}(X)$ , noting that this value is  $\text{Pic}(X)$  unconditionally since  $\text{Pic}(\text{Spec } k) = 0$ ; the  $k$ -rational-point hypothesis of definition 7.4.1 is what the *sheafification* uses, not this presheaf value. Then state precisely which line bundles on  $X \times_k T$  the quotient by  $\text{pr}_T^* \text{Pic}(T)$  discards, and explain in one sentence why they carry no information about the way line bundles vary along  $X$ .

2. Let  $E$  be an elliptic curve over  $k$  with origin  $O$ . Using the description of  $\mathrm{Pic}_{E/k}^0$  in example 7.4.4, verify directly that the map  $P \mapsto [\mathcal{O}_E(P - O)]$  sends  $O$  to the identity of  $\mathrm{Pic}_{E/k}^0$ , and identify the class  $[\mathcal{O}_E(P - O)]$  as an element of  $\mathrm{Pic}_{E/k}^0$  by computing  $\deg \mathcal{O}_E(P - O)$ .
3. In Step 1 of the proof of the seesaw principle (theorem 7.4.5) the completeness of the fibre  $X_t$  is used to prove that a line bundle  $N$  on  $X_t$  is trivial if and only if both  $N$  and  $N^{-1}$  carry a nonzero global section. The “only if” direction is formal and survives on any variety; the “if” direction consumes completeness. This exercise isolates that dependence, and shows why the theorem also requires  $k$  algebraically closed. Take  $A = k[x, y]/(x^2 + y^2 - 1)$  with  $\mathrm{char} k \neq 2$  and  $X = \mathrm{Spec} A$  the affine circle.
  1. Over an algebraically closed  $k$  the “if” direction still holds, but only because the bundle is forced trivial. Assuming  $\sqrt{-1} \in k$ , put  $u = x + \sqrt{-1}y$  and  $v = x - \sqrt{-1}y$ . Show  $A \cong k[u, v]/(uv - 1) = k[u, u^{-1}]$ , the coordinate ring of  $\mathbb{G}_m$ , a principal ideal domain. Deduce that every rank-one projective  $A$ -module is free, so  $\mathrm{Pic}(X) = 0$ ; in particular the ideal  $M = (1 - y, x)$  is principal (identify it with  $(u - \sqrt{-1})$ ), and every line bundle on  $X$  is trivial. Conclude that no *nontrivial* counterexample to the “if” direction exists over  $k = \bar{k}$ : the completeness hypothesis cannot be dropped by moving to the affine circle while keeping  $k$  algebraically closed.
  2. Over  $\mathbb{R}$  the “if” direction genuinely fails. Now take  $k = \mathbb{R}$ , so  $\sqrt{-1} \notin k$  and the change of variables of part (a) is unavailable. Let  $N$  be the line bundle associated to  $M = (1 - y, x) \subseteq A$ . Show that  $M \otimes_A M \cong A$ , so  $N^{-1} \cong N$ , yet  $M$  is *not* principal, so  $N$  is nontrivial. (For non-principality, parametrize the real circle by  $x = \sin \theta$ ,  $y = \cos \theta$  and argue that a generator would have to vanish to odd order at the single point  $(0, 1)$ , impossible for a global function on  $S^1(\mathbb{R})$ .) Exhibit nonzero global sections of  $N$  and of  $N^{-1}$  whose tensor product is a global function that vanishes somewhere, so the “if” direction fails on  $X$ .
  3. Explain which step of the completeness argument breaks, and why the seesaw principle needs *both* completeness and  $k = \bar{k}$ .
4. Let  $X$  be a complete variety over an algebraically closed field  $k$  and  $T$  a variety over  $k$ , and let  $L$  be a line bundle on  $X \times T$  that is trivial on every fibre  $X \times \{t\}$ . The seesaw principle (theorem 7.4.5) gives  $L \cong \mathrm{pr}_T^* M$  with  $M = \mathrm{pr}_{T*} L$ . Prove that the line bundle  $M$  is unique up to isomorphism: if  $\mathrm{pr}_T^* M \cong \mathrm{pr}_T^* M'$  for line bundles  $M, M'$  on  $T$ , then  $M \cong M'$ . (You may use that  $X$  is complete and geometrically connected, so that  $H^0(X, \mathcal{O}_X) = k$ .)
5. Let  $X$  and  $Y$  be complete geometrically connected varieties over an algebraically closed field  $k$ , each with a  $k$ -rational point  $x_0 \in X$ ,  $y_0 \in Y$ . Prove the following strengthening of proposition 7.4.6: if  $L$  is a line bundle on  $X \times Y$  whose restrictions to  $X \times \{y_0\}$  and to  $\{x_0\} \times Y$  are both trivial and whose restriction to  $X \times \{y\}$  is trivial for all  $y$  in a dense subset  $S \subseteq Y$ , then  $L$  is trivial on  $X \times Y$ . Then deduce proposition 7.4.6 as the special case  $S = Y$ .





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## *The Theorem of the Cube*

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Part I bought every theorem with a transcendental input: a Hermitian form on  $\mathbb{C}^g$ , the Riemann relations, the convergence of a theta series. Over an arbitrary field none of that is available, and yet the same structural facts hold. The instrument that replaces the analysis is a single statement about line bundles on a triple product, the *theorem of the cube*: a line bundle trivial on each of the three coordinate slices through a base point is trivial. It carries no metric and no series, only the properness assembled in chapter 7, and it runs on the seesaw principle (theorem 7.4.5) proved there.

From this one theorem the algebraic theory unfolds by pure formalism. The theorem of the square follows at once, and with it the fact that translating a line bundle moves its class inside  $\text{Pic}^0$  by a homomorphism  $\varphi_L$ ; the finite group  $K(L)$  measuring the failure of  $\varphi_L$  to be injective is the algebraic shadow of the type of a polarization. The same formalism computes the pullback of a line bundle under multiplication by  $n$ , and hence that  $[n]$  is an isogeny of degree  $n^{2g}$ , the algebraic count that chapter 5 obtained analytically from the volume of a fundamental domain. Chapter 10 will promote  $\varphi_L$  to a morphism into the dual variety, and chapter 9 takes up the torsion  $X[n]$  whose size this chapter first pins down.

### 8.1 The Theorem of the Cube

This section proves that a line bundle on a triple product of complete varieties is trivial as soon as it is trivial on the three coordinate slices through a base point. The proof reduces the triple product to its base by one application of the seesaw principle (theorem 7.4.5), after showing that  $L$  is trivial on every slice of the complete first factor. Two coordinate slices alone would not force this (a line bundle on a product trivial on the two coordinate axes through a point need not be trivial); the third factor supplies the missing rigidity. The locus over the base where a horizontal slice is trivial is closed and contains two coordinate axes, and the theorem's core is that this forces the locus to be dense, hence everything; that step is established by reducing to the case of a curve.

**Theorem 8.1.1: The Theorem of the Cube**

Let  $X$  and  $Y$  be complete geometrically connected varieties over a field  $k$ , and let  $Z$  be a variety over  $k$ , with  $k$ -rational points  $x_0 \in X(k)$ ,  $y_0 \in Y(k)$ , and  $z_0 \in Z(k)$ . Let  $L$  be a line bundle on  $X \times Y \times Z$  whose restrictions to the three coordinate slices through  $(x_0, y_0, z_0)$ ,

$$L|_{\{x_0\} \times Y \times Z}, \quad L|_{X \times \{y_0\} \times Z}, \quad L|_{X \times Y \times \{z_0\}}$$

are all trivial. Then  $L$  is trivial on  $X \times Y \times Z$ .

**Proof:**

We may assume the ground field is algebraically closed: the bundle  $L$  is rigidified by its trivializations along the slices through the  $k$ -point  $(x_0, y_0, z_0)$ , and for such a rigidified bundle triviality descends from  $\bar{k}$  to  $k$  (*Algebraic Geometry* [13]). Completeness and geometric connectedness of  $X$  then give  $H^0(X, \mathcal{O}) = \bar{k}$ , the input the seesaw principle, the semicontinuity theorem, and the theorem on cohomology and base change all require.

*Overview.* Group the factors as  $X \times (Y \times Z)$  with  $X$  the complete fibre and  $Y \times Z$  the base, and write  $p: X \times Y \times Z \rightarrow Y \times Z$  for the projection. The seesaw principle (theorem 7.4.5) reduces the theorem to a single fact: that  $L$  is trivial on every horizontal slice  $X \times \{t\}$ ,  $t \in Y \times Z$ . Granting that, theorem 7.4.5 gives  $L \cong p^*M$  for a line bundle  $M$  on  $Y \times Z$ , and restricting the isomorphism to  $\{x_0\} \times Y \times Z$  (which  $p$  maps isomorphically onto  $Y \times Z$ ) identifies  $M$  with  $L|_{\{x_0\} \times Y \times Z}$ , trivial by the first hypothesis; hence  $M$ , and with it  $L$ , is trivial. Everything comes down to the triviality of every horizontal slice. The locus  $T$  of good slices is closed (Step 1, by semicontinuity) and contains the two coordinate axes cut out by the second and third hypotheses; the theorem's core (Step 2) is that a closed slice-triviality locus containing those two axes is dense, hence, the base being irreducible, all of it. The two axes alone would not force triviality, so the third slice is essential; it is brought to bear by reducing the density to the case of a curve.

*Step 1: The horizontal-triviality locus is closed.* For a point  $t \in Y \times Z$  write  $L_t = L|_{X \times \{t\}}$ , a line bundle on the complete geometrically connected variety  $X$ , so that  $H^0(X, \mathcal{O}_X) = k$ . As established in the proof of theorem 7.4.5, completeness makes triviality of  $L_t$  equivalent to the pair of inequalities  $\dim_k H^0(X, L_t) \geq 1$  and  $\dim_k H^0(X, L_t^{-1}) \geq 1$ , since a section of  $L_t$  and a section of  $L_t^{-1}$  multiply to a nonzero constant exactly when  $L_t$  is trivial. The projection  $p$  is proper with  $L$  and  $L^{-1}$  flat over the base, so by the semicontinuity theorem for cohomology and base change (*Algebraic Geometry* [13]) the two functions

$$t \mapsto \dim_k H^0(X, L_t), \quad t \mapsto \dim_k H^0(X, L_t^{-1})$$

are upper semicontinuous. Hence the horizontal-triviality locus

$$T := \{t \in Y \times Z : L_t \text{ is trivial}\} = \{t : \dim_k H^0(X, L_t) \geq 1\} \cap \{t : \dim_k H^0(X, L_t^{-1}) \geq 1\}$$

is closed in  $Y \times Z$ .

*Step 2: The locus  $T$  contains the two axes and is dense.* By the third hypothesis  $L|_{X \times Y \times \{z_0\}}$  is trivial, so each slice  $L_{(y, z_0)}$ , being a restriction of it, is trivial, and the axis  $Y \times \{z_0\}$  lies in  $T$ . By the second hypothesis  $L|_{X \times \{y_0\} \times Z}$  is trivial, so  $\{y_0\} \times Z \subseteq T$  as well. Two axes alone

would not force triviality (the Poincaré bundle on  $E \times \hat{E}$  is trivial on both axes through  $(e, 0)$  yet is nontrivial, its restrictions to the  $E$ -fibres running through all of  $\text{Pic}^0(E)$ ), so the third hypothesis, contributing the entire trivial slice  $X \times Y \times \{z_0\}$ , must be brought to bear. It is, through the density of  $T$ .

That the closed locus  $T$ , containing the two coordinate axes  $Y \times \{z_0\}$  and  $\{y_0\} \times Z$ , is in fact all of the base is the technical core of the theorem. It is proved by reducing to the case of a curve: any two points of an irreducible variety over an algebraically closed field lie on an irreducible curve through them (cut the variety down by Chow's lemma and repeated general hyperplane sections, *Algebraic Geometry* [13]), so it is enough to show that  $T$  meets a dense set of such connecting curves in a dense subset, and on the product of the complete  $X$  with a nonsingular complete curve the triviality of the horizontal slices is a Riemann-Roch computation run through cohomology and base change, in which the three slice hypotheses enter together. This is the Weil-Murre argument (Mumford, *Abelian Varieties*, II.6), and its cohomology-and-base-change input is imported from *Algebraic Geometry* [13]; it is the one substantial fact about line bundles in families that the cube consumes, every other step being elementary given the seesaw. Granting it,  $T = Y \times Z$ : every horizontal slice  $L_t$  is trivial.

*Step 3: Descend to the base.* By Step 2 the full-strength hypothesis of the seesaw principle (theorem 7.4.5) holds for  $p: X \times (Y \times Z) \rightarrow Y \times Z$ , so there is a line bundle  $M$  on  $Y \times Z$  with

$$L \cong p^*M$$

Restricting to  $\{x_0\} \times Y \times Z$ , on which  $p^*M$  becomes  $M$  while  $L$  becomes the trivial bundle  $L|_{\{x_0\} \times Y \times Z}$  (first hypothesis), shows  $M$  trivial on  $Y \times Z$ . Therefore  $L \cong p^*M$  is trivial on  $X \times Y \times Z$ , as we sought to show.

The theorem converts properness into a rigidity statement about line bundles: on a product of complete varieties, a bundle is pinned down by its behaviour on a handful of coordinate slices. Nothing about the group law has been used yet; the hypotheses are purely those of theorem 7.4.5, completeness and geometric connectedness, applied one factor at a time. It becomes a tool once the factors are copies of an abelian variety  $A$  (definition 7.1.2) and the maps into  $A$  are the group operations. Then the three slice-triviality hypotheses become identities about  $\text{pr}_i^*$  of a bundle, automatically satisfied, and the conclusion becomes a nontrivial relation among the pullbacks of a single bundle along sums of morphisms. Every later result of chapter 8 is a specialization of that relation: the theorem of the square (theorem 8.2.1), the fact that translation moves a bundle's class by a homomorphism  $\varphi_L$ , and the pullback law  $[n]^*L \cong L^{\otimes n^2}$  underlying  $\deg[n] = n^{2g}$ , each results from feeding particular morphisms  $T \rightarrow A$  into the cube.

The clean packaging of that specialization is a triviality statement for a seven-fold alternating combination of pullbacks. It is the form in which the cube is invoked throughout the sequel.

**Proposition 8.1.2: The Cube Relation for Morphisms into an Abelian Variety**

Let  $A$  be an abelian variety over  $k$ , let  $L$  be a line bundle on  $A$ , and let  $T$  be any variety over  $k$  with three morphisms  $f, g, h: T \rightarrow A$ . Then the line bundle

$$(f + g + h)^*L \otimes (f + g)^*L^{-1} \otimes (g + h)^*L^{-1} \otimes (f + h)^*L^{-1} \otimes f^*L \otimes g^*L \otimes h^*L$$

is trivial on  $T$ , where  $f + g$  denotes the morphism  $x \mapsto f(x) + g(x)$  obtained by composing  $(f, g): T \rightarrow A \times A$  with the group law, and similarly for the other sums.

**Proof:**

Consider first the universal case  $T = A \times A \times A$  with  $f, g, h$  the three projections  $\text{pr}_1, \text{pr}_2, \text{pr}_3$ . Define the line bundle

$$\Theta := m_{123}^*L \otimes m_{12}^*L^{-1} \otimes m_{23}^*L^{-1} \otimes m_{13}^*L^{-1} \otimes \text{pr}_1^*L \otimes \text{pr}_2^*L \otimes \text{pr}_3^*L$$

on  $A \times A \times A$ , where  $m_{ij}$  adds the  $i$ th and  $j$ th coordinates ( $m_{ij} = m \circ (\text{pr}_i, \text{pr}_j)$ ) and  $m_{123}$  adds all three. The claim is that  $\Theta$  is trivial; the proposition then follows by pulling  $\Theta$  back along  $(f, g, h): T \rightarrow A \times A \times A$ , since each factor of  $\Theta$  pulls back to the corresponding factor of the displayed bundle (for instance  $m_{123} \circ (f, g, h) = f + g + h$ ).

*Step 1: Verify the three slice-triviality hypotheses.* Let  $e \in A(k)$  be the identity. Apply the theorem of the cube (theorem 8.1.1) to  $X = Y = Z = A$  with base point  $(x_0, y_0, z_0) = (e, e, e)$ ; each factor is a complete geometrically connected variety with the  $k$ -point  $e$ . It remains to check that  $\Theta$  is trivial on each of the three coordinate slices through  $(e, e, e)$ . Consider the slice  $\{e\} \times A \times A$ , on which the first coordinate is fixed to  $e$ . There  $m_{123}$  becomes  $m_{23}$  (adding  $e$  changes nothing),  $m_{12}$  becomes  $\text{pr}_2$ ,  $m_{13}$  becomes  $\text{pr}_3$ ,  $m_{23}$  stays  $m_{23}$ , and  $\text{pr}_1^*L$  restricts to the trivial bundle  $e^*L$  (a bundle on a point, pulled back, hence trivial on the slice). Substituting, the restriction of  $\Theta$  to this slice is

$$m_{23}^*L \otimes \text{pr}_2^*L^{-1} \otimes m_{23}^*L^{-1} \otimes \text{pr}_3^*L^{-1} \otimes \mathcal{O} \otimes \text{pr}_2^*L \otimes \text{pr}_3^*L$$

in which  $m_{23}^*L \otimes m_{23}^*L^{-1}$  cancel,  $\text{pr}_2^*L^{-1} \otimes \text{pr}_2^*L$  cancel, and  $\text{pr}_3^*L^{-1} \otimes \text{pr}_3^*L$  cancel, leaving the trivial bundle. The same cancellation, by the symmetry of  $\Theta$  in the three coordinates, shows the restriction to  $A \times \{e\} \times A$  and to  $A \times A \times \{e\}$  is trivial as well.

*Step 2: Conclude.* All three coordinate-slice restrictions of  $\Theta$  are trivial, so by theorem 8.1.1 the bundle  $\Theta$  is trivial on  $A \times A \times A$ . Pulling back along  $(f, g, h)$  gives the triviality of the displayed seven-fold bundle on  $T$ , completing the proof.

The alternating pattern of exponents is the reason for the name: the seven signs  $+, -, -, -, +, +, +$  attached to the three-, two-, and one-fold sums are the vertices of a cube with the empty sum omitted, and the relation says that this alternating combination of pullbacks cancels to the trivial bundle. Read as a statement about the class of  $L$  in  $\text{Pic}^0$  or in  $\text{Pic}(A)$ , it is a bilinearity: the failure of  $L$  to be additive under the group law is itself additive. Specializing  $h$  to a constant map recovers the theorem of the square, and specializing all three to multiplication maps recovers the  $[n]$ -pullback formula, both taken up in the following sections.

The relation is most transparent in the genus-one case, where line bundles are divisors and the group law is addition of points on the curve.

**Example 8.1.3: The Cube Relation on an Elliptic Curve**

Let  $E$  be an elliptic curve over  $k$  with origin  $O$ , and take  $L = \mathcal{O}_E(O)$ , the line bundle of the degree-one divisor  $[O]$ . On  $E$  a line bundle of degree zero is trivial exactly when its associated divisor class is zero, and the group law on  $E$  is Abel's theorem in the form

$$[P] - [O] \sim [Q] - [O] \iff P = Q, \quad [P + Q] - [O] \sim [P] + [Q] - 2[O]$$

the second relation expressing that  $P \mapsto [\mathcal{O}_E(P - O)]$  is the group isomorphism  $E \xrightarrow{\sim} \text{Pic}^0(E)$  of *Elliptic Curves* [4]. Take  $T = E \times E \times E$  and  $f, g, h$  the three projections, sending a triple  $(P, Q, R)$  to  $P, Q, R$ . Then proposition 8.1.2 asserts the triviality of the bundle whose divisor class at the point  $(P, Q, R)$  is the alternating sum

$$[P + Q + R] - [P + Q] - [Q + R] - [P + R] + [P] + [Q] + [R]$$

where each bracket denotes the corresponding translate of the divisor  $[O]$  pulled back by the relevant sum, that is,  $(f + g + h)^*[O]$  is the divisor  $\{P + Q + R = O\}$  and so on. Before invoking the point-sum isomorphism one must know the class is degree zero on each coordinate fibre, since  $\text{Pic}^0(E) \cong E$  classifies only degree-zero classes. Restricting to a fibre in the  $P$ -direction (fixing  $Q, R$ ), the seven translates that meet it have degrees  $+1, -1, 0, -1, +1$  (the terms  $[P + Q + R]$ ,  $[P + Q]$ ,  $[Q + R]$ ,  $[P + R]$ ,  $[P]$ ; the constant  $[Q]$  and  $[R]$  do not vary with  $P$ ), summing to  $+1 - 1 - 0 - 1 + 1 = 0$ ; the same holds in the  $Q$ - and  $R$ -directions by symmetry, so the class is fibrewise degree zero and the isomorphism applies. Evaluating the class in  $\text{Pic}^0(E) \cong E$  by the isomorphism above, the seven translates sum to

$$(P + Q + R) - (P + Q) - (Q + R) - (P + R) + P + Q + R = O$$

in the group  $E$ . The coefficient of  $P$  tallies to zero: it appears with sign  $+1$  in the three-fold sum  $P + Q + R$ , with sign  $-1$  in each of the two two-fold sums  $P + Q$  and  $P + R$  that contain it, with sign  $0$  in  $Q + R$ , and with sign  $+1$  in the one-fold sum  $P$ , giving  $1 - 1 - 1 + 1 = 0$ ; symmetry gives the same for  $Q$  and  $R$ . The alternating combination is therefore the identity of  $\text{Pic}^0(E)$ , so the seven-fold bundle is trivial, exactly as the cube predicts. For  $g = 1$  the theorem of the cube reduces to the Abel theorem identifying the degree-zero divisor class group with the curve itself; the higher-dimensional theorem is its replacement when no such explicit divisor calculus is available.

**Exercise 8.1.1**

1. State the three slice-triviality hypotheses of the theorem of the cube (theorem 8.1.1) explicitly for a line bundle  $L$  on  $X \times Y \times Z$  with base point  $(x_0, y_0, z_0)$ , and identify which coordinate slice is used at which step of the proof (the descent to the base, and the two axes of the locus  $T$ ). Then explain in one sentence why the conclusion fails if only the hypotheses on  $L|_{X \times Y \times \{z_0\}}$  and  $L|_{X \times \{y_0\} \times Z}$  are assumed.
2. In the proof of proposition 8.1.2, the bundle  $\Theta$  on  $A \times A \times A$  was shown trivial on the slice  $\{e\} \times A \times A$ . Carry out the analogous computation on the slice  $A \times A \times \{e\}$ : restrict each of the seven factors of  $\Theta$  to this slice and verify that they cancel to the trivial bundle.

3. Let  $A$  be an abelian variety over  $k$ , write  $t_a: A \rightarrow A$  for translation by a point  $a \in A(\bar{k})$ , and let  $L$  be a line bundle on  $A$ . Deduce the theorem of the square

$$t_{x+y}^* L \otimes L \cong t_x^* L \otimes t_y^* L$$

from proposition 8.1.2 by specializing *two* of the three morphisms to constants: take  $T = A$ ,  $f = \text{id}_A$ , and  $g \equiv x$ ,  $h \equiv y$  the constant maps at  $x$  and  $y$ . Identify each of the seven factors as a translate or a constant pullback, and explain why setting only one morphism to a constant does *not* produce the square: with  $h \equiv e$  alone the seven-fold bundle degenerately cancels to  $e^* L$ , yielding no relation among translates.

4. Take  $E$  an elliptic curve with origin  $O$ ,  $L = \mathcal{O}_E(O)$ , and specialize example 8.1.3 to  $f = g = h = \text{id}_E$  (so  $T = E$  and each morphism is the identity). Write down the seven-fold divisor class the cube produces as a function of  $P \in E$ , compute its degree, and evaluate it in  $\text{Pic}^0(E) \cong E$ ; reconcile the outcome with the pullback law  $[n]^* L \cong L^{\otimes n^2}$  at  $n = 2$ .
5. Prove that on a product  $X \times Y$  of two complete geometrically connected varieties over an algebraically closed field, a line bundle  $L$  trivial on both axes  $\{x_0\} \times Y$  and  $X \times \{y_0\}$  through a base point  $(x_0, y_0)$  need not be trivial, yet becomes trivial as soon as, in addition to the axis triviality on  $\{x_0\} \times Y$ , its restriction to each fibre  $X \times \{y\}$  is trivial. Prove the second statement using the seesaw principle (theorem 7.4.5), noting where the axis hypothesis on  $\{x_0\} \times Y$  is used, and cite the section's own witness to the failure of the first.
6. Show that the bundle  $\Theta$  defined in the proof of proposition 8.1.2 is symmetric under any permutation of the three factors of  $A \times A \times A$ , and use this symmetry together with proposition 8.1.2 to prove that  $L$  and  $L^{-1}$  produce the same cube relation up to inversion, that is, that the cube bundle for  $L^{-1}$  is the inverse of the cube bundle for  $L$ .

## 8.2 The Theorem of the Square

The theorem of the cube governs a line bundle on a triple product; its first payoff is a statement about a single abelian variety. Specialize the three maps of the cube to the identity and a pair of translations on one  $X$ , and the trilinear identity collapses to a bilinear one: the pullback of  $L$  along a sum of two translations is determined by the two translations separately. This is the theorem of the square. It is the identity that turns the set of translates of  $L$  into a group and produces the homomorphism  $\varphi_L$  studied in section 8.3.

Throughout,  $X$  is an abelian variety over  $k$  with  $\bar{k}$  an algebraic closure, and for a point  $a \in X(\bar{k})$  we write  $t_a: X \rightarrow X$  for translation by  $a$ , so  $t_a(x) = x + a$ . The theorem of the cube in its morphism form (proposition 8.1.2) is stated for morphisms  $f, g, h: T \rightarrow X$  from an arbitrary variety  $T$ ; taking  $T = X$  and choosing the three morphisms among the identity and constant maps is what produces the square.

**Theorem 8.2.1: Theorem of the Square**

Let  $X$  be an abelian variety over  $k$ , let  $L \in \text{Pic}(X)$ , and let  $x, y \in X(\bar{k})$ . Then

$$t_{x+y}^* L \otimes L \cong t_x^* L \otimes t_y^* L$$

of line bundles on  $X_{\bar{k}}$ .

**Proof:**

Work over  $\bar{k}$ , where the points  $x, y$  are morphisms  $\text{Spec } \bar{k} \rightarrow X$ . Apply the morphism form of the theorem of the cube (proposition 8.1.2) to the three morphisms

$$f = \text{id}_X, \quad g = c_x, \quad h = c_y$$

from  $T = X_{\bar{k}}$  to  $X$ , where  $c_x, c_y: X \rightarrow X$  denote the constant morphisms at the points  $x$  and  $y$ . For a constant morphism  $c_a$  and any line bundle  $M$ , the pullback  $c_a^* M$  is the trivial bundle  $\mathcal{O}_X$  (a constant map factors through a point, and every line bundle on a point is trivial). The morphism form asserts that

$$(f+g+h)^* L \otimes (f+g)^* L^{-1} \otimes (g+h)^* L^{-1} \otimes (f+h)^* L^{-1} \otimes f^* L \otimes g^* L \otimes h^* L$$

is trivial. Evaluate each of the seven factors on a point of  $X$ , reading  $f + g + h$  and the pairwise sums as morphisms  $X \rightarrow X$ :

- $f + g + h$  sends  $z \mapsto z + x + y$ , so it is  $t_{x+y}$  and  $(f+g+h)^* L = t_{x+y}^* L$ ;
- $f + g$  sends  $z \mapsto z + x$ , so it is  $t_x$  and  $(f+g)^* L = t_x^* L$ ; likewise  $f + h = t_y$  gives  $(f+h)^* L = t_y^* L$ ;
- $g + h$  sends  $z \mapsto x + y$ , the constant morphism at  $x + y$ , so  $(g+h)^* L = \mathcal{O}_X$ ;
- $f = \text{id}_X$  gives  $f^* L = L$ ; and  $g = c_x, h = c_y$  are constant, so  $g^* L = h^* L = \mathcal{O}_X$ .

Substituting these into the seven-fold product, the trivial factors drop out and the triviality reads

$$t_{x+y}^* L \otimes t_x^* L^{-1} \otimes \mathcal{O}_X \otimes t_y^* L^{-1} \otimes L \otimes \mathcal{O}_X \otimes \mathcal{O}_X \cong \mathcal{O}_X$$

Rearranging the tensor product,  $t_{x+y}^* L \otimes L \cong t_x^* L \otimes t_y^* L$ , as we sought to show.

Setting  $y = -x$  recovers a sanity check: the left side becomes  $t_0^* L \otimes L = L^{\otimes 2}$ , while the right is  $t_x^* L \otimes t_{-x}^* L$ , so the two translates  $t_x^* L$  and  $t_{-x}^* L$  are inverse to one another modulo  $L^{\otimes 2}$ . More usefully, the theorem says that the assignment  $x \mapsto t_x^* L$  fails to be a homomorphism into  $\text{Pic}(X)$  by exactly the constant factor  $L$ : it is  $L$  that reappears on the left, obstructing additivity. The correct additive object is the class  $t_x^* L \otimes L^{-1}$ , which strips that constant away, and section 8.3 runs on the resulting corollary.

The theorem of the square is often stated without the ambient theorem of the cube, as a fact about how a fixed divisor moves under translation. If  $D$  is a divisor with  $L = \mathcal{O}_X(D)$ , then linear equivalence of divisors  $D_{x+y} + D \sim D_x + D_y$  (writing  $D_a$  for the translate  $D + a$ ) is exactly the statement above; the cube supplies the uniform reason.

**Corollary 8.2.2: The Translation Homomorphism**  $x \mapsto t_x^* L \otimes L^{-1}$ 

Let  $L \in \text{Pic}(X)$ . The map

$$\varphi_L: X(\bar{k}) \longrightarrow \text{Pic}^0(X)(\bar{k}), \quad x \longmapsto [t_x^* L \otimes L^{-1}]$$

is a well-defined group homomorphism. In particular, for every  $x \in X(\bar{k})$  the class  $t_x^* L \otimes L^{-1}$  is algebraically trivial.

**Proof:**

*Step 1: the homomorphism property.* For  $x, y \in X(\bar{k})$ , the theorem of the square (theorem 8.2.1) gives  $t_{x+y}^* L \otimes L \cong t_x^* L \otimes t_y^* L$ . Tensor both sides with  $L^{-2}$ :

$$t_{x+y}^* L \otimes L^{-1} \cong (t_x^* L \otimes L^{-1}) \otimes (t_y^* L \otimes L^{-1})$$

which in  $\text{Pic}(X)$  reads  $\varphi_L(x+y) = \varphi_L(x) + \varphi_L(y)$ . Thus  $\varphi_L$  is a homomorphism of abstract groups (once the target is known to contain the image, established next); note  $\varphi_L(0) = [t_0^* L \otimes L^{-1}] = [\mathcal{O}_X] = 0$  is forced.

*Step 2: the image lands in  $\text{Pic}^0$ .* The class of  $t_x^* L \otimes L^{-1}$  must be shown algebraically trivial, that is, in the identity component  $\text{Pic}^0(X)$  (definition 7.4.3). Consider the difference bundle

$$M := m^* L \otimes \text{pr}_1^* L^{-1} \otimes \text{pr}_2^* L^{-1}$$

on  $X \times X$ , where  $m: X \times X \rightarrow X$  is the group law and  $\text{pr}_1, \text{pr}_2$  are the projections. Its restriction to the slice  $X \times \{x\}$  is, along the identification  $X \cong X \times \{x\}$ , the bundle  $(m|_{X \times \{x\}})^* L \otimes L^{-1} \otimes (\text{constant}) = t_x^* L \otimes L^{-1}$ , since  $m(z, x) = z + x = t_x(z)$  and  $\text{pr}_2^* L^{-1}$  restricts to a trivial bundle on the slice. As  $x$  varies over  $X(\bar{k})$ , these restrictions are the fibres of the single family  $M$  over the second factor. Restricting  $M$  to  $X \times \{0\}$  gives  $t_0^* L \otimes L^{-1} = \mathcal{O}_X$ , the trivial bundle. The members of an algebraic family of line bundles on  $X$  parametrized by the connected base  $X$  all lie in a single connected component of  $\text{Pic}(X)$ ; since the member over 0 is trivial, that component is the identity component  $\text{Pic}^0(X)$ , and every  $t_x^* L \otimes L^{-1}$  lies in it. Hence  $\varphi_L(x) \in \text{Pic}^0(X)(\bar{k})$  for all  $x$ , completing the proof.

The corollary is the algebraic incarnation of a map the analytic theory already produced: over  $\mathbb{C}$  the assignment  $x \mapsto t_x^* L \otimes L^{-1}$  is the polarization homomorphism  $\varphi_L: X \rightarrow \hat{X}$  built from the Hermitian form (definition 5.3.1), and the two agree under the comparison of the two Parts. Here nothing analytic is available: the theorem of the square, and behind it the theorem of the cube, is what forces the map to be additive, and the family  $M$  is what forces its values to be algebraically trivial. That  $\varphi_L$  is not just a set map but the  $\bar{k}$ -points of a morphism of group schemes  $X \rightarrow \text{Pic}^0(X)$  is taken up in section 8.3, where its kernel  $K(L)$  enters.

Two features deserve emphasis. First,  $\varphi_L$  depends on  $L$  only through its class modulo  $\text{Pic}^0(X)$ : if  $L' = L \otimes N$  with  $N \in \text{Pic}^0(X)$ , then  $t_x^* N \otimes N^{-1}$  is trivial (a translation-invariant bundle, since  $N$  is already in the identity component), so  $\varphi_{L'} = \varphi_L$ . Thus  $\varphi_L$  depends only on the class of  $L$  in the Néron-Severi group. Second, when  $L$  is trivial the map  $\varphi_L$  is identically zero, so  $\varphi_L$  measures the failure of  $L$  to be translation-invariant.



**Corollary 8.2.3: Translates and Tensor Powers**

Let  $L \in \text{Pic}(X)$ , let  $x \in X(\bar{k})$ , and let  $n \geq 1$  be an integer. Then

$$t_{nx}^* L \otimes L^{-1} \cong (t_x^* L \otimes L^{-1})^{\otimes n}$$

and consequently, for points  $x_1, \dots, x_m \in X(\bar{k})$  with  $x_1 + \dots + x_m = 0$ ,

$$t_{x_1}^* L \otimes t_{x_2}^* L \otimes \dots \otimes t_{x_m}^* L \cong L^{\otimes m}$$

**Proof:**

The first isomorphism is  $\varphi_L(nx) = n\varphi_L(x)$ , which is immediate from corollary 8.2.2 because  $\varphi_L$  is a homomorphism and  $nx = x + \dots + x$  ( $n$  summands). For the second, apply the additivity of  $\varphi_L$  to the sum  $x_1 + \dots + x_m$ :

$$\varphi_L(x_1) + \dots + \varphi_L(x_m) = \varphi_L(x_1 + \dots + x_m) = \varphi_L(0) = 0$$

which in  $\text{Pic}(X)$  reads  $\bigotimes_{i=1}^m (t_{x_i}^* L \otimes L^{-1}) \cong \mathcal{O}_X$ . Tensoring both sides with  $L^{\otimes m}$  and cancelling gives  $\bigotimes_{i=1}^m t_{x_i}^* L \cong L^{\otimes m}$ , as we sought to show.

The second identity is the form most often invoked downstream: any finite family of translates of  $L$  whose centres sum to zero recombines into a plain tensor power of  $L$ . The special case  $m = 2$ ,  $x_2 = -x_1$ , returns the theorem of the square at  $y = -x$ ; the case where all  $x_i$  are torsion of order dividing  $m$  underlies the computation of  $[n]^* L$  and the degree of multiplication by  $n$  in section 8.4. The first identity says that along the arithmetic progression  $x, 2x, 3x, \dots$  the translation classes grow linearly, a fact that makes  $\varphi_L$  visible on cyclic subgroups.

The elliptic curve makes both the theorem and its homomorphism concrete, and reconciles them with the classical picture from *Elliptic Curves* [4].

**Example 8.2.4: The Abel-Jacobi Map on an Elliptic Curve**

Let  $E$  be an elliptic curve over  $k$  with origin  $O$ , and take  $L = \mathcal{O}_E(O)$ , the degree-one line bundle attached to the divisor  $(O)$ . For a point  $P \in E(\bar{k})$ , translation  $t_P$  sends  $O$  to  $P$ , so  $t_P^* \mathcal{O}_E(O) = \mathcal{O}_E(t_{-P}(O)) = \mathcal{O}_E((-P))$ , the bundle of the divisor  $(-P)$ . Hence the translation class of corollary 8.2.2 is

$$t_P^* L \otimes L^{-1} = \mathcal{O}_E((-P) - (O))$$

Replacing  $P$  by  $-P$  (a harmless reindexing of the map, or apply  $[-1]^*$ ), the homomorphism  $\varphi_L$  becomes

$$\varphi_L: E(\bar{k}) \longrightarrow \text{Pic}^0(E)(\bar{k}), \quad P \longmapsto [\mathcal{O}_E((P) - (O))]$$

which is the degree-zero Abel-Jacobi map: it sends a point  $P$  to the class of the divisor  $(P) - (O)$  of degree zero. That this is a group homomorphism is precisely the statement that

$$(P) - (O) + (Q) - (O) \sim (P+Q) - (O)$$

for the group law on  $E$ , that is,  $(P) + (Q) \sim (P+Q) + (O)$  under linear equivalence. This linear equivalence is the definition of the group law on an elliptic curve as recalled in *Elliptic Curves* [4]: three points sum to the origin iff they are collinear, iff their divisor sum is linearly equivalent to

3( $O$ ). So the theorem of the square, specialized to  $g = 1$ , encodes the chord-and-tangent group law, and  $\varphi_L$  is the classical isomorphism

$$E(\bar{k}) \xrightarrow{\sim} \text{Pic}^0(E)(\bar{k}) = \text{Pic}^0(E)(\bar{k}), \quad P \mapsto [(P) - (O)]$$

identifying  $E$  with its own degree-zero Picard group. Here  $\varphi_L$  is a bijection because  $L = \mathcal{O}_E(O)$  is a principal polarization: its class in  $\text{NS}(E)$  has degree one, and  $K(L)$  is trivial. The higher-genus analogue is that  $\varphi_L$  for a principally polarized  $X$  is an isomorphism onto the dual  $\hat{X}$ , the content of chapter 10.

### Exercise 8.2.1

1. Let  $L \in \text{Pic}(X)$  and  $x \in X(\bar{k})$ . Using the theorem of the square (theorem 8.2.1), show that

$$t_{-x}^* L \cong L^{\otimes 2} \otimes (t_x^* L)^{-1}$$

and deduce that the class  $[t_x^* L \otimes L^{-1}]$  and the class  $[t_{-x}^* L \otimes L^{-1}]$  are inverse to one another in  $\text{Pic}(X)$ .

2. Let  $L \in \text{Pic}(X)$  and let  $x_1, x_2, x_3 \in X(\bar{k})$  satisfy  $x_1 + x_2 + x_3 = 0$ . Using the homomorphism  $\varphi_L$  of corollary 8.2.2, prove the linear-equivalence identity

$$t_{x_1}^* L \otimes t_{x_2}^* L \otimes t_{x_3}^* L \cong L^{\otimes 3}$$

and check that specializing  $x_3 = 0$  (so  $x_2 = -x_1$ ) recovers the theorem of the square at  $y = -x$ .

3. Suppose  $N \in \text{Pic}(X)$  is *translation-invariant*:  $t_x^* N \cong N$  for every  $x \in X(\bar{k})$ . Prove that  $\varphi_N = 0$ , and use this to show that  $\varphi_L$  depends on  $L$  only through its class in  $\text{NS}(X)$ , provided one grants the characterization of  $\text{Pic}^0(X)$  as the translation-invariant classes, so that  $L$  and  $L'$  have the same Néron-Severi class exactly when they are related by a translation-invariant twist. (That every  $N \in \text{Pic}^0(X)$  is in fact translation-invariant is a genuine theorem lying beyond the four results of this section; here it is taken as the input.)
4. Let  $E$  be an elliptic curve over  $k$  with origin  $O$  and let  $L = \mathcal{O}_E(O)$ , as in example 8.2.4. Take  $k = \bar{k}$ , let  $P, Q \in E(\bar{k})$ , and set  $R = -(P + Q)$ , so that  $P + Q + R = O$  in the group law. Compute the class  $\varphi_L(P) + \varphi_L(Q) + \varphi_L(R)$  directly from the divisor description

$$\varphi_L(P) = [\mathcal{O}_E((P) - (O))]$$

and confirm that the answer is consistent with  $R = -(P + Q)$  and with  $\varphi_L$  being a homomorphism.

5. Let  $L \in \text{Pic}(X)$  and let  $x \in X(\bar{k})$  be a torsion point of order  $m$ , so  $mx = 0$ . Using corollary 8.2.3, prove that the translation class  $t_x^* L \otimes L^{-1}$  is an  $m$ -torsion element of  $\text{Pic}^0(X)$ ; that is, its  $m$ -th tensor power is trivial. Give an example on an elliptic curve  $E$  with  $L = \mathcal{O}_E(O)$  where this class is nontrivial, exhibiting a divisor representing it.
6. The theorem of the square as stated (theorem 8.2.1) is derived from the theorem of the cube by setting two of the three cube-maps to constant morphisms. Reverse the logic on a fixed abelian variety: assuming only that the assignment  $x \mapsto [t_x^* L \otimes L^{-1}]$  is a homomorphism  $X(\bar{k}) \rightarrow \text{Pic}(X)$ , recover the isomorphism  $t_{x+y}^* L \otimes L \cong t_x^* L \otimes t_y^* L$  for all  $x, y \in X(\bar{k})$ . This

shows the theorem of the square and the homomorphism property of  $\varphi_L$  are equivalent formulations.

### 8.3 The Homomorphism $\varphi_L$ and the Group $K(L)$

The theorem of the square produced, from a line bundle  $L$  on an abelian variety  $X$ , a homomorphism sending a point  $x$  to the class of  $t_x^*L \otimes L^{-1}$  in  $\text{Pic}^0(X)$  (corollary 8.2.2). This section names that homomorphism  $\varphi_L$ , records that it is a morphism of varieties and not just a map on points, and studies its kernel. The kernel is a closed subgroup scheme  $K(L)$  of  $X$ , and when  $L$  is ample  $K(L)$  is finite, which is what makes  $\varphi_L$  an isogeny onto the dual. Over  $\mathbb{C}$  this finiteness was the nondegeneracy of the Hermitian form; here it is a consequence of ampleness.

The homomorphism  $x \mapsto t_x^*L \otimes L^{-1}$  is already a map on  $\bar{k}$ -points. To call it a morphism of varieties requires that the family of line bundles  $\{t_x^*L \otimes L^{-1}\}_{x \in X}$  vary algebraically with  $x$ , which is exactly the kind of statement the Picard functor was built to express: the family is a single line bundle on  $X \times X$ , and the morphism to  $\text{Pic}^0(X)$  is the one it classifies.

#### Definition 8.3.1: The Homomorphism $\varphi_L$

Let  $X$  be an abelian variety over  $k$  and  $L$  a line bundle on  $X$ . The *Mumford bundle* of  $L$  is the line bundle

$$\mathcal{M}(L) := m^*L \otimes \text{pr}_1^*L^{-1} \otimes \text{pr}_2^*L^{-1}$$

on  $X \times X$ , where  $m: X \times X \rightarrow X$  is the group law and  $\text{pr}_1, \text{pr}_2$  are the projections. Its restriction to the fibre  $\{x\} \times X$  is  $t_x^*L \otimes L^{-1}$ , a line bundle algebraically trivial on  $X$ ; so  $\mathcal{M}(L)$  is a family of classes in  $\text{Pic}^0(X)$  parametrized by the first factor, and by the universal property of the Picard scheme it classifies a morphism

$$\varphi_L: X \longrightarrow \text{Pic}^0(X), \quad x \longmapsto t_x^*L \otimes L^{-1}$$

This  $\varphi_L$  is a homomorphism of group functors, whose induced map on  $\bar{k}$ -points is the homomorphism of corollary 8.2.2.

Two things are packaged here. First,  $\varphi_L$  is a genuine morphism of varieties, because the Mumford bundle  $\mathcal{M}(L)$  realizes the pointwise assignment  $x \mapsto t_x^*L \otimes L^{-1}$  as an algebraic family: restricting  $\mathcal{M}(L)$  to  $\{x\} \times X$  returns  $t_x^*L \otimes L^{-1}$ , since  $m|_{\{x\} \times X} = t_x$ , while  $\text{pr}_1^*L^{-1}$  restricts to the constant bundle  $L^{-1}(x)$  and  $\text{pr}_2^*L^{-1}$  restricts to  $L^{-1}$ . Each such restriction is algebraically trivial, so the family lands in  $\text{Pic}^0(X)$ , and the universal property of the Picard scheme (definition 7.4.3) converts the family into a morphism into  $\text{Pic}^0(X)$ . Second, that this morphism is a homomorphism is the theorem of the square: additivity in  $x$  was the content of corollary 8.2.2, and it persists at the level of the morphism because a morphism of varieties agreeing with a homomorphism on the dense set of  $\bar{k}$ -points is a homomorphism. The bundle  $\mathcal{M}(L)$  is symmetric in its two arguments,  $\mathcal{M}(L) \cong s^*\mathcal{M}(L)$  for the swap  $s: X \times X \rightarrow X \times X$ , which foreshadows the symmetry of  $\varphi_L$  under biduality proved in chapter 10.

This  $\varphi_L$  is the algebraic counterpart of the analytic polarization homomorphism  $\varphi_L: X \rightarrow \hat{X}$  of definition 5.3.1 in Part I, where over  $\mathbb{C}$  it was induced by the  $\mathbb{R}$ -linear map  $v \mapsto H(v, -)$  built from the Hermitian form  $H$ ; the two constructions describe the same map on the same object viewed through the two Parts, and the analytic version is not re-derived here.

The kernel of  $\varphi_L$  measures how far  $L$  is from being rigid under translation: a point  $x$  lies in it precisely when translating  $L$  by  $x$  leaves its class unchanged.

**Definition 8.3.2: The Group  $K(L)$**

Let  $L$  be a line bundle on an abelian variety  $X$  over  $k$ . The group

$$K(L) := \ker \varphi_L$$

is the kernel of the homomorphism  $\varphi_L: X \rightarrow \text{Pic}^0(X)$ , a closed subgroup scheme of  $X$ . On  $\bar{k}$ -points it is the set

$$K(L)(\bar{k}) = \{ x \in X(\bar{k}) : t_x^* L \cong L \}$$

of points whose translation fixes the isomorphism class of  $L$ .

As the kernel of a homomorphism of group schemes,  $K(L)$  is a closed subgroup scheme of  $X$ , cut out as the preimage of the identity section of  $\text{Pic}^0(X)$  under the morphism  $\varphi_L$ . Its  $\bar{k}$ -points are the  $x$  with  $t_x^* L \otimes L^{-1}$  trivial, that is, with  $t_x^* L \cong L$ : the translations that preserve  $L$ . In characteristic zero  $K(L)$  is reduced and is determined by its  $\bar{k}$ -points, but in characteristic  $p$  it may carry a nonreduced (infinitesimal) part, which is why  $K(L)$  is defined as a subgroup scheme and not just as a finite set of points. The size and shape of  $K(L)$  are the algebraic invariants that replace the type of a polarization from the analytic theory.

**Definition 8.3.3: Nondegenerate Line Bundle**

A line bundle  $L$  on an abelian variety  $X$  over  $k$  is *nondegenerate* if the group scheme  $K(L)$  is finite, equivalently if  $\varphi_L: X \rightarrow \text{Pic}^0(X)$  has finite kernel.

Nondegeneracy is the condition that  $\varphi_L$  collapses only a finite part of  $X$ , so that its image has the full dimension  $g$ . Since  $\varphi_L$  is a homomorphism of abelian varieties (its target  $\text{Pic}^0(X)$  is an abelian variety of dimension  $g$ , built in chapter 10), finiteness of the kernel is equivalent to  $\varphi_L$  being surjective onto a  $g$ -dimensional image, hence to  $\varphi_L$  being an isogeny. Degeneracy, by contrast, means  $K(L)$  contains a positive-dimensional abelian subvariety of  $X$  along which  $L$  is translation-invariant. The next result is the reason the theory is not empty: an ample line bundle is always nondegenerate.

An ample bundle should be nondegenerate because ampleness is a strong positivity, and positivity is incompatible with translation-invariance along a positive-dimensional subvariety. If some abelian subvariety  $Y \subseteq K(L)^0$  had positive dimension, then  $L$  restricted to  $Y$  would be simultaneously ample (a restriction of an ample bundle to a closed subvariety) and translation-invariant along  $Y$ ; but a translation-invariant bundle on an abelian variety has vanishing top self-intersection by the theorem of the cube, whereas an ample bundle on a positive-dimensional variety has positive top self-intersection, and no bundle can be both.

**Proposition 8.3.4: Ample Line Bundles Are Nondegenerate**

Let  $X$  be an abelian variety over  $k$  and  $L$  an ample line bundle on  $X$ . Then  $K(L)$  is finite; equivalently,  $\varphi_L: X \rightarrow \text{Pic}^0(X)$  is an isogeny.

**Proof:**

It suffices to show that the identity component  $K(L)^0$  of the group scheme  $K(L)$  is trivial. A group scheme of finite type over  $k$  is finite precisely when its identity component has dimension zero; since  $K(L)$  is a closed subgroup scheme of the abelian variety  $X$ , its identity component  $K(L)^0$  is a connected group subvariety of  $X$ , hence, working over  $\bar{k}$  where the reduction is a subgroup scheme, an abelian subvariety  $Y := K(L)_{\text{red}}^0 \subseteq X_{\bar{k}}$  (a connected proper group variety is an abelian variety, definition 7.1.2). The claim, a geometric one, is that  $\dim Y = 0$ .

*Step 1:  $L$  restricted to  $Y$  is translation-invariant.* Every  $\bar{k}$ -point  $y$  of  $Y \subseteq K(L)$  satisfies  $t_y^*L \cong L$  by definition 8.3.2, so translation by  $y$  preserves the class of  $L$  on  $X$ , and a fortiori translation by  $y$  within  $Y$  preserves the class of the restriction  $L|_Y$ : writing  $\iota: Y \hookrightarrow X$  for the inclusion and using  $\iota \circ t_y = t_y \circ \iota$  on  $Y$ , one gets  $t_y^*(L|_Y) = t_y^*\iota^*L = \iota^*t_y^*L \cong \iota^*L = L|_Y$  for every  $y \in Y(\bar{k})$ . Thus  $L|_Y$  is fixed by all translations of  $Y$ , which is to say  $\varphi_{L|_Y}: Y \rightarrow \text{Pic}^0(Y)$  is the zero map, so  $K(L|_Y) = Y$ . It is this translation-invariance, not any triviality statement, that Step 3 consumes.

*Step 2:  $L$  restricted to  $Y$  is ample.* The restriction of an ample line bundle to a closed subvariety is ample; this is a general fact of projective geometry, that the pullback of an ample bundle along a closed immersion is ample (*Algebraic Geometry* [13]). Applied to the closed immersion  $\iota: Y \hookrightarrow X$  and the ample  $L$ , the restriction  $L|_Y = \iota^*L$  is ample on  $Y$ .

*Step 3: An abelian variety of positive dimension carries no ample translation-invariant bundle.* Suppose  $\dim Y = d > 0$ , and write  $M := L|_Y$ . By Step 1 the bundle  $M$  is translation-invariant on the abelian variety  $Y$ , and the theorem of the cube (chapter 8) forces its top self-intersection to vanish. First, translation-invariance makes  $M$  additive under the group law. The Mumford bundle  $m^*M \otimes \text{pr}_1^*M^{-1} \otimes \text{pr}_2^*M^{-1}$  on  $Y \times Y$  restricts on each slice  $\{y\} \times Y$  to  $t_y^*M \otimes M^{-1}$ , trivial for every  $y$  by Step 1, so the seesaw principle (theorem 7.4.5) makes it a pullback from the first factor, and its triviality on  $Y \times \{e\}$  then makes it trivial outright; that is,  $m^*M \cong \text{pr}_1^*M \otimes \text{pr}_2^*M$ . Pulling this isomorphism back along  $(\text{id}, [-1]): Y \rightarrow Y \times Y$ ,  $y \mapsto (y, -y)$ , whose composite with  $m$  is the constant map to  $e$ , gives  $(-1)^*M \cong M^{-1}$ . Feeding this anti-symmetry into the general multiplication-by- $n$  pullback formula (established in the next section, proposition 8.4.3) collapses it to  $[n]^*M \cong M^{\otimes(n^2+n)/2} \otimes (-1)^*M^{\otimes(n^2-n)/2} \cong M^{\otimes n}$  for every  $n$ . Comparing top self-intersection numbers across the finite surjection  $[n]: Y \rightarrow Y$ , of degree  $n^{2d}$  (also theorem 8.4.5, whose proof rests on the ample-bundle self-intersection formula and not on this proposition, so the appeal is non-circular), yields

$$n^{2d} (M^d) = \deg[n] \cdot (M^d) = ([n]^*M)^d = (M^{\otimes n})^d = n^d (M^d)$$

so  $(n^{2d} - n^d)(M^d) = 0$  for all  $n$ ; taking  $n = 2$ , where  $d > 0$  makes  $2^{2d} \neq 2^d$ , forces  $(M^d) = 0$ . But by Step 2 the bundle  $M$  is ample on the projective variety  $Y$  of dimension  $d > 0$ , so its top self-intersection is strictly positive,  $(M^d) > 0$  (the degree of  $Y$  under the projective embedding furnished by a power of  $M$ ). These two facts contradict one another, so the assumption  $\dim Y > 0$  is impossible.

Therefore  $\dim Y = 0$ , so  $K(L)^0$  is zero-dimensional and  $K(L)$  is finite. A homomorphism  $\varphi_L: X \rightarrow \text{Pic}^0(X)$  of abelian varieties of the same dimension  $g$  with finite kernel is surjective and finite, hence an isogeny, completing the proof.

The proposition is the crossing point at which the algebraic theory catches up with the analytic. Over  $\mathbb{C}$  a polarization was a positive Hermitian form, its nondegeneracy was manifest, and  $\varphi_L$  was an isogeny of degree  $\det E = |K(L)|$  (proposition 5.3.4); here positivity of the Hermitian form is replaced by ampleness of the bundle, and the same conclusion, that  $\varphi_L$  is an isogeny with finite kernel  $K(L)$ , is recovered by a purely algebraic intersection-theoretic argument valid over any field. The finiteness of  $K(L)$  for ample  $L$  is what makes the dual abelian variety accessible: it guarantees that  $\varphi_L$  has an image of the full dimension  $g$ , the first step toward proving that  $\text{Pic}^0(X)$  is itself an abelian variety.

Two structural remarks record how  $\varphi_L$  and  $K(L)$  depend on  $L$ . The construction is additive: since  $t_x^*(L \otimes L') = t_x^*L \otimes t_x^*L'$ , one has  $\varphi_{L \otimes L'} = \varphi_L + \varphi_{L'}$  as homomorphisms  $X \rightarrow \text{Pic}^0(X)$ , so  $L \mapsto \varphi_L$  is a homomorphism from  $\text{Pic}(X)$  to the group of homomorphisms  $X \rightarrow \text{Pic}^0(X)$ . Its kernel is  $\text{Pic}^0(X)$  itself: a bundle  $L \in \text{Pic}^0(X)$  is translation-invariant, so  $\varphi_L = 0$ , because an algebraically trivial bundle satisfies  $t_x^*L \cong L$  for all  $x$ . Consequently  $\varphi_L$ , and with it  $K(L)$ , depends only on the class of  $L$  in  $\text{NS}(X) = \text{Pic}(X)/\text{Pic}^0(X)$  (definition 10.5.1). The homomorphism  $L \mapsto \varphi_L$  thus descends to an injection of  $\text{NS}(X)$  into the homomorphisms  $X \rightarrow \text{Pic}^0(X)$ , the algebraic incarnation of the Hermitian-forms description of  $\text{NS}(X)$  from the analytic theory.

For  $L$  ample, chapter 10 promotes  $\varphi_L$  from a homomorphism into the group  $\text{Pic}^0(X)$  to a morphism into the dual variety  $\hat{X} = \text{Pic}_{X/k}^0$ , an isogeny of degree  $|K(L)|$ ; the finiteness proved here is what licenses that promotion, and the group scheme  $K(L)$  becomes the kernel of that isogeny, self-dual under the Weil pairing.

#### Example 8.3.5: $K(L)$ for a Line Bundle on an Elliptic Curve

Let  $E$  be an elliptic curve over  $k$  and  $L$  a line bundle of degree  $d$  on  $E$ . Under the autoduality  $E \xrightarrow{\sim} \text{Pic}^0(E)$  (example 7.4.4) the homomorphism  $\varphi_L: E \rightarrow \text{Pic}^0(E) \cong E$  is multiplication by the degree,

$$\varphi_L = [d]: E \longrightarrow E$$

To see this, take first  $L = \mathcal{O}_E(O)$  of degree one, where  $O$  is the origin. For  $x \in E$  the identity

$$t_x^*\mathcal{O}_E(O) \otimes \mathcal{O}_E(O)^{-1} = \mathcal{O}_E([-x]) \otimes \mathcal{O}_E([O])^{-1} = \mathcal{O}_E([-x] - [O])$$

holds, since translation by  $x$  moves the divisor  $[O]$  to  $[-x]$  (a point  $p$  satisfies  $p + x = O$  exactly when  $p = -x$ ); and under the autoduality identification  $E \xrightarrow{\sim} \text{Pic}^0(E)$ ,  $P \mapsto [\mathcal{O}_E([O] - [P])]$ , the class  $[\mathcal{O}_E([-x] - [O])] = [\mathcal{O}_E([O] - [x])]$  corresponds to the point  $x$ . So on the degree-one bundle  $\varphi_L$  is  $x \mapsto x = [1]$ , and a general degree- $d$  bundle is  $\mathcal{O}_E(O)^{\otimes d}$  modulo  $\text{Pic}^0(E)$ , on which  $\varphi_L = d \cdot \varphi_{\mathcal{O}_E(O)} = [d]$  by additivity. The kernel is therefore the  $d$ -torsion subgroup scheme,

$$K(L) = \ker[d] = E[d]$$

of order  $d^2$  when  $d$  is prime to  $\text{char } k$  (and  $|K(L)| = \deg[d] = d^2$  as group schemes in every characteristic). Thus  $L$  is nondegenerate exactly when  $d \neq 0$ , and ample exactly when  $d > 0$ , matching proposition 8.3.4: an ample bundle on  $E$  has  $d > 0$ , hence finite  $K(L) = E[d]$ . The group law and the  $d$ -torsion of  $E$  are developed for  $g = 1$  in *Elliptic Curves* [4], and  $|K(L)| = d^2$  reproduces there the general formula  $|K(L)| = \chi(L)^2$  specialized to  $\chi(L) = \deg L = d$  on a curve.

#### Exercise 8.3.1

1. Let  $X$  be an abelian variety and  $L$  a line bundle on  $X$ . Using the definition of the Mumford

- bundle in definition 8.3.1, verify directly that the restriction of  $\mathcal{M}(L)$  to the fibre  $\{x\} \times X$  is  $t_x^*L \otimes L^{-1}$ , and to the fibre  $X \times \{e\}$  (where  $e$  is the identity) is trivial. Deduce that  $\varphi_L(e) = 0$ .
2. Let  $E$  be an elliptic curve and  $L$  a line bundle of degree  $d$  on  $E$ . Using example 8.3.5, determine  $K(L)$  and its order for  $d = 3$ , and for  $d = 0$ . In each case state whether  $L$  is nondegenerate in the sense of definition 8.3.3, and whether it can be ample.
  3. Prove that the map  $L \mapsto \varphi_L$  is additive: for line bundles  $L, L'$  on an abelian variety  $X$  one has  $\varphi_{L \otimes L'} = \varphi_L + \varphi_{L'}$  as homomorphisms  $X \rightarrow \text{Pic}^0(X)$ . Deduce that if  $L \in \text{Pic}^0(X)$  then  $\varphi_L = 0$ , and hence that  $\varphi_L$  depends only on the class of  $L$  in  $\text{NS}(X)$ .
  4. Let  $L$  be an ample line bundle on an abelian variety  $X$ , and let  $Y \subseteq X$  be an abelian subvariety with  $Y \subseteq K(L)(\bar{k})$ . Using the definitions in definition 8.3.2 and definition 8.3.3, show that  $Y$  is a single point. State which result of this section this specializes.
  5. Let  $L$  be an ample line bundle on an abelian variety  $X$  of dimension  $g$ , and let  $n \geq 1$  be an integer prime to  $\text{char } k$ . Consider the bundle  $L^{\otimes n}$ . Show that  $K(L) \subseteq K(L^{\otimes n})$ , and that  $L^{\otimes n}$  is again nondegenerate. Then, for  $E$  an elliptic curve and  $L = \mathcal{O}_E(O)$  of degree one, compute  $K(L^{\otimes n})$  explicitly and compare with  $K(L)$ , using example 8.3.5.
  6. Let  $L$  be a nondegenerate line bundle on an abelian variety  $X$ , and let  $M \in \text{Pic}^0(X)$ . Prove that  $L$  and  $L \otimes M$  have the same homomorphism,  $\varphi_{L \otimes M} = \varphi_L$ , and hence  $K(L \otimes M) = K(L)$  as group schemes. Explain, using the description of  $\varphi_L$  as an isogeny in proposition 8.3.4, why this shows nondegeneracy of an ample  $L$  is unaffected by twisting by an algebraically trivial bundle.

## 8.4 Multiplication by $n$ and Its Degree

The homomorphism  $\varphi_L$  of section 8.3 measured how a line bundle moves under translation by a point. The endomorphisms  $[n]$  ask a different question: how does a line bundle behave when the whole variety is dilated by an integer? The answer is the pullback formula  $[n]^*L \cong L^{\otimes n^2}$  for symmetric  $L$ , and from it drops the degree of  $[n]$ , the size of the torsion  $X[n]$ , and the algebraic proof that  $\deg[n] = n^{2g}$ . Over  $\mathbb{C}$  this count was immediate from the lattice (proposition 5.2.4:  $X[n] = \frac{1}{n}\Lambda/\Lambda$ ); here there is no lattice to reduce, and the cube supplies what the lattice gave for free.

### Definition 8.4.1: Multiplication by $n$

Let  $X$  be an abelian variety over  $k$  and  $n \in \mathbb{Z}$ . The *multiplication-by- $n$*  map is the endomorphism

$$[n]: X \longrightarrow X, \quad x \longmapsto \underbrace{x + \cdots + x}_n$$

for  $n \geq 1$ , with  $[0]$  the constant map to the identity  $e$ ,  $[-n] = (-1) \circ [n]$ , and  $[1] = \text{id}_X$ . Each  $[n]$  is a homomorphism of abelian varieties, and  $[m] \circ [n] = [mn]$ ,  $[m] + [n] = [m+n]$  inside  $\text{End}(X)$ .

That  $[n]$  is a homomorphism is immediate from commutativity of the group law (corollary 7.2.4): the diagonal  $x \mapsto (x, x)$  followed by the  $n$ -fold addition is a composite of homomorphisms. The map  $[-1] = \iota$  is the inversion of the group variety, and the sign  $(-1)^*L$  of a line bundle will play a role



throughout, because a line bundle need not be invariant under inversion.

**Definition 8.4.2: Symmetric Line Bundle**

A line bundle  $L$  on an abelian variety  $X$  is *symmetric* if  $(-1)^*L \cong L$ , where  $(-1) = \iota: X \rightarrow X$  is the inversion. It is *antisymmetric* if  $(-1)^*L \cong L^{-1}$ .

Every line bundle decomposes into a symmetric and an antisymmetric part after passing to a suitable power, and the two behave differently under  $[n]$ : this is the reason the pullback formula has two cases. A polarization can always be replaced by a symmetric one, so the symmetric case is the one that carries the geometry, and it is the one to establish first.

The single computation that drives the section is the effect of  $[n]$  on  $\text{Pic}(X)$ . It follows from the theorem of the cube (equivalently, from iterating the theorem of the square, theorem 8.2.1), applied to the three maps  $[n]$ ,  $[1]$ ,  $[-1]$  from  $X$  to itself.

**Proposition 8.4.3: The Pullback of a Line Bundle under  $[n]$**

Let  $X$  be an abelian variety over  $k$  and  $L \in \text{Pic}(X)$ . For every  $n \in \mathbb{Z}$ ,

$$[n]^*L \cong L^{\otimes(n^2+n)/2} \otimes (-1)^*L^{\otimes(n^2-n)/2}$$

In particular, if  $L$  is symmetric ( $(-1)^*L \cong L$ ), then

$$[n]^*L \cong L^{\otimes n^2}$$

and if  $L$  is antisymmetric ( $(-1)^*L \cong L^{-1}$ ), then  $[n]^*L \cong L^{\otimes n}$ .

**Proof:**

*Step 1: The cube recursion.* Apply the morphism form of the theorem of the cube (proposition 8.1.2) to the three homomorphisms  $f = [n]$ ,  $g = [1]$ ,  $h = [-1]$  from  $X$  to  $X$ . The seven translates that appear are the maps

$$f + g + h = [n], \quad f + g = [n + 1], \quad g + h = [0], \quad f + h = [n - 1]$$

together with  $f = [n]$ ,  $g = [1]$ ,  $h = [-1]$ . Proposition 8.1.2 asserts that

$$[n]^*L \otimes [n + 1]^*L^{-1} \otimes [0]^*L^{-1} \otimes [n - 1]^*L^{-1} \otimes [n]^*L \otimes [1]^*L \otimes [-1]^*L$$

is trivial. Now  $[0]^*L \cong \mathcal{O}_X$  (the map  $[0]$  is constant, so its pullback of any bundle is trivial),  $[1]^*L = L$ , and  $[-1]^*L = (-1)^*L$ . Rearranging the triviality into an isomorphism yields the recursion

$$[n + 1]^*L \cong [n]^*L^{\otimes 2} \otimes [n - 1]^*L^{-1} \otimes L \otimes (-1)^*L$$

valid for every  $n \in \mathbb{Z}$ .

*Step 2: Solving the recursion.* Write  $M_n := [n]^*L$ , so that the recursion of Step 1 reads

$$M_{n+1} \cong M_n^{\otimes 2} \otimes M_{n-1}^{-1} \otimes L \otimes (-1)^*L$$



with base cases  $M_0 = [0]^*L \cong \mathcal{O}_X$  and  $M_1 = L$ . Let  $N_n := L^{\otimes(n^2+n)/2} \otimes (-1)^*L^{\otimes(n^2-n)/2}$ ; the exponents  $(n^2 \pm n)/2$  are integers, being  $\binom{n+1}{2}$  and  $\binom{n}{2}$ . Then  $N_0 = \mathcal{O}_X$  and  $N_1 = L$ , matching the base cases, and  $N_n$  satisfies the same recursion. To see this, compare exponents. The exponent of  $L$  in  $N_{n+1}$  is  $\frac{(n+1)^2 + (n+1)}{2}$ , and the second difference of the quadratic  $n \mapsto \frac{n^2+n}{2}$  is its leading coefficient doubled,

$$\frac{(n+1)^2 + (n+1)}{2} - 2 \cdot \frac{n^2 + n}{2} + \frac{(n-1)^2 + (n-1)}{2} = 1$$

so the  $L$ -exponent of  $N_{n+1}$  equals twice that of  $N_n$ , minus that of  $N_{n-1}$ , plus 1, which is exactly the exponent produced on the right-hand side of the recursion (the  $M_n^{\otimes 2}$ , the  $M_{n-1}^{-1}$ , and the single  $L$ ). The exponent of  $(-1)^*L$  obeys the identical identity, with  $\frac{n^2-n}{2}$  in place of  $\frac{n^2+n}{2}$  and the single  $(-1)^*L$  supplying the +1. Two solutions of a second-order linear recursion that agree at  $n = 0$  and  $n = 1$  agree for all  $n \in \mathbb{Z}$ , by induction upward and downward, so  $M_n \cong N_n$ , which is the asserted formula.

*Step 3: The symmetric and antisymmetric cases.* If  $(-1)^*L \cong L$ , the two exponents add:  $(n^2 + n)/2 + (n^2 - n)/2 = n^2$ , so  $[n]^*L \cong L^{\otimes n^2}$ . If  $(-1)^*L \cong L^{-1}$ , they subtract:  $(n^2 + n)/2 - (n^2 - n)/2 = n$ , so  $[n]^*L \cong L^{\otimes n}$ , completing the proof.

The formula is the algebraic engine of the section, and its two faces are worth separating. For a symmetric bundle the exponent is  $n^2$ : pulling back under  $[n]$  multiplies the class of  $L$  in the Néron-Severi group by  $n^2$ , exactly as an alternating form  $E$  scales to  $n^2E$  under the dilation  $\lambda \mapsto n\lambda$  of a lattice. This quadratic growth is the source of the degree  $n^{2g}$ : an ample symmetric  $L$  has  $[n]^*L = L^{\otimes n^2}$  ample, and the Euler characteristic, which scales as the  $g$ -th power of the class, picks up  $(n^2)^g = n^{2g}$ . For an antisymmetric bundle the exponent is only  $n$ , reflecting that  $\text{Pic}^0$  classes (the antisymmetric part, since  $(-1)^*$  acts as  $-1$  on  $\text{Pic}^0$ ) are pulled back linearly. The decomposition into a symmetric part carrying  $L^2$  and an antisymmetric part carrying  $L \otimes (-1)^*L^{-1}$  is precisely the splitting  $\text{Pic}(X) \otimes \mathbb{Q} = \text{NS}(X)_{\mathbb{Q}} \oplus \text{Pic}^0(X)_{\mathbb{Q}}$  under the involution  $(-1)^*$ .

**8.4.4 Note: symmetrizing a line bundle** For any  $L$ , the bundle  $L \otimes (-1)^*L$  is symmetric, and  $L \otimes (-1)^*L^{-1}$  is antisymmetric; their product is  $L^{\otimes 2}$ . Thus  $L^{\otimes 2}$  is the product of a symmetric and an antisymmetric bundle, and any  $L$  becomes symmetric after tensoring with a suitable degree-zero bundle: if  $L$  is ample, so is  $L \otimes (-1)^*L$  (a tensor product of two ample bundles), and it is symmetric. This is the replacement used below to reduce the degree computation to the symmetric case.

With the pullback formula in hand, the degree of  $[n]$  is a Riemann-Roch computation. The strategy is the one the note records: choose an ample symmetric  $L$ , so that  $[n]^*L = L^{\otimes n^2}$  is again ample; this forces  $[n]$  to be finite and surjective, hence an isogeny, and its degree is read off by comparing Euler characteristics.

**Theorem 8.4.5: The Degree of Multiplication by  $n$** 

Let  $X$  be an abelian variety of dimension  $g$  over  $k$ , and let  $n \geq 1$ . Then  $[n]: X \rightarrow X$  is an isogeny of degree

$$\deg[n] = n^{2g}$$

In particular  $[n]$  is surjective with finite kernel  $X[n]$ , and  $[n]$  is nonzero in  $\text{End}(X)$  for  $n \neq 0$ .

**Proof:**

*Step 1:  $[n]$  is an isogeny.* By theorem 7.3.3,  $X$  carries an ample line bundle  $L_0$ ; replacing  $L_0$  by  $L := L_0 \otimes (-1)^* L_0$  produces an ample symmetric line bundle (box 8.4.4). By proposition 8.4.3,

$$[n]^* L \cong L^{\otimes n^2}$$

which is ample, being a positive tensor power of an ample bundle. A morphism  $f: X \rightarrow X$  with  $f^*(\text{ample})$  ample is finite: if some positive-dimensional subvariety  $Z$  were contracted by  $[n]$  to a point, then  $[n]^* L$  would restrict to a trivial (hence non-ample) bundle on  $Z$ , contradicting ampleness of  $[n]^* L = L^{\otimes n^2}$ . A finite morphism of projective varieties of the same dimension is surjective, so  $[n]$  is a surjective homomorphism with finite kernel, that is, an isogeny.

*Step 2: Euler characteristics scale.* For a finite flat morphism  $f: X \rightarrow Y$  of degree  $d$  and any line bundle  $M$  on  $Y$ ,

$$\chi(f^* M) = d \cdot \chi(M)$$

The map  $[n]$  is finite by Step 1 and flat, being a finite morphism between smooth varieties of the same dimension. This finite-flat identity  $\chi(f^* M) = (\deg f) \chi(M)$  is proved in *Algebraic Geometry* [13]. Applying it to  $f = [n]$  and  $M = L$ ,

$$\chi([n]^* L) = \deg[n] \cdot \chi(L)$$

*Step 3: The class scales as a  $g$ -th power.* On the other hand, for any line bundle  $L$  on an abelian variety and any  $m \geq 0$ ,

$$\chi(L^{\otimes m}) = m^g \chi(L)$$

Indeed  $\chi(L^{\otimes m}) = ((L^{\otimes m})^g)/g! = m^g (L^g)/g! = m^g \chi(L)$ , where  $\chi(L) = (L^g)/g!$  is the abelian-variety Riemann-Roch formula and  $((L^{\otimes m})^g) = m^g (L^g)$  is the multilinearity of the top self-intersection number in the class of  $L$ ; both facts are cited to *Algebraic Geometry* [13] (the intersection number  $(L^g)$  and its behavior under tensor powers, and cohomology and base change for  $\chi$ ).

*Step 4: Comparison.* Combine the two computations for the ample symmetric  $L$  of Step 1, using  $[n]^* L \cong L^{\otimes n^2}$ :

$$\deg[n] \cdot \chi(L) = \chi([n]^* L) = \chi(L^{\otimes n^2}) = (n^2)^g \chi(L) = n^{2g} \chi(L)$$

Since  $L$  is ample,  $\chi(L) = \dim_k H^0(X, L) > 0$  (the higher cohomology of an ample line bundle on an abelian variety vanishes, so  $\chi(L) = h^0(L) = (L^g)/g! > 0$ ; see *Algebraic Geometry* [13]). Cancelling the nonzero  $\chi(L)$  gives  $\deg[n] = n^{2g}$ . Finiteness of the degree makes  $[n]$  an isogeny, and  $\deg[n] = n^{2g} \neq 0$  shows  $[n] \neq 0$  in  $\text{End}(X)$  for  $n \neq 0$ , as we sought to show.

The number  $n^{2g}$  is the algebraic derivation of a count chapter 5 obtained analytically: over  $\mathbb{C}$  the kernel  $X[n] = \frac{1}{n}\Lambda/\Lambda$  has order  $n^{2g}$  because the lattice  $\Lambda$  has rank  $2g$ , and  $\deg[n] = |X[n]|$  was the volume of a fundamental domain for  $\frac{1}{n}\Lambda$  inside one for  $\Lambda$  (proposition 5.2.4). The proof here replaces the lattice with the line bundle: the exponent  $2g$  is the product of the 2 from the quadratic growth  $L \mapsto L^{\otimes n^2}$  and the  $g$  from the self-intersection  $(L^g)$ . No transcendental input enters, only the theorem of the cube and Riemann-Roch. The degree is the first invariant that the cube computes, and it is the number every later count of torsion and every isogeny degree will be measured against. The degree  $n^{2g}$  counts the kernel of  $[n]$  with multiplicity, that is, scheme-theoretically. Whether that count equals the number of geometric torsion points depends on the separability of  $[n]$ , which in turn depends on the characteristic of  $k$ .

#### Corollary 8.4.6: The Count of Torsion Points

Let  $X$  be an abelian variety of dimension  $g$  over  $k$ , and let  $n \geq 1$  be prime to  $\text{char } k$  (any  $n$  if  $\text{char } k = 0$ ). Then  $[n]$  is a separable isogeny, that is, étale, and the group of geometric  $n$ -torsion points is

$$X[n](\bar{k}) \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$$

of order  $n^{2g} = \deg[n]$ . If  $\text{char } k = p > 0$  and  $p|n$ , then  $[n]$  is not separable, and  $|X[n](\bar{k})| < n^{2g}$  strictly.

#### Proof:

*Step 1: Separability when  $n$  is prime to the characteristic.* The differential of  $[n]$  at the identity is multiplication by  $n$  on the tangent space  $T_e X$ , a  $g$ -dimensional  $k$ -vector space. When  $n$  is prime to  $\text{char } k$ , this scalar  $n$  is invertible, so  $d[n]_e$  is an isomorphism; by translation homogeneity (proposition 7.1.3)  $d[n]_x$  is an isomorphism at every point, so  $[n]$  is étale. An étale isogeny is separable, and its degree equals the number of geometric points in a fibre, so  $|X[n](\bar{k})| = \deg[n] = n^{2g}$  by theorem 8.4.5.

*Step 2: The group structure.* The finite abelian group  $G := X[n](\bar{k})$  has order  $n^{2g}$  and is killed by  $n$ . Fix a prime  $\ell|n$ ; the étale argument of Step 1 applies verbatim to every  $[\ell^j]$  with  $\ell^j|n$ , so

$$|X[\ell^j](\bar{k})| = \deg[\ell^j] = \ell^{2jg} \quad (0 \leq j \leq v := v_\ell(n))$$

Let  $G_\ell$  be the  $\ell$ -primary part of  $G$ , a finite abelian  $\ell$ -group, and write it as  $\bigoplus_i \mathbb{Z}/\ell^{e_i}$ . For each  $j$ ,  $X[\ell^j](\bar{k}) = G_\ell[\ell^j]$  is the  $\ell^j$ -torsion of  $G_\ell$ , of order  $\prod_i \ell^{\min(e_i, j)}$ ; the count  $\ell^{2jg}$  for every  $j \leq v$  forces  $\sum_i \min(e_i, j) = 2jg$  for all such  $j$ . Taking  $j = 1$  gives exactly  $2g$  nonzero exponents  $e_i$ , and taking successive differences  $j$  to  $j + 1$  forces every  $e_i \geq v$ ; since  $G_\ell$  is killed by  $\ell^v$ , every  $e_i = v$ . Thus  $G_\ell \cong (\mathbb{Z}/\ell^v)^{2g}$ . Assembling over the primes dividing  $n$  by the Chinese remainder theorem gives  $G \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$ .

*Step 3: The inseparable case.* If  $\text{char } k = p > 0$  and  $p|n$ , write  $n = p^a m$  with  $p \nmid m$ . The differential  $d[p]_e$  is multiplication by  $p = 0$  on  $T_e X$ , so  $[p]$  is inseparable; hence  $[n] = [p^a] \circ [m]$  is inseparable, and its separable degree, which equals  $|X[n](\bar{k})|$ , is a proper divisor of  $\deg[n] = n^{2g}$ . Thus  $|X[n](\bar{k})| < n^{2g}$  strictly. The exact value depends on the  $p$ -rank of  $X$ ; the full analysis is deferred to chapter 9, completing the proof.

The corollary is the algebraic form of the torsion count that over  $\mathbb{C}$  was read straight off the lattice: for  $n$  invertible in the field, the geometric  $n$ -torsion is  $(\mathbb{Z}/n)^{2g}$ , exactly as  $\frac{1}{n}\Lambda/\Lambda$  was (proposition 5.2.4). The characteristic- $p$  subtlety has no analytic counterpart, because over  $\mathbb{C}$  every  $[n]$  is separable. In characteristic  $p$  the map  $[p]$  is inseparable: its scheme-theoretic kernel  $X[p]$  still has order  $p^{2g}$ , but part of that order is carried by an infinitesimal (non-reduced) group scheme with no geometric points, so the count of geometric  $\bar{k}$ -points drops. The precise statement is that  $X[p](\bar{k}) \cong (\mathbb{Z}/p)^f$  with  $0 \leq f \leq g$  the  $p$ -rank of  $X$ , with  $f = g$  for ordinary abelian varieties and  $f = 0$  for supersingular ones. This is the first place the arithmetic of the characteristic intrudes on the geometry, and unwinding it is the business of finite group schemes and the Tate module in chapter 9.

The  $g = 1$  case recovers everything the reader already knows about elliptic curves, and it is the sanity check for the general formula.

**Example 8.4.7: Multiplication by  $n$  on an Elliptic Curve**

Let  $E$  be an elliptic curve over  $k$ , so  $g = 1$ . Theorem 8.4.5 gives

$$\deg[n] = n^2$$

and for  $n$  prime to  $\text{char } k$ , corollary 8.4.6 gives

$$E[n](\bar{k}) \cong (\mathbb{Z}/n\mathbb{Z})^2$$

These are the classical facts of *Elliptic Curves* [4], where  $\deg[n] = n^2$  is proved through the division polynomials: the polynomial  $\psi_n$  cutting out the  $n$ -torsion has degree  $(n^2 - 1)/2$  in  $x$  (for  $n$  odd), and counting its roots with the two points above each  $x$ -value, plus the point at infinity, gives  $n^2$  points in  $E[n]$ . The pullback formula proposition 8.4.3 for a symmetric  $L$  of degree 1 (say  $L = \mathcal{O}_E(O)$ , symmetric because  $(-1)^*O = O$ ) reads  $[n]^*L \cong L^{\otimes n^2}$ , that is,  $[n]^*\mathcal{O}_E(O) \cong \mathcal{O}_E(n^2O)$ , which says the divisor  $[n]^{-1}(O) = E[n]$  has degree  $n^2$  over  $O$  as a fibre of the degree- $n^2$  map  $[n]$ . In characteristic  $p$ , writing  $p|n$ , the count  $E[p](\bar{k})$  is  $\mathbb{Z}/p$  when  $E$  is ordinary and 0 when  $E$  is supersingular, the  $g = 1$  instance of the  $p$ -rank dichotomy, and  $\deg[p] = p^2$  throughout as a scheme-theoretic degree.

**Exercise 8.4.1**

1. Using proposition 8.4.3, write out the pullback  $[2]^*L$  for an arbitrary  $L \in \text{Pic}(X)$  in terms of  $L$  and  $(-1)^*L$ , and specialize to the symmetric and antisymmetric cases. Then verify directly from the definition that  $[-1]^*L \cong (-1)^*L$  is consistent with the formula.
2. Let  $L$  be antisymmetric, so that  $(-1)^*L \cong L^{-1}$ . Show that  $[n]^*L \cong L^{\otimes n}$  for all  $n \in \mathbb{Z}$  directly from proposition 8.4.3, and deduce that the map  $L \mapsto [n]^*L$  acts on the class of  $L$  in the group  $\text{Pic}(X)$  as multiplication by  $n$ . Contrast this with the symmetric case.
3. Fix  $L \in \text{Pic}(X)$  and set  $S := L \otimes (-1)^*L$  and  $A := L \otimes (-1)^*L^{-1}$ . Verify the claims of box 8.4.4: that  $S$  is symmetric, that  $A$  is antisymmetric, and that  $S \otimes A \cong L^{\otimes 2}$ . Using proposition 8.4.3, compute  $[n]^*S$  and  $[n]^*A$ , and check that their product recovers  $[n]^*(L^{\otimes 2})$  as given by the general pullback formula.
4. Let  $X$  have dimension  $g$  and let  $L$  be an ample symmetric line bundle on  $X$ . Combine theorem 8.4.5 with the Riemann-Roch identities of its proof to compute  $h^0(X, [n]^*L)$  in terms of  $h^0(X, L)$ , and then in terms of the top self-intersection number  $(L^g)$ . Verify that the answer

is consistent with  $[n]^*L$  being the pullback of  $L$  under a degree- $n^{2g}$  isogeny.

5. Let  $X$  be an abelian variety of dimension  $g$  over a field  $k$  with  $\text{char } k = p > 0$ , and suppose  $[p]$  is inseparable with  $|X[p](\bar{k})| = p^f$  for the  $p$ -rank  $0 \leq f \leq g$ . Let  $\ell \neq p$  be a prime. Compute the order of the geometric torsion group  $X[p^a \ell^b](\bar{k})$  for  $a, b \geq 1$ , assuming that the inseparable degree of  $[p]$  is  $p^{2g-f}$  and that it multiplies across composites. Compare the total  $|X[p^a \ell^b](\bar{k})|$  against the scheme-theoretic degree  $\deg[p^a \ell^b]$  from theorem 8.4.5.
6. Prove that for an abelian variety  $X$  over a field  $k$  of characteristic 0 (or of characteristic prime to  $n$ ), the multiplication-by- $n$  map  $[n]$  is surjective on  $\bar{k}$ -points, without invoking the group structure  $(\mathbb{Z}/n\mathbb{Z})^{2g}$  of corollary 8.4.6. Then use surjectivity together with theorem 8.4.5 to show that every abelian variety over an algebraically closed field is a divisible group under its addition law.



# 9

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## *Isogenies, Torsion, and Tate Modules*

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Multiplication by  $n$  is an isogeny of degree  $n^{2g}$  (chapter 8); the maps it exemplifies (surjections with finite kernel) are the natural morphisms between abelian varieties, and this chapter makes them the object of study. An isogeny has a degree that splits into separable and inseparable parts, and in characteristic  $p$  the inseparable part is where the arithmetic hides. Away from  $p$  the kernels  $X[\ell^n]$  assemble into the *Tate module*  $T_\ell X \cong \mathbb{Z}_\ell^{2g}$ , a free module of rank  $2g$  carrying a continuous action of the absolute Galois group.

The Tate module is the algebraic replacement for the first homology lattice of chapter 1: it linearizes the abelian variety without any recourse to  $\mathbb{C}$ . Its first payoff is that  $\text{End}(X) \otimes \mathbb{Z}_\ell$  embeds in  $\text{End}(T_\ell X)$ , so  $\text{End}(X)$  is finitely generated and  $\text{End}^0(X)$  is a finite-dimensional  $\mathbb{Q}$ -algebra, the input Albert's classification will need in chapter 11. The  $\ell$ -adic representations  $\rho_\ell$  this produces are the bridge to chapter 13, where the characteristic polynomial of Frobenius acting on  $T_\ell X$  governs the arithmetic over a finite field. The chapter closes with the part of the picture the Tate module cannot see: the finite group schemes at  $p$ , the connected-étale sequence, and the ordinary/supersingular dichotomy.

### 9.1 Isogenies and Their Degrees

Multiplication by  $n$  is the model isogeny. It is surjective, its kernel  $X[n]$  is finite, and its degree  $n^{2g}$  was computed in theorem 8.4.5 as the top self-intersection of an ample bundle. Every construction of this section takes  $[n]$  as its template: a general isogeny is a surjective homomorphism with finite kernel, and its degree measures how far it is from an isomorphism. The task here is to make that degree precise over an arbitrary field  $k$  (where the kernel is a finite group *scheme*, not just a finite

set of points), to split it into a separable and an inseparable factor, and to show that isogeny is an equivalence relation whose inverse is furnished, up to a multiplication map, by  $[n]$  itself. Throughout,  $X$  and  $Y$  are abelian varieties over  $k$ , of dimensions  $g$  and  $g'$ , and  $\bar{k}$  is a fixed algebraic closure.

The definition names the class of maps that dominate the theory of abelian varieties: the surjective homomorphisms with finite kernel. Three formulations coincide, and it is worth having all three at hand, since different arguments make different ones convenient.

**Definition 9.1.1: Isogeny**

A homomorphism  $f: X \rightarrow Y$  of abelian varieties over  $k$  is an *isogeny* if it satisfies any (equivalently all) of the following:

1.  $f$  is surjective with finite kernel  $\ker f$ ;
2.  $\dim X = \dim Y$  and  $f$  is surjective;
3.  $f$  is finite and surjective.

The equivalence of the three conditions rests on the homogeneity of an abelian variety. Since  $f$  is a homomorphism, all of its fibres are translates of the fibre  $\ker f$  over the identity, so they share a common dimension  $\dim \ker f$ . The fibre-dimension theorem of *Algebraic Geometry* [13] then gives  $\dim X = \dim Y + \dim \ker f$  whenever  $f$  is surjective. Thus  $\ker f$  is finite ( $\dim \ker f = 0$ ) exactly when  $\dim X = \dim Y$ , which is condition (2); and a surjective homomorphism with all fibres finite between varieties of the same dimension is proper (the source is complete) with finite fibres, hence finite, which is condition (3). Conversely a finite morphism has finite fibres and, being proper with finite fibres, is closed with 0-dimensional image components, so a finite surjective homomorphism has finite kernel. A non-surjective homomorphism, by contrast, cannot be an isogeny: its image is a proper abelian subvariety of strictly smaller dimension.

This is the algebraic counterpart of the analytic isogeny of definition 5.2.1, where over  $\mathbb{C}$  an isogeny  $V/\Lambda \rightarrow V/\Lambda'$  was a surjection with finite kernel  $\Lambda'/F(\Lambda)$ ; the present definition drops the lattice and works over any field, at the cost that the kernel is now a group scheme whose  $\bar{k}$ -points may undercount its length in characteristic  $p$ .

The degree of an isogeny is the size of a generic fibre, measured as a field-extension degree. Because  $f$  is finite and surjective onto a variety of the same dimension, the function field  $k(X)$  is a finite extension of  $f^*k(Y) = k(Y)$ , and the degree of that extension is the invariant we want.



**Definition 9.1.2: Degree of an isogeny**

Let  $f: X \rightarrow Y$  be an isogeny. Its *degree* is the degree of the finite field extension  $k(X)/f^*k(Y)$ ,

$$\deg f := [k(X) : k(Y)]$$

where  $k(Y)$  is identified with its image  $f^*k(Y)$  in  $k(X)$ . Writing  $k(X)/k(Y)$  as a separable extension over a purely inseparable one, the *separable degree*  $\deg_s f := [k(X) : k(Y)]_s$  and the *inseparable degree*  $\deg_i f := [k(X) : k(Y)]_i$  satisfy

$$\deg f = \deg_s f \cdot \deg_i f$$

The isogeny  $f$  is *separable* if  $\deg_i f = 1$  (so  $\deg f = \deg_s f$ ) and *purely inseparable* if  $\deg_s f = 1$ .

The splitting  $\deg f = \deg_s f \cdot \deg_i f$  is the multiplicativity of separable and inseparable degrees for a finite extension, proved in *Commutative Algebra* [2]: any finite extension  $k(Y) \subseteq k(X)$  factors as a separable extension  $k(Y) \subseteq k(X)_s$  followed by a purely inseparable one  $k(X)_s \subseteq k(X)$ , and the two factors have degrees  $\deg_s f$  and  $\deg_i f$ . The inseparable degree is always a power of the characteristic:  $\deg_i f = 1$  in characteristic 0, and  $\deg_i f = p^m$  for some  $m \geq 0$  in characteristic  $p$ .

The separable degree is exactly what the kernel measures on  $\bar{k}$ -points. A finite morphism  $f: X \rightarrow Y$  is separable if and only if the extension  $k(X)/k(Y)$  is separable, and for such an  $f$  every fibre over a  $\bar{k}$ -point of  $Y$  has exactly  $[k(X) : k(Y)]_s$  points. Since the fibres of the homomorphism  $f$  are translates of  $\ker f$ , and translation by a  $\bar{k}$ -point is a bijection on  $\bar{k}$ -points, the generic fibre and the identity fibre have the same  $\bar{k}$ -point count. Thus

$$\deg_s f = |\ker f(\bar{k})|$$

the number of geometric points of the kernel, while  $\deg_i f$  records the non-reduced, infinitesimal part of  $\ker f$  that those points fail to see. When  $k$  is perfect and  $f$  is separable,  $\ker f$  is an étale group scheme and  $\deg f = |\ker f(\bar{k})|$  outright, recovering the naive count  $\deg f = |\ker f|$  of the analytic proposition 5.2.2. In characteristic  $p$  the two can genuinely differ, as the Frobenius isogeny below shows.

An immediate consequence is that an isogeny of degree 1 is an isomorphism:  $\deg f = 1$  forces  $k(X) = k(Y)$ , so  $f$  is birational, and a birational finite morphism between smooth varieties is an isomorphism. Thus the degree detects isomorphisms, and this observation is the seed of the next result.

Isogenous abelian varieties should be interchangeable for many purposes: they have the same dimension, isogenous Tate modules, and isogenous endomorphism algebras. For that to be a usable notion, “isogenous” must be an equivalence relation, and in particular symmetric: an isogeny in one direction must be matched by one in the other. The mechanism is that every isogeny divides a multiplication map.

**Proposition 9.1.3: Isogeny is an equivalence relation**

Call  $X$  and  $Y$  *isogenous*, written  $X \sim Y$ , if there exists an isogeny  $f: X \rightarrow Y$ . This is an equivalence relation on abelian varieties over  $k$ . Symmetry holds in the strong form: for every isogeny  $f: X \rightarrow Y$  there is a *complementary isogeny*  $g: Y \rightarrow X$  with

$$g \circ f = [n]_X \quad \text{and} \quad f \circ g = [n]_Y$$

where  $n = \deg f$ .

**Proof:**

Reflexivity is witnessed by the identity  $\text{id}_X$ , an isogeny of degree 1. Transitivity will follow once the degree is shown multiplicative (proposition 9.1.4): the composite of two surjective homomorphisms with finite kernels is again one, since  $\ker(g \circ f)$  is an extension of a subscheme of  $\ker g$  by  $\ker f$ , hence finite. The content is symmetry, which the complementary isogeny supplies.

*Step 1: The kernel of  $f$  lies in  $X[n]$ .* Let  $n = \deg f$  and set  $N = \ker f$ , a finite commutative group scheme of order  $n$  (its order, the length of its coordinate ring, equals  $\deg f$ ; over  $\bar{k}$  this refines the  $\bar{k}$ -point count  $\deg_s f$ ). By the theorem that a finite commutative group scheme of order  $n$  is annihilated by  $n$  (Deligne's theorem, cited to *Algebraic Geometry* [13]; on  $\bar{k}$ -points it is the order-of-a-group Lagrange statement, and the scheme-theoretic refinement covers the infinitesimal part in characteristic  $p$ ), multiplication by  $n$  is zero on  $N$ . Thus  $N$  is a closed subgroup scheme of  $X$  with  $nN = 0$ , so

$$N = \ker f \subseteq \ker [n]_X = X[n]$$

*Step 2: Factor  $[n]_X$  through  $f$ .* The inclusion  $\ker f \subseteq \ker [n]_X$  means that  $[n]_X: X \rightarrow X$  kills  $\ker f$ . Since  $f: X \rightarrow Y$  is a faithfully flat (finite surjective) homomorphism with kernel  $\ker f$ , it is the quotient of  $X$  by  $\ker f$ : any homomorphism out of  $X$  that vanishes on  $\ker f$  factors uniquely through  $f$ . Applying this to  $[n]_X$  produces a unique homomorphism  $g: Y \rightarrow X$  with

$$g \circ f = [n]_X$$

*Step 3:  $g$  is an isogeny, and  $f \circ g = [n]_Y$ .* The composite  $g \circ f = [n]_X$  is an isogeny (theorem 8.4.5), so its image is all of  $X$ ; since that image is contained in the image of  $g$ , the map  $g$  is surjective. For finiteness of  $\ker g$ , note that  $f$  maps  $\ker g$  into  $\ker(g \circ f) = \ker [n]_X = X[n]$ , which is finite, and  $f$  has finite kernel, so  $\ker g$  is finite. Hence  $g$  is an isogeny. For the reverse relation, compute

$$f \circ g \circ f = f \circ [n]_X = [n]_Y \circ f$$

using that  $f$ , being a homomorphism, commutes with multiplication by  $n$ . Cancelling the surjection  $f$  on the right gives  $f \circ g = [n]_Y$ , as we sought to show.

The complementary isogeny is the abelian-variety substitute for a genuine inverse: one cannot in general invert  $f$ , but one can invert it up to the harmless distortion  $[n]$ . This is exactly what makes

isogeny a workable equivalence, a  $\mathbb{Q}$ -linear notion, and it is the reason the endomorphism algebra  $\text{End}^0(X) = \text{End}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$ , in which every isogeny becomes invertible, is the natural object rather than  $\text{End}(X)$  itself. For  $g = 1$  this is the familiar dual isogeny of an elliptic curve, where  $\hat{f} \circ f = [\deg f]$ ; the general dual isogeny, defined through the dual abelian variety, is taken up in chapter 10.

Transitivity of the relation, and much else, needs the degree to be multiplicative in composites. This is the field-theoretic tower law wearing a geometric disguise.

**Proposition 9.1.4: Multiplicativity of the degree**

Let  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  be isogenies. Then  $g \circ f$  is an isogeny and

$$\deg(g \circ f) = \deg g \cdot \deg f$$

and likewise  $\deg_s(g \circ f) = \deg_s g \cdot \deg_s f$  and  $\deg_i(g \circ f) = \deg_i g \cdot \deg_i f$ .

**Proof:**

That  $g \circ f$  is a surjective homomorphism is immediate, both being surjective. Its kernel sits in an exact sequence

$$0 \rightarrow \ker f \rightarrow \ker(g \circ f) \rightarrow \ker g$$

the right-hand map being the restriction of  $f$ ; since  $\ker f$  and  $\ker g$  are finite, so is  $\ker(g \circ f)$ , and  $g \circ f$  is an isogeny.

For the degree, pull back function fields along the tower  $Z \rightarrow Y \rightarrow X$ , which contravariantly gives the tower of fields

$$k(Z) \subseteq k(Y) \subseteq k(X)$$

The multiplicativity of field-extension degrees in a tower (*Commutative Algebra* [2]) reads

$$[k(X) : k(Z)] = [k(X) : k(Y)] \cdot [k(Y) : k(Z)]$$

which is precisely  $\deg(g \circ f) = \deg f \cdot \deg g$ . The separable and inseparable degrees are individually multiplicative in towers by the same reference, giving the two refined identities, completing the proof.

Multiplicativity closes the loop on proposition 9.1.3: combined with  $\deg id = 1$ , it makes isogeny transitive, and it lets the degree of  $[n]_X = g \circ f$  be read off as  $\deg g \cdot \deg f = n^{2g}$ , recovering theorem 8.4.5 and pinning  $\deg f \cdot \deg g = n^{2g}$  for the complementary pair. The degree is thus a homomorphism from the monoid of isogenies out of  $X$  to the positive integers, and it descends to a well-behaved invariant of the isogeny category. The same tower law splits each degree into its separable and inseparable parts, so separability of a composite is governed factor by factor.

The one phenomenon with no analytic shadow is a nontrivial inseparable degree, and its source is the Frobenius morphism. Over a field of characteristic  $p$  it produces an isogeny that is bijective on geometric points yet has degree  $p^g$ : the entire degree is inseparable.

**Example 9.1.5: The Frobenius isogeny**

Let  $k$  have characteristic  $p > 0$  and let  $X$  be an abelian variety over  $k$  of dimension  $g$ . The *relative Frobenius* is the homomorphism

$$F: X \rightarrow X^{(p)}$$

where  $X^{(p)}$  is the base change of  $X$  along the absolute Frobenius  $a \mapsto a^p$  of  $k$ , and  $F$  is, on coordinate rings, the  $p$ -power map relative to  $k$ . Because  $F$  raises local coordinates to the  $p$ -th power, the induced extension of function fields  $k(X^{(p)}) \subseteq k(X)$  satisfies  $k(X)^p \subseteq k(X^{(p)})$ , so it is purely inseparable; a separating transcendence basis of  $g$  local coordinates contributes one factor of  $p$  each, giving degree  $p^g$ . Hence  $F$  is an isogeny with

$$\deg F = p^g, \quad \deg_i F = p^g, \quad \deg_s F = 1$$

In particular  $F$  is purely inseparable: its kernel  $\ker F$  is an infinitesimal group scheme of order  $p^g$  with  $\ker F(\bar{k}) = \{0\}$ , so on  $\bar{k}$ -points  $F$  is a bijection even though it is far from an isomorphism of varieties. This is the extreme case in which the naive point-count  $\deg_s F = |\ker F(\bar{k})| = 1$  misses the whole degree.

When  $k = \mathbb{F}_q$  with  $q = p^r$ , iterating the relative Frobenius  $r$  times gives the  $q$ -power Frobenius. The  $r$ -fold composite

$$F_X: X \rightarrow X^{(p^r)} = X$$

is an *endomorphism* of  $X$ , because  $X$  is defined over  $\mathbb{F}_q$  and so  $X^{(q)} = X$  canonically; it raises coordinates to the  $q$ -th power. By proposition 9.1.4 its degree is  $\deg F_X = (p^g)^r = q^g$ , again purely inseparable. Over  $\mathbb{F}_p$  this  $F_X$  is the Frobenius endomorphism whose characteristic polynomial governs the arithmetic of  $X$ , the point of departure for the Weil conjectures in chapter 13. For  $g = 1$  this is the  $q$ -power Frobenius on an elliptic curve  $E/\mathbb{F}_q$ , of degree  $q$ , treated in *Elliptic Curves* [4].

The elliptic-curve case, where  $g = 1$  and every isogeny is a map of curves, grounds the general definitions and reconciles them with a theory the reader has already met.

**Example 9.1.6: Isogenies of elliptic curves**

Take  $g = 1$ , so  $X = E$  and  $Y = E'$  are elliptic curves over  $k$  and an isogeny  $f: E \rightarrow E'$  is a nonconstant homomorphism. Nonconstant already forces surjectivity (the image is an abelian subvariety of the 1-dimensional  $E'$ , hence all of it) and forces  $\ker f$  finite, so the three conditions of definition 9.1.1 reduce to the single requirement that  $f$  be nonconstant, matching the definition in *Elliptic Curves* [4]. The degree  $\deg f = [k(E) : k(E')]$  is the degree of the corresponding finite map of curves.

Consider the standard families over a field of characteristic  $\neq 2, 3$ .

1. Multiplication  $[n]: E \rightarrow E$  has degree  $n^2$ , the  $g = 1$  case of  $n^{2g}$  (theorem 8.4.5); if  $p = \text{char } k$  does not divide  $n$  it is separable with  $E[n](\bar{k}) \cong (\mathbb{Z}/n)^2$ , so  $\deg_s [n] = n^2 = \deg [n]$ .
2. A 2-isogeny  $E \rightarrow E'$  with kernel a chosen order-2 subgroup  $\{0, P\}$  has degree 2; in characteristic  $\neq 2$  it is separable, and its complementary isogeny (proposition 9.1.3) is the dual isogeny  $\hat{f}$  with  $\hat{f} \circ f = [2]$ .

3. In characteristic  $p$ , multiplication  $[p]: E \rightarrow E$  has degree  $p^2$  and factors as  $\hat{F} \circ F$ , with  $F: E \rightarrow E^{(p)}$  the Frobenius of example 9.1.5 (degree  $p$ , inseparable) and  $\hat{F}$  the Verschiebung. When  $E$  is ordinary,  $\hat{F}$  is separable and  $E[p](\bar{k}) \cong \mathbb{Z}/p$ ; when  $E$  is supersingular,  $[p]$  is purely inseparable and  $E[p](\bar{k}) = \{0\}$ .

Every degree computation here is a special case of the field-extension definition, and the separable/inseparable split reproduces the ordinary versus supersingular dichotomy that chapter 9 develops in dimension  $g$ .

### Exercise 9.1.1

1. Let  $X$  be an abelian variety of dimension  $g$  over a field  $k$ , and let  $n$  be a nonzero integer. Using definition 9.1.2 and theorem 8.4.5, compute  $\deg[n]_X$ . When  $k$  has characteristic 0, or characteristic  $p \nmid n$ , determine  $\deg_s[n]_X$  and  $\deg_i[n]_X$  from the fact that  $[n]_X$  is then separable with  $X[n](\bar{k}) \cong (\mathbb{Z}/n)^{2g}$ .
2. Let  $k$  have characteristic  $p > 0$  and let  $X$  be an abelian variety of dimension  $g$  over  $k$ . Consider the relative Frobenius  $F: X \rightarrow X^{(p)}$  of example 9.1.5. Compute the three degrees  $\deg F$ ,  $\deg_s F$ , and  $\deg_i F$ , and count the geometric points  $|\ker F(\bar{k})|$ . Reconcile the fact that  $F$  is a bijection on  $\bar{k}$ -points with the statement that  $F$  is not an isomorphism.
3. Let  $f: X \rightarrow Y$  be an isogeny with  $n = \deg f$ , and let  $g: Y \rightarrow X$  be the complementary isogeny of proposition 9.1.3, so that  $g \circ f = [n]_X$  and  $f \circ g = [n]_Y$ . Show that  $\deg g = n^{2g-1}$ , where  $g = \dim X$ . (Take care to distinguish the complementary isogeny  $g$  from the dimension  $g$ .)
4. Let  $f: X \rightarrow Y$  be a separable isogeny of abelian varieties over a perfect field  $k$ , and suppose  $\deg f = n$ . Prove that  $\ker f$ , viewed as a finite group of  $\bar{k}$ -points, is annihilated by  $n$ , and deduce that  $\ker f \subseteq X[n]$ . State precisely which result of the section this reproduces, and explain what the scheme-theoretic version adds in characteristic  $p$ .
5. Let  $E$  be an ordinary elliptic curve over a perfect field  $k$  of characteristic  $p > 0$ . Using the factorization  $[p] = \hat{F} \circ F$  of example 9.1.6, together with proposition 9.1.4, determine  $\deg F$ ,  $\deg \hat{F}$ , and their separable and inseparable parts. Deduce  $|E[p](\bar{k})|$  and confirm it agrees with the value stated for the ordinary case in example 9.1.6.
6. Let  $f: X \rightarrow Y$  be an isogeny of degree  $n = \deg f$ , and let  $\ell$  be a prime with  $\ell \nmid n$  and  $\ell \neq \text{char } k$ . Prove that  $f$  restricts to an isomorphism  $X[\ell] \rightarrow Y[\ell]$  on  $\ell$ -torsion subgroup schemes. Use the complementary isogeny of proposition 9.1.3 and the multiplicativity of degree.

## 9.2 Torsion and the Tate Module

The degree computation of chapter 8 showed that  $[n]: X \rightarrow X$  is an isogeny of degree  $n^{2g}$ , and corollary 8.4.6 read off the prime-to- $p$  torsion: for  $n$  prime to  $\text{char } k$  the group  $X[n](\bar{k})$  has exactly  $n^{2g}$  points and is abstractly  $(\mathbb{Z}/n\mathbb{Z})^{2g}$ . Each such  $X[n]$  is a snapshot of the torsion at a single level. The organizing idea of this section is to stop looking at these groups one  $n$  at a time and instead assemble a whole tower of  $\ell$ -power torsion,  $X[\ell] \subseteq X[\ell^2] \subseteq \cdots$ , into a single module over the  $\ell$ -adic integers. The result, the Tate module  $T_\ell X$ , is a free  $\mathbb{Z}_\ell$ -module of rank  $2g$  carrying a continuous action of the absolute Galois group. It is the algebraic replacement for the homology lattice  $H_1(X, \mathbb{Z}) = \Lambda$  of the

complex-analytic theory: where the lattice recorded the topology of the torus  $V/\Lambda$ , the Tate module records the arithmetic of the abelian variety, and the two agree exactly where the theories overlap.

The starting point is the algebraic torsion count, the structural statement standing behind the numerical corollary of chapter 8. Over a field of characteristic zero, or in characteristic  $p$  for  $n$  prime to  $p$ , the  $n$ -torsion is as large and as simple as it can possibly be.

**Proposition 9.2.1: Structure of the Prime-to- $p$  Torsion**

Let  $X$  be an abelian variety of dimension  $g$  over a field  $k$ , and let  $n \geq 1$  be an integer prime to  $\text{char } k$ . Then

$$X[n](\bar{k}) \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$$

as abstract groups, where  $\bar{k}$  is an algebraic closure of  $k$ .

**Proof:**

By theorem 8.4.5 the multiplication map  $[n]: X \rightarrow X$  is an isogeny of degree  $n^{2g}$ . Because  $n$  is prime to  $\text{char } k$ , this isogeny is separable, so its kernel is an étale group scheme and the number of geometric points equals the degree:

$$|X[n](\bar{k})| = \deg[n] = n^{2g}$$

which is corollary 8.4.6. It remains to identify the group structure. The group  $X[n](\bar{k})$  is a finite abelian group annihilated by  $n$ , so it is a module over  $\mathbb{Z}/n\mathbb{Z}$ . For any prime  $\ell \mid n$ , write  $\ell^a$  for the exact power of  $\ell$  dividing  $n$ ; the  $\ell$ -primary part of  $X[n](\bar{k})$  is  $X[\ell^a](\bar{k})$ , of order  $\ell^{2ag}$ .

The claim is that  $X[\ell^a](\bar{k}) \cong (\mathbb{Z}/\ell^a\mathbb{Z})^{2g}$ . Consider the subgroup  $X[\ell](\bar{k})$ : it has order  $\ell^{2g}$  and is annihilated by  $\ell$ , hence is a vector space over the field  $\mathbb{F}_\ell$  of dimension  $2g$ . Now multiplication by  $\ell$  carries  $X[\ell^{j+1}](\bar{k})$  surjectively onto  $X[\ell^j](\bar{k})$ : given  $y \in X[\ell^j](\bar{k})$ , separability of  $[\ell]$  (again  $\ell \neq \text{char } k$ ) makes  $[\ell]$  surjective on  $\bar{k}$ -points, so  $y = \ell x$  for some  $x \in X(\bar{k})$ , and then  $\ell^{j+1}x = \ell^j y = 0$  shows  $x \in X[\ell^{j+1}](\bar{k})$ . Thus each  $X[\ell^{j+1}](\bar{k}) \rightarrow X[\ell^j](\bar{k})$  is surjective with kernel  $X[\ell](\bar{k})$ , of order  $\ell^{2g}$ . A finite abelian  $\ell$ -group  $A$  annihilated by  $\ell^a$ , with  $|A[\ell]| = \ell^{2g}$  and  $|A| = \ell^{2ag}$ , has exactly  $2g$  cyclic factors, each of order  $\ell^a$ : the number of invariant factors is  $\dim_{\mathbb{F}_\ell} A[\ell] = 2g$ , and matching the order forces every factor to be  $\mathbb{Z}/\ell^a\mathbb{Z}$ . Hence  $X[\ell^a](\bar{k}) \cong (\mathbb{Z}/\ell^a\mathbb{Z})^{2g}$ . Assembling the  $\ell$ -primary parts over the primes  $\ell \mid n$  and applying the Chinese remainder theorem gives  $X[n](\bar{k}) \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$ , as we sought to show.

This is the algebraic counterpart of the analytic torsion computation of proposition 5.2.4, where  $X = V/\Lambda$  gave  $X[n] = \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$  directly from the lattice; the analytic argument was a one-line quotient computation, whereas over an arbitrary field the same answer emerges only after the degree of  $[n]$  and its separability have been established. The restriction to  $n$  prime to  $\text{char } k$  is essential and not a mere convenience of the proof: in characteristic  $p$  the map  $[p]$  is inseparable, its kernel carries infinitesimal structure invisible to  $\bar{k}$ -points, and  $X[p](\bar{k})$  can be strictly smaller than  $(\mathbb{Z}/p\mathbb{Z})^{2g}$ . That phenomenon is the subject of section 9.4; for the rest of this section the prime  $\ell$  is always distinct from  $\text{char } k$ , and the torsion behaves exactly as it does over  $\mathbb{C}$ .

Fixing a single prime  $\ell \neq \text{char } k$ , the groups  $X[\ell^n](\bar{k})$  for  $n = 1, 2, 3, \dots$  form a tower, each mapped to the previous one by multiplication by  $\ell$ . The proof above showed these maps are surjective with kernel  $X[\ell](\bar{k})$ ; they realize  $X[\ell^n](\bar{k})$  as the  $\ell^n$ -torsion inside a single divisible group. Rather than

choose one level, the Tate module records the entire compatible system at once by taking the limit of the tower.

**Definition 9.2.2: The Tate Module**

Let  $X$  be an abelian variety over  $k$  and let  $\ell$  be a prime distinct from  $\text{char } k$ . The  $\ell$ -adic Tate module of  $X$  is the limit

$$T_\ell X := \varprojlim_n X[\ell^n](\bar{k})$$

over the system with transition maps  $X[\ell^{n+1}](\bar{k}) \xrightarrow{[\ell]} X[\ell^n](\bar{k})$  given by multiplication by  $\ell$ . A point of  $T_\ell X$  is a sequence  $(x_n)_{n \geq 1}$  with  $x_n \in X[\ell^n](\bar{k})$  and  $\ell x_{n+1} = x_n$  for all  $n$ .

Because each  $X[\ell^n](\bar{k})$  is a  $\mathbb{Z}/\ell^n\mathbb{Z}$ -module and the transition maps are  $\mathbb{Z}$ -linear, the limit  $T_\ell X$  is a module over the ring of  $\ell$ -adic integers  $\mathbb{Z}_\ell = \varprojlim_n \mathbb{Z}/\ell^n\mathbb{Z}$ . The next result identifies it completely: it is free of rank  $2g$ , the exact  $\ell$ -adic shadow of the rank- $2g$  lattice one meets analytically.

**Proposition 9.2.3: The Tate Module is Free of Rank  $2g$**

With notation as in definition 9.2.2,  $T_\ell X$  is a free  $\mathbb{Z}_\ell$ -module of rank  $2g$ :

$$T_\ell X \cong \mathbb{Z}_\ell^{2g}$$

**Proof:**

By proposition 9.2.1, for each  $n$  there is an isomorphism  $X[\ell^n](\bar{k}) \cong (\mathbb{Z}/\ell^n\mathbb{Z})^{2g}$ , and the transition map  $[\ell]: X[\ell^{n+1}](\bar{k}) \rightarrow X[\ell^n](\bar{k})$  is surjective (as shown in the proof of proposition 9.2.1). The Tate module is thus the limit of the tower

$$\cdots \longrightarrow (\mathbb{Z}/\ell^{n+1}\mathbb{Z})^{2g} \xrightarrow{\times \ell} (\mathbb{Z}/\ell^n\mathbb{Z})^{2g} \longrightarrow \cdots \longrightarrow (\mathbb{Z}/\ell\mathbb{Z})^{2g}$$

so it suffices to show that the reduction maps here are, up to a choice of compatible bases, the standard maps  $(\mathbb{Z}/\ell^{n+1}\mathbb{Z})^{2g} \rightarrow (\mathbb{Z}/\ell^n\mathbb{Z})^{2g}$  whose limit is  $\mathbb{Z}_\ell^{2g}$ .

*Step 1: choosing compatible bases.* Compatible bases are built by induction up the tower. Start with any  $\mathbb{Z}/\ell\mathbb{Z}$ -basis  $e_1^{(1)}, \dots, e_{2g}^{(1)}$  of  $X[\ell](\bar{k})$ , a  $2g$ -dimensional  $\mathbb{F}_\ell$ -vector space. Suppose a  $\mathbb{Z}/\ell^n\mathbb{Z}$ -basis  $e_1^{(n)}, \dots, e_{2g}^{(n)}$  of  $X[\ell^n](\bar{k})$  has been chosen. Since  $[\ell]: X[\ell^{n+1}](\bar{k}) \rightarrow X[\ell^n](\bar{k})$  is surjective, choose  $e_i^{(n+1)} \in X[\ell^{n+1}](\bar{k})$  with  $\ell e_i^{(n+1)} = e_i^{(n)}$  for each  $i$ . Their images under  $[\ell]$  span  $X[\ell^n](\bar{k})$ , so the  $e_i^{(n+1)}$  generate  $X[\ell^{n+1}](\bar{k})$  modulo  $\ker[\ell] = X[\ell](\bar{k})$ . Since  $\ker[\ell] = \ell^n X[\ell^{n+1}](\bar{k}) \subseteq \ell X[\ell^{n+1}](\bar{k})$  (as  $n \geq 1$ ), they in particular generate modulo  $\ell X[\ell^{n+1}](\bar{k})$ , so by Nakayama's lemma over the local ring  $\mathbb{Z}/\ell^{n+1}\mathbb{Z}$  (whose maximal ideal is  $(\ell)$ ) they generate  $X[\ell^{n+1}](\bar{k})$  outright. A set of  $2g$  generators of a module isomorphic to  $(\mathbb{Z}/\ell^{n+1}\mathbb{Z})^{2g}$  is a basis, so  $e_1^{(n+1)}, \dots, e_{2g}^{(n+1)}$  is a  $\mathbb{Z}/\ell^{n+1}\mathbb{Z}$ -basis. The induction yields, for every  $n$ , a basis  $e_1^{(n)}, \dots, e_{2g}^{(n)}$  of  $X[\ell^n](\bar{k})$  with  $\ell e_i^{(n+1)} = e_i^{(n)}$ .

*Step 2: identifying the limit.* With these compatible bases the transition map  $X[\ell^{n+1}](\bar{k}) \rightarrow X[\ell^n](\bar{k})$  becomes the reduction  $(\mathbb{Z}/\ell^{n+1}\mathbb{Z})^{2g} \rightarrow (\mathbb{Z}/\ell^n\mathbb{Z})^{2g}$  sending the  $i$ -th basis vector to the  $i$ -th basis vector, coordinatewise the canonical surjection  $\mathbb{Z}/\ell^{n+1}\mathbb{Z} \rightarrow \mathbb{Z}/\ell^n\mathbb{Z}$ . The limit of this

system commutes with the finite direct sum, so

$$T_\ell X = \varprojlim_n (\mathbb{Z}/\ell^n \mathbb{Z})^{2g} = \left( \varprojlim_n \mathbb{Z}/\ell^n \mathbb{Z} \right)^{2g} = \mathbb{Z}_\ell^{2g}$$

and the compatible basis vectors  $e_i = (e_i^{(n)})_n \in T_\ell X$  furnish a free  $\mathbb{Z}_\ell$ -basis. Hence  $T_\ell X$  is free of rank  $2g$ , completing the proof.

The rank  $2g$  is exactly the rank of the homology lattice  $H_1(X, \mathbb{Z}) = \Lambda$  from chapter 1, and this is no coincidence. Over  $k = \mathbb{C}$  the comparison of chapter 5 identifies  $X[\ell^n] = \frac{1}{\ell^n} \Lambda / \Lambda$ , and passing to the limit gives  $T_\ell X \cong \Lambda \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$ : the Tate module is the  $\ell$ -adic completion of the period lattice. Over a general field there is no lattice to complete, yet  $T_\ell X$  survives as the piece of arithmetic data the lattice would have carried. This is the sense in which the Tate module is the algebraic replacement for singular homology; it is available over any field, it has the right rank, and, as the next definitions record, it comes with an action of the arithmetic symmetries of  $k$  that the topological lattice never saw.

For many arguments one wants to work over a field rather than the ring  $\mathbb{Z}_\ell$ , discarding torsion phenomena and keeping only the underlying  $2g$ -dimensional linear-algebra object. Inverting  $\ell$  accomplishes this.

#### Definition 9.2.4: The Rational Tate Module

With  $X$  and  $\ell$  as in definition 9.2.2, the *rational Tate module* is

$$V_\ell X := T_\ell X \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$$

a  $\mathbb{Q}_\ell$ -vector space of dimension  $2g$ , where  $\mathbb{Q}_\ell$  is the field of  $\ell$ -adic numbers.

By proposition 9.2.3,  $T_\ell X \cong \mathbb{Z}_\ell^{2g}$ , so  $V_\ell X \cong \mathbb{Q}_\ell^{2g}$ ; the Tate module sits inside as the  $\mathbb{Z}_\ell$ -lattice  $T_\ell X \subset V_\ell X$ . The rational module loses the information of the individual torsion levels but retains the dimension, and its being a vector space over a field lets one apply linear algebra, characteristic polynomials, and eigenvalue arguments without the bookkeeping of  $\mathbb{Z}_\ell$ -modules. The distinction between  $T_\ell X$  and  $V_\ell X$  mirrors the distinction between the lattice  $\Lambda$  and the vector space  $V = \Lambda \otimes_{\mathbb{Z}} \mathbb{R}$  of the analytic theory, with  $\mathbb{Q}_\ell$  in place of  $\mathbb{R}$ .

Everything so far used  $\bar{k}$ -points, and the passage from  $k$  to  $\bar{k}$  introduces a symmetry the analytic theory has no analogue for: the absolute Galois group  $\text{Gal}(\bar{k}/k)$  of field automorphisms of  $\bar{k}$  fixing  $k$ . It acts on the geometric torsion, and this action is the source of the arithmetic content of the Tate module.

Each  $\sigma \in \text{Gal}(\bar{k}/k)$  acts on  $\bar{k}$ -points of  $X$  by acting on coordinates. Because  $X$ , its group law, and the map  $[\ell^n]$  are all defined over  $k$ , this action carries  $X[\ell^n](\bar{k})$  to itself and commutes with the transition maps  $[\ell]$ ; a torsion point maps to a torsion point of the same order, and  $\sigma(\ell x) = \ell \sigma(x)$ . Thus  $\text{Gal}(\bar{k}/k)$  acts on each  $X[\ell^n](\bar{k})$  by group automorphisms, compatibly with the tower, hence acts on the limit  $T_\ell X$ . The action is  $\mathbb{Z}_\ell$ -linear: it fixes the integers  $\mathbb{Z} \subset \mathbb{Z}_\ell$  acting by multiplication, and  $\mathbb{Z}_\ell$ -linearity follows by continuity from  $\mathbb{Z}$ -linearity. Tensoring with  $\mathbb{Q}_\ell$  gives a  $\mathbb{Q}_\ell$ -linear action on  $V_\ell X$ .



**9.2.5 Note: Continuity of the Galois action** The group  $\text{Gal}(\bar{k}/k)$  carries its profinite (Krull) topology and  $T_\ell X \cong \mathbb{Z}_\ell^{2g}$  its  $\ell$ -adic topology. The action  $\text{Gal}(\bar{k}/k) \times T_\ell X \rightarrow T_\ell X$  is continuous: the action on each finite quotient  $X[\ell^n](\bar{k})$  factors through the finite Galois group  $\text{Gal}(k_n/k)$  of the finite extension  $k_n$  generated by the coordinates of the  $\ell^n$ -torsion points, and a compatible system of continuous actions on finite discrete groups has continuous limit. This continuity is what makes  $T_\ell X$  a genuine  $\ell$ -adic Galois representation; the general formalism of continuous cohomology of profinite groups is developed in *Galois Cohomology* [6], and is not needed here beyond the statement that the action is continuous.

The Tate module is therefore not only a  $\mathbb{Z}_\ell$ -module of rank  $2g$  but a continuous  $\mathbb{Z}_\ell$ -linear representation of  $\text{Gal}(\bar{k}/k)$ , and it is this representation that encodes the arithmetic of  $X$  over  $k$ . The associated homomorphism  $\text{Gal}(\bar{k}/k) \rightarrow \text{GL}(T_\ell X)$  is the  $\ell$ -adic representation studied in section 9.3; over a number field or a finite field it is the primary tool for extracting arithmetic invariants, and it is the feature that has no counterpart in the purely topological lattice  $H_1(X, \mathbb{Z})$  of chapter 1.

The case  $g = 1$  grounds the construction and reconciles it with the elliptic curves already in hand.

**Example 9.2.6: The Tate Module of an Elliptic Curve**

Let  $E$  be an elliptic curve over a field  $k$  and  $\ell$  a prime distinct from  $\text{char} k$ . Here  $g = 1$ , so proposition 9.2.1 gives  $E[\ell^n](\bar{k}) \cong (\mathbb{Z}/\ell^n \mathbb{Z})^2$  for every  $n$ : the  $\ell^n$ -torsion is generated by two points  $P_n, Q_n$  each of order  $\ell^n$ . Choosing them compatibly, so that  $\ell P_{n+1} = P_n$  and  $\ell Q_{n+1} = Q_n$ , the pair  $(P_n)_n, (Q_n)_n$  is a  $\mathbb{Z}_\ell$ -basis of the Tate module, and by proposition 9.2.3

$$T_\ell E \cong \mathbb{Z}_\ell^2 \quad \text{and} \quad V_\ell E \cong \mathbb{Q}_\ell^2$$

This is exactly the object constructed for elliptic curves in *Elliptic Curves* [4], where the Galois representation  $\text{Gal}(\bar{k}/k) \rightarrow \text{GL}_2(\mathbb{Z}_\ell)$  on  $T_\ell E$  is the gateway to the arithmetic of  $E$ : for  $k$  a number field it governs the  $\ell$ -adic behaviour of the Frobenius elements, and for  $k = \mathbb{F}_q$  its characteristic polynomial recovers the point count of  $E$ . The dimension  $2 = 2g$  is the same rank the analytic construction produced from the lattice  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ : over  $\mathbb{C}$  the  $\ell^n$ -torsion is  $\frac{1}{\ell^n} \Lambda / \Lambda$  by proposition 5.2.4, and its limit is  $\Lambda \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \cong \mathbb{Z}_\ell^2$ . The Tate module of an elliptic curve is the  $\ell$ -adic replacement for the homology lattice of its complex points, and  $T_\ell X$  generalizes this replacement to every dimension.

### Exercise 9.2.1

1. Let  $X$  be an abelian variety of dimension  $g = 3$  over a field  $k$  with  $\text{char} k \neq 5$ . Using proposition 9.2.1, determine the order and the abstract group structure of  $X[25](\bar{k})$ , and state the  $\mathbb{F}_5$ -dimension of the subgroup  $X[5](\bar{k})$ .
2. Fix  $g$  and a prime  $\ell \neq \text{char} k$ . For each  $n \geq 1$  compute the number of  $\mathbb{Z}_\ell$ -linear surjections  $T_\ell X \rightarrow \mathbb{Z}/\ell^n \mathbb{Z}$ , and deduce the number of index- $\ell$  submodules of  $T_\ell X$ . Use proposition 9.2.3 to reduce to a count over the residue field  $\mathbb{F}_\ell$ .
3. Let  $E$  be an elliptic curve over  $k$  and  $\ell \neq \text{char} k$ . In the notation of example 9.2.6, an element of  $T_\ell E$  is a compatible sequence  $(x_n)_n$  with  $x_n \in E[\ell^n](\bar{k})$  and  $\ell x_{n+1} = x_n$ . Show that reduction modulo  $\ell^m$ , sending  $(x_n)_n$  to  $x_m$ , induces an isomorphism  $T_\ell E / \ell^m T_\ell E \cong E[\ell^m](\bar{k})$  of  $\mathbb{Z}/\ell^m \mathbb{Z}$ -modules.

4. A point  $t = (x_n)_n \in T_\ell X$  is fixed by every  $\sigma \in \text{Gal}(\bar{k}/k)$  if and only if each  $x_n$  lies in  $X[\ell^n](k)$ , the  $k$ -rational  $\ell^n$ -torsion. Using this, show that the fixed submodule  $(T_\ell X)^{\text{Gal}(\bar{k}/k)}$  is a free  $\mathbb{Z}_\ell$ -module, and that if  $X(k)$  has no  $\ell$ -power torsion then this fixed submodule is 0.
5. Let  $\ell \neq \text{char } k$ . Working directly from definition 9.2.2, prove two things about  $T_\ell X$ . First, it has no nonzero infinitely  $\ell$ -divisible element: if  $t \in T_\ell X$  satisfies  $t \in \ell^m T_\ell X$  for every  $m \geq 1$ , then  $t = 0$ . Second,  $T_\ell X$  is torsion-free as a  $\mathbb{Z}_\ell$ -module.

### 9.3 The $\ell$ -adic Representation

The Tate module  $T_\ell X$  of section 9.2 is a free  $\mathbb{Z}_\ell$ -module of rank  $2g$ , and two structures act on it: the endomorphisms of  $X$ , which are  $\mathbb{Z}_\ell$ -linear because they commute with  $[\ell^n]$  and so preserve every  $X[\ell^n]$ , and the absolute Galois group  $\text{Gal}(\bar{k}/k)$ , which permutes the geometric torsion points. This section makes both actions faithful linear algebra. The endomorphism action gives an embedding  $\text{End}(X) \otimes \mathbb{Z}_\ell \hookrightarrow \text{End}_{\mathbb{Z}_\ell}(T_\ell X)$ , whose injectivity forces  $\text{End}(X)$  to be a finitely generated free  $\mathbb{Z}$ -module and  $\text{End}^0(X)$  a finite-dimensional  $\mathbb{Q}$ -algebra. The Galois action is the  $\ell$ -adic representation  $\rho_\ell$  attached to  $X$ , a  $2g$ -dimensional representation of  $\text{Gal}(\bar{k}/k)$  over  $\mathbb{Z}_\ell$  that carries the arithmetic of  $X$  over  $k$ .

With no lattice available,  $T_\ell X$  plays the role that  $\Lambda \otimes \mathbb{Q}$  played over  $\mathbb{C}$  (proposition 6.1.3), and the injectivity below is the algebraic analogue of the faithfulness used there. The mechanism is that an endomorphism is detected by its action on torsion. An endomorphism  $f \in \text{End}(X)$  acts on each finite group  $X[\ell^n](\bar{k})$ , compatibly with the transition maps, hence on the limit  $T_\ell X$ . If that action is zero,  $f$  annihilates every  $\ell$ -power torsion point; the following lemma turns this into  $f = 0$  by density.

#### Lemma 9.3.1: $\ell$ -power Torsion is Zariski Dense

Let  $X$  be an abelian variety of dimension  $g \geq 1$  over  $k$ , and let  $\ell$  be a prime different from  $\text{char } k$ . The union of the  $\ell$ -power torsion subgroups,

$$X[\ell^\infty](\bar{k}) = \bigcup_{n \geq 1} X[\ell^n](\bar{k})$$

is Zariski dense in  $X_{\bar{k}}$ .

#### Proof:

Let  $Z$  be the Zariski closure of  $X[\ell^\infty](\bar{k})$  in  $X_{\bar{k}}$ , and let  $Y$  be the connected component of  $Z$  containing the identity  $e$ . Because  $X[\ell^\infty](\bar{k})$  is a subgroup of  $X(\bar{k})$  and the group operations are morphisms,  $Z$  is a closed algebraic subgroup of  $X_{\bar{k}}$ , and  $Y$  is a connected closed subgroup, hence an abelian subvariety of  $X_{\bar{k}}$ . It suffices to show  $Y = X_{\bar{k}}$ , for then  $Z \supseteq Y = X_{\bar{k}}$  and the closure is everything.

The group  $X[\ell^\infty](\bar{k})$  is  $\ell$ -divisible: since each  $X[\ell^n](\bar{k}) \cong (\mathbb{Z}/\ell^n)^{2g}$  (proposition 9.2.1), the map  $[\ell]$  carries  $X[\ell^{n+1}](\bar{k})$  onto  $X[\ell^n](\bar{k})$ , so every  $\ell$ -power torsion point is  $[\ell]$  of another. Consider the composite  $X[\ell^\infty](\bar{k}) \rightarrow Z(k)/Y(k)$ . Every element of  $X[\ell^\infty](\bar{k})$  has  $\ell$ -power order, so its image is a finite  $\ell$ -group; as a quotient of an  $\ell$ -divisible group it is itself  $\ell$ -divisible, and a finite  $\ell$ -divisible  $\ell$ -group is trivial. Thus the image is 0, meaning  $X[\ell^\infty](\bar{k}) \subseteq Y(k)$ . Therefore  $Z = Y$

is connected:  $Z$  is an abelian subvariety of  $X_{\bar{k}}$ .

It remains to see  $\dim Z = g$ . The subgroup  $X[\ell^n](\bar{k}) \subseteq Z(\bar{k})$  has order  $\ell^{2ng}$  for every  $n$  (proposition 9.2.1). If  $\dim Z = d$ , then the  $\ell^n$ -torsion of the abelian variety  $Z$  has order  $\ell^{2nd}$ , again by proposition 9.2.1 applied to  $Z$ . But  $X[\ell^n](\bar{k}) \subseteq Z[\ell^n](\bar{k})$ , so  $\ell^{2ng} \leq \ell^{2nd}$ , giving  $d \geq g$ ; as  $Z \subseteq X_{\bar{k}}$  forces  $d \leq g$ , we get  $d = g$  and  $Z = X_{\bar{k}}$ , completing the proof.

Density is the algebraic substitute for a fact that over  $\mathbb{C}$  was visible in the lattice: the points  $\frac{1}{\ell^n}\Lambda/\Lambda$  accumulate everywhere in the real torus, so no nonzero holomorphic condition can hold on all of them. Here the accumulation is Zariski, not analytic, and the argument is purely group-theoretic: the closure of an infinite torsion subgroup is a subgroup, its identity component is an abelian subvariety, and the torsion count pins its dimension to  $g$ . The lemma fails at  $\ell = \text{char } k$ , where  $X[\ell^n](\bar{k})$  can be much smaller than  $\ell^{2ng}$  and need not be dense, which is one reason the Tate module is defined only for  $\ell \neq \text{char } k$ .

With density established, the embedding is immediate. The map records how an endomorphism acts on  $T_\ell X$ , extended  $\mathbb{Z}_\ell$ -linearly from  $\text{End}(X)$ .

### Theorem 9.3.2: Faithfulness on the Tate Module

Let  $X$  be an abelian variety over  $k$  and  $\ell$  a prime different from  $\text{char } k$ . The natural homomorphism of  $\mathbb{Z}_\ell$ -algebras

$$\text{End}(X) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \longrightarrow \text{End}_{\mathbb{Z}_\ell}(T_\ell X)$$

sending  $f \otimes 1$  to the map induced by  $f$  on  $T_\ell X$  is injective.

#### Proof:

*Step 1: An endomorphism killing  $T_\ell X$  is zero.* Suppose  $f \in \text{End}(X)$  acts as 0 on  $T_\ell X$ . For each  $n$  the action of  $f$  on  $X[\ell^n](\bar{k})$  is the reduction of its action on  $T_\ell X = \varprojlim_n X[\ell^n](\bar{k})$  modulo  $\ell^n$  (definition 9.2.2), so  $f$  acts as 0 on every  $X[\ell^n](\bar{k})$ . Thus  $f$  vanishes on  $X[\ell^\infty](\bar{k})$ . The set  $\{x \in X(\bar{k}) : f(x) = e\} = \ker f(\bar{k})$  is Zariski closed in  $X_{\bar{k}}$ , being the geometric points of the closed subgroup scheme  $\ker f$ . Since this closed set contains the dense set  $X[\ell^\infty](\bar{k})$  (lemma 9.3.1), it is all of  $X_{\bar{k}}$ , so  $f$  maps every geometric point to  $e$ . A homomorphism of abelian varieties that is zero on all geometric points is the zero homomorphism, so  $f = 0$  in  $\text{End}(X)$ .

*Step 2:  $\text{End}(X)$  is torsion free.* Suppose  $[m] \circ f = 0$  for some integer  $m \geq 1$  and  $f \in \text{End}(X)$ . Then for every geometric point  $x$  the point  $f(x)$  satisfies  $[m]f(x) = e$ , so  $\text{im}(f) \subseteq X[m]$ , a finite group scheme. But  $\text{im}(f)$  is the image of the connected variety  $X_{\bar{k}}$  under a morphism, hence connected, and it contains  $f(e) = e$ ; a connected subset of the finite set  $X[m](\bar{k})$  containing  $e$  is  $\{e\}$ . Thus  $f$  maps every geometric point to  $e$ , so  $f = 0$  in  $\text{End}(X)$ . Hence no nonzero element of  $\text{End}(X)$  is killed by a positive integer, and  $\text{End}(X)$  is torsion free as an abelian group.

*Step 3: Injectivity after tensoring with  $\mathbb{Z}_\ell$ .* Let  $\xi \in \text{End}(X) \otimes \mathbb{Z}_\ell$  map to 0 in  $\text{End}_{\mathbb{Z}_\ell}(T_\ell X)$ . The element  $\xi$  is a  $\mathbb{Z}_\ell$ -combination of finitely many endomorphisms, so it lies in  $M_0 \otimes \mathbb{Z}_\ell$  for some finitely generated subgroup  $M_0 \subseteq \text{End}(X)$ . Let  $M := (M_0 \otimes \mathbb{Q}) \cap \text{End}(X)$  be the *saturation* of  $M_0$ ; it contains  $M_0$ , hence  $\xi \in M \otimes \mathbb{Z}_\ell$ , and, crucially, the quotient  $\text{End}(X)/M$  is torsion

free. The naive route, tensoring the  $\mathbb{Z}$ -injection  $M_0 \hookrightarrow N := \text{End}_{\mathbb{Z}_\ell}(T_\ell X)$  by the flat ring  $\mathbb{Z}_\ell$ , only yields injectivity of  $M_0 \otimes \mathbb{Z}_\ell \rightarrow N \otimes \mathbb{Z}_\ell$ ; composing with the  $\mathbb{Z}_\ell$ -module multiplication  $\mu: N \otimes \mathbb{Z}_\ell \rightarrow N$  back into  $N$  destroys it, since  $\mu$  has a large kernel (for instance  $\alpha \otimes 1 - 1 \otimes \alpha \mapsto 0$  for  $\alpha \in \mathbb{Z}_\ell \setminus \mathbb{Q}$ ). Saturation is what repairs the argument.

First, on any *saturated* subgroup  $M$  the reduction map is injective at every finite level: for each  $n$ ,

$$M/\ell^n M \longrightarrow \text{End}(X[\ell^n](\bar{k})), \quad f \bmod \ell^n M \longmapsto f|_{X[\ell^n]},$$

is injective. Indeed, suppose  $f \in M$  acts as 0 on  $X[\ell^n](\bar{k})$ . Then  $\ker[\ell^n] = X[\ell^n] \subseteq \ker f$ , and since  $[\ell^n]$  is an isogeny (theorem 8.4.5) the endomorphism  $f$  factors through it:  $f = [\ell^n] \circ h = \ell^n h$  for a unique  $h \in \text{End}(X)$ . Now  $\ell^n h = f \in M$ , and  $\text{End}(X)/M$  is torsion free, so  $h \in M$ ; hence  $f = \ell^n h \in \ell^n M$ . For an arbitrary finitely generated  $M_0$  this fails: the reductions of a  $\mathbb{Z}$ -basis can become dependent modulo  $\ell$ , and it is exactly that dependence the flatness argument overlooks.

This saturated  $M$  is finitely generated: since  $\text{End}(X)$  is itself a free  $\mathbb{Z}$ -module of finite rank (corollary 9.3.3), its subgroup  $M$  is free of some finite rank  $s$ . The forward appeal is not circular: the proof of corollary 9.3.3 uses only Steps 1 and 2 above together with the saturated level- $n$  injectivity just established, never the present tensor-injectivity. Being free of finite rank,  $M$  has  $M \otimes \mathbb{Z}_\ell \cong \mathbb{Z}_\ell^s$ , computed as the limit of the finite quotients  $M/\ell^n M$ .

Now pass to the limit over  $n$ . The injections  $M/\ell^n M \hookrightarrow \text{End}(X[\ell^n])$  are compatible with the transition maps, so

$$M \otimes \mathbb{Z}_\ell = \varprojlim_n M/\ell^n M \hookrightarrow \varprojlim_n \text{End}(X[\ell^n]) = \text{End}_{\mathbb{Z}_\ell}(T_\ell X)$$

is injective, the outer equalities being the definitions of  $T_\ell X = \varprojlim_n X[\ell^n](\bar{k})$  (definition 9.2.2) and of  $M \otimes \mathbb{Z}_\ell$  for the finitely generated free  $M$ , and an limit of injections being injective. Finally  $\xi$  itself vanishes: tensoring the inclusion  $M_0 \hookrightarrow M$  of  $\mathbb{Z}$ -modules by the flat ring  $\mathbb{Z}_\ell$  gives an injection  $M_0 \otimes \mathbb{Z}_\ell \hookrightarrow M \otimes \mathbb{Z}_\ell$  (this time landing in  $M \otimes \mathbb{Z}_\ell$ , the correct target, so flatness applies cleanly), and  $\xi$  maps through it to an element of  $M \otimes \mathbb{Z}_\ell$  with zero image in  $\text{End}_{\mathbb{Z}_\ell}(T_\ell X)$ ; by the injectivity just proved that image is 0, so  $\xi = 0$ . Therefore the full map  $\text{End}(X) \otimes \mathbb{Z}_\ell \rightarrow \text{End}_{\mathbb{Z}_\ell}(T_\ell X)$  is injective, as we sought to show.

The theorem bounds  $\text{End}(X)$  by transporting it into a  $\mathbb{Z}_\ell$ -module of known rank, the step that makes finite generation and finite dimensionality accessible. Its proof isolates two ingredients that recur: the density of torsion, which says an endomorphism is determined by its action on  $T_\ell X$ , and a divisibility principle, which says that an endomorphism killing the  $\ell^n$ -torsion is divisible by  $\ell^n$  inside  $\text{End}(X)$  because it factors through the isogeny  $[\ell^n]$ . The first is geometry, the second is the arithmetic of the isogeny  $[\ell^n]$ ; saturation is the bookkeeping that lets the second promote a torsion-level injection to an integral one, a point where the softer appeal to flatness alone fails. Together they transport a statement about the group  $\text{End}(X)$  into a statement about a  $\mathbb{Z}_\ell$ -module of known rank. The action itself is  $\mathbb{Z}_\ell$ -linear precisely because an endomorphism commutes with multiplication by  $\ell$ , so it respects the tower  $X[\ell] \subseteq X[\ell^2] \subseteq \cdots$  and passes to the limit.

The payoff is a bound on  $\text{End}(X)$  that no direct argument supplies over a general field.

**Corollary 9.3.3: Finite Generation of the Endomorphism Ring**

Let  $X$  be an abelian variety of dimension  $g$  over  $k$ . Then  $\text{End}(X)$  is a free  $\mathbb{Z}$ -module of finite rank,

$$\text{rank}_{\mathbb{Z}} \text{End}(X) \leq 4g^2$$

and  $\text{End}^0(X) = \text{End}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$  is a  $\mathbb{Q}$ -algebra of finite dimension at most  $4g^2$ .

**Proof:**

*Step 1: The rank is bounded.* By proposition 9.2.3 the Tate module  $T_{\ell}X$  is a free  $\mathbb{Z}_{\ell}$ -module of rank  $2g$ , so

$$\text{End}_{\mathbb{Z}_{\ell}}(T_{\ell}X) \cong M_{2g}(\mathbb{Z}_{\ell})$$

is a free  $\mathbb{Z}_{\ell}$ -module of rank  $(2g)^2 = 4g^2$ . Let  $M \subseteq \text{End}(X)$  be any finitely generated subgroup; it is free of some rank  $s$ , since  $\text{End}(X)$  is torsion free (theorem 9.3.2, Step 2), and its saturation  $\widetilde{M} = (M \otimes \mathbb{Q}) \cap \text{End}(X)$  has the same rank  $s$ . The rank bound comes from the *saturated* case established inside theorem 9.3.2, Step 3: for a finitely generated saturated subgroup the level- $n$  reductions  $\widetilde{M}/\ell^n \widetilde{M} \hookrightarrow \text{End}(X[\ell^n])$  are injective, so passing to the limit  $\widetilde{M} \otimes \mathbb{Z}_{\ell} \hookrightarrow \text{End}_{\mathbb{Z}_{\ell}}(T_{\ell}X) \cong M_{2g}(\mathbb{Z}_{\ell})$ ; hence  $\mathbb{Z}_{\ell}^s \cong \widetilde{M} \otimes \mathbb{Z}_{\ell}$  embeds in  $\mathbb{Z}_{\ell}^{4g^2}$ , forcing  $s \leq 4g^2$ . (We invoke here only the saturated level- $n$  injectivity, not the full tensor-injectivity of theorem 9.3.2, which is what makes the present corollary a legitimate input to that theorem's final step rather than a consequence of it.) Every finitely generated subgroup of  $\text{End}(X)$  therefore has rank at most  $4g^2$ , so  $\text{End}(X)$  has finite rank  $r \leq 4g^2$ .

*Step 2:  $\text{End}(X)$  is free of finite rank.* Fix a maximal  $\mathbb{Z}$ -linearly independent set  $f_1, \dots, f_r \in \text{End}(X)$ , with  $r \leq 4g^2$  the rank from Step 1, and let  $M_0 = \bigoplus_i \mathbb{Z}f_i$ . Then  $\text{End}^0(X) = \text{End}(X) \otimes \mathbb{Q} = M_0 \otimes \mathbb{Q}$  is a  $\mathbb{Q}$ -algebra of dimension  $r \leq 4g^2$ , so the finite-dimensionality is already secured. It remains to see that  $\text{End}(X)$  itself is finitely generated over  $\mathbb{Z}$ ; being torsion free (Step 2 of theorem 9.3.2), it is then free of rank  $r$ .

Endow  $\text{End}^0(X)$  with the symmetric  $\mathbb{Q}$ -bilinear *trace form*  $\langle f, g \rangle := \text{Tr}(f \circ g)$ , the trace being that of the action on  $T_{\ell}X \cong \mathbb{Z}_{\ell}^{2g}$ . This trace is a *rational integer*, independent of  $\ell$ . For  $h \in \text{End}(X)$  the characteristic polynomial  $P_h(t) = \det(t \cdot \text{id} - h|T_{\ell}X) \in \mathbb{Q}_{\ell}[t]$  is a priori only  $\ell$ -adic and monic of degree  $2g$ . That it has *rational-integer* coefficients, the same for every  $\ell \neq \text{char } k$ , is the theorem on the characteristic polynomial of an endomorphism proved in chapter 13: its mechanism is the identity

$$b^{2g} P_h\left(\frac{a}{b}\right) = \det(a \cdot \text{id} - bh|T_{\ell}X) = \deg([a] - [b]h)$$

for all integers  $a, b$  with  $b \neq 0$ , the degree of the isogeny  $[a] - [b]h$  (or 0) being a nonnegative rational integer independent of  $\ell$ , equal by that theorem to the determinant of the endomorphism's action on  $T_{\ell}X$ . Taking  $h = 0$  recovers  $\deg[a] = a^{2g}$  (theorem 8.4.5). We import the integrality and  $\ell$ -independence of  $P_h$  from there as the one nontrivial input to this corollary; the cited result covers both the Frobenius and a general endomorphism, and its argument does not depend on the present result. Taking  $h = f \circ g$ , the trace  $\text{Tr}(h)$  is a coefficient of  $P_h$ , hence a rational integer, so  $\langle -, - \rangle$  is a  $\mathbb{Z}$ -valued form on  $\text{End}(X) \times \text{End}(X)$ . It is nondegenerate on  $\text{End}^0(X)$ : that algebra is semisimple, a product of matrix algebras over division algebras (chapter 11, from the isogeny decomposition), and over a field of characteristic 0 the trace form of a faithful module over a

semisimple algebra is nondegenerate; here  $\text{End}^0(X)$  acts faithfully on  $V_\ell X = T_\ell X \otimes \mathbb{Q}_\ell$ . This faithfulness is again the saturated case of theorem 9.3.2, Step 3: taking the saturation  $\widetilde{M}$  of  $M_0$  (a finitely generated saturated subgroup with  $\widetilde{M} \otimes \mathbb{Q} = \text{End}^0(X)$ ), the level- $n$  injectivity gives  $\widetilde{M} \otimes \mathbb{Q}_\ell \hookrightarrow \text{End}_{\mathbb{Q}_\ell}(V_\ell X)$ , so  $\text{End}^0(X)$  acts faithfully. Two forward appeals enter here, neither circular: the semisimplicity of chapter 11 rests only on the finite-dimensionality of  $\text{End}^0(X)$  already secured above from the Step-1 rank bound, and the faithfulness uses only the saturated level- $n$  injectivity, not the tensor-injectivity that theorem 9.3.2 deduces from this corollary. Let  $D = \det(\langle f_i, f_j \rangle)_{i,j} \in \mathbb{Z}$ ; nondegeneracy gives  $D \neq 0$ .

Now take any  $f \in \text{End}(X)$  and write  $f = \sum_i q_i f_i$  with  $q_i \in \mathbb{Q}$ . Pairing against each basis vector,  $\langle f, f_j \rangle = \sum_i q_i \langle f_i, f_j \rangle$  for  $j = 1, \dots, r$ , a linear system with integer right-hand sides  $\langle f, f_j \rangle \in \mathbb{Z}$  (as  $f, f_j \in \text{End}(X)$ ) and integer coefficient matrix  $(\langle f_i, f_j \rangle)$  of nonzero determinant  $D$ . By Cramer's rule each  $q_i$  lies in  $\frac{1}{D}\mathbb{Z}$ , so  $f \in \frac{1}{D}M_0$ . Thus  $\text{End}(X) \subseteq \frac{1}{D}M_0$ . A subgroup of the finitely generated abelian group  $\frac{1}{D}M_0 \cong \mathbb{Z}^r$  is finitely generated, and being torsion free it is free:  $\text{End}(X) \cong \mathbb{Z}^r$  with  $r \leq 4g^2$ . Finally  $\text{End}^0(X) = \text{End}(X) \otimes \mathbb{Q} \cong \mathbb{Q}^r$  has  $\dim_{\mathbb{Q}} \text{End}^0(X) = r \leq 4g^2$ , completing the proof.

Finite dimensionality of  $\text{End}^0(X)$  is the hypothesis under which the whole structure theory operates, and the Tate module is what secures it over an arbitrary field. Over  $\mathbb{C}$  the same bound  $4g^2 = (2g)^2$  came from the faithful action on rational homology (proposition 6.1.3); the two arguments are the same argument in different coefficients, with  $T_\ell X$  over  $\mathbb{Z}_\ell$  in the role that  $\Lambda$  over  $\mathbb{Z}$  played analytically. With finite dimensionality in hand,  $\text{End}^0(X)$  is a finite-dimensional  $\mathbb{Q}$ -algebra, and once it is shown semisimple with a positive involution the Albert classification of chapter 11 applies. The bound  $4g^2$  is attained only in the extreme case of a power of a supersingular elliptic curve over  $\overline{\mathbb{F}}_p$ , where  $\text{End}^0(X)$  fills the ambient matrix algebra; for a simple ordinary abelian variety it is far smaller.

The second structure on  $T_\ell X$  is the Galois action. A point of  $X[\ell^n](\bar{k})$  has coordinates in  $\bar{k}$ , and an element  $\sigma \in \text{Gal}(\bar{k}/k)$  acts on those coordinates, sending a torsion point to a torsion point of the same order because  $\sigma$  fixes the equations defining  $X$  and the group law over  $k$ .

#### Definition 9.3.4: The $\ell$ -adic Representation

Let  $X$  be an abelian variety over  $k$ , fix a separable closure  $\bar{k}$  with absolute Galois group  $\Gamma := \text{Gal}(\bar{k}/k)$ , and let  $\ell \neq \text{char } k$  be prime. The action of  $\Gamma$  on each  $X[\ell^n](\bar{k})$  is  $\mathbb{Z}/\ell^n$ -linear and commutes with the transition maps, so it induces a  $\mathbb{Z}_\ell$ -linear action on the Tate module  $T_\ell X$ . The resulting homomorphism

$$\rho_\ell: \Gamma \longrightarrow \text{GL}(T_\ell X) \cong \text{GL}_{2g}(\mathbb{Z}_\ell)$$

is the  $\ell$ -adic representation attached to  $X$ . It is continuous for the profinite topology on  $\Gamma$  and the  $\ell$ -adic topology on  $\text{GL}_{2g}(\mathbb{Z}_\ell)$ , and it commutes with the action of  $\text{End}(X)$  on  $T_\ell X$ , so its image lies in the centralizer of  $(\text{End}(X) \otimes \mathbb{Z}_\ell)$  inside  $\text{End}_{\mathbb{Z}_\ell}(T_\ell X)$ .

The isomorphism  $\text{GL}(T_\ell X) \cong \text{GL}_{2g}(\mathbb{Z}_\ell)$  follows from a choice of  $\mathbb{Z}_\ell$ -basis of  $T_\ell X$  (proposition 9.2.3); the representation is well defined up to conjugacy independently of the basis. Continuity is the statement that the action on each finite quotient  $X[\ell^n](\bar{k})$  factors through a finite quotient of  $\Gamma$ , namely  $\text{Gal}(k(X[\ell^n])/k)$ , where  $k(X[\ell^n])$  is the finite extension of  $k$  generated by the coordinates of the  $\ell^n$ -torsion points; the full action on  $T_\ell X$  is the limit of these, so  $\rho_\ell$  is continuous with image in



the profinite group  $\mathrm{GL}_{2g}(\mathbb{Z}_\ell)$ . The Galois and endomorphism actions commute: an endomorphism defined over  $k$  is  $\Gamma$ -equivariant, so  $\rho_\ell$  takes values in  $\mathrm{GL}_{\mathrm{End}(X)}(T_\ell X)$ , and the interplay between a large endomorphism algebra and the Galois image constrains both: when  $k = \mathbb{F}_q$  the endomorphism-commutant restricts the conjugacy class of  $\rho_\ell(\mathrm{Frob}_q)$ , and hence the characteristic polynomial that chapter 13 studies. The precise foundations of the continuous Galois action are developed in *Galois Cohomology* [6]; here the action is recorded and used, not rebuilt.

The representation  $\rho_\ell$  is the object that governs the arithmetic of  $X$  over  $k$ . When  $k$  is a number field it packages the reduction data and the  $L$ -function of  $X$ ; when  $k = \mathbb{F}_q$  is finite, the Galois group is topologically generated by the Frobenius, and the characteristic polynomial of  $\rho_\ell(\mathrm{Frob}_q)$  acting on  $T_\ell X$  is the object the Weil conjectures control in chapter 13. The functoriality is exact: an isogeny  $f: X \rightarrow Y$  (definition 9.1.1) induces a  $\mathbb{Z}_\ell$ -linear,  $\Gamma$ -equivariant isomorphism  $V_\ell X \rightarrow V_\ell Y$  on the rational Tate modules, so isogenous abelian varieties have isomorphic  $\ell$ -adic representations after tensoring with  $\mathbb{Q}_\ell$ ; the representation  $\rho_\ell \otimes \mathbb{Q}_\ell$  is an isogeny invariant, exactly as  $\mathrm{End}^0(X)$  is.

The elliptic case is the model on which the general construction is built, and it is where the reader has already met both actions.

### Example 9.3.5: The $\ell$ -adic Representation of an Elliptic Curve

Let  $E$  be an elliptic curve over  $k$  and  $\ell \neq \mathrm{char} k$  a prime, so  $g = 1$ . The Tate module  $T_\ell E$  is free of rank  $2g = 2$  over  $\mathbb{Z}_\ell$  (example 9.2.6), and theorem 9.3.2 embeds

$$\mathrm{End}(E) \otimes \mathbb{Z}_\ell \hookrightarrow \mathrm{End}_{\mathbb{Z}_\ell}(T_\ell E) \cong M_2(\mathbb{Z}_\ell)$$

so corollary 9.3.3 bounds  $\mathrm{rank}_{\mathbb{Z}} \mathrm{End}(E) \leq 4$ . The three cases of example 6.1.5 sit inside  $M_2(\mathbb{Z}_\ell)$  as follows: when  $\mathrm{End}(E) = \mathbb{Z}$ , the image is the scalar matrices  $\mathbb{Z}_\ell \cdot I_2$ ; when  $E$  has complex multiplication by an order in an imaginary quadratic field  $K$ , the image is a  $\mathbb{Z}_\ell$ -order in the commutative algebra  $K \otimes \mathbb{Q}_\ell$ , of rank 2; and for a supersingular curve over an algebraically closed field of characteristic  $p$  (with  $\ell \neq p$ ), where  $\mathrm{End}(E)$  attains its maximal rank 4, the quaternion order embeds and fills  $M_2(\mathbb{Z}_\ell)$  up to finite index. The  $\ell$ -adic representation is

$$\rho_\ell: \mathrm{Gal}(\bar{k}/k) \longrightarrow \mathrm{GL}(T_\ell E) \cong \mathrm{GL}_2(\mathbb{Z}_\ell)$$

the two-dimensional representation whose reductions  $\rho_{\ell,n}: \mathrm{Gal}(\bar{k}/k) \rightarrow \mathrm{GL}_2(\mathbb{Z}/\ell^n)$  describe the Galois action on the  $\ell^n$ -torsion  $E[\ell^n](\bar{k}) \cong (\mathbb{Z}/\ell^n)^2$ . This is precisely the Galois representation of *Elliptic Curves* [4], where for  $k$  a number field the openness of the image (Serre's theorem, in the non-CM case) and the behavior of  $\rho_\ell(\mathrm{Frob}_v)$  at primes of good reduction are the entry point to the arithmetic. In the CM case the commuting action of  $\mathrm{End}(E) \otimes \mathbb{Z}_\ell$  forces the image of  $\rho_\ell$  to be abelian, landing in the Cartan subgroup of  $\mathrm{GL}_2(\mathbb{Z}_\ell)$  that centralizes the CM order, the two-dimensional shadow of the general principle that the endomorphism algebra cuts down the Galois image.

### Exercise 9.3.1

1. Let  $X$  be an abelian variety of dimension  $g$  over  $k$  and  $\ell \neq \mathrm{char} k$  a prime. Using only the definitions and the rank statement, determine each of the following as an explicit free module over the appropriate ring, giving its rank: (a) the target  $\mathrm{End}_{\mathbb{Z}_\ell}(T_\ell X)$  of the embedding of theorem 9.3.2; (b) the abelian group  $\mathrm{End}(X)$ ; (c) the  $\mathbb{Q}$ -algebra  $\mathrm{End}^0(X)$ . State the numerical bound from corollary 9.3.3 in each case and evaluate all ranks for  $g = 2$ .

2. Let  $X$  be an abelian variety over  $k$  and  $\ell \neq \text{char } k$ . Suppose  $f \in \text{End}(X)$  acts on  $T_\ell X$  as multiplication by a scalar  $c \in \mathbb{Z}_\ell$ . Show that  $c \in \mathbb{Z}$  and that  $f = [c]$  in  $\text{End}(X)$ . (Reduce to the injectivity of theorem 9.3.2.)
3. Fix a  $\mathbb{Z}_\ell$ -basis of  $T_\ell E$  for an elliptic curve  $E$  over  $k$  with  $\ell \neq \text{char } k$ , so that  $\rho_\ell: \text{Gal}(\bar{k}/k) \rightarrow \text{GL}_2(\mathbb{Z}_\ell)$  of example 9.3.5. Suppose  $E$  has complex multiplication by an order in an imaginary quadratic field  $K$ , and let  $\alpha \in \text{End}(E)$  act on  $T_\ell E$  with two eigenvalues  $\lambda_1, \lambda_2 \in \mathbb{Z}_\ell$  that are distinct *modulo*  $\ell$  (the split-Cartan case, e.g.  $\ell$  split in the CM order and prime to the conductor). Prove that the image of  $\rho_\ell$  is abelian, and, after passing to an eigenbasis of  $\alpha$ , identify it as a subgroup of the diagonal torus of  $\text{GL}_2(\mathbb{Z}_\ell)$ . (Use that  $\rho_\ell$  commutes with the endomorphism action, as recorded in definition 9.3.4.)
4. Let  $\sigma \in \text{Gal}(\bar{k}/k)$  and let  $\rho_\ell(\sigma) \in \text{GL}(T_\ell X)$  be its image under definition 9.3.4. Let  $P_\sigma(t) = \det(t \cdot \text{id} - \rho_\ell(\sigma)|_{T_\ell X})$  be its characteristic polynomial. Show that  $P_\sigma(t) \in \mathbb{Z}_\ell[t]$  is monic of degree  $2g$  with constant term  $\det \rho_\ell(\sigma) \in \mathbb{Z}_\ell^\times$ , and that the reduction of  $P_\sigma$  modulo  $\ell^n$  is the characteristic polynomial of the action of  $\sigma$  on  $X[\ell^n](\bar{k}) \cong (\mathbb{Z}/\ell^n)^{2g}$ .
5. Prove that  $\ell$ -power torsion is Zariski dense in  $X_{\bar{k}}$  (lemma 9.3.1) fails at  $\ell = \text{char } k = p$  in the following precise sense. Let  $E$  be a supersingular elliptic curve over an algebraically closed field  $k$  of characteristic  $p$ , so  $E[p^n](\bar{k}) = 0$  for all  $n$ . Show that  $E[p^\infty](\bar{k})$  is not Zariski dense in  $E$ , and show that the endomorphism  $[p] \in \text{End}(E)$  nonetheless acts as 0 on the group  $E[p^\infty](\bar{k})$  while being nonzero in  $\text{End}(E)$ . Explain in one sentence which step of the proof of theorem 9.3.2 this would break if  $\ell$  were allowed to equal  $p$ .
6. Let  $f: X \rightarrow Y$  be an isogeny of abelian varieties over  $k$  (definition 9.1.1) and  $\ell \neq \text{char } k$ . Prove that the induced map  $V_\ell f: V_\ell X \rightarrow V_\ell Y$  on rational Tate modules  $V_\ell X = T_\ell X \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$  is a  $\text{Gal}(\bar{k}/k)$ -equivariant isomorphism, and conclude that the representations  $\rho_\ell^X \otimes \mathbb{Q}_\ell$  and  $\rho_\ell^Y \otimes \mathbb{Q}_\ell$  are isomorphic. (Use that  $f$  has a quasi-inverse and that  $T_\ell$  is functorial in isogenies.)

## 9.4 Finite Group Schemes and the Characteristic- $p$ Picture

The Tate module  $T_\ell X$  records the  $\ell$ -power torsion for every prime  $\ell$  different from the characteristic, and over an algebraically closed field it is a free  $\mathbb{Z}_\ell$ -module of rank  $2g$  (proposition 9.2.3). One prime is missing from this picture. When  $\text{char } k = p > 0$ , the multiplication map  $[p]$  is inseparable (theorem 8.4.5), so  $X[p](\bar{k})$  has strictly fewer than  $p^{2g}$  points, and the naive count  $(\mathbb{Z}/p)^{2g}$  fails. The  $p$ -power torsion is not captured by its  $\bar{k}$ -points at all: it is a finite group scheme with an infinitesimal part that sees nilpotents, invisible to the point-set functor. This section develops the vocabulary for that part of the torsion. Finite group schemes organize the kernels of isogenies in characteristic  $p$ ; the connected-etale sequence splits each such kernel into an infinitesimal piece and an etale piece; and the  $p$ -power torsion, assembled into the  $p$ -divisible group, carries a discrete invariant, the  $p$ -rank, whose two extreme values name the ordinary and supersingular abelian varieties. The full classification of these objects is Dieudonné theory, which this book does not develop; the arithmetic payoff over a finite field waits for chapter 13.

The kernel of an isogeny is a finite closed subscheme of  $X$  that inherits a group structure. Over a field of characteristic zero, or for the prime-to- $p$  torsion, such a kernel is determined by its finitely many points; in characteristic  $p$  it is not, and the correct object is a group scheme whose coordinate ring may fail to be reduced. The measure of size is the dimension of that ring, not the number of points.



**Definition 9.4.1: Finite Group Scheme**

Let  $k$  be a field. A *finite group scheme* over  $k$  is a group scheme  $G = \operatorname{Spec} A$  over  $k$  whose coordinate ring  $A$  is a finite-dimensional  $k$ -algebra. Its *order* is

$$|G| := \dim_k A$$

Equivalently,  $G$  is a group object in the category of affine schemes over  $k$  that is finite and flat over  $\operatorname{Spec} k$  (finiteness over a field forces flatness, since every  $k$ -vector space is free).

The comultiplication  $A \rightarrow A \otimes_k A$ , counit  $A \rightarrow k$ , and antipode  $A \rightarrow A$  encoding  $m, e, \iota$  make  $A$  a commutative Hopf algebra over  $k$ , and a finite group scheme is the same datum as a finite-dimensional commutative Hopf algebra. The order is multiplicative in short exact sequences and agrees with the number of  $\bar{k}$ -points precisely when  $G$  is étale, that is, when  $A \otimes_k \bar{k}$  is reduced. The representability of these kernels as finite group schemes, together with the Cartier duality  $G \mapsto G^\vee$  that exchanges the roles of the algebra and its dual, is developed in *Algebraic Geometry* [13]; the results are stated and used here rather than re-derived. The two invariants to keep in view are the order (a dimension count, insensitive to reducedness) and the number of geometric points (which the infinitesimal part cannot see).

The kernel of multiplication by  $n$  furnishes the primary example, and the separable-inseparable split of section 9.1 reads off its order.

**Example 9.4.2: The Kernels  $X[n]$  and  $\mu_p, \alpha_p$** 

Let  $X$  be an abelian variety of dimension  $g$  over  $k$ . The kernel  $X[n] = \ker([n]: X \rightarrow X)$  is a finite group scheme of order  $\deg[n] = n^{2g}$  (theorem 8.4.5), independent of  $\operatorname{char} k$ .

1. If  $n$  is prime to  $\operatorname{char} k$ , then  $[n]$  is étale (section 8.4), so  $X[n]$  is an étale group scheme and  $|X[n]| = |X[n](\bar{k})| = n^{2g}$ , recovering  $X[n](\bar{k}) \cong (\mathbb{Z}/n)^{2g}$  (corollary 8.4.6).
2. Suppose  $\operatorname{char} k = p$ . The group scheme  $\mu_p = \operatorname{Spec} k[t]/(t^p - 1)$  of  $p$ -th roots of unity has order  $p$ . Since  $t^p - 1 = (t - 1)^p$  in characteristic  $p$ , its coordinate ring is nonreduced, so  $\mu_p(\bar{k}) = \{1\}$  is a single point:  $\mu_p$  is a finite group scheme of order  $p$  with only the trivial geometric point. The additive analogue  $\alpha_p = \operatorname{Spec} k[t]/(t^p)$ , the kernel of Frobenius on the additive group  $\mathbb{G}_a$ , likewise has order  $p$  and a single geometric point.

Both  $\mu_p$  and  $\alpha_p$  arise as subgroup schemes of  $X[p]$  for suitable  $X$ ; they are the char- $p$  phenomena the point count misses.

A finite group scheme over an arbitrary field mixes an infinitesimal part, concentrated at the identity and detected only by nilpotents in the coordinate ring, with an étale part, spread over honest geometric points. Over a perfect field these two parts separate cleanly, and the separation is functorial. This is the structural tool that reduces every finite group scheme to a nonreduced local piece and a Galois module.

**Proposition 9.4.3: The Connected-Etale Sequence**

Let  $k$  be a perfect field and  $G$  a finite group scheme over  $k$ . Let  $G^0$  be the connected component of the identity and  $G^{\text{et}} := G/G^0$ . Then there is a short exact sequence of finite group schemes

$$0 \longrightarrow G^0 \longrightarrow G \longrightarrow G^{\text{et}} \longrightarrow 0$$

in which  $G^0$  is connected (equivalently, local: its coordinate ring is a local  $k$ -algebra) and  $G^{\text{et}}$  is étale. The sequence is functorial in  $G$ , and over a perfect field it splits functorially, so  $G \cong G^0 \times G^{\text{et}}$ . The étale quotient  $G^{\text{et}}$  corresponds, under the equivalence between étale group schemes and finite Galois modules, to the finite  $\text{Gal}(\bar{k}/k)$ -module  $G(\bar{k}) = G^{\text{et}}(\bar{k})$ .

**Proof:**

The identity component  $G^0 = \text{Spec } A_0$  is the spectrum of the local factor of  $A$  at the augmentation ideal, and the projections onto the connected components of  $\text{Spec } A$  realize  $G^{\text{et}}$  as the maximal étale quotient; that  $G^0$  is a normal subgroup scheme and  $G^{\text{et}}$  a group scheme is the general structure of finite group schemes over a field, for which we cite *Algebraic Geometry* [13]. The functorial splitting over a perfect field is the point requiring perfectness: writing  $A = A_0 \times A_{\text{et}}$  as a product of  $k$ -algebras, the étale factor  $A_{\text{et}}$  becomes, after base change to  $\bar{k}$ , a product of copies of  $\bar{k}$  indexed by  $G(\bar{k})$ , and the reduced subscheme  $G_{\text{red}} \subseteq G$  is a subgroup scheme exactly when  $k$  is perfect, since the property of being reduced is then stable under the base change  $k \rightarrow \bar{k}$  (over an imperfect field  $G_{\text{red}}$  need not be a subgroup). The composite  $G_{\text{red}} \hookrightarrow G \twoheadrightarrow G^{\text{et}}$  is an isomorphism, giving the splitting, and it is functorial because  $G_{\text{red}}$  and  $G^0$  are. The identification of étale group schemes with finite Galois modules via  $G \mapsto G(\bar{k})$  is the group-scheme form of the equivalence between finite étale  $k$ -schemes and finite continuous  $\text{Gal}(\bar{k}/k)$ -sets, cited to *Algebraic Geometry* [13] and *Commutative Algebra* [2] for the separability underlying it, completing the proof.

The sequence sorts a finite group scheme into the part the Tate module already saw and the part it did not. The étale quotient  $G^{\text{et}}$  is a Galois module: for  $G = X[\ell^n]$  with  $\ell \neq p$  it is all of  $X[\ell^n]$ , and its limit is  $T_\ell X$  with its Galois action (section 9.2). The connected part  $G^0$  is where characteristic  $p$  lives: it is infinitesimal, its only geometric point is the identity, and it contributes to the order without contributing to the point count. For  $G = X[p^n]$  both parts can be nontrivial, and the balance between them is the invariant this section is building toward. Over an imperfect field the splitting can fail, which is why the standing hypothesis here, and in the applications of chapter 13, is that the base field is perfect.

The connected part is governed by two mutually inverse-up-to- $[p]$  homomorphisms intrinsic to characteristic  $p$ . Frobenius raises coordinates to the  $p$ -th power; Verschiebung is its dual under Cartier duality. Their composite is multiplication by  $p$ , and this single relation controls the inseparability of  $[p]$ .

**Definition 9.4.4: Frobenius and Verschiebung**

Let  $k$  be a field of characteristic  $p > 0$  and  $G$  a commutative finite group scheme (or, more generally, a commutative group scheme) over  $k$ . The *relative Frobenius*

$$F = F_{G/k}: G \longrightarrow G^{(p)}$$

is the  $k$ -morphism induced on coordinate rings by  $a \mapsto a^p$ , where  $G^{(p)}$  is the base change of  $G$  along the absolute Frobenius  $x \mapsto x^p$  of  $k$ . The *Verschiebung*

$$V = V_{G/k}: G^{(p)} \longrightarrow G$$

is the Cartier dual of the Frobenius of the dual group scheme, and it satisfies

$$V \circ F = [p]_G, \quad F \circ V = [p]_{G^{(p)}}$$

Frobenius is the source of all inseparability: for an abelian variety  $X$  over a perfect field, the relative Frobenius  $F: X \rightarrow X^{(p)}$  is a purely inseparable isogeny of degree  $p^g$ , and the factorization  $[p] = V \circ F$  (example 9.1.5) splits multiplication by  $p$  into its inseparable and remaining factors. The identity  $VF = [p]$  pins the degree bookkeeping: since  $\deg[p] = p^{2g}$  and  $\deg F = p^g$ , the Verschiebung has degree  $p^g$  as well, but its separable degree is what varies, and that variation is the  $p$ -rank. On the additive group  $\mathbb{G}_a$ , Frobenius is  $t \mapsto t^p$  with kernel  $\alpha_p$  and Verschiebung is zero; on the multiplicative group  $\mathbb{G}_m$ , Frobenius has kernel  $\mu_p$ . These two kernels are the connected-etale extremes among order- $p$  group schemes:  $\alpha_p$  is local with local dual,  $\mu_p$  is local with etale dual, and  $\mathbb{Z}/p$  is etale with local dual.

Assembling the  $p$ -power torsion across all  $n$  produces a single object that plays, for the prime  $p$ , the role the Tate module plays for  $\ell \neq p$ . Because  $[p]$  is an isogeny of degree  $p^{2g}$ , the tower  $X[p] \subseteq X[p^2] \subseteq \cdots$  grows by a factor  $p^{2g}$  at each step, and the colimit is a  $p$ -divisible group of height  $2g$ .

**Definition 9.4.5: The  $p$ -Divisible Group**

Let  $X$  be an abelian variety of dimension  $g$  over a field  $k$  of characteristic  $p > 0$ . The  *$p$ -divisible group* (or *Barsotti-Tate group*) of  $X$  is the colimit of finite group schemes

$$X[p^\infty] := \varinjlim_n X[p^n]$$

taken along the inclusions  $X[p^n] \hookrightarrow X[p^{n+1}]$ . Each  $X[p^n]$  is a finite group scheme of order  $p^{2ng}$ , and the system  $(X[p^n])_n$  is a  $p$ -divisible group of *height*  $2g$ : the maps  $[p]: X[p^{n+1}] \rightarrow X[p^n]$  are surjective with kernel  $X[p]$  of order  $p^{2g}$ .

The  $p$ -divisible group is the characteristic- $p$  companion of the Tate module: it packages the entire  $p$ -power torsion, including its infinitesimal part, into one pro-object. For  $\ell \neq p$  the analogous colimit  $\varinjlim_n X[\ell^n]$  recovers  $T_\ell X$  up to the passage from  $\bar{k}$ -points to the etale group scheme, but its group scheme is etale, so nothing is lost by taking points; for  $p$  the group scheme structure is essential, and  $X[p^\infty]$  cannot be replaced by its geometric points. Its height  $2g$  matches the  $\mathbb{Z}_\ell$ -rank of  $T_\ell X$ , reflecting that both objects are  $2g$ -dimensional shadows of the same  $2g$ -dimensional torsion. The

connected-etale sequence of each  $X[p^n]$  assembles into a sequence of  $p$ -divisible groups

$$0 \longrightarrow X[p^\infty]^0 \longrightarrow X[p^\infty] \longrightarrow X[p^\infty]^{\text{et}} \longrightarrow 0$$

whose etale quotient has a height equal to the dimension of the  $\bar{k}$ -points of  $X[p]$ , and that height is the invariant named next.

**Definition 9.4.6: Ordinary and Supersingular Abelian Varieties**

Let  $X$  be an abelian variety of dimension  $g$  over an algebraically closed field  $k$  of characteristic  $p > 0$ . The  $p$ -rank of  $X$  is the integer  $f = f(X)$  determined by

$$X[p](\bar{k}) \cong (\mathbb{Z}/p)^f$$

The abelian variety  $X$  is *ordinary* if  $f = g$ , the maximal value (so  $X[p](\bar{k}) \cong (\mathbb{Z}/p)^g$ ), and *supersingular* if  $f = 0$  (so  $X[p](\bar{k}) = 0$ , and  $X[p]$  is connected with local dual).

The  $p$ -rank measures how much of the  $p$ -torsion is etale, that is, visible on geometric points. It equals the height of the etale quotient  $X[p^\infty]^{\text{et}}$ , since  $X[p]^{\text{et}}(\bar{k}) = X[p](\bar{k}) \cong (\mathbb{Z}/p)^f$ . The ordinary case is the generic one: the  $p$ -torsion is as etale as it can be,  $X[p^\infty]$  splits as an extension of an etale  $p$ -divisible group of height  $g$  by a connected one of height  $g$  (a copy of  $(\mu_{p^\infty})^g$  up to isogeny), and the situation most resembles the prime-to- $p$  picture. The supersingular case is the opposite extreme: no  $p$ -torsion is etale,  $X[p]$  is entirely infinitesimal, and the  $p$ -divisible group is as far from etale as possible. The intermediate values  $0 < f < g$  interpolate between them. That  $f$  cannot exceed  $g$  is not obvious from the definition, since  $X[p](\bar{k})$  sits inside  $X[p]$  of order  $p^{2g}$ , which would allow rank up to  $2g$  a priori; the bound is a consequence of the Frobenius-Verschiebung factorization.

**Proposition 9.4.7: The  $p$ -Rank Bound**

Let  $X$  be an abelian variety of dimension  $g$  over an algebraically closed field  $k$  of characteristic  $p > 0$ , with  $p$ -rank  $f$  defined by  $X[p](\bar{k}) \cong (\mathbb{Z}/p)^f$ . Then

$$0 \leq f \leq g$$

**Proof:**

The lower bound is immediate, since  $f$  is by definition the rank of a finite abelian  $p$ -group, hence nonnegative. For the upper bound, factor multiplication by  $p$  through the relative Frobenius. By definition 9.4.4,  $[p] = V \circ F$  with  $F: X \rightarrow X^{(p)}$  the relative Frobenius and  $V: X^{(p)} \rightarrow X$  the Verschiebung, so

$$X[p] = \ker[p] = \ker(V \circ F)$$

Passing to  $\bar{k}$ -points, a geometric point of  $X[p]$  is a geometric point  $x$  with  $V(F(x)) = 0$ . Frobenius is purely inseparable, so  $F$  is injective on  $\bar{k}$ -points:  $F(x) = F(y)$  forces  $x = y$  because the field extension  $k(X) \supseteq k(X^{(p)})$  under  $F$  is purely inseparable and a purely inseparable isogeny induces a bijection on geometric points. Hence

$$F: X[p](\bar{k}) \longrightarrow \ker(V)(\bar{k})$$

is injective, so  $|X[p](\bar{k})| \leq |\ker(V)(\bar{k})|$ .

*Bounding the points of  $\ker V$ .* The Verschiebung  $V: X^{(p)} \rightarrow X$  is an isogeny of degree  $p^g$ , since  $\deg F = p^g$  and  $\deg[p] = \deg V \cdot \deg F = p^{2g}$  (theorem 8.4.5). Its kernel  $\ker(V)$  is therefore a

finite group scheme of order  $p^g$ , and the number of its geometric points is bounded by its order:

$$|\ker(V)(\bar{k})| \leq |\ker(V)| = p^g$$

Combining the two inequalities,  $|X[p](\bar{k})| \leq p^g$ , that is,  $p^f \leq p^g$ , whence  $f \leq g$ , as we sought to show.

The proof isolates the mechanism cleanly: Frobenius injects the geometric  $p$ -torsion into the kernel of Verschiebung, an order- $p^g$  group scheme, so at most  $p^g$  of the  $p^{2g}$  points of  $X[p]$  can be étale. Equivalently,  $X[p](\bar{k})$  maps injectively under  $F$  into a group of order at most  $p^g$ , and the  $p$ -rank is the rank of the étale part of  $\ker(V)$ . The extremes  $f = g$  and  $f = 0$  are the ordinary and supersingular cases; the ordinary case is the one where  $F$  carries  $X[p](\bar{k})$  isomorphically onto the whole étale part of  $\ker(V)$ .

The elliptic-curve case, where  $g = 1$ , is the origin of the terminology and the setting in which the two extremes are classically distinguished by a single modular quantity.

#### Example 9.4.8: Ordinary and Supersingular Elliptic Curves

Let  $E$  be an elliptic curve over an algebraically closed field  $k$  of characteristic  $p > 0$ , so  $g = 1$  and the  $p$ -rank  $f \in \{0, 1\}$  by proposition 9.4.7. The two cases are:

1.  $E$  is *ordinary*:  $f = 1$ , so  $E[p](\bar{k}) \cong \mathbb{Z}/p$ . The full  $p$ -torsion group scheme  $E[p]$ , of order  $p^2$ , is an extension of the étale  $\mathbb{Z}/p$  by the connected  $\mu_p$ ; thus  $E[p] \cong \mu_p \times \mathbb{Z}/p$  over the perfect field  $\bar{k}$ , and  $E[p^\infty]$  is an extension of an étale height-1 group by  $\mu_{p^\infty}$ .
2.  $E$  is *supersingular*:  $f = 0$ , so  $E[p](\bar{k}) = 0$ . The group scheme  $E[p]$  of order  $p^2$  is connected with connected dual (a nonsplit self-dual extension of  $\alpha_p$  by  $\alpha_p$ ), and  $E[p^\infty]$  is the unique connected  $p$ -divisible group of height 2 and dimension 1.

The two cases are separated by the *Hasse invariant*, the scalar  $A(E) \in k$  by which the relative Frobenius acts on the one-dimensional space  $H^0(E, \Omega_E^1)$  of regular differentials (equivalently, the coefficient of  $x^{p-1}$  in  $f(x)^{(p-1)/2}$  for a Weierstrass model  $y^2 = f(x)$ ): the curve is ordinary when  $A(E) \neq 0$  and supersingular when  $A(E) = 0$ . Supersingular curves are finite in number over  $\mathbb{F}_p$  (for example  $y^2 = x^3 - x$  when  $p \equiv 3 \pmod{4}$ , and  $y^2 = x^3 - 1$  when  $p \equiv 2 \pmod{3}$ ), and every one is defined over  $\mathbb{F}_{p^2}$ ; the ordinary curves are the generic case. The full  $g = 1$  theory of the Hasse invariant, the endomorphism rings (an order in an imaginary quadratic field in the ordinary case, a maximal order in the quaternion algebra ramified at  $p$  and  $\infty$  in the supersingular case), and the supersingular count is developed in *Elliptic Curves* [4].

The dichotomy generalizes only partly to higher dimension. For  $g \geq 2$  the  $p$ -rank  $f$  takes every value in  $\{0, 1, \dots, g\}$ , so ordinary ( $f = g$ ) and supersingular ( $f = 0$ ) are the two extremes of a finer stratification rather than a strict dichotomy, and supersingularity for  $g \geq 2$  is usually defined by the stronger requirement that  $X$  be isogenous to a product of supersingular elliptic curves, which for  $g = 1$  coincides with  $f = 0$  but is stronger than  $f = 0$  in general. The full invariant that classifies  $X[p^\infty]$  up to isogeny over  $\bar{k}$  is its Newton polygon, and up to isomorphism its Dieudonné module; these are the province of Dieudonné theory, which translates  $p$ -divisible groups into semilinear algebra over the Witt vectors. That theory is not developed here. The arithmetic use of the ordinary-supersingular distinction, through the action of Frobenius on the Tate module and the Newton polygon of the characteristic polynomial of Frobenius, is taken up for abelian varieties over finite fields in chapter 13, and the further reading on Dieudonné theory and  $p$ -divisible groups is collected in the closing chapter.

### Exercise 9.4.1

1. Let  $k$  be a field of characteristic  $p > 0$ . Compute the order of each of the following finite group schemes, and in each case decide whether the group scheme is étale by determining  $\dim_k A$  and the number of  $\bar{k}$ -points:  $\mu_p = \operatorname{Spec} k[t]/(t^p - 1)$ ,  $\alpha_p = \operatorname{Spec} k[t]/(t^p)$ , and the constant group scheme  $\mathbb{Z}/p = \operatorname{Spec} (k^{\times p})$  (a product of  $p$  copies of  $k$ ). Which of the three are étale?
2. Let  $X$  be an abelian variety of dimension  $g$  over an algebraically closed field  $k$  with  $\operatorname{char} k = p > 0$ . Using the definition of order (definition 9.4.1) and example 9.4.2, compute the order  $|X[p]|$  and, in terms of the  $p$ -rank  $f$ , the number of geometric points  $|X[p](\bar{k})|$ . For  $g = 2$  and  $f = 1$ , give both numbers explicitly.
3. Let  $k$  be a perfect field of characteristic  $p > 0$  and let  $G$  be a finite group scheme over  $k$  with connected-étale sequence

$$0 \longrightarrow G^0 \longrightarrow G \longrightarrow G^{\text{ét}} \longrightarrow 0$$

as in proposition 9.4.3. Suppose  $|G| = p^3$  and  $|G(\bar{k})| = p$ . Determine the orders  $|G^0|$  and  $|G^{\text{ét}}|$ , and explain why  $G^0$  contributes to the order but not to the point count.

4. Let  $X$  be an abelian variety of dimension  $g$  over a perfect field  $k$  with  $\operatorname{char} k = p > 0$ , and let  $F: X \rightarrow X^{(p)}$  and  $V: X^{(p)} \rightarrow X$  be the relative Frobenius and Verschiebung (definition 9.4.4). Given that  $\deg F = p^g$  and  $\deg[p] = p^{2g}$  (theorem 8.4.5), prove that  $\deg V = p^g$ . Then explain why the étale part of  $\ker(V)$  has order at most  $p^g$ , and identify what the  $p$ -rank  $f$  of  $X$  equals in terms of this étale part.
5. Let  $E$  be an ordinary elliptic curve over an algebraically closed field  $k$  of characteristic  $p > 0$  (example 9.4.8). Using the connected-étale sequence and the isomorphism  $E[p] \cong \mu_p \times \mathbb{Z}/p$  over  $k$ , verify directly that the order and the geometric-point count of  $E[p]$  agree with the general formulas of exercise 9.4.1 (2) for  $g = 1$ ,  $f = 1$ . Then show that a supersingular  $E$  has  $E[p]$  connected, and explain why this forces  $|E[p](\bar{k})| = 1$ .
6. Prove the upper bound  $f \leq g$  of proposition 9.4.7 for the  $p$ -rank of an abelian variety  $X$  of dimension  $g$  over an algebraically closed field  $k$  of characteristic  $p > 0$ , using the factorization  $[p] = V \circ F$ . Your proof should show that Frobenius injects  $X[p](\bar{k})$  into  $\ker(V)(\bar{k})$  and that the target has at most  $p^g$  points.

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## *The Dual Abelian Variety*

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Over  $\mathbb{C}$  the dual of an abelian variety was another torus,  $\hat{X}$ , built from the dual lattice (chapter 5). The construction spent the complex structure freely, and the problem now is to recover the same object over an arbitrary field. The answer is that  $\hat{X}$  is a moduli space: it parametrizes the algebraically trivial line bundles on  $X$ , and so it is the identity component  $\text{Pic}_{X/k}^0$  of the Picard scheme assembled in chapter 7. That  $\text{Pic}_{X/k}^0$  is again a proper connected group variety of dimension  $g$  is what makes duality an operation on abelian varieties rather than a passage out of the category.

A single line bundle on  $X \times \hat{X}$ , the *Poincaré bundle*, realizes the universal property and forces biduality:  $X$  is canonically its own double dual. Against this backdrop the map  $\varphi_L$  of chapter 8 becomes a morphism  $X \rightarrow \hat{X}$ , an isogeny exactly when  $L$  is ample, and a *polarization* is such a map. The Weil pairing reappears as the duality between  $X[n]$  and  $\hat{X}[n]$  with values in  $\mu_n$ , now a statement about finite group schemes valid in every characteristic, and the dual of an isogeny exchanges kernel for Cartier dual. The chapter ends by reading ampleness off  $\varphi_L$  and recovering the Riemann-Roch count  $\dim H^0(X, L) = \sqrt{\deg \varphi_L}$  of chapter 3 with no theta series in sight. The symmetric homomorphisms  $X \rightarrow \hat{X}$  collected here are the raw material for the Rosati involution and Albert's classification in chapter 11.

### 10.1 The Picard Scheme and the Dual Abelian Variety

This section constructs the dual abelian variety over an arbitrary field as the identity component of the Picard scheme, parametrizing the algebraically trivial line bundles on  $X$ , and shows the resulting object is again a  $g$ -dimensional abelian variety, functorial in  $X$ . It is the algebraic replacement for the analytic dual torus, built by hand from antilinear functionals (chapter 5), which no longer has a lattice to dualize.

Recall from chapter 7 that for  $X$  projective over  $k$  the functor  $\text{Pic}_{X/k}$  is representable by a  $k$ -

scheme (definition 7.4.1, theorem 7.4.2), and that its identity component  $\text{Pic}_{X/k}^0$  collects the algebraically trivial classes (definition 7.4.3). The dual is nothing more than this identity component.

**Definition 10.1.1: The Dual Abelian Variety**

Let  $X$  be an abelian variety of dimension  $g$  over a field  $k$ . The *dual abelian variety* of  $X$  is

$$\hat{X} := \text{Pic}_{X/k}^0 = \text{Pic}_{X/k}^0$$

the identity component of the Picard scheme of  $X$  (definition 7.4.3, theorem 7.4.2). Its  $\bar{k}$ -points are the algebraically trivial line bundle classes on  $X_{\bar{k}}$ : for a  $k$ -scheme  $T$ , the group  $\hat{X}(T) = \text{Pic}_{X/k}^0(T)$  consists of the  $T$ -families of degree-zero (algebraically trivial) line bundles on  $X$ , rigidified along the identity section.

The dual is a moduli space, and one should read its points that way: a point of  $\hat{X}$  valued in a field extension  $K/k$  is an algebraically trivial line bundle on  $X_K$ . This is the exact algebraic shadow of the analytic picture, where  $\hat{X} \cong \text{Pic}^0(X)$  was the group of line bundles with trivial first Chern class (proposition 5.1.3). The contrast with the analytic construction is worth stating once: over  $\mathbb{C}$  the dual torus  $\hat{X} = \hat{V}/\hat{\Lambda}$  (definition 5.1.1) was assembled from the  $\mathbb{C}$ -antilinear functionals on  $V$ , an object of transcendental origin, whereas over a general field the same group is extracted purely algebraically as  $\text{Pic}_{X/k}^0$ , with no vector space or lattice in sight. That the two agree over  $\mathbb{C}$  is the analytic/algebraic comparison drawn in chapter 11.

Nothing in definition 10.1.1 yet guarantees that  $\hat{X}$  deserves the name *abelian variety*: a priori  $\text{Pic}_{X/k}^0$  is only a group scheme, the identity component of a scheme whose representability is granted but whose geometry must still be pinned down. The following theorem supplies the three properties an abelian variety requires (it is proper, geometrically connected, and smooth) and computes its dimension.

**Theorem 10.1.2: The Dual Is an Abelian Variety**

Let  $X$  be an abelian variety of dimension  $g$  over  $k$ . Then  $\hat{X} = \text{Pic}_{X/k}^0$  is an abelian variety of dimension  $g$  over  $k$ .

**Proof:**

By theorem 7.4.2 the Picard functor  $\text{Pic}_{X/k}$  is representable by a separated  $k$ -scheme, locally of finite type, and  $\text{Pic}_{X/k}^0$  is by definition its identity component (definition 7.4.3): a connected group scheme, of finite type over  $k$  because it is a single component. To call it an abelian variety is to verify that it is smooth, proper, and geometrically connected, and then to read off its dimension. We argue in four steps.

*Step 1: Geometric connectedness.* The identity component of a group scheme locally of finite type over a field is geometrically connected: base change to  $\bar{k}$  preserves the identity component, since the identity section is a  $k$ -point and a connected component through a rational point is geometrically connected (*Algebraic Geometry* [13]). Thus  $\text{Pic}_{X/k}^0 \times_k \bar{k} = \text{Pic}_{X_{\bar{k}}/\bar{k}}^0$  is connected, which is the assertion.

*Step 2: Smoothness.* A group scheme of finite type over a field is smooth if and only if its dimension equals the dimension of its tangent space at the identity (equivalently, if and only if



it is geometrically reduced); over a perfect field reducedness is automatic, but in general this is a cohomological condition. The tangent space to  $\mathrm{Pic}_{X/k}$  at the identity is  $H^1(X, \mathcal{O}_X)$ , so smoothness of  $\mathrm{Pic}_{X/k}^0$  amounts to the equality  $\dim \mathrm{Pic}_{X/k}^0 = \dim_k H^1(X, \mathcal{O}_X)$ . We establish the equality by bracketing  $\dim \mathrm{Pic}_{X/k}^0$  between two independent bounds.

The upper bound is the tangent-space inequality that holds for any finite-type group scheme:  $\dim \mathrm{Pic}_{X/k}^0 \leq \dim_k T_e \mathrm{Pic}_{X/k}^0 = \dim_k H^1(X, \mathcal{O}_X)$ . Here the right-hand side is computed by the one characteristic-free input this proof uses: the cohomology of the structure sheaf of an abelian variety is the exterior algebra  $H^\bullet(X, \mathcal{O}_X) = \bigwedge^\bullet H^1(X, \mathcal{O}_X)$  on a  $g$ -dimensional space (*Algebraic Geometry* [13]), so  $\dim_k H^1(X, \mathcal{O}_X) = g$  and the upper bound reads  $\dim \mathrm{Pic}_{X/k}^0 \leq g$ .

The lower bound is the content, and it comes from a map out of  $X$  rather than from cohomology. Fix an ample line bundle  $M$  on  $X$ , which exists because  $X$  is projective (theorem 7.3.3). The homomorphism  $\varphi_M: X \rightarrow \mathrm{Pic}_{X/k}^0$ ,  $x \mapsto t_x^* M \otimes M^{-1}$  (definition 8.3.1), has finite kernel  $K(M)$  because  $M$  is ample (proposition 8.3.4). A homomorphism of group schemes with finite kernel has image of dimension equal to  $\dim X = g$ , so the image  $\varphi_M(X)$  is a closed subgroup of dimension  $g$  inside  $\mathrm{Pic}_{X/k}^0$ ; hence  $\dim \mathrm{Pic}_{X/k}^0 \geq g$ . Together with the upper bound this forces  $\dim \mathrm{Pic}_{X/k}^0 = g = \dim_k H^1(X, \mathcal{O}_X)$  in every characteristic, and (since  $\varphi_M(X)$  is then a  $g$ -dimensional closed subgroup of the connected  $g$ -dimensional  $\mathrm{Pic}_{X/k}^0$ )  $\varphi_M$  is surjective. The equality of dimension and tangent dimension dispels the naive characteristic- $p$  non-reducedness worry:  $\mathrm{Pic}_{X/k}^0$  is smooth, a group variety and not just a group scheme.

*Step 3: Properness.* By Steps 1 and 2,  $\mathrm{Pic}_{X/k}^0$  is a smooth, geometrically connected group scheme of finite type over  $k$ , hence separated; properness must be established, and here the completeness of  $X$  is load-bearing. Because  $\mathrm{Pic}_{X/k}^0$  is of finite type, the valuative criterion need only be checked over discrete valuation rings (*Algebraic Geometry* [13]): given a discrete valuation ring  $R$  over  $k$  with fraction field  $K$  and a  $K$ -point  $\xi: \mathrm{Spec} K \rightarrow \mathrm{Pic}_{X/k}^0$ , one must show  $\xi$  extends (uniquely) to an  $R$ -point  $\mathrm{Spec} R \rightarrow \mathrm{Pic}_{X/k}^0$ . Separatedness gives uniqueness of any such extension, so only existence is at issue. The point  $\xi$  is an algebraically trivial line bundle  $L_K$  on  $X_K = X \times_k K$ , rigidified along the identity section; the task is to extend  $L_K$  to a rigidified algebraically trivial line bundle  $L_R$  on  $X_R = X \times_k R$ , whose generic fibre recovers  $L_K$ . The extension is produced from  $L_K$  itself, not from any translate of the ample bundle  $M$ : there is in general no  $K$ -rational point  $x \in X(K)$  realizing the class  $L_K$  as a translate difference  $t_x^* M \otimes M^{-1}$ , since surjectivity of  $\varphi_M$  (Step 2) is surjectivity of schemes, hence only on  $\bar{K}$ -points, and the resulting  $\bar{K}$ -point need not descend to  $K$  (the obstruction is a class in  $H^1(K, K(M))$ , typically nonzero over the fraction field of a DVR). Instead one extends the line bundle directly, and this is where the completeness of  $X$  enters.

Because  $X$  is smooth over  $k$ , the total space  $X_R = X \times_k R$  is a regular integral scheme, flat and proper over  $\mathrm{Spec} R$ , with generic fibre  $X_K$  (*Algebraic Geometry* [13]). On a regular scheme every Weil divisor is Cartier and the restriction  $\mathrm{Pic}(X_R) \rightarrow \mathrm{Pic}(X_K)$  is surjective: a line bundle on the open subscheme  $X_K \subset X_R$  is  $\mathcal{O}(D)$  for a Cartier divisor  $D$ , whose Zariski closure  $\bar{D}$  in  $X_R$  is again a Cartier divisor (regularity), and  $\mathcal{O}_{X_R}(\bar{D})$  restricts to  $L_K$  on the generic fibre. Fix such an extension and call it  $L_R$ ; its generic fibre recovers  $L_K$  by construction. It remains to check two things: that  $L_R$  can be arranged to lie in  $\mathrm{Pic}^0$  on the closed fibre, and that it can be rigidified along the identity section.

For the first, the restriction  $L_R$  defines a morphism  $\mathrm{Spec} R \rightarrow \mathrm{Pic}_{X/k}$  sending the generic point to  $\xi \in \mathrm{Pic}_{X/k}^0$ , hence sending the closed point into the closure of  $\{\xi\}$ . Since  $\mathrm{Pic}_{X/k}^0$  is a connected component of  $\mathrm{Pic}_{X/k}$ , it is open and closed (*Algebraic Geometry* [13]); a specialization of a

point of  $\text{Pic}_{X/k}^0$  therefore remains in  $\text{Pic}_{X/k}^0$ , so the closed fibre of  $L_R$  is again algebraically trivial and the whole  $R$ -point factors through  $\text{Pic}_{X/k}^0$ . Here completeness of  $X$  is what makes  $X_R \rightarrow \text{Spec } R$  proper, so that the flat family of line bundles it carries has a genuine closed fibre and the specialization argument applies. For the second, write  $\pi: X_R \rightarrow \text{Spec } R$  for the structure map and  $e: \text{Spec } R \rightarrow X_R$  for the identity section; replacing  $L_R$  by  $L_R \otimes \pi^*(e^*L_R)^{-1}$  makes the restriction along  $e$  canonically trivial, which is exactly a rigidification, and the twist, being pulled back from the base, does not change the class in  $\text{Pic}^0$  on any fibre. The rigidified bundle is the required  $R$ -point extending  $\xi$ . Hence  $\text{Pic}_{X/k}^0$  satisfies the valuative criterion and is proper. A smooth, geometrically connected, proper group variety over  $k$  is precisely an abelian variety.

*Step 4: Dimension.* The dimension was already pinned down in Step 2, where the isogeny  $\varphi_M$  supplied the lower bound  $\dim \text{Pic}_{X/k}^0 \geq g$  and the tangent space the upper bound  $\dim \text{Pic}_{X/k}^0 \leq \dim_k H^1(X, \mathcal{O}_X) = g$ , so

$$\dim \hat{X} = \dim \text{Pic}_{X/k}^0 = \dim_k H^1(X, \mathcal{O}_X) = g$$

the Hodge number  $h^{0,1}$  of  $X$ . Both bounds are independent of characteristic, so  $\dim \hat{X} = g$  over any field.

Combining the four steps,  $\hat{X} = \text{Pic}_{X/k}^0$  is a smooth, proper, geometrically connected group variety over  $k$  of dimension  $g$ , an abelian variety of dimension  $g$ , as we sought to show.

The theorem is the algebraic reincarnation of the analytic fact that  $\hat{X}$  was again a  $g$ -dimensional complex torus (definition 5.1.1). Both proofs isolate the same numerical coincidence: the dual has the same dimension as  $X$  because the space  $H^1(X, \mathcal{O}_X)$  that measures the dual's dimension is itself  $g$ -dimensional. Analytically this was the statement  $\dim_{\mathbb{C}} \hat{V}^* = g$ ; algebraically it is  $h^{0,1}(X) = g$ . The equality  $\dim \hat{X} = \dim X$  is the first sign that  $X$  and  $\hat{X}$  are dual, not just paired, a symmetry made precise by the biduality theorem of section 10.2, where the Poincaré bundle exhibits  $X$  as the dual of  $\hat{X}$  in turn.

Two inputs cooperate to make  $\text{Pic}_{X/k}^0$  smooth of dimension  $g$ . The exterior-algebra structure of  $H^\bullet(X, \mathcal{O}_X)$  caps the tangent dimension at  $g$ , and the polarization isogeny  $\varphi_M: X \rightarrow \text{Pic}_{X/k}^0$  pushes the dimension back up to  $g$ ; the two bounds meeting is what forces both smoothness and the dimension, in every characteristic. Neither bound alone suffices: the tangent computation is only an upper bound, and it is the map out of  $X$  that supplies the matching lower bound. The construction needs no transcendental input. The representability granted by theorem 7.4.2 supplies the scheme; the exterior-algebra bound, the isogeny  $\varphi_M$ , and the completeness of  $X$  together make that scheme an abelian variety.

The dual is not just an object attached to  $X$  but a functor: a homomorphism of abelian varieties should induce a homomorphism of their duals, in the opposite direction, by pulling back line bundles. This is the algebraic content of contravariance, and it makes  $X \mapsto \hat{X}$  a duality on the category of abelian varieties.

**Proposition 10.1.3: Functoriality of the Dual**

Let  $f: X \rightarrow Y$  be a homomorphism of abelian varieties over  $k$ . Pullback of line bundles induces a homomorphism of abelian varieties

$$\hat{f}: \hat{Y} \rightarrow \hat{X}, \quad [L] \mapsto [f^*L]$$

the *dual homomorphism* of  $f$ . The assignment is contravariantly functorial:

1.  $\widehat{\text{id}_X} = \text{id}_{\hat{X}}$ ;
2.  $\widehat{g \circ f} = \hat{g} \circ \hat{f}$  for homomorphisms  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$ .

**Proof:**

Pullback along  $f$  carries line bundles on  $Y$  to line bundles on  $X$  and respects tensor product, so it is a homomorphism  $f^*: \text{Pic}(Y) \rightarrow \text{Pic}(X)$ . It preserves algebraic triviality: if  $L$  lies in a connected family of line bundles on  $Y$  through  $\mathcal{O}_Y$ , then  $f^*L$  lies in the pulled-back family on  $X$  through  $f^*\mathcal{O}_Y = \mathcal{O}_X$ , so  $f^*L$  is algebraically trivial whenever  $L$  is. Hence  $f^*$  carries  $\text{Pic}_{Y/k}^0$  into  $\text{Pic}_{X/k}^0$ . The construction is uniform in families: pullback along  $f \times \text{id}_T$  carries a  $T$ -family of algebraically trivial line bundles on  $Y$  to a  $T$ -family on  $X$ , compatibly with base change in  $T$ , so it defines a natural transformation of the functors  $\hat{Y}$  and  $\hat{X}$  (definition 10.1.1); by Yoneda this natural transformation is a morphism of  $k$ -schemes  $\hat{f}: \hat{Y} \rightarrow \hat{X}$ . It sends the identity class to the identity class and respects tensor product, so it is a homomorphism of the group schemes, and by theorem 10.1.2 a homomorphism of abelian varieties.

For functoriality we compute on classes. The identity map on  $X$  pulls back every line bundle to itself, so  $\widehat{\text{id}_X}[L] = [\text{id}_X^*L] = [L]$ , giving  $\widehat{\text{id}_X} = \text{id}_{\hat{X}}$ . For the composite, functoriality of pullback gives  $(g \circ f)^*L = f^*(g^*L)$  for every line bundle  $L$  on  $Z$ , so

$$\widehat{g \circ f}[L] = [(g \circ f)^*L] = [f^*(g^*L)] = \hat{f}[g^*L] = (\hat{f} \circ \hat{g})[L]$$

and hence  $\widehat{g \circ f} = \hat{g} \circ \hat{f}$ , completing the proof.

Contravariance reverses arrows: a map  $X \rightarrow Y$  produces a map  $\hat{Y} \rightarrow \hat{X}$ , exactly as the analytic dual  $\hat{f}: \hat{Y} \rightarrow \hat{X}$  did over  $\mathbb{C}$  (the analytic proposition 5.1.4). The two identities say that dualization is a contravariant functor from abelian varieties to abelian varieties. Applied twice it returns a covariant functor  $X \mapsto \hat{\hat{X}}$ , and biduality (section 10.2) is that this double dual is canonically  $X$  itself, so dualization is very nearly an involution on the category.

The smallest case reconciles the construction with the elliptic-curve theory the reader already knows.

**Example 10.1.4: The Dual of an Elliptic Curve**

Let  $E$  be an elliptic curve over  $k$ , an abelian variety of dimension  $g = 1$ . Its dual is

$$\hat{E} = \text{Pic}_{E/k}^0 = \text{Pic}_{E/k}^0$$

the connected component of the Picard scheme parametrizing degree-zero line bundles on  $E$ . By theorem 10.1.2 it is an abelian variety of dimension 1, that is, an elliptic curve. The classical

identification of the degree-zero divisor class group with the points of  $E$  (sending a point  $P$  to the class  $[P] - [O]$ , where  $O$  is the origin) is an isomorphism of groups, and in fact an isomorphism of abelian varieties

$$E \xrightarrow{\sim} \hat{E}, \quad P \mapsto [(P) - (O)]$$

developed in full for  $g = 1$  in *Elliptic Curves* [4]. Thus an elliptic curve is canonically isomorphic to its own dual: it is *principally polarized*, the isomorphism  $E \cong \hat{E}$  being the principal polarization  $\varphi_L$  attached to the degree-one line bundle  $L = \mathcal{O}_E(O)$  (section 10.3).

This recovers, over an arbitrary field, the analytic statement  $\hat{E} \cong E$  obtained over  $\mathbb{C}$  from the dual torus of chapter 5. Over  $\mathbb{C}$  the isomorphism came from the antilinear-functional description of  $\hat{V}$  collapsing, in dimension one, to  $V$  itself; here it comes from the Abel-Jacobi identification of  $E$  with its degree-zero Picard group. The self-duality visible already for elliptic curves persists for all Jacobians (chapter 12), where the theta divisor supplies a canonical principal polarization; a general abelian variety, by contrast, need not be isomorphic to its dual.

### Exercise 10.1.1

1. Let  $X$  be an abelian variety of dimension  $g$  over  $k$ . Using definition 10.1.1 and theorem 10.1.2, state what  $\dim \hat{X}$  is and identify the tangent space  $T_e \hat{X}$  at the identity with a cohomology group of  $X$ . Then, for  $X$  an abelian surface ( $g = 2$ ), compute  $\dim_k H^\bullet(X, \mathcal{O}_X)$  in each degree and read off  $\dim_k H^1(X, \mathcal{O}_X)$ , confirming it equals  $\dim \hat{X}$ .
2. In Step 2 of the proof of theorem 10.1.2, smoothness of  $\text{Pic}_{X/k}^0$  is deduced from the equality  $\dim \text{Pic}_{X/k}^0 = \dim_k H^1(X, \mathcal{O}_X)$ . Explain why, for a finite-type group scheme over  $k$ , the inequality  $\dim \text{Pic}_{X/k}^0 \leq \dim_k T_e \text{Pic}_{X/k}^0$  always holds, and why the reverse inequality (hence equality, hence smoothness) is precisely what fails in general for a group scheme that is not smooth. Give the standard example over a field of characteristic  $p$  of a finite-type group scheme whose dimension is strictly less than its tangent dimension at the identity.
3. The lower bound  $\dim \text{Pic}_{X/k}^0 \geq g$  in Step 2 comes from the homomorphism  $\varphi_M: X \rightarrow \text{Pic}_{X/k}^0$  attached to an ample  $M$ , which has finite kernel. Prove in general that a homomorphism  $f: X \rightarrow G$  from an abelian variety of dimension  $g$  to a group scheme of finite type over  $k$ , with finite kernel, has image a closed subgroup of dimension  $g$ ; conclude that if  $G$  is connected of dimension  $g$  then  $f$  is surjective. Then explain why this last surjectivity of  $\varphi_M$  is nonetheless not enough, on its own, to extend a  $K$ -point of  $\hat{X}$  over a discrete valuation ring in the properness argument of Step 3.
4. Let  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  be homomorphisms of abelian varieties over  $k$ , and suppose  $f$  is an isomorphism. Using proposition 10.1.3, prove that  $\hat{f}: \hat{Y} \rightarrow \hat{X}$  is an isomorphism with inverse  $\widehat{f^{-1}}$ . Deduce that if  $g \circ f$  is an isomorphism then  $\hat{g}$  is a split injection and  $\hat{f}$  a split surjection, and identify the splitting.
5. Let  $E$  be an elliptic curve over  $k$  with origin  $O$ , and let  $f: E \rightarrow E$  be the multiplication-by- $n$  endomorphism,  $n \geq 1$ . Under the self-duality isomorphism  $E \xrightarrow{\sim} \hat{E}$ ,  $P \mapsto [(P) - (O)]$  of example 10.1.4, prove that the dual homomorphism  $\hat{f}: \hat{E} \rightarrow \hat{E}$  corresponds again to multiplication by  $n$  on  $E$ . (Compute  $\hat{f}$  on classes  $[(P) - (O)]$  using proposition 10.1.3 and the description of pullback of degree-zero classes along an isogeny.)

## 10.2 The Poincaré Bundle and Biduality

The moduli space  $\hat{X} = \text{Pic}_{X/k}^0$  of section 10.1 carries a universal family, an object on the product  $X \times \hat{X}$  whose restriction to each slice  $X \times \{\xi\}$  is the algebraically trivial line bundle that  $\xi$  names. That family is the Poincaré bundle, and everything this section proves is read off it. Its universal property recovers the moduli description in the language of maps into  $\hat{X}$ , and applying that property to the family on  $X \times X$  produced by  $\hat{X}$  itself forces biduality: the canonical map  $X \rightarrow \hat{X}$  is an isomorphism, so passing to the dual is an involution on abelian varieties, matching the analytic  $\hat{\hat{X}} = X$  of theorem 5.1.5 without any lattice.

The bundle is not quite unique as stated, because a line bundle on  $X \times \hat{X}$  can be twisted by a bundle pulled back from  $\hat{X}$  without changing any of its slices over  $\hat{X}$ . Rigidifying along  $\{e\} \times \hat{X}$  removes exactly this ambiguity and pins the universal family down.

### Definition 10.2.1: The Poincaré Bundle

Let  $X$  be an abelian variety over  $k$  with dual  $\hat{X} = \text{Pic}_{X/k}^0$  (definition 10.1.1), and let  $e \in X(k)$  be the identity. The *Poincaré bundle* of  $X$  is the line bundle  $\mathcal{P}$  on  $X \times \hat{X}$  characterized by two properties:

1. for each point  $\xi \in \hat{X}$  with residue field  $k(\xi)$ , the restriction  $\mathcal{P}|_{X \times \{\xi\}}$  is the line bundle on  $X_{k(\xi)}$  whose class in  $\text{Pic}^0(X_{k(\xi)})$  is  $\xi$ ;
2.  $\mathcal{P}$  is *rigidified along*  $\{e\} \times \hat{X}$ , that is, equipped with a trivialization of  $\mathcal{P}|_{\{e\} \times \hat{X}}$ .

Equivalently,  $\mathcal{P}$  is the universal line bundle representing the functor  $\text{Pic}_{X/k}^0$ : it is the line bundle on  $X \times \hat{X}$  corresponding to the identity morphism  $\text{id}_{\hat{X}} \in \hat{X}(\hat{X}) = \text{Pic}_{X/k}^0(\hat{X})$ .

The two clauses do different work. The first says  $\mathcal{P}$  is a family: as  $\xi$  ranges over  $\hat{X}$  its slices run through every algebraically trivial bundle on  $X$ , each occurring exactly once, so  $\mathcal{P}$  is the tautological family over the moduli space. The second clause removes the slack. Any line bundle  $N$  on  $\hat{X}$  pulls back to  $\text{pr}_2^*N$  on  $X \times \hat{X}$ , whose restriction to each slice  $X \times \{\xi\}$  is trivial, so  $\mathcal{P}$  and  $\mathcal{P} \otimes \text{pr}_2^*N$  have identical slices over  $\hat{X}$  and clause (1) alone cannot tell them apart. Demanding a trivialization along  $\{e\} \times \hat{X}$  fixes the choice, because a bundle trivial on every  $X$ -slice and on the section  $\{e\} \times \hat{X}$  is trivial. That  $\mathcal{P}$  exists is the representability of  $\text{Pic}_{X/k}^0$  (theorem 7.4.2, imported from *Algebraic Geometry* [13]): a fine moduli space is precisely one carrying a universal object, and the universal object is  $\mathcal{P}$ .

The counterpart over  $\mathbb{C}$  is concrete. There the dual torus is  $\hat{X} = \hat{V}/\hat{\Lambda}$  and the algebraically trivial bundles are the  $L(0, \chi)$  of the Appell-Humbert classification (theorem 2.2.8); the Poincaré bundle is assembled by letting the character  $\chi$  vary holomorphically with  $\xi \in \hat{X}$ , giving a single bundle on  $X \times \hat{X}$  restricting to  $L(0, \chi_\xi)$  on each slice. The algebraic definition packages the same idea with no reference to characters.

The defining feature of the Poincaré bundle is that it converts families of trivial bundles into morphisms to  $\hat{X}$ . This is the universal property, and it drives the duality theory: constructing a map  $T \rightarrow \hat{X}$  is the same as producing a suitable line bundle on  $X \times T$ .

**Proposition 10.2.2: Universal Property of the Poincaré Bundle**

Let  $X$  be an abelian variety over  $k$  with Poincaré bundle  $\mathcal{P}$  on  $X \times \hat{X}$ . For every  $k$ -scheme  $T$  and every line bundle  $L$  on  $X \times T$  whose restriction  $L|_{X \times \{t\}}$  lies in  $\text{Pic}^0(X_{k(t)})$  for all  $t \in T$ , there is a unique morphism  $g: T \rightarrow \hat{X}$  and a line bundle  $N$  on  $T$  with

$$L \cong (\text{id}_X \times g)^* \mathcal{P} \otimes \text{pr}_T^* N$$

If in addition  $L$  is rigidified along  $\{e\} \times T$ , then  $N$  is trivial and the isomorphism  $L \cong (\text{id}_X \times g)^* \mathcal{P}$  is unique.

**Proof:**

A line bundle  $L$  on  $X \times T$  all of whose  $X$ -slices are algebraically trivial is, by definition of the Picard functor (definition 7.4.1), a  $T$ -point of  $\text{Pic}_{X/k}^0$ , that is, an element of  $\text{Pic}_{X/k}^0(T) = \hat{X}(T)$ , at least once  $L$  is normalized so that its class is well defined; the ambiguity in the normalization is exactly a line bundle pulled back from  $T$ .

*Step 1: The rigidified case.* Suppose first that  $L$  is rigidified along  $\{e\} \times T$ , meaning  $L|_{\{e\} \times T}$  is trivialized. Then the class of  $L$  in  $\text{Pic}_{X/k}^0(T)$  is a well-defined element with no residual ambiguity, because the Picard functor is defined as line bundles rigidified along the identity section modulo isomorphism (definition 7.4.1). By representability (theorem 7.4.2) this element is a unique morphism  $g: T \rightarrow \hat{X}$ , and the defining property of the universal bundle is that  $g$  is the morphism for which  $(\text{id}_X \times g)^* \mathcal{P} \cong L$  as rigidified bundles. Uniqueness of  $g$  is the injectivity half of representability: two morphisms  $T \rightarrow \hat{X}$  pulling  $\mathcal{P}$  back to isomorphic rigidified bundles represent the same  $T$ -point of  $\text{Pic}_{X/k}^0$ , hence coincide.

*Step 2: The general case.* If  $L$  is not rigidified, choose any trivialization of  $L|_{\{e\} \times T}$  after twisting: the restriction  $M := L|_{\{e\} \times T}$  is a line bundle on  $\{e\} \times T \cong T$ , and the twisted bundle  $L' := L \otimes \text{pr}_T^* M^{-1}$  satisfies

$$L'|_{\{e\} \times T} = M \otimes M^{-1} \cong \mathcal{O}_T$$

so  $L'$  is rigidified along  $\{e\} \times T$ . Each  $X$ -slice of  $L'$  differs from that of  $L$  by the pullback of a bundle from a point, so  $L'|_{X \times \{t\}} \cong L|_{X \times \{t\}}$  lies in  $\text{Pic}^0$ ; Step 1 applies to  $L'$  and yields a unique  $g: T \rightarrow \hat{X}$  with  $L' \cong (\text{id}_X \times g)^* \mathcal{P}$ . Setting  $N := M$  gives

$$L \cong L' \otimes \text{pr}_T^* M \cong (\text{id}_X \times g)^* \mathcal{P} \otimes \text{pr}_T^* N$$

as claimed. The morphism  $g$  is unique because it is determined by the rigidified bundle  $L'$ , and  $L'$  is determined by  $L$  up to the pullback  $\text{pr}_T^* M$ , which does not change any  $X$ -slice; the seesaw principle (theorem 7.4.5) then identifies the discrepancy between two such representations as a pullback from  $T$ , absorbed into  $N$ . This completes the proof.

The universal property is the statement that  $\hat{X}$  represents the functor  $T \mapsto \{\text{line bundles on } X \times T \text{ trivial on each } X\text{-slice}\} / (\text{pullbacks from } T)$ , with  $\mathcal{P}$  the universal element. Read from right to left it is a recipe: to build a morphism into the dual, produce a line bundle on  $X \times T$  whose slices are algebraically trivial, and the morphism falls out. This is how every map to  $\hat{X}$  in the sequel is constructed,  $\varphi_L: X \rightarrow \hat{X}$  among them, whose defining bundle  $m^* L \otimes \text{pr}_1^* L^{-1} \otimes \text{pr}_2^* L^{-1}$  on  $X \times X$

(definition 8.3.1) is exactly a family of the required kind. The interchangeability of families and maps is the algebraic residue of the analytic fact that  $\hat{X}$  parametrizes characters of  $\Lambda$ .

Applying the universal property to  $\hat{X}$  in place of  $T$ , with the Poincaré bundle's own transpose as the family, produces the map that biduality asserts to be an isomorphism. The Poincaré bundle lives on  $X \times \hat{X}$ , a product symmetric in its two factors up to swapping them; feeding the swapped bundle into the universal property of the dual of  $\hat{X}$  gives a canonical map from  $X$  to  $\hat{\hat{X}}$ , and the symmetry of  $\mathcal{P}$  is what makes it an isomorphism.

### Theorem 10.2.3: Biduality

Let  $X$  be an abelian variety over  $k$ . The canonical homomorphism

$$\kappa: X \longrightarrow \hat{\hat{X}}$$

that sends a point  $x \in X$  to the class in  $\hat{\hat{X}} = \text{Pic}_{\hat{X}/k}^0$  of the line bundle  $\mathcal{P}|_{\{x\} \times \hat{X}}$  on  $\hat{X}$  is an isomorphism of abelian varieties. Under  $\kappa$  the Poincaré bundle of  $\hat{X}$  is identified with the swap  $s^*\mathcal{P}$  of the Poincaré bundle of  $X$ , where  $s: \hat{X} \times X \rightarrow X \times \hat{X}$  interchanges the factors.

#### Proof:

*Step 1:  $\kappa$  is a well-defined homomorphism.* The slices  $\mathcal{P}|_{\{x\} \times \hat{X}}$  assemble into a single line bundle on the product  $\hat{X} \times X$ , namely the swap  $s^*\mathcal{P}$  viewed as a family over the second factor  $X$ . Each of its slices over  $X$  is a line bundle on  $\hat{X}$ ; that these slices are algebraically trivial, hence lie in  $\text{Pic}_{\hat{X}/k}^0 = \hat{\hat{X}}$ , follows because  $\mathcal{P}|_{\{e\} \times \hat{X}}$  is trivial by the rigidification (definition 10.2.1), so the slice over  $x = e$  is trivial and the whole family is a deformation of the trivial bundle, staying inside the identity component  $\text{Pic}_{\hat{X}/k}^0$ . By the universal property of the Poincaré bundle of  $\hat{X}$  (proposition 10.2.2 applied with  $X$  replaced by  $\hat{X}$  and  $T = X$ ), this family determines the unique morphism  $\kappa: X \rightarrow \hat{\hat{X}}$  pulling the Poincaré bundle of  $\hat{X}$  back to  $s^*\mathcal{P}$ , and  $\kappa(e) = e$  because the slice over  $e$  is trivial. A morphism of abelian varieties fixing the identity is a homomorphism (corollary 7.2.2), so  $\kappa$  is a homomorphism.

*Step 2: Nondegeneracy of  $\mathcal{P}$ .* The argument turns on  $\mathcal{P}$  pairing the two factors without degeneracy: the only  $\xi \in \hat{X}(\bar{k})$  with  $\mathcal{P}|_{X \times \{\xi\}}$  trivial is  $\xi = 0$ , and the only  $x \in X(\bar{k})$  with  $\mathcal{P}|_{\{x\} \times \hat{X}}$  trivial is  $x = e$ . The first statement is immediate from clause (1) of definition 10.2.1, since  $\mathcal{P}|_{X \times \{\xi\}}$  represents the class  $\xi$ , trivial exactly when  $\xi = 0$ . The second is the substance. Let  $x \in \ker \kappa(\bar{k})$ , so  $\mathcal{P}|_{\{x\} \times \hat{X}}$  is trivial. The set  $\ker \kappa$  is a closed subgroup scheme of  $X$ ; were it positive-dimensional then, working over  $\bar{k}$  so that the reduced identity component  $(\ker \kappa)_{\text{red}}^0$  is a smooth connected subgroup (a subvariety over the perfect field  $\bar{k}$ ), it would contain a nonzero abelian subvariety  $X_0$ , and  $\mathcal{P}|_{\{y\} \times \hat{X}}$  would be trivial for every  $y \in X_0$ . View  $\mathcal{P}|_{X_0 \times \hat{X}}$  as a family on  $(\hat{X}) \times X_0$  with complete first factor  $\hat{X}$  and base  $X_0$ ; its slice over each  $y \in X_0$  is  $\mathcal{P}|_{\{y\} \times \hat{X}}$ , trivial for all  $y$ . The seesaw principle (theorem 7.4.5), whose complete factor is  $\hat{X}$  here, then gives  $\mathcal{P}|_{X_0 \times \hat{X}} \cong \text{pr}_{X_0}^* N$  for a line bundle  $N$  on  $X_0$ , and restricting to  $X_0 \times \{0\}$  identifies  $N$  with  $\mathcal{P}|_{X_0 \times \{0\}}$ , which is trivial; so  $\mathcal{P}|_{X_0 \times \hat{X}}$  is trivial. But then, restricting to slices in the other direction,  $\mathcal{P}|_{X_0 \times \{\xi\}}$  is trivial for every  $\xi$ , so every algebraically trivial class of  $X$  restricts trivially to  $X_0$ . Now the restriction of algebraically trivial classes from  $X$  to  $X_0$  is the map



$\hat{\iota}: \hat{X} \rightarrow \hat{X}_0$  dual to the inclusion  $\iota: X_0 \hookrightarrow X$  (proposition 10.1.3), and this dual is surjective. To see it directly, form the quotient abelian variety  $Q := X/X_0$  with projection  $\pi: X \rightarrow Q$ , so that  $0 \rightarrow X_0 \xrightarrow{\iota} X \xrightarrow{\pi} Q \rightarrow 0$  is exact; dualizing sends this to  $\hat{Q} \xrightarrow{\hat{\pi}} \hat{X} \xrightarrow{\hat{\iota}} \hat{X}_0$  (proposition 10.1.3). The map  $\hat{\pi}$  is injective, because a class on  $Q$  pulling back trivially along the surjection  $\pi$  is trivial, so  $\dim(\text{im } \hat{\pi}) = \dim \hat{Q} = \dim Q = g - \dim X_0$ . Since  $\hat{\iota} \circ \hat{\pi} = \widehat{\pi \circ \iota} = 0$ , the image of  $\hat{\pi}$  lies in  $\ker \hat{\iota}$ ; but  $\ker \hat{\iota}$  can be no larger, as dualizing the exact sequence of abelian varieties is exact (*Algebraic Geometry* [13], the Picard scheme of a quotient), whence  $\ker \hat{\iota} = \text{im } \hat{\pi}$  has dimension  $g - \dim X_0$ . Its image therefore has dimension  $\dim \hat{X} - (g - \dim X_0) = \dim X_0 = \dim \hat{X}_0$ , so  $\hat{\iota}$  is surjective. Thus every algebraically trivial class on  $X_0$  is the restriction of one on  $X$ . But  $X_0$ , being a nonzero abelian variety, carries a nontrivial algebraically trivial bundle (for instance  $t_a^*M \otimes M^{-1}$  for  $M$  ample and  $a$  generic, nontrivial by definition 8.3.1), so some class on  $X$  restricts nontrivially to  $X_0$ , contradicting the previous sentence. Hence  $\ker \kappa$  is finite.

*Step 3:  $\kappa$  is an isogeny, then an isomorphism.* Both  $X$  and  $\hat{\hat{X}}$  are abelian varieties of dimension  $g$ , the latter because  $\hat{X}$  is (theorem 10.1.2) and dualizing preserves dimension. A homomorphism of abelian varieties of the same dimension with finite kernel is surjective, hence an isogeny; by Step 2 this applies to  $\kappa = \kappa_X$ . It remains to see its degree is 1. Write  $\kappa_Y$  for the canonical biduality map  $Y \rightarrow \hat{\hat{Y}}$  of an abelian variety  $Y$ ; the construction is functorial, so  $\kappa_Y$  is defined for every  $Y$  and is an isogeny by Steps 1 and 2 applied to  $Y$ . Dualizing the isogeny  $\kappa = \kappa_X: X \rightarrow \hat{X}$  gives  $\hat{\kappa}: \hat{\hat{X}} \rightarrow \hat{X}$  (proposition 10.1.3), a homomorphism with target  $\hat{X}$  and itself an isogeny (the dual of an isogeny is an isogeny); the biduality map of  $\hat{X}$  is the isogeny  $\kappa_{\hat{X}}: \hat{X} \rightarrow \hat{\hat{X}}$ . These compose as  $\hat{X} \xrightarrow{\kappa_{\hat{X}}} \hat{\hat{X}} \xrightarrow{\hat{\kappa}} \hat{X}$ , and the triangular identity  $\hat{\kappa} \circ \kappa_{\hat{X}} = \text{id}_{\hat{X}}$  holds. To verify it, evaluate the composite on a  $\bar{k}$ -point  $\eta \in \hat{X}$ ; since both sides are homomorphisms of abelian varieties, agreement on  $\bar{k}$ -points forces equality. By its definition (Step 1 applied to  $\hat{X}$ ),  $\kappa_{\hat{X}}(\eta)$  is the class in  $\hat{\hat{X}} = \text{Pic}_{\hat{X}/k}^0$  of the slice  $\mathcal{P}_{\hat{X}}|_{\{\eta\} \times \hat{X}}$ , where  $\mathcal{P}_{\hat{X}}$  is the Poincaré bundle of  $\hat{X}$  on  $\hat{X} \times \hat{X}$ . Its image under  $\hat{\kappa} = \widehat{\kappa_X}$ , which is pullback of line bundles along  $\kappa = \kappa_X: X \rightarrow \hat{X}$  (proposition 10.1.3), is the class on  $X$  of  $\kappa_X^*(\mathcal{P}_{\hat{X}}|_{\{\eta\} \times \hat{X}})$ , that is, the slice over  $\{\eta\}$  of the bundle  $(\text{id}_{\hat{X}} \times \kappa_X)^*\mathcal{P}_{\hat{X}}$  on  $\hat{X} \times X$ . But the symmetry established in Step 1 is exactly  $(\text{id}_{\hat{X}} \times \kappa_X)^*\mathcal{P}_{\hat{X}} \cong s^*\mathcal{P}$ , so this slice is  $s^*\mathcal{P}|_{\{\eta\} \times X} = \mathcal{P}|_{X \times \{\eta\}}$ , whose class is  $\eta$  itself by clause (1) of definition 10.2.1. Hence  $\hat{\kappa}(\kappa_{\hat{X}}(\eta)) = \eta$  for every  $\eta$ , so  $\hat{\kappa} \circ \kappa_{\hat{X}} = \text{id}_{\hat{X}}$ . Taking degrees of  $\hat{\kappa} \circ \kappa_{\hat{X}} = \text{id}_{\hat{X}}$  gives  $\deg \hat{\kappa} \cdot \deg \kappa_{\hat{X}} = 1$ , so both factors are 1; in particular  $\deg \hat{\kappa} = 1$ . But the dual of an isogeny has the same degree as the isogeny (*Algebraic Geometry* [13], via the Cartier duality  $\ker \hat{f} = (\ker f)^\vee$  taken up in section 10.4), so  $\deg \kappa_X = \deg \hat{\kappa} = 1$ . An isogeny of degree 1 is an isomorphism, so  $\kappa = \kappa_X$  is an isomorphism of abelian varieties, as we sought to show. The symmetry statement is then read off Step 1: under the isomorphism  $\kappa$ , the Poincaré bundle of  $\hat{X}$  pulls back along  $\text{id}_{\hat{X}} \times \kappa$  to  $s^*\mathcal{P}$ .

Biduality is the structural fact that makes duality a symmetry rather than a one-way construction:  $X$  and  $\hat{\hat{X}}$  are canonically the same abelian variety, and the identification is compatible with every homomorphism, so one dualizes freely and writes  $\hat{\hat{X}} = X$  throughout. Its content is the nondegeneracy of the Poincaré bundle, which pairs  $X$  against  $\hat{X}$  with neither side carrying a kernel; the two nondegeneracy statements of Step 2 are the two halves of that pairing, and the theorem is their symmetric consequence. Under  $\hat{\hat{X}} = X$  the dual of a homomorphism  $f: X \rightarrow Y$  has  $\hat{\hat{f}} = f$ , and the



dual of  $\varphi_L: X \rightarrow \hat{X}$  is again a map  $X \rightarrow \hat{X}$  that can be compared with  $\varphi_L$  itself, the comparison that will single out polarizations among all maps to the dual in section 10.3. This reproduces algebraically the analytic theorem 5.1.5, where  $\hat{X} = X$  came from the two layers of antilinearity in  $\hat{V} = \bar{V}^*$ ; the projective proof replaces that transcendental input with the seesaw principle and the universal family. The construction is most transparent for an elliptic curve, where the dual is the curve itself and the Poincaré bundle is a bundle on  $E \times E$  already familiar from the theorem of the cube.

**Example 10.2.4: The Poincaré Bundle on an Elliptic Curve**

Let  $E$  be an elliptic curve over  $k$  with origin  $O$ . Its dual is  $\hat{E} = \text{Pic}^0(E) \cong E$ , the isomorphism sending a point  $Q \in E$  to the class  $[\mathcal{O}_E(Q - O)]$  of the degree-zero divisor  $Q - O$  (*Elliptic Curves* [4]); under it the identity  $O$  of  $E$  corresponds to the trivial class  $0 \in \hat{E}$ . The Poincaré bundle on  $E \times \hat{E} \cong E \times E$  is

$$\mathcal{P} \cong \mathcal{O}_{E \times E}(\Delta - (\{O\} \times E) - (E \times \{O\}))$$

where  $\Delta \subseteq E \times E$  is the diagonal. Its restriction to the slice  $E \times \{Q\}$  is  $\mathcal{O}_E(\Delta|_{E \times \{Q\}} - \{O\}) = \mathcal{O}_E(Q - O)$ , precisely the degree-zero bundle that the point  $Q$  names in  $\text{Pic}^0(E)$ , so clause (1) of definition 10.2.1 holds; and its restriction to  $\{O\} \times E$  is  $\mathcal{O}_E(\{O\} - \{O\}) = \mathcal{O}_E$ . Here the two components  $\Delta$  (entering with  $+$ ) and  $E \times \{O\}$  (entering with  $-$ ) each meet  $\{O\} \times E$  transversally in the single point  $(O, O)$ , contributing  $\mathcal{O}_E(\{O\})$  and  $\mathcal{O}_E(\{O\})^{-1}$ ; the component  $\{O\} \times E$  meets the slice non-transversally, being the slice itself, so its contribution is not a naive divisor pullback but the degree-zero normal-bundle term  $\mathcal{O}_E$  (the self-intersection  $(\{O\} \times E)^2 = 0$  on  $E \times E$ ). The three combine to  $\mathcal{O}_E(\{O\}) \otimes \mathcal{O}_E(\{O\})^{-1} \otimes \mathcal{O}_E \cong \mathcal{O}_E$ , so  $\mathcal{P}$  is rigidified along  $\{O\} \times \hat{E}$ , giving clause (2). This is the same bundle whose nontriviality was noted in example 8.1.3 to show that two coordinate axes do not force triviality:  $\mathcal{P}$  is trivial on both  $\{O\} \times E$  and  $E \times \{O\}$  yet nontrivial overall, its  $E$ -fibres running through all of  $\text{Pic}^0(E)$ . This is why the theorem of the cube (theorem 8.1.1) needs all three coordinate restrictions to be trivial, not two.

Biduality here is the statement that the canonical map  $\kappa: E \rightarrow \hat{\hat{E}}$  is an isomorphism, and it is transparent from the symmetry of the divisor  $\Delta - (\{O\} \times E) - (E \times \{O\})$  under interchanging the two factors of  $E \times E$ : the swap  $s$  fixes  $\Delta$  and exchanges the two axes, so  $s^*\mathcal{P} \cong \mathcal{P}$ , and the map  $\kappa$  reading a point  $x$  off the slice  $\mathcal{P}|_{\{x\} \times \hat{E}} = \mathcal{O}_E(x - O)$  is the identity isomorphism  $E \xrightarrow{\sim} \hat{\hat{E}} \cong E$ . The nondegeneracy of  $\mathcal{P}$  is the fact that  $\mathcal{O}_E(Q - O)$  is trivial only for  $Q = O$ , which is the injectivity of the Abel-Jacobi map  $E \rightarrow \text{Pic}^0(E)$ . The whole biduality theorem is, for  $g = 1$ , the statement that this map is a bijection, recovering the  $\hat{E} \cong E$  of example 5.1.2 and the analytic  $\hat{E} = E$  of theorem 5.1.5.

**Exercise 10.2.1**

1. State the two properties characterizing the Poincaré bundle  $\mathcal{P}$  of an abelian variety  $X$  (definition 10.2.1). Show that the second property is not redundant: exhibit, for any nonzero line bundle  $N$  on  $\hat{X}$ , a line bundle on  $X \times \hat{X}$  that satisfies the first property but is not isomorphic to  $\mathcal{P}$ , and explain why it fails the second.
2. Let  $E$  be an elliptic curve with origin  $O$ , and use the description of its Poincaré bundle in example 10.2.4. Compute the restriction of  $\mathcal{P} \cong \mathcal{O}_{E \times E}(\Delta - (\{O\} \times E) - (E \times \{O\}))$  to

- the slice  $\{O\} \times E$ , and verify directly that it is trivial. Then compute the restriction to the diagonal  $\Delta$ , identified with  $E$  via the first projection, and determine its degree.
3. Let  $L$  be a line bundle on  $X \times T$  whose slice  $L|_{X \times \{t\}}$  is algebraically trivial for every  $t \in T$ , and suppose  $L$  is rigidified along  $\{e\} \times T$ ; let  $g: T \rightarrow \hat{X}$  be the classifying morphism proposition 10.2.2 produces. For any morphism  $\tau: T' \rightarrow T$ , show that the pulled-back bundle  $(\text{id}_X \times \tau)^*L$  on  $X \times T'$  is again a rigidified family with algebraically trivial slices, and that its classifying morphism is  $g \circ \tau$ . Deduce that the assignment  $L \mapsto g$  is natural in  $T$ .
  4. Let  $f: X \rightarrow Y$  be an isogeny of abelian varieties, with dual isogeny  $\hat{f}: \hat{Y} \rightarrow \hat{X}$ . Granting biduality (theorem 10.2.3) and the functoriality of dualization (proposition 10.1.3), prove that  $\hat{\hat{f}} = f$  under the identifications  $\kappa_X: X \xrightarrow{\sim} \hat{\hat{X}}$  and  $\kappa_Y: Y \xrightarrow{\sim} \hat{\hat{Y}}$ ; that is, prove  $\kappa_Y \circ f = \hat{\hat{f}} \circ \kappa_X$ .
  5. In Step 2 of the proof of theorem 10.2.3 the key input is that  $\mathcal{P}$  pairs  $X$  against  $\hat{X}$  without degeneracy: the only  $\xi \in \hat{X}(\bar{k})$  with  $\mathcal{P}|_{X \times \{\xi\}}$  trivial is  $\xi = 0$ . Prove this half of the nondegeneracy directly from clause (1) of definition 10.2.1. Then, assuming the other half (that  $\mathcal{P}|_{\{x\} \times \hat{X}}$  is trivial only for  $x = e$ ), show that the canonical map  $\kappa: X \rightarrow \hat{X}$  has finite kernel.

### 10.3 Polarizations and the Map $\varphi_L$

The homomorphism  $\varphi_L$  was constructed in chapter 8 at the level of group functors: for a line bundle  $L$  on  $X$ , the assignment  $x \mapsto [t_x^*L \otimes L^{-1}]$  is a homomorphism  $X \rightarrow \text{Pic}^0(X)$ , with kernel the group scheme  $K(L)$ . At that stage  $\text{Pic}^0(X)$  was available only as an abstract group, so  $\varphi_L$  was a map of group functors. Now that the dual  $\hat{X} = \text{Pic}_{X/k}^0$  is an abelian variety in its own right (definition 10.1.1), nothing about  $\varphi_L$  as a map of points changes but its target upgrades:  $\varphi_L$  becomes a morphism  $X \rightarrow \hat{X}$  of abelian varieties, and for  $L$  ample it is an isogeny. This packages the geometry of an ample line bundle into a single self-dual isogeny, the algebraic polarization.

#### Theorem 10.3.1: The Isogeny $\varphi_L$ for $L$ Ample

Let  $X$  be an abelian variety over  $k$  and  $L$  an ample line bundle on  $X$ . Then

$$\varphi_L: X \rightarrow \hat{X}, \quad x \mapsto [t_x^*L \otimes L^{-1}]$$

is an isogeny of abelian varieties with kernel  $K(L)$ ; in particular  $\deg \varphi_L = |K(L)|$ . Moreover  $\varphi_L$  is *symmetric*: under the biduality isomorphism  $\kappa: X \xrightarrow{\sim} \hat{\hat{X}}$  of theorem 10.2.3, the dual homomorphism satisfies

$$\hat{\varphi}_L \circ \kappa = \varphi_L$$

#### Proof:

*Step 1:*  $\varphi_L$  is a homomorphism of abelian varieties. The map  $\varphi_L$  was defined in definition 8.3.1 as a homomorphism of group functors  $X \rightarrow \text{Pic}^0(X)$ , sending a point  $x$  to the class  $[t_x^*L \otimes L^{-1}]$ , which lies in  $\text{Pic}^0(X)$  by the theorem of the square (corollary 8.2.2). Now that the dual  $\hat{X} = \text{Pic}_{X/k}^0$  is an abelian variety (definition 10.1.1), the target is a group variety, and a homomorphism of group functors into a group variety is a morphism of varieties. Hence  $\varphi_L: X \rightarrow \hat{X}$  is a morphism, and since it respects the group laws it is a homomorphism of abelian varieties.

*Step 2:  $\varphi_L$  has finite kernel, hence is an isogeny.* By definition  $\ker \varphi_L = K(L)$  (definition 8.3.2), the closed subgroup scheme of  $X$  on which  $t_x^*L \otimes L^{-1}$  is trivial. Since  $L$  is ample,  $K(L)$  is finite (proposition 8.3.4). A homomorphism of abelian varieties with finite kernel is surjective: its image is a closed subvariety (the image of a complete variety is complete, hence closed) and a subgroup, and  $\dim \operatorname{im} \varphi_L = \dim X - \dim K(L) = \dim X = g$ , so  $\operatorname{im} \varphi_L = \hat{X}$  because  $\hat{X}$  is connected of dimension  $g$  (theorem 10.1.2). A surjective homomorphism with finite kernel is an isogeny, and  $\deg \varphi_L = |K(L)|$ , the order of the kernel as a group scheme.

*Step 3: symmetry.* Write  $\sigma: X \times X \rightarrow X \times X$  for the coordinate swap  $(x, y) \mapsto (y, x)$ ,  $m: X \times X \rightarrow X$  for the group law, and  $p_1, p_2: X \times X \rightarrow X$  for the two projections. Recall the Mumford bundle

$$\mathcal{M}(L) := m^*L \otimes p_1^*L^{-1} \otimes p_2^*L^{-1}$$

on  $X \times X$  (definition 8.3.1). Its restriction to the slice  $X \times \{x\}$  is  $t_x^*L \otimes L^{-1} \otimes (\text{constant bundle})$ ; rigidifying along  $\{e\} \times X$  (where it is canonically trivial, since  $m(e, y) = y$  forces  $\mathcal{M}(L)|_{\{e\} \times X} \cong L \otimes L^{-1} \cong \mathcal{O}$ ) removes the constant factor, so  $\mathcal{M}(L)$  is exactly the line bundle on  $X \times X$  that classifies, over its second factor, the morphism  $x \mapsto [t_x^*L \otimes L^{-1}] = \varphi_L(x)$  into  $\hat{X} = \operatorname{Pic}_{X/k}^0$ . By the universal property of the Poincaré bundle (proposition 10.2.2), which represents  $\operatorname{Pic}_{X/k}^0$ , this rigidified family is the pullback along the classifying map:

$$\mathcal{M}(L) \cong (\operatorname{id}_X \times \varphi_L)^*\mathcal{P}$$

where  $\mathcal{P}$  is the Poincaré bundle on  $X \times \hat{X}$  of definition 10.2.1, and  $\operatorname{id}_X \times \varphi_L: X \times X \rightarrow X \times \hat{X}$  sends  $(y, x) \mapsto (y, \varphi_L(x))$ .

Now unwind the dual and biduality. Fix any homomorphism  $f: X \rightarrow \hat{X}$  and set  $P_f := (\operatorname{id}_X \times f)^*\mathcal{P}$ , the pullback of  $\mathcal{P}$  along  $(y, x) \mapsto (y, f(x))$ . Its slice over the second factor is  $P_f|_{X \times \{x\}} = \mathcal{P}|_{X \times \{f(x)\}}$ , the line bundle on  $X$  whose class in  $\operatorname{Pic}^0(X) = \hat{X}$  is  $f(x)$ ; and  $P_f$  is rigidified along  $\{e\} \times X$  because  $\mathcal{P}$  is rigidified along  $\{e\} \times \hat{X}$  (definition 10.2.1). So, by the universal property (proposition 10.2.2),  $P_f$  classifies  $f$  itself over the second factor. The claim is that the swapped bundle  $\sigma^*P_f$  classifies, over its second factor, the composite  $\hat{f} \circ \kappa: X \rightarrow \hat{X}$ .

This is a computation of slices. Restrict  $\sigma^*P_f$  to the slice  $X \times \{x\}$  (first coordinate free, second fixed at  $x$ ): for a point  $y$  of the free factor,

$$(\sigma^*P_f)|_{(y,x)} = P_f|_{\sigma(y,x)} = P_f|_{(x,y)} = \mathcal{P}|_{(x,f(y))} = (f^*(\mathcal{P}|_{\{x\} \times \hat{X}}))|_y$$

so  $\sigma^*P_f|_{X \times \{x\}} = f^*(\mathcal{P}|_{\{x\} \times \hat{X}})$ , the pullback along  $f$  of the line bundle  $\mathcal{P}|_{\{x\} \times \hat{X}}$  on  $\hat{X}$ . Now  $\mathcal{P}|_{\{x\} \times \hat{X}}$  is, by the definition of biduality (theorem 10.2.3), a bundle on  $\hat{X}$  representing the point  $\kappa(x) \in \hat{X} = \operatorname{Pic}_{\hat{X}/k}^0$ ; and the dual homomorphism  $\hat{f}: \hat{X} \rightarrow \hat{X}$  is, by definition, the map on Picard schemes induced by pullback of line bundles along  $f$  (proposition 10.1.3), so  $\hat{f}(\kappa(x))$  is the class of  $f^*(\mathcal{P}|_{\{x\} \times \hat{X}})$  in  $\operatorname{Pic}^0(X) = \hat{X}$ . Thus the slice of  $\sigma^*P_f$  over  $x$  represents exactly  $(\hat{f} \circ \kappa)(x)$ . Moreover  $\sigma^*P_f$  is rigidified along  $\{e\} \times X$ : its restriction there is  $\mathcal{P}|_{(x,f(e))} = \mathcal{P}|_{X \times \{0\}}$ , trivial because  $f(e) = 0$ . By the universal property (proposition 10.2.2) again,  $\sigma^*P_f$  classifies the homomorphism  $\hat{f} \circ \kappa$  over its second factor.

Two rigidified bundles on  $X \times X$  classifying maps into  $\hat{X}$  over the second factor are isomorphic if and only if those maps coincide (the classifying morphism is unique, proposition 10.2.2). Applied to  $P_f$  and  $\sigma^*P_f$ , which classify  $f$  and  $\hat{f} \circ \kappa$  respectively, this gives:  $\hat{f} \circ \kappa = f$  holds precisely when  $P_f$  is symmetric, that is when  $\sigma^*(\text{id}_X \times f)^*\mathcal{P} \cong (\text{id}_X \times f)^*\mathcal{P}$ .

It therefore suffices to check that  $\mathcal{M}(L)$  is symmetric, and this is a direct computation. Since  $m \circ \sigma = m$  and  $\sigma$  interchanges  $p_1$  and  $p_2$  (that is  $p_1 \circ \sigma = p_2$  and  $p_2 \circ \sigma = p_1$ ), pulling back gives

$$\sigma^*\mathcal{M}(L) = (m \circ \sigma)^*L \otimes (p_1 \circ \sigma)^*L^{-1} \otimes (p_2 \circ \sigma)^*L^{-1} = m^*L \otimes p_2^*L^{-1} \otimes p_1^*L^{-1} = \mathcal{M}(L)$$

the middle two factors swapping. Thus  $\sigma^*(\text{id}_X \times \varphi_L)^*\mathcal{P} \cong \sigma^*\mathcal{M}(L) \cong \mathcal{M}(L) \cong (\text{id}_X \times \varphi_L)^*\mathcal{P}$ , so  $\varphi_L$  is symmetric:  $\hat{\varphi}_L \circ \kappa = \varphi_L$ . (The seesaw principle (theorem 7.4.5) is what underlies the classification of homomorphisms  $X \rightarrow \hat{X}$  by bundles on  $X \times X$  trivial along the axes, but no further slice computation is needed here.) This completes the proof.

The symmetry relation is the algebraic incarnation of a fact already visible in the analytic theory: over  $\mathbb{C}$  the map  $\varphi_L$  is induced by  $v \mapsto H(v, -)$  for a Hermitian form  $H$ , and  $H(v, w) = \overline{H(w, v)}$  is exactly the statement that  $\varphi_L$  equals its own dual under biduality (proposition 5.3.3). A homomorphism  $X \rightarrow \hat{X}$  satisfying  $\hat{f} \circ \kappa = f$  is one that comes from a class in  $\text{NS}(X)$  rather than an arbitrary map to the dual; the polarizations are the positive such classes, and the symmetry is what makes the Rosati involution of chapter 11 an anti-involution rather than a  $\mathbb{Q}$ -linear map.

With  $\varphi_L$  upgraded to an isogeny, a polarization can be defined without reference to a specific line bundle over the base field. The subtlety is that ampleness is a geometric condition: a line bundle on  $X$  may fail to be ample over  $k$  yet become ample after base change to  $\bar{k}$ , and it is the isogeny to the dual, together with geometric ampleness, that carries the intrinsic content.

#### Definition 10.3.2: Polarization

Let  $X$  be an abelian variety over  $k$ . A *polarization* of  $X$  is an isogeny  $\lambda: X \rightarrow \hat{X}$  such that  $\lambda_{\bar{k}} = \varphi_L$  for some ample line bundle  $L$  on  $X_{\bar{k}}$ . A polarization  $\lambda$  is *principal* if it is an isomorphism.

Two features of this definition deserve emphasis. First, a polarization is a homomorphism to the dual, not a line bundle: passing from  $L$  to  $\varphi_L$  discards the algebraically trivial part of  $L$  and retains only its class in  $\text{NS}(X)$ , and distinct ample bundles differing by an element of  $\text{Pic}^0(X)$  induce the same  $\varphi_L$  because both  $\varphi_L$  and  $K(L)$  depend only on the class of  $L$  in  $\text{NS}(X)$  (definition 8.3.1). Second, the condition is imposed after base change to  $\bar{k}$ : over a non-closed field an abelian variety may carry a polarization defined over  $k$  whose underlying ample bundle exists only over  $\bar{k}$ , and the definition is arranged so that a polarization is always a  $k$ -morphism  $X \rightarrow \hat{X}$  even when no single  $k$ -rational ample bundle induces it. Every polarization is automatically symmetric,  $\hat{\lambda} \circ \kappa = \lambda$ , by theorem 10.3.1 applied over  $\bar{k}$ ; the polarizations are thus a distinguished cone of symmetric isogenies to the dual.

Principality is the strongest form:  $\lambda$  an isomorphism means  $K(L)$  is trivial, which forces  $\deg \varphi_L = 1$  and, in the language of the analytic theory, a polarization of type  $(1, \dots, 1)$ . Principally polarized abelian varieties are the ones with an autoduality  $X \cong \hat{X}$  respecting the group structure, and they are exactly the abelian varieties that arise as Jacobians of curves (chapter 12) and as the moduli points of the Siegel modular variety.

The degree of a polarization refines to a finer invariant, its type, recording not just the order of the

kernel but its structure as a finite group scheme.

**Definition 10.3.3: Type of a Polarization**

Let  $\lambda: X \rightarrow \hat{X}$  be a polarization with  $\lambda_{\bar{k}} = \varphi_L$ . The kernel  $\ker \lambda = K(L)$  is a finite commutative group scheme over  $\bar{k}$ ; the invariant it records is the *order as a group scheme* (the rank of its coordinate algebra as a  $\bar{k}$ -vector space), which is the degree  $\deg \varphi_L = |K(L)|$  of theorem 10.3.1. The pairing  $e^L$  on  $K(L)$  is a nondegenerate alternating form, the algebraic counterpart of the analytic commutator pairing (proposition 4.1.7); it realizes  $K(L)$  as its own Cartier dual (its perfectness is taken up as the Weil pairing in the next section, section 10.4). This self-duality makes the order a perfect square and equips  $K(L)$  with a canonical list of invariant factors that occur in equal pairs, yielding uniquely determined integers

$$d_1 | d_2 | \cdots | d_g, \quad d_i \geq 1$$

with  $|K(L)| = (d_1 \cdots d_g)^2$ , so  $d_1 \cdots d_g = \sqrt{\deg \varphi_L}$ . The tuple  $(d_1, \dots, d_g)$  is the *type* of  $\lambda$ , the trivial entries dropped. When  $\text{char } k$  does not divide  $\deg \varphi_L$  the group scheme  $K(L)$  is étale, and the  $d_i$  are exactly the elementary divisors of one Lagrangian (maximal isotropic) half of the geometric-point group,

$$K(L)(\bar{k}) \cong G \oplus G, \quad G = \bigoplus_{i=1}^g \mathbb{Z}/d_i \mathbb{Z}$$

the  $e^L$ -orthogonal splitting into a hyperbolic (symplectic) group; the invariant factors of  $K(L)(\bar{k})$  are then the doubled list  $(d_1, d_1, \dots, d_g, d_g)$ . In characteristic  $p$  dividing  $\deg \varphi_L$  the geometric points may undercount the scheme order, and the  $d_i$  are read from  $|K(L)|$  and the Cartier self-duality rather than from  $K(L)(\bar{k})$ ; see the discussion below. The polarization is principal precisely when  $\ker \lambda$  is trivial as a group scheme, equivalently  $\deg \varphi_L = 1$ , equivalently the type is empty (or  $(1, \dots, 1)$ ).

The type is the algebraic shadow of the analytic type  $(d_1, \dots, d_g)$  of definition 3.2.3. Over  $\mathbb{C}$  the kernel  $K(L) = \Lambda(L)/\Lambda$  is the finite abelian group  $(\bigoplus_i \mathbb{Z}/d_i)^2$ , always étale, so the analytic type is precisely the length- $g$  list read off one Lagrangian factor, matching the char-coprime case above; the degree  $\deg \varphi_L = (d_1 \cdots d_g)^2$  is the analytic count  $\det E$ . The subtlety appears only in characteristic  $p$  dividing the degree, where part of  $K(L)$  is a nonétale (infinitesimal) group scheme: then  $|K(L)(\bar{k})| < |K(L)|$ , so the geometric points no longer see the full order, and the type must be read from the scheme order  $|K(L)|$  and the Cartier self-duality supplied by  $e^L$ . A supersingular factor already exhibits this, with  $K(L)$  containing an  $\alpha_p \times \alpha_p$  of scheme order  $p^2$  but a single geometric point; the length- $g$  tuple is recovered from the group-scheme structure through Cartier duality (chapter 10), not from  $K(L)(\bar{k})$ .

The type sorts polarizations into families of fixed kernel size, and the principal ones sit at the bottom. The clearest instance is the elliptic curve, where every ample line bundle of degree one is a principal polarization and the isogeny  $\varphi_L$  is an isomorphism identifying  $E$  with its dual.

**Example 10.3.4: Principal Polarization of an Elliptic Curve**

Let  $E$  be an elliptic curve over  $k$ , with origin the identity point  $O \in E(k)$ , and let  $L = \mathcal{O}_E(O)$  be the line bundle attached to the divisor consisting of the single point  $O$ . This bundle has degree 1,

so by Riemann-Roch on the curve  $\dim H^0(E, L) = 1$ ; it is ample because it has positive degree. The associated homomorphism is

$$\varphi_L: E \rightarrow \hat{E}, \quad x \mapsto [t_x^* \mathcal{O}_E(O) \otimes \mathcal{O}_E(O)^{-1}] = [\mathcal{O}_E([-x] - [O])]$$

the last equality because, with the convention  $t_x(y) = y + x$  fixing  $\varphi_L$  in definition 8.3.1, the pullback  $t_x^* \mathcal{O}_E(O) = \mathcal{O}_E(t_x^{-1}(O)) = \mathcal{O}_E([-x])$ , since  $t_x^{-1}(O) = O - x = [-x]$ . Since  $\ker \varphi_L = K(L)$  and  $|K(L)| = (\deg L)^2 = 1$ , the kernel is trivial and  $\varphi_L$  is an isomorphism:  $L = \mathcal{O}_E(O)$  is a principal polarization, of empty type. This is the canonical identification  $E \cong \hat{E}$  of example 5.1.2: under the Abel-Jacobi isomorphism  $E \xrightarrow{\sim} \hat{E} = \text{Pic}^0(E)$ ,  $z \mapsto [\mathcal{O}_E([z] - [O])]$ , the map  $\varphi_L$  becomes  $x \mapsto [-x]$ , so  $\varphi_L$  corresponds to  $-\text{id}$ ; the sign is that of the translation convention in definition 8.3.1, and the identification with  $E$  itself is what matters for principality. The full development of the group law, the isomorphism  $E \cong \text{Pic}^0(E)$ , and this canonical polarization for  $g = 1$  is in *Elliptic Curves* [4].

The contrast with the analytic polarization isogeny of proposition 5.3.3 is exact. Over  $\mathbb{C}$  the curve is  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ , and the principal polarization comes from the positive Hermitian form  $H(v, w) = v\bar{w}/\text{Im}\tau$ , whose imaginary part  $E = \text{Im}H$  is the standard integral symplectic form on  $\Lambda$  of type (1). The map  $\varphi_L$  is then  $v \mapsto H(v, -)$ , an isomorphism  $E \xrightarrow{\sim} \hat{E}$  of complex tori, and the algebraic  $\varphi_L$  above is its scheme-theoretic counterpart: the same self-duality of  $E$ , now realized over an arbitrary field without recourse to a lattice or a Hermitian form.

### Exercise 10.3.1

1. Let  $L$  be an ample line bundle on an abelian variety  $X$  of dimension  $g$  over  $k$ , and suppose  $|K(L)| = 36$ . Using definition 10.3.3, determine  $\deg \varphi_L$ , the product  $d_1 \cdots d_g$ , and list every type  $(d_1, \dots, d_g)$  (trivial entries dropped) compatible with this order for  $g = 2$ . Which of these types, if any, is principal?
2. Let  $\lambda: X \rightarrow \hat{X}$  be a polarization, so  $\lambda_{\bar{k}} = \varphi_L$  for some ample  $L$  on  $X_{\bar{k}}$ , and let  $n$  be a positive integer. Show that  $n\lambda$  (the composite of  $\lambda$  with multiplication by  $n$  on  $\hat{X}$ ) is again a polarization, and express its degree in terms of  $\deg \lambda$ ,  $n$ , and  $g = \dim X$ . *You may use that  $[n]_{\hat{X}}$  is an isogeny of degree  $n^{2g}$  and that  $n\varphi_L = \varphi_{L^n}$  with  $L^n$  ample.*
3. Let  $E$  be an elliptic curve over  $k$  with origin  $O$ , and let  $L = \mathcal{O}_E(O)$  as in example 10.3.4. Fix a point  $a \in E(k)$  and set  $L_a = t_a^* L = \mathcal{O}_E([-a])$ . Compute  $\varphi_{L_a}$  as a map  $E \rightarrow \hat{E}$  and compare it to  $\varphi_L$ . What does this illustrate about the dependence of  $\varphi_L$  on the class of  $L$  in  $\text{NS}(X)$ ?
4. Prove that the map  $L \mapsto \varphi_L$  is additive on line bundles: for line bundles  $L, M$  on  $X$ ,

$$\varphi_{L \otimes M} = \varphi_L + \varphi_M$$

as homomorphisms  $X \rightarrow \hat{X}$ , where the sum is taken in the group  $\text{Hom}(X, \hat{X})$ . Deduce that  $\varphi_{L^{-1}} = -\varphi_L$  and that if  $L$  is algebraically trivial then  $\varphi_L = 0$ .

5. Prove that the coordinate swap  $\sigma: X \times X \rightarrow X \times X$ ,  $(x, y) \mapsto (y, x)$ , fixes the Mumford bundle:  $\sigma^* \mathcal{M}(L) \cong \mathcal{M}(L)$ . Then explain, using the classification of homomorphisms  $X \rightarrow \hat{X}$  by rigidified bundles on  $X \times X$  over the second factor (proposition 10.2.2), why this single computation is what forces the symmetry  $\varphi_L \circ \kappa = \varphi_L$  of theorem 10.3.1.

6. Let  $\lambda: X \rightarrow \hat{X}$  be a polarization of type  $(d_1, \dots, d_g)$  over  $\bar{k}$ , and suppose  $\text{char } k$  does not divide  $\deg \lambda$ . Show that the abstract group  $K(\lambda)(\bar{k}) = \ker(\lambda)(\bar{k})$  determines the type, and describe the type of the elliptic curve example example 10.3.4. If instead a  $g = 2$  abelian variety has  $\ker(\lambda)(\bar{k}) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ , what is its type, and is it principal?

## 10.4 The Weil Pairing and Duality of Isogenies

The dual  $\hat{X} = \text{Pic}_{X/k}^0$  of section 10.1 pairs an abelian variety with its line bundles; the polarization  $\varphi_L$  of section 10.3 folds that pairing into a self-duality. This section extracts the arithmetic residue of the duality on torsion. The torsion subgroup schemes  $X[n]$  and  $\hat{X}[n]$  turn out to be Cartier dual to one another, and the pairing witnessing that duality is the Weil pairing  $e_n: X[n] \times \hat{X}[n] \rightarrow \mu_n$ . Unlike the analytic construction of section 5.2, which read the pairing off the alternating form  $E = \text{Im} H$ , the present pairing is manufactured from the interplay of an isogeny and its dual and so exists in every characteristic. Two further duality statements follow it: the dual of an isogeny exchanges a kernel for its Cartier dual, and the pairing attached to a polarization is alternating.

The starting observation is that multiplication by  $n$  is compatible with duality in the sharpest possible way. Recall from section 10.1 that a homomorphism  $f: X \rightarrow Y$  induces  $\hat{f}: \hat{Y} \rightarrow \hat{X}$ , contravariantly; applied to  $f = [n]: X \rightarrow X$  this gives  $\widehat{[n]} = [n]: \hat{X} \rightarrow \hat{X}$ , since duality is additive and sends  $\text{id}$  to  $\text{id}$ . Dualizing the exact sequence  $0 \rightarrow X[n] \rightarrow X \xrightarrow{[n]} X \rightarrow 0$  of  $\bar{k}$ -groups (for  $n$  prime to  $\text{char } k$ , where  $[n]$  is separable) therefore relates  $X[n]$  to a kernel inside  $\hat{X}$ , and this relation is the Weil pairing, stated below as the duality between a finite group scheme and its Cartier dual.

### Definition 10.4.1: The Weil Pairing

Let  $X$  be an abelian variety over  $k$ , let  $n \geq 1$  be an integer prime to  $\text{char } k$ , and write  $\mu_n$  for the group scheme of  $n$ -th roots of unity. The *Weil pairing* is the bilinear pairing of finite group schemes

$$e_n: X[n] \times \hat{X}[n] \longrightarrow \mu_n$$

defined as follows. A point of  $\hat{X}[n] = (\text{Pic}_{X/k}^0)[n]$  is the class  $[M]$  of a line bundle  $M \in \text{Pic}^0(X)$  with  $M^{\otimes n} \cong \mathcal{O}_X$ . Because  $M$  is algebraically trivial, the pullback of  $M$  along  $[n]: X \rightarrow X$  is  $[n]^*M \cong M^{\otimes n}$  (the multiplication-by- $n$  pullback formula proposition 8.4.3, using  $(-1)^*M \cong M^{-1}$  for  $M \in \text{Pic}^0$ ), and  $M^{\otimes n} \cong \mathcal{O}_X$ ; so  $[n]^*M$  is trivial. Now  $[n]: X \rightarrow X$  is, for  $n$  prime to  $\text{char } k$ , a finite étale Galois cover with group  $X[n]$  (theorem 8.4.5), and a line bundle on  $X$  whose pullback to the cover is trivial is given by a descent datum for the trivial bundle  $[n]^*M \cong \mathcal{O}_X$ , that is, by a character of the deck group. Thus  $M$  determines a homomorphism

$$\chi_M: X[n] \longrightarrow \mathbb{G}_m, \quad \text{equivalently} \quad \chi_M \in \underline{\text{Hom}}(X[n], \mathbb{G}_m),$$

and since  $X[n]$  is killed by  $n$  its values are  $n$ -th roots of unity,  $\chi_M \in \underline{\text{Hom}}(X[n], \mu_n) = X[n]^\vee$ . Set

$$e_n(x, [M]) := \chi_M(x) \in \mu_n$$

the value of this character at  $x$ . The assignment is independent of the choices made (the trivializations of  $[n]^*M$  and of  $M^{\otimes n}$  enter only through the descent datum  $\chi_M$ , which they do not change), so  $e_n(x, [M]) \in \mu_n$  is well defined.



The pairing is the algebraic counterpart of the analytic Weil pairing definition 5.2.7, which over  $\mathbb{C}$  was the explicit exponential  $e_n(\bar{v}, \bar{\ell}) = \exp(2\pi i n \operatorname{Im} \ell(v))$  read from the dual lattice; here there is no lattice, and the pairing is built from an  $n$ -torsion bundle and the covering  $[n]$ . The construction is worth unwinding. A class  $[M] \in \hat{X}[n]$  is an  $n$ -torsion line bundle in  $\operatorname{Pic}^0$ , so it becomes trivial after pullback along  $[n]$ , for the two reasons that combine in the definition:  $[n]^*M \cong M^{\otimes n}$  because  $M$  is algebraically trivial, and  $M^{\otimes n}$  is itself trivial because  $[M]$  is  $n$ -torsion. A bundle trivialized on an étale Galois cover is exactly the datum of a character of the deck group  $X[n]$ , and  $e_n(x, [M]) = \chi_M(x)$  reads off the value of that character. It lands in  $\mu_n$  for a structural reason, not by a separate computation: a character of the group scheme  $X[n]$ , which is killed by  $n$ , automatically takes  $n$ -th-root-of-unity values, so  $e_n(x, [M])^n = \chi_M(x)^n = \chi_M(nx) = 1$ . The theorem of the square supplies the geometric picture behind this: it makes  $x \mapsto [t_x^*M \otimes M^{-1}]$  a homomorphism  $X \rightarrow \operatorname{Pic}^0(X)$  killing  $x$  (theorem 8.2.1), so translation by an  $n$ -torsion point carries  $M$  back to itself, and  $\chi_M$  records how the trivialization of  $[n]^*M$  transforms under the deck action of that translation. This is precisely the mechanism the theta group gave analytically in section 5.2, now phrased so that nothing but line bundles and the covering  $[n]$  enters, and so surviving into positive characteristic. For  $g = 1$  this is the classical Weil pairing  $e_n: E[n] \times \hat{E}[n] \rightarrow \mu_n$  constructed in *Elliptic Curves* [4] from the divisor-and-function description of  $E[n]$ ; the  $g = 1$  case is not reproved here.

The definition already builds bilinearity into the pairing, since  $\chi_M$  is a character and  $[M] \mapsto \chi_M$  is additive in the bundle; the substantive fact is that  $e_n$  is perfect, and the clean reason is Cartier duality of the two torsion group schemes.

**Proposition 10.4.2: The Weil Pairing is Perfect**

Let  $X$  be an abelian variety over  $k$  and  $n$  prime to  $\operatorname{char} k$ . The Weil pairing  $e_n: X[n] \times \hat{X}[n] \rightarrow \mu_n$  of definition 10.4.1 is bilinear and perfect: it induces an isomorphism of finite group schemes

$$\hat{X}[n] \xrightarrow{\sim} X[n]^\vee = \underline{\operatorname{Hom}}(X[n], \mu_n)$$

identifying  $\hat{X}[n]$  with the Cartier dual of  $X[n]$ . It is Galois-equivariant: for  $\sigma \in \operatorname{Gal}(\bar{k}/k)$ ,  $x \in X[n](\bar{k})$  and  $\xi \in \hat{X}[n](\bar{k})$ ,  $e_n(\sigma x, \sigma \xi) = \sigma e_n(x, \xi)$ .

**Proof:**

*Step 1: Bilinearity.* By construction  $e_n(x, [M]) = \chi_M(x)$ , where  $\chi_M: X[n] \rightarrow \mu_n$  is the character classifying the trivialization of  $[n]^*M$ . Linearity in the first variable is then immediate,  $\chi_M$  being a homomorphism:  $e_n(x + x', [M]) = \chi_M(x + x') = \chi_M(x) \chi_M(x') = e_n(x, [M]) e_n(x', [M])$ . Linearity in the second variable is the additivity of  $[M] \mapsto \chi_M$ : for  $[M], [M'] \in \hat{X}[n]$  the covering  $[n]$  trivializes  $M \otimes M'$  by the tensor product of the two descent data, so  $\chi_{M \otimes M'} = \chi_M \chi_{M'}$  and  $e_n(x, [M] \otimes [M']) = e_n(x, [M]) e_n(x, [M'])$ . Thus  $e_n$  is a homomorphism in each variable, and it induces a homomorphism of finite group schemes

$$\Psi: \hat{X}[n] \longrightarrow X[n]^\vee = \underline{\operatorname{Hom}}(X[n], \mu_n), \quad [M] \longmapsto \chi_M$$

*Step 2:  $\Psi$  is the Cartier-duality isomorphism.* The multiplication map  $[n]: X \rightarrow X$  is an isogeny of degree  $n^{2g}$  (theorem 8.4.5) with kernel the finite group scheme  $X[n]$ . Dualizing,  $\widehat{[n]} = [n]: \hat{X} \rightarrow \hat{X}$  is again an isogeny of degree  $n^{2g}$ , with kernel  $\hat{X}[n]$ . By theorem 10.4.3, applied to  $f = [n]$  (whose dual is again  $[n]$ ), the kernel of the dual isogeny is the Cartier dual of



the kernel of the original: there is a canonical isomorphism of finite group schemes

$$\hat{X}[n] = \ker \widehat{[n]} \xrightarrow{\sim} (\ker [n])^\vee = X[n]^\vee = \underline{\mathrm{Hom}}(X[n], \mu_n)$$

It remains to check that this canonical isomorphism is exactly  $\Psi$ . Both send  $[M]$  to a character of  $X[n]$ , and both characters are constructed from the same datum, so it suffices to unwind them. The isomorphism of theorem 10.4.3 sends a class  $[M] \in \ker \widehat{[n]}$  - that is,  $[M] \in \mathrm{Pic}^0(X)$  with  $\widehat{[n]}[M] = [n]^*M$  trivial - to the character of the deck group  $X[n] = \ker [n]$  classifying the trivialized pullback  $[n]^*M$  along the Galois cover  $[n]$  (the mechanism given in the proof of theorem 10.4.3, Step 2, with  $f = [n]$ ). That character is precisely  $\chi_M$  of definition 10.4.1. Hence  $\Psi$  coincides with the Cartier-duality isomorphism, so  $\Psi$  is an isomorphism and  $e_n$  is perfect.

*Step 3: Galois equivariance.* Every ingredient of definition 10.4.1 is defined over  $k$ : the group law, the isogeny  $[n]$  with its deck group  $X[n]$ , and the bundle  $\mathcal{O}_X$ . An element  $\sigma \in \mathrm{Gal}(\bar{k}/k)$  therefore carries the trivialization of  $[n]^*M$  to a trivialization of  $[n]^*(\sigma M)$ , so it carries the classifying character  $\chi_M$  to  $\sigma\chi_M = \chi_{\sigma M}$ , its values acted on by  $\sigma$  and its argument transported along  $\sigma$ . Evaluating at  $\sigma x$  gives  $e_n(\sigma x, \sigma[M]) = \chi_{\sigma M}(\sigma x) = \sigma\chi_M(x) = \sigma e_n(x, [M])$ , since  $\sigma$  is a field automorphism and  $\chi_M(x)$  is a scalar. This is the stated equivariance, completing the proof.

Perfectness is what upgrades the Weil pairing from a mere bilinear map to an identification:  $\hat{X}[n]$  is not just abstractly of the same size as  $X[n]$  but is canonically its Cartier dual, so any structure on the  $n$ -torsion of  $X$  is reflected, contravariantly, on the  $n$ -torsion of  $\hat{X}$ . Passing to the limit over powers of a prime  $\ell \neq \mathrm{char} k$ , the pairings  $e_{\ell^m}$  assemble into a perfect  $\mathbb{Z}_\ell$ -bilinear pairing  $T_\ell X \times T_\ell \hat{X} \rightarrow \mathbb{Z}_\ell(1)$  on the Tate modules of definition 9.2.2, where  $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^m}$  is the Tate twist; it identifies  $T_\ell \hat{X}$  with the  $\mathbb{Z}_\ell$ -dual of  $T_\ell X$  twisted by  $\mathbb{Z}_\ell(1)$ . The Galois equivariance then says this identification respects the action of  $\mathrm{Gal}(\bar{k}/k)$ , so the Galois representation on  $T_\ell \hat{X}$  is determined by that on  $T_\ell X$ ; the continuous formalism underlying the Galois action is that of *Galois Cohomology* [6], used here only to record that the pairing is equivariant. This is the  $\ell$ -adic form of the duality, and it is the mechanism by which a polarization will constrain the Frobenius eigenvalues in chapter 13.

The Cartier-duality step invoked a general fact about dualizing an isogeny, which deserves a statement and proof of its own, since it governs how kernels transform under duality and is the source of the perfectness just used.

#### Theorem 10.4.3: Duality of Isogenies

Let  $f: X \rightarrow Y$  be an isogeny of abelian varieties over  $k$ , with kernel the finite group scheme  $\ker f$ . Then the dual homomorphism  $\hat{f}: \hat{Y} \rightarrow \hat{X}$  is an isogeny, and its kernel is the Cartier dual of  $\ker f$ : there is a canonical short exact sequence of finite group schemes

$$0 \longrightarrow (\ker f)^\vee \longrightarrow \hat{Y} \xrightarrow{\hat{f}} \hat{X} \longrightarrow 0$$

In particular  $\deg \hat{f} = \deg f$ .

**Proof:**

*Step 1:  $\hat{f}$  is an isogeny.* By proposition 9.1.3 there is an isogeny  $g: Y \rightarrow X$  with  $g \circ f = [n]_X$  and  $f \circ g = [n]_Y$ , where  $n = \deg f$ . Dualizing and using contravariant functoriality (proposition 10.1.3) together with  $\widehat{[n]} = [n]$  gives  $\hat{f} \circ \hat{g} = [n]_{\hat{X}}$  and  $\hat{g} \circ \hat{f} = [n]_{\hat{Y}}$ . Since  $[n]$  is an isogeny (theorem 8.4.5),  $\hat{f}$  is surjective with finite kernel, hence an isogeny; its degree is finite and  $\dim \hat{Y} = \dim \hat{X} = g$ .

*Step 2: Identifying  $\ker \hat{f}$  with  $(\ker f)^\vee$ .* Write  $K = \ker f$ , a finite group scheme, and consider the exact sequence  $0 \rightarrow K \rightarrow X \xrightarrow{f} Y \rightarrow 0$ . Dualizing a short exact sequence of abelian varieties and finite group schemes interchanges the roles of sub and quotient and replaces the finite piece by its Cartier dual; concretely, the exact sequence induces, on Picard schemes, the sequence

$$0 \longrightarrow K^\vee \longrightarrow \hat{Y} \xrightarrow{\hat{f}} \hat{X} \longrightarrow 0$$

The identification  $\ker \hat{f} \cong K^\vee$  is the content of Cartier duality for finite group schemes, which we take from *Algebraic Geometry* [13]. We give the mechanism. A geometric point of  $\ker \hat{f}$  is a class  $[M] \in \text{Pic}^0(Y)$  with  $f^*M \cong \mathcal{O}_X$ ; such an  $M$  is trivialized after pullback along  $f$ , so it is classified by a homomorphism from the deck group  $K = \ker f$  of the isogeny  $f$  (which is a  $K$ -torsor over  $Y$ ) to the multiplicative group, that is, by an element of  $\underline{\text{Hom}}(K, \mathbb{G}_m) = K^\vee$ . Conversely a character  $\chi: K \rightarrow \mathbb{G}_m$  produces, by descent along the  $K$ -torsor  $f: X \rightarrow Y$ , a line bundle  $M_\chi$  on  $Y$  with  $f^*M_\chi \cong \mathcal{O}_X$ , and  $M_\chi \in \text{Pic}^0(Y)$  because it is killed by the isogeny  $f$ . These two assignments are mutually inverse and compatible with the group structures, giving the isomorphism  $\ker \hat{f} \cong K^\vee$  as group schemes.

*Step 3: The degree.* Cartier duality preserves the order of a finite group scheme,  $|K^\vee| = |K|$ , so

$$\deg \hat{f} = |\ker \hat{f}| = |K^\vee| = |K| = \deg f$$

which is the last assertion, completing the proof.

The theorem is the structural reason duality is degree-preserving and, more than that, the reason it exchanges the two kinds of finite group scheme that populate characteristic  $p$ . If  $f$  is separable, so  $\ker f$  is étale, then  $(\ker f)^\vee$  can be connected: dualizing an étale kernel produces a possibly-infinitesimal one, exactly as  $(\mathbb{Z}/p)^\vee = \mu_p$  in the language of section 9.4. This is why the dual of a separable isogeny need not be separable in characteristic  $p$ , and it is invisible over  $\mathbb{C}$ , where every finite group scheme is étale and the sequence just says  $\deg \hat{f} = \deg f$  (which section 5.2 obtained directly from the transpose of a lattice inclusion). The exact sequence also refines the biduality  $\hat{\hat{X}} = X$  of theorem 10.2.3: dualizing the sequence a second time returns the original  $0 \rightarrow \ker f \rightarrow X \rightarrow Y \rightarrow 0$ , consistent with  $(\ker f)^{\vee\vee} = \ker f$  and  $\hat{\hat{f}} = f$ .

A polarization supplies a self-map of the torsion by composing the Weil pairing with  $\varphi_L$ , and the resulting pairing on  $X[n]$  alone carries the signature of the polarization: it is alternating. This is the algebraic residue of the fact that  $\varphi_L$  is symmetric (theorem 10.3.1), and it is the pairing that the Rosati involution and Albert's classification will exploit.

**Proposition 10.4.4: The Polarization Pairing is Alternating**

Let  $L$  be an ample line bundle on the abelian variety  $X$  over  $k$ , with polarization isogeny  $\varphi_L: X \rightarrow \hat{X}$  (theorem 10.3.1), and let  $n$  be prime to  $\text{char } k$ . Define

$$e_n^L: X[n] \times X[n] \longrightarrow \mu_n, \quad e_n^L(x, y) := e_n(x, \varphi_L(y))$$

where  $e_n$  is the Weil pairing of definition 10.4.1. Then  $e_n^L$  is bilinear and *alternating*:  $e_n^L(x, x) = 1$  for all  $x \in X[n]$ , and consequently  $e_n^L(y, x) = e_n^L(x, y)^{-1}$ . It is nondegenerate on  $X[n]/(X[n] \cap \ker \varphi_L)$ , and perfect on all of  $X[n]$  when  $L$  is a principal polarization.

**Proof:**

Bilinearity is immediate from that of  $e_n$  (proposition 10.4.2) and the fact that  $\varphi_L$  is a homomorphism, so  $\varphi_L(y + y') = \varphi_L(y) + \varphi_L(y')$ ; thus  $e_n^L$  is a homomorphism in each variable. The map  $\varphi_L$  carries  $X[n]$  into  $\hat{X}[n]$  because it is a homomorphism and hence commutes with  $[n]$ : if  $ny = 0$  then  $n\varphi_L(y) = \varphi_L(ny) = 0$ , so  $\varphi_L(y) \in \hat{X}[n]$  and the composite  $e_n^L$  is defined.

*Antisymmetry from the symmetry of  $\varphi_L$ .* Under the biduality identification  $\kappa: X \xrightarrow{\sim} \hat{\hat{X}}$  of theorem 10.2.3, the polarization isogeny satisfies  $\hat{\varphi}_L \circ \kappa = \varphi_L$  (theorem 10.3.1), the statement that  $\varphi_L$  equals its own dual. The compatibility of the Weil pairing with dual isogenies, which is the adjunction  $e_n^Y(fa, \eta) = e_n^X(a, \hat{f}\eta)$  for an isogeny  $f: X \rightarrow Y$ , with  $a \in X[n]$  and  $\eta \in \hat{Y}[n]$  (each side written in the canonical order,  $X[n] \times \hat{X}[n]$  and  $Y[n] \times \hat{Y}[n]$ ; the identity is inherited from the perfectness and the duality of isogenies, proposition 10.4.2, theorem 10.4.3), together with the antisymmetry of the Cartier-duality pairing under  $\kappa$  (reading the pairing of  $X[n]$  with its dual in the two orders gives inverse values), yields for  $x, y \in X[n]$

$$e_n^L(x, y) = e_n(x, \varphi_L(y)) = e_n(x, \hat{\varphi}_L(\kappa y)) = e_n(\varphi_L(x), \kappa y) = e_n(\varphi_L(x), y)^{-1} = e_n^L(y, x)^{-1}$$

so  $e_n^L(x, y) e_n^L(y, x) = 1$ : the pairing is antisymmetric. Setting  $y = x$  gives  $e_n^L(x, x)^2 = 1$ .

*The strict alternating property.* Antisymmetry alone leaves open the possibility  $e_n^L(x, x) = -1$  for 2-torsion  $x$  when  $n$  is even; the strict alternating property  $e_n^L(x, x) = 1$  for all  $x$  is a finer fact, reflecting that  $\varphi_L$  comes from an actual line bundle  $L$  and not just from a symmetric homomorphism. It is the algebraic form of the theta-group commutator computation: the self-pairing  $e_n^L(x, x)$  is the commutator of a lift of  $x$  with itself in the Heisenberg extension attached to  $L$ , hence trivial. This is the general-characteristic analogue of the analytic identity  $e_n^L(x, y) = \exp(2\pi i n E(y, x))$  of section 5.2, whose skew-symmetry  $E(x, x) = 0$  is manifest; we take the strict alternating property from Mumford, *Abelian Varieties*, since the commutator computation is a general fact of the theta group rather than special to this pairing.

*Nondegeneracy.* Since  $e_n$  is perfect (proposition 10.4.2),  $e_n^L(x, y) = e_n(x, \varphi_L(y))$  is trivial for all  $x$  if and only if  $\varphi_L(y) = 0$ , that is,  $y \in X[n] \cap \ker \varphi_L = X[n] \cap K(L)$ . Thus the radical of  $e_n^L$  is  $X[n] \cap K(L)$ , and  $e_n^L$  descends to a nondegenerate alternating pairing on the quotient. When  $L$  is principal,  $\varphi_L$  is an isomorphism (definition 10.3.2), so  $K(L) = 0$ , the radical is trivial, and  $e_n^L$  is a perfect alternating pairing on all of  $X[n]$ , completing the proof.

The alternating pairing  $e_n^L$  is the arithmetic incarnation of a principal polarization: it equips  $X[n] \cong (\mathbb{Z}/n)^{2g}$  with a perfect symplectic form, canonically, and this symplectic structure is what makes the group  $\mathrm{Sp}(2g, \mathbb{Z}/n)$  the natural symmetry group of the torsion of a principally polarized abelian variety. The passage to the Tate module turns  $e_n^L$  into a perfect alternating  $\mathbb{Z}_\ell$ -pairing  $T_\ell X \times T_\ell X \rightarrow \mathbb{Z}_\ell(1)$ , the *Weil pairing of the polarization*, and its Galois equivariance forces the image of the  $\ell$ -adic representation into the group of symplectic similitudes  $\mathrm{GSp}(2g, \mathbb{Z}_\ell)$  rather than all of  $\mathrm{GL}(2g, \mathbb{Z}_\ell)$ . Over  $\mathbb{C}$  this recovers the alternating form  $E = \mathrm{Im} H$  on  $X[n]$  that section 5.2 obtained by composing the analytic Weil pairing with  $\varphi_L$ , and over a finite field it is the constraint that, together with the Weil “Riemann hypothesis”, pins down the possible characteristic polynomials of Frobenius in chapter 13.

The elliptic-curve case makes the two duality statements concrete and reconciles them with the pairing already in hand for  $g = 1$ .

#### Example 10.4.5: The Weil Pairing on an Elliptic Curve

Let  $E$  be an elliptic curve over  $k$  and  $n$  prime to  $\mathrm{char} k$ . Here  $g = 1$ , so  $E[n](\bar{k}) \cong (\mathbb{Z}/n)^2$  by proposition 9.2.1. The dual  $\hat{E}$  is canonically isomorphic to  $E$  itself: an elliptic curve is principally polarized by the class of the origin, so  $\varphi_L: E \xrightarrow{\sim} \hat{E}$  is an isomorphism (example 5.1.2), and under it the abstract Weil pairing  $e_n: E[n] \times \hat{E}[n] \rightarrow \mu_n$  of definition 10.4.1 becomes a pairing

$$e_n^L: E[n] \times E[n] \longrightarrow \mu_n$$

on the  $n$ -torsion of  $E$  alone. By proposition 10.4.4 it is alternating and, the polarization being principal, perfect: a symplectic form on the rank-2 module  $(\mathbb{Z}/n)^2$ , hence given in any basis  $\{P, Q\}$  of  $E[n]$  by  $e_n^L(aP + bQ, cP + dQ) = \zeta^{ad-bc}$  for a primitive  $n$ -th root of unity  $\zeta = e_n^L(P, Q)$ . This is exactly the classical Weil pairing on  $E[n]$  constructed in *Elliptic Curves* [4] from functions with prescribed divisors; there it is shown alternating, Galois-equivariant, and nondegenerate, and its surjectivity onto  $\mu_n$  is what forces  $\mu_n \subseteq k$  whenever  $E[n] \subseteq E(k)$ . The duality of isogenies (theorem 10.4.3) specializes for  $g = 1$  to the statement that an isogeny  $f: E \rightarrow E'$  and its dual  $\hat{f}: E' \rightarrow E$  have equal degree and  $\hat{f} \circ f = [\deg f]$ , the dual isogeny of the  $g = 1$  theory; the Cartier-dual kernel  $(\ker f)^\vee$  is, over  $\mathbb{C}$  or in the separable case, abstractly  $\ker f$  again, but in characteristic  $p$  the sequence  $0 \rightarrow (\ker f)^\vee \rightarrow \hat{E}' \rightarrow \hat{E} \rightarrow 0$  records the connected-étale exchange of section 9.4, for instance sending the étale kernel  $\mathbb{Z}/p$  of a separable  $p$ -isogeny to the infinitesimal  $\mu_p$ .

#### Exercise 10.4.1

1. Let  $X$  be an abelian variety over  $k$  and  $n$  prime to  $\mathrm{char} k$ . Using only the definition of the Weil pairing (definition 10.4.1), show directly that  $e_n(x, [M])^n = 1$  for every  $x \in X[n]$  and every  $[M] \in \hat{X}[n]$ , so that  $e_n$  indeed takes values in  $\mu_n$ . Where does the hypothesis  $nx = 0$  enter?
2. Deduce from the perfectness of the Weil pairing (proposition 10.4.2) that the two finite group schemes  $X[n]$  and  $\hat{X}[n]$  have the same order, and that, under the standing hypothesis  $n$  prime to  $\mathrm{char} k$ , if  $X[n]$  is étale then so is its Cartier dual  $X[n]^\vee = \hat{X}[n]$ . (Why is the hypothesis needed? Recall  $(\mathbb{Z}/p)^\vee = \mu_p$ .) What is the order of  $X[n]$ ?
3. Let  $f: X \rightarrow Y$  be an isogeny of abelian varieties over  $k$ . Using theorem 10.4.3, show that  $\hat{f}$  is separable if and only if  $\ker f$  has étale Cartier dual. Then explain why, over  $\mathbb{C}$ , separability of  $f$  forces separability of  $\hat{f}$ , whereas in characteristic  $p$  a separable isogeny of degree  $p$  can have

inseparable dual; give the standard example.

4. Let  $L$  be a principal polarization on  $X$ , so  $\varphi_L: X \rightarrow \hat{X}$  is an isomorphism, and let  $e_n^L$  be the alternating pairing of proposition 10.4.4. For  $g = 1$ , an elliptic curve  $E$  with basis  $\{P, Q\}$  of  $E[n]$  has  $e_n^L(aP + bQ, cP + dQ) = \zeta^{ad-bc}$  where  $\zeta = e_n^L(P, Q)$  is a primitive  $n$ -th root of unity (example 10.4.5). Let  $\alpha: E[n] \rightarrow E[n]$  be the endomorphism given in the basis  $\{P, Q\}$  by a matrix  $A \in M_2(\mathbb{Z}/n)$ . Show that  $e_n^L(\alpha u, \alpha v) = e_n^L(u, v)^{\det A}$  for all  $u, v \in E[n]$ , and conclude that  $e_n^L(\alpha u, \alpha v) = e_n^L(u, v)^{\deg \alpha}$  when  $\alpha$  is multiplication by an integer.
5. Let  $f: X \rightarrow Y$  be an isogeny and  $\hat{f}: \hat{Y} \rightarrow \hat{X}$  its dual. Prove the adjunction

$$e_n^Y(fx, \eta) = e_n^X(x, \hat{f}\eta)$$

for  $x \in X[n]$  and  $\eta \in \hat{Y}[n]$ , where  $e_n^X$  and  $e_n^Y$  denote the Weil pairings of  $X$  and  $Y$ . Work from the character description of the pairing in definition 10.4.1 and the mechanism of theorem 10.4.3.

6. Let  $L$  be an ample line bundle on  $X$  with polarization isogeny  $\varphi_L: X \rightarrow \hat{X}$ , and let  $e_n^L$  be as in proposition 10.4.4. Prove that the radical of  $e_n^L$  on  $X[n]$ , that is the set of  $y \in X[n]$  with  $e_n^L(x, y) = 1$  for all  $x \in X[n]$ , equals  $X[n] \cap \ker \varphi_L$ . Conclude that  $e_n^L$  is nondegenerate on  $X[n]$  precisely when  $\ker \varphi_L$  contains no nonzero  $n$ -torsion, and that this holds for all  $n$  prime to  $\text{char } k$  exactly when the prime-to- $p$  part of  $\ker \varphi_L$  vanishes, equivalently when its maximal étale quotient is a  $p$ -group (with  $p = \text{char } k$ ; connectedness of  $\ker \varphi_L$  is only the special case where the étale quotient is trivial). Deduce that if  $\varphi_L$  is an isomorphism then  $e_n^L$  is nondegenerate on every  $X[n]$  with  $n$  prime to  $\text{char } k$ , but that in characteristic  $p$  the converse fails: exhibit an  $L$  for which  $e_n^L$  is nondegenerate on all such  $X[n]$  while  $\varphi_L$  is not an isomorphism.

## 10.5 The Neron-Severi Group and Ampleness

This section reads the numerical geometry of a line bundle  $L$  off the isogeny  $\varphi_L: X \rightarrow \hat{X}$ . The degree of  $\varphi_L$  is a single integer attached to the class of  $L$ , and it computes the two invariants a line bundle has: the size of  $K(L)$ , and, through the square-root identity  $\chi(L)^2 = \deg \varphi_L$ , the Euler characteristic. When  $L$  is ample the Euler characteristic is the dimension of  $H^0(X, L)$ , so the count of sections is recovered from  $\varphi_L$  alone, with no theta functions written down. The classes  $L$  can occupy form the Neron-Severi group.

The construction  $L \mapsto \varphi_L$  factors through the class of  $L$  modulo  $\text{Pic}^0(X)$ , since an algebraically trivial bundle is translation-invariant and so has  $\varphi_L = 0$ . That quotient is the algebraic Neron-Severi group, and  $L \mapsto \varphi_L$  realizes it inside the homomorphisms  $X \rightarrow \hat{X}$ .

**Definition 10.5.1: The Neron-Severi Group**

Let  $X$  be an abelian variety over  $k$ . The *Neron-Severi group* of  $X$  is the quotient

$$\mathrm{NS}(X) := \mathrm{Pic}(X) / \mathrm{Pic}^0(X)$$

of the group of line bundles by the algebraically trivial ones. The assignment  $L \mapsto \varphi_L$  (definition 8.3.1) descends to an injective homomorphism

$$\mathrm{NS}(X) \hookrightarrow \mathrm{Hom}(X, \hat{X})$$

whose image is the group of *symmetric* homomorphisms, those  $\lambda: X \rightarrow \hat{X}$  satisfying  $\hat{\lambda} \circ \kappa = \lambda$  under the biduality isomorphism  $\kappa: X \xrightarrow{\sim} \hat{\hat{X}}$  of theorem 10.2.3.

Three facts are packaged in the definition. That  $L \mapsto \varphi_L$  is a homomorphism  $\mathrm{Pic}(X) \rightarrow \mathrm{Hom}(X, \hat{X})$  was recorded when  $\varphi_L$  was built:  $t_x^*(L \otimes L') = t_x^*L \otimes t_x^*L'$  forces  $\varphi_{L \otimes L'} = \varphi_L + \varphi_{L'}$ . That its kernel is exactly  $\mathrm{Pic}^0(X)$  is the content of descent to  $\mathrm{NS}(X)$ : a bundle with  $\varphi_L = 0$  has  $t_x^*L \cong L$  for every  $x$ , and a translation-invariant line bundle on an abelian variety is algebraically trivial (*Algebraic Geometry* [13]), so  $L \in \mathrm{Pic}^0(X)$ ; conversely  $\mathrm{Pic}^0(X)$  maps to zero. The homomorphism is therefore injective on the quotient. That its image is the symmetric homomorphisms is the algebraic form of the symmetry  $\hat{\varphi}_L \circ \kappa = \varphi_L$ , recorded for an ample  $L$  in section 10.3 (theorem 10.3.1) and extending to every class because every class is a difference of ample ones and  $L \mapsto \varphi_L$  is a homomorphism; a homomorphism  $X \rightarrow \hat{X}$  arises from a line bundle precisely when it is symmetric in this sense, and the polarizations are the symmetric homomorphisms that come from an ample bundle.

This is the algebraic counterpart of the analytic Neron-Severi group  $\mathrm{NS}(X) = \{H: \mathrm{Im}H \text{ integral on } \Lambda\}$  of Hermitian forms (definition 2.3.1 in chapter 2): there a class was a Hermitian form and  $\varphi_L$  was the map  $v \mapsto H(v, -)$ , symmetric because  $H$  is Hermitian; here a class is a bundle modulo  $\mathrm{Pic}^0$  and  $\varphi_L$  is the translation homomorphism, symmetric under biduality. Over  $\mathbb{C}$  the two descriptions agree, the Hermitian form  $H$  being the first Chern class of the bundle and the symmetry of  $v \mapsto H(v, -)$  matching the symmetry of  $\varphi_L$ . The name is inherited from the theorem of the base, which asserts that  $\mathrm{NS}(X)$  is a finitely generated abelian group for any projective variety (*Algebraic Geometry* [13]); for an abelian variety this is the statement that the symmetric homomorphisms  $X \rightarrow \hat{X}$  form a finitely generated group, of rank the Picard number  $\rho(X)$ .

The degree of  $\varphi_L$  is the numerical content of a class in  $\mathrm{NS}(X)$ . It is finite exactly when  $L$  is nondegenerate, and its value is the order of the group scheme  $K(L)$ , the kernel of  $\varphi_L$ . The identity that turns this into the theory of sections is the algebraic Riemann-Roch theorem, which equates  $\deg \varphi_L$  with the square of the Euler characteristic.

**Theorem 10.5.2: Ampleness and the Riemann-Roch Identity**

Let  $X$  be an abelian variety of dimension  $g$  over  $k$  and  $L$  a line bundle on  $X$ . Then

$$\chi(L)^2 = \deg \varphi_L = |K(L)|$$

where  $\chi(L)$  is the Euler characteristic and  $\deg \varphi_L$  is set to 0 when  $\varphi_L$  is not an isogeny. Moreover the following are equivalent:

1.  $L$  is ample;
2.  $\varphi_L: X \rightarrow \hat{X}$  is an isogeny and  $H^0(X, L) \neq 0$ ;
3.  $L$  is nondegenerate and of index 0, that is,  $K(L)$  is finite and  $H^i(X, L) = 0$  for all  $i > 0$ .

**Proof:**

*Step 1: The identity  $\chi(L)^2 = \deg \varphi_L$ .* The proof runs through the line bundle on  $X \times X$  that defines  $\varphi_L$ , the *Mumford bundle*

$$\Lambda(L) := m^*L \otimes \mathrm{pr}_1^*L^{-1} \otimes \mathrm{pr}_2^*L^{-1}$$

with  $m: X \times X \rightarrow X$  the addition and  $\mathrm{pr}_1, \mathrm{pr}_2$  the projections. Its restriction to the slice  $X \times \{x\}$  is  $t_x^*L \otimes L^{-1}$ , whose class in  $\mathrm{Pic}^0(X)$  is  $\varphi_L(x)$  (definition 8.3.1); so, as a family over the second factor,  $\Lambda(L)$  has algebraically trivial  $X$ -slices and is rigidified along  $\{e\} \times X$ . The universal property of the Poincaré bundle (proposition 10.2.2) identifies it as the pullback of  $\mathcal{P}$  along  $\mathrm{id}_X \times \varphi_L$ :

$$\Lambda(L) \cong (\mathrm{id}_X \times \varphi_L)^*\mathcal{P}$$

the family of trivial bundles it names being exactly  $\varphi_L$ . The identity is now the comparison of  $\chi(\Lambda(L))$  computed two ways.

Fix  $L$  nondegenerate, so  $\varphi_L$  is an isogeny (theorem 10.3.1) with finite kernel  $K(L)$  and  $\deg \varphi_L = |K(L)|$ . First,  $\mathrm{id}_X \times \varphi_L: X \times X \rightarrow X \times \hat{X}$  is finite flat of degree  $\deg \varphi_L$ , so pulling  $\mathcal{P}$  back along it multiplies the Euler characteristic by that degree:

$$\chi(\Lambda(L)) = \deg \varphi_L \cdot \chi(\mathcal{P})$$

The Poincaré bundle has  $\chi(\mathcal{P}) = (-1)^g$ : its cohomology on  $X \times \hat{X}$  is concentrated in the single degree  $g$  and is one dimensional there, the mixing of the two dual factors killing all other degrees (*Algebraic Geometry* [13]; Mumford, *Abelian Varieties* Sec. 16). Hence  $\chi(\Lambda(L)) = (-1)^g \deg \varphi_L$ . Second, computing  $\chi(\Lambda(L))$  directly on  $X \times X$  by expanding  $m^*L$  through the theorem of the cube (theorem 8.1.1) and running the resulting alternating sum against the Künneth decomposition of  $H^\bullet(X \times X, -)$  gives

$$\chi(\Lambda(L)) = (-1)^g \chi(L)^2$$

the two factors of  $\chi(L)$  coming from the two copies of  $X$  (*Algebraic Geometry* [13] for cohomology and base change; Mumford, *Abelian Varieties* Sec. 16 for the cube expansion). These two computations of  $\chi(\Lambda(L))$  are the cohomology-and-base-change input the section defers to *Algebraic Geometry* [13]; the reduction of the headline identity to them is the content above. Equating them and cancelling the common sign  $(-1)^g$  yields

$$\chi(L)^2 = \deg \varphi_L$$



When  $\varphi_L$  is not an isogeny,  $\text{id}_X \times \varphi_L$  is not finite,  $K(L)$  is positive-dimensional, and  $\chi(L) = 0$ : a translation-invariant direction kills the top self-intersection ( $L^g$ ), hence the leading term of  $\chi$ , so both sides are 0.

*Step 2: (1) implies (3).* Let  $L$  be ample. Then  $K(L)$  is finite (proposition 8.3.4), so  $L$  is nondegenerate. Ampleness gives  $H^i(X, L) = 0$  for  $i > 0$ : an ample line bundle on an abelian variety has index 0, meaning its cohomology is concentrated in degree 0 (*Algebraic Geometry* [13]; the vanishing is standard, proved there by the Mumford-bundle vanishing theorem, and it is the cohomology-and-base-change input deferred by this section). Thus  $L$  is nondegenerate of index 0.

*Step 3: (3) implies (2).* If  $L$  is nondegenerate then  $\varphi_L$  is an isogeny (theorem 10.3.1). If moreover  $H^i(X, L) = 0$  for all  $i > 0$ , then  $\chi(L) = \dim_k H^0(X, L)$ ; and  $\chi(L) \neq 0$  because  $\chi(L)^2 = \deg \varphi_L = |K(L)| > 0$  by Step 1. Hence  $\dim_k H^0(X, L) = \chi(L) \neq 0$ , so  $H^0(X, L) \neq 0$ .

*Step 4: (2) implies (1).* Suppose  $\varphi_L$  is an isogeny and  $H^0(X, L) \neq 0$ . A nonzero section gives an effective divisor  $D$  with  $L \cong \mathcal{O}_X(D)$ . Nondegeneracy of  $L$  means  $K(L)$  is finite, so no positive-dimensional abelian subvariety  $Y$  has  $L|_Y$  translation-invariant; equivalently,  $D$  has no positive-dimensional subvariety of translations fixing it. For an effective divisor  $D$  on an abelian variety, the finiteness of  $K(\mathcal{O}_X(D))$  is exactly the condition that  $\mathcal{O}_X(D)$  is ample: the linear system of  $\mathcal{O}_X(2D)$  separates points and tangent vectors once  $D$  is nondegenerate and effective, so  $\mathcal{O}_X(2D)$ , hence  $L$ , is ample (the Lefschetz-style argument, *Algebraic Geometry* [13]; the analytic template is chapter 4). Thus  $L$  is ample, closing the cycle  $(1) \Rightarrow (3) \Rightarrow (2) \Rightarrow (1)$ , completing the proof.

The theorem is the algebraic replacement for the Riemann relations. Over  $\mathbb{C}$  ampleness meant a positive definite Hermitian form and the count of sections was the Pfaffian of its imaginary part; here there is no Hermitian form, and ampleness is instead the conjunction of two conditions read off  $\varphi_L$ : that it is an isogeny (nondegeneracy, so  $K(L)$  is finite) and that  $L$  has a section (index 0, the positivity that in the analytic theory was the sign of  $H$ ). The identity  $\chi(L)^2 = \deg \varphi_L$  converts the degree of the isogeny, a purely geometric quantity, into the square of a cohomological one. It is the mechanism by which the dual abelian variety measures the size of linear systems:  $\varphi_L$  knows  $|K(L)|$ , and  $|K(L)|$  knows  $\chi(L)^2$ . The equivalence  $(1) \Leftrightarrow (3)$  is the one used in practice, since nondegeneracy is checked on  $\varphi_L$  and the index-0 vanishing is the standard vanishing for an ample bundle on an abelian variety.

For an ample bundle the vanishing of higher cohomology removes the ambiguity of sign in the square root, and the Euler characteristic becomes a dimension.

**Proposition 10.5.3: Algebraic Riemann-Roch**

Let  $X$  be an abelian variety of dimension  $g$  over  $k$  and  $L$  an ample line bundle on  $X$ . Then

$$\dim_k H^0(X, L) = \chi(L) = \sqrt{\deg \varphi_L} = \frac{(L^g)}{g!}$$

and  $\dim_k H^0(X, L)^2 = |K(L)|$ .



**Proof:**

Since  $L$  is ample, theorem 10.5.2 gives  $H^i(X, L) = 0$  for  $i > 0$ , so

$$\dim_k H^0(X, L) = \chi(L)$$

The asymptotic Riemann-Roch expansion on a smooth projective variety computes  $\chi(L)$  as the degree- $g$  part of  $\int_X \text{ch}(L) \text{td}(X)$  (*Algebraic Geometry* [13]); the tangent bundle of an abelian variety is trivial, so its Todd class is 1 and the expression collapses to the top self-intersection

$$\chi(L) = \frac{(L^g)}{g!}$$

with no lower-order corrections. Finally theorem 10.5.2 gives  $\chi(L)^2 = \deg \varphi_L = |K(L)|$ , and  $\chi(L) > 0$  for  $L$  ample (it is the positive dimension  $\dim_k H^0(X, L)$ ), so taking the positive square root yields  $\dim_k H^0(X, L) = \chi(L) = \sqrt{\deg \varphi_L}$ , as we sought to show.

The proposition is the algebraic Riemann-Roch theorem for an abelian variety, and it recovers the Pfaffian count of chapter 3 without a single theta series. There  $\dim H^0(X, L) = \text{Pf}(E) = d_1 \cdots d_g$  was proved analytically by a Fourier decomposition of theta functions (theorem 3.4.2); the two agree because  $\deg \varphi_L = \det E = (d_1 \cdots d_g)^2$ , so the algebraic square root  $\sqrt{\deg \varphi_L}$  is exactly the Pfaffian  $\sqrt{\det E}$ . The type  $(d_1, \dots, d_g)$  of the analytic polarization reappears algebraically as the group scheme  $K(L) \cong \prod_i (\mathbb{Z}/d_i)^2$ , whose order  $(d_1 \cdots d_g)^2$  is  $\deg \varphi_L$ . The self-intersection formula  $\chi(L) = (L^g)/g!$  says the number of sections is a volume, the top self-intersection number of the ample class divided by  $g!$ . This is what makes the projective embedding of Part II quantitative in the same way theorem 3.4.2 did for Part I, and it does so over any field, with no complex analysis available.

The whole apparatus specializes to the one case where every quantity is a degree.

**Example 10.5.4: The Neron-Severi Group of an Elliptic Curve**

Let  $E$  be an elliptic curve over  $k$ , and use the autoduality  $E \xrightarrow{\sim} \hat{E}$  of example 5.1.2. A line bundle  $L$  of degree  $d$  has  $\varphi_L = [d]: E \rightarrow \hat{E} \cong E$  (example 8.3.5), and two line bundles have the same  $\varphi_L$  exactly when they differ by a degree-0 bundle, that is, by an element of  $\text{Pic}^0(E)$ . Hence the degree is a complete invariant of a class modulo  $\text{Pic}^0$ , and

$$\text{NS}(E) \cong \mathbb{Z}$$

the isomorphism sending the class of  $L$  to  $\deg L$ , with  $\varphi_L = [d]$  the symmetric homomorphism  $E \rightarrow \hat{E}$  attached to the integer  $d$ . This is the algebraic reading of the analytic  $\text{NS}(E) \cong \mathbb{Z}$  of definition 2.3.1, where a class was a Hermitian form  $H_m = mH_1$  and the integer was again the degree.

The Riemann-Roch identities are transparent here. For  $g = 1$  the Euler characteristic of a degree- $d$  bundle is its degree,  $\chi(L) = \deg L = d$  (Riemann-Roch for a genus-one curve gives  $\chi(L) = \deg L$ ; *Elliptic Curves* [4]), while

$$\deg \varphi_L = \deg [d] = d^2$$

since multiplication by  $d$  on an elliptic curve has degree  $d^2$ . The identity of theorem 10.5.2 reads

$$\chi(L)^2 = d^2 = \deg \varphi_L$$

and for  $L$  ample, meaning  $d > 0$ , proposition 10.5.3 gives  $\dim_k H^0(E, L) = \chi(L) = \sqrt{d^2} = d$ , the classical count of sections of a positive-degree line bundle on an elliptic curve. The kernel  $K(L) = E[d]$  has order  $d^2 = \deg \varphi_L = \chi(L)^2$ , so all three invariants, the size of  $K(L)$ , the degree of  $\varphi_L$ , and the square of the Euler characteristic, coincide with  $d^2$ . This is the  $g = 1$  shadow of the general Pfaffian count of chapter 3: the tuple  $(d_1, \dots, d_g)$  collapses to the single elementary divisor  $d$ , the Pfaffian  $d_1 \cdots d_g$  to  $d$ , and  $\det E$  to  $d^2$ .

### Exercise 10.5.1

1. Let  $E$  be an elliptic curve over  $k$  and  $L$  a line bundle of degree  $d$  on  $E$ . Using example 10.5.4, compute  $\deg \varphi_L$ , the order of  $K(L)$ , and  $\chi(L)$ , and verify the identity of theorem 10.5.2 for this  $L$ . For which  $d$  is  $L$  ample, and what is  $\dim_k H^0(E, L)$  in that case?
2. Show directly from the definition of  $\text{NS}(X)$  (definition 10.5.1) that  $L \mapsto \varphi_L$  is a homomorphism  $\text{Pic}(X) \rightarrow \text{Hom}(X, \hat{X})$ , and that an algebraically trivial bundle maps to 0. Deduce that  $\varphi_L$  depends only on the class of  $L$  in  $\text{NS}(X)$ .
3. Let  $X$  be an abelian variety of dimension  $g$  and  $L$  an ample line bundle on  $X$  with  $\dim_k H^0(X, L) = n$ . Express  $\deg \varphi_L$ ,  $|K(L)|$ , and the top self-intersection  $(L^g)$  in terms of  $n$  and  $g$ , citing the results of this section. Now let  $M = L^{\otimes 2}$ ; compute the same three quantities for  $M$  in terms of  $n$  and  $g$ .
4. Suppose  $L$  is a nondegenerate line bundle on an abelian variety  $X$  with  $\chi(L) = 0$ . Explain why this is impossible, that is, show that a nondegenerate bundle has  $\chi(L) \neq 0$ . Conversely, if  $\chi(L) = 0$ , what can be said about  $\varphi_L$  and  $K(L)$ ?
5. Prove that for a line bundle  $L$  on an abelian variety  $X$  of dimension  $g$ , ampleness is equivalent to the conjunction “ $\varphi_L$  is an isogeny and  $H^0(X, L) \neq 0$ ”, deriving this equivalence from the three-way statement of theorem 10.5.2. Use it to show that if  $L$  is ample then so is  $L^{\otimes m}$  for every integer  $m \geq 1$ .
6. Let  $X = E_1 \times E_2$  be a product of two elliptic curves, and let  $L = \text{pr}_1^* L_1 \otimes \text{pr}_2^* L_2$  with  $L_i$  a line bundle of degree  $d_i > 0$  on  $E_i$ . Assuming  $\chi(L) = \chi(L_1)\chi(L_2)$  and that  $\varphi_L$  is the product  $\varphi_{L_1} \times \varphi_{L_2}$  under the identification  $\widehat{E_1 \times E_2} \cong \hat{E}_1 \times \hat{E}_2$ , compute  $\deg \varphi_L$ ,  $|K(L)|$ , and  $\dim_k H^0(X, L)$ , and confirm the identity  $\chi(L)^2 = \deg \varphi_L = |K(L)|$  directly.

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## *Polarizations, Rosati, and Endomorphism Algebras*

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The endomorphism algebra, the Rosati involution, and Albert's classification were already assembled in chapter 6, but entirely over  $\mathbb{C}$ , where a polarization was a positive Hermitian form and its positivity was visible to the eye. This chapter rebuilds the same structure over an arbitrary field, with the algebraic polarization  $\varphi_L: X \rightarrow \hat{X}$  of chapter 10 in place of the Hermitian form. The single input is one inequality: the Rosati involution attached to a polarization is *positive*,  $\text{Tr}(ff') > 0$  for every nonzero endomorphism  $f$ .

That inequality does the work of the whole chapter. It forces  $\text{End}^0(X)$  to be a semisimple algebra, which closes the finite-dimensionality left open in chapter 9; it pins down the possible endomorphism algebras through Albert's classification, now sensitive to the characteristic in a way the analytic theory never saw; and it splits every abelian variety, up to isogeny, into simple pieces (Poincaré reducibility). The chapter ends by making the parallel with Part I exact: over  $\mathbb{C}$  the analytic abelian variety and the algebraic one are the same object, and the two developments of the endomorphism theory agree under that identification. The positivity proved here returns in chapter 13 as the Riemann hypothesis for abelian varieties over a finite field.

### 11.1 The Rosati Involution and Its Positivity

A polarization does more than embed  $X$  in projective space: the dual construction of chapter 10 attaches to each  $f: X \rightarrow X$  its dual  $\hat{f}: \hat{X} \rightarrow \hat{X}$ , and a polarization  $\lambda: X \rightarrow \hat{X}$  is precisely the isomorphism (over  $\mathbb{Q}$ ) that lets one transport  $\hat{f}$  back to an endomorphism of  $X$ . This section builds the resulting operation  $f \mapsto f'$ , the Rosati involution, algebraically over an arbitrary field  $k$ , in terms of the degree and the trace on the Tate module rather than the Hodge-theoretic trace of chapter 6, and

proves its defining positivity property,  $\text{Tr}(ff') > 0$  for  $f \neq 0$ . The downstream uses of that positivity (the semisimplicity of  $\text{End}^0(X)$ , chapter 9; Albert's classification, section 11.2; and the Weil bound, chapter 13) are the business of the rest of the chapter. The analytic involution of definition 6.2.1 is its complex counterpart.

Fix once and for all a polarization  $\lambda = \varphi_L: X \rightarrow \hat{X}$  (definition 10.3.2) attached to an ample line bundle  $L$ . Since  $\lambda$  is an isogeny (theorem 10.3.1), it becomes invertible in  $\text{Hom}(\hat{X}, X) \otimes \mathbb{Q}$ , so the formula below lands in  $\text{End}^0(X)$  even though  $\lambda^{-1}$  is not itself a morphism.

**Definition 11.1.1: Rosati Involution (algebraic)**

Let  $X$  be an abelian variety over  $k$  with polarization  $\lambda = \varphi_L: X \rightarrow \hat{X}$ . The *Rosati involution* associated to  $\lambda$  is the  $\mathbb{Q}$ -linear map

$$': \text{End}^0(X) \rightarrow \text{End}^0(X), \quad f' = \lambda^{-1} \circ \hat{f} \circ \lambda$$

where  $\hat{f}: \hat{X} \rightarrow \hat{X}$  is the dual endomorphism (proposition 10.1.3) and  $\lambda^{-1} \in \text{Hom}(\hat{X}, X) \otimes \mathbb{Q}$  is the inverse of  $\lambda$  in the isogeny category. One calls  $f'$  the *Rosati adjoint* of  $f$ .

The composite  $f' = \lambda^{-1} \circ \hat{f} \circ \lambda$  traverses the square with vertical edges  $\lambda: X \rightarrow \hat{X}$  and horizontal edges  $f$  on top,  $\hat{f}$  on the bottom, running down the left, across the bottom, and back up the right:  $f$  and  $f'$  are conjugate under  $\lambda$  up to replacing  $f$  by its dual. When  $\lambda$  is a principal polarization it is an isomorphism and  $f'$  is already an integral endomorphism; in general  $\lambda^{-1}$  is only rational, so the involution is defined on  $\text{End}^0(X) = \text{End}(X) \otimes \mathbb{Q}$  and need not preserve  $\text{End}(X)$ . This is the exact algebraic transcription of the analytic recipe  $f' = \varphi_L^{-1} \circ \hat{f} \circ \varphi_L$  from definition 6.2.1, now with  $\hat{f}$  the dual of a morphism of abelian varieties rather than the transpose of a  $\mathbb{C}$ -linear map of the dual torus. Before extracting any structure from  $'$  one must know it deserves the name involution: that it reverses products and squares to the identity. Both facts come directly from the functoriality of the dual and the symmetry of a polarization.

**Proposition 11.1.2: The Rosati involution is a  $\mathbb{Q}$ -linear anti-involution**

The map  $f \mapsto f'$  of definition 11.1.1 satisfies, for all  $f, g \in \text{End}^0(X)$  and  $a \in \mathbb{Q}$ ,

1.  $(f + g)' = f' + g'$  and  $(af)' = af'$ ;
2.  $(fg)' = g'f'$ ;
3.  $f'' = f$ .

Thus  $'$  is a  $\mathbb{Q}$ -linear anti-involution of  $\text{End}^0(X)$ . Moreover  $[n]' = [n]$  for every  $n \in \mathbb{Z}$ .

**Proof:**

Throughout, write  $\lambda: X \rightarrow \hat{X}$  for the fixed polarization and  $\kappa: X \xrightarrow{\sim} \hat{\hat{X}}$  for the biduality isomorphism (theorem 10.2.3). The formation of the dual is additive and contravariant on the isogeny category (proposition 10.1.3):  $\widehat{\hat{f} + g} = \hat{f} + \hat{g}$ ,  $\widehat{af} = a\hat{f}$ , and  $\widehat{fg} = \hat{g}\hat{f}$ . The one further input is that a polarization is *symmetric*: under the identification  $\kappa$ , one has  $\hat{\lambda} \circ \kappa = \lambda$  (theorem 10.3.1), which we abbreviate by saying  $\hat{\lambda} = \lambda$  under biduality.

1.  *$\mathbb{Q}$ -linearity.* Since  $f \mapsto \hat{f}$  is  $\mathbb{Q}$ -linear and  $\lambda, \lambda^{-1}$  are fixed,  $(f + g)' = \lambda^{-1}(\widehat{f + g})\lambda = \lambda^{-1}(\hat{f} + \hat{g})\lambda = f' + g'$ , and likewise  $(af)' = af'$ .

2. *Anti-multiplicativity.* Using contravariance of the dual,

$$(fg)' = \lambda^{-1} \widehat{fg} \lambda = \lambda^{-1} \hat{g} \hat{f} \lambda = (\lambda^{-1} \hat{g} \lambda)(\lambda^{-1} \hat{f} \lambda) = g' f'$$

where the middle equality inserts  $\lambda\lambda^{-1} = \text{id}$  between  $\hat{g}$  and  $\hat{f}$ .

3. *Involutivity.* By definition  $f' = \lambda^{-1} \hat{f} \lambda$ , so  $(f')' = \lambda^{-1} \widehat{f'} \lambda$ . Compute the dual of  $f'$  using contravariance and  $\hat{\hat{f}} = \kappa f \kappa^{-1}$  (the standard identity relating the double dual of a morphism to biduality):

$$\widehat{f'} = \widehat{\lambda^{-1} \hat{f} \lambda} = \hat{\lambda} \hat{\hat{f}} \widehat{\lambda^{-1}} = \hat{\lambda} (\kappa f \kappa^{-1}) (\hat{\lambda})^{-1}$$

Now substitute  $\hat{\lambda} = \lambda \kappa^{-1}$  (the symmetry  $\hat{\lambda} \circ \kappa = \lambda$  solved for  $\hat{\lambda}$ ). Then  $\hat{\lambda} \kappa = \lambda$  and  $\kappa^{-1}(\hat{\lambda})^{-1} = \lambda^{-1}$ , so

$$\widehat{f'} = (\hat{\lambda} \kappa) f (\kappa^{-1}(\hat{\lambda})^{-1}) = \lambda f \lambda^{-1}$$

Therefore  $(f')' = \lambda^{-1}(\lambda f \lambda^{-1})\lambda = f$ .

For the final assertion,  $[n]$  is central in  $\text{End}^0(X)$  and its dual is  $\widehat{[n]} = [n]$  on  $\hat{X}$ ; hence  $[n]' = \lambda^{-1}[n]\lambda = [n]$ . All three parts together say  $'$  is a  $\mathbb{Q}$ -linear anti-involution, as we sought to show.

The proposition makes  $(\text{End}^0(X), ')$  an algebra with anti-involution over  $\mathbb{Q}$ , exactly the datum Albert's classification takes as input (section 11.2). Two features distinguish the Rosati involution from an abstract anti-involution and make it powerful. First, it fixes  $\mathbb{Z}$  pointwise ( $[n]' = [n]$ ), so it is  $\mathbb{Q}$ -linear over the centre in a way that pins down its action. Second, it is *positive*: the associated trace form is positive-definite. Positivity is what promotes the formal structure of Wedderburn theory into the rigid Albert list; without it, any finite-dimensional  $\mathbb{Q}$ -algebra with anti-involution could occur.

To state positivity one needs a trace on  $\text{End}^0(X)$ . The rational representation on the Tate module supplies it: for a prime  $\ell \neq \text{char } k$ , an endomorphism  $f$  acts  $\mathbb{Q}_\ell$ -linearly on  $V_\ell X \cong \mathbb{Q}_\ell^{2g}$  (definition 9.2.4), and  $\text{Tr}(f)$  denotes the trace of this action. This trace is a rational number independent of  $\ell$  (it agrees with the trace of  $f$  on the  $2g$ -dimensional rational homology / the characteristic polynomial  $P_f$ , whose  $\ell$ -independent integrality follows from the degree mechanism  $P_h(n) = \deg([n] - h)$  developed in chapter 13), and it extends  $\mathbb{Q}$ -linearly to  $\text{End}^0(X)$ . The positivity statement is the one inequality announced at the outset; it does not rest on this integrality, since Step 3 below exhibits  $\text{Tr}(ff')$  as an  $\ell$ -independent intersection number directly.

### Theorem 11.1.3: Positivity of the Rosati involution

Let  $X$  be an abelian variety of dimension  $g$  over  $k$  with polarization  $\lambda = \varphi_L$ ,  $L$  ample, and let  $'$  be the associated Rosati involution. The symmetric  $\mathbb{Q}$ -bilinear form

$$\langle f, g \rangle = \text{Tr}(fg'), \quad f, g \in \text{End}^0(X)$$

is positive-definite. Equivalently,  $\text{Tr}(ff') > 0$  for every  $f \in \text{End}^0(X)$  with  $f \neq 0$ .

**Proof:**

*Step 1: the form is symmetric.* The trace is invariant under the Rosati involution:  $f' = \lambda^{-1} \hat{f} \lambda$  is conjugate in  $\text{End}^0(X)$  to  $\hat{f}$ , and  $\hat{f}$  acts on  $V_\ell \hat{X}$  as the transpose of the action of  $f$  on  $V_\ell X$  (the Tate module of the dual is the  $\mathbb{Q}_\ell$ -dual of  $V_\ell X$  under the Weil pairing, definition 10.4.1), so  $\text{Tr}(f') = \text{Tr}(\hat{f}) = \text{Tr}(f)$ . Applying this to  $gf'$  and using  $(gf')' = f''g' = fg'$  from proposition 11.1.2,

$$\langle g, f \rangle = \text{Tr}(gf') = \text{Tr}((gf')') = \text{Tr}(fg') = \langle f, g \rangle$$

so the form is symmetric.

*Step 2: reduction to a polarization homomorphism.* By cyclicity of the trace,  $\text{Tr}(ff') = \text{Tr}(f'f)$ , so it suffices to analyse  $f'f$ . Unwinding the definition of the Rosati adjoint,  $\lambda \circ f' = \hat{f} \circ \lambda$ , hence

$$\lambda \circ (f'f) = \hat{f} \circ \lambda \circ f = \hat{f} \circ \varphi_L \circ f$$

The composite  $\hat{f} \circ \varphi_L \circ f$  is exactly the polarization homomorphism  $\varphi_{f^*L}$  of the pulled-back bundle  $f^*L$ : this is the functoriality  $\varphi_{f^*L} = \hat{f} \circ \varphi_L \circ f$  of the map  $L \mapsto \varphi_L$  under pullback (definition 10.5.1). Therefore

$$\lambda \circ (f'f) = \varphi_{f^*L}$$

Both  $f'f$  and its image under  $\lambda$  are symmetric:  $f'f$  is fixed by the Rosati involution, since  $(f'f)' = f'f'' = f'f$ , and correspondingly  $\varphi_{f^*L}$  is a symmetric homomorphism  $X \rightarrow \hat{X}$ , so it is the polarization homomorphism of a genuine class  $f^*L \in \text{NS}(X) \otimes \mathbb{Q}$  (definition 10.5.1; the rational coefficients are explained just below). No ampleness of  $f^*L$  is claimed or needed: a general nonzero  $f \in \text{End}^0(X)$  need not be an isogeny (on a non-simple  $X$  its image can have dimension  $< g$ , so  $\varphi_{f^*L}$  need not be an isogeny), but  $f^*L$  is nonetheless a well-defined class whose first Chern class is the pullback of an ample class. That is all Step 3 requires: it computes the trace as an intersection number against  $D = c_1(L)$ , and the sign will come from ampleness of  $D$  together with nefness of the pullback, not from any ampleness of  $f^*L$ .

Two points of bookkeeping keep this rigorous. First,  $f \in \text{End}^0(X) = \text{End}(X) \otimes \mathbb{Q}$  is only a  $\mathbb{Q}$ -endomorphism; choose an integer  $N \geq 1$  with  $Nf \in \text{End}(X)$ , so that  $(Nf)^*L$  and  $c_1((Nf)^*L) = (Nf)^*D$  are honest, and read  $f^*D := N^{-2}(Nf)^*D \in \text{NS}(X) \otimes \mathbb{Q}$ . Since  $\text{Tr}((Nf)(Nf)') = N^2 \text{Tr}(ff')$  and  $N^2 > 0$ , the sign of  $\text{Tr}(ff')$  is unaffected, and one may argue with  $Nf$  in place of  $f$ . Second, replacing  $f$  by  $Nf$  leaves the vanishing  $f = 0 \Leftrightarrow Nf = 0$  intact, so “ $f \neq 0$ ” is genuinely the hypothesis in play.

*Step 3: the trace is a positive multiple of a top intersection number.* Only the sign now remains. Write  $D = c_1(L)$ , an ample class. The trace of the composite  $f'f = \lambda^{-1} \varphi_{f^*L}$  is computed by the standard identity relating the trace of  $\lambda^{-1} \varphi_M$  to the intersection of the class  $M$  with  $L$ : for any  $M \in \text{NS}(X) \otimes \mathbb{Q}$ ,

$$\text{Tr}(\lambda^{-1} \varphi_M) = 2g \frac{(D^{g-1} \cdot c_1(M))}{(D^g)}$$

whose input, the expression of the trace of a symmetric endomorphism and of  $\chi$  through top self-intersection numbers  $(D^g) = g! \chi(L)$ , is cited to *Algebraic Geometry* [13]. Taking  $M = f^*L$ ,

so  $c_1(M) = f^*D$  (a class in  $\mathrm{NS}(X) \otimes \mathbb{Q}$ , read through  $Nf$  as above), gives

$$\mathrm{Tr}(ff') = \mathrm{Tr}(f'f) = 2g \frac{(D^{g-1} \cdot f^*D)}{(D^g)}$$

Now  $D$  is ample, and  $f^*D$  is *nef*, being the pullback of the ample class  $D$  under a morphism (pullback of a nef class is nef, and an ample class is nef); moreover  $f^*D \neq 0$  in  $\mathrm{NS}(X) \otimes \mathbb{Q}$ , since  $f \neq 0$  forces  $Nf$  to be a nonconstant morphism, whose pullback of an ample class is nonzero. The numerator  $(D^{g-1} \cdot f^*D)$  is then strictly positive: pairing a nonzero nef class against a product of ample classes gives a positive number (the positivity part of the Nakai-Moishezon / Kleiman criterion, an ample class having strictly positive intersection with any nonzero nef class, cited to *Algebraic Geometry* [13]). The denominator  $(D^g) = g! \chi(L) > 0$  is positive because  $L$  is ample. Therefore  $\mathrm{Tr}(ff') > 0$  whenever  $f \neq 0$ .

The form  $\langle f, g \rangle = \mathrm{Tr}(fg')$  is symmetric (Step 1) and satisfies  $\langle f, f \rangle = \mathrm{Tr}(ff') > 0$  for  $f \neq 0$  (Steps 2 and 3), so it is positive-definite, completing the proof.

The positivity of  $\mathrm{Tr}(ff')$  is an arithmetic fact wearing a geometric proof: the trace of a purely algebraic operation on  $\mathrm{End}^0(X)$  turns out to count how the ample cone meets an effective cycle, and ampleness supplies the sign. Over  $\mathbb{C}$  this is theorem 6.2.3, where the same number is  $2 \operatorname{Re}$  of a Hermitian inner product of  $L$  against  $f^*L$ ; the algebraic proof replaces the Hermitian form by intersection theory and thereby survives into characteristic  $p$ , where no Hermitian form is available. The inequality is what makes the endomorphism algebra tractable, as the next corollary shows.

Positivity of the trace form is exactly the hypothesis under which a finite-dimensional algebra with anti-involution cannot have nilpotents in its radical: a nonzero element of the radical would be nilpotent, and its trace form would vanish identically against itself. This is the algebraic content that finishes the structural claim about  $\mathrm{End}^0(X)$  left dangling in chapter 9.

#### Corollary 11.1.4: $\mathrm{End}^0(X)$ is semisimple

The  $\mathbb{Q}$ -algebra  $\mathrm{End}^0(X)$  is finite-dimensional and semisimple.

#### Proof:

Finite-dimensionality is now in hand: the integral injectivity  $\mathrm{End}(X) \hookrightarrow \mathrm{End}(T_\ell X)$  of theorem 9.3.2 is native, and the positive form of theorem 11.1.3 supplies the rank bound left open there, so  $\mathrm{End}^0(X)$  is a finite-dimensional  $\mathbb{Q}$ -algebra (this is the discharge recorded in corollary 9.3.3). It remains to show the Jacobson radical  $J$  of  $\mathrm{End}^0(X)$  is zero. Suppose, for contradiction, that  $J$  contains a nonzero element  $f$ . Since  $J$  is a two-sided ideal,  $ff'$  again lies in  $J$ . The radical of a finite-dimensional algebra is a nilpotent ideal, so  $ff'$  is nilpotent; a nilpotent endomorphism acts nilpotently on  $V_\ell X$  and therefore has trace zero, giving  $\mathrm{Tr}(ff') = 0$ . But theorem 11.1.3 gives  $\mathrm{Tr}(ff') > 0$  for  $f \neq 0$ , a contradiction. Hence  $J = 0$  and  $\mathrm{End}^0(X)$  is semisimple, as we sought to show.

By Wedderburn-Artin,  $\mathrm{End}^0(X)$  is therefore a finite product of matrix algebras over division rings,  $\mathrm{End}^0(X) \cong \prod_i M_{n_i}(D_i)$ ; the shape of the factors, together with the constraint the Rosati involution

imposes on each, is the subject of section 11.2. This corollary discharges the forward deferral recorded in chapter 9: the semisimplicity of  $\text{End}^0(X)$  (and with it the finite-generation bound  $\dim_{\mathbb{Q}} \text{End}^0(X) \leq 4g^2$ ) rested on the existence of a positive anti-involution, which theorem 11.1.3 now provides.

In dimension one the three possibilities for  $\text{End}^0(E)$  already display all three of Albert's positive-involution types, and one can read off the Rosati involution and its positivity by hand.

**Example 11.1.5: The Rosati involution on an elliptic curve**

Let  $E$  be an elliptic curve over  $k$ , principally polarized by the class of the origin, so  $\lambda = \varphi_{\mathcal{O}_E([0])}: E \xrightarrow{\sim} \hat{E}$  is an isomorphism (the autoduality  $\text{Jac}(E) \cong E$ , previewed in chapter 12). Because  $\lambda$  is a principal polarization, the Rosati involution preserves  $\text{End}(E)$  itself. The endomorphism algebra  $\text{End}^0(E) = \text{End}(E) \otimes \mathbb{Q}$  is one of three algebras, developed for  $g = 1$  in *Elliptic Curves* [4], and in each the Rosati involution is a classical conjugation:

1.  $\text{End}^0(E) = \mathbb{Q}$  (the generic case). The only endomorphisms are the multiplications  $[n]$ , and  $\lambda = [1]$  up to a positive scalar; the Rosati involution is the identity. Positivity reads  $\text{Tr}([n][n]') = \text{Tr}([n]^2) = 2n^2 > 0$  for  $n \neq 0$ , since  $[n]$  acts as  $n$  on the 2-dimensional  $V_\ell E$ .
2.  $\text{End}^0(E) = K$ , an imaginary quadratic field (complex multiplication). An endomorphism  $f$  acts on  $V_\ell E$  with characteristic polynomial  $t^2 - \text{Tr}(f)t + \deg f$ ; the Rosati adjoint  $f'$  is the complex conjugate  $\bar{f}$ , so  $ff' = f\bar{f} = N_{K/\mathbb{Q}}(f) = \deg f$  is the norm. Positivity  $\text{Tr}(ff') = 2 \deg f > 0$  is the positivity of the norm on the imaginary quadratic field, and the trace form is the trace form of  $K/\mathbb{Q}$  scaled by the polarization.
3.  $\text{End}^0(E) = D$ , a definite quaternion algebra over  $\mathbb{Q}$  (the supersingular case in characteristic  $p$ ). Here  $f'$  is the canonical (main) involution of the quaternion algebra,  $f \mapsto \bar{f}$  with  $f\bar{f} = \text{Nrd}(f)$  the reduced norm, and  $\text{Tr}(ff') = 2 \text{Nrd}(f)$ . Positivity is the definiteness of the reduced norm:  $\text{Nrd}(f) > 0$  for  $f \neq 0$  exactly because  $D$  is a definite (i.e. ramified at the real place) quaternion algebra. A positive Rosati involution on  $\text{End}^0(E)$  forces  $D$  to be definite, which is why the indefinite rational quaternion algebra does not arise as the endomorphism algebra of an elliptic curve.

In all three cases  $\text{Tr}(ff')$  is twice the appropriate norm or degree, always strictly positive on nonzero  $f$ , as theorem 11.1.3 demands. Over  $\mathbb{C}$  these are the identity, complex conjugation on  $K$ , and the quaternion conjugation of theorem 6.2.3; the algebraic account here adds the third, definite-quaternion, case, which lives only in characteristic  $p$ .

**Exercise 11.1.1**

1. Let  $\lambda = \varphi_L: X \rightarrow \hat{X}$  be a polarization and  $'$  the associated Rosati involution (definition 11.1.1). Using only the definition  $f' = \lambda^{-1} \circ \hat{f} \circ \lambda$  and the functoriality of the dual, show directly that  $[n]' = [n]$  for every  $n \in \mathbb{Z}$  and that  $(\text{id}_X)' = \text{id}_X$ . Deduce that a scalar  $a \in \mathbb{Q} \subseteq \text{End}^0(X)$  satisfies  $a' = a$ .
2. An endomorphism  $f \in \text{End}^0(X)$  is called *symmetric* (with respect to  $\lambda$ ) if  $f' = f$ , and *anti-symmetric* if  $f' = -f$ . Using proposition 11.1.2, prove that for every  $f \in \text{End}^0(X)$  both  $f'f$  and  $f + f'$  are symmetric while  $f - f'$  is anti-symmetric. Conclude that every  $f \in \text{End}^0(X)$  is uniquely a sum of a symmetric and an anti-symmetric element.



3. Recall from Step 1 of the proof of theorem 11.1.3 that the Tate-module trace is Rosati-invariant,  $\text{Tr}(f') = \text{Tr}(f)$ . Show that the bilinear form  $\langle f, g \rangle = \text{Tr}(fg')$  of theorem 11.1.3 is *associative* for the involution, in the sense that

$$\langle hf, g \rangle = \langle f, h'g \rangle \quad \text{for all } f, g, h \in \text{End}^0(X)$$

and that  $'$  is self-adjoint for this form,  $\langle f', g \rangle = \langle g', f \rangle$ .

4. Let  $E$  be an elliptic curve with complex multiplication by an imaginary quadratic field  $K$ , principally polarized as in example 11.1.5(2), so that  $f' = \bar{f}$  is complex conjugation on  $K = \text{End}^0(E)$ . Take  $f = \alpha \in K$  with  $\text{Tr}_{K/\mathbb{Q}}(\alpha) = s$  and  $N_{K/\mathbb{Q}}(\alpha) = n$ . Compute  $\text{Tr}(ff')$ ,  $\text{Tr}(f)$ , and the Gram matrix of the form  $\langle, \rangle$  on the  $\mathbb{Q}$ -basis  $\{1, \alpha\}$  of  $K$ , and verify directly that this matrix is positive-definite, in accordance with theorem 11.1.3.
5. Prove that the Rosati involution is *faithful* in the following sense: an element  $f \in \text{End}^0(X)$  satisfies  $f'f = 0$  if and only if  $f = 0$ . Deduce that  $\text{End}^0(X)$  has no nonzero two-sided ideal  $I$  on which the trace form  $\langle f, f \rangle$  vanishes identically, and explain in one sentence how this recovers the semisimplicity of corollary 11.1.4.
6. Let  $\mu = \varphi_M: X \rightarrow \hat{X}$  be a *second* polarization, with Rosati involution  $'_\mu$ , and write  $u = \lambda^{-1}\mu \in \text{End}^0(X)$ . Show that  $u$  is symmetric for the  $\lambda$ -Rosati involution ( $u' = u$ ) and that the two involutions are related by

$$f'_\mu = u^{-1} f' u \quad \text{for all } f \in \text{End}^0(X)$$

Show moreover that  $\text{Tr}(f f'_\mu) > 0$  for  $f \neq 0$  directly from theorem 11.1.3 applied to  $\mu$ , and conclude that  $u$  is a symmetric element with  $\text{Tr}(u g' g) > 0$  for every  $g \neq 0$  (positivity of  $u$  for the  $\lambda$ -form).

## 11.2 Albert's Classification

The positivity of the Rosati involution, together with the semisimplicity it forced in corollary 11.1.4, pins down  $\text{End}^0(X)$  far more tightly than semisimplicity alone. For  $X$  simple the algebra is a division algebra, and it carries a positive anti-involution; the finite-dimensional division algebras over  $\mathbb{Q}$  that admit a positive involution form a short list, enumerated by Albert. This section states that list over an arbitrary field and works out the numerical constraints tying it to  $\dim X$ . The pure-algebra core, the classification of division algebras with positive involution, is a theorem about algebras and is cited to *Core Algebra* [12]; what is native here is the passage from the geometry to such an algebra, and the way the admissible invariants depend on the characteristic of  $k$ .

The object to classify is the pair: the division algebra  $D$  together with the involution  $'$  it carries, sorted by the centre of  $D$  and by the totally real field the involution fixes.

**Definition 11.2.1: The Four Albert Types**

Let  $D$  be a finite-dimensional division algebra over  $\mathbb{Q}$  whose centre  $K$  is a field, equipped with a positive involution  $'$ : a  $\mathbb{Q}$ -linear anti-involution for which the trace form  $(a, b) \mapsto \text{Trd}_{D/\mathbb{Q}}(ab')$  is positive definite. Write  $F = \{a \in K \mid a' = a\}$  for the fixed field of  $'$  inside the centre; positivity forces  $F$  to be a totally real number field, and  $K$  to be either  $F$  itself or a totally imaginary quadratic extension of  $F$  (a CM field). Set  $e_0 = [F : \mathbb{Q}]$  and  $d = [D : K]^{1/2}$  (the degree of  $D$ ). The pair  $(D, ')$  is of one of four types.

1. *Type I.*  $D = K = F$  is a totally real field and  $'$  is the identity. The involution is of the *first kind* (trivial on the centre).
2. *Type II.*  $K = F$  is totally real and  $D$  is a quaternion algebra over  $F$  ( $d = 2$ ) that is *totally indefinite*, split at every real place ( $D \otimes_{F, \sigma} \mathbb{R} \cong M_2(\mathbb{R})$  for every embedding  $\sigma : F \hookrightarrow \mathbb{R}$ ); the involution is of the first kind, of *orthogonal type*.
3. *Type III.*  $K = F$  is totally real and  $D$  is a quaternion algebra over  $F$  ( $d = 2$ ) that is *totally definite*, ramified at every real place ( $D \otimes_{F, \sigma} \mathbb{R}$  is the Hamilton quaternions  $\mathbb{H}$  for every  $\sigma$ ); the involution is of the first kind, of *symplectic type*.
4. *Type IV.*  $K$  is a CM field, a totally imaginary quadratic extension of the totally real  $F$ , and  $D$  is a division algebra with centre  $K$  of degree  $d$ ; the involution is of the *second kind*, restricting to complex conjugation on  $K$ .

The four types are read off two binary invariants and one further split. First, whether the centre is totally real ( $K = F$ , involution of the first kind) or CM ( $K \neq F$ , involution of the second kind) separates Types I, II, III from Type IV. Second, among the first-kind cases, whether  $D$  is commutative ( $d = 1$ , Type I) or a quaternion algebra ( $d = 2$ ) separates Type I from the quaternionic Types II and III. The final split, orthogonal versus symplectic, distinguishes Types II and III, and positivity ties it to the local invariant of  $D$ : a positive first-kind involution on a quaternion algebra is orthogonal exactly when the algebra is totally indefinite and symplectic exactly when it is totally definite, so the two cases cannot mix. The definition is the algebraic counterpart of the analytic definition 6.3.3, which recorded the same four families over  $\mathbb{C}$ ; the substance below is that over a field of characteristic  $p$  the same four types occur but with a wider range of admissible invariants.

The smallest instance makes the four types concrete before the general theorem.

**Example 11.2.2: The Types in Low Degree**

Take  $F = \mathbb{Q}$ , so  $e_0 = 1$ . Type I is then  $D = \mathbb{Q}$  with the trivial involution, the endomorphism algebra of a generic abelian variety. Type IV with  $D = K$  a field and  $e_0 = 1$  is an imaginary quadratic field  $K = \mathbb{Q}(\sqrt{-m})$  ( $m > 0$ ) with complex conjugation, the CM case in dimension one. Type II with  $e_0 = 1$  is an indefinite quaternion algebra over  $\mathbb{Q}$ , for instance the algebra  $\left(\frac{a, b}{\mathbb{Q}}\right)$  with  $a > 0$ , split at the real place; its standard-conjugation-twisted positive involution is orthogonal. Type III with  $e_0 = 1$  is a definite quaternion algebra over  $\mathbb{Q}$ , for instance the Hamilton quaternions' rational form  $\left(\frac{-1, -1}{\mathbb{Q}}\right)$ , ramified at  $\infty$ ; its standard conjugation  $x \mapsto \bar{x}$  is symplectic and positive because  $\text{Trd}(x\bar{x}) = 2\text{Nrd}(x) > 0$ . These realize the four Albert types over  $\mathbb{Q}$  with  $F = \mathbb{Q}$ ; Type IV additionally admits division algebras of degree  $d > 1$  over the imaginary quadratic centre, not only the field  $K$  itself.

With the list fixed, the theorem asserts that these four exhaust the endomorphism algebras of simple abelian varieties, and that the pure-algebra classification is inherited from *Core Algebra*.

### Theorem 11.2.3: Albert's Classification over a Field

Let  $X$  be a simple abelian variety over a field  $k$ , and let  $'$  be the Rosati involution attached to a polarization of  $X$  (definition 10.3.2). Then  $\text{End}^0(X)$  is a finite-dimensional division algebra over  $\mathbb{Q}$  whose centre is a field, and  $(\text{End}^0(X), ')$  is a division algebra with positive involution of exactly one of the four types of definition 11.2.1. The type is determined by the centre  $K$ , its totally real subfield  $F$ , and the kind of the involution:

- Type I when  $\text{End}^0(X) = F$  is totally real;
- Types II and III when  $\text{End}^0(X)$  is a quaternion algebra over  $F = K$ , according as the positive involution is orthogonal (totally indefinite) or symplectic (totally definite);
- Type IV when the centre  $K$  is a CM field.

#### Proof:

*Step 1:  $\text{End}^0(X)$  is a division algebra with positive involution.* Since  $X$  is simple,  $\text{End}^0(X)$  is a division algebra: it is semisimple by corollary 11.1.4, and a nonzero endomorphism of a simple abelian variety is an isogeny, hence invertible in  $\text{End}^0(X)$ , so the algebra has no zero divisors and every nonzero element is a unit. A finite-dimensional  $\mathbb{Q}$ -algebra that is a domain in which every nonzero element is invertible is a division algebra, and its centre  $K$  is a field. The Rosati involution  $'$  attached to a polarization is a  $\mathbb{Q}$ -linear anti-involution with  $\text{Tr}(ff') > 0$  for every  $f \neq 0$  (theorem 11.1.3), where  $\text{Tr}$  is the trace on the rational Tate module  $V_\ell X$ ; this trace is a positive multiple of the reduced trace form, so  $'$  is positive in the sense of definition 11.2.1. Thus  $(\text{End}^0(X), ')$  is a division algebra with positive involution.

*Step 2: the pure-algebra classification.* A finite-dimensional division algebra over  $\mathbb{Q}$  with a positive involution, and centre a field, is of exactly one of the four types of definition 11.2.1. This is a statement about algebras with involution, independent of abelian varieties, and it is proved in *Core Algebra* [12]: positivity forces the fixed field  $F$  to be totally real and the centre  $K$  to be  $F$  or a CM field; a positive involution of the first kind on a noncommutative  $D$  forces  $D$  to be a quaternion algebra (Albert's degree bound  $d \leq 2$ , from the interaction of positivity at every real place with the global division-algebra structure); and the orthogonal or symplectic kind is pinned to the totally indefinite or totally definite case. Applying this to  $D = \text{End}^0(X)$  from Step 1 yields the four listed cases, and the invariants  $(K, F, \text{kind})$  are mutually exclusive, so  $X$  falls into exactly one type, completing the proof.

The theorem separates cleanly into a geometric half and an algebraic half. That  $\text{End}^0(X)$  is a division algebra with a positive involution is where the abelian variety enters: it is Poincaré reducibility and the Rosati form of chapter 6 rebuilt over  $k$ , the content this book owns. That such algebras form exactly four families is a fact about division algebras with involution over  $\mathbb{Q}$ , proved once in *Core Algebra* and cited here rather than reproved. The same partition of labour governs the analytic theorem 6.3.4, which reached the same list over  $\mathbb{C}$ ; what the algebraic statement adds is that it holds over any field, and, as proposition 11.2.4 shows, that the admissible numerical invariants are larger in positive characteristic.

The types are constrained by how  $D$  acts on the homology of  $X$ . Over  $\mathbb{C}$  that homology was  $H_1(X, \mathbb{Q})$ ; over a general field its role is played by the rational Tate module  $V_\ell X$ , a  $\mathbb{Q}_\ell$ -vector space of dimension  $2g$  on which  $\text{End}^0(X)$  acts faithfully.

**Proposition 11.2.4: Numerical Constraints on the Type**

Let  $X$  be a simple abelian variety of dimension  $g$  over a field  $k$ , with  $D = \text{End}^0(X)$ , centre  $K$ , totally real fixed field  $F$  of the Rosati involution,  $e_0 = [F : \mathbb{Q}]$ , and degree  $d = [D : K]^{1/2}$ . In characteristic zero the divisibility  $[D : \mathbb{Q}] \mid 2g$  holds, giving

- Type I:  $D = F$ , so  $[D : \mathbb{Q}] = e_0$  and  $e_0 \mid g$ ;
- Type II:  $K = F$ ,  $d = 2$ , so  $[D : \mathbb{Q}] = 4e_0$  and  $2e_0 \mid g$ ;
- Type III:  $K = F$ ,  $d = 2$ , so  $[D : \mathbb{Q}] = 4e_0$  and  $2e_0 \mid g$ ;
- Type IV:  $[K : F] = 2$ , so  $[D : \mathbb{Q}] = 2e_0 d^2$  and  $e_0 d^2 \mid g$ .

Which types occur, and with which invariants, depends on  $\text{char } k$ . Over  $\mathbb{C}$ , and more generally in characteristic zero, Type III does not occur in dimension one and Type IV is confined by the multiplicity of  $D$  on the tangent space  $H^{1,0}$ . In characteristic  $p$  the divisibility  $[D : \mathbb{Q}] \mid 2g$  can fail, and only the weaker reduced-degree bound (that  $2g$  is a multiple of the  $\mathbb{Q}_\ell$ -dimension of a simple  $D \otimes \mathbb{Q}_\ell$ -module,  $\ell \neq p$ ) survives: Type III occurs already for a supersingular abelian variety, where  $[D : \mathbb{Q}]$  may exceed  $2g$ , and the Type IV range is enlarged, so a definite quaternion algebra and larger CM division algebras appear that are absent over  $\mathbb{C}$ .

**Proof:**

*Step 1: the divisibility  $[D : \mathbb{Q}] \mid 2g$  in characteristic zero.* Suppose first  $\text{char } k = 0$ . Choosing an embedding of a finitely generated field of definition of  $X$  into  $\mathbb{C}$  (under which  $\text{End}^0(X)$  injects into  $\text{End}^0(X_{\mathbb{C}})$ ), one obtains the Betti homology  $H_1(X, \mathbb{Q})$ , a  $\mathbb{Q}$ -vector space of dimension  $2g$  on which  $\text{End}^0(X)$  acts faithfully and  $\mathbb{Q}$ -linearly. Because  $X$  is simple,  $D = \text{End}^0(X)$  is a division algebra, so  $H_1(X, \mathbb{Q})$  is a module over  $D$ . A module over a division algebra is free, so  $H_1(X, \mathbb{Q}) \cong D^r$  for some  $r$ , whence  $\dim_{\mathbb{Q}} H_1(X, \mathbb{Q}) = 2g$  is a multiple of  $\dim_{\mathbb{Q}} D = [D : \mathbb{Q}]$ ; thus  $[D : \mathbb{Q}] \mid 2g$ .

*Step 2: reading off each type.* Substituting  $[D : \mathbb{Q}] = e_0 d^2 [K : F]$  into  $[D : \mathbb{Q}] \mid 2g$  gives the per-type divisibilities. Type I ( $d = 1$ ,  $[K : F] = 1$ ) yields  $e_0 \mid 2g$ , refined to  $e_0 \mid g$  because the totally real field  $F$  acts on the  $g$ -dimensional tangent space  $H^{1,0}$  through a rank- $g$  representation, so  $e_0 = [F : \mathbb{Q}]$  divides  $g$ . Types II and III ( $d = 2$ ,  $[K : F] = 1$ ) yield  $4e_0 \mid 2g$ , that is  $2e_0 \mid g$ . Type IV ( $[K : F] = 2$ ) yields  $2e_0 d^2 \mid 2g$ , that is  $e_0 d^2 \mid g$ . These are the stated constraints.

*Step 3: dependence on the characteristic.* Step 1 used that  $H_1(X, \mathbb{Q})$ , a characteristic-zero object, is a free  $D$ -module; the analogous statement can fail when  $\text{char } k = p > 0$ , where the substitute for  $H_1(X, \mathbb{Q})$  is the rational Tate module  $V_\ell X$  ( $\ell \neq p$ ), a  $\mathbb{Q}_\ell$ -module over  $A_\ell := D \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$  of dimension  $2g$  (theorem 9.3.2, proposition 9.2.3). Over  $\mathbb{C}$  the freeness came with the Hodge decomposition  $H_1(X, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$ , in which  $H^{1,0}$  is a  $D \otimes_{\mathbb{Q}} \mathbb{C}$ -module of  $\mathbb{C}$ -dimension  $g$  whose multiplicities are pinned by the requirement that  $H^{1,0}$  and its conjugate assemble the full

$2g$ -dimensional representation; this is what excludes Type III in low dimension and confines Type IV. In characteristic  $p$  there is no such decomposition,  $V_\ell X$  carries only the  $\mathbb{Q}_\ell$ -linear action, and the multiplicity restrictions lapse. The only surviving constraint is that  $2g = \dim_{\mathbb{Q}_\ell} V_\ell X$  is a multiple of the  $\mathbb{Q}_\ell$ -dimension of a simple  $A_\ell$ -module. The quaternion algebra  $D$  over  $\mathbb{Q}$  ramified exactly at  $p$  and  $\infty$  splits at every  $\ell \neq p$ , so  $A_\ell \cong M_2(\mathbb{Q}_\ell)$  and its simple module has  $\mathbb{Q}_\ell$ -dimension 2; a supersingular abelian variety (definition 9.4.6) can carry such a  $D$  with  $V_\ell X$  that simple module, even though  $[D: \mathbb{Q}] = 4$  exceeds  $2g$  when  $g = 1$ . The full divisibility  $[D: \mathbb{Q}] | 2g$  therefore fails, while the reduced-degree bound  $2 | 2$  holds, so Type III occurs where it could not over  $\mathbb{C}$ ; the Type IV range is enlarged for the same reason. The precise realizability table in each characteristic is finite but intricate and is cited to Mumford's *Abelian Varieties* and Oort's work on the supersingular locus; what is proved here is the qualitative content, the characteristic-zero divisibilities and the characteristic- $p$  enlargement, which is what the applications require, completing the proof.

The proposition sharpens the classification from a list of four families to a constraint on how large each can be relative to  $g$ , and it exposes the one place where the algebraic theory diverges from the analytic. Over  $\mathbb{C}$  the tangent space carries a complex structure, and the way  $D$  must act on it excludes several combinations that the raw divisibilities permit; in particular the definite quaternion Type III never occurs for a simple abelian variety of small dimension, and the elliptic case sees only two of the four types. In characteristic  $p$  that obstruction disappears. The supersingular abelian varieties, whose  $p$ -divisible groups are as far from ordinary as possible (section 9.4), are exactly the ones on which the quaternionic and enlarged CM cases materialize. The elliptic curve already displays both faces of this dichotomy.

#### Example 11.2.5: Endomorphism Algebras of Elliptic Curves

Let  $X = E$  be a simple abelian variety of dimension  $g = 1$ , so every  $E$  qualifies. In characteristic zero the divisibility  $[D: \mathbb{Q}] | 2g = 2$  of proposition 11.2.4 leaves  $[D: \mathbb{Q}] \in \{1, 2\}$ ; in characteristic  $p$  that bound is replaced by the reduced-degree constraint, which permits  $[D: \mathbb{Q}] = 4$ . The three possibilities are:

1. *Type I.*  $\text{End}^0(E) = \mathbb{Q}$ , with  $F = \mathbb{Q}$ ,  $e_0 = 1 = g$ . This is the generic elliptic curve, whose only endomorphisms are the multiplications  $[n]$ .
2. *Type IV.*  $\text{End}^0(E) = K$  an imaginary quadratic field, with  $F = \mathbb{Q}$ ,  $e_0 = 1$ ,  $d = 1$ , and  $[K: F] = 2$ , so  $e_0 d^2 = 1 | g$ . The curve  $E$  has complex multiplication by  $K$ , and the Rosati involution is complex conjugation on  $K$ ; the CM order inside  $K$  is recovered in *Elliptic Curves* [4]. This case occurs in every characteristic, and for an ordinary elliptic curve over  $\overline{\mathbb{F}}_p$  it is the whole endomorphism algebra.
3. *Type III.*  $\text{End}^0(E) = D$  a definite quaternion algebra over  $\mathbb{Q}$ , with  $F = \mathbb{Q}$ ,  $e_0 = 1$ ,  $d = 2$ , so  $[D: \mathbb{Q}] = 4$ . This case is impossible over  $\mathbb{C}$ , where  $\text{End}^0(E)$  is always commutative, and impossible in characteristic zero, where the full divisibility  $[D: \mathbb{Q}] | 2g$  would force  $4 | 2$ . It occurs *only* in characteristic  $p$ , for a supersingular elliptic curve, where  $\text{End}^0(E)$  is the quaternion algebra over  $\mathbb{Q}$  ramified exactly at  $p$  and  $\infty$  (example 9.4.8). The characteristic-zero bound is superseded exactly as in Step 3 of proposition 11.2.4:  $D$  ramifies at  $\infty$  but splits at every  $\ell \neq p$ , so  $V_\ell E$  is the standard 2-dimensional module of  $D \otimes \mathbb{Q}_\ell \cong M_2(\mathbb{Q}_\ell)$  rather than a free  $D$ -module, and only the reduced-degree constraint  $e_0 d [K: F] = 2 | 2$  survives.

The three cases are the  $g = 1$  shadow of the full classification. The analytic theorem 6.3.4 and its definition 6.3.3 saw only the first two, since a 1-dimensional complex torus has commutative

endomorphism algebra; the definite quaternion Type III is invisible over  $\mathbb{C}$  in dimension one and appears only once the theory is rebuilt over a field of positive characteristic. The algebraic development is thus not a transcription of the analytic one: it detects endomorphism algebras that the complex-analytic theory cannot produce.

### Exercise 11.2.1

1. For each of the four Albert types of definition 11.2.1, state the centre  $K$ , the fixed field  $F$ , the degree  $d = [D: K]^{1/2}$ , the kind of the involution (first or second), and, for the quaternionic types, whether the involution is orthogonal or symplectic. Then identify which type is realized by each of  $D = \mathbb{Q}$ ,  $D = \mathbb{Q}(\sqrt{-5})$  with complex conjugation, and  $D = \left(\frac{-1, -1}{\mathbb{Q}}\right)$  with  $x \mapsto \bar{x}$ , in each case with  $F = \mathbb{Q}$ .
2. Verify that the standard conjugation  $x \mapsto \bar{x}$  on the rational Hamilton quaternions  $D = \left(\frac{-1, -1}{\mathbb{Q}}\right)$  is a positive involution in the sense of definition 11.2.1, and determine its Albert type. Here  $\bar{x} = \text{Trd}(x) - x$ , so that  $\text{Trd}(x\bar{x}) = 2 \text{Nrd}(x)$ .
3. Let  $X$  be a simple abelian variety of dimension  $g = 2$  in characteristic zero, and suppose  $\text{End}^0(X)$  is not equal to  $\mathbb{Q}$ . Using the divisibility of proposition 11.2.4, list every Albert type that can occur together with the admissible values of  $e_0 = [F: \mathbb{Q}]$  and (for Type IV) the degree  $d$ . Which quaternionic type is excluded by the analytic constraints recalled in proposition 11.2.4, and why does the divisibility alone not exclude it?
4. Prove the divisibility  $[D: \mathbb{Q}] \mid 2g$  of proposition 11.2.4 in characteristic zero, and deduce the per-type constraint  $e_0 d^2 \mid g$  for Type IV. State precisely where the argument uses that  $\text{char } k = 0$ , and give the reason the same conclusion fails for a supersingular elliptic curve.
5. Let  $E$  be a supersingular elliptic curve over  $\overline{\mathbb{F}}_p$ , so that  $D = \text{End}^0(E)$  is the quaternion algebra over  $\mathbb{Q}$  ramified exactly at  $p$  and  $\infty$  (example 11.2.5). Show that  $[D: \mathbb{Q}] \nmid 2g$  for  $g = 1$ , yet the reduced-degree bound of proposition 11.2.4 is satisfied. Explain, in terms of the  $D \otimes_{\mathbb{Q}} \mathbb{Q}_{\ell}$ -module structure of  $V_{\ell}E$  for  $\ell \neq p$ , why this does not contradict the characteristic-zero divisibility.
6. A first-kind positive involution on a quaternion algebra  $D$  over a totally real field  $F$  is orthogonal exactly when  $D$  is totally indefinite and symplectic exactly when  $D$  is totally definite (definition 11.2.1). Explain why, for a fixed such  $D$ , Types II and III cannot both arise, and use this to show that a totally definite quaternion algebra over  $F$  can never carry a positive involution of orthogonal type. Contrast this with the situation over  $\mathbb{C}$  recalled in example 11.2.5, where Type III is invisible in dimension one.

## 11.3 Poincaré Complete Reducibility

Beyond semisimplicity, the positivity of the Rosati form (theorem 11.1.3) splits an abelian variety, up to isogeny, into indivisible pieces. Over  $\mathbb{C}$  the mechanism was orthogonality for a positive Hermitian form (theorem 5.4.3); here the positive Rosati form on  $\text{End}^0(X)$  takes over that role, producing for every abelian subvariety a complementary one purely algebraically, over an arbitrary field  $k$  and in every characteristic. The indivisible pieces are the simple abelian varieties, and once  $X$  is written up

to isogeny as a product of powers of them, the endomorphism algebra falls apart into a product of matrix algebras over division algebras.

The atoms of the theory are the abelian varieties that cannot be split at all.

### Definition 11.3.1: Simple Abelian Variety

An abelian variety  $X$  of positive dimension over a field  $k$  is *simple* if it has no abelian subvariety other than  $0$  and  $X$  itself. Here an *abelian subvariety* of  $X$  is a closed subgroup variety  $Y \subseteq X$  that is connected (equivalently, an abelian variety in its own right, by definition 7.1.2).

This is the same notion as the analytic definition 5.4.5, now read off the algebraic structure rather than from complex subtori: there the subobjects were the complex subtori  $W/(\Lambda \cap W)$ , here they are the connected closed subgroup varieties. Simplicity is a strong condition. Every elliptic curve is simple, because a nonzero proper abelian subvariety would have dimension strictly between  $0$  and  $1$ ; but in higher dimension an abelian variety can fail to be simple even when it is not visibly a product, being merely isogenous to one. The point of the reducibility theorem is that these atoms genuinely generate everything.

### Example 11.3.2: Simplicity Fails for a Product

A product  $X = A \times B$  of abelian varieties of positive dimension is never simple: the factor  $A \times \{0\}$  is a nonzero proper abelian subvariety, being the image of the closed immersion  $A \hookrightarrow A \times B$ ,  $a \mapsto (a, 0)$ , and a connected closed subgroup variety. Concretely, for  $A = B = E$  an elliptic curve over  $k$ , the product  $E \times E$  has the two coordinate copies of  $E$ , the diagonal  $\Delta = \{(x, x)\}$ , and the antidiagonal  $\{(x, -x)\}$  all as abelian subvarieties, each isomorphic to  $E$ ; the diagonal and antidiagonal meet in  $\{(x, x) : 2x = 0\} \cong E[2]$ , a finite group scheme. This finiteness of the intersection is exactly what the reducibility theorem turns into a splitting up to isogeny.

To split off an abelian subvariety  $Y \subseteq X$  one produces a complementary abelian subvariety  $Z$  with  $Y + Z = X$  and  $Y \cap Z$  finite. Over  $\mathbb{C}$  this was the  $H$ -orthogonal complement of the subspace defining  $Y$ ; the algebraic replacement is a symmetric idempotent in  $\text{End}^0(X)$  built from the polarization, and the positivity of the Rosati form is precisely what guarantees the idempotent exists.

### Theorem 11.3.3: Poincaré Complete Reducibility

Let  $X$  be an abelian variety over a field  $k$ , equipped with a polarization  $\lambda = \varphi_L : X \rightarrow \hat{X}$  (definition 10.3.2), and let  $Y \subseteq X$  be an abelian subvariety. Then there is an abelian subvariety  $Z \subseteq X$ , the *complementary abelian subvariety* of  $Y$  with respect to  $\lambda$ , with

1.  $Y + Z = X$ ;
2.  $Y \cap Z$  finite.

Consequently the addition map  $Y \times Z \rightarrow X$ ,  $(y, z) \mapsto y + z$ , is an isogeny, and  $X$  is isogenous to  $Y \times Z$ .



**Proof:**

Let  $\iota: Y \hookrightarrow X$  be the inclusion, with dual  $\hat{\iota}: \hat{X} \rightarrow \hat{Y}$  (proposition 10.1.3), which is surjective because  $\iota$  is a closed immersion of abelian varieties. Restricting the polarization  $\lambda$  to  $Y$  gives a polarization on  $Y$ : the composite

$$\lambda_Y = \hat{\iota} \circ \lambda \circ \iota: Y \longrightarrow \hat{Y}$$

is the polarization  $\varphi_{\iota^*L}$  attached to the ample restriction  $\iota^*L$  (an ample bundle restricts to an ample bundle on a closed subvariety, *Algebraic Geometry* [13]), hence is an isogeny (theorem 10.3.1).

*Step 1: A symmetric idempotent.* Consider the endomorphism

$$u = \iota \circ \lambda_Y^{-1} \circ \hat{\iota} \circ \lambda \in \text{End}^0(X),$$

where  $\lambda_Y^{-1} \in \text{Hom}(\hat{Y}, Y) \otimes \mathbb{Q}$  is the isogeny quasi-inverse of  $\lambda_Y$ . On  $Y$  this map is the identity:  $u \circ \iota = \iota \circ \lambda_Y^{-1} \circ (\hat{\iota} \circ \lambda \circ \iota) = \iota \circ \lambda_Y^{-1} \circ \lambda_Y = \iota$ . Hence  $u \circ u = u$  (both compositions send  $X$  into  $Y$ , on which  $u$  acts as the identity), so  $u$  is an idempotent in  $\text{End}^0(X)$ . Moreover  $u$  is fixed by the Rosati involution  $'$  attached to  $\lambda$ . Dualization reverses composites, so  $\hat{u} = \hat{\lambda} \circ \hat{\iota} \circ \widehat{\lambda_Y^{-1}} \circ \hat{\iota}$ ; using biduality  $\kappa: X \xrightarrow{\sim} \hat{\hat{X}}$  (theorem 10.2.3) to identify  $\hat{\iota}$  with  $\iota$ , and the symmetry of the polarizations  $\lambda$  and  $\lambda_Y$  ( $\hat{\lambda} = \lambda$ ,  $\hat{\lambda}_Y = \lambda_Y$ , hence  $\widehat{\lambda_Y^{-1}} = \lambda_Y^{-1}$ ), this reads  $\hat{u} = \lambda \circ \iota \circ \lambda_Y^{-1} \circ \hat{\iota}$ . Therefore

$$u' = \lambda^{-1} \circ \hat{u} \circ \lambda = \lambda^{-1} \circ \lambda \circ \iota \circ \lambda_Y^{-1} \circ \hat{\iota} \circ \lambda = \iota \circ \lambda_Y^{-1} \circ \hat{\iota} \circ \lambda = u$$

Thus  $u = u' = u^2$  is a symmetric idempotent, and  $e = 1 - u$  is the complementary symmetric idempotent.

*Step 2: The complement.* The idempotents  $u, e \in \text{End}^0(X)$  are not in general integral endomorphisms, but a single positive integer  $n$  clears the denominators of both, so  $nu, ne \in \text{End}(X)$ . Set

$$Z = ((ne)(X))^0,$$

the identity component of the image of  $ne$ : a connected closed subgroup variety of  $X$ , hence an abelian subvariety. Since  $u$  is the identity on  $Y$ , we have  $e \circ \iota = (1 - u) \circ \iota = 0$ , so  $ne$  kills  $Y$ . The two remaining properties are verified directly.

*Step 3:  $Y \cap Z$  is finite.* The endomorphism  $ne$  vanishes on  $Y$  (Step 2), so  $Y \subseteq \ker(ne)$ . On  $Z$  the map  $ne$  acts as multiplication by  $n$ : for a point  $z = (ne)(x)$  in the image, the idempotent relation  $e^2 = e$  gives  $(ne)(z) = (ne)(ne)(x) = n^2e^2(x) = n(ne)(x) = nz$ , and by continuity  $ne$  restricts to  $[n]$  on the whole identity component  $Z$ . Thus  $ne|_Z = [n]$  is an isogeny (theorem 8.4.5) with finite kernel  $Z[n]$ , whence  $Y \cap Z \subseteq \ker(ne) \cap Z = \ker(ne|_Z) = Z[n]$  is finite.

*Step 4:  $Y + Z = X$  and the addition map is an isogeny.* The sum  $Y + Z$  is the image of the addition homomorphism  $\mu: Y \times Z \rightarrow X$ , a closed abelian subvariety of  $X$ . Its dimension is  $\dim Y + \dim Z - \dim(Y \cap Z) = \dim Y + \dim Z$ , since  $Y \cap Z$  is finite by Step 3. On the other hand  $u + e = 1$  in  $\text{End}^0(X)$ , so  $[n] = n(u + e) = nu + ne$ ; every  $x \in X$  satisfies  $[n]x = (nu)(x) + (ne)(x)$  with  $(nu)(x) \in Y$  and  $(ne)(x) \in Z$ , whence  $[n]X \subseteq Y + Z$ . Since  $[n]: X \rightarrow X$  is surjective (theorem 8.4.5),  $Y + Z = X$ , giving (1) and forcing  $\dim Y + \dim Z = \dim X$ . Therefore  $\mu: Y \times Z \rightarrow X$  is a surjective homomorphism between abelian varieties of the same dimension with finite kernel  $\{(t, -t) : t \in Y \cap Z\}$ , that is, an isogeny (definition 9.1.1). Thus  $X \sim Y \times Z$ , as claimed.



The theorem says an abelian subvariety is, up to isogeny, a direct factor. This fails for a general commutative group variety without a polarization: a nonsplit extension of one elliptic curve by another has no complementary subvariety, precisely because there is no polarization to build the symmetric idempotent from. What forces the splitting is the positive Rosati form, exactly as over  $\mathbb{C}$  the positive Hermitian form did (theorem 5.4.3): positivity is what makes the idempotent  $u = u' = u^2$  available. The complement depends on the polarization: a different  $\lambda$  produces a different idempotent and hence a different  $Z$  for the same  $Y$ . The passage from a genuine direct-sum decomposition (of the algebra by its idempotents) to a mere isogeny of abelian varieties is the price of working with the group  $X$  rather than a vector space: the idempotents  $u, e$  live in  $\text{End}^0(X)$ , not  $\text{End}(X)$ , and clearing denominators is what introduces the finite kernel.

**Example 11.3.4: Splitting the Diagonal in  $E \times E$**

Let  $E$  be an elliptic curve over  $k$  with its canonical principal polarization, and give  $X = E \times E$  the product polarization  $\lambda = \lambda_E \times \lambda_E$ . Take  $Y = \Delta$ , the diagonal, with inclusion  $\iota: E \rightarrow E \times E$ ,  $x \mapsto (x, x)$ . The restricted polarization  $\lambda_\Delta$  is (up to the identification  $\Delta \cong E$ ) the isogeny [2]-scaled principal polarization, since  $\iota^*(\mathcal{O}(\{0\}) \boxtimes \mathcal{O}(\{0\}))$  has degree 2 on  $E$ . Carrying out the idempotent computation of the proof, one finds  $u(x, y) = \frac{1}{2}(x + y, x + y)$  and  $e(x, y) = \frac{1}{2}(x - y, y - x)$  in  $\text{End}^0(E \times E) = M_2(\text{End}^0 E)$ . Clearing the denominator,  $2e(x, y) = (x - y, y - x)$ , whose image is the antidiagonal

$$Z = \{(u, -u) : u \in E\} \cong E$$

The complement of the diagonal is thus the antidiagonal, exactly as in the analytic computation. Their intersection  $\Delta \cap Z = \{(x, x) : 2x = 0\} \cong E[2]$  is finite of order 4, and the addition map  $\Delta \times Z \rightarrow E \times E$  is an isogeny of degree 4.

Before iterating, one records the field-independent input that makes the atoms rigid: over any  $k$ , a nonzero homomorphism between simple abelian varieties is already an isogeny, and hence the endomorphism algebra of a simple abelian variety is a division algebra. The argument is purely algebraic, resting only on the fact that the image and the identity component of the kernel of a homomorphism are abelian subvarieties; it is the algebraic counterpart of the analytic proposition 5.4.7, which reached the same conclusion over  $\mathbb{C}$  from the semisimplicity of the rational representation.

**Lemma 11.3.5: Homomorphisms of Simple Abelian Varieties**

Let  $S, T$  be simple abelian varieties over  $k$  and  $f: S \rightarrow T$  a nonzero homomorphism. Then  $f$  is an isogeny; in particular  $S$  and  $T$  are isogenous. Consequently, for a single simple abelian variety  $S$ , every nonzero element of  $\text{End}^0(S)$  is invertible, so  $\text{End}^0(S)$  is a division algebra.

**Proof:**

The image  $f(S)$  is a closed connected subgroup variety of  $T$ , hence an abelian subvariety; it is nonzero because  $f \neq 0$ , so  $f(S) = T$  by simplicity of  $T$ , and  $f$  is surjective. The identity component  $(\ker f)^0$  is an abelian subvariety of  $S$ , proper because  $f \neq 0$ , so  $(\ker f)^0 = 0$  by simplicity of  $S$ ; thus  $\ker f$  is finite. A surjective homomorphism of abelian varieties with finite kernel is an isogeny (definition 9.1.1), and an isogeny has a two-sided quasi-inverse in  $\text{Hom}^0$ . Taking  $T = S$ , every nonzero  $f \in \text{End}(S)$  is an isogeny, hence a unit of  $\text{End}^0(S)$ ; an associative  $\mathbb{Q}$ -algebra in which every nonzero element is a unit is a division algebra, so  $\text{End}^0(S)$  is one.

Iterating the splitting reduces every abelian variety to its simple atoms.

**Corollary 11.3.6: Isotypic Decomposition up to Isogeny**

Every abelian variety  $X$  of positive dimension over  $k$  is isogenous to a product

$$X \sim X_1^{n_1} \times X_2^{n_2} \times \cdots \times X_r^{n_r}$$

of powers of pairwise non-isogenous simple abelian varieties  $X_1, \dots, X_r$ . The simple factors  $X_i$  (up to isogeny) and their multiplicities  $n_i$  are uniquely determined by  $X$ , up to permutation.

**Proof:**

*Existence.* Induct on  $\dim X$ . If  $X$  is simple, take  $r = 1$ ,  $n_1 = 1$ . Otherwise choose an abelian subvariety  $Y$  with  $0 \neq Y \neq X$  of minimal positive dimension, which is then simple (a smaller nonzero abelian subvariety of  $Y$  would be one of  $X$ ). Fix a polarization on  $X$  and let  $Z$  be the complement of theorem 11.3.3, so  $X \sim Y \times Z$  with  $\dim Z < \dim X$ ; apply induction to  $Z$  and collect isogenous factors into common powers.

*Uniqueness.* For a simple abelian variety  $S$  and any abelian variety  $A$  with simple isogeny factors  $A_k$ , a homomorphism from the simple  $S$  to a simple  $T$  is 0 unless  $S \sim T$  (lemma 11.3.5: a nonzero such homomorphism is an isogeny). Hence  $\text{Hom}(S, A) \otimes \mathbb{Q} = \bigoplus_k (\text{Hom}(S, A_k) \otimes \mathbb{Q})$ , and viewed as a left vector space over the division algebra  $\text{End}^0(S)$  (division by lemma 11.3.5),  $\dim_{\text{End}^0(S)}(\text{Hom}(S, A) \otimes \mathbb{Q})$  counts the multiplicity of the isogeny class of  $S$  in  $A$ . An isogeny  $A \rightarrow A'$  induces an isomorphism  $\text{Hom}(S, A) \otimes \mathbb{Q} \xrightarrow{\sim} \text{Hom}(S, A') \otimes \mathbb{Q}$  (compose with the isogeny and a quasi-inverse), so this multiplicity is an isogeny invariant. Taking  $S = X_i$  shows the two decompositions agree up to permutation.

Up to isogeny an abelian variety is a formal monomial  $\prod_i X_i^{n_i}$  in simple factors, so the isogeny category of abelian varieties over  $k$  is a semisimple category whose simple objects are the simple abelian varieties. This is the exact analogue, for abelian varieties, of the decomposition of a semisimple module into isotypic components, and it is why the endomorphism algebra assembles into a product of matrix algebras over division rings. That last assembly is the corollary the rest of the chapter runs on.

**Corollary 11.3.7: Product Structure of the Endomorphism Algebra**

Let  $X \sim \prod_i X_i^{n_i}$  be the isotypic decomposition of corollary 11.3.6, with the  $X_i$  pairwise non-isogenous simple abelian varieties. Then

$$\text{End}^0(X) \cong \prod_i M_{n_i}(D_i), \quad D_i = \text{End}^0(X_i) \text{ a division algebra,}$$

and  $\text{Hom}^0(X_i, X_j) = 0$  for  $i \neq j$ . This is the explicit Wedderburn form of the semisimple algebra  $\text{End}^0(X)$  of corollary 11.1.4.

**Proof:**

Isogenous abelian varieties have isomorphic  $\text{End}^0$ , so  $\text{End}^0(X) \cong \text{End}^0(\prod_i X_i^{n_i})$ . For  $i \neq j$  the simple abelian varieties  $X_i, X_j$  are non-isogenous, so  $\text{Hom}^0(X_i, X_j) = 0$ : a nonzero homomorphism between them would be an isogeny (lemma 11.3.5), contradicting non-isogeny.

Hence  $\text{End}^0(\prod_i X_i^{n_i}) = \prod_i \text{End}^0(X_i^{n_i})$ , and an endomorphism of the power  $X_i^{n_i}$  is an  $n_i \times n_i$  matrix of elements of  $\text{End}^0(X_i)$ , giving  $\text{End}^0(X_i^{n_i}) \cong M_{n_i}(D_i)$  with  $D_i = \text{End}^0(X_i)$ . Each  $D_i$  is a division algebra because  $X_i$  is simple (lemma 11.3.5). The product  $\prod_i M_{n_i}(D_i)$  is a semisimple  $\mathbb{Q}$ -algebra in Wedderburn form, matching corollary 11.1.4, as we sought to show.

The corollary reduces the endomorphism theory of abelian varieties to two pieces: the division algebras  $D_i = \text{End}^0(X_i)$  of the simple factors, and the matrix bookkeeping of multiplicities. Which division algebras can occur is the content of Albert's classification (theorem 11.2.3), which uses the Rosati involution and its positivity to cut the possibilities down to four types. The next observation is that the endomorphism algebra is the same object over  $\mathbb{C}$  whether one computes it analytically or algebraically, and the following example makes both the product structure and the analytic comparison concrete in the smallest case.

**Example 11.3.8: Endomorphism Algebra of  $E \times E$**

Let  $E$  be an elliptic curve over  $k$  without complex multiplication, so  $\text{End}^0(E) = \mathbb{Q}$ . The product  $X = E \times E$  is isogenous to  $E^2$ , a power of the single simple abelian variety  $E$ ; its isotypic decomposition (corollary 11.3.6) has one simple factor of multiplicity 2. By corollary 11.3.7,

$$\text{End}^0(E \times E) \cong M_2(\text{End}^0(E)) = M_2(\mathbb{Q}),$$

a semisimple  $\mathbb{Q}$ -algebra of dimension 4, well within the bound  $\dim_{\mathbb{Q}} \text{End}^0(X) \leq (2g)^2 = 16$  for  $g = 2$ . Concretely an endomorphism sends  $(x, y) \mapsto (ax + by, cx + dy)$  with  $a, b, c, d \in \mathbb{Q}$ , and composition is matrix multiplication; the element  $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$  is nonzero and non-invertible, the algebraic shadow of  $E \times E$  not being simple.

Contrast two non-isogenous elliptic curves. If  $E$  and  $E'$  are non-isogenous (say two curves over  $k$  with  $j(E) \neq j(E')$  and no isogeny between them, such as CM by distinct imaginary quadratic fields), then  $\text{Hom}^0(E, E') = 0$  by corollary 11.3.7, and  $E \times E'$  has two distinct simple factors each of multiplicity 1:

$$\text{End}^0(E \times E') \cong \text{End}^0(E) \times \text{End}^0(E') = \mathbb{Q} \times \mathbb{Q}$$

The two  $\mathbb{Q}$  factors are the two idempotents projecting to the coordinate elliptic curves; there is no matrix mixing because there are no nonzero homomorphisms between the factors. For  $g = 1$  these endomorphism computations reduce to the corresponding statements for elliptic curves in *Elliptic Curves* [4]. Over  $\mathbb{C}$  this reproduces the analytic theorem 5.4.3 and proposition 5.4.7: the same  $M_2(\mathbb{Q})$  and  $\mathbb{Q} \times \mathbb{Q}$  arise, whether  $\text{End}^0$  is computed from complex subtori or from the algebraic group structure.

### Exercise 11.3.1

1. Let  $E$  be an elliptic curve over  $k$  without complex multiplication, and consider  $X = E \times E \times E$ . Using corollary 11.3.6 and corollary 11.3.7, identify the isotypic decomposition of  $X$  and compute  $\text{End}^0(X)$  as a matrix algebra. What is  $\dim_{\mathbb{Q}} \text{End}^0(X)$ , and how does it compare to the bound  $(2g)^2$ ?
2. In the notation of the proof of theorem 11.3.3, let  $u = \iota \circ \lambda_Y^{-1} \circ \hat{\iota} \circ \lambda$  and  $e = 1 - u$ . Verify directly that  $e$  is idempotent and symmetric for the Rosati involution, that is,  $e^2 = e$  and  $e' = e$ , given that  $u^2 = u$  and  $u' = u$ .
3. Let  $E$  be an elliptic curve over  $k$  with its canonical principal polarization, and give  $X = E \times E$

the product polarization. Following example 11.3.4, take  $Y = \Delta$  the diagonal. Compute the complement of the antidiagonal  $Z = \{(u, -u) : u \in E\}$  with respect to the same polarization (that is, apply the theorem to  $Y = Z$ ), and check that the resulting complement is  $\Delta$ . Confirm that the complement operation is symmetric here.

4. Show that if  $X$  is a simple abelian variety over  $k$ , then  $\text{End}^0(X)$  has no nontrivial idempotents, that is, the only  $u \in \text{End}^0(X)$  with  $u^2 = u$  are  $u = 0$  and  $u = 1$ . Deduce that a simple abelian variety is not isogenous to a product  $A \times B$  of abelian varieties of positive dimension.
5. Prove that two abelian varieties  $X$  and  $X'$  over  $k$  are isogenous if and only if their isotypic decompositions (corollary 11.3.6) have the same simple factors up to isogeny with the same multiplicities. Then use this to decide whether  $E \times E$  and  $E \times E'$  can be isogenous, where  $E, E'$  are non-isogenous elliptic curves without complex multiplication.
6. Let  $S$  be a simple abelian variety over  $k$  and  $X = S^n$ . Prove that every abelian subvariety  $Y \subseteq X$  with  $\dim Y = \dim S$  that is simple is isogenous to  $S$ , and that  $\text{Hom}^0(S, X)$  is a left vector space over  $D = \text{End}^0(S)$  of dimension  $n$ . Explain how this recovers  $\text{End}^0(S^n) \cong M_n(D)$  of corollary 11.3.7.

## 11.4 The Analytic and Algebraic Theories over $\mathbb{C}$

The book has built the theory of abelian varieties twice: analytically over  $\mathbb{C}$  in Part I and by scheme theory over an arbitrary field in Part II. Over  $\mathbb{C}$  these two developments describe the same objects. This section makes the identification precise and turns it into a bridge, so that a computation carried out on one side transports verbatim to the other.

The identification is a consequence of two classical theorems that this book does not reprove: Chow's theorem, that a closed analytic subvariety of projective space is algebraic, and Serre's GAGA comparison, that on a projective complex variety the analytic and algebraic coherent cohomologies agree. Both belong to *Complex Analysis* [16] and *Algebraic Geometry* [13]; they are quoted here and applied, not established.

### Theorem 11.4.1: Analytic Equals Algebraic over $\mathbb{C}$

Let  $k = \mathbb{C}$ . The analytification functor, sending a projective variety over  $\mathbb{C}$  to its associated complex-analytic space, restricts to an equivalence of categories

$$\{\text{complex abelian varieties}\} \xrightarrow{\sim} \{\text{polarizable compact complex tori}\}$$

from the abelian varieties of Part II (projective connected group varieties over  $\mathbb{C}$ , definition 7.1.2) to the abelian varieties of Part I (polarizable complex tori, definition 3.2.1). It is an equivalence of categories with products and with the group-variety structure: a homomorphism on either side corresponds to a homomorphism on the other, and the two notions of abelian variety agree under it.

**Proof:**

The functor and its inverse are supplied by Chow's theorem and GAGA, quoted from *Complex Analysis* [16] and *Algebraic Geometry* [13].

*Step 1: The functor is well defined.* Let  $A$  be a complex abelian variety in the algebraic sense, a projective connected group variety over  $\mathbb{C}$  (definition 7.1.2). Its analytification  $A^{\text{an}}$  is a compact complex manifold, connected, with a holomorphic group law induced by the algebraic one, hence a compact complex Lie group, that is, a complex torus  $V/\Lambda$ . The algebraic embedding  $A \hookrightarrow \mathbb{P}^N$  analytifies to a holomorphic embedding  $A^{\text{an}} \hookrightarrow (\mathbb{P}^N)^{\text{an}} = \mathbb{P}^N(\mathbb{C})$ , so  $A^{\text{an}}$  is a projective complex torus, and a projective complex torus is polarizable: the hyperplane class of the embedding is a positive line bundle, whose first Chern class is a polarization (definition 3.1.3). Thus  $A^{\text{an}}$  is an abelian variety in the analytic sense of definition 3.2.1.

*Step 2: Essential surjectivity.* Let  $X = V/\Lambda$  be a polarizable compact complex torus. A polarization furnishes, through the theta functions of Part I, a holomorphic embedding  $X \hookrightarrow \mathbb{P}^N$  realizing  $X$  as a closed analytic subvariety of projective space. By Chow's theorem this subvariety is algebraic: it is the analytification of a projective variety  $A$  over  $\mathbb{C}$ . The group law  $m: X \times X \rightarrow X$  is a holomorphic map between projective varieties, so by Chow's theorem again (applied to its graph, a closed analytic subvariety of  $X \times X \times X$ ) it is a morphism of algebraic varieties, and likewise the inverse. Hence  $A$  is a projective group variety over  $\mathbb{C}$  with  $A^{\text{an}} \cong X$ , an algebraic abelian variety, so the functor of Step 1 hits  $X$  up to isomorphism.

*Step 3: Fully faithful.* Let  $A, B$  be complex abelian varieties. A holomorphic homomorphism  $f: A^{\text{an}} \rightarrow B^{\text{an}}$  has, for its graph, a closed analytic subvariety  $\Gamma_f \subseteq (A \times B)^{\text{an}}$ ;  $A \times B$  is projective, so Chow's theorem makes  $\Gamma_f$  algebraic, and an algebraic subvariety that is the graph of a set-theoretic map is the graph of a morphism  $A \rightarrow B$  of varieties. This morphism is a homomorphism because  $f$  is, so analytification is bijective on homomorphisms. That it respects composition and identities is immediate, so the functor is fully faithful. Combined with Step 2 it is an equivalence, and Step 1 shows it matches the two notions of abelian variety, as we sought to show.

The theorem is the precise form of the slogan that an abelian variety over  $\mathbb{C}$  has an analytic face and an algebraic face and that these are the same object. The two constructions of Part I and Part II start from unrelated data, a lattice with a Hermitian form on one side and a projective group scheme on the other, and arrive at equivalent categories. Chow's theorem is what forbids the analytic side from being larger: no compact complex submanifold of projective space escapes the algebraic world. GAGA is what forbids the invariants from disagreeing, and it is GAGA that the comparison of line bundles and cohomology below rests on. The two theorems together license the reader to move a construction from one Part to the other without redoing it, and the rest of this section carries out that transport for the structures this chapter cares about: the endomorphism algebra, the Rosati involution and Albert type, the polarization, and the dual.

The first structure to transport is the endomorphism algebra. Chapter 9 built  $\text{End}^0(X)$  algebraically from the action on the Tate module, while chapter 6 built it analytically from the action on  $V$ ; the equivalence forces these to coincide, because a holomorphic homomorphism is algebraic.

**Proposition 11.4.2: The Two Endomorphism Algebras Agree**

Let  $X$  be an abelian variety over  $\mathbb{C}$ , regarded through theorem 11.4.1 as both a projective group variety and a complex torus. Then the algebraic endomorphism ring  $\text{End}(X)$  of Part II and the analytic endomorphism ring  $\text{End}(X)$  of Part I coincide, and hence so do the endomorphism algebras  $\text{End}^0(X) = \text{End}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$  of definition 6.1.1. Under this identification the analytic Rosati involution of definition 6.2.1 agrees with the algebraic Rosati involution of section 11.1, and the analytic Albert type of theorem 6.3.4 agrees with the algebraic Albert type of section 11.2.

**Proof:**

*Step 1: The endomorphism rings coincide.* An algebraic endomorphism of  $X$  analytifies to a holomorphic endomorphism of the complex torus  $X^{\text{an}}$ , giving an inclusion of the algebraic  $\text{End}(X)$  into the analytic one. Conversely, a holomorphic endomorphism  $f: X^{\text{an}} \rightarrow X^{\text{an}}$  is, by the full faithfulness of theorem 11.4.1 (applied with  $A = B = X$ ), the analytification of a unique algebraic endomorphism: its graph is a closed analytic subvariety of  $(X \times X)^{\text{an}}$ , hence algebraic by Chow's theorem. The two inclusions are mutually inverse, so the rings are equal, and tensoring with  $\mathbb{Q}$  gives  $\text{End}^0(X)$  the same either way.

*Step 2: The Rosati involutions agree.* Fix a polarization  $L$  on  $X$ . Both Rosati involutions are defined by the identical formula  $f \mapsto \varphi_L^{-1} \circ \hat{f} \circ \varphi_L$  (definition 6.2.1 analytically, section 11.1 algebraically), built from the polarization isogeny  $\varphi_L: X \rightarrow \hat{X}$  and the dual homomorphism. By proposition 11.4.3 below the analytic and algebraic  $\varphi_L$  coincide under the identification  $\hat{X}^{\text{an}} = \hat{X}$ , and by Step 1 the dual homomorphism  $\hat{f}$  is the same map on either side. Since  $f \mapsto f'$  is the same composite of the same maps, the two involutions agree on  $\text{End}^0(X)$ .

*Step 3: The Albert types agree.* The Albert type of a simple  $X$  is an invariant of the pair  $(\text{End}^0(X), ')$ , the endomorphism algebra together with its Rosati involution, read off the centre, the degree, and the kind of the involution (theorem 6.3.4, section 11.2). Steps 1 and 2 identify this pair on the two sides, so the invariants extracted from it coincide, completing the proof.

The proposition says that the arithmetic of  $X$ , packaged in  $\text{End}^0(X)$  with its Rosati involution and Albert type, is insensitive to which Part built it. This is what makes the char-zero specialization of the algebraic Albert classification of section 11.2 consistent with the analytic classification of chapter 6: over  $\mathbb{C}$  only Types I through IV in their char-zero forms occur, exactly the list theorem 6.3.4 produced from the positive Hermitian form. The mechanism is one clause of Chow's theorem, that a holomorphic homomorphism of projective complex tori is algebraic; everything else is bookkeeping. The agreement is a phenomenon of characteristic zero. Over a field of characteristic  $p$  the algebraic theory sees endomorphisms, such as Frobenius, and Albert types, such as the totally definite quaternion algebras of supersingular abelian varieties, that have no analytic counterpart, which is why section 11.2 records constraints depending on char  $k$  that theorem 6.3.4 did not.

The second structure to transport is the polarization together with the dual it maps into. Here the comparison is a statement about line bundles, and it is GAGA rather than Chow's theorem that does the work: the analytic and algebraic Picard groups of a projective complex variety agree, so the two dual tori and the two maps  $\varphi_L$  are literally the same.

**Proposition 11.4.3: The Two Polarizations Agree**

Let  $X$  be an abelian variety over  $\mathbb{C}$ . Then:

1. The analytic dual torus  $\hat{X} = \hat{V}/\hat{\Lambda}$  of definition 5.1.1 and the algebraic dual abelian variety  $\hat{X} = \text{Pic}_{X/\mathbb{C}}^0$  of definition 10.1.1 are canonically isomorphic, compatibly with analytification.
2. Under this identification the analytic polarization homomorphism  $\varphi_L: X \rightarrow \hat{X}$  of definition 5.3.1 equals the algebraic one of definition 8.3.1, theorem 10.3.1; the two have the same kernel, degree, and type.
3. The analytic Neron-Severi group  $\text{NS}(X)$  of Hermitian forms (definition 2.3.1) equals the algebraic  $\text{NS}(X) = \text{Pic}(X)/\text{Pic}^0(X)$  of definition 10.5.1, and a class is a polarization for one exactly when it is for the other.

**Proof:**

*Step 1: The Picard groups agree.* The analytic Picard group  $\text{Pic}(X^{\text{an}})$  classifies holomorphic line bundles on  $X^{\text{an}}$ , and the algebraic  $\text{Pic}(X)$  classifies algebraic line bundles. A line bundle is a rank-one locally free coherent sheaf, so by GAGA (*Complex Analysis* [16], *Algebraic Geometry* [13]) the analytification  $\text{Pic}(X) \rightarrow \text{Pic}(X^{\text{an}})$  is an isomorphism on a projective variety. This isomorphism respects algebraic triviality: a family of line bundles is algebraic exactly when it is holomorphic, again by GAGA, so it carries  $\text{Pic}^0(X)$  isomorphically onto the analytic degree-zero part, the bundles with trivial first Chern class.

*Step 2: The dual tori agree.* By Step 1 the two dual groups coincide as sets of degree-zero bundles. The analytic dual  $\hat{X} = \hat{V}/\hat{\Lambda}$  was identified with  $\text{Pic}^0(X)$  analytically in chapter 5, and the algebraic dual  $\hat{X} = \text{Pic}_{X/\mathbb{C}}^0$  is  $\text{Pic}^0(X)$  by construction (definition 10.1.1). Both carry the complex structure of the moduli of degree-zero bundles, and GAGA matches the analytic and algebraic families, so the identity map on the underlying group is an isomorphism of complex tori  $\hat{V}/\hat{\Lambda} \cong (\text{Pic}_{X/\mathbb{C}}^0)^{\text{an}}$  compatible with analytification. This is the promised identification  $\hat{X}^{\text{an}} = \hat{X}$ .

*Step 3: The maps  $\varphi_L$  agree.* Both  $\varphi_L$  are the same assignment on points,  $x \mapsto [t_x^* L \otimes L^{-1}]$ : analytically definition 5.3.1, algebraically definition 8.3.1. Since Step 1 identifies the bundles  $t_x^* L \otimes L^{-1}$  and Step 2 identifies the target dual tori, the two homomorphisms  $X \rightarrow \hat{X}$  are the same map. Consequently they have the same kernel  $K(L)$ , the same degree  $\deg \varphi_L = |K(L)|$ , and the same type: the analytic type  $(d_1, \dots, d_g)$  read off  $K(L) = \Lambda(L)/\Lambda$  (definition 3.2.3) and the algebraic type read off the group scheme  $K(L)$  agree because  $K(L)$  is étale in characteristic zero, so its scheme order equals its number of geometric points.

*Step 4: The Neron-Severi groups and polarizations agree.* By Step 1,  $\text{NS}(X) = \text{Pic}(X)/\text{Pic}^0(X)$  is the same group analytically and algebraically. Analytically a class is a Hermitian form  $H$  with  $\text{Im}H$  integral on  $\Lambda$ , its first Chern class (definition 2.3.1); algebraically it is a symmetric homomorphism  $\varphi_L: X \rightarrow \hat{X}$  (definition 10.5.1); and Step 3 matches  $H$  with  $\varphi_L$  through  $\varphi_L = (v \mapsto H(v, -))$ . A polarization is a positive class on both sides: analytically  $H > 0$  (definition 3.1.3), algebraically the symmetric isogeny  $\varphi_L$  attached to an ample  $L$  (definition 10.3.2), and over  $\mathbb{C}$  ampleness of  $L$  is positivity of  $H = c_1(L)$  by the Riemann relations. Hence the two



notions of polarization, and their types, coincide, completing the proof.

The proposition is the line-bundle half of the dictionary, and it closes the loop opened in Part II, where the dual abelian variety was introduced as the algebraic replacement for the analytic dual torus with the promise that the two agree over  $\mathbb{C}$ . They agree because a degree-zero line bundle is the same object holomorphically and algebraically (GAGA), so the moduli space of such bundles is one complex torus wearing two descriptions. The map  $\varphi_L$  is then forced to match, being the same formula  $x \mapsto [t_x^* L \otimes L^{-1}]$  evaluated in the same Picard group, and with it the whole numerical apparatus: the kernel  $K(L)$ , the degree  $(d_1 \cdots d_g)^2 = \det E$ , and the Pfaffian count of sections  $\dim H^0(X, L) = d_1 \cdots d_g$ , which proposition 10.5.3 recovered algebraically as  $\sqrt{\deg \varphi_L}$  and Part I obtained analytically from theta functions. That the two counts agree is a consequence of the two  $\varphi_L$  being the same isogeny.

A last remark places the comparison in the wider setting of Hodge theory, where it is the weight-one instance of a general correspondence. The analytic abelian variety carries on  $H^1(X, \mathbb{Z}) = \Lambda^\vee$  (theorem 1.3.2) the weight-one Hodge structure  $H^1(X, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$  of definition 1.3.3, and a polarization is exactly a polarization of that Hodge structure. From this vantage the equivalence of theorem 11.4.1 is the statement that the category of complex abelian varieties is equivalent to the category of polarizable integral Hodge structures of weight one, the linear-algebra data  $(\Lambda, H^{1,0}, H)$  recovering the torus and its polarization; the general dictionary between polarizable Hodge structures and the geometry they come from is developed in *Hodge Theory* [7], of which the present weight-one case is the part visible without transcendence.

The comparison specializes, as every construction in this book does, to the elliptic curve, where all three faces, the analytic uniformization, the Hermitian form, and the theta embedding, reproduce the algebraic curve and its principal polarization.

#### Example 11.4.4: The Elliptic Curve Reconciled

Let  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ ,  $\tau \in \mathbb{H}$ , the analytic elliptic curve of Part I. Its polarization is the positive Hermitian form  $H(v, w) = v\bar{w}/\text{Im}\tau$ , whose imaginary part  $E_0 = \text{Im}H$  is the standard integral symplectic form on  $\Lambda$ , of type (1) (example 5.3.7). The theta functions of Part I embed  $E$  in  $\mathbb{P}^2$  as a smooth cubic, and by Chow's theorem (theorem 11.4.1) this analytic embedding is algebraic:  $E$  is the projective plane curve cut out by a Weierstrass equation, the algebraic elliptic curve of Part II. The full passage from the lattice  $\Lambda$  and the Weierstrass  $\wp$ -function to the affine equation  $y^2 = 4x^3 - g_2x - g_3$  is developed in *Elliptic Curves* [4], and *Complex Analysis* [16] supplies the analytic uniformization it rests on.

The dual and the polarization reconcile term by term. Analytically  $\hat{E} = \hat{V}/\hat{\Lambda} \cong \text{Pic}^0(E) \cong E$ , the self-duality of example 5.1.2; algebraically  $\hat{E} = \text{Pic}_{E/\mathbb{C}}^0 \cong E$  through the Abel-Jacobi map  $P \mapsto [(P) - (O)]$ . By proposition 11.4.3 these are the same isomorphism  $E \cong \hat{E}$ . The analytic polarization isogeny  $\varphi_L$ , induced by  $v \mapsto H(v, -)$  and of degree  $(d_1)^2 = 1$  (example 5.3.7), equals the algebraic  $\varphi_{\mathcal{O}_E(O)}$ , of degree  $|K(L)| = (\deg \mathcal{O}_E(O))^2 = 1$ : both are the principal polarization, an isomorphism  $E \xrightarrow{\sim} \hat{E}$ . The kernel is trivial and the type is (1) on both sides.

The endomorphism theory matches likewise. Whether computed analytically as a subalgebra of  $\text{End}_{\mathbb{C}}(V) = \mathbb{C}$  (example 6.2.5) or algebraically from the Tate module,  $\text{End}^0(E)$  is either  $\mathbb{Q}$  or an imaginary quadratic field  $K$ , and in the CM case  $E$  has complex multiplication by  $K$  in the sense of both definition 6.4.5 and its algebraic counterpart; the CM elliptic curves  $\mathbb{C}/\mathcal{O}_K$  of theorem 6.4.6



and example 6.4.11 are the same curves whose algebraic CM is developed in *Elliptic Curves* [4]. The Rosati involution is complex conjugation on  $K$  analytically (example 6.2.5) and its algebraic namesake, the same anti-involution, and the Albert type is IV on both sides. For  $\text{End}^0(E) = \mathbb{Q}$  the involution is the identity and the type is I. Every quantity attached to  $E$  is thus computed identically in the two Parts, the elliptic curve being the case where the dictionary can be checked by hand in full.

### Exercise 11.4.1

1. State precisely which of the two classical inputs to theorem 11.4.1, Chow's theorem or GAGA, is used at each of the following steps, and in one sentence say why: (a) essential surjectivity of the analytification functor; (b) full faithfulness on homomorphisms; (c) the agreement of the analytic and algebraic Picard groups in proposition 11.4.3.
2. For the elliptic curve  $E = \mathbb{C}/\Lambda$  with  $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$  of example 11.4.4, the polarization is  $H(v, w) = v\bar{w}/\text{Im}\tau$ . Verify that  $E_0 = \text{Im}H$  takes integer values on the basis  $\{1, \tau\}$  of  $\Lambda$  and compute the matrix of  $E_0$  in this basis. Confirm that the resulting polarization has type (1), so that  $\varphi_L$  has degree 1.
3. Let  $A$  be a compact complex torus  $V/\Lambda$  of dimension  $g$  that admits no polarization. Explain why  $A$  need not be the analytification of any projective variety over  $\mathbb{C}$ , and reconcile this with the fact that *every* such  $A$  carries plenty of holomorphic functions locally. For  $g = 2$ , describe in one sentence where a generic  $A$  fails the hypotheses of theorem 11.4.1.
4. Let  $X$  be an abelian variety over  $\mathbb{C}$  of dimension  $g$  with a polarization  $L$  of type  $(d_1, \dots, d_g)$ . Using proposition 11.4.3, show that the number of holomorphic global sections of  $L$ , computed analytically from theta functions, equals the algebraic count  $\sqrt{\deg \varphi_L}$ , and identify both with  $d_1 \cdots d_g$ .
5. Prove that if  $X$  and  $Y$  are abelian varieties over  $\mathbb{C}$ , then every holomorphic homomorphism  $f: X^{\text{an}} \rightarrow Y^{\text{an}}$  is the analytification of a unique algebraic homomorphism  $X \rightarrow Y$ , and deduce that the analytic and algebraic endomorphism rings  $\text{End}(X)$  coincide. Then explain why this identification is a phenomenon of characteristic zero, naming one endomorphism the algebraic theory sees in characteristic  $p$  that has no analytic counterpart.
6. Let  $E$  be an elliptic curve over  $\mathbb{C}$  with  $\text{End}^0(E) = K$  an imaginary quadratic field (the CM case of example 11.4.4). Show that the analytic Rosati involution, complex conjugation on  $K$ , agrees with the algebraic Rosati involution, and conclude that the Albert type of  $E$  is IV computed either way. Contrast this with the case  $\text{End}^0(E) = \mathbb{Q}$ .



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## *Jacobian Varieties*

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Abelian varieties have so far been studied one at a time; this chapter supplies the mechanism that manufactures them in bulk. Every smooth projective curve  $C$  of genus  $g$  carries one, its *Jacobian*  $\text{Jac}(C) = \text{Pic}^0(C)$ , the abelian variety of dimension  $g$  parametrizing the degree-zero line bundles on  $C$ . It is built from the same Picard machinery as the dual (chapter 10), and it comes with more structure than a bare abelian variety: the Abel-Jacobi map embeds  $C$  into  $\text{Jac}(C)$ , and the theta divisor swept out by the symmetric powers of  $C$  furnishes a canonical *principal* polarization.

Jacobians are the richest source of examples in the subject and the point where it rejoins the geometry of curves. The theta divisor remembers enough of the curve that the Torelli theorem can recover  $C$  from the polarized pair  $(\text{Jac}(C), \Theta)$  alone, so the passage from a curve to its Jacobian loses nothing. In genus one the construction closes a loop:  $\text{Jac}(E) \cong E$ , and the principal polarization is the autoduality already met for elliptic curves in *Elliptic Curves* [4]. The principally polarized Jacobians cut out a distinguished locus inside the moduli of all abelian varieties, the thread the closing chapter follows toward Shimura varieties.

### 12.1 The Picard Variety of a Curve

The dual abelian variety of chapter 10 was extracted from a single abelian variety  $X$  as the identity component  $\text{Pic}_{X/k}^0$  of its Picard scheme. The same construction, applied not to an abelian variety but to a smooth projective curve, manufactures abelian varieties out of thin air: every curve of genus  $g$  carries a Picard scheme whose identity component is an abelian variety of dimension  $g$ , its *Jacobian*. Elliptic curves already show the phenomenon in dimension one, where the curve is its own Jacobian; in higher genus the Jacobian is a genuinely new object, a  $g$ -dimensional abelian variety attached functorially to a 1-dimensional curve, and it carries a principal polarization (the business of section 12.3). This section constructs  $\text{Jac}(C)$  and proves it is an abelian variety of the stated dimension; the maps between the curve and its Jacobian, and the polarization, follow in later

sections.

The construction reuses the Picard-scheme machinery of chapter 10 verbatim. The only change is the object it is applied to: a curve in place of an abelian variety. Representability of the Picard functor was granted for projective varieties in general (theorem 7.4.2), so it applies here without amendment; what must be checked is that, for a curve, the identity component of the resulting Picard scheme is proper, smooth, geometrically connected, and of the right dimension.

**Definition 12.1.1: The Jacobian of a Curve**

Let  $C$  be a smooth projective geometrically connected curve of genus  $g$  over a field  $k$ . The *Jacobian variety* of  $C$  is

$$\mathrm{Jac}(C) := \mathrm{Pic}_{C/k}^0 = \mathrm{Pic}_{C/k}^0$$

the identity component of the Picard scheme of  $C$  (definition 10.1.1, theorem 7.4.2). Its  $\bar{k}$ -points are the degree-zero line bundle classes on  $C_{\bar{k}}$ : for a  $k$ -scheme  $T$ , the group  $\mathrm{Jac}(C)(T) = \mathrm{Pic}_{C/k}^0(T)$  consists of the  $T$ -families of degree-zero line bundles on  $C$ , rigidified along a chosen base point.

The definition is the exact analogue of the dual abelian variety, and deliberately so:  $\mathrm{Jac}(C)$  is  $\mathrm{Pic}^0$  of  $C$  just as  $\hat{X}$  was  $\mathrm{Pic}^0$  of  $X$ . The difference is entirely in the input. For an abelian variety  $X$ , degree was measured by the map  $\mathrm{Pic}(X) \rightarrow \mathrm{NS}(X)$  and  $\mathrm{Pic}_{X/k}^0$  collected the algebraically trivial classes; for a curve, the degree of a line bundle is a single integer (the sum of the multiplicities of zeros minus poles of any rational section), the degree map  $\mathrm{Pic}(C_{\bar{k}}) \rightarrow \mathbb{Z}$  has  $\mathrm{Jac}(C)(\bar{k})$  as its kernel, and  $\mathrm{Pic}_{C/k}^0$  collects exactly the degree-zero classes. The name *Picard variety* is the older synonym, emphasizing that the object parametrizes line bundles; *Jacobian* is the name that stuck, after Jacobi's inversion problem for the periods of a genus- $g$  curve, which the next section makes precise. Both names denote the same scheme  $\mathrm{Pic}_{C/k}^0$ .

That the construction lands in the world of abelian varieties, rather than just producing a group scheme, is the theorem the section exists to prove. It is the algebraic incarnation of a classical analytic fact: over  $\mathbb{C}$  the Jacobian is the complex torus  $H^0(C, \Omega_C^1)^*/H_1(C, \mathbb{Z})$ , a  $g$ -dimensional torus that carries a polarization and is therefore an abelian variety, exactly as foreshadowed in example 3.2.8. The algebraic statement below reproduces this over an arbitrary field, with no transcendental input.

**Theorem 12.1.2: The Jacobian Is an Abelian Variety**

Let  $C$  be a smooth projective geometrically connected curve of genus  $g$  over a field  $k$ . Then  $\mathrm{Jac}(C) = \mathrm{Pic}_{C/k}^0$  is an abelian variety of dimension  $g$  over  $k$ .

**Proof:**

The Picard functor  $\mathrm{Pic}_{C/k}$  is representable by a separated  $k$ -scheme, locally of finite type, because  $C$  is a smooth projective variety with a rational point after a separable extension (*Algebraic Geometry* [13]); this is the same representability theorem (theorem 7.4.2) invoked for abelian varieties, and  $\mathrm{Pic}_{C/k}^0$  is by definition its identity component, a connected group scheme of finite type over  $k$ . To call it an abelian variety is to check that it is proper, smooth, and geometrically connected, and then to read off the dimension. We argue in four steps.

*Step 1: Properness.* The claim is that the single component  $\mathrm{Pic}_{C/k}^0$  is proper over  $k$ ; the full

Picard scheme  $\text{Pic}_{C/k} = \coprod_{d \in \mathbb{Z}} \text{Pic}_{C/k}^d$  is not, being an infinite disjoint union of components indexed by degree, hence not even quasi-compact. What is proper is each individual component, and  $\text{Pic}_{C/k}^0$  is one of them. As the identity component,  $\text{Pic}_{C/k}^0$  is of finite type over  $k$  (*Algebraic Geometry* [13]), and it is separated because  $\text{Pic}_{C/k}$  is; properness thus reduces to universal closedness, which the valuative criterion supplies. A family of degree-zero line bundles over the punctured spectrum of a discrete valuation ring  $R$  extends across the puncture: the total space  $C_R$  is a regular scheme, so the Weil divisor defining the generic line bundle takes closure to a Cartier divisor over all of  $C_R$ , supplying the required extension, and the extended bundle again has degree zero because degree is locally constant in a family. Universal closedness of the finite-type separated  $\text{Pic}_{C/k}^0$  is exactly the criterion for properness, so  $\text{Pic}_{C/k}^0$  is proper over  $k$ .

*Step 2: Geometric connectedness.* The identity component of a group scheme locally of finite type over a field is geometrically connected, because the identity section is a  $k$ -point and a connected component through a rational point survives base change to  $\bar{k}$  (*Algebraic Geometry* [13]). Thus  $\text{Pic}_{C/k}^0 \times_k \bar{k} = \text{Pic}_{C_{\bar{k}}/\bar{k}}^0$  is connected, which is the assertion. This is where the hypothesis that  $C$  is geometrically connected enters: it guarantees  $C_{\bar{k}}$  is a single curve, so its Picard scheme has a single identity component rather than a product over several geometric pieces.

*Step 3: Smoothness and dimension.* A group scheme of finite type over a field is smooth if and only if its dimension equals the dimension of its tangent space at the identity. The tangent space to  $\text{Pic}_{C/k}$  at the identity is  $H^1(C, \mathcal{O}_C)$  (*Algebraic Geometry* [13]), whose  $k$ -dimension is by definition the genus:

$$\dim_k H^1(C, \mathcal{O}_C) = g$$

so smoothness of  $\text{Pic}_{C/k}^0$  amounts to the equality  $\dim \text{Pic}_{C/k}^0 = g$ . That equality is granted by the same source: for a smooth projective curve the Picard scheme is smooth of dimension  $h^1(\mathcal{O}_C) = g$ , a theorem of the general theory (*Algebraic Geometry* [13]), so  $\text{Pic}_{C/k}^0$  is a smooth group variety of dimension  $g$ , not just a group scheme. The equality of dimension and tangent dimension dispels the naive characteristic- $p$  non-reducedness worry, exactly as it did for the dual (theorem 10.1.2): unlike the Picard scheme of a general variety, that of a curve is always reduced.

*Step 4: Assembling the conclusion.* By Steps 1 through 3,  $\text{Pic}_{C/k}^0$  is a proper, smooth, geometrically connected group scheme of finite type over  $k$ , of dimension  $g$ . A proper smooth connected group variety is an abelian variety (definition 7.1.2), so  $\text{Jac}(C) = \text{Pic}_{C/k}^0$  is an abelian variety of dimension  $g$ , completing the proof.

The proof is a citation-and-assembly argument: representability, properness of  $\text{Pic}_{C/k}^0$  for a curve, and the identification  $\dim \text{Pic}^0 = h^1(\mathcal{O}_C)$  are the three inputs drawn from the general theory of the Picard scheme in *Algebraic Geometry* [13], while the abelian-variety conclusion is assembled here from the definition (definition 7.1.2). The three inputs are precisely the ones the parallel result for the dual (theorem 10.1.2) also needed, applied now to a curve rather than an abelian variety; the arguments run in step. What the theorem delivers is a machine: feed it any smooth projective curve, and out comes an abelian variety whose dimension is the genus.

The coincidence of the two meanings of  $g$  deserves to be recorded on its own; it ties the theory of curves to the theory of abelian varieties.

**Proposition 12.1.3: Dimension of the Jacobian**

Let  $C$  be a smooth projective geometrically connected curve over  $k$ . Then  $\dim \text{Jac}(C) = g$ , the genus of  $C$ . Equivalently, the dimension of the Jacobian as an abelian variety equals the dimension  $\dim_k H^0(C, \Omega_C^1) = \dim_k H^1(C, \mathcal{O}_C)$  of the space of regular differentials on  $C$ .

**Proof:**

By theorem 12.1.2,  $\dim \text{Jac}(C) = \dim \text{Pic}_{C/k}^0 = \dim_k H^1(C, \mathcal{O}_C)$ . Serre duality on the smooth projective curve  $C$  identifies  $H^1(C, \mathcal{O}_C)$  with the dual of  $H^0(C, \Omega_C^1)$  (*Algebraic Geometry* [13]), so the two cohomology dimensions agree, and each equals the genus  $g$  by the definition of the genus of a curve as  $\dim_k H^0(C, \Omega_C^1)$ . Hence  $\dim \text{Jac}(C) = g$ , as we sought to show.

The two occurrences of the letter  $g$  now denote the same integer for a reason, not by coincidence of notation. The genus of  $C$  is a 1-dimensional invariant, the number of independent holomorphic differentials the curve carries, equivalently (over  $\mathbb{C}$ ) the number of handles on the Riemann surface; the dimension of  $\text{Jac}(C)$  is a  $g$ -dimensional invariant, the number of independent coordinates on the parameter space of degree-zero line bundles. These agree because the tangent space to the moduli of line bundles at the trivial bundle is  $H^1(C, \mathcal{O}_C)$ , and Serre duality makes that space the dual of the differentials. Every construction to come reads one of these two faces of  $g$  against the other, and the Abel-Jacobi map of section 12.2 is the geometric bridge between them.

The smallest case is the one the reader has already met, where the Jacobian machine returns the curve unchanged. It is worth running the construction explicitly in genus one to see the two meanings of  $g$  collapse to a single object.

**Example 12.1.4: The Jacobian of an Elliptic Curve**

Let  $E$  be an elliptic curve over  $k$ , a smooth projective geometrically connected curve of genus 1 equipped with a rational point  $O \in E(k)$ . Then  $\text{Jac}(E) = \text{Pic}_{E/k}^0$  is an abelian variety of dimension  $\dim \text{Jac}(E) = g = 1$  by theorem 12.1.2 and proposition 12.1.3. In fact  $\text{Jac}(E)$  is canonically isomorphic to  $E$  itself. The isomorphism sends a point  $P \in E$  to the class of the degree-zero line bundle  $\mathcal{O}_E(P - O)$ , and its inverse recovers the point of  $E$  from a degree-zero bundle by the Riemann-Roch theorem on  $E$ , which supplies a unique effective degree-one divisor in the class of any degree-one bundle. This is exactly the identification  $E \cong \text{Pic}^0(E)$  already computed for elliptic curves in *Elliptic Curves* [4], where the group law on  $E$  is defined so that  $P + Q = R$  precisely when  $\mathcal{O}_E(P + Q - R - O)$  is trivial; that definition is engineered so that  $P \mapsto [\mathcal{O}_E(P - O)]$  is a group homomorphism, hence an isomorphism  $E \xrightarrow{\sim} \text{Jac}(E)$ .

Here the two meanings of  $g$  coincide in the most transparent way: the genus of  $E$  is 1, the dimension of  $\text{Jac}(E)$  is 1, and the curve and its Jacobian are the same 1-dimensional abelian variety. In higher genus the two objects part ways: a curve of genus  $g \geq 2$  is a 1-dimensional variety, while its Jacobian is a  $g$ -dimensional abelian variety, and the curve sits inside its Jacobian as a 1-dimensional subvariety through the Abel-Jacobi map of section 12.2. The full statement of the genus-one case, with the canonical principal polarization matching the autoduality of  $E$ , is section 12.5.

**Exercise 12.1.1**

1. Let  $C$  be a smooth projective geometrically connected curve of genus  $g$  over  $k$ . State the dimension of  $\text{Jac}(C)$  as an abelian variety, and identify the tangent space to  $\text{Pic}_{C/k}$  at the identity as a cohomology group of  $C$ . Using these, compute  $\dim \text{Jac}(C)$  for a smooth plane curve of degree 4 in  $\mathbb{P}^2$ , whose genus is  $g = 3$ .
2. The Jacobian is defined as  $\text{Jac}(C) = \text{Pic}_{C/k}^0$ , a single component of the Picard scheme, rather than the full  $\text{Pic}_{C/k}$ . Explain why the full Picard scheme  $\text{Pic}_{C/k} = \coprod_{d \in \mathbb{Z}} \text{Pic}_{C/k}^d$  is not proper, and hence not an abelian variety, whereas  $\text{Pic}_{C/k}^0$  is. Which  $\bar{k}$ -points of  $\text{Pic}_{C/k}$  lie in  $\text{Jac}(C)$ ?
3. Two of the four steps in the proof of theorem 12.1.2 use hypotheses on  $C$  in an essential way. Identify where the hypothesis that  $C$  is *geometrically connected* is used, and explain what could fail without it. Separately, explain how Step 3 rules out the characteristic- $p$  non-reducedness that afflicts the Picard scheme of a general variety.
4. Let  $E$  be an elliptic curve over  $k$  with origin  $O \in E(k)$ . Following example 12.1.4, verify directly that the map  $\varphi: E \rightarrow \text{Jac}(E)$ ,  $P \mapsto [\mathcal{O}_E(P - O)]$ , is a bijection on  $\bar{k}$ -points: that it is injective, and that every degree-zero line bundle class on  $E_{\bar{k}}$  is of this form. State precisely which classical theorem on  $E$  supplies surjectivity.
5. Prove the properness of  $\text{Pic}_{C/k}^0$  (Step 1 of theorem 12.1.2) in detail via the valuative criterion. That is, let  $R$  be a discrete valuation ring with fraction field  $K$ , and suppose given a degree-zero line bundle on  $C_K$ ; produce an extension to a degree-zero line bundle on  $C_R$ , and explain why this yields universal closedness and hence properness of  $\text{Pic}_{C/k}^0$ . You may use that  $\text{Pic}_{C/k}^0$  is separated and of finite type over  $k$ , and that  $C_R$  is a regular scheme.
6. Combine proposition 12.1.3 with Serre duality to explain, in a self-contained way, why the “two meanings of  $g$ ” coincide: the genus of  $C$  (the number of independent regular differentials) and the dimension of  $\text{Jac}(C)$  (the number of coordinates on the moduli of degree-zero line bundles). Then determine  $\dim_k H^1(C, \mathcal{O}_C)$  for a curve of genus  $g$ , and state what  $\text{Jac}(C)$  is when  $g = 0$ .

## 12.2 The Abel-Jacobi Map

The Jacobian  $\text{Jac}(C) = \text{Pic}_{C/k}^0$  (definition 12.1.1) is a  $g$ -dimensional abelian variety assembled out of the degree-zero line bundles on the curve, but so far it stands apart from  $C$  itself. This section builds the map that ties the two together: a morphism  $C \rightarrow \text{Jac}(C)$  that embeds the curve inside its own Jacobian and identifies the group  $\text{Jac}(C)$  with the group of degree-zero divisor classes on  $C$ . The identification is Abel’s theorem, and the surjectivity of the induced map on symmetric powers is Jacobi inversion; together they are the reason the Jacobian deserves its name. The embedding realizes  $C$  as the geometric generator of  $\text{Jac}(C)$ , the source from which every point of the Jacobian is a sum of points of the curve.

A degree-zero divisor on  $C$  is a formal  $\mathbb{Z}$ -combination  $\sum_P n_P(P)$  of closed points with  $\sum_P n_P = 0$ , and two such divisors are linearly equivalent when their difference is the divisor of a rational function. The group of degree-zero divisors modulo linear equivalence is the degree-zero divisor class group; on a smooth projective curve it is exactly  $\text{Pic}^0(C)$ , the degree-zero part of the Picard group, since a line bundle of degree zero is  $\mathcal{O}_C(D)$  for a degree-zero divisor  $D$  unique up to linear equivalence. The construction below turns a single point of  $C$  into such a class by measuring it against a fixed base point.

**Definition 12.2.1: The Abel-Jacobi Map**

Let  $C/k$  be a smooth projective geometrically connected curve of genus  $g$ , and fix a base point  $P_0 \in C(k)$ . The *Abel-Jacobi map* is the morphism

$$\alpha: C \longrightarrow \text{Jac}(C), \quad P \longmapsto [\mathcal{O}_C(P - P_0)]$$

sending a point  $P$  to the class of the degree-zero line bundle  $\mathcal{O}_C(P - P_0)$  in  $\text{Jac}(C) = \text{Pic}_{C/k}^0$ . For each  $d \geq 1$  it extends to the  $d$ -th *symmetric power*  $\text{Sym}^d C = C^d/S_d$ , the variety parametrizing effective divisors of degree  $d$  on  $C$ , by

$$\alpha_d: \text{Sym}^d C \longrightarrow \text{Jac}(C), \quad (P_1 + \cdots + P_d) \longmapsto [\mathcal{O}_C(\sum_{i=1}^d P_i - dP_0)]$$

a morphism of varieties, with  $\alpha_1 = \alpha$ . On  $k$ -points  $\alpha_d$  sends an effective divisor  $D$  of degree  $d$  to the class  $[\mathcal{O}_C(D - dP_0)] \in \text{Jac}(C)(k)$ .

That  $\alpha$  is a morphism, not just a map of sets, is exactly the universal property of the Jacobian: the line bundle  $\mathcal{O}_{C \times C}(\Delta - C \times \{P_0\})$  on  $C \times C$ , restricted to a slice  $\{P\} \times C$ , is a family of degree-zero bundles on  $C$  parametrized by  $P \in C$ , and the moduli property of  $\text{Pic}_{C/k}^0$  (*Algebraic Geometry* [13]) turns that family into the morphism  $\alpha$ . The extension  $\alpha_d$  arises the same way from a family over  $\text{Sym}^d C$ , using that  $\text{Sym}^d C$  is a smooth projective variety of dimension  $d$  for a smooth curve  $C$ . The definition depends on  $P_0$  only up to translation: changing the base point from  $P_0$  to  $P'_0$  replaces  $\alpha_d$  by  $\alpha_d$  composed with translation by the fixed class  $d[\mathcal{O}_C(P_0 - P'_0)]$ , an automorphism of  $\text{Jac}(C)$ , so the images and the fibre structure are unchanged for every  $d$ .

The fibres of  $\alpha_d$  record precisely the linear systems on  $C$ . Two effective divisors  $D, D'$  of degree  $d$  have  $\alpha_d(D) = \alpha_d(D')$  if and only if  $\mathcal{O}_C(D - dP_0) \cong \mathcal{O}_C(D' - dP_0)$ , that is, if and only if  $D$  and  $D'$  are linearly equivalent; so the fibre of  $\alpha_d$  through  $D$  is the complete linear system  $|D|$ , a projective space  $\mathbb{P}(H^0(C, \mathcal{O}_C(D)))$  of dimension  $\dim |D|$ . For  $d = 1$  this dimension is zero for every  $D = (P)$  when  $g \geq 1$ , since a nonconstant rational function with a single simple pole would exhibit  $C$  as  $\mathbb{P}^1$ ; this is the first hint that  $\alpha$  is injective for positive genus, made precise in proposition 12.2.4.

Naming the group of degree-zero divisor classes and comparing it with the  $k$ -points of the Jacobian is the substance of Abel's theorem. The map  $\alpha$  extended linearly to all of  $\text{Div}^0(C)$  sends a degree-zero divisor  $\sum n_P(P)$  to  $[\mathcal{O}_C(\sum n_P(P))]$ , forgetting the base point entirely; its kernel is the obstruction to a degree-zero divisor being principal, and Abel's theorem computes that kernel exactly. Jacobi inversion is the complementary statement that  $\alpha_g$  already exhausts the Jacobian: every class is a sum of  $g$  points.



**Theorem 12.2.2: Abel's Theorem and Jacobi Inversion**

Let  $C/k$  be a smooth projective geometrically connected curve of genus  $g \geq 1$  with a base point  $P_0 \in C(k)$ .

1. (*Abel's theorem*) The homomorphism

$$\text{cl}: \text{Div}^0(C) \longrightarrow \text{Jac}(C)(k), \quad D \longmapsto [\mathcal{O}_C(D)]$$

is surjective, and its kernel is exactly the group of principal divisors. It therefore induces an isomorphism of groups

$$\text{Pic}^0(C) \xrightarrow{\sim} \text{Jac}(C)(k)$$

between the degree-zero divisor class group of  $C$  and the  $k$ -points of the Jacobian.

2. (*Jacobi inversion*) The morphism  $\alpha_g: \text{Sym}^g C \rightarrow \text{Jac}(C)$  is surjective and birational.

**Proof:**

*Step 1: The class map and its kernel.* By construction  $\text{Jac}(C) = \text{Pic}_{C/k}^0$  represents the functor of degree-zero line bundles, so on  $k$ -points  $\text{Jac}(C)(k) = \text{Pic}^0(C)$ , the group of isomorphism classes of degree-zero line bundles on  $C$  (*Algebraic Geometry* [13]; the identification of  $k$ -points with the group of classes uses that  $C$  has the rational point  $P_0$ , so every class is represented by a  $k$ -rational bundle). On a smooth projective curve every line bundle is  $\mathcal{O}_C(D)$  for a Weil divisor  $D$ , unique up to linear equivalence, and  $\deg \mathcal{O}_C(D) = \deg D$ ; restricting to degree zero, the assignment  $D \mapsto [\mathcal{O}_C(D)]$  is a surjective homomorphism  $\text{Div}^0(C) \rightarrow \text{Pic}^0(C) = \text{Jac}(C)(k)$ . Its kernel is the set of degree-zero divisors  $D$  with  $\mathcal{O}_C(D) \cong \mathcal{O}_C$ , and an isomorphism  $\mathcal{O}_C(D) \cong \mathcal{O}_C$  is a rational function  $f$  with  $\text{div}(f) = D$ ; so the kernel is exactly the group of principal divisors. This is the content of part (1): the induced map  $\text{Pic}^0(C) = \text{Div}^0(C)/\sim \rightarrow \text{Jac}(C)(k)$  is an isomorphism.

*Step 2: Reduction of inversion to a dimension count.* For part (2) every fibre of  $\alpha_g$  is a linear system, as computed above: the fibre through an effective divisor  $D$  of degree  $g$  is  $|D| = \mathbb{P}(H^0(C, \mathcal{O}_C(D)))$ , of dimension  $\dim |D| = \dim_k H^0(C, \mathcal{O}_C(D)) - 1$ . By the Riemann-Roch theorem on  $C$  (*Algebraic Geometry* [13]),

$$\dim_k H^0(C, \mathcal{O}_C(D)) - \dim_k H^1(C, \mathcal{O}_C(D)) = \deg D - g + 1 = 1$$

for  $\deg D = g$ , so  $\dim_k H^0(C, \mathcal{O}_C(D)) = 1 + \dim_k H^1(C, \mathcal{O}_C(D)) \geq 1$ , with equality  $\dim_k H^0 = 1$  precisely when  $H^1(C, \mathcal{O}_C(D)) = 0$ . When  $\dim_k H^0(C, \mathcal{O}_C(D)) = 1$  the linear system  $|D|$  is a single point, so  $\alpha_g$  is injective over the class of  $D$ ; when it is larger, the fibre is a positive-dimensional projective space.

*Step 3: A dense open locus of one-point fibres.* The special divisors, those with  $H^1(C, \mathcal{O}_C(D)) = H^0(C, \Omega_C^1(-D))^\vee \neq 0$  by Serre duality (*Algebraic Geometry* [13]), are constrained: a degree- $g$  divisor  $D$  is special exactly when it is contained in the zero divisor of some nonzero holomorphic differential, and the differentials form the  $g$ -dimensional space  $H^0(C, \Omega_C^1)$ , each with a divisor of degree  $2g-2$ . The effective special divisors of degree  $g$  therefore sweep out a subvariety of  $\text{Sym}^g C$  of dimension strictly less than  $g$ : a nonzero differential ranges over the  $(g-1)$ -dimensional

projective space  $\mathbb{P}(H^0(C, \Omega_C^1))$ , and for each the  $g$  points are chosen among its finitely many  $2g-2$  zeros, so the special locus has dimension at most  $g-1 < g$  and cannot fill the  $g$ -dimensional  $\text{Sym}^g C$ . (For  $g=1$  a nonzero differential has no zeros, so the special locus is empty and the bound holds vacuously.) Hence the non-special divisors form a dense open subset  $U \subseteq \text{Sym}^g C$ , and over  $\alpha_g(U)$  the map  $\alpha_g$  has one-point fibres.

*Step 4: Surjectivity and birationality.* The variety  $\text{Jac}(C)$  is irreducible projective of dimension  $g$  (proposition 12.1.3), and  $\text{Sym}^g C$  is irreducible projective of dimension  $g$ , being a quotient of the irreducible  $C^g$ . The morphism  $\alpha_g$  is injective on the dense open  $U$  by Step 3, so its image has dimension  $g$ ; being a morphism of projective varieties, its image is closed, and a closed irreducible subvariety of the irreducible  $\text{Jac}(C)$  of full dimension  $g$  is all of  $\text{Jac}(C)$ . Thus  $\alpha_g$  is surjective, which is Jacobi inversion on the level of points: every class in  $\text{Jac}(C)(k)$ , being  $[\mathcal{O}_C(D - gP_0)]$  for some effective  $D$  of degree  $g$  once  $k = \bar{k}$ , is a sum of  $g$  points of  $C$ . Finally, over the dense open  $\alpha_g(U)$  the fibre of  $\alpha_g$  is the linear system  $|D| = \mathbb{P}(H^0(C, \mathcal{O}_C(D)))$  computed in Step 2, a single *reduced* point when  $h^0(\mathcal{O}_C(D)) = 1$ ; so  $\alpha_g$  is an isomorphism there in every characteristic, hence birational. This establishes both parts, completing the proof.

Abel's theorem is the precise sense in which the Jacobian *is* the divisor class group: the abstract abelian variety  $\text{Pic}_{C/k}^0$  and the concrete group  $\text{Div}^0(C)/(\text{principal})$  built from functions on  $C$  are one and the same, functorially. The gain is that the second, defined by generators and relations, now carries the geometry of an abelian variety. Jacobi inversion says this variety is not larger than it needs to be:  $g$  points of  $C$  already produce every class, and generically in exactly one way, so  $\text{Jac}(C)$  is birationally the space  $\text{Sym}^g C$  of effective degree- $g$  divisors. The failure of injectivity is measured by special divisors, and the locus where it fails is the geometry that the theta divisor of section 12.3 will organize. The name of the subject is justified here: the map  $\alpha_g$  inverts the passage from divisors to their classes, up to the linear systems that Riemann-Roch counts.

The genus-one case is the one the reader has already met, and there the whole apparatus collapses to the group law of an elliptic curve.

### Example 12.2.3: The Abel-Jacobi Map in Genus One

Let  $E$  be an elliptic curve over  $k$  with origin  $O \in E(k)$ , taken as the base point  $P_0 = O$ . Here  $g=1$ , so  $\text{Sym}^1 E = E$  and the Abel-Jacobi map is

$$\alpha: E \longrightarrow \text{Jac}(E), \quad P \longmapsto [\mathcal{O}_E(P - O)]$$

By Abel's theorem  $\alpha$  is the isomorphism  $E \xrightarrow{\sim} \text{Pic}^0(E) = \text{Jac}(E)(k)$  already recorded in the algebraic theory (example 5.1.2): an elliptic curve is canonically its own Jacobian,  $\text{Jac}(E) = E$ , and  $\alpha$  is the identity under this identification, since  $[\mathcal{O}_E(P - O)]$  is by definition the point of  $\text{Pic}^0(E)$  that the classical divisor-class isomorphism attaches to  $P$ . Jacobi inversion  $\alpha_1 = \alpha$  is then simply this isomorphism.

Abel's theorem in this case is exactly the group law. Three points  $P, Q, R \in E$  satisfy  $P+Q+R = O$  in the group  $E$  if and only if

$$(P) + (Q) + (R) - 3(O) = \text{div}(f)$$

is principal for some rational function  $f$ , that is, if and only if the three points lie on a common line in the Weierstrass plane model,  $f$  being the linear form cutting out that line divided by the

appropriate power of the coordinate at infinity. Collinearity of  $P, Q, R$  is therefore equivalent to  $[\mathcal{O}_E((P) + (Q) + (R) - 3(O))] = 0$  in  $\text{Jac}(E)$ , which under  $\alpha$  reads  $\alpha(P) + \alpha(Q) + \alpha(R) = 0$ : the chord-and-tangent group law is Abel's theorem for  $g = 1$ . The full development of this group law, and the verification that the geometric addition agrees with addition of divisor classes, is the starting point of *Elliptic Curves* [4].

For genus one the map  $\alpha$  is an isomorphism, hence certainly a closed immersion; the same embedding statement holds for every positive genus, though  $\alpha$  is far from surjective once  $g \geq 2$ . The curve embeds in its Jacobian as a closed subvariety, and this embedding is what lets the geometry of  $\text{Jac}(C)$  be studied through the curve.

#### Proposition 12.2.4: The Curve Embeds in Its Jacobian

Let  $C/k$  be a smooth projective geometrically connected curve of genus  $g \geq 1$  with a base point  $P_0 \in C(k)$ . The Abel-Jacobi map  $\alpha: C \rightarrow \text{Jac}(C)$  is a closed immersion. In particular  $C$  is isomorphic to a closed subvariety of its Jacobian.

#### Proof:

Since  $C$  is a projective curve and  $\text{Jac}(C)$  is projective,  $\alpha$  is a proper morphism, so its image is closed and it suffices to show  $\alpha$  is a closed immersion onto its image; equivalently, that  $\alpha$  is injective on  $\bar{k}$ -points and injective on tangent vectors (an unramified proper monomorphism onto its image is a closed immersion, *Algebraic Geometry* [13]).

*Step 1: Injectivity on points.* Suppose  $\alpha(P) = \alpha(Q)$  for points  $P, Q \in C(\bar{k})$ . Then  $\mathcal{O}_C(P - P_0) \cong \mathcal{O}_C(Q - P_0)$ , so  $\mathcal{O}_C(P - Q) \cong \mathcal{O}_C$  and the divisor  $(P) - (Q)$  is principal: there is a rational function  $f$  on  $C_{\bar{k}}$  with  $\text{div}(f) = (P) - (Q)$ . If  $P \neq Q$  this  $f$  has a single simple pole and a single simple zero, hence defines a degree-one morphism  $f: C_{\bar{k}} \rightarrow \mathbb{P}^1$ , which is an isomorphism; but then  $C$  has genus 0, contradicting  $g \geq 1$ . Therefore  $P = Q$ , and  $\alpha$  is injective on  $\bar{k}$ -points.

*Step 2: Injectivity on tangent vectors.* It remains to check that the differential  $d\alpha_P$  is injective at each point  $P$ , so that  $\alpha$  does not collapse a tangent direction. The cotangent space of  $\text{Jac}(C) = \text{Pic}_{C/k}^0$  at the identity is  $H^1(C, \mathcal{O}_C)^\vee \cong H^0(C, \Omega_C^1)$  by Serre duality, and translating to the point  $\alpha(P)$ , the codifferential  $\alpha^*$  is identified with the evaluation

$$H^0(C, \Omega_C^1) \longrightarrow \Omega_{C,P}^1, \quad \omega \longmapsto \omega(P)$$

of a global differential at  $P$  (*Algebraic Geometry* [13]; this is the standard identification of the pullback along  $\alpha$  of invariant differentials on the Jacobian with holomorphic differentials on  $C$ ). The differential  $d\alpha_P$  is injective if and only if this evaluation onto the one-dimensional cotangent fibre is surjective, that is, if and only if some global differential is nonzero at  $P$ . The failure of surjectivity means every  $\omega \in H^0(C, \Omega_C^1)$  vanishes at  $P$ , i.e.  $P$  is a base point of the canonical system  $|K_C|$ . For  $g \geq 1$  the canonical system is base-point-free at every point: by Riemann-Roch,  $\dim_k H^0(C, \Omega_C^1(-P)) = \dim_k H^0(C, \Omega_C^1) - 1 = g - 1$  for every  $P$  (the point  $P$  imposes one independent condition on the  $g$ -dimensional space of differentials whenever  $g \geq 1$ ), so there is always a differential nonzero at  $P$ . Hence the evaluation is surjective,  $d\alpha_P$  is injective, and  $\alpha$  is unramified at  $P$ .

Combining the two steps,  $\alpha$  is a proper injective unramified morphism, hence a closed immersion

onto its image, and  $C \cong \alpha(C)$  is a closed subvariety of  $\text{Jac}(C)$ , as we sought to show.

The proposition embeds the geometry: a curve of positive genus lives inside an abelian variety, its own Jacobian, as a closed curve generating that variety under the group law. The two conditions in the proof match the two ways an embedding can fail. Injectivity on points fails only if  $C$  carries a degree-one map to  $\mathbb{P}^1$ , which forces  $g = 0$ ; injectivity on tangent vectors fails only if the canonical differentials all vanish at a point, which again forces  $g = 0$ . Positive genus rules out both, and the same Riemann-Roch computation that governed the fibres of  $\alpha_g$  governs the tangent behaviour here. The image  $\alpha(C)$  is a curve that, together with the group structure it inherits from being inside  $\text{Jac}(C)$ , recovers all of  $\text{Jac}(C)$ : the sums  $\alpha(P_1) + \cdots + \alpha(P_g)$  range over the whole Jacobian by theorem 12.2.2, so  $C$  is a generating subvariety and not just a subvariety.

### Exercise 12.2.1

1. Let  $C/k$  be a smooth projective geometrically connected curve of genus  $g \geq 1$  with base point  $P_0 \in C(k)$ , and let  $\alpha_d: \text{Sym}^d C \rightarrow \text{Jac}(C)$  be the maps of definition 12.2.1. For effective divisors  $D$  of degree  $d$  and  $D'$  of degree  $d'$ , express  $\alpha_{d+d'}(D + D')$  in terms of  $\alpha_d(D)$  and  $\alpha_{d'}(D')$ , using the group law on  $\text{Jac}(C)$ . Deduce that  $\alpha_d(D) = \alpha_1(P_1) + \cdots + \alpha_1(P_d)$  for  $D = P_1 + \cdots + P_d$ .
2. definition 12.2.1 fixes a base point  $P_0 \in C(k)$ . Let  $P'_0 \in C(k)$  be a second base point, with associated maps  $\alpha'_d$ . Show that  $\alpha'_d = t \circ \alpha_d$ , where  $t: \text{Jac}(C) \rightarrow \text{Jac}(C)$  is translation by the fixed class  $d[\mathcal{O}_C(P_0 - P'_0)]$ . Conclude that the fibres of  $\alpha_d$  and  $\alpha'_d$  coincide.
3. Fix a divisor  $D$  of degree  $d$  on  $C$ , and let  $\bar{D} \in \text{Sym}^d C$  denote a point mapping to it. Using definition 12.2.1, prove that the fibre  $\alpha_d^{-1}(\alpha_d(D))$  is exactly the complete linear system  $|D|$ , a projective space of dimension  $\dim_k H^0(C, \mathcal{O}_C(D)) - 1$ . Then compute this fibre dimension when  $\deg D = g$  and  $H^1(C, \mathcal{O}_C(D)) = 0$ .
4. Let  $E/k$  be an elliptic curve with origin  $O$ , taken as base point. Three points  $P, Q, R \in E(k)$  lie on a common line in a Weierstrass plane model. Using the genus-one description in example 12.2.3, show that  $[\mathcal{O}_E((P) + (Q) + (R) - 3(O))] = 0$  in  $\text{Jac}(E)$ , and translate this into a relation among  $\alpha(P), \alpha(Q), \alpha(R)$  under the identification  $\text{Jac}(E) = E$ .
5. A curve  $C/k$  of genus  $g \geq 1$  is called *hyperelliptic* when it admits a degree-two morphism  $\varphi: C \rightarrow \mathbb{P}^1$ . Fix such a  $\varphi$  and a base point  $P_0$ . Show that the map  $\alpha_2: \text{Sym}^2 C \rightarrow \text{Jac}(C)$  is not injective. Where in the proof of injectivity for  $\alpha = \alpha_1$  (proposition 12.2.4) does the corresponding argument break down for  $\alpha_2$ ?
6. Prove that for  $g \geq 1$  the Abel-Jacobi map  $\alpha: C \rightarrow \text{Jac}(C)$  is unramified: at every point  $P \in C$  there is a global differential  $\omega \in H^0(C, \Omega_C^1)$  with  $\omega(P) \neq 0$ . Then explain how the failure of this statement would force  $g = 0$ , connecting your argument to proposition 12.2.4.

## 12.3 The Theta Divisor and the Canonical Principal Polarization

The Jacobian was built as an abelian variety of dimension  $g$  (theorem 12.1.2), and the Abel-Jacobi map embedded the curve inside it (proposition 12.2.4). This section produces the theta divisor  $\Theta$ ,

the image of the  $(g - 1)$ -st symmetric power of  $C$  under the extended Abel-Jacobi map, and proves that the homomorphism  $\varphi_{\mathcal{O}(\Theta)}$  it induces to the dual is an isomorphism, so  $\text{Jac}(C)$  is a principally polarized abelian variety, canonically self-dual.

The Abel-Jacobi map of definition 12.2.1 extends from  $C$  to each symmetric power: a degree- $d$  effective divisor  $P_1 + \cdots + P_d$  maps to the class  $[\mathcal{O}_C(P_1 + \cdots + P_d - dP_0)]$ , giving a morphism  $\alpha_d: \text{Sym}^d C \rightarrow \text{Jac}(C)$ . For  $d = g$  this is the Jacobi inversion of theorem 12.2.2, surjective and birational. One step below the top, at  $d = g - 1$ , the map is no longer surjective: its image is a divisor, and that divisor is the theta divisor.

### Definition 12.3.1: The Theta Divisor of a Jacobian

Let  $C$  be a smooth projective geometrically connected curve of genus  $g \geq 1$  over  $k$ , with Jacobian  $\text{Jac}(C) = \text{Pic}_{C/k}^0$  (definition 12.1.1) and Abel-Jacobi map  $\alpha: C \rightarrow \text{Jac}(C)$  based at  $P_0 \in C(k)$  (definition 12.2.1). The *theta divisor* is the image

$$\Theta := \alpha_{g-1}(\text{Sym}^{g-1} C) \subseteq \text{Jac}(C)$$

of the  $(g - 1)$ -st symmetric power under the extended Abel-Jacobi map  $\alpha_{g-1}: \text{Sym}^{g-1} C \rightarrow \text{Jac}(C)$ ,  $\sum_{i=1}^{g-1} P_i \mapsto [\mathcal{O}_C(\sum_i P_i - (g-1)P_0)]$ . It is an effective divisor on  $\text{Jac}(C)$ , a reduced irreducible subvariety of dimension  $g - 1$ , well defined up to translation by a point of  $\text{Jac}(C)$  (a change of base point  $P_0$  or of the chosen representative translates  $\Theta$ ).

The definition is a purely algebraic one, made over any field: no theta function is written down, only the image of a symmetric power. This is the algebraic incarnation of the analytic theta divisor of definition 4.5.1, which was the zero locus of the Riemann theta function on a principally polarized complex torus; the two agree over  $\mathbb{C}$  because the vanishing locus of  $\theta$  is exactly the locus of classes  $[\mathcal{O}_C(D - (g-1)P_0)]$  with  $D$  effective of degree  $g - 1$ , the content of the Riemann vanishing theorem. Here the construction is taken as primary and the polarization is read off from it, the reverse of the analytic order of chapter 4, where the polarization came first and  $\Theta$  was its zero divisor.

Two features of definition 12.3.1 carry the weight. First,  $\Theta$  has dimension  $g - 1$ , one less than  $\text{Jac}(C)$ , so it is a divisor and not a lower-dimensional cycle; this is because  $\alpha_{g-1}$  is generically injective (a generic effective divisor  $D$  of degree  $g - 1$  has  $l(D) = 1$ , equivalently  $\dim |D| = 0$ , so it is the unique effective divisor in its complete linear system  $|D|$  and the fiber  $\alpha_{g-1}^{-1}([D]) = |D|$  is a single point, by the Abel-Jacobi machinery of theorem 12.2.2), so  $\dim \Theta = \dim \text{Sym}^{g-1} C = g - 1$ . Second,  $\Theta$  is only well defined up to translation, exactly as the analytic theta divisor was (definition 4.5.1 produced  $\Theta$  from a section unique only up to scalar, itself pinned only after choosing a bundle in the polarization class). The translation ambiguity is harmless: a polarization is a homomorphism to the dual, and translating the divisor does not change the homomorphism it induces, since  $t_a^* \mathcal{O}(\Theta) \otimes \mathcal{O}(\Theta)^{-1}$  is algebraically trivial. It is the associated class in  $\text{NS}(\text{Jac}(C))$ , not the divisor itself, that is canonical.

The dimension count that makes  $\Theta$  a divisor also delivers the one input the principality argument rests on: that the space of global sections of  $\mathcal{O}(\Theta)$  is one-dimensional.

**Proposition 12.3.2: Ampleness of the Theta Divisor**

Let  $\Theta = \alpha_{g-1}(\text{Sym}^{g-1}C)$  be the theta divisor of definition 12.3.1 on  $\text{Jac}(C)$ . Then  $\mathcal{O}(\Theta)$  is ample, and

$$\dim_k H^0(\text{Jac}(C), \mathcal{O}(\Theta)) = 1$$

so  $\Theta$  is the unique effective divisor in its complete linear system.

**Proof:**

*Step 1:  $\mathcal{O}(\Theta)$  is ample.* The divisor  $\Theta$  is effective and, by the dimension count above, an irreducible subvariety of codimension one (a hypersurface for  $g \geq 2$ ; a point when  $g = 1$ ). The one curve-specific input is the top self-intersection of the Jacobian theta divisor,

$$(\Theta^g) = g! \neq 0$$

the *Poincaré formula* for  $\Theta = W_{g-1} = \alpha_{g-1}(\text{Sym}^{g-1}C)$ , whose computation is the Riemann-vanishing / Poincaré-formula content of the intersection theory of *Algebraic Geometry* [13]: it counts the effective representatives of a generic class of degree  $g - 1$ , and holds over any field. (This is the algebraic incarnation of the analytic  $(\Theta^g) = g!$  of proposition 4.5.2, which was proved for the theta divisor of an abstract principal polarization over  $\mathbb{C}$ , a different object; here the number is established for  $W_{g-1}$  itself.) The nonvanishing  $(\Theta^g) \neq 0$  makes  $\mathcal{O}(\Theta)$  nondegenerate, so its homomorphism  $\varphi_{\mathcal{O}(\Theta)}: \text{Jac}(C) \rightarrow \widehat{\text{Jac}(C)}$  has finite kernel and is an isogeny. An effective nondegenerate bundle whose  $\varphi$  is an isogeny is ample: this is the ampleness criterion for abelian varieties (theorem 10.5.2), the effective section cutting out  $\Theta$  supplying the required positivity.

*Step 2:  $h^0(\mathcal{O}(\Theta)) = 1$ .* The Riemann-vanishing theorem identifies  $\mathcal{O}(\Theta)$ , up to translation, with the line bundle whose unique (up to scalar) global section is cut out by the theta function of the curve, so its space of sections is one-dimensional and  $\Theta = W_{g-1}$  is the only effective divisor in the complete linear system:

$$\dim_k H^0(\text{Jac}(C), \mathcal{O}(\Theta)) = 1$$

This is the curve-specific content of Riemann-Roch and the Riemann-vanishing theorem on  $C$ , cited to *Algebraic Geometry* [13]. Algebraic Riemann-Roch on the abelian variety agrees: for the ample bundle  $L = \mathcal{O}(\Theta)$ ,

$$\dim_k H^0(\text{Jac}(C), L) = \chi(L) = \frac{(L^g)}{g!} = \frac{(\Theta^g)}{g!}$$

by proposition 10.5.3 (the higher cohomology vanishing being part of that statement), and substituting the Poincaré formula  $(\Theta^g) = g!$  of Step 1 recovers  $\dim_k H^0 = g!/g! = 1$ . A one-dimensional space of sections has a single effective divisor, namely  $\Theta$ , completing the proof.

The proposition isolates what makes the theta divisor special: it is an ample divisor whose linear system is a single point. A general ample bundle  $L$  on an abelian variety has  $\dim H^0(X, L) = \sqrt{\deg \varphi_L}$ , which exceeds 1 as soon as  $\varphi_L$  has degree greater than one; the theta divisor is precisely the case where this count collapses to 1, and by proposition 10.5.3 that forces  $\deg \varphi_{\mathcal{O}(\Theta)} = 1$ . This is the numerical fact the next theorem promotes to a statement about the dual abelian variety.

**Theorem 12.3.3: The Canonical Principal Polarization of a Jacobian**

Let  $C$  be a smooth projective geometrically connected curve of genus  $g \geq 1$  over  $k$ , with Jacobian  $\text{Jac}(C)$  and theta divisor  $\Theta$  (definition 12.3.1). Then  $\mathcal{O}(\Theta)$  is ample and the associated homomorphism

$$\varphi_{\mathcal{O}(\Theta)}: \text{Jac}(C) \longrightarrow \widehat{\text{Jac}(C)}, \quad x \mapsto [t_x^* \mathcal{O}(\Theta) \otimes \mathcal{O}(\Theta)^{-1}]$$

is an *isomorphism*. In particular  $\varphi_{\mathcal{O}(\Theta)}$  is a principal polarization (definition 10.3.2), so  $\text{Jac}(C)$  is a principally polarized abelian variety, and it is canonically autodual:

$$\text{Jac}(C) \cong \widehat{\text{Jac}(C)}$$

The class of  $\Theta$  in  $\text{NS}(\text{Jac}(C))$  is independent of the base point and the representative, so this principal polarization is canonical.

**Proof:**

By proposition 12.3.2,  $\mathcal{O}(\Theta)$  is ample, so theorem 10.3.1 applies:  $\varphi_{\mathcal{O}(\Theta)}: \text{Jac}(C) \rightarrow \widehat{\text{Jac}(C)}$  is an isogeny with kernel  $K(\mathcal{O}(\Theta))$ , and

$$\deg \varphi_{\mathcal{O}(\Theta)} = |K(\mathcal{O}(\Theta))|$$

It remains to identify this degree. Algebraic Riemann-Roch for the ample bundle  $L = \mathcal{O}(\Theta)$  reads  $\dim_k H^0(\text{Jac}(C), L)^2 = |K(L)| = \deg \varphi_L$  (proposition 10.5.3). By proposition 12.3.2 the section count is  $\dim_k H^0(\text{Jac}(C), L) = 1$ , so

$$\deg \varphi_{\mathcal{O}(\Theta)} = 1^2 = 1$$

An isogeny of degree one is an isomorphism: its kernel  $K(\mathcal{O}(\Theta))$  is the trivial group scheme, and a homomorphism of abelian varieties that is bijective with trivial kernel is an isomorphism. Hence  $\varphi_{\mathcal{O}(\Theta)}$  is an isomorphism, and by definition 10.3.2 a polarization whose associated homomorphism is an isomorphism is principal; the isomorphism  $\text{Jac}(C) \xrightarrow{\sim} \widehat{\text{Jac}(C)}$  it furnishes is the autoduality.

For canonicity, recall from definition 12.3.1 that changing the base point or the representative translates  $\Theta$  by a point  $a \in \text{Jac}(C)$ . Under such a translation  $\mathcal{O}(\Theta)$  is replaced by  $t_a^* \mathcal{O}(\Theta) = \mathcal{O}(\Theta) \otimes P$  with  $P = t_a^* \mathcal{O}(\Theta) \otimes \mathcal{O}(\Theta)^{-1} \in \text{Pic}^0(\text{Jac}(C))$  algebraically trivial, so the class of  $\Theta$  in  $\text{NS}(\text{Jac}(C)) = \text{Pic}(\text{Jac}(C)) / \text{Pic}^0(\text{Jac}(C))$  is unchanged. Since  $\varphi_L$  depends only on the class of  $L$  in  $\text{NS}$ , the homomorphism  $\varphi_{\mathcal{O}(\Theta)}$  is independent of these choices, so the principal polarization is canonical, completing the proof.

Every smooth projective curve manufactures, with no extra data, a principally polarized abelian variety  $(\text{Jac}(C), \Theta)$ , and the polarization is canonical, forced by the curve rather than chosen. Two consequences shape the theory that follows. First, the autoduality  $\text{Jac}(C) \cong \widehat{\text{Jac}(C)}$  means the Jacobian is its own dual in a canonical way, which is why divisor classes on  $C$  and the extensions they classify can be handled symmetrically; principally polarized abelian varieties are exactly the ones with such an autoduality (definition 10.3.2). Second, the principality identifies  $(\text{Jac}(C), \Theta)$  as a point of the moduli space of principally polarized abelian varieties, and the map  $C \mapsto (\text{Jac}(C), \Theta)$  from curves to that moduli space is the Torelli map. That this map is injective, that the pair recovers the



curve, is the Torelli theorem (section 12.4), and the theta divisor is precisely the geometric structure from which the reconstruction proceeds. The principality is what makes  $\Theta$  a single divisor.

A concrete instance sharpens the count. For a curve of genus 2 the theta divisor is the curve itself, sitting inside its Jacobian as the defining divisor of the polarization.

**Example 12.3.4: The Theta Divisor of a Genus-Two Curve**

Let  $C$  be a smooth projective genus-2 curve over  $k$ , so  $g = 2$  and  $\text{Jac}(C)$  is an abelian surface. The theta divisor is the image of the  $(g - 1) = 1$ -st symmetric power,

$$\Theta = \alpha_1(\text{Sym}^1 C) = \alpha_1(C) = \alpha(C) \subseteq \text{Jac}(C)$$

Since  $\text{Sym}^1 C = C$  and  $\alpha_1$  is the Abel-Jacobi map  $\alpha: C \rightarrow \text{Jac}(C)$  itself, the theta divisor is the embedded image of the curve. For  $g \geq 1$  the Abel-Jacobi map is a closed immersion (proposition 12.2.4), so  $\Theta \cong C$  as a subvariety: the genus-2 curve is a divisor on its own Jacobian, a curve of self-intersection

$$(\Theta^2) = g! = 2! = 2$$

inside the abelian surface  $\text{Jac}(C)$ . The count of sections is  $\dim_k H^0(\text{Jac}(C), \mathcal{O}(\Theta)) = (\Theta^2)/2! = 1$ , matching proposition 12.3.2, so  $\Theta = \alpha(C)$  is the unique effective divisor in its class and  $\varphi_{\mathcal{O}(\Theta)}: \text{Jac}(C) \xrightarrow{\sim} \widehat{\text{Jac}(C)}$  is the canonical principal polarization of the abelian surface. The genus-2 case is thus the first where the theta divisor is a genuine curve rather than a single point: in genus 1 one has  $\text{Sym}^{g-1} C = \text{Sym}^0 C$  a single point, the empty divisor, so  $\Theta$  is the 0-dimensional divisor  $\alpha_0(\text{Sym}^0 C) = \{0\}$ , the origin of  $\text{Jac}(C) \cong C$  (matching the elliptic theta divisor of example 4.5.5), while in genus 2 it is the curve  $C$  itself, and its geometry inside  $\text{Jac}(C)$ , in particular whether  $C$  is hyperelliptic, is already enough to recover  $C$  by the Torelli theorem. The intersection theory here is the algebraic form of the analytic self-intersection  $(\Theta^g) = g!$  computed by the Pfaffian in *Algebraic Geometry* [13].

### Exercise 12.3.1

1. Let  $C$  be a smooth projective geometrically connected curve of genus  $g$ , with theta divisor  $\Theta = \alpha_{g-1}(\text{Sym}^{g-1} C)$  of definition 12.3.1. Compute  $\dim \Theta$  and the top self-intersection number  $(\Theta^g)$ , and use proposition 10.5.3 to read off  $\dim_k H^0(\text{Jac}(C), \mathcal{O}(\Theta))$ . State what these three numbers become when  $g = 3$ .
2. A general ample line bundle  $L$  on an abelian variety of dimension  $g$  satisfies  $\dim_k H^0(X, L) = \sqrt{\deg \varphi_L}$ . Suppose  $L = \mathcal{O}(\Theta)^{\otimes 2}$  is the second tensor power of the theta divisor of  $\text{Jac}(C)$ . Using the numerical facts of this section and proposition 10.5.3, compute  $\deg \varphi_L$  and  $\dim_k H^0(\text{Jac}(C), L)$ . Is  $\varphi_L$  an isomorphism? Justify.
3. Let  $E$  be an elliptic curve, viewed as its own Jacobian  $\text{Jac}(E) \cong E$  via the Abel-Jacobi map based at  $P_0$ . Identify the theta divisor  $\Theta$  explicitly as a subvariety of  $E$ , using definition 12.3.1 and example 12.3.4. Verify directly that  $\mathcal{O}(\Theta)$  has  $\dim_k H^0(E, \mathcal{O}(\Theta)) = 1$  and that  $\varphi_{\mathcal{O}(\Theta)}: E \rightarrow \widehat{E}$  is the canonical principal polarization, checking consistency with the section's degree count.
4. The class of  $\Theta$  in  $\text{NS}(\text{Jac}(C))$  is asserted in definition 12.3.1 and theorem 12.3.3 to be independent of the base point  $P_0$ . Let  $\alpha, \alpha': C \rightarrow \text{Jac}(C)$  be the Abel-Jacobi maps based at two points  $P_0, P'_0 \in C(k)$ , with theta divisors  $\Theta, \Theta'$ . Show that  $\Theta' = t_a^* \Theta$  for an explicit



$a \in \text{Jac}(C)$  that you determine, and deduce that  $\mathcal{O}(\Theta')$  and  $\mathcal{O}(\Theta)$  define the same class in  $\text{NS}(\text{Jac}(C))$  and hence the same homomorphism  $\varphi_{\mathcal{O}(\Theta)} = \varphi_{\mathcal{O}(\Theta')}$ .

5. Prove that  $\varphi_{\mathcal{O}(\Theta)}: \text{Jac}(C) \rightarrow \widehat{\text{Jac}(C)}$  is an isomorphism, assembling the argument of theorem 12.3.3 from its inputs. Specifically: cite the ampleness and section count of proposition 12.3.2, invoke theorem 10.3.1 to see that  $\varphi_{\mathcal{O}(\Theta)}$  is an isogeny with  $\deg \varphi_{\mathcal{O}(\Theta)} = |K(\mathcal{O}(\Theta))|$ , use algebraic Riemann-Roch (proposition 10.5.3) in the form  $\dim_k H^0(\text{Jac}(C), L)^2 = \deg \varphi_L$  to pin the degree, and conclude that a degree-one isogeny of abelian varieties is an isomorphism.
6. Let  $(X, \varphi)$  be a principally polarized abelian variety of dimension  $g$  over  $k$ , with  $\varphi = \varphi_L$  for an ample  $L$ . Show that  $\dim_k H^0(X, L) = 1$ , so  $L$  has a unique effective divisor  $D$  in its complete linear system, and that  $(D^g) = g!$ . Then explain, referring to theorem 12.3.3, why  $(\text{Jac}(C), \varphi_{\mathcal{O}(\Theta)})$  is an instance of this situation and why the theta divisor is forced to be a single divisor rather than a positive-dimensional linear system.

## 12.4 The Torelli Theorem

The construction of the previous sections attaches to a curve  $C$  a principally polarized abelian variety  $(\text{Jac}(C), \Theta)$ : the Jacobian (definition 12.1.1) carries a canonical principal polarization furnished by the theta divisor (theorem 12.3.3). The passage is functorial and geometric, but it also looks lossy. A curve of genus  $g$  has moduli of dimension  $3g - 3$  (for  $g \geq 2$ ), while the principally polarized abelian varieties of dimension  $g$  have moduli of dimension  $g(g+1)/2$ ; the two counts agree only at  $g = 2, 3$ , and for large  $g$  the Jacobian lives in a space vastly larger than the curve came from. One might expect the map to forget something. The Torelli theorem says it does not: the polarized Jacobian remembers the curve completely.

The precision matters. It is not enough that  $\text{Jac}(C)$  determine  $C$  as an abstract abelian variety, for that is false already at  $g = 2$ , where non-isomorphic curves can have isomorphic (unpolarized) Jacobians. The polarization is the missing datum. An abelian variety underlying a Jacobian typically admits many principal polarizations, and only the one coming from the theta divisor sees the curve. What Torelli asserts is that the *pair*  $(\text{Jac}(C), \Theta)$ , remembered up to isomorphism of polarized abelian varieties, pins  $C$  down exactly.

### Theorem 12.4.1: The Torelli Theorem

Let  $C$  and  $C'$  be smooth projective curves of genus  $\geq 1$  over an algebraically closed field  $k$ , with Jacobians  $\text{Jac}(C)$  and  $\text{Jac}(C')$  carrying their canonical principal polarizations  $\Theta$  and  $\Theta'$  (theorem 12.3.3). If

$$(\text{Jac}(C), \Theta) \cong (\text{Jac}(C'), \Theta')$$

as principally polarized abelian varieties, then  $C \cong C'$ . In other words, a curve of genus  $\geq 1$  is determined up to isomorphism by its principally polarized Jacobian.

The theorem is the injectivity statement for the Torelli map, the assignment  $C \mapsto (\text{Jac}(C), \Theta)$  from isomorphism classes of curves to isomorphism classes of principally polarized abelian varieties. Read that way it is a statement about a map of moduli: the Torelli map is injective on points. It is not surjective for  $g \geq 4$ , and describing its image is the Schottky problem, one of the open frontiers of the subject. Injectivity alone is already the assertion that the entire geometry of  $C$  can be reconstructed from the linear-algebraic and divisor-theoretic data of its Jacobian.

Two clarifications guard against over-reading the statement. First, the hypothesis of an isomorphism of *polarized* abelian varieties is indispensable, as noted above. Second, at  $g = 1$  the theorem is a tautology worth recording: a genus-one curve  $C$  over  $k$  with a rational point is its own Jacobian,  $\text{Jac}(C) \cong C$  (chapter 12 develops this in section 12.5), and the canonical polarization is the class of a single point, so  $(\text{Jac}(C), \Theta)$  literally contains  $C$ . The content lives entirely at  $g \geq 2$ .

The proof of theorem 12.4.1 is longer and more geometry-specific than the constructions this chapter has otherwise carried out in full, and it does not reduce to the abelian-variety machinery assembled so far: it requires a genuine geometric analysis of the theta divisor and its singularities. For that reason the full argument is deferred; this is a sanctioned deferral, not a claim proved. What can be given cleanly, and what conveys the mechanism, is the geometric reconstruction that drives every proof of the theorem.

#### Proposition 12.4.2: Key Ideas of the Torelli Proof

Let  $C$  be a smooth projective curve of genus  $g \geq 2$  over  $k$ , with Jacobian  $J = \text{Jac}(C)$ , theta divisor  $\Theta \subseteq J$  (definition 12.3.1), and Abel-Jacobi map  $\alpha: C \rightarrow J$  (definition 12.2.1). The pair  $(J, \Theta)$  reconstructs  $C$  by the following mechanism.

1. The theta divisor  $\Theta = \alpha_{g-1}(\text{Sym}^{g-1}C)$  is the image of the symmetric power under the extended Abel-Jacobi map; its smooth locus carries a Gauss map  $\gamma: \Theta^{\text{sm}} \rightarrow \mathbb{P}(T_0 J^\vee) = \mathbb{P}^{g-1}$  sending a smooth point to the translate of its tangent hyperplane. The image and ramification of  $\gamma$  encode the canonical embedding of  $C$ .
2. The singular locus  $\Theta_{\text{sing}}$  measures the special linear series on  $C$ : by the Riemann singularity theorem, the multiplicity of  $\Theta$  at a point corresponding to a divisor class  $[D]$  of degree  $g - 1$  equals  $\dim H^0(C, \mathcal{O}_C(D))$ , so  $\Theta$  is singular exactly where  $C$  carries a nontrivial special series.
3. From these data the canonical model of  $C$  in  $\mathbb{P}^{g-1}$  (for  $C$  non-hyperelliptic) or the hyperelliptic pencil (for  $C$  hyperelliptic) is recovered intrinsically from  $(J, \Theta)$ , and hence so is  $C$  itself.

#### Proof Key ideas:

The reconstruction rests on reading  $C$  off the geometry of  $\Theta$ . The Abel-Jacobi map (theorem 12.2.2) identifies  $\Theta$  with the image of  $\text{Sym}^{g-1}C$ , so a general point of  $\Theta$  is the class of an effective divisor  $D = P_1 + \cdots + P_{g-1}$  with the  $P_i$  distinct; at such a point  $\Theta$  is smooth and its tangent hyperplane, translated back to the origin of  $J$ , is the hyperplane in  $T_0 J^\vee = H^0(C, \Omega_C^1)$  spanned by the canonical images of  $P_1, \dots, P_{g-1}$ . The Gauss map  $\gamma$  therefore records, for a general divisor, the hyperplane through its canonical points, and the closure of the branch behavior of  $\gamma$  recovers the canonical curve  $\varphi_K(C) \subseteq \mathbb{P}^{g-1}$  as the locus of canonical points. For a non-hyperelliptic curve the canonical map is an embedding, so this returns  $C$ ; for a hyperelliptic curve the canonical map is two-to-one onto a rational normal curve, and the branch locus of that double cover returns the  $2g + 2$  Weierstrass points and hence  $C$ .

The singularities of  $\Theta$  supply the same information from a complementary direction and are what make the two cases visible intrinsically. By the Riemann singularity theorem, the multiplicity of  $\Theta$  at the class  $[D] \in \text{Pic}^{g-1}(C)$  is  $h^0(C, \mathcal{O}_C(D))$ ; thus  $\Theta$  is smooth in codimension one precisely when  $C$  has no special divisors of the relevant degree, and the dimension and structure of  $\Theta_{\text{sing}}$  distinguish the hyperelliptic locus from the generic one. Assembling the Gauss map, the singular

locus, and the polarization class produces the canonical model, and the curve is recovered.

The remaining work, turning this reconstruction into a proof that an isomorphism of polarized pairs comes from an isomorphism of curves, requires care about the geometry of the reconstruction and a separate treatment of the two exceptional signs by which a curve and its Weierstrass-point involution can enter. The complete argument, in both the complex-analytic and the algebraic settings, is given in Milne's *Abelian Varieties* notes and, in exhaustive geometric detail, in Arbarello, Cornalba, Griffiths, and Harris, *Geometry of Algebraic Curves*; the ampleness and cohomological inputs are those cited to *Algebraic Geometry* [13] in section 12.3. The mechanism above is the whole of it in outline.

Proposition 12.4.2 isolates the single geometric fact that powers the theorem: the theta divisor is not an opaque ample divisor but a highly structured object whose smooth points remember the canonical map and whose singular points remember the special linear series. Everything about  $C$  that matters, its canonical model, its gonality, its Weierstrass points, is legible in  $\Theta$ . The polarization enters because it is the polarization that fixes  $\Theta$  up to translation; without it one would not know which ample divisor to read.

The hyperelliptic case makes the reconstruction fully explicit and is the right place to see the mechanism in operation, since there the special series is a single well-understood pencil.

#### Example 12.4.3: Reconstruction for a Hyperelliptic Curve

Let  $C$  be a hyperelliptic curve of genus  $g \geq 2$  over  $k$  ( $\text{char } k \neq 2$ ), presented as the smooth projective model of  $y^2 = f(x)$  with  $f$  separable of degree  $2g + 1$  or  $2g + 2$ ; the branch points of the degree-two map  $C \rightarrow \mathbb{P}^1$  are the  $2g + 2$  Weierstrass points, the ramification points of the hyperelliptic involution (section 12.5 treats  $g = 1$ ; the general hyperelliptic model is developed in *Algebraic Geometry* [13]). The canonical series is  $|K_C| = (g - 1)g_2^1$ , the  $(g - 1)$ -fold multiple of the hyperelliptic pencil  $g_2^1$ , so the canonical map  $\varphi_K: C \rightarrow \mathbb{P}^{g-1}$  factors as

$$C \xrightarrow{2:1} \mathbb{P}^1 \xrightarrow{\nu} \mathbb{P}^{g-1}$$

with  $\nu$  the degree- $(g - 1)$  rational normal (Veronese) curve. The canonical image is thus a rational normal curve, doubly covered by  $C$ .

Now watch the reconstruction of proposition 12.4.2 recover the Weierstrass points from  $(J, \Theta)$  alone. A divisor class  $[D]$  of degree  $g - 1$  is a singular point of  $\Theta$  exactly when  $h^0(\mathcal{O}_C(D)) \geq 2$ , that is, when  $D$  moves in a pencil, and for a hyperelliptic curve this happens precisely when  $D$  contains a fiber of  $g_2^1$ , that is, a pair  $P + \iota P$  of conjugate points under the involution  $\iota$ . By the Riemann singularity theorem the multiplicity of  $\Theta$  at  $[D]$  is  $h^0(\mathcal{O}_C(D))$ , so  $\Theta_{\text{sing}}$  is governed by how many conjugate pairs  $D$  contains; the Weierstrass points, the fixed points of  $\iota$ , are exactly the points  $W$  where a conjugate pair degenerates to  $2W$ . Tracing the Gauss map to the canonical rational normal curve and its branch locus therefore pinpoints the  $2g + 2$  Weierstrass points on  $\mathbb{P}^1$ , and  $C$  is the double cover of  $\mathbb{P}^1$  branched there, recovered up to isomorphism. The singular-theta mechanism first bites at  $g \geq 3$ : for the smallest case  $g = 2$  the theta divisor  $\Theta = \alpha_1(\text{Sym}^1 C) = \alpha(C)$  is the Abel-Jacobi image of  $C$  itself, hence a smooth genus-two curve inside the abelian surface  $J$ . Every effective divisor  $D = P$  of degree  $g - 1 = 1$  has  $h^0(\mathcal{O}_C(P)) = 1$  (a single point does not move on a curve of genus  $\geq 1$ ), so by the Riemann singularity theorem  $\Theta$  has multiplicity 1 at every point and  $\Theta_{\text{sing}} = \emptyset$ . Here  $C$  is therefore read off directly:  $\Theta$  is a copy of  $C$ , and  $(J, \Theta)$  returns the genus-two curve as  $\Theta$  up to translation and the sign of the Weierstrass-point involution.

Torelli is, in the end, a statement about information. The Jacobian construction of chapter 12 manufactures a principally polarized abelian variety from every curve, and one might have feared that the manufacturing discards the fine geometry of  $C$  in favor of the coarse linear algebra of a polarized lattice. The theorem says the opposite: the map from curves to principally polarized abelian varieties is injective on isomorphism classes, so nothing is lost. The curve is fully present inside  $(\text{Jac}(C), \Theta)$ , waiting to be read off the singularities of the theta divisor. What is genuinely lost, or rather never gained, is surjectivity: not every principally polarized abelian variety is a Jacobian. The Jacobians trace out the Torelli locus, a  $(3g - 3)$ -dimensional subvariety of the  $g(g + 1)/2$ -dimensional moduli of principally polarized abelian varieties, and characterizing that locus, the Schottky problem, is the boundary of what Jacobians can be. It is the natural next question, and one the reader now has the vocabulary to pose.

### Exercise 12.4.1

1. State theorem 12.4.1 precisely, and explain why the hypothesis that the isomorphism  $\text{Jac}(C) \cong \text{Jac}(C')$  respect the polarizations  $\Theta$  and  $\Theta'$  cannot be dropped. Give the smallest genus at which an unpolarized isomorphism of Jacobians fails to force  $C \cong C'$ .
2. Let  $C$  be a smooth projective curve of genus  $g$  over  $k$ . Compute  $\dim H^0(C, \mathcal{O}_C(D))$  for a general effective divisor  $D$  of degree  $g - 1$ , and use the Riemann singularity theorem, as invoked in proposition 12.4.2, to determine the multiplicity of  $\Theta$  at the corresponding point  $[D]$ . Conclude that the general point of  $\Theta$  is a smooth point.
3. In the setting of example 12.4.3, let  $C$  be hyperelliptic of genus  $g \geq 2$  with involution  $\iota$ . Show that a divisor class  $[D]$  of degree  $g - 1$  is a singular point of  $\Theta$  if and only if the effective divisor  $D$  contains a conjugate pair  $P + \iota P$ , and identify the largest possible multiplicity of  $\Theta$  along  $\Theta_{\text{sing}}$ .
4. Explain the case  $g = 2$  of example 12.4.3 in full. Show that  $\Theta_{\text{sing}} = \emptyset$ , that  $\Theta$  is a smooth genus-two curve inside the abelian surface  $J = \text{Jac}(C)$ , and describe exactly how  $C$  is recovered from  $(J, \Theta)$ . Reconcile this with the dimension count  $3g - 3 = g(g + 1)/2 = 3$ .
5. The canonical map of a non-hyperelliptic curve of genus  $g \geq 3$  is an embedding, while that of a hyperelliptic curve is two-to-one onto a rational normal curve. Using proposition 12.4.2, explain how the Gauss map  $\gamma: \Theta^{\text{sm}} \rightarrow \mathbb{P}^{g-1}$  distinguishes the two cases and, in each, recovers  $C$  from its canonical image.
6. Prove that the Torelli map is not surjective for  $g \geq 4$ , by a dimension count. Identify the resulting locus of Jacobians inside the moduli of principally polarized abelian varieties of dimension  $g$ , name the problem of characterizing it, and explain why the count is an equality precisely at  $g = 2, 3$ .

## 12.5 The Genus-One Case: $\text{Jac}(E) = E$

The Jacobian construction runs over a curve  $C$  of any genus, and  $g = 1$  is the boundary case where the curve is itself an abelian variety. Here the machinery of the chapter must return something already familiar: an elliptic curve is a genus-one curve with a marked rational point, and it carries a group law of its own, put there by hand in *Elliptic Curves* [4]. The Jacobian offers a second, intrinsic group law, defined without any choice of coordinates. The local task is to check that these two group

laws agree and that the whole apparatus closes back on the curve it started from.

We first record the identification itself, then extract the polarization statement from it.

### Theorem 12.5.1: The Jacobian of an Elliptic Curve

Let  $E$  be an elliptic curve over a field  $k$ , that is a smooth projective geometrically connected curve of genus 1 with a marked point  $O \in E(k)$ . The Abel-Jacobi map

$$\alpha: E \longrightarrow \text{Jac}(E) = \text{Pic}^0(E), \quad P \longmapsto [\mathcal{O}_E(P - O)]$$

is an isomorphism of abelian varieties. Under it the canonical principal polarization of theorem 12.3.3 corresponds to the autoduality  $\hat{E} \cong E$  of example 5.1.2, and the group law transported from  $\text{Jac}(E)$  to  $E$  is the group law of  $E$ .

#### Proof:

By theorem 12.1.2 the Jacobian  $\text{Jac}(E) = \text{Pic}_{E/k}^0$  is an abelian variety of dimension  $g = 1$ , and by proposition 12.2.4 the Abel-Jacobi map  $\alpha$  (with base point  $O$ ) is a closed immersion for  $g \geq 1$ . A closed immersion between smooth projective curves of the same dimension whose source is irreducible is an isomorphism onto a closed subvariety of the same dimension; since  $\dim E = \dim \text{Jac}(E) = 1$  and  $\text{Jac}(E)$  is irreducible, the image  $\alpha(E)$  is all of  $\text{Jac}(E)$ , and  $\alpha$  is an isomorphism of varieties.

It remains to see that  $\alpha$  is a homomorphism of abelian varieties and that the transported group law is the standard one. Because  $\alpha(O) = [\mathcal{O}_E(O - O)] = [\mathcal{O}_E] = 0$ , the map  $\alpha$  sends the identity of  $E$  (the marked point  $O$ ) to the identity of  $\text{Jac}(E)$ ; by corollary 7.2.3 a morphism of abelian varieties fixing the identity is a homomorphism, so  $\alpha$  is an isomorphism of groups once  $E$  is given its own group structure with  $O$  as identity. That group structure is exactly the one for which  $\alpha$  is additive, and by construction

$$\alpha(P) + \alpha(Q) = [\mathcal{O}_E(P - O)] + [\mathcal{O}_E(Q - O)] = [\mathcal{O}_E(P + Q - 2O)] \quad \text{in } \text{Pic}^0(E),$$

so  $P \oplus Q$  (the sum in  $E$ ) is the unique point  $R$  with  $\mathcal{O}_E(R - O) \cong \mathcal{O}_E(P + Q - 2O)$ , that is with  $R - O \sim P + Q - 2O$  as divisors. This is precisely the addition law on an elliptic curve read through  $\text{Pic}^0$ : two points add to the point linearly equivalent to  $P + Q - O$ . Thus the transported group law is the group law of  $E$ .

Finally, the identification of the polarization. The dual  $\hat{E}$  is  $\text{Pic}_{E/k}^0 = \text{Jac}(E)$  by definition (definition 10.1.1), and example 5.1.2 exhibits the resulting autoduality  $\hat{E} \cong E$ . The canonical principal polarization of theorem 12.3.3 is the isogeny  $\varphi_{\mathcal{O}(\Theta)}: \text{Jac}(E) \rightarrow \widehat{\text{Jac}(E)}$  attached to the theta divisor; that it is the same isomorphism as the autoduality  $\hat{E} \cong E$ , up to the identifications already made, is the content of proposition 12.5.2, proved next.

An elliptic curve is thus its own Jacobian, and the abstract  $\text{Pic}^0$  group law is the concrete chord-and-tangent law. The theorem says two things at once. First, the correspondence  $P \mapsto [\mathcal{O}_E(P - O)]$  turns points of  $E$  into degree-zero divisor classes and does so bijectively: every degree-zero class on a genus-one curve is  $[\mathcal{O}_E(P - O)]$  for a unique  $P$ , which is Abel's theorem (theorem 12.2.2) specialized to  $g = 1$ . Second, the group law the Jacobian carries as an abelian variety is not a new structure but the one already on  $E$ . The choice of base point  $O$  is the only input, and it is exactly the choice

that makes  $E$  an elliptic curve rather than a bare genus-one curve. This is why a curve of genus one has to be fixed with a rational point before it has a group law: the Jacobian always exists, but the isomorphism  $\alpha$  needs  $O$  to be written down.

The polarization statement is the residue of the general theory of section 12.3 in the degenerate case  $g = 1$ , where the theta divisor shrinks to a point.

**Proposition 12.5.2: The Theta Divisor in Genus One**

Let  $E$  be an elliptic curve. The theta divisor  $\Theta = \alpha_{g-1}(\text{Sym}^{g-1} E)$  of definition 12.3.1 is, for  $g = 1$ , the single point  $0 \in \text{Jac}(E)$ , an effective divisor of degree 1. Its associated homomorphism  $\varphi_{\mathcal{O}(\Theta)}: \text{Jac}(E) \rightarrow \widehat{\text{Jac}(E)}$  is an isomorphism, so the canonical polarization is principal, and it agrees with the polarization  $\varphi_{\mathcal{O}_E(O)}$  of example 10.3.4.

**Proof:**

For a curve of genus  $g$  the theta divisor is the image of the  $(g-1)$ -st symmetric power  $\text{Sym}^{g-1} E$  under the Abel-Jacobi map  $\alpha_{g-1}$  (definition 12.3.1). When  $g = 1$  the exponent is  $g-1 = 0$ , and  $\text{Sym}^0 E = \text{Spec } k$  is a single reduced point: the empty effective divisor on  $E$ , of degree 0, whose class in  $\text{Pic}^0(E)$  is 0. Hence  $\Theta = \alpha_0(\text{Sym}^0 E) = \{0\}$ , the origin of  $\text{Jac}(E)$ , an effective divisor of degree 1 on the one-dimensional abelian variety  $\text{Jac}(E)$ .

Now  $\mathcal{O}_{\text{Jac}(E)}(\Theta) = \mathcal{O}(\{0\})$  is the line bundle of a single point on a genus-one curve, so  $\deg \mathcal{O}(\Theta) = 1$  and, by Riemann-Roch on  $\text{Jac}(E)$ ,  $\chi(\mathcal{O}(\Theta)) = 1$ . The algebraic Riemann-Roch identity  $\chi(L)^2 = \deg \varphi_L$  of theorem 10.5.2 gives

$$\deg \varphi_{\mathcal{O}(\Theta)} = \chi(\mathcal{O}(\Theta))^2 = 1$$

so  $\varphi_{\mathcal{O}(\Theta)}$  is an isogeny of degree 1, hence an isomorphism. A polarization whose associated isogeny is an isomorphism is principal (definition 10.3.2), which is the assertion of theorem 12.3.3 made explicit in this case.

Finally, on  $E$  itself the point  $O$  is an effective divisor of degree 1, and  $\mathcal{O}_E(O)$  is ample with  $\chi(\mathcal{O}_E(O)) = 1$ ; transporting along the isomorphism  $\alpha$  of theorem 12.5.1, the origin  $0 = \alpha(O)$  carries  $\mathcal{O}(\Theta) = \mathcal{O}(\{0\})$  to  $\mathcal{O}_E(O)$ , so  $\varphi_{\mathcal{O}(\Theta)}$  and  $\varphi_{\mathcal{O}_E(O)}$  are the same homomorphism under  $\alpha$ . This is the polarization  $\varphi_{\mathcal{O}_E(O)}$  recorded in example 10.3.4, and being of degree 1 it is principal there as well.

That  $\Theta$  is a single point is the whole degeneration: on a curve of genus  $g$  the theta divisor is a  $(g-1)$ -dimensional subvariety, whose geometry (its singularities, its Gauss map) carries enough information to reconstruct the curve by Torelli (theorem 12.4.1). In genus one there is nothing left of that geometry, since a point has no singularities and no Gauss map, and correspondingly Torelli is vacuous for  $g = 1$ : all elliptic curves have the same 0-dimensional theta divisor, and it is the  $j$ -invariant, not  $\Theta$ , that separates them. The principality is automatic because a degree-one line bundle on a genus-one curve has  $\chi = 1$ , matching the Pfaffian count  $\dim H^0 = d_1 \cdots d_g = 1$  of a principal polarization from chapter 3 in its lowest-dimensional instance.

A worked pass through the identification makes concrete what each abstract map does.

**Example 12.5.3: Abel's Theorem as the Chord-and-Tangent Law**

Take  $E$  a smooth plane cubic in Weierstrass form over  $k$ , with the inflection point at infinity as  $O$ ; the classical group law adds  $P$  and  $Q$  by drawing the line through them, taking its third intersection  $P * Q$  with  $E$ , and reflecting through  $O$  to get  $P \oplus Q$  ([4]). We read this off from  $\text{Jac}(E)$ .

The line  $\ell$  through  $P$  and  $Q$  meets the cubic in the three points  $P$ ,  $Q$ ,  $P * Q$ , and a line is the zero locus of a section of  $\mathcal{O}_E(1) = \mathcal{O}_E(3O)$  (three collinear points cut out by a hyperplane on a cubic), so as divisors

$$P + Q + (P * Q) \sim 3O$$

that is  $P + Q + (P * Q) - 3O \sim 0$  in  $\text{Div}^0(E)$ . The vertical line through  $P * Q$  and  $O$  meets  $E$  in  $P * Q$ ,  $O$ , and  $P \oplus Q$ , giving likewise  $(P * Q) + O + (P \oplus Q) \sim 3O$ , hence  $(P * Q) + (P \oplus Q) - 2O \sim 0$ . Subtracting,

$$(P \oplus Q) - O \sim (P - O) + (Q - O)$$

which is exactly  $\alpha(P \oplus Q) = \alpha(P) + \alpha(Q)$  in  $\text{Pic}^0(E)$ . The chord-and-tangent recipe is therefore nothing but the additivity of  $P \mapsto [\mathcal{O}_E(P - O)]$ : two points sum, in the group law, to the unique point whose class is  $[\mathcal{O}_E(P + Q - 2O)]$ . Abel's theorem (theorem 12.2.2) for  $g = 1$  is the statement that this recipe has a solution and only one, that is that  $\text{Div}^0(E)/\sim = \text{Pic}^0(E)$  is represented bijectively by  $\{P - O : P \in E\}$ ; the surjectivity is Jacobi inversion in dimension one, and the injectivity is that a degree-zero principal divisor  $P - O$  forces  $P = O$  on a genus-one curve. Every step of the  $g = 1$  theory, the group law, isogenies, the Weil pairing, complex multiplication, is developed in full in *Elliptic Curves* [4]; here it is recovered as the  $g = 1$  shadow of the Jacobian.

Genus one is the case where every construction of this chapter collapses onto an object the reader already owns. The Jacobian, a device for manufacturing an abelian variety out of a curve, returns the curve unchanged when the curve was already an abelian variety; the Abel-Jacobi embedding becomes the identity map; the canonical principal polarization becomes the autoduality  $\hat{E} \cong E$ ; the theta divisor becomes a point; and Abel's theorem becomes the chord-and-tangent law. This is the sanity check on the general theory. A construction that failed to reproduce  $\text{Jac}(E) = E$  would be reproducing the wrong group, and the fact that the higher-genus machinery specializes cleanly to the one case computable by elementary means is the evidence that it is the correct generalization of *Elliptic Curves* [4].

**Exercise 12.5.1**

1. Let  $E$  be an elliptic curve with marked point  $O$ , and let  $\alpha: E \rightarrow \text{Jac}(E) = \text{Pic}^0(E)$  be the Abel-Jacobi map of theorem 12.5.1. Compute  $\alpha(O)$ , and describe explicitly the divisor class  $\alpha(P)$  attached to a point  $P \in E$ . Using the definition of  $\alpha$ , verify directly that  $\alpha(P) = 0$  if and only if  $P = O$ .
2. In proposition 12.5.2 the theta divisor of  $E$  is computed. State what  $\Theta \subseteq \text{Jac}(E)$  is for  $g = 1$ , give its degree as a divisor on  $\text{Jac}(E)$ , and compute  $\chi(\mathcal{O}(\Theta))$  and  $\deg \varphi_{\mathcal{O}(\Theta)}$ . Which of these numbers is the reason the polarization is principal?
3. Fix three collinear points  $P, Q, R$  on a plane cubic  $E$  in Weierstrass form, cut out by a line  $\ell$ , with  $O$  the inflection point at infinity. Using only the linear equivalence  $P + Q + R \sim 3O$  from example 12.5.3, show that  $\alpha(P) + \alpha(Q) + \alpha(R) = 0$  in  $\text{Pic}^0(E)$ . Deduce the value of  $P \oplus Q \oplus R$  in the group law of  $E$ .



4. Let  $P \in E$  be a point of order 2 in the group law, that is  $P \oplus P = O$  with  $P \neq O$ . Using the additivity of  $\alpha$  established in theorem 12.5.1, show that the divisor  $2P - 2O$  is principal, and conversely that any  $P \neq O$  with  $2P - 2O$  principal has order 2. Interpret this as the statement  $\alpha(P) \in \text{Jac}(E)[2]$ .
5. Prove that the Abel-Jacobi map  $\alpha$  of theorem 12.5.1 is a homomorphism of abelian varieties, given as known: (i)  $\alpha$  is a morphism of varieties, (ii)  $\alpha(O) = 0$ , and (iii) a morphism of abelian varieties fixing the identity is a homomorphism (corollary 7.2.3). Then show that the group law transported from  $\text{Jac}(E)$  to  $E$  through  $\alpha$  makes  $P \oplus Q$  the unique point  $S$  with  $S - O \sim P + Q - 2O$ , and explain why such an  $S$  exists and is unique on a genus-one curve.
6. The section observes that Torelli is vacuous for  $g = 1$ . Explain, using proposition 12.5.2, why the theta divisor cannot distinguish two elliptic curves  $E$  and  $E'$ , and name the invariant that does. Contrast this with the higher-genus situation in one or two sentences.



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# *Abelian Varieties over Finite Fields*

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Over a finite field the geometry of the previous chapters becomes arithmetic. An abelian variety  $X$  over  $\mathbb{F}_q$  carries a distinguished endomorphism the other base fields do not supply: the Frobenius  $\pi_X$ , raising coordinates to the  $q$ -th power. It is a genuine element of  $\text{End}(X)$ , and almost everything about  $X$  over  $\mathbb{F}_q$  is read from it. Its characteristic polynomial on the Tate module has integer coefficients (the fact chapter 9 deferred to here), and its value at 1 counts the rational points,  $\#X(\mathbb{F}_q) = P_\pi(1)$ .

The depth of the subject is that Frobenius is tightly constrained. The Riemann hypothesis for abelian varieties, a theorem of Weil, confines every eigenvalue of  $\pi_X$  to the circle  $|\alpha| = q^{1/2}$ ; its proof is the Rosati positivity of chapter 11 applied to Frobenius, one inequality doing global work. Tate's isogeny theorem then makes the Tate module a faithful record of the isogeny class, and the Honda-Tate theorem completes the picture, matching isogeny classes of simple abelian varieties over  $\mathbb{F}_q$  with Weil numbers. The chapter closes on the ordinary-supersingular dichotomy, now read off the Newton polygon of Frobenius, and the arithmetic developed here for the  $g = 1$  case throughout *Elliptic Curves* [4] is its constant reference point.

## 13.1 The Frobenius Endomorphism and Its Characteristic Polynomial

Over a finite field an abelian variety carries an endomorphism that no abelian variety over  $\mathbb{C}$  or over a number field possesses: the map raising coordinates to the  $q$ -th power. Almost every arithmetic invariant of  $X/\mathbb{F}_q$  is read off a single monic polynomial of degree  $2g$  attached to it, the characteristic polynomial of Frobenius. This section introduces that endomorphism, defines its characteristic polynomial through the action on a Tate module, and proves the fact on which the rest of the chapter

rests: the coefficients are rational integers, independent of the auxiliary prime  $\ell$  used to define them. The Frobenius endomorphism was already met in chapter 9 as the extreme example of a purely inseparable isogeny (example 9.1.5): iterating the relative Frobenius over  $\mathbb{F}_q$  produced a degree- $q^g$  endomorphism raising coordinates to the  $q$ -th power. Here it is promoted from an example to the organizing object of the arithmetic theory, and named.

### Definition 13.1.1: The Frobenius endomorphism

Let  $X$  be an abelian variety of dimension  $g$  over the finite field  $\mathbb{F}_q$ ,  $q = p^r$ . The *Frobenius endomorphism*  $\pi_X \in \text{End}(X)$  is the  $q$ -power Frobenius: the  $r$ -fold composite of the relative Frobenius, which on the coordinate ring is the  $q$ -th power map and which is an endomorphism of  $X$  because  $X$  is defined over  $\mathbb{F}_q$ , so that  $X^{(q)} = X$  canonically (example 9.1.5). It is an isogeny, purely inseparable, of degree

$$\deg \pi_X = q^g$$

On  $\bar{\mathbb{F}}_q$ -points  $\pi_X$  acts as  $(x_0 : \cdots : x_N) \mapsto (x_0^q : \cdots : x_N^q)$ , and its fixed points are exactly the  $\mathbb{F}_q$ -rational points of  $X$ .

This is not the same object as the Frobenius  $F$  of finite group schemes from chapter 9 (definition 9.4.4): that  $F$  is the relative Frobenius  $X \rightarrow X^{(p)}$  landing in a base-changed variety, paired with the Verschiebung  $V$  under  $VF = [p]$ , and is defined over any field of characteristic  $p$ ; the Frobenius endomorphism  $\pi_X$  is the arithmetic  $q$ -power map, an honest endomorphism of  $X$  that exists precisely because the field of definition is finite. When  $q = p$  the two coincide with  $X^{(p)} = X$ , but in general  $\pi_X$  is the  $r$ -fold composite and lives in  $\text{End}(X)$ , whereas  $F$  does not.

The defining feature, that the  $\mathbb{F}_q$ -points of  $X$  are the fixed points of  $\pi_X$ , is the classical observation that a point of  $\mathbb{P}^N(\bar{\mathbb{F}}_q)$  has coordinates in  $\mathbb{F}_q$  if and only if it is unchanged by raising every coordinate to the  $q$ -th power, since  $\mathbb{F}_q = \{a \in \bar{\mathbb{F}}_q : a^q = a\}$ . This is the hook by which a geometric endomorphism controls an arithmetic count, and it is made precise in proposition 13.1.6.

**13.1.2 Convention: Frobenius on the Tate module** Because  $\pi_X$  is defined over  $\mathbb{F}_q$ , it commutes with the Galois action on torsion points, and its action on the Tate module  $T_\ell X$  coincides with that of the canonical topological generator of  $\text{Gal}(\bar{\mathbb{F}}_q/\mathbb{F}_q)$ , the arithmetic Frobenius  $\text{Frob}_q$ . Throughout, “the action of Frobenius on  $T_\ell X$ ” means the  $\mathbb{Z}_\ell$ -linear map  $\rho_\ell(\pi_X)$  induced by the endomorphism  $\pi_X$ , which is the same operator as the Galois generator.

Since  $\pi_X$  is an endomorphism, it acts  $\mathbb{Z}_\ell$ -linearly on the Tate module  $T_\ell X \cong \mathbb{Z}_\ell^{2g}$  of definition 9.2.2, for any prime  $\ell \neq p$ , and hence  $\mathbb{Q}_\ell$ -linearly on the rational Tate module  $V_\ell X = T_\ell X \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ . A  $2g$ -dimensional linear operator has a characteristic polynomial, and that polynomial is the invariant this section is built around.

**Definition 13.1.3: The characteristic polynomial of Frobenius**

Let  $\ell \neq p$  be a prime and let  $\pi_X$  act on the rational Tate module  $V_\ell X = T_\ell X \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell \cong \mathbb{Q}_\ell^{2g}$  (definition 9.2.2). The *characteristic polynomial of Frobenius* is

$$P_\pi(t) := \det(t \cdot \text{id} - \pi_X|_{V_\ell X}) \in \mathbb{Q}_\ell[t]$$

a monic polynomial of degree  $2g$ . Its roots (in  $\bar{\mathbb{Q}}_\ell$ , or in  $\mathbb{C}$  after a chosen embedding) are the *Frobenius eigenvalues* of  $X$ .

As stated, the definition depends on two choices: the prime  $\ell$  and the identification of  $V_\ell X$  with  $\mathbb{Q}_\ell^{2g}$ . The second is harmless, since the characteristic polynomial of an operator is independent of the basis in which it is computed. The first is not obviously harmless: for each  $\ell$  one obtains a monic degree- $2g$  polynomial with coefficients in the field  $\mathbb{Q}_\ell$ , and a priori these polynomials could differ from prime to prime. The central theorem of this section removes both worries at once, by showing that  $P_\pi$  has rational-integer coefficients that do not depend on  $\ell$ . Only once that is known is it legitimate to speak of  $P_\pi$  as a single polynomial in  $\mathbb{Z}[t]$  attached to  $X/\mathbb{F}_q$ , and to compare its roots across embeddings, as chapter 13 does throughout.

The mechanism is a bridge between two things that look unrelated: the  $\ell$ -adic determinant of an operator, and the degree of an isogeny. The bridge is the observation that evaluating  $P_\pi$  at an integer  $n$  computes the determinant of  $n - \pi_X$  on  $T_\ell X$ , and that this determinant is exactly the degree of the endomorphism  $[n] - \pi_X$ . Degrees of isogenies are honest nonnegative integers computed geometrically, with no reference to  $\ell$ ; so if  $P_\pi(n)$  equals such a degree for every  $n$ , the polynomial is pinned down to be  $\ell$ -independent and integral. The following lemma is the general form of the bridge (it holds for any endomorphism, not only for  $\pi_X$ ), and it is exactly the mechanism whose integrality was forward-deferred from chapter 9.

**Lemma 13.1.4: The degree formula for the characteristic polynomial**

Let  $X$  be an abelian variety of dimension  $g$  over any field  $k$ , let  $h \in \text{End}(X)$ , and fix a prime  $\ell \neq \text{char } k$ . Let  $P_h(t) = \det(t \cdot \text{id} - h|_{V_\ell X}) \in \mathbb{Q}_\ell[t]$ , a monic polynomial of degree  $2g$ . Then for every pair of integers  $a, b$  with  $b \neq 0$ ,

$$b^{2g} P_h\left(\frac{a}{b}\right) = \det(a \cdot \text{id} - bh|_{T_\ell X}) = \deg([a] - [b]h) \quad (13.1)$$

where  $\deg$  denotes the degree of the isogeny  $[a] - [b]h$  (interpreted as 0 when  $[a] - [b]h$  is not an isogeny). In particular  $\deg([a] - [b]h)$  is a nonnegative rational integer, independent of  $\ell$ .

**Proof:**

*Step 1: The determinant identity.* For an operator  $M$  on a free  $\mathbb{Z}_\ell$ -module of rank  $2g$  and scalars  $a, b$ , one has  $\det(a \cdot \text{id} - bM) = b^{2g} \det\left(\frac{a}{b} \cdot \text{id} - M\right) = b^{2g} P_M\left(\frac{a}{b}\right)$ , purely by multilinearity of the determinant in the  $2g$  rows. Applying this to  $M = h$  acting on  $T_\ell X \cong \mathbb{Z}_\ell^{2g}$  gives the first equality of (13.1); the determinant is the same computed on  $T_\ell X$  or on  $V_\ell X = T_\ell X \otimes \mathbb{Q}_\ell$ , since the operator is defined over  $\mathbb{Z}_\ell$ .

*Step 2: The determinant is the degree.* The endomorphism  $\psi := [a] - [b]h \in \text{End}(X)$  acts on  $T_\ell X$  as the operator  $a \cdot \text{id} - bh$ , since  $\ell$ -adic realization  $\text{End}(X) \rightarrow \text{End}_{\mathbb{Z}_\ell}(T_\ell X)$  is a ring

homomorphism (theorem 9.3.2) carrying  $[n]$  to  $n \cdot \text{id}$ . The claim is that for any  $\varphi \in \text{End}(X)$ ,

$$\det(\varphi|V_\ell X) = \deg \varphi$$

with the convention  $\deg \varphi = 0$  when  $\varphi$  is not an isogeny. Both sides are multiplicative in  $\varphi$ : the determinant because it is multiplicative on composites, and the degree by proposition 9.1.4. They agree on  $[n]$ , where  $\det(n \cdot \text{id}|V_\ell X) = n^{2g} = \deg[n]$  (theorem 8.4.5). For a general isogeny  $\varphi$  the identity is the standard fact that the degree of  $\varphi$  equals the  $\ell$ -adic determinant of its action, the  $\ell$ -part of  $\deg \varphi$  being the index  $[T_\ell X : \varphi T_\ell X] = |\det(\varphi|T_\ell X)|_\ell^{-1}$  and the various  $\ell$  assembling by multiplicativity to the full degree (the degree theory of isogenies is section 9.1; the intersection-theoretic degree is cited to *Algebraic Geometry* [13]). If  $\varphi$  is not an isogeny its kernel is infinite, so  $\varphi$  has nontrivial kernel on  $V_\ell X$  and  $\det(\varphi|V_\ell X) = 0 = \deg \varphi$ . Applying this to  $\varphi = \psi = [a] - [b]h$  and combining with Step 1 gives (13.1).

*Step 3: Nonnegativity and  $\ell$ -independence.* The degree of an isogeny is a positive integer, and 0 in the degenerate case, so the right side of (13.1) is a nonnegative rational integer computed geometrically, purely in terms of the map  $[a] - [b]h$  on  $X$ , with no reference to  $\ell$ . Hence  $\deg([a] - [b]h)$  is independent of the prime  $\ell$  chosen to form  $T_\ell X$ , completing the proof.

The lemma is the engine. It says that the  $\ell$ -adic polynomial  $P_h$ , evaluated at any rational number  $a/b$  and cleared of denominators, computes a geometric quantity that has nothing to do with  $\ell$ . Taking  $b = 1$ , the values  $P_h(a) = \deg([a] - h)$  are nonnegative integers, the same for every  $\ell$ ; and a monic polynomial is determined by enough of its values. The next step is to extract from this that the coefficients themselves lie in  $\mathbb{Z}$  and agree across  $\ell$ . Specializing  $h = \pi_X$  is the payload for the arithmetic to come.

#### Theorem 13.1.5: Integrality of the characteristic polynomial

Let  $X$  be an abelian variety of dimension  $g$  over  $\mathbb{F}_q$  and  $\pi_X$  its Frobenius endomorphism. The characteristic polynomial  $P_\pi(t) = \det(t \cdot \text{id} - \pi_X|V_\ell X)$  has coefficients in  $\mathbb{Z}$ , is monic of degree  $2g$ , and is independent of the prime  $\ell \neq p$  used to define it. The same holds with  $\pi_X$  replaced by any  $h \in \text{End}(X)$ , over any field  $k$ : the characteristic polynomial  $P_h \in \mathbb{Q}_\ell[t]$  lies in  $\mathbb{Z}[t]$  and is independent of  $\ell \neq \text{char} k$ .

#### Proof:

Prove the general statement; the Frobenius case is the specialization  $h = \pi_X$ . Write  $P_h(t) = t^{2g} + c_{2g-1}t^{2g-1} + \cdots + c_1t + c_0$  with  $c_i \in \mathbb{Q}_\ell$ ; the claim is that every  $c_i$  is a rational integer independent of  $\ell$ .

*Step 1: The values are  $\ell$ -independent integers.* By lemma 13.1.4 with  $b = 1$ , for every integer  $n$

$$P_h(n) = \det(n \cdot \text{id} - h|T_\ell X) = \deg([n] - h)$$

which is a nonnegative rational integer, and is the same for every  $\ell \neq \text{char} k$  because the right side is the geometric degree of  $[n] - h$ . Thus the function  $n \mapsto P_h(n)$  from  $\mathbb{Z}$  to  $\mathbb{Z}$  takes  $\ell$ -independent integer values.

*Step 2: The coefficients are rational and  $\ell$ -independent.* Since  $P_h$  is monic of degree  $2g$ , it is

determined by its values at any  $2g + 1$  distinct points, and Lagrange interpolation through the  $2g + 1$  points  $(j, P_h(j))$ ,  $j = 0, 1, \dots, 2g$ , expresses each coefficient  $c_i$  as a rational-coefficient combination of the values  $P_h(0), \dots, P_h(2g)$ . By Step 1 those values are rational integers not depending on  $\ell$ , so each  $c_i$  is a fixed rational number, the same for every  $\ell \neq \text{char } k$ . In particular the polynomial  $P_h$  itself is independent of  $\ell$ : two admissible primes give monic degree- $2g$  polynomials agreeing at all integers, hence agreeing as polynomials.

*Step 3: The roots are algebraic integers, so the coefficients are rational integers.* Each root  $\alpha_i$  of  $P_h$  is an eigenvalue of the operator  $h$  acting on  $T_\ell X$ , and  $h$  preserves the lattice  $T_\ell X \cong \mathbb{Z}_\ell^{2g}$  (theorem 9.3.2), being represented in a  $\mathbb{Z}_\ell$ -basis by a matrix in  $M_{2g}(\mathbb{Z}_\ell)$ ; so  $\alpha_i$  is a root of the monic polynomial  $P_h \in \mathbb{Z}_\ell[t]$  and is therefore integral over  $\mathbb{Z}_\ell$ , that is,  $|\alpha_i|_w \leq 1$  at every place  $w$  of  $\mathbb{Q}$  above a prime  $\ell \neq p$ . To see that  $\alpha_i$  is a genuine algebraic integer one more input is needed at the prime  $p$ , and it is the classical fact that the eigenvalues of an endomorphism on the Tate module are algebraic integers, proved in Mumford, *Abelian Varieties*, §19: the  $\alpha_i$  are integral over  $\mathbb{Z}$ , with  $\prod_i \alpha_i = P_h(0) = \deg h$ . Granting this, each coefficient  $c_i$  is  $\pm$  an elementary symmetric function of the algebraic integers  $\alpha_1, \dots, \alpha_{2g}$ , hence itself an algebraic integer; being also rational by Step 2, it lies in  $\mathbb{Z}$ . Therefore  $P_h \in \mathbb{Z}[t]$ , and it is the single  $\ell$ -independent monic polynomial whose values are the degrees  $\deg([n] - h)$ , as we sought to show.

The theorem is the license for everything that follows. It says that although  $P_\pi$  was manufactured  $\ell$ -adically (as the characteristic polynomial of an operator on a  $\mathbb{Q}_\ell$ -vector space), it is in truth a monic integer polynomial intrinsic to  $X/\mathbb{F}_q$ , with no dependence on the auxiliary prime. Its roots may therefore be regarded as a well-defined multiset of algebraic numbers, the same whether viewed  $\ell$ -adically or complex-analytically, and it is legitimate to ask (as chapter 13 does) for their complex absolute values. The degree formula  $P_\pi(n) = \deg([n] - \pi_X)$  that drives the proof is also the computational tool: it reduces the coefficients of  $P_\pi$  to a finite list of isogeny degrees, and in the reverse direction it turns knowledge of  $P_\pi$  into point counts, as proposition 13.1.6 makes explicit.

This result discharges the forward reference left open in chapter 9. There the finite-generation corollary 9.3.3 needed, as an input, that the characteristic polynomial  $P_h(n) = \deg([n] - h)$  of an endomorphism has rational-integer coefficients independent of  $\ell$ , and imported it from “the theorem on the characteristic polynomial of an endomorphism proved in chapter 13”. That theorem is theorem 13.1.5 in its general- $h$  form, whose proof rests only on the degree  $\deg[n] = n^{2g}$  of multiplication (theorem 8.4.5) and the  $\ell$ -adic realization of endomorphisms (theorem 9.3.2), not on the finite-generation corollary; the deferral is therefore non-circular, and it is now settled.

The reason  $\pi_X$  deserves this attention is that its characteristic polynomial computes the number of rational points, the first arithmetic invariant of  $X/\mathbb{F}_q$ . The link is the fixed-point description of  $X(\mathbb{F}_q)$  and the separability of the isogeny  $1 - \pi_X$ .

#### Proposition 13.1.6: Frobenius and the point count

Let  $X$  be an abelian variety of dimension  $g$  over  $\mathbb{F}_q$  with Frobenius endomorphism  $\pi_X$ . Then  $[1] - \pi_X$  is a separable isogeny, its kernel is the group  $X(\mathbb{F}_q)$  of  $\mathbb{F}_q$ -rational points, and

$$\#X(\mathbb{F}_q) = \deg([1] - \pi_X) = P_\pi(1)$$

**Proof:**

*Step 1: The rational points are the fixed points of  $\pi_X$ .* A geometric point  $x \in X(\bar{\mathbb{F}}_q)$  has coordinates in  $\mathbb{F}_q$  if and only if it is fixed by the  $q$ -power map on coordinates, that is, if and only if  $\pi_X(x) = x$ , since  $\mathbb{F}_q = \{a \in \bar{\mathbb{F}}_q : a^q = a\}$  (definition 13.1.1). Thus  $X(\mathbb{F}_q) = \{x \in X(\bar{\mathbb{F}}_q) : (\pi_X - \text{id})(x) = 0\} = \ker([1] - \pi_X)(\bar{\mathbb{F}}_q)$ , the geometric points of the kernel of the endomorphism  $[1] - \pi_X$ .

*Step 2:  $[1] - \pi_X$  is a separable isogeny.* The differential of  $\pi_X$  at the identity is 0:  $\pi_X$  raises coordinates to the  $q$ -th power, and  $d(x^q) = q x^{q-1} dx = 0$  in characteristic  $p$ , so  $\pi_X$  acts as the zero map on the tangent space  $T_0 X$ . Hence  $[1] - \pi_X$  acts on  $T_0 X$  as  $\text{id} - 0 = \text{id}$ , which is invertible, so  $[1] - \pi_X$  is étale at the identity, hence étale (a homomorphism of group varieties has constant rank), hence a separable isogeny. Its kernel is therefore a finite étale group scheme, and its order equals the number of geometric points,  $\deg([1] - \pi_X) = \#\ker([1] - \pi_X)(\bar{\mathbb{F}}_q)$  (definition 9.1.2).

*Step 3: Assemble.* Combining Steps 1 and 2,

$$\#X(\mathbb{F}_q) = \#\ker([1] - \pi_X)(\bar{\mathbb{F}}_q) = \deg([1] - \pi_X)$$

and by the degree formula (13.1) with  $h = \pi_X$ ,  $a = 1$ ,  $b = 1$ , this degree is  $P_\pi(1)$ , completing the proof.

The proposition converts a geometric count into an evaluation of one polynomial. Writing  $P_\pi(t) = \prod_{i=1}^{2g} (t - \alpha_i)$  with the  $\alpha_i$  the Frobenius eigenvalues, the point count becomes  $\#X(\mathbb{F}_q) = \prod_{i=1}^{2g} (1 - \alpha_i)$ , and the same argument applied to the base extension  $X_{\mathbb{F}_{q^m}}$ , whose Frobenius is  $\pi_X^m$  with eigenvalues  $\alpha_i^m$ , gives  $\#X(\mathbb{F}_{q^m}) = \prod_i (1 - \alpha_i^m)$  for every  $m \geq 1$ . These are the numbers whose generating function is the zeta function of  $X$ , and the constraint  $|\alpha_i| = q^{1/2}$  that section 13.2 establishes next is what pins the point counts to their expected size. The whole arithmetic of  $X$  over the tower  $\mathbb{F}_{q^m}$  is thereby encoded in the  $2g$  algebraic numbers  $\alpha_i$ .

The  $g = 1$  case is the one the reader has already met, and it fixes the normalization of  $P_\pi$  against the familiar trace of Frobenius on an elliptic curve.

**Example 13.1.7: Frobenius on an elliptic curve**

Let  $E$  be an elliptic curve over  $\mathbb{F}_q$ , so  $g = 1$  and  $T_\ell E \cong \mathbb{Z}_\ell^2$  (definition 9.2.2). The characteristic polynomial of Frobenius is monic of degree 2,

$$P_\pi(t) = t^2 - a t + q$$

with  $a \in \mathbb{Z}$  the *trace of Frobenius*. The two coefficients are read off by the degree formula. The constant term is  $P_\pi(0) = \det(-\pi_E|V_\ell E) = \deg(-\pi_E) = \deg \pi_E = q$  (definition 13.1.1), and the middle coefficient by evaluating at  $t = 1$ : since  $\#E(\mathbb{F}_q) = P_\pi(1) = 1 - a + q$  (proposition 13.1.6),

$$a = q + 1 - \#E(\mathbb{F}_q)$$

So the trace of Frobenius is the defect of the point count from the naive value  $q + 1$ . This is exactly the quantity that governs the arithmetic of  $E/\mathbb{F}_q$  in *Elliptic Curves* [4], where  $a_E = q + 1 - \#E(\mathbb{F}_q)$  and the two Frobenius eigenvalues  $\alpha, \bar{\alpha}$  are the roots of  $t^2 - at + q$  with  $\alpha\bar{\alpha} = q$ ; the Hasse bound  $|a| \leq 2\sqrt{q}$  there is the  $g = 1$  case of the absolute-value estimate  $|\alpha| = q^{1/2}$  proved in general in section 13.2. For instance, the curve  $y^2 = x^3 + x$  over  $\mathbb{F}_3$  has  $\#E(\mathbb{F}_3) = 4$  (the points

$\infty, (0, 0), (2, 1), (2, 2)$ , so  $a = 3 + 1 - 4 = 0$  and  $P_\pi(t) = t^2 + 3$ , with  $\alpha = \sqrt{-3}$ ,  $|\alpha| = \sqrt{3}$ , the supersingular case.

### Exercise 13.1.1

1. Let  $X/\mathbb{F}_q$  have dimension  $g$  with Frobenius endomorphism  $\pi_X$ . Using definition 13.1.1, compute  $P_\pi(0)$ , the constant term of the characteristic polynomial, as an isogeny degree, and deduce  $\det(\pi_X|V_\ell X)$ . What is  $\prod_{i=1}^{2g} \alpha_i$  in terms of  $q$  and  $g$ , where the  $\alpha_i$  are the Frobenius eigenvalues?
2. Let  $X/\mathbb{F}_q$  be an abelian surface ( $g = 2$ ) whose Frobenius eigenvalues are  $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ , so that  $P_\pi(t) = \prod_{i=1}^4 (t - \alpha_i)$  is monic of degree 4. Using proposition 13.1.6 and the point count of the base extension, write  $\#X(\mathbb{F}_q)$  and  $\#X(\mathbb{F}_{q^2})$  as products in the  $\alpha_i$ , and express both as evaluations of a polynomial built from  $P_\pi$ .
3. Let  $E/\mathbb{F}_5$  be an elliptic curve with  $\#E(\mathbb{F}_5) = 8$ . Following example 13.1.7, determine  $P_\pi(t)$ , the trace of Frobenius  $a$ , and  $\#E(\mathbb{F}_{25})$ . (For the last, use that the Frobenius eigenvalues over  $\mathbb{F}_{25}$  are  $\alpha^2, \bar{\alpha}^2$ .)
4. The proof of theorem 13.1.5 extracts the coefficients of  $P_h$  from finitely many of its values. Carry this out explicitly for  $g = 1$  and  $h = \pi_E$  an elliptic-curve Frobenius: express the two nonleading coefficients of the monic quadratic  $P_\pi(t) = t^2 + c_1 t + c_0$  in terms of the degrees  $\deg([n] - \pi_E) = P_\pi(n)$  for suitable integers  $n$ , and check the result against example 13.1.7.
5. Let  $X/\mathbb{F}_q$  have dimension  $g$ , and let  $h \in \text{End}(X)$  be an endomorphism satisfying  $h^2 = [q]$  in  $\text{End}(X)$  (for instance  $h = \pi_X$  when  $\pi_X^2 = [q]$ ). Using lemma 13.1.4, prove that every Frobenius-style eigenvalue of  $h$  on  $V_\ell X$  has square  $q$ , and conclude that the characteristic polynomial  $P_h$  factors over  $\mathbb{Q}$  into linear factors whose roots satisfy  $\alpha_i^2 = q$ .
6. Prove that the characteristic polynomial  $P_\pi \in \mathbb{Z}[t]$  of definition 13.1.3 is a *monic* polynomial whose value at every integer  $n$  is a nonnegative integer, and that  $P_\pi(n) = 0$  if and only if  $[n] - \pi_X$  fails to be an isogeny, i.e. has infinite kernel. Use lemma 13.1.4 and theorem 13.1.5, and interpret the vanishing geometrically in terms of  $n$  being an eigenvalue of  $\pi_X$ .

## 13.2 The Riemann Hypothesis for Abelian Varieties

The Frobenius endomorphism  $\pi_X$  and its characteristic polynomial  $P_\pi$  (definition 13.1.3) are integer objects:  $P_\pi \in \mathbb{Z}[t]$  is monic of degree  $2g$  and independent of  $\ell$ , and its value at 1 counts the rational points,  $\#X(\mathbb{F}_q) = P_\pi(1)$  (proposition 13.1.6). Knowing the point count therefore amounts to knowing where the roots of  $P_\pi$  sit in the complex plane. This section locates them exactly: every root has absolute value  $q^{1/2}$ . The roots are pinned to a single circle, and the point count is forced into a narrow window around  $q^g$ .

The constraint comes from a source outside the arithmetic of  $\mathbb{F}_q$  altogether. A polarization equips  $\text{End}^0(X)$  with the Rosati involution, whose trace form is positive definite (theorem 11.1.3); Frobenius interacts with that involution through one clean algebraic identity,  $\pi'_X \pi_X = q$ , and positivity does the rest. The pattern is the one the reader has already met for  $g = 1$ : the trace  $a = \pi + \bar{\pi}$  of Frobenius on an elliptic curve is bounded by  $2q^{1/2}$  precisely because  $\pi\bar{\pi} = q$  and  $\pi, \bar{\pi}$  are complex conjugate.



The name is deliberate. For a curve  $C/\mathbb{F}_q$  the zeta function  $Z(C, t)$  has numerator  $P_\pi(t)$  (the characteristic polynomial of Frobenius on  $H^1$ , which for the Jacobian is  $V_\ell$ ), and the statement  $|\alpha| = q^{1/2}$  is the assertion that the zeros of  $Z(C, q^{-s})$  lie on the line  $\operatorname{Re}(s) = \frac{1}{2}$ . It is the function-field analogue of the classical Riemann hypothesis, and for abelian varieties it is a theorem, proved by Weil.

**Theorem 13.2.1: The Riemann Hypothesis for Abelian Varieties**

Let  $X$  be an abelian variety of dimension  $g$  over  $\mathbb{F}_q$ , and let  $P_\pi \in \mathbb{Z}[t]$  be the characteristic polynomial of the Frobenius endomorphism  $\pi_X$  (definition 13.1.3). Every root  $\alpha$  of  $P_\pi$  satisfies

$$|\iota(\alpha)| = q^{1/2}$$

for every embedding  $\iota: \mathbb{Q}(\alpha) \hookrightarrow \mathbb{C}$ . Equivalently, writing  $P_\pi(t) = \prod_{i=1}^{2g} (t - \alpha_i)$  over  $\mathbb{C}$ , the roots come in pairs  $\alpha \leftrightarrow q/\alpha$ , and each root satisfies

$$\alpha \bar{\alpha} = q$$

under every complex embedding.

The two formulations agree once one knows  $P_\pi$  has real (indeed integer) coefficients: the non-real roots then occur in complex-conjugate pairs  $\alpha, \bar{\alpha}$ , and  $|\alpha|^2 = \alpha \bar{\alpha} = q$  says the conjugate partner  $\bar{\alpha}$  equals  $q/\alpha$ , so complex conjugation and the involution  $\alpha \mapsto q/\alpha$  act the same way on the roots. The whole statement reduces to a single identity in  $\operatorname{End}^0(X)$  together with the positivity of the Rosati form.

The engine is the following description of the Rosati adjoint of Frobenius. It is the algebraic incarnation of the relation  $\pi \bar{\pi} = q$  familiar from the elliptic case.

**Proposition 13.2.2: The Rosati adjoint of Frobenius**

Fix a polarization  $\lambda = \varphi_L: X \rightarrow \hat{X}$  (definition 10.3.2), and let  $f \mapsto f' = \lambda^{-1} \circ \hat{f} \circ \lambda$  be the associated Rosati involution on  $\operatorname{End}^0(X)$ . Then the Frobenius endomorphism satisfies

$$\pi'_X = q \pi_X^{-1} \quad \text{in } \operatorname{End}^0(X)$$

equivalently  $\pi'_X \pi_X = \pi_X \pi'_X = q$ .

**Proof:**

An ample line bundle exists on the projective variety  $X$ , so a polarization  $\lambda = \varphi_L$  is available (theorem 10.3.1); fix one. Write  $\pi = \pi_X$  for brevity. The map  $L \mapsto \varphi_L$  (definition 8.3.1) is a homomorphism  $\operatorname{NS}(X) \rightarrow \operatorname{Hom}(X, \hat{X})$ , and it is functorial in the sense that for any endomorphism  $f: X \rightarrow X$  and any line bundle  $L$ ,

$$\varphi_{f^*L} = \hat{f} \circ \varphi_L \circ f$$

where  $\hat{f}: \hat{X} \rightarrow \hat{X}$  is the dual of  $f$ . This pullback-compatibility is a consequence of the theorem of the square (theorem 8.2.1), which governs how  $x \mapsto [t_x^* L \otimes L^{-1}]$  behaves under  $f$ : the translation  $t_{f(x)}$  pulls back along  $f$  to  $t_x$ , so  $\varphi_{f^*L} = \hat{f} \circ \varphi_L \circ f$ .



The one arithmetic input is the action of Frobenius on line bundles. Because  $\pi$  is the identity on the underlying topological space of  $X_{\bar{\mathbb{F}}_q}$  and raises regular functions to the  $q$ -th power, it acts on  $\text{Pic}(X)$  by multiplication by  $q$ : for every line bundle  $L$ ,

$$\pi^* L \cong L^{\otimes q}$$

(see *Algebraic Geometry* [13] for the action of the  $q$ -power Frobenius on the Picard group). Combining the two displays with additivity of  $L \mapsto \varphi_L$  (which sends  $L^{\otimes q}$  to  $q\varphi_L$ ) gives

$$\hat{\pi} \circ \lambda \circ \pi = \varphi_{\pi^* L} = \varphi_{L^{\otimes q}} = q\varphi_L = q\lambda$$

Now  $\lambda: X \rightarrow \hat{X}$  is an isogeny, hence invertible in the isogeny category, and  $\pi' = \lambda^{-1}\hat{\pi}\lambda$  is a genuine element of  $\text{End}^0(X)$ ; multiplying  $\hat{\pi}\lambda\pi = q\lambda$  on the left by  $\lambda^{-1}$ ,

$$\pi' \pi = (\lambda^{-1}\hat{\pi}\lambda)\pi = \lambda^{-1}(\hat{\pi}\lambda\pi) = \lambda^{-1}(q\lambda) = q$$

Since  $q$  is central and  $\pi$  is invertible in  $\text{End}^0(X)$  (it is an isogeny, of degree  $q^g$ ), this reads  $\pi' = q\pi^{-1}$ , whence  $\pi\pi' = \pi'\pi = q$ , completing the proof.

The identity  $\pi\pi' = q$  says Frobenius and its Rosati adjoint are inverse up to the scalar  $q$ . That  $q$  is exactly the modulus-squared the theorem predicts; positivity of the Rosati form converts the algebraic identity into the metric statement  $|\alpha|^2 = q$ . The mechanism is that a positive involution acts as complex conjugation on the commutative subalgebra generated by any element it fixes setwise, and  $\pi$  generates such an algebra because  $\pi' = q\pi^{-1}$  lies in  $\mathbb{Q}[\pi]$  (in general an étale  $\mathbb{Q}$ -algebra, a product of number fields, and a single number field precisely when  $X$  is isotypic, that is, isogenous to a power of a single simple abelian variety).

**Proof of theorem 13.2.1, the Riemann hypothesis:**

Fix a polarization and let  $f \mapsto f'$  be the Rosati involution, positive definite by theorem 11.1.3:  $\text{Tr}(ff') > 0$  for every  $f \neq 0$  in  $\text{End}^0(X)$ , where  $\text{Tr}$  is the trace on  $V_\ell X$ . By proposition 13.2.2,  $\pi' = q\pi^{-1}$ , so  $\pi' \in \mathbb{Q}[\pi]$  and the commutative  $\mathbb{Q}$ -subalgebra  $E := \mathbb{Q}[\pi] \subseteq \text{End}^0(X)$  is stable under the involution.

*Step 1: The Rosati involution is complex conjugation on  $E$ .* The Rosati trace form restricts to a positive-definite symmetric form on the subalgebra  $E$ : for  $x \in E$  nonzero,  $\text{Tr}(xx') > 0$ . Extend scalars to  $\mathbb{R}$  and set  $E_{\mathbb{R}} := E \otimes_{\mathbb{Q}} \mathbb{R}$ , a finite-dimensional commutative  $\mathbb{R}$ -algebra carrying the  $\mathbb{R}$ -linear extension of the involution, with  $\text{Tr}(xx') > 0$  still for every nonzero  $x \in E_{\mathbb{R}}$ . A nondegenerate trace form forces  $E_{\mathbb{R}}$  to be semisimple, so

$$E_{\mathbb{R}} \cong \mathbb{R}^a \times \mathbb{C}^b$$

a product of copies of  $\mathbb{R}$  and  $\mathbb{C}$ , these being the only finite field extensions of  $\mathbb{R}$ . The one point to fix is how the Rosati trace  $\text{Tr}$  (the trace on  $V_\ell X$ ) sees this product, since  $\text{Tr}$  is *not* the field trace of any factor. Write  $E = \mathbb{Q}[\pi] \cong \prod_j K_j$  as a product of number fields (an étale  $\mathbb{Q}$ -algebra: one field  $K_j$  for each distinct irreducible factor of the minimal polynomial  $m_\pi$ ). The Tate module splits accordingly,  $V_\ell X = \bigoplus_j V_j$  with  $V_j$  a module over  $K_j \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$ , and each  $V_j$  is *free* over  $K_j \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$  of some rank  $m_j \geq 1$ ; hence on each factor  $\text{Tr}$  restricts to  $m_j$  times the field trace,  $\text{Tr}|_{K_j} = m_j \text{Tr}_{K_j/\mathbb{Q}}$ . The multiplicities  $m_j$  may differ across the factors, so  $\text{Tr}|_E$  is *not* a single

uniform multiple of  $\text{Tr}_{E/\mathbb{Q}}$ ; what matters is only that on each factor  $\text{Tr}$  is a *positive* multiple of the field trace. Extending scalars to  $\mathbb{R}$ ,  $\text{Tr}$  therefore restricts on each factor of  $E_{\mathbb{R}}$  to a positive multiple  $c > 0$  of that factor's field trace. Positivity of the constant needs no computation: for the idempotent  $e$  cutting out a factor preserved by the involution,  $e' = e$  gives  $\text{Tr}(e) = \text{Tr}(ee') > 0$ . The involution permutes the factors and restricts to each. It cannot exchange two distinct factors: if  $e, e^*$  are the idempotents of factors swapped by the involution, then  $e' = e^*$  gives  $ee' = 0$  with  $e \neq 0$ , so  $\text{Tr}(ee') = 0$ , contradicting positivity. Hence the involution preserves each factor. On an  $\mathbb{R}$ -factor the only algebra involution is the identity, which is positive because there  $\text{Tr}(xx') = cx^2 > 0$  for  $x \neq 0$  (the field trace on  $\mathbb{R}$  is the identity). On a  $\mathbb{C}$ -factor the involution is the identity or complex conjugation; the identity is excluded, since it would give  $\text{Tr}(xx') = c \text{Tr}_{\mathbb{C}/\mathbb{R}}(x^2)$ , negative at  $x = i$  where  $c \text{Tr}_{\mathbb{C}/\mathbb{R}}(i \cdot i) = c \text{Tr}_{\mathbb{C}/\mathbb{R}}(-1) = -2c < 0$ . Thus the involution is complex conjugation on every factor.

Every root  $\alpha$  of  $P_{\pi}$  is a root of the minimal polynomial  $m_{\pi}$  of  $\pi$  over  $\mathbb{Q}$ ; since  $E = \mathbb{Q}[\pi] \cong \mathbb{Q}[t]/(m_{\pi})$ , sending  $t \mapsto \alpha$  descends to a  $\mathbb{Q}$ -algebra homomorphism  $\iota_{\alpha}: E \rightarrow \mathbb{C}$  with  $\iota_{\alpha}(\pi) = \alpha$  precisely because  $m_{\pi}(\alpha) = 0$ . Such a homomorphism factors through the projection of  $E_{\mathbb{R}}$  onto one of its  $\mathbb{R}$ - or  $\mathbb{C}$ -factors; on that factor the involution is complex conjugation, so

$$\iota_{\alpha}(x') = \overline{\iota_{\alpha}(x)} \quad \text{for all } x \in E$$

*Step 2: The modulus.* Apply  $\iota_{\alpha}$  to the identity  $\pi\pi' = q$  of proposition 13.2.2. Since  $\iota_{\alpha}$  is a ring homomorphism and  $\iota_{\alpha}(\pi') = \overline{\iota_{\alpha}(\pi)} = \bar{\alpha}$  by Step 1,

$$\alpha \bar{\alpha} = \iota_{\alpha}(\pi) \iota_{\alpha}(\pi') = \iota_{\alpha}(\pi\pi') = \iota_{\alpha}(q) = q$$

Therefore  $|\alpha|^2 = \alpha \bar{\alpha} = q$ , that is  $|\alpha| = q^{1/2}$ , and this holds for every embedding  $\iota_{\alpha}$  and hence for every root of  $P_{\pi}$ .

*Step 3: The pairing.* From  $\alpha \bar{\alpha} = q$  the conjugate  $\bar{\alpha} = q/\alpha$  is again a root of  $P_{\pi}$ , since  $P_{\pi} \in \mathbb{Z}[t]$  has real coefficients and its roots are stable under complex conjugation (theorem 13.1.5). The involution  $\alpha \mapsto q/\alpha$  thus permutes the roots, pairing each  $\alpha$  with  $q/\alpha = \bar{\alpha}$ ; a real root can only be  $\pm q^{1/2}$ , self-paired. This is the claimed pairing, and every root lies on the circle of radius  $q^{1/2}$ , as we sought to show.

The proof isolates precisely where each ingredient enters. The identity  $\pi\pi' = q$  is pure algebra, valid over any polarization and requiring only that Frobenius pulls a line bundle back to its  $q$ -th power. Positivity of the Rosati form, imported from theorem 11.1.3, is what pins  $\pi$  to a number field on which the involution is honest complex conjugation. Neither ingredient mentions  $\ell$ : the conclusion holds for the  $\ell$ -independent polynomial  $P_{\pi}$ , so the eigenvalues of Frobenius on  $V_{\ell}X$  have absolute value  $q^{1/2}$  for every  $\ell \neq p$  at once.

Locating the roots bounds the number of rational points. Since  $\#X(\mathbb{F}_q) = P_{\pi}(1)$  is the value of a polynomial whose roots all lie on a known circle, the count cannot stray far from its leading term.

**Corollary 13.2.3: The Weil bound on rational points**

Let  $X/\mathbb{F}_q$  be an abelian variety of dimension  $g$ , with  $P_\pi(t) = \prod_{i=1}^{2g} (t - \alpha_i)$ . Then

$$\#X(\mathbb{F}_q) = \prod_{i=1}^{2g} (1 - \alpha_i)$$

and, since  $|\alpha_i| = q^{1/2}$  for every  $i$ ,

$$(q^{1/2} - 1)^{2g} \leq \#X(\mathbb{F}_q) \leq (q^{1/2} + 1)^{2g}$$

In particular  $\#X(\mathbb{F}_q) = q^g + O(q^{g-1/2})$ . For  $g = 1$  this is the Hasse bound

$$|\#E(\mathbb{F}_q) - q - 1| \leq 2q^{1/2}$$

**Proof:**

The first display is  $\#X(\mathbb{F}_q) = \deg(1 - \pi_X) = P_\pi(1)$  (proposition 13.1.6) together with the factorization  $P_\pi(t) = \prod_i (t - \alpha_i)$ , evaluated at  $t = 1$ . The product  $\prod_i (1 - \alpha_i)$  is a nonnegative integer, hence equals its own modulus:

$$\#X(\mathbb{F}_q) = \left| \prod_{i=1}^{2g} (1 - \alpha_i) \right| = \prod_{i=1}^{2g} |1 - \alpha_i|$$

Each factor obeys  $||1| - |\alpha_i|| \leq |1 - \alpha_i| \leq 1 + |\alpha_i|$  by the triangle inequality, and  $|\alpha_i| = q^{1/2}$  by theorem 13.2.1, so

$$q^{1/2} - 1 \leq |1 - \alpha_i| \leq q^{1/2} + 1$$

Taking the product over the  $2g$  factors gives the two-sided bound. The leading term is  $q^g$ : the top-order contribution to  $\prod_i (1 - \alpha_i)$  is  $\prod_i (-\alpha_i) = \prod_i \alpha_i = P_\pi(0) = \deg \pi_X = q^g$ , and every other term is a sum of products of at most  $2g - 1$  of the  $\alpha_i$ , of modulus at most  $\binom{2g}{j} q^{j/2}$  with  $j \leq 2g - 1$ , so  $\#X(\mathbb{F}_q) - q^g = O(q^{g-1/2})$ .

For  $g = 1$  write  $P_\pi(t) = (t - \alpha)(t - \bar{\alpha}) = t^2 - at + q$  with  $a = \alpha + \bar{\alpha} = \text{Tr}(\pi_E) \in \mathbb{Z}$ . Then

$$\#E(\mathbb{F}_q) = P_\pi(1) = 1 - a + q$$

and  $|a| = |\alpha + \bar{\alpha}| \leq |\alpha| + |\bar{\alpha}| = 2q^{1/2}$ , so  $|\#E(\mathbb{F}_q) - q - 1| = |a| \leq 2q^{1/2}$ , completing the proof.

The bound says the point count of an abelian variety over  $\mathbb{F}_q$  is  $q^g$  to within an error of order  $q^{g-1/2}$ : the naive expectation  $q^g$  for a  $g$ -dimensional variety, corrected by a term no larger than the square root of the main term. The width of the window is governed entirely by the radius  $q^{1/2}$  from theorem 13.2.1. For  $g = 1$  the correction is the trace of Frobenius, an integer in  $[-2q^{1/2}, 2q^{1/2}]$ , and the bound is sharp: every such value of  $a$  occurs for some elliptic curve over  $\mathbb{F}_q$ .

**Example 13.2.4: Frobenius eigenvalues of an elliptic curve**

Let  $E/\mathbb{F}_q$  be an elliptic curve, so  $g = 1$  and  $\pi_E$  is the  $q$ -power Frobenius. The characteristic polynomial of  $\pi_E$  on  $V_\ell E \cong \mathbb{Q}_\ell^2$  is

$$P_\pi(t) = t^2 - at + q, \quad a = \text{Tr}(\pi_E) = q + 1 - \#E(\mathbb{F}_q) \in \mathbb{Z}$$

the second equality from  $\#E(\mathbb{F}_q) = P_\pi(1) = 1 - a + q$ . Its two roots are

$$\alpha, \bar{\alpha} = \frac{a \pm \sqrt{a^2 - 4q}}{2}$$

By theorem 13.2.1 they have absolute value  $q^{1/2}$ , so  $\alpha\bar{\alpha} = q$  (consistent with the constant term of  $P_\pi$ ) and  $\alpha + \bar{\alpha} = a$  (the linear coefficient). Two cases arise. When  $a^2 < 4q$  the discriminant is negative and  $\alpha, \bar{\alpha}$  are genuine complex conjugates on the circle  $|z| = q^{1/2}$ ; this is the generic situation, of non-real Frobenius eigenvalues. When  $a^2 = 4q$  (possible only if  $q$  is a square) both roots equal  $q^{1/2}$  in absolute value and are real,  $\alpha = \bar{\alpha} = \pm q^{1/2}$ . In every case  $a^2 \leq 4q$ , which is exactly the Hasse bound  $|a| \leq 2q^{1/2}$  of *Elliptic Curves* [4]. The identity  $\pi_E \bar{\pi}_E = q$  underlying the proof is the same relation that governs Frobenius on an elliptic curve there:  $\pi_E$  and its dual isogeny  $\bar{\pi}_E$  multiply to  $[q]$ .

**Exercise 13.2.1**

1. Let  $E/\mathbb{F}_9$  be an elliptic curve with  $\#E(\mathbb{F}_9) = 16$ . Using the description of the Frobenius characteristic polynomial in example 13.2.4, compute the trace  $a = \text{Tr}(\pi_E)$ , write down  $P_\pi(t)$ , and find the two Frobenius eigenvalues  $\alpha, \bar{\alpha}$ . Verify directly that  $|\alpha| = q^{1/2}$  as theorem 13.2.1 requires, and determine whether the eigenvalues are real or genuinely complex.
2. For an abelian variety  $X/\mathbb{F}_q$  of dimension  $g$ , use corollary 13.2.3 to write  $\#X(\mathbb{F}_q)$  as a value of  $P_\pi$  and to give the two-sided bound. For  $g = 2$  and  $q = 4$ , compute the numerical interval  $[(q^{1/2} - 1)^{2g}, (q^{1/2} + 1)^{2g}]$  in which  $\#X(\mathbb{F}_4)$  must lie, and compare its width to the main term  $q^g$ .
3. The identity  $\pi'_X \pi_X = q$  of proposition 13.2.2 forces the roots of  $P_\pi$  to pair as  $\alpha \leftrightarrow q/\alpha$ . Show that this pairing is compatible with the constant term: prove that  $P_\pi(0) = q^g$ , and explain how this follows both from the root pairing and from  $\deg \pi_X = q^g$ .
4. Two abelian varieties  $X, Y$  over  $\mathbb{F}_q$  are isogenous if and only if  $P_{\pi_X} = P_{\pi_Y}$ . Granting this, prove that if  $X$  and  $Y$  are isogenous of dimension  $g$ , then  $\#X(\mathbb{F}_{q^n}) = \#Y(\mathbb{F}_{q^n})$  for every  $n \geq 1$ . (Use that the Frobenius eigenvalues of  $X$  over  $\mathbb{F}_{q^n}$  are the  $n$ -th powers of those over  $\mathbb{F}_q$ , together with corollary 13.2.3.)
5. Prove the following case of theorem 13.2.1 directly, without invoking the general theorem. Let  $\lambda$  be a polarization on  $X$  with Rosati involution  $f \mapsto f'$ , and suppose that  $E := \mathbb{Q}[\pi_X]$  is a field on which the involution acts as complex conjugation. Show that for every embedding  $\iota: E \hookrightarrow \mathbb{C}$  one has  $|\iota(\pi_X)|^2 = q$ . Identify precisely which two facts from the section your argument uses.

### 13.3 Tate's Isogeny Theorem

Over  $\mathbb{F}_q$  the Galois action on the Tate module is governed by a single element, the Frobenius, and the task of this section is to show that the Tate module together with its Galois action is a *faithful* invariant of the isogeny class. Tate's theorem says more than that homomorphisms are detected on the Tate module; it says every Galois-equivariant map of Tate modules comes from an honest homomorphism, after tensoring with  $\mathbb{Z}_\ell$ . The forward half of this statement, that a homomorphism is recovered from its action on the Tate module, is the  $\ell$ -adic faithfulness of chapter 9 applied to Hom; it is proved here in full. The converse, the surjectivity, is Tate's theorem proper, and rests on a finiteness input specific to finite fields; its proof is sketched and cited.

Throughout,  $X$  and  $Y$  are abelian varieties over  $\mathbb{F}_q$ ,  $p = \text{char}\mathbb{F}_q$ , and  $\ell \neq p$  is a prime. Write  $\Gamma = \text{Gal}(\bar{\mathbb{F}}_q/\mathbb{F}_q)$  for the absolute Galois group; it acts  $\mathbb{Z}_\ell$ -linearly and continuously on the Tate modules  $T_\ell X$  and  $T_\ell Y$  through the  $\ell$ -adic representation (definition 9.3.4). A homomorphism  $f: X \rightarrow Y$  of abelian varieties is defined over  $\mathbb{F}_q$ , hence commutes with the Galois action, and so the map  $T_\ell f: T_\ell X \rightarrow T_\ell Y$  it induces on Tate modules is  $\Gamma$ -equivariant. Tensoring the resulting map  $\text{Hom}(X, Y) \rightarrow \text{Hom}_\Gamma(T_\ell X, T_\ell Y)$  with  $\mathbb{Z}_\ell$  gives the comparison the theorem concerns.

#### Theorem 13.3.1: Tate's Isogeny Theorem

Let  $X$  and  $Y$  be abelian varieties over the finite field  $\mathbb{F}_q$ , and let  $\ell$  be a prime different from  $p = \text{char}\mathbb{F}_q$ . The natural map of  $\mathbb{Z}_\ell$ -modules

$$\text{Hom}(X, Y) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \longrightarrow \text{Hom}_\Gamma(T_\ell X, T_\ell Y), \quad \Gamma = \text{Gal}(\bar{\mathbb{F}}_q/\mathbb{F}_q)$$

sending  $f \otimes 1$  to the induced  $\Gamma$ -equivariant map  $T_\ell f$ , is an isomorphism.

The theorem has two halves of very different character. Injectivity is a statement about a single homomorphism: a nonzero  $f$  acts nontrivially on the Tate module, so no information is lost in passing to  $T_\ell$ . This half holds over any field, not only over  $\mathbb{F}_q$ , and is the Hom-version of the faithfulness already proved for End. Surjectivity is the harder half and is genuinely arithmetic: it asserts that the map has enough homomorphisms in its source to account for every Galois-equivariant map of Tate modules, and this is false over a general field. Over  $\mathbb{F}_q$  it is true, and true because the isogeny classes over a finite field are severely constrained. The injectivity is proved first, in full.

#### **Proof of injectivity:**

The source and target are functorial in the pair  $(X, Y)$ , and the map is compatible with the identification  $\text{Hom}(X, Y) \subseteq \text{End}(X \times Y)$  that sends  $f$  to the endomorphism  $(x, y) \mapsto (0, f(x))$  of the product. Concretely, decompose

$$\text{End}(X \times Y) \cong \text{End}(X) \oplus \text{Hom}(X, Y) \oplus \text{Hom}(Y, X) \oplus \text{End}(Y)$$

by projecting an endomorphism of  $X \times Y$  onto its four matrix blocks; the summand  $\text{Hom}(X, Y)$  is the image of the corner embedding just described. Correspondingly  $T_\ell(X \times Y) \cong T_\ell X \oplus T_\ell Y$ , and  $\text{End}_{\mathbb{Z}_\ell}(T_\ell(X \times Y))$  decomposes into the four blocks  $\text{Hom}_{\mathbb{Z}_\ell}(T_\ell X, T_\ell Y)$  and its companions, compatibly with the summand decomposition of  $\text{End}(X \times Y)$ . Under these identifications the map of the theorem for  $(X, Y)$  is the restriction to one block of the map

$$\text{End}(X \times Y) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \longrightarrow \text{End}_{\mathbb{Z}_\ell}(T_\ell(X \times Y))$$

of theorem 9.3.2. That map is injective, being an instance of faithfulness on the Tate module. A block of an injective block-diagonal comparison is injective, so  $\mathrm{Hom}(X, Y) \otimes \mathbb{Z}_\ell \rightarrow \mathrm{Hom}_{\mathbb{Z}_\ell}(T_\ell X, T_\ell Y)$  is injective; a fortiori its corestriction to the  $\Gamma$ -equivariant maps is injective, as we sought to show.

The mechanism of the injective half is the density of torsion. By theorem 9.3.2, an endomorphism killing  $T_\ell X$  vanishes on all  $\ell$ -power torsion, and the  $\ell$ -power torsion is Zariski-dense in  $X_{\overline{\mathbb{F}}_q}$ , so the endomorphism is zero. A homomorphism  $f: X \rightarrow Y$  is folded into the product  $X \times Y$  and the same argument applies. Nothing in this half uses that the ground field is finite: over any field, a homomorphism of abelian varieties is faithfully recorded by its action on  $T_\ell$  for any  $\ell \neq \mathrm{char} k$ . The finiteness of the ground field enters only in the reverse direction.

**13.3.2 Key Ideas: Tate's surjectivity** The surjectivity of the map in theorem 13.3.1 is Tate's theorem, and it does not hold over an arbitrary field. Its proof rests on two inputs available over  $\mathbb{F}_q$ .

*The finiteness input.* Over  $\mathbb{F}_q$  the abelian varieties isogenous to a fixed  $X$ , with a bounded degree of isogeny, form a finite set: an abelian variety over a finite field has only finitely many isomorphism classes of abelian varieties  $\ell^n$ -isogenous to it, because there are only finitely many abelian varieties of a given dimension over  $\mathbb{F}_q$  with polarization of bounded degree. This is the finiteness that replaces the transcendental input available over  $\mathbb{C}$ , and it is special to finite base fields.

*The semisimplicity and the counting argument.* The Galois action  $\rho_\ell$  on  $V_\ell X = T_\ell X \otimes \mathbb{Q}_\ell$  is semisimple, and Tate's structural theorem identifies  $\mathrm{End}^0(X) \otimes \mathbb{Q}_\ell$  with the commutant of the Galois action,  $\mathrm{End}_{\mathbb{Q}_\ell[\Gamma]}(V_\ell X)$ : the endomorphism algebra is exactly what commutes with Frobenius. Given a  $\Gamma$ -equivariant map  $\varphi: T_\ell X \rightarrow T_\ell Y$ , one uses the finiteness to produce, from the graph of  $\varphi$  inside  $T_\ell(X \times Y)$ , a sequence of abelian subvarieties whose stabilization forces  $\varphi$  into the image of  $\mathrm{Hom}(X, Y) \otimes \mathbb{Z}_\ell$ . A dimension count then matches the rank of the commutant against the rank of  $\mathrm{End}^0(X \times Y)$ , forcing the injection of the theorem to be a bijection.

The full argument for  $g = 1$  is carried out in *Elliptic Curves* [4]; the general case is Tate's theorem, and its foundations on the continuous Galois action are those of *Galois Cohomology* [6].

The surjectivity is deferred rather than proved: its finiteness input is a theorem about abelian varieties over finite fields whose own proof rests on the theory of heights and reduction that this book does not develop, and the  $g = 1$  case is treated in full in *Elliptic Curves* [4]. What is proved here in full, the injective half, already carries the payoff most often used: a homomorphism is determined by its action on the Tate module.

The immediate consequence is a criterion for isogeny in terms of Frobenius. Recall that the Galois group  $\Gamma = \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$  is topologically generated by the  $q$ -power Frobenius, and that the geometric Frobenius acts on  $T_\ell X$  as the Tate-module incarnation of the Frobenius endomorphism  $\pi_X$  (definition 13.1.1). Its characteristic polynomial on  $V_\ell X$  is  $P_{\pi_X}$  (definition 13.1.3), an element of  $\mathbb{Z}[t]$  independent of  $\ell$  (theorem 13.1.5). Two abelian varieties over  $\mathbb{F}_q$  carry the same Frobenius data exactly when these polynomials agree.

**Corollary 13.3.3: Isogeny and the Frobenius Characteristic Polynomial**

Let  $X$  and  $Y$  be abelian varieties over  $\mathbb{F}_q$ . Then  $X$  and  $Y$  are isogenous over  $\mathbb{F}_q$  if and only if their Frobenius characteristic polynomials coincide:

$$X \sim Y \iff P_{\pi_X} = P_{\pi_Y}$$

Equivalently,  $X$  and  $Y$  are isogenous over  $\mathbb{F}_q$  if and only if the rational Tate modules  $V_\ell X$  and  $V_\ell Y$  are isomorphic as  $\Gamma$ -modules for one (equivalently every) prime  $\ell \neq p$ .

**Proof:**

*Forward direction.* Suppose there is an isogeny  $f: X \rightarrow Y$  over  $\mathbb{F}_q$ . Then  $f$  is defined over  $\mathbb{F}_q$ , hence  $\Gamma$ -equivariant, and it induces a  $\mathbb{Z}_\ell$ -linear  $\Gamma$ -equivariant map  $T_\ell f: T_\ell X \rightarrow T_\ell Y$  that becomes an isomorphism  $V_\ell f: V_\ell X \rightarrow V_\ell Y$  after tensoring with  $\mathbb{Q}_\ell$  (chapter 9). Because  $f$  is defined over  $\mathbb{F}_q$  it commutes with the geometric Frobenius, so  $V_\ell f$  intertwines the action of  $\pi_X$  on  $V_\ell X$  with that of  $\pi_Y$  on  $V_\ell Y$ . Conjugate operators have equal characteristic polynomials, so

$$P_{\pi_X}(t) = \det(t - \pi_X|V_\ell X) = \det(t - \pi_Y|V_\ell Y) = P_{\pi_Y}(t)$$

and  $P_{\pi_X} = P_{\pi_Y}$  as claimed. This direction is unconditional and uses only the functoriality of the Tate module.

*Reverse direction.* Suppose  $P_{\pi_X} = P_{\pi_Y}$ . The rational Tate module  $V_\ell X$  is a  $\mathbb{Q}_\ell$ -vector space of dimension  $2g$  on which  $\Gamma$  acts through the single operator  $\pi_X$  (up to the dense image of Frobenius), and likewise for  $V_\ell Y$ . By the semisimplicity of the Frobenius action (itself part of Tate's package, deferred alongside the surjectivity), a  $\mathbb{Q}_\ell[\pi]$ -module is determined up to isomorphism by the characteristic polynomial of  $\pi$ ; equality  $P_{\pi_X} = P_{\pi_Y}$  therefore yields a  $\Gamma$ -equivariant isomorphism  $V_\ell X \cong V_\ell Y$ . Tate's theorem (theorem 13.3.1), whose surjective half supplies the homomorphisms, then produces a  $\mathbb{Q}_\ell$ -point of  $\text{Hom}(X, Y) \otimes \mathbb{Q}_\ell$  inducing this isomorphism; clearing denominators gives an isogeny  $X \rightarrow Y$  defined over  $\mathbb{F}_q$ . Hence  $X \sim Y$ , completing the proof.

The forward direction is elementary and unconditional, and it is what one uses in practice to prove two abelian varieties are *not* isogenous: compute the two Frobenius polynomials and compare. The reverse direction uses the surjective half of Tate's theorem, and it is the reason the theorem is stated as an isomorphism rather than an injection. The equivalence turns the classification of abelian varieties over  $\mathbb{F}_q$  up to isogeny into a question about characteristic polynomials of Frobenius, a purely arithmetic object, and this reframing sets up the Honda-Tate theorem of section 13.4, which describes exactly which polynomials arise.

The  $g = 1$  case is where the reader has met the criterion before.

**Example 13.3.4: Isogenous Elliptic Curves over a Finite Field**

Let  $E$  and  $E'$  be elliptic curves over  $\mathbb{F}_q$ , so  $g = 1$ . The Frobenius characteristic polynomial of  $E$  is

$$P_{\pi_E}(t) = t^2 - at + q, \quad a = a_E := q + 1 - \#E(\mathbb{F}_q)$$

a monic integer quadratic of degree  $2g = 2$ ; the integer  $a$  is the *trace of Frobenius*, and the point count is recovered as  $\#E(\mathbb{F}_q) = P_{\pi_E}(1) = q + 1 - a$  (proposition 13.1.6). Because the



polynomial has degree two with fixed constant term  $q$ , it is determined by the single integer  $a$ . Thus corollary 13.3.3 specializes to the classical statement:

$$E \sim E' \iff P_{\pi_E} = P_{\pi_{E'}} \iff a_E = a_{E'} \iff \#E(\mathbb{F}_q) = \#E'(\mathbb{F}_q)$$

Two elliptic curves over  $\mathbb{F}_q$  are isogenous over  $\mathbb{F}_q$  if and only if they have the same number of  $\mathbb{F}_q$ -rational points, if and only if they have the same trace of Frobenius. This is the isogeny criterion of *Elliptic Curves* [4], where it is proved directly from Tate's theorem for  $g = 1$ ; the Weil bound (corollary 13.2.3) confines  $a$  to the Hasse interval  $|a| \leq 2\sqrt{q}$ , so the isogeny classes over  $\mathbb{F}_q$  are indexed by the finitely many integers in that range that arise as a trace.

Concretely, take  $E: y^2 = x^3 + 1$  over  $\mathbb{F}_5$ . Since  $5 \equiv 2 \pmod{3}$  the cube map is a bijection on  $\mathbb{F}_5$ , and evaluating  $x^3 + 1$  over  $x = 0, \dots, 4$  gives the values 1, 2, 4, 3, 0; of these 1, 4, 0 are squares, contributing  $2 + 2 + 1 = 5$  affine points, so with the point at infinity  $\#E(\mathbb{F}_5) = 6$  and  $a_E = 5 + 1 - 6 = 0$ . The Frobenius polynomial is  $P_{\pi_E}(t) = t^2 + 5$ , whose roots  $\pm\sqrt{-5} = \pm\sqrt{5}i$  have absolute value  $\sqrt{5}$ , in accordance with the Weil bound. The quadratic twist  $E'$  of  $E$  by a non-square in  $\mathbb{F}_5^\times$  has trace  $a_{E'} = -a_E = 0$ , hence again  $\#E'(\mathbb{F}_5) = 6$ ; by the criterion  $E$  and  $E'$  are isogenous over  $\mathbb{F}_5$ , yet the nontrivial quadratic twist is not isomorphic to  $E$  over  $\mathbb{F}_5$ .

### Exercise 13.3.1

1. Let  $E: y^2 = x^3 + x$  over  $\mathbb{F}_5$ . Count  $\#E(\mathbb{F}_5)$  directly, then write down the Frobenius characteristic polynomial  $P_{\pi_E}$  and the trace of Frobenius  $a_E$ . Verify that the roots of  $P_{\pi_E}$  have absolute value  $\sqrt{5}$ , in accordance with the Weil bound (corollary 13.2.3).
2. Fix an abelian variety  $X$  over  $\mathbb{F}_q$  and a prime  $\ell \neq p$ . Using theorem 13.3.1 for the pair  $(X, X)$ , describe  $\text{End}(X) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$  as a subring of  $\text{End}_{\mathbb{Z}_\ell}(T_\ell X)$ , and state precisely which endomorphisms of  $T_\ell X$  lie in the image. What does the injective half of the theorem say about the  $\mathbb{Z}_\ell$ -rank of  $\text{End}(X)$  relative to  $(2g)^2$ ?
3. Let  $E$  and  $E'$  be elliptic curves over  $\mathbb{F}_9$  with  $\#E(\mathbb{F}_9) = 4$  and  $\#E'(\mathbb{F}_9) = 16$ . Decide whether  $E$  and  $E'$  are isogenous over  $\mathbb{F}_9$ , citing the criterion you use. For each curve, compute the trace of Frobenius and the roots of the Frobenius polynomial, and confirm both lie in the Hasse interval.
4. Prove the injectivity of the map in theorem 13.3.1: the natural map  $\text{Hom}(X, Y) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \rightarrow \text{Hom}_\Gamma(T_\ell X, T_\ell Y)$  is injective. You may use theorem 9.3.2 (faithfulness on the Tate module for endomorphisms) as a black box. Then explain in one sentence why this half does not use finiteness of the ground field.
5. Prove the forward direction of corollary 13.3.3: if  $X$  and  $Y$  are isogenous over  $\mathbb{F}_q$ , then  $P_{\pi_X} = P_{\pi_Y}$ . Then explain why this direction, unlike the reverse one, is unconditional, and describe how it is used in practice to show two abelian varieties are *not* isogenous.
6. The reverse direction of corollary 13.3.3 is stated in terms of the rational Tate module: equal Frobenius polynomials produce a  $\Gamma$ -equivariant isomorphism  $V_\ell X \cong V_\ell Y$ , and Tate's theorem then produces an isogeny. Identify precisely the two deferred inputs of box 13.3.2 that this step relies on, and explain why each is genuinely needed, that is, why the reverse direction fails over a general field where these inputs are unavailable.



## 13.4 The Honda-Tate Theorem

The isogeny theorem (corollary 13.3.3) and Poincaré reducibility (theorem 11.3.3) reduce the isogeny category over  $\mathbb{F}_q$  to its simple objects; the task of this section is to describe those simple objects intrinsically. For a simple  $X/\mathbb{F}_q$  the Frobenius  $\pi_X$  generates a field  $\mathbb{Q}(\pi_X) \subseteq \text{End}^0(X)$ , the char poly  $P_{\pi_X}$  is a power of the minimal polynomial of  $\pi_X$  over  $\mathbb{Q}$ , and the Weil bound (theorem 13.2.1) constrains  $\pi_X$  sharply. The Honda-Tate theorem says these constraints are the whole story: the simple isogeny classes over  $\mathbb{F}_q$  are classified, without residue, by a single algebraic invariant, the conjugacy class of the Frobenius eigenvalue.

The invariant is an algebraic number of a very restricted kind, isolated by the Weil bound.

### Definition 13.4.1: Weil $q$ -number

Let  $q$  be a power of a prime  $p$ . A *Weil  $q$ -number* is an algebraic integer  $\pi$  such that

$$|\iota(\pi)| = q^{1/2}$$

for every embedding  $\iota: \mathbb{Q}(\pi) \hookrightarrow \mathbb{C}$ . Two Weil  $q$ -numbers  $\pi, \pi'$  are *conjugate* if they have the same minimal polynomial over  $\mathbb{Q}$ , equivalently if some field isomorphism  $\mathbb{Q}(\pi) \rightarrow \mathbb{Q}(\pi')$  carries  $\pi$  to  $\pi'$ .

The condition packages the Weil “Riemann hypothesis” as a property of a single number rather than of a family of eigenvalues:  $\pi$  is a Weil  $q$ -number precisely when all of its Galois conjugates have complex absolute value  $q^{1/2}$ . Conjugacy is exactly the equivalence a minimal polynomial detects, so a conjugacy class of Weil  $q$ -numbers is the same datum as an irreducible monic polynomial in  $\mathbb{Z}[t]$  all of whose complex roots have absolute value  $q^{1/2}$ . The Frobenius endomorphism of a simple abelian variety supplies such a number: by theorem 13.1.5 the char poly  $P_{\pi_X}$  lies in  $\mathbb{Z}[t]$ , so  $\pi_X$  is an algebraic integer, and by theorem 13.2.1 each of its conjugates has absolute value  $q^{1/2}$ . The definition is engineered to name exactly the numbers that can arise this way.

### Example 13.4.2: Weil Numbers for $q = p$

Take  $q = p$  prime. The rational integer  $\pi = p$  is not a Weil  $p$ -number:  $|\iota(p)| = p \neq p^{1/2}$ . Nor is  $\pi = 1$ . A quadratic Weil  $p$ -number is a root of  $t^2 - at + p$  with  $a \in \mathbb{Z}$  and both roots of absolute value  $p^{1/2}$ ; writing the roots as  $\alpha, \bar{\alpha}$  with  $\alpha\bar{\alpha} = p$  and  $\alpha + \bar{\alpha} = a$ , the roots are complex conjugates (rather than real) exactly when the discriminant  $a^2 - 4p$  is negative, that is when  $|a| < 2\sqrt{p}$ , and then  $|\alpha|^2 = \alpha\bar{\alpha} = p$  as required. So the conjugacy classes of imaginary quadratic Weil  $p$ -numbers are indexed exactly by the integers  $a$  with  $|a| < 2\sqrt{p}$ ; the boundary values  $a = \pm 2\sqrt{p}$  are excluded because  $p$  is not a square, and no real  $a$  with  $|a| > 2\sqrt{p}$  qualifies, its roots then having absolute values  $\neq \sqrt{p}$ . For  $p \geq 5$  these integers  $a$  are precisely the traces of Frobenius on elliptic curves over  $\mathbb{F}_p$ : the only in-range multiple of  $p$  is  $a = 0$ , the supersingular trace, which is realized, so every  $a$  with  $|a| < 2\sqrt{p}$  occurs. At  $p = 2$  and  $p = 3$  the equivalence fails at the interior supersingular values,  $a = \pm 2$  for  $p = 2$  and  $a = \pm 3$  for  $p = 3$ : these are genuine imaginary quadratic Weil  $p$ -numbers, yet by the Deuring-Waterhouse classification the only supersingular trace realized over the prime field is  $a = 0$ , so they are not elliptic-curve traces. The coincidence for  $p \geq 5$ , and its failure at  $p = 2, 3$ , is what example 13.4.5 explains.

With the invariant named, the classification can be stated. The assignment  $X \mapsto \pi_X$  sends a simple abelian variety to its Frobenius endomorphism, an element of the field  $\mathbb{Q}(\pi_X)$ ; the theorem

asserts that this assignment is a bijection between geometry and arithmetic, with no isogeny class unaccounted for and no Weil number unrealized.

**Theorem 13.4.3: Honda-Tate**

Let  $q$  be a power of a prime. The map

$$X \mapsto \pi_X$$

sending a simple abelian variety  $X$  over  $\mathbb{F}_q$  to its Frobenius endomorphism induces a bijection

$$\left\{ \begin{array}{l} \text{isogeny classes of simple} \\ \text{abelian varieties over } \mathbb{F}_q \end{array} \right\} \xrightarrow{\sim} \left\{ \begin{array}{l} \text{conjugacy classes of} \\ \text{Weil } q\text{-numbers} \end{array} \right\}$$

The bijection has two halves of very different character. Injectivity, together with the fact that every simple  $X$  does produce a Weil  $q$ -number, is Tate's contribution and rests on the isogeny theorem already in hand; it is proved below, with one step drawn from Tate's computation of the endomorphism algebra. Surjectivity, that every Weil  $q$ -number is the Frobenius of some simple abelian variety, is Honda's contribution and is the harder half: it constructs an abelian variety with prescribed Frobenius, and the construction runs through the theory of complex multiplication and the reduction of CM abelian varieties, which sit beyond this book's scope. That half is stated and sketched, with its full proof cited to class field theory.

The injectivity half is the following.

**Proposition 13.4.4: Injectivity of the Frobenius invariant**

The map  $X \mapsto \pi_X$  of theorem 13.4.3 is well defined and injective:

1. for every simple abelian variety  $X/\mathbb{F}_q$ , the Frobenius  $\pi_X$  is a Weil  $q$ -number, and its conjugacy class depends only on the isogeny class of  $X$ ;
2. if  $X$  and  $Y$  are simple abelian varieties over  $\mathbb{F}_q$  with  $\pi_X$  conjugate to  $\pi_Y$ , then  $X$  is isogenous to  $Y$ .

**Proof:**

We prove the two assertions in turn.

*Assertion (1):*  $\pi_X$  is a Weil  $q$ -number depending only on the isogeny class. By theorem 13.1.5 the characteristic polynomial  $P_{\pi_X}(t) = \det(t - \pi_X | V_\ell X)$  lies in  $\mathbb{Z}[t]$  and is monic; hence  $\pi_X$ , being a root of a monic integer polynomial, is an algebraic integer. The minimal polynomial  $m_\pi$  of  $\pi_X$  over  $\mathbb{Q}$  divides  $P_{\pi_X}$ , so every root of  $m_\pi$  is a root of  $P_{\pi_X}$ , and by theorem 13.2.1 every root  $\alpha$  of  $P_{\pi_X}$  satisfies  $|\alpha| = q^{1/2}$  under every complex embedding. The roots of  $m_\pi$  are exactly the images  $\iota(\pi_X)$  as  $\iota$  ranges over the embeddings  $\mathbb{Q}(\pi_X) \hookrightarrow \mathbb{C}$ , so  $|\iota(\pi_X)| = q^{1/2}$  for every  $\iota$ . Thus  $\pi_X$  is a Weil  $q$ -number.

To see that its conjugacy class is an isogeny invariant, let  $u: X \rightarrow Y$  be an isogeny of abelian varieties over  $\mathbb{F}_q$ . Frobenius commutes with every morphism defined over  $\mathbb{F}_q$ : since  $u$  is  $\mathbb{F}_q$ -rational,  $u \circ \pi_X = \pi_Y \circ u$ . Tensoring the induced map  $V_\ell(u): V_\ell X \rightarrow V_\ell Y$  with  $\mathbb{Q}_\ell$  gives an isomorphism intertwining the actions of  $\pi_X$  and  $\pi_Y$ , so  $\pi_X$  and  $\pi_Y$  have the same characteristic polynomial on the Tate module, whence the same minimal polynomial over  $\mathbb{Q}$ . Therefore  $\pi_X$

and  $\pi_Y$  are conjugate, and the conjugacy class of  $\pi_X$  depends only on the isogeny class of  $X$ .

*Assertion (2): conjugate Frobenii force an isogeny.* Suppose  $X$  and  $Y$  are simple over  $\mathbb{F}_q$  with  $\pi_X$  conjugate to  $\pi_Y$ , so  $m_{\pi_X} = m_{\pi_Y}$  as polynomials in  $\mathbb{Q}[t]$ . It suffices to show  $P_{\pi_X} = P_{\pi_Y}$ , for then corollary 13.3.3 gives  $X \sim Y$  directly.

Because  $X$  is simple,  $\text{End}^0(X)$  is a division algebra (corollary 11.3.7), and the subfield  $\mathbb{Q}(\pi_X) \subseteq \text{End}^0(X)$  is a field of degree  $\deg m_{\pi_X}$  over  $\mathbb{Q}$ . The characteristic polynomial  $P_{\pi_X}$  has degree  $2g_X$ , where  $g_X = \dim X$ , and is a power of the minimal polynomial:

$$P_{\pi_X} = m_{\pi_X}^{e_X}, \quad e_X = \frac{2g_X}{\deg m_{\pi_X}}$$

Concretely,  $V_\ell X$  is a free module over  $\mathbb{Q}(\pi_X) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$  (the subalgebra through which  $\pi_X$  acts on  $V_\ell X$ ), and  $\pi_X$  acts through the factor  $\mathbb{Q}(\pi_X)$ , so its characteristic polynomial on  $V_\ell X$  is the  $e_X$ -th power of  $m_{\pi_X}$ , where  $e_X$  is that rank; equating degrees gives  $e_X = 2g_X / \deg m_{\pi_X}$ . The same holds for  $Y$ :  $P_{\pi_Y} = m_{\pi_Y}^{e_Y}$  with  $e_Y = 2g_Y / \deg m_{\pi_Y}$ .

Since  $m_{\pi_X} = m_{\pi_Y}$ , the two char polys agree once the exponents do, and it remains to see  $e_X = e_Y$ . Here  $e_X$  is the *reduced degree*, or Schur index, of the division algebra  $\text{End}^0(X)$  over its centre  $\mathbb{Q}(\pi_X)$ : it is the rank of  $V_\ell X$  as a free module over  $\mathbb{Q}(\pi_X) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$ , so that  $P_{\pi_X} = m_{\pi_X}^{e_X}$ , and  $2g_X = e_X \cdot [\mathbb{Q}(\pi_X) : \mathbb{Q}]$  with  $e_X = [\text{End}^0(X) : \mathbb{Q}(\pi_X)]^{1/2}$ . That  $e_X$  depends only on the conjugacy class of  $\pi_X$  is the one point in the injectivity argument that is not a formal consequence of the isogeny theorem: it needs the computation of the endomorphism division algebra  $\text{End}^0(X)$  itself, which is Tate's endomorphism theorem (Tate, *Endomorphisms of abelian varieties over finite fields*, Invent. Math. 1966). That theorem determines the local invariants of  $\text{End}^0(X)$  purely in terms of  $\pi_X$ : at a finite place  $v$  of  $\mathbb{Q}(\pi_X)$  above  $p$  the invariant is  $\text{ord}_v(\pi_X) / \text{ord}_v(q) \cdot [\mathbb{Q}(\pi_X)_v : \mathbb{Q}_p]$ , at the real places it is  $\frac{1}{2}$  or 0 according to sign, and it vanishes at the places above  $\ell \neq p$ . Being a Schur index,  $e_X$  is the least common multiple of the denominators of these invariants, so it too is a function of the conjugacy class of  $\pi_X$  alone. Its proof rests on the local class field theory governing the structure of  $\text{End}^0(X)$  over  $\mathbb{Q}(\pi_X)$ , which lies beyond the isogeny theorem and beyond this book; the argument is given in Tate's paper and, for  $g = 1$ , in *Elliptic Curves* [4]. Granting it,  $\pi_X$  and  $\pi_Y$  conjugate forces  $e_X = e_Y$ , and therefore  $P_{\pi_X} = m_{\pi_X}^{e_X} = m_{\pi_Y}^{e_Y} = P_{\pi_Y}$ . By corollary 13.3.3,  $X$  is isogenous to  $Y$ , completing the proof.

The two assertions together say that the map of theorem 13.4.3 is a well-defined injection: it lands in conjugacy classes of Weil  $q$ -numbers by assertion (1), and it separates distinct isogeny classes by assertion (2). Assertion (1) and the reduction of assertion (2) to the identity  $P_{\pi_X} = P_{\pi_Y}$  are direct consequences of corollary 13.3.3 and the Weil bound. The one ingredient drawn from outside is that the exponent  $e_X$  depends only on  $\pi_X$ , which rests on Tate's computation of  $\text{End}^0(X)$  cited above.

What injectivity does not supply is that the map hits every conjugacy class. That is the existence problem: given a Weil  $q$ -number  $\pi$ , produce a simple abelian variety over  $\mathbb{F}_q$  whose Frobenius is conjugate to  $\pi$ .

#### **Proof Key ideas:**

The surjectivity of theorem 13.4.3 is Honda's theorem, and its full proof rests on the theory of complex multiplication and the good reduction of CM abelian varieties, which lie beyond this book; the construction is carried out in *Class Field Theory* [15]. The idea is as follows. Given a Weil  $q$ -number  $\pi$ , the field  $K = \mathbb{Q}(\pi)$  is a CM field (or a totally real field or  $\mathbb{Q}$  in the

degenerate real cases), and the local invariants prescribed by Tate's formula assemble into an admissible collection: their sum over all places of  $K$  is an integer, so by the fundamental exact sequence of class field theory they are the invariants of a division algebra  $D$  over  $K$ . One then constructs, over a number field, a CM abelian variety  $A$  with complex multiplication by a CM field containing  $K$ , whose reduction modulo a suitable prime above  $p$  is an abelian variety over a finite field  $\mathbb{F}_{q^n}$  with Frobenius a power of  $\pi$ ; Honda's descent argument produces from it a simple abelian variety over  $\mathbb{F}_q$  with Frobenius conjugate to  $\pi$  itself. The construction of CM abelian varieties and their reductions, and the class field theory that governs the reflex field and the reduction, are developed in *Class Field Theory* [15]; the CM types and the abelian varieties  $X_\Phi$  were introduced in section 6.4, and the reciprocity that pins down the field of definition is the CM main theorem deferred there.

The two halves of theorem 13.4.3 draw on different inputs. Tate's injectivity rests on the isogeny theorem and the Weil bound, together with the computation of the endomorphism algebra  $\text{End}^0(X)$  that identifies the exponent  $e_X$  with an invariant of  $\pi_X$ . Honda's existence imports class field theory and the reduction of CM abelian varieties, and connects the classification over  $\mathbb{F}_q$  to complex multiplication and, through the reflex field, to Shimura varieties. Together they make the passage  $X \mapsto \pi_X$  a translation: an arithmetic question about simple abelian varieties over  $\mathbb{F}_q$  up to isogeny, their endomorphism algebras, their dimensions, their fields of definition, translates into a question about Weil  $q$ -numbers and the division algebras their local invariants define.

The genus-one case is the historical source of the theorem and makes the dictionary explicit.

**Example 13.4.5: Isogeny Classes of Elliptic Curves over  $\mathbb{F}_q$**

For  $g = 1$  the Frobenius  $\pi_E$  of an elliptic curve  $E/\mathbb{F}_q$  is a root of  $P_{\pi_E}(t) = t^2 - at + q$ , where  $a = q + 1 - \#E(\mathbb{F}_q)$  is the trace of Frobenius (proposition 13.1.6). The Weil bound (theorem 13.2.1) is here the Hasse bound  $|a| \leq 2q^{1/2}$ , and  $E$  is simple, being one-dimensional, so theorem 13.4.3 says the isogeny classes of elliptic curves over  $\mathbb{F}_q$  are the conjugacy classes of Weil  $q$ -numbers  $\pi$  with  $[\mathbb{Q}(\pi) : \mathbb{Q}] \leq 2$ . These are indexed by the trace  $a \in \mathbb{Z}$ , and Honda-Tate specializes to the Deuring-Waterhouse classification of which values of  $a$  occur.

Write  $p$  for the characteristic and  $q = p^n$ . The classification says that an integer  $a$  with  $|a| \leq 2q^{1/2}$  is the trace of Frobenius of an elliptic curve over  $\mathbb{F}_q$  exactly in the following cases:

1.  $\gcd(a, p) = 1$  (the *ordinary* case): every such  $a$  with  $|a| \leq 2q^{1/2}$  occurs, and the curve is ordinary;
2.  $p|a$  (the *supersingular* case), where the possibilities depend on whether the exponent  $n$  is even or odd; in particular the trace  $a = 0$  occurs for every odd  $n$ , and for even  $n$  exactly when  $p \not\equiv 1 \pmod{4}$ .

The ordinary values  $a$  with  $p \nmid a$  give a Weil  $q$ -number generating an imaginary quadratic field (as in example 13.4.2), and the reduction of a CM elliptic curve realizes the isogeny class. The supersingular values, where  $\pi\bar{\pi} = q$  forces  $\pi/\sqrt{q}$  to be a root of unity, are the constrained cases: whenever  $a = 0$  is realized one has  $\pi = \pm\sqrt{-q}$  and the resulting curve is supersingular, whereas the boundary traces  $a = \pm 2q^{1/2}$  occur only when  $q$  is a perfect square (so that  $2q^{1/2} \in \mathbb{Z}$ ). The Deuring-Waterhouse tables enumerating the supersingular traces, and the count of isogeny classes, are developed for  $g = 1$  in *Elliptic Curves* [4]; they are the one-dimensional shadow of the local-invariant bookkeeping that drives the existence half of theorem 13.4.3.

**Exercise 13.4.1**

1. For  $q = 2$ , decide which of the following algebraic numbers are Weil 2-numbers in the sense of definition 13.4.1, computing the minimal polynomial and the absolute values of the conjugates in each case:
  1.  $\pi = 2$ ;
  2.  $\pi = 1 + i$ ;
  3.  $\pi = \sqrt{2}$ ;
  4.  $\pi = 1 + \sqrt{2}$ .
2. Following example 13.4.2, classify the conjugacy classes of imaginary quadratic Weil 7-numbers. List the integers  $a$  that occur, identify the unique supersingular value, and explain why the boundary values are absent. Which of these classes arise as traces of Frobenius of elliptic curves over  $\mathbb{F}_7$ ?
3. Let  $X$  be a simple abelian variety over  $\mathbb{F}_q$  whose Frobenius  $\pi_X$  has minimal polynomial over  $\mathbb{Q}$  of degree  $d$ , and let  $g = \dim X$ . Using the identity  $P_{\pi_X} = m_{\pi_X}^{e_X}$  from proposition 13.4.4, express the exponent  $e_X$  and the dimension  $[\text{End}^0(X) : \mathbb{Q}]$  in terms of  $d$  and  $g$ . Apply this to the two cases: (i)  $g = 1$  with  $m_{\pi_X}$  of degree 2; (ii)  $g = 2$  with  $m_{\pi_X}$  of degree 2. In case (ii), what kind of algebra is  $\text{End}^0(X)$ ?
4. Let  $\pi$  be a Weil  $q$ -number of degree  $d = [\mathbb{Q}(\pi) : \mathbb{Q}]$ , with minimal polynomial  $m_\pi(t) = t^d + c_{d-1}t^{d-1} + \cdots + c_0$  over  $\mathbb{Q}$ .
  1. Show that  $q/\pi$  lies in  $\mathbb{Q}(\pi)$  and is again a Weil  $q$ -number.
  2. Show that  $|c_0| = q^{d/2}$ , and deduce that if  $q$  is not a perfect square then  $d$  is even.
5. Let  $E/\mathbb{F}_q$  be an ordinary elliptic curve, so its trace  $a = q + 1 - \#E(\mathbb{F}_q)$  satisfies  $\gcd(a, p) = 1$ , where  $p$  is the characteristic. Using example 13.4.5, prove that  $\mathbb{Q}(\pi_E)$  is an imaginary quadratic field, that the exponent  $e_E$  of proposition 13.4.4 equals 1, and hence that  $\text{End}^0(E) = \mathbb{Q}(\pi_E)$ .

**13.5 Ordinary and Supersingular Abelian Varieties**

The characteristic- $p$  dichotomy of chapter 9 separated abelian varieties by their  $p$ -rank: ordinary when  $X[p](\bar{\mathbb{F}}_q)$  is as large as possible, supersingular when it is as small as possible. That definition lived inside the finite group scheme  $X[p]$ , invisible to the  $\ell$ -adic Tate module for  $\ell \neq p$ . Over a finite field there is a second reading of the same dichotomy, and it sits on the surface of the Frobenius. The characteristic polynomial  $P_\pi$  (definition 13.1.3) is an integer polynomial whose roots carry an archimedean size, fixed by the Weil bound  $|\alpha| = q^{1/2}$  of theorem 13.2.1; those same roots also carry a  $p$ -adic size, and it is the  $p$ -adic sizes that record whether  $X$  is ordinary or supersingular. This section reads the dichotomy off the Newton polygon of  $P_\pi$  and closes the arithmetic thread of the chapter.

**Definition 13.5.1: Ordinary and Supersingular over a Finite Field**

Let  $X$  be an abelian variety of dimension  $g$  over  $\mathbb{F}_q$ ,  $q = p^a$ , with Frobenius characteristic polynomial  $P_\pi \in \mathbb{Z}[t]$  of degree  $2g$  (definition 13.1.3). Fix an embedding of  $\mathbb{Q}$  into an algebraic closure of  $\mathbb{Q}_p$  and let  $v$  be the associated valuation, normalized so that  $v(q) = 1$ . Then  $X$  is:

- *ordinary* if exactly  $g$  of the  $2g$  roots of  $P_\pi$  (counted with multiplicity) are  $p$ -adic units, that is, satisfy  $v(\alpha) = 0$ ;
- *supersingular* if every root of  $P_\pi$  satisfies  $v(\alpha) = \frac{1}{2}$ .

The two clauses are the endpoints of the range of possible  $p$ -adic behavior, and each connects to the group scheme definition inherited from chapter 9. A root that is a  $p$ -adic unit contributes to the  $p$ -divisible étale part of  $X$ , and the count of unit roots equals the  $p$ -rank; so exactly  $g$  unit roots is precisely the maximal  $p$ -rank  $g$  of definition 9.4.6, and the definitions of ordinary agree. At the other end, all roots having slope  $\frac{1}{2}$  leaves no unit root at all, hence  $p$ -rank 0; so a supersingular  $X$  in the sense of definition 13.5.1 has  $p$ -rank 0, the supersingular condition of definition 9.4.6. This implication is one-directional in general: for  $g \geq 3$  there are abelian varieties of  $p$ -rank 0 that are not supersingular, for instance a threefold with Newton slopes  $\{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{2}{3}, \frac{2}{3}, \frac{2}{3}\}$ , which has no unit root yet no slope equal to  $\frac{1}{2}$ . The two notions of supersingularity coincide exactly when  $g \leq 2$ ; for  $g \geq 3$  the supersingular locus (all slopes  $\frac{1}{2}$ ) sits strictly inside the  $p$ -rank-0 locus. The valuation  $v$  depends on the chosen embedding, but the multiset of values  $\{v(\alpha)\}$  does not:  $P_\pi \in \mathbb{Z}[t]$  is a fixed integer polynomial and  $\mathbb{Q}$  has a single place above  $p$ , so any two embeddings  $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$  differ by an automorphism of  $\mathbb{Q}_p/\mathbb{Q}_p$ , which preserves the valuation and permutes the roots.

That last relation already constrains the slopes severely, and organizing it through the Newton polygon turns the constraint into a complete description.

**Proposition 13.5.2: The Newton Polygon of Frobenius**

Let  $X/\mathbb{F}_q$  have dimension  $g$  and Frobenius characteristic polynomial

$$P_\pi(t) = t^{2g} + c_{2g-1}t^{2g-1} + \cdots + c_1t + c_0, \quad c_i \in \mathbb{Z}$$

with  $v$  normalized so that  $v(q) = 1$ . The Newton polygon of  $P_\pi$  with respect to  $v$  is the lower convex hull of the points  $(i, v(c_{2g-i}))$  for  $0 \leq i \leq 2g$  (with  $c_{2g} = 1$ , so the polygon runs from  $(0, 0)$  to  $(2g, g)$ ). Then:

1. the slopes of the polygon are exactly the values  $v(\alpha)$  of the roots  $\alpha$  of  $P_\pi$ , each slope occurring on a horizontal segment whose length is the number of roots with that valuation;
2. every slope lies in the interval  $[0, 1]$ , and the slopes are symmetric about  $\frac{1}{2}$ : if  $s$  is a slope of multiplicity  $m$ , then  $1 - s$  is a slope of multiplicity  $m$ ;
3.  $X$  is ordinary if and only if the slopes are 0 with multiplicity  $g$  and 1 with multiplicity  $g$ ;  $X$  is supersingular if and only if every slope equals  $\frac{1}{2}$ .

**Proof:**

1. The identification of the slopes of the Newton polygon of a polynomial over a valued field with the valuations of its roots is the standard Newton polygon theorem, taken as input from the valuation theory of  $\mathbb{Q}_p$  developed in *Core Algebra* [12]. In the form used here the polygon is built from the reversed coefficient sequence: if the roots are  $\alpha_1, \dots, \alpha_{2g}$  then  $c_{2g-j} = (-1)^j e_j(\alpha_1, \dots, \alpha_{2g})$  is the  $j$ th elementary symmetric function, and the lower convex hull of  $(i, v(c_{2g-i}))$  has, on the segment from abscissa  $i$  to  $i+m$ , slope equal to the common valuation of the  $m$  roots accounted for there. Reading off the segments therefore recovers each root valuation  $v(\alpha)$  as a slope, with horizontal length the number of roots of that valuation, giving statement (1).
2. By theorem 13.2.1 the roots come in pairs with  $\alpha\bar{\alpha} = q$ , where  $\bar{\alpha}$  is the complex conjugate; more precisely  $P_\pi$  satisfies the functional equation  $t^{2g}P_\pi(q/t) = q^gP_\pi(t)$ , so  $\alpha \mapsto q/\alpha$  permutes the roots. Applying  $v$  to  $\alpha \cdot (q/\alpha) = q$  gives

$$v(\alpha) + v(q/\alpha) = v(q) = 1$$

so the map  $\alpha \mapsto q/\alpha$  sends a root of valuation  $s$  to a root of valuation  $1-s$ , establishing the symmetry of the multiset of slopes about  $\frac{1}{2}$ . Each root  $\alpha$  is an algebraic integer, since  $P_\pi$  is monic with integer coefficients, so  $v(\alpha) \geq 0$ ; and  $q/\alpha$  is likewise a root, hence an algebraic integer, so  $v(q/\alpha) = 1 - v(\alpha) \geq 0$ . Thus  $0 \leq v(\alpha) \leq 1$  for every root, proving statement (2).

3. Suppose  $X$  is ordinary, so exactly  $g$  roots have  $v(\alpha) = 0$  by definition 13.5.1. Their images under  $\alpha \mapsto q/\alpha$  are  $g$  roots of valuation 1, and these account for all  $2g$  roots because a unit root and its partner exhaust distinct roots (a root of valuation 0 cannot also have valuation 1). Hence the slopes are 0 and 1, each with multiplicity  $g$ ; conversely, that slope profile forces exactly  $g$  unit roots, which is ordinarity. Now suppose every slope equals  $\frac{1}{2}$ : then by definition 13.5.1  $X$  is supersingular, and conversely if  $X$  is supersingular every root has  $v(\alpha) = \frac{1}{2}$ , so the single slope  $\frac{1}{2}$  occurs with multiplicity  $2g$ . This proves statement (3), completing the proof.

The proposition converts a definition in terms of  $p$ -adic root sizes into a picture that can be drawn from the integer coefficients of  $P_\pi$  alone. The Newton polygon of an ordinary abelian variety is the broadest possible: a segment of slope 0 of horizontal length  $g$ , followed by a segment of slope 1 of length  $g$ , a polygon that dips as low as the symmetry allows. The supersingular polygon is the tautest: a single straight segment of slope  $\frac{1}{2}$  spanning the whole width  $2g$ , the unique polygon compatible with the symmetry that has no vertices in its interior. Between these two lie the abelian varieties of intermediate  $p$ -rank, whose polygons have several slopes strictly between 0 and 1; the isogeny invariant recorded by the polygon is the reason the dichotomy is stable under isogeny, since isogenous abelian varieties share the same  $P_\pi$  (corollary 13.3.3) and hence the same Newton polygon.

The elliptic curve makes both polygons concrete and reconciles this reading with the one from *Elliptic Curves* [4].



**Example 13.5.3: Ordinary and Supersingular Elliptic Curves**

Let  $E/\mathbb{F}_q$  be an elliptic curve,  $q = p^n$ , with  $\#E(\mathbb{F}_q) = q + 1 - a$  where  $a = \text{tr}(\pi) \in \mathbb{Z}$  is the trace of Frobenius. Here  $g = 1$  and

$$P_\pi(t) = t^2 - at + q$$

by definition 13.1.3 and proposition 13.1.6, with the two roots  $\alpha, \bar{\alpha}$  satisfying  $\alpha + \bar{\alpha} = a$  and  $\alpha\bar{\alpha} = q$ . The valuations satisfy  $v(\alpha) + v(\bar{\alpha}) = 1$ , so the only possibilities compatible with  $0 \leq v \leq \frac{1}{2} \leq 1 - v$  are  $\{0, 1\}$  and  $\{\frac{1}{2}, \frac{1}{2}\}$ .

The two cases are read off  $a$  directly. If  $p \nmid a$ , then  $v(a) = 0$ , and from  $\alpha + \bar{\alpha} = a$  one of the roots is a unit while the other has valuation 1: the slopes are 0 and 1, so  $E$  is ordinary. If  $p \mid a$ , then  $v(a) > 0$ , and since  $\alpha + \bar{\alpha} = a$  cannot have a summand of valuation 0 balanced against one of valuation 1 (their sum would then be a unit, contradicting  $v(a) > 0$ ), both roots must have valuation  $\frac{1}{2}$ : the slopes are  $\frac{1}{2}, \frac{1}{2}$ , so  $E$  is supersingular. Equivalently,  $E$  is ordinary if and only if  $a = \text{tr}(\pi)$  is prime to  $p$ , and supersingular if and only if  $p \mid a$ . When  $q = p$  with  $p \geq 5$  the Hasse bound  $|a| \leq 2\sqrt{p}$  (corollary 13.2.3) forces  $p \mid a$  to mean  $a = 0$ , since  $\sqrt{p} > 2$  leaves no nonzero multiple of  $p$  inside the bound, so a supersingular curve over such an  $\mathbb{F}_p$  has  $\#E(\mathbb{F}_p) = p + 1$ ; for  $p \in \{2, 3\}$  one has  $p \leq 2\sqrt{p}$  and supersingular curves with  $a \neq 0$  occur. This is exactly the criterion developed for  $g = 1$  in *Elliptic Curves* [4].

The elliptic case exhibits both the reading of ordinarity as “ $a$  prime to  $p$ ” and the reading of supersingularity as the single slope  $\frac{1}{2}$ , and it supplies the building block for the general supersingular abelian variety. A supersingular elliptic curve has Newton polygon the segment of slope  $\frac{1}{2}$  of width 2; a product of  $g$  of them has, by additivity of the characteristic polynomial under products, the segment of slope  $\frac{1}{2}$  of width  $2g$ . The converse also holds: every supersingular abelian variety arises this way up to isogeny.

**Theorem 13.5.4: Supersingular Abelian Varieties are Isogenous to Products**

Let  $X$  be an abelian variety of dimension  $g$  over  $\bar{\mathbb{F}}_q$ . Then  $X$  is supersingular if and only if  $X$  is isogenous over  $\bar{\mathbb{F}}_q$  to a product  $E_1 \times \cdots \times E_g$  of supersingular elliptic curves.

**Proof Key ideas:**

The reverse implication is the additivity used above: a product of supersingular elliptic curves has Frobenius characteristic polynomial the product of the factors’ polynomials, so its Newton polygon is the concatenation of  $g$  slope- $\frac{1}{2}$  segments, again a single slope- $\frac{1}{2}$  segment of width  $2g$ ; by proposition 13.5.2 the product is supersingular, and by corollary 13.3.3 supersingularity is invariant under the isogeny.

The forward implication is the substantive half. Supersingularity says every Frobenius slope is  $\frac{1}{2}$ , equivalently that the  $p$ -divisible group  $X[p^\infty]$  is isoclinic of slope  $\frac{1}{2}$ . The classification of  $p$ -divisible groups over  $\bar{\mathbb{F}}_q$  up to isogeny by their Newton slopes, together with the fact that a supersingular elliptic curve’s  $p$ -divisible group is the unique isoclinic- $\frac{1}{2}$  group of height 2, forces  $X[p^\infty]$  to be isogenous to the  $g$ -fold self-product of that group; a Tate-module and  $p$ -divisible-group argument then lifts this to an isogeny of the abelian varieties themselves. This rests on the Dieudonné theory of  $p$ -divisible groups, which is beyond the scope of this book; the



full argument is developed in *Algebraic Geometry* [13], and the statement is taken as established, completing the discussion.

The supersingular locus is thus a single isogeny class of  $p$ -divisible group, realized by products of one fixed curve; the ordinary case sits at the other end, its Frobenius eigenvalues ranging over the slopes 0 and 1. Between the two the Newton polygon stratifies the isogeny classes over  $\mathbb{F}_q$  into finitely many types, one for each admissible slope profile. With the Newton polygon, the Frobenius characteristic polynomial (definition 13.1.3) has now been made to carry the full arithmetic content of  $X/\mathbb{F}_q$ : the point counts of section 13.1, the archimedean sizes of theorem 13.2.1, the isogeny classification of section 13.3 and section 13.4, and now the  $p$ -adic dichotomy of definition 9.4.6.

### Exercise 13.5.1

1. Let  $E/\mathbb{F}_q$  be an elliptic curve with  $q = p^2$  and trace of Frobenius  $a = \text{tr}(\pi) = p$ . Write down  $P_\pi$ , compute the valuations of its two roots with  $v$  normalized so that  $v(q) = 1$ , and decide whether  $E$  is ordinary or supersingular using definition 13.5.1 and example 13.5.3.
2. Let  $X/\mathbb{F}_q$  be an abelian surface ( $g = 2$ ) whose Frobenius characteristic polynomial is  $P_\pi(t) = t^4 + pt^3 + p^3t + p^4$ , with  $q = p^2$ . Determine the slopes of the Newton polygon of  $P_\pi$  (proposition 13.5.2) and classify  $X$  as ordinary, supersingular, or neither.
3. Consider the abelian threefold whose Newton slopes are  $\{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{2}{3}, \frac{2}{3}, \frac{2}{3}\}$ , cited in the discussion after definition 13.5.1. Verify that this multiset is symmetric about  $\frac{1}{2}$  and lies in  $[0, 1]$  as required by proposition 13.5.2(2), compute the  $p$ -rank (the number of unit roots), and explain why  $X$  is not supersingular yet has  $p$ -rank 0.
4. Prove that the two notions of supersingularity, all Frobenius slopes equal to  $\frac{1}{2}$  and  $p$ -rank 0, coincide for every abelian variety with  $g \leq 2$ . That is, show that if  $g \leq 2$  and  $X/\mathbb{F}_q$  has  $p$ -rank 0 (no root of  $P_\pi$  is a  $p$ -adic unit), then every root has valuation  $\frac{1}{2}$ .
5. Let  $X$  and  $Y$  be abelian varieties over  $\mathbb{F}_q$  that are isogenous over  $\mathbb{F}_q$ . Using corollary 13.3.3 and proposition 13.5.2, prove that  $X$  is ordinary if and only if  $Y$  is ordinary, and  $X$  is supersingular if and only if  $Y$  is supersingular. Then deduce from theorem 13.5.4 that over  $\mathbb{F}_q$  every simple supersingular abelian variety of dimension  $g \geq 2$  is isogenous to a power of a single supersingular elliptic curve, hence is not simple for  $g \geq 2$ .



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## *What Next?*

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An abelian variety has been made to wear its three faces at once: the complex torus of Part I, the projective group scheme of Part II, and, over  $\mathbb{C}$ , the polarized Hodge structure that chapter 11 showed the first two share. From that single object the book has reached the Rosati involution and Albert's classification, the Jacobian of a curve, and the Frobenius eigenvalues that govern the arithmetic over a finite field. Each of these is the doorway to a subject of its own, and this closing chapter maps the roads that leave from here.

It proves nothing. For each direction it names the next subject, the companion EYNTKA volume that develops it, and one or two standard external references. The organizing observation is that almost every construction in the book has a moduli, an arithmetic, or a categorical continuation that the book stops just short of, and these continuations are where abelian varieties become a tool rather than an object.

A single principally polarized abelian variety of dimension  $g$  is a point of a space, the Siegel moduli space  $\mathcal{A}_g$ , whose analytic model is the quotient  $\mathfrak{H}_g/\mathrm{Sp}_{2g}(\mathbb{Z})$  of the Siegel upper half-space met in chapter 3. Its points are the objects of this book, its functions are Siegel modular forms, and its geometry, as a variety over  $\mathbb{Q}$  and as an algebraic stack, is the subject of *Moduli, Deformation Theory, and Stacks* [10]. The Torelli map of chapter 12 embeds the moduli of curves inside  $\mathcal{A}_g$ , and the question of which principally polarized abelian varieties are Jacobians is the Schottky problem. The standard references are Faltings and Chai, *Degeneration of Abelian Varieties*, and van der Geer's survey of Siegel modular forms.

The complex multiplication of chapter 6 is the first case of a general phenomenon: the moduli spaces  $\mathcal{A}_g$ , and their variants carrying level structure and extra endomorphisms, are *Shimura varieties*, and their CM points are governed by class field theory [15]. These varieties are the geometric arena of the Langlands program, where the Galois representation on the Tate module (chapter 9) is matched with an automorphic form; the theory is developed in *Shimura Varieties* [11], to which the CM theory of this book is the natural prelude. Milne's notes on Shimura varieties and Deligne's Bourbaki reports are the entry points.

Over a number field an abelian variety can be reduced modulo a prime, and the correct integral model, the Néron model, converts the arithmetic of chapter 13 into a local invariant at each place. For  $g = 1$  the theory is carried out in *Elliptic Curves* [4], and the local machinery it rests on is the province of *Local Field Theory* [9]. The good-reduction case feeds the Tate module into a global Galois representation; the bad-reduction case is where the finite group schemes of section 9.4 return. The reference is Bosch, Lütkebohmert, and Raynaud, *Néron Models*.

The rational points of an abelian variety over a number field form a finitely generated group, the Mordell-Weil theorem, and measuring them requires the theory of heights, a quadratic form refining the degree function of chapter 8. Heights are the engine of the deepest finiteness theorems in the subject, including Faltings's theorem that a curve of genus at least two has finitely many rational points, proved through the moduli space of chapter 12. Hindry and Silverman, *Diophantine Geometry*, and Bombieri and Gubler, *Heights in Diophantine Geometry*, are the standard treatments.

The characteristic- $p$  torsion that the Tate module could not see (section 9.4) is organized by the  $p$ -divisible group  $X[p^\infty]$ , and the ordinary-supersingular dichotomy of chapter 13 is the coarsest cut of its finer invariant, the Newton polygon. Dieudonné theory translates  $p$ -divisible groups into semilinear algebra over the Witt vectors, making the classification of  $X[p^\infty]$  up to isogeny a matter of slopes; the crystalline cohomology that globalizes it borders *Étale Cohomology* [5]. Demazure's *Lectures on  $p$ -Divisible Groups* and the survey of Manin are the classical sources.

The Poincaré bundle of chapter 10, read as a kernel, defines an equivalence between the derived category of coherent sheaves on  $X$  and that on its dual  $\hat{X}$ , the Fourier-Mukai transform. It turns the biduality  $\hat{\hat{X}} = X$  into a genuine symmetry of homological invariants and reinterprets the Weil pairing and the isogenies of chapter 10 as operations on complexes. Mukai's original paper and Polishchuk's *Abelian Varieties, Theta Functions and the Fourier Transform* develop it in full, with Huybrechts's *Fourier-Mukai Transforms in Algebraic Geometry* the standard textbook.

Each road above begins at a construction the reader now owns, and each leaves the category of abelian varieties for a wider one, of moduli, of automorphic forms, of derived categories, in which an abelian variety is a single well-understood point. That is the sense in which this book is foundational: it is not the end of a subject but the equipment for entering several.

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